

# A Reflexive Root-Polytope Model for Preorder $h$ -Polynomials

Simplicial-polytopal realization, palindromicity, and unimodality

With Part II (added 28 September 2026): height-two preorders — matching supports, nonreal-root families, and crown obstructions

With Part III (added 29 September 2026): cactus rigidity for matching-support determinants

With Part IV (added 29 September 2026): gamma-positivity for every finite preorder

With Parts V–VII (added 29 September 2026): ultra-log-concave support polynomials; stability of signed matching supports; the transversal matroid of a preorder

With Part VIII (added 30 September 2026): marked-tree stability — arbitrary apex neighbourhoods, enumeration and repair

With Part IX (added 30 September 2026): two-flow bijections — full-coordinate sampling of demand vectors and preorder lattice points

With Part X (added 30 September 2026): rare stable markings on random trees — sparse transitions and annealed–quenched separation

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## Abstract

For a finite preorder  $\tau$  on  $[n]$ , let  $h_\tau(t)$  count the integer points of its preorder polytope by the number of nonzero coordinates. We give a proposed general proof that this polynomial is the  $h$ -polynomial of an  $n$ -dimensional simplicial polytope, establishing Conjecture 5.1(c), and hence (a) and (b), of Athanasiadis and Chapoton. The principal construction is the explicit  $n$ -dimensional lattice polytope

$$C_\tau = \text{conv}(\{\pm \mathbf{e}_i : 1 \leq i \leq n\} \cup \{\mathbf{e}_i - \mathbf{e}_j : j \leq_\tau i, i \neq j\}).$$

We prove  $h^*(C_\tau, t) = h_\tau(t)$  by an integer-fiber decomposition of an augmented bipartite root polytope. Every boundary pulling triangulation of  $C_\tau$ , coned from the origin, is unimodular; its boundary is combinatorially polytopal. We also prove reflexivity, an exact formula for the gauge of  $C_\tau$ , and a Gorenstein translation formula for the larger root polytope. The higher-dimensional root-polytope identity and preorder duality are already results of Dai, Hou, Liu, Thawinrak, and Wang, and are not claimed as new here. A separate hypertree argument reconstructs that known identity. Exact verification code and complete small-case data accompany the manuscript.

*Addition of 28 September 2026 (batch 39).* Part II, from a later manuscript, proves that for a poset of height at most two with bipartite comparability graph  $G$  the support polynomial is  $(1+t)^n p_G(t/(1+t)^2)$ , where  $p_G$  counts perfectly matchable vertex sets; this gives  $\gamma$ -positivity (Conjecture 5.2) in height two. It records that real-rootedness (Conjecture 5.3) fails, crediting the first counterexample, an eight-element poset, to a posting by Shivam Patel, and proves an exact real-root count for an infinite family extending it. It shows that an induced crown on  $2r \geq 6$  elements forces a minimal nonface of size  $r$  in every lattice triangulation of  $C_\tau$ ; Conjecture 5.1(d) itself remains open.

*Addition of 29 September 2026 (batch 42).* Part III, from a further manuscript, proves that a bipartite graph carries a complex matrix on its edges whose square minors have squared modulus exactly the perfect-matchability indicator if and only if it is a cactus, and that this is strictly stronger than real stability of the signed matching-support polynomial; noncacti miss the exact model by the sharp universal defect  $(\sqrt{17} - 3)/2$ .

*Addition of 29 September 2026 (batch 44).* Part IV, from two further manuscripts, proves Conjecture 5.2 for every finite preorder: the  $k$ th  $\gamma$ -coefficient of  $h_\tau$  counts ordered pairs of disjoint  $k$ -element sets joined by a  $\tau$ -increasing bijection. It also shows that for a reflexive relation the doubled matching-support polynomial is palindromic exactly when the relation is transitive.

*Addition of 29 September 2026 (batch 55).* Part V, from two further manuscripts, proves that every bipartite demand-support enumerator, divided by binomial coefficients of the smaller shore, is log-concave (Röhrle and Ulirsch's bimatroid inequality transferred through the support-counting bridge), so every  $h_\tau$  and every height-two  $\gamma$ -vector is ultra-log-concave; it classifies the palindromic support polynomials of all bipartite graphs, gives approximation schemes for partition functions and single coefficients, and shows exact counting #P-complete already in height two. Part VI shows that signed matching-support stability is block-local, holds for a tree's incidence–apex graph exactly for the finite and affine ADE trees, and is classified for books and multi-theta graphs; it is strictly stronger than real-rootedness. Part VII shows that the transversal matroid of a preorder determines the preorder up to isomorphism and that height-two preorder matroids generate the gammoids under minors.

*Addition of 30 September 2026 (batch 63).* Part VIII, from a further manuscript, removes the terminal convention of Part VI's tree criterion for one apex: for a tree and an arbitrary set of apex neighbours, the signed support polynomial is stable exactly when every branching vertex of the spanned subtree is marked and the suppressed skeleton is a finite or affine ADE tree. This follows from Part VI's theorems; the manuscript adds a direct proof, linear-time recognition, a sharp diameter window, exact polynomial-time counting, optimization and repair of stable neighbourhoods, and shows that their size-counting polynomial is real-rooted only for paths.

*Addition of 30 September 2026 (batch 64).* Part IX, from a further manuscript, gives explicit polynomial-time inverse bijections between the demand vectors of a bipartite graph with a prescribed support and the supplier sets matchable to that support: two minimum-cost transportation problems with integer margins share one optimal spanning tree, whose supplier degrees and supplier leaves encode the two sides. The existence of such a correspondence is due to Oh. Composed with the matroid basis walk, the bijection samples actual lattice points of every finite preorder polytope almost uniformly, not only their supports, which answers Part V's sampling boundary; its disjoint union is an elementary proof, without Ehrhart theory, of the support-counting identity of Parts II, IV and V.

*Addition of 30 September 2026 (batch 66).* Part X, from a further manuscript, answers Part VIII's question on random trees for the uniform labelled tree with independent marks of probability  $p$ : from Part VIII's classification it derives exact counts, generating functions and a Poisson identity, shows that the probability of a stable marking is asymptotic to  $\frac{9}{8}\sqrt{2\pi}p^3n^{5/2}e^{-pn}$  whenever  $p\sqrt{n} \rightarrow \infty$ , describes the window  $p\sqrt{n} \rightarrow \lambda$  by an explicit entire function and the sparse regime by at most two marks, and proves conditional Gaussian laws and a linear conditioned diameter at fixed  $p$ . For every fixed tree the probability is determined by the diameter up to a polynomial factor, which separates the typical-tree exponential rate  $\log(1 - p)$  from the averaged rate  $-p$ .

**Scope and status.** The intended new result is the general simplicial-polytopal realization, not the already established duality theorem. The proof uses explicitly identified results on support-enumerators and pulling triangulations. All steps specific to the construction are proved below; the external results are identified explicitly. The manuscript has not been independently refereed or checked by a proof assistant. The finite computations test the constructions; they do not replace the proof. Flag realizability,  $\gamma$ -positivity, and real-rootedness are not proved. [Added 28 September 2026, batch 39: this sentence describes Part I. Part II proves  $\gamma$ -positivity for posets of height at most two, records counterexamples to real-rootedness (the first credited to prior work), and obstructs flag lattice triangulations of  $C_\tau$ ; see Section 11.3.] [Added 29 September 2026, batch 44: Part IV proves  $\gamma$ -positivity for every finite preorder; see Section 46.3.] [Added 29 September 2026, batch 55: Part V proves ultra-log-concavity of every  $h_\tau$  and classifies palindromic bipartite support polynomials; Part VI proves that stability of the signed support polynomial is strictly stronger than real-rootedness; see Sections 66.3, 83.3 and 100.3.] [Added 30 September 2026, batch 63: Part VIII classifies stability for every apex neighbourhood on a tree; see Section 116.3.] [Added 30 September 2026, batch 64: Part IX constructs the demand/support bijection and a full-coordinate lattice-point sampler; see Section 132.3.] [Added 30 September 2026, batch 66: Part X determines the probability that an independently marked random labelled tree is stable; see Section 152.3.]

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**Reading route.** Sections 1–5 contain the main argument. Section 6 gives a second geometric check. Sections 8–10 provide examples, exact experiments, and limitations. Appendix A gives an alternative proof of the known higher-dimensional identity; Appendix B records the proof dependencies and the main normalization checks. Part II (Sections 11–25), added on 28 September 2026, treats height-two preorders; it precedes the appendices of Part I. Part III (Sections 26–45), added on 29 September 2026, answers Part II’s question on exact cactus-style matrices; it follows Part II. Part IV (Sections 46–65), added on 29 September 2026, proves  $\gamma$ -positivity for every finite preorder; it follows Part III. Parts V (Sections 66–82), VI (Sections 83–99) and VII (Sections 100–115), added on 29 September 2026, treat ultra-log-concavity and counting, stability, and the transversal matroid of a preorder; they follow Part IV. Part VIII (Sections 116–131), added on 30 September 2026, classifies marked-tree stability; it follows Part VII and precedes the appendices of Part I. Part IX (Sections 132–151), added on 30 September 2026, constructs demand/support bijections and a full-coordinate sampler; it follows Part VIII and precedes the appendices of Part I. Part X (Sections 152–171), added on 30 September 2026, treats stable markings of random trees; it follows Part IX and precedes the appendices of Part I.

# 1 The problem and the result

A *preorder* is a reflexive, transitive binary relation; antisymmetry is not required. For a preorder  $\tau$  on  $E = [n]$ , write  $\mathcal{J}(\tau)$  for its order ideals, including  $\emptyset$  and  $E$ . For a vector  $x$  and a subset  $A \subseteq E$ , write  $x(A) = \sum_{i \in A} x_i$ . The preorder polytope and its support polynomial are

$$\mathcal{Q}_\tau = \{x \in \mathbb{R}_{\geq 0}^n : x(I) \leq |I| \text{ for every } I \in \mathcal{J}(\tau)\}, \quad (1.1)$$

$$P_\tau = \mathcal{Q}_\tau \cap \mathbb{Z}^n, \quad h_\tau(t) = \sum_{x \in P_\tau} t^{|\text{supp}(x)|} = \sum_{k=0}^n h_k(\tau) t^k. \quad (1.1)$$

Here  $\text{supp}(x) = \{i : x_i \neq 0\}$ . These are the objects of Athanasiadis and Chapoton [1]. Their Conjecture 5.1(c) asks whether  $(h_0(\tau), \dots, h_n(\tau))$  is the  $h$ -vector of an  $n$ -dimensional simplicial polytope. Parts (a) and (b) ask, respectively, for palindromicity and palindromic unimodality. The stronger part (d) requires a flag simplicial polytope.

## 1.1 What was already known when this problem was selected

The arXiv version of [1] is dated 26 May 2026. A subsequent paper by Dai, Hou, Liu, Thawinrak, and Wang [2], in its 27 August 2026 revision, proves that the support-enumerator of a bipartite graph is the Ehrhart numerator of its augmented root polytope. Its Theorem 1.3 explicitly settles Conjecture 5.4 of [1], the identity  $h_\tau = h_{\tau^*}$  under reversal of the preorder. These facts must therefore be treated as prior results, not as open problems. Problem 5.3 of [2] still discusses the coefficient-shape conjectures for preorders. [Added 29 September 2026, batch 55: its unimodality and palindromicity parts are answered for all bipartite graphs in Part V (Corollary 70.2, Theorem 67.4).] Menon's August 2026 paper [6] gives further special cases of  $\gamma$ -positivity for arbors.

The target of the present manuscript is the *general* assertion in Conjecture 5.1(c). The advance proposed here is a dimension-reducing lattice fiber decomposition and the resulting explicit root-polytope realization. The literature check establishes these version-specific statements; it does not certify absolute priority against every unpublished or unindexed argument.

**Theorem 1.1** (Main theorem). *Let  $\tau$  be any preorder on  $[n]$ , where  $n \geq 1$ , and put*

$$C_\tau = \text{conv}(\{\pm \mathbf{e}_i : i \in [n]\} \cup \{\mathbf{e}_i - \mathbf{e}_j : j \leq_\tau i, i \neq j\}) \subset \mathbb{R}^n. \quad (1.2)$$

*Then  $C_\tau$  is reflexive and*

$$h^*(C_\tau, t) = h_\tau(t). \quad (1.3)$$

*Moreover, any pulling triangulation of  $\partial C_\tau$  is combinatorially the boundary of an  $n$ -dimensional simplicial polytope  $S_\tau$  satisfying*

$$h(S_\tau, t) = h_\tau(t).$$

*Consequently  $h_k(\tau) = h_{n-k}(\tau)$  for all  $k$ , and*

$$1 = h_0(\tau) \leq h_1(\tau) \leq \dots \leq h_{\lfloor n/2 \rfloor}(\tau).$$

*In fact the full  $g$ -theorem conditions hold for  $h(\tau)$ .*

A pulling triangulation is obtained by fixing an order on the vertices and recursively triangulating faces by cones from their first vertex. The actual root polytope  $C_\tau$  need not be simplicial. The theorem asserts that its triangulated boundary has a simplicial convex realization with the desired  $h$ -vector. These two assertions should not be conflated.

The empty preorder is harmless: its support polynomial is 1, and both root models degenerate to a point. We treat  $n \geq 1$  below to avoid unnecessary zero-dimensional boundary conventions.

## 1.2 A few useful sanity checks

Since  $E$  is an ideal, every  $x \in P_\tau$  has  $x(E) \leq n$ . Hence  $P_\tau$  is finite,  $h_0 = 1$ , and  $h_n = 1$ : the only point with all coordinates positive is  $\mathbf{1}$ . For a point supported just at  $i$ , the largest allowed value of its coordinate is  $|\downarrow i|$ , where  $\downarrow i = \{j : j \leq_\tau i\}$ . Therefore

$$h_1(\tau) = \sum_{i=1}^n |\downarrow i| = |\{(j, i) : j \leq_\tau i\}|. \quad (1.4)$$

Elements in the same equivalence class of the preorder remain separate coordinates throughout. Replacing a preorder by its quotient poset without retaining class sizes would change the problem.

## 2 The augmented bipartite root polytope

Let  $G_\tau$  have left vertices  $1_L, \dots, n_L$ , right vertices  $1_R, \dots, n_R$ , and an edge  $i_L j_R$  precisely when  $j \leq_\tau i$ . Adjoin  $0_L$  and  $0_R$ , with  $0_L$  adjacent to every right vertex and  $0_R$  adjacent to every left vertex. Denote the resulting graph by  $\widehat{G}_\tau$ . In particular  $0_L 0_R$  is an edge. Its root polytope is

$$B_\tau = \text{conv}\{\mathbf{e}_i + \mathbf{f}_j : i_L j_R \in E(\widehat{G}_\tau)\} \subset \mathbb{R}^{n+1} \oplus \mathbb{R}^{n+1}. \quad (2.1)$$

The two coordinate blocks will be denoted  $(u, v)$ . We use the plus-sign convention for bipartite roots. Negating the second block converts this into the difference convention and preserves the lattice.

The affine span of  $B_\tau$  is

$$H_1 = \{(u, v) : \sum_{i=1}^n u_i = \sum_{j=1}^n v_j = 1\}. \quad (2.2)$$

Indeed, the edges  $i_L 0_R$  and  $0_L j_R$  form a spanning tree, whose  $2n + 1$  root vertices are affinely independent. Thus  $\dim B_\tau = 2n$ . At dilation  $m$ , the lattice is the full integer lattice in  $H_m = \{\sum u_i = \sum v_j = m\}$ , not a smaller lattice generated by selected coordinates.

### 2.1 The known support-enumerator theorem

For a bipartite graph with left neighborhoods  $D_1, \dots, D_n$  in an  $n$ -element right shore, its draconian vectors are the nonnegative integer vectors  $x$  satisfying

$$x(A) \leq \left| \bigcup_{i \in A} D_i \right| \quad (A \subseteq [n]). \quad (2.3)$$

Theorem 1.2 of [2] identifies their support-enumerator with the  $h^*$ -polynomial of the graph augmented by the two universal vertices just used in (2.1).

**Lemma 2.1.** *For  $D_i = \downarrow i$ , the draconian vectors in (2.3) are exactly  $P_\tau$ .*

*Proof.* If  $x \in P_\tau$  and  $A \subseteq E$ , transitivity makes  $\downarrow A$  an ideal. Nonnegativity gives

$$x(A) \leq x(\downarrow A) \leq |\downarrow A| = \left| \bigcup_{i \in A} D_i \right|.$$

Conversely, if (2.3) holds and  $I$  is an ideal, reflexivity and downward closure give  $\bigcup_{i \in I} D_i = I$ . Taking  $A = I$  gives  $x(I) \leq |I|$ .  $\square$

**Proposition 2.2** (Known higher-dimensional identity). *For every preorder on  $[n]$ ,*

$$\text{Ehr}_{B_\tau}(t) := \sum_{m \geq 0} |mB_\tau \cap \mathbb{Z}^{2n+2}| t^m = \frac{h_\tau(t)}{(1-t)^{2n+1}}. \quad (2.4)$$

*Proof.* Apply the support-enumerator theorem of [2] and Lemma 2.1.  $\square$

Appendix A supplies another proof of this proposition using the Kálmán–Postnikov theorem [3]. Its elementary matching construction also explains why a support statistic, rather than a sum-of-coordinates statistic, occurs. Neither proof makes the already known proposition a novelty claim of this manuscript.

### 3 A complete inequality description

The special feature of a preorder is that all relevant neighborhood sets can be reduced to ideals. We need the reduction over the reals, not only for integer degree vectors.

**Lemma 3.1** (Weighted Hall criterion). *Let  $G$  be a bipartite graph with shores  $L, R$ . Nonnegative margins  $u$  and  $v$  with common total  $m$  are the row and column sums of a nonnegative matrix supported on the edges of  $G$  if and only if*

$$u(A) \leq v(N(A)) \quad (A \subseteq L). \quad (3.1)$$

*If the margins are integral, the matrix can be chosen integral.*

*Proof.* Necessity follows by summing the entries in the rows of  $A$ . For sufficiency, make a network with source-to-left capacities  $u_i$ , right-to-sink capacities  $v_j$ , and capacity  $m + 1$  on each allowed left-to-right edge. A cut of capacity at most  $m$  cannot cut an allowed left-to-right edge. If its source side contains the left set  $A$ , it must therefore contain  $N(A)$  on the right; its capacity is at least  $m - u(A) + v(N(A)) \geq m$ . A flow of value  $m$  follows from the max-flow/min-cut argument. For completeness, that argument chooses a maximum feasible flow, which exists by compactness; if its residual graph contains a source-to-sink path, a positive augmentation contradicts maximality. Otherwise the vertices reachable from the source give a cut with capacity equal to the flow value. Integral capacities allow integral augmentations from the zero flow, so an integral maximum flow exists in the integral case.  $\square$

**Proposition 3.2** (Ideal inequalities for  $B_\tau$ ). *For every real  $m \geq 0$ , a pair  $(u, v) \in H_m$  belongs to  $mB_\tau$  exactly when all its coordinates are nonnegative and*

$$u(I) - v(I) \leq v_0 \quad (I \in \mathcal{J}(\tau)). \quad (3.2)$$

*Here  $u(I), v(I)$  involve only the coordinates indexed by  $[n]$ .*

*Proof.* A point of  $mB_\tau$  is precisely the pair of margins of a nonnegative matrix of total  $m$  supported on  $\widehat{G}_\tau$ . For a left subset containing  $0_L$ , its neighborhood is the entire right shore, so (3.1) follows from nonnegativity and the total-sum condition. For  $A \subseteq [n]$ ,

$$N(A) = \{0_R\} \cup (\downarrow A)_R.$$

Thus the remaining Hall conditions are  $u(A) \leq v_0 + v(\downarrow A)$ . If (3.2) holds, then

$$u(A) \leq u(\downarrow A) \leq v_0 + v(\downarrow A),$$

so all Hall conditions hold. Conversely, taking  $A = I$  for an ideal gives  $\downarrow I = I$  and recovers (3.2). The case  $m = 0$  is immediate.  $\square$

The empty-ideal inequality is just  $v_0 \geq 0$ , and the full-ideal inequality is equivalent to  $u_0 \geq 0$ . They may be redundant, but retaining them makes later formulas uniform. We do not assert that every inequality in this presentation defines a distinct facet.

## 4 The dimension-reducing lattice fibers

Define the projection

$$\pi(u, v) = (u_i - v_i)_{i=1}^n. \quad (4.1)$$

On the root vertices in (2.1), it has values 0,  $\pm \mathbf{e}_i$ , and  $\mathbf{e}_i - \mathbf{e}_j$  for  $j \leq_\tau i$ . Therefore

$$\pi(B_\tau) = C_\tau. \quad (4.2)$$

Its kernel on the direction space of  $H_1$  is spanned by  $\mathbf{e}_i + \mathbf{f}_i - \mathbf{e}_0 - \mathbf{f}_0$ ,  $1 \leq i \leq n$ . The dimension loss is exactly  $n$ . A linear projection alone does *not* usually preserve an Ehrhart numerator. The following explicit fibers are what make this one work.

For  $y \in \mathbb{R}^n$ , put

$$\begin{aligned} y_i^+ &= \max(y_i, 0), & y_i^- &= \max(-y_i, 0), \\ p(y) &= \sum_i y_i^+, & q(y) &= \sum_i y_i^-, \end{aligned} \quad (4.3)$$

$$a_\tau(y) = \max_{I \in \mathcal{J}(\tau)} y(I), \quad g_\tau(y) = q(y) + a_\tau(y). \quad (4.4)$$

The empty and full ideals imply  $a_\tau(y) \geq 0$  and  $a_\tau(y) \geq p(y) - q(y)$ .

**Theorem 4.1** (Exact fibers and an integral gauge). *For  $y \in \mathbb{R}^n$ , define*

$$\begin{aligned} u^{\min}(y) &= (a_\tau(y) + q(y) - p(y), y_1^+, \dots, y_n^+), \\ v^{\min}(y) &= (a_\tau(y), y_1^-, \dots, y_n^-). \end{aligned} \quad (4.5)$$

*Every point of  $mB_\tau$  projecting to  $y$  has a unique expression*

$$(u, v) = (u^{\min}(y), v^{\min}(y)) + \sum_{i=0}^n w_i(\mathbf{e}_i + \mathbf{f}_i), \quad (4.6)$$

*where  $w_i \geq 0$  and*

$$\sum_{i=0}^n w_i = m - g_\tau(y). \quad (4.7)$$

*Consequently,*

$$y \in mC_\tau \iff g_\tau(y) \leq m. \quad (4.8)$$

*For integral  $y, m$ , all terms in (4.5) are integral, and the lattice points in the fiber correspond exactly to nonnegative integer vectors  $w$  satisfying (4.7).*

*Proof.* The two vectors in (4.5) are nonnegative. Their sums are both  $a_\tau(y) + q(y) = g_\tau(y)$ , and their difference on  $[n]$  is  $y$ . Moreover,

$$u^{\min}(I) - v^{\min}(I) = y(I) \leq a_\tau(y) = v_0^{\min}.$$

Proposition 3.2 puts this pair in  $g_\tau(y)B_\tau$ . Adding a nonnegative multiple of a diagonal root leaves every difference  $u(I) - v(I)$  unchanged. For the root indexed by 0, the right side  $v_0$  also

increases, so all inequalities remain valid. This proves that every expression in (4.6) with (4.7) is an admissible point.

Conversely, suppose  $(u, v) \in mB_\tau$  projects to  $y$ . For  $1 \leq i \leq n$ , nonnegativity and  $u_i - v_i = y_i$  imply uniquely

$$u_i = y_i^+ + w_i, \quad v_i = y_i^- + w_i, \quad w_i = \min(u_i, v_i) \geq 0.$$

The ideal inequalities give  $v_0 \geq \max_I y(I) = a_\tau(y)$ . Set  $w_0 = v_0 - a_\tau(y) \geq 0$ . The equality of the two total sums then forces

$$u_0 = v_0 + q(y) - p(y) = u_0^{\min}(y) + w_0.$$

All coordinates in (4.6) are now determined. Summing either block yields (4.7). Projection proves (4.8). If  $y$  is integral, the maximum defining  $a_\tau(y)$  is an integer, which establishes the final assertion without any lattice-saturation assumption.  $\square$

**Corollary 4.2** (Equality of Ehrhart numerators). *For every finite preorder,*

$$\text{Ehr}_{B_\tau}(t) = \frac{\text{Ehr}_{C_\tau}(t)}{(1-t)^n}, \quad h^*(B_\tau, t) = h^*(C_\tau, t) = h_\tau(t). \quad (4.9)$$

*Proof.* For integral  $y$ , the least admissible integral dilation is  $g_\tau(y)$ . Theorem 4.1 gives

$$\text{Ehr}_{B_\tau}(t) = \frac{\sum_{y \in \mathbb{Z}^n} t^{g_\tau(y)}}{(1-t)^{n+1}}, \quad \text{Ehr}_{C_\tau}(t) = \frac{\sum_{y \in \mathbb{Z}^n} t^{g_\tau(y)}}{1-t}.$$

These are legitimate formal power series: each coefficient receives only finitely many contributions because  $C_\tau$  is bounded. Dividing the two identities proves the first equality in (4.9). The dimensions are  $2n$  and  $n$ , respectively, so their Ehrhart numerators coincide. Finally apply Proposition 2.2.  $\square$

There is a useful coefficient-level form. The fiber over an integral  $y$  at dilation  $m$  contains

$$\binom{m - g_\tau(y) + n}{n} \quad \text{if } g_\tau(y) \leq m, \quad 0 \quad \text{otherwise.} \quad (4.10)$$

Thus the factor  $(1-t)^{-n}$  is not a heuristic cancellation of dimensions; it comes from a concrete, uniquely parametrized family of integer fibers.

## 4.1 An order-theoretic interpretation of the gauge

The gauge can also be written

$$g_\tau(y) = \max_{I \in \mathcal{J}(\tau)} \left( \sum_{i \in I} y_i^+ + \sum_{i \notin I} y_i^- \right). \quad (4.11)$$

Thus testing membership in a dilation of  $C_\tau$  is a maximum-weight-ideal problem. One need not enumerate ideals in a practical implementation: put source capacity  $y_i^+$  at  $i$ , sink capacity  $y_i^-$  at  $i$ , and sufficiently large capacities on arcs  $i \rightarrow j$  for  $j \leq_\tau i$ . A finite minimum cut has a downward-closed source side  $I$ , and its capacity is  $p(y) - y(I)$ . Hence  $a_\tau(y)$ , and therefore  $g_\tau(y)$ , can be obtained by a network-flow calculation. The accompanying small-case verifier deliberately uses exhaustive ideal enumeration instead, keeping the check independent of a flow library.



## 5 Reflexivity and simplicial-polytopal realization

We now finish the proof of Theorem 1.1. The required geometry is particularly simple because all nonzero vertices of  $C_\tau$  are type- $A$  roots with one coordinate deleted.

### 5.1 The polar is integral

Since  $C_\tau$  contains  $\text{conv}\{\pm \mathbf{e}_i\}$ , it is full-dimensional and contains 0 in its interior. Its polar is

$$C_\tau^\circ = \{z \in \mathbb{R}^n : -1 \leq z_i \leq 1, \quad z_i - z_j \leq 1 \text{ whenever } j \leq_\tau i\}. \quad (5.1)$$

**Lemma 5.1** (Root determinants). *Put  $\mathbf{e}_0 = 0 \in \mathbb{Z}^n$ . Any  $n$  linearly independent vectors of the form  $\mathbf{e}_i - \mathbf{e}_j$ , with  $i, j \in \{0, \dots, n\}$  and  $i \neq j$ , form a basis of  $\mathbb{Z}^n$ .*

*Proof.* Associate the vector  $\mathbf{e}_i - \mathbf{e}_j$  to a directed edge from  $j$  to  $i$  on  $\{0, \dots, n\}$ . A cycle in the underlying undirected graph would give a linear dependence, with signs determined by the edge directions. Consequently  $n$  independent vectors give a forest with  $n$  edges on  $n + 1$  vertices, hence a spanning tree. The matrix of these vectors is a directed incidence matrix with the row indexed by 0 deleted. Remove a leaf different from 0; its row has one nonzero entry, equal to 1 or  $-1$ . Expanding the determinant along that row and repeating reduces the determinant to 1 or  $-1$ .  $\square$

**Proposition 5.2.** *The polar (5.1) is a lattice polytope. In fact all its vertices have coordinates in  $\{-1, 0, 1\}$ . Thus  $C_\tau$  is reflexive.*

*Proof.* At a vertex of the full-dimensional polar, choose  $n$  linearly independent active constraints. With  $z_0 = 0$ , each selected equality is of the form  $z_i - z_j = 1$ . Its coefficient row is a root as in Lemma 5.1. The resulting square matrix has determinant  $\pm 1$ , so its solution is integral. The bounds  $-1 \leq z_i \leq 1$  restrict these integers to  $-1, 0, 1$ . Both the original polytope and its polar are therefore integral, with the origin in the interior, which is precisely reflexivity.  $\square$

This proof also supplies a finite exact way to test lattice-point membership: one may evaluate against all ternary vectors satisfying (5.1), since that set contains every vertex of the polar. The verifier uses this method as a check independent of the gauge formula.

### 5.2 Unimodular cones over the boundary

Recall that a lattice simplex with vertices  $0, r_1, \dots, r_n$  is unimodular when  $r_1, \dots, r_n$  form a lattice basis. A triangulation is unimodular if all its full-dimensional simplices are unimodular.

**Proposition 5.3.** *Let  $\Delta$  be any triangulation of  $\partial C_\tau$  using only vertices of  $C_\tau$ . Then the star triangulation  $0 * \Delta$  of  $C_\tau$  is unimodular.*

*Proof.* Every boundary simplex is contained in a supporting hyperplane not containing 0. The  $n$  vertices of a maximal boundary simplex are therefore linearly independent: an affine dependence is already excluded by being a simplex, and a linear dependence with nonzero coefficient sum would place 0 in that supporting hyperplane. All its vertices are roots  $\mathbf{e}_i - \mathbf{e}_j$ , allowing index 0. Lemma 5.1 shows that the cone simplex from 0 is unimodular. Since 0 is interior, these cones form a triangulation of all of  $C_\tau$ .  $\square$

### 5.3 Why the triangulated boundary is polytopal

We use the following standard pulling fact, stated as Lemma 2.1 in Athanasiadis [4]: a pulling (reverse lexicographic) triangulation of the boundary of a convex  $n$ -polytope is combinatorially the boundary of a simplicial convex  $n$ -polytope. The geometric mechanism is successive small outward perturbations of the ordered vertices; at each step one chooses the perturbation small enough not to change the already fixed incidences. The resulting convex boundary realizes the prescribed pulling subdivision. This is an actual polytopality assertion, not merely shellability or the assertion that the complex is a sphere.

Here is the Ehrhart connection, with its normalization made explicit. If  $K$  is a  $d$ -dimensional unimodular triangulation of a lattice polytope  $P$ , then

$$\text{Ehr}_P(t) = \sum_{\sigma \in K} \left( \frac{t}{1-t} \right)^{|\sigma|} = \frac{h(K, t)}{(1-t)^{d+1}}, \quad (5.2)$$

where the empty face contributes 1. Indeed, the relative interior of an  $r$ -dimensional unimodular simplex contributes  $\binom{m-1}{r}$  lattice points at positive dilation  $m$ , with generating series  $t^{r+1}/(1-t)^{r+1}$ . Relative interiors partition the triangulation, giving the first equality. The second equality is the definition of its  $h$ -polynomial. For a cone, the face generating function gains a factor  $1 + t/(1-t)$ , while the denominator exponent increases by one; hence

$$h(0 * \Delta, t) = h(\Delta, t). \quad (5.3)$$

*Completion of the proof of Theorem 1.1.* Choose a pulling triangulation  $\Delta$  of  $\partial C_\tau$ . By the pulling fact,  $\Delta$  is the boundary of an  $n$ -dimensional simplicial polytope  $S_\tau$ . Proposition 5.3 and (5.2)–(5.3) give

$$h(S_\tau, t) = h(\Delta, t) = h^*(C_\tau, t) = h_\tau(t),$$

where the last equality is Corollary 4.2. Reflexivity was proved in Proposition 5.2. The Dehn–Sommerville relations for simplicial polytopes give palindromicity, and the  $g$ -theorem gives the stated inequalities and its stronger numerical conditions; see also the polytopal conclusions summarized in [4, Theorem 3.5].  $\square$

## 6 A second geometric check in dimension $2n$

The larger polytope  $B_\tau$  supplies two independent checks on the preceding argument: an interior-point translation and an explicit special simplex. They are useful safeguards against a sign error or a mismatch of dimensions.

### 6.1 An exact interior translation

Let  $\mathbf{c} = (1, \dots, 1; 1, \dots, 1) \in \mathbb{Z}^{2n+2}$ . For every  $m \geq n+1$ ,

$$(\text{relint}(mB_\tau)) \cap \mathbb{Z}^{2n+2} = \mathbf{c} + ((m-n-1)B_\tau \cap \mathbb{Z}^{2n+2}). \quad (6.1)$$

To see this, note that every coordinate inequality is nonconstant on  $B_\tau$ , and each ideal inequality has strict slack at  $\mathbf{e}_0 + \mathbf{f}_0$ . Consequently relative interior is characterized by positive coordinates and strict versions of (3.2). At integer points these strict inequalities are

$$u(I) - v(I) \leq v_0 - 1.$$

Subtracting 1 from every coordinate reduces each block sum by  $n + 1$ . The terms  $|I|$  cancel on the left, and the right side becomes the new  $v_0$ . This is exactly the weak inequality description of  $(m - n - 1)B_\tau$ . Conversely, adding  $\mathbf{c}$  makes every defining inequality strict. There are no interior integer points for  $1 \leq m < n + 1$ , since  $n + 1$  positive coordinates in either block already require total at least  $n + 1$ .

In particular,  $B_\tau$  has codegree  $n + 1$ , with unique interior integer point  $\mathbf{c}$  in its first nonempty interior dilation. It is Gorenstein of index  $n + 1$ . Ehrhart reciprocity translates (6.1) into

$$t^{2n+1}h^*(B_\tau, t^{-1}) = t^{n+1}h^*(B_\tau, t),$$

which again gives  $h_\tau(t) = t^n h_\tau(t^{-1})$ . This symmetry proof does not use the  $g$ -theorem or polytopality of a sphere.

## 6.2 The diagonal special simplex

A simplex with  $r$  vertices inside a polytope is *special* if each facet contains exactly  $r - 1$  of those vertices. Consider

$$\Sigma_\tau = \text{conv}\{d_i : 0 \leq i \leq n\}, \quad d_i = \mathbf{e}_i + \mathbf{f}_i. \quad (6.2)$$

Reflexivity of the preorder makes these root vertices available, and they are affinely independent. A coordinate hyperplane  $u_i = 0$  or  $v_i = 0$  contains exactly the diagonal vertices other than  $d_i$ . On an ideal hyperplane  $u(I) - v(I) = v_0$ , every  $d_i$  with  $i > 0$  lies on the hyperplane and  $d_0$  does not. Every facet is furnished by at least one nonredundant inequality of Proposition 3.2; thus  $\Sigma_\tau$  is special.

Every triangulation of a bipartite root polytope using its vertices is unimodular. Indeed, affine dependence of edge roots is equivalent to a cycle in the supporting bipartite graph; a full-dimensional simplex therefore corresponds to a spanning tree. Leaf elimination in its incidence matrix shows that its edge vectors form a basis of the integer lattice  $\{(u, v) : \sum u_i = \sum v_j\}$ . Intersecting with the height-one hyperplane shows affine unimodularity. In particular every pulling ordering is compressed in the terminology of [4].

Athanasiadis's special-simplex theorem [4, Theorem 3.5], applied with dimension  $2n$  and  $n + 1$  special vertices, therefore realizes  $h^*(B_\tau, t)$  as the  $h$ -polynomial of a simplicial polytope of dimension  $2n - (n + 1) + 1 = n$ . This recovers the main theorem by another route. The broader Gorenstein/unimodular-triangulation framework of Bruns and Römer [5] provides a further compatible interpretation, but is not needed for our main proof.

## 7 Consequences and quantitative formulas

### 7.1 Duality, with its prior attribution

Swapping the coordinate blocks identifies  $B_\tau$  with  $B_{\tau^*}$ . This is the already established duality mechanism of [2]. In the smaller model it becomes the especially transparent equality

$$C_{\tau^*} = -C_\tau. \quad (7.1)$$

Thus  $h_{\tau^*} = h_\tau$ . The gauge gives a third check. Ideals of  $\tau^*$  are complements of ideals of  $\tau$ , so

$$a_{\tau^*}(-y) = a_\tau(y) - y(E), \quad g_{\tau^*}(-y) = g_\tau(y).$$

This rederivation is included to connect the new lower-dimensional model with the known theorem, not to reclassify Conjecture 5.4 as open.

## 7.2 Ehrhart polynomials and volumes

The ordinary Ehrhart polynomial of  $C_\tau$  is now explicit in terms of the preorder support counts:

$$|mC_\tau \cap \mathbb{Z}^n| = \sum_{k=0}^n h_k(\tau) \binom{m+n-k}{n}, \quad m \in \mathbb{Z}_{\geq 0}. \quad (7.2)$$

Similarly,

$$|mB_\tau \cap \mathbb{Z}^{2n+2}| = \sum_{k=0}^n h_k(\tau) \binom{m+2n-k}{2n}. \quad (7.3)$$

A binomial term with nonnegative upper argument smaller than its lower argument is zero. These formulas also extend as polynomial identities in  $m$ . Normalized volume, meaning dimension factorial times volume in the relevant lattice, satisfies

$$\text{Vol}_{\mathbb{Z}}(C_\tau) = \text{Vol}_{\mathbb{Z}}(B_\tau) = h_\tau(1) = |P_\tau|. \quad (7.4)$$

The two volumes in this equality live in different dimensions and are both *normalized* volumes.

Remark 5.6 of [1] considers  $h_\tau(t)/(1-t)^{n+1}$ . Equation (7.2) identifies its coefficients as the Ehrhart polynomial of  $C_\tau$ . This gives that series a concrete geometric interpretation. It does not prove that its polynomial has all complex roots on a particular vertical line; no such root-location assertion follows just from reflexivity.

## 7.3 Vertices, elementary bounds, and the $g$ -vector

Every vector displayed in (1.2) is a vertex. To expose  $\mathbf{e}_i$ , choose a linear functional taking value 1 at coordinate  $i$  and a fixed  $\varepsilon \in (0, 1)$  at all other coordinates. The negative case is analogous. To expose  $\mathbf{e}_i - \mathbf{e}_j$ , give coordinates  $i, j$  the values 1,  $-1$  and all others the value 0; this root is the unique displayed vector with value 2. Thus

$$f_0(C_\tau) = n + |\{(j, i) : j \leq_\tau i\}| = n + h_1(\tau). \quad (7.5)$$

Pulling adds no vertices, so the same number occurs in  $S_\tau$ .

The inclusions

$$[0, 1]^n \subseteq Q_\tau \subseteq \{x \geq 0 : x(E) \leq n\}$$

give the useful coefficientwise bounds

$$\binom{n}{k} \leq h_k(\tau) \leq \binom{n}{k}^2 \quad (0 \leq k \leq n). \quad (7.6)$$

For the upper bound, choose the support in  $\binom{n}{k}$  ways; the positive integer values on it with total at most  $n$  can be chosen in  $\binom{n}{k}$  ways by stars and bars. The lower bound uses the corresponding 0–1 points.

For completeness, the full numerical strengthening supplied by polytopality can be stated as follows. Put  $g_0 = 1$  and  $g_i = h_i - h_{i-1}$  for  $1 \leq i \leq \lfloor n/2 \rfloor$ . The  $g$ -theorem says this sequence is an  $M$ -sequence. In particular  $g_i \geq 0$  and  $g_{i+1} \leq g_i^{\langle i \rangle}$  whenever both sides are relevant. Here, if

$$a = \binom{b_i}{i} + \binom{b_{i-1}}{i-1} + \cdots + \binom{b_j}{j}, \quad b_i > b_{i-1} > \cdots > b_j \geq j \geq 1,$$

is the Macaulay expansion, then  $a^{\langle i \rangle} = \binom{b_{i+1}}{i+1} + \cdots + \binom{b_{j+1}}{j+1}$ , with  $0^{\langle i \rangle} = 0$ . These inequalities are stronger than unimodality and follow here from an actual simplicial-polytope realization.

## 8 Examples and the role of transitivity

### 8.1 Antichains and a single equivalence class

For an antichain,  $\mathcal{Q}_\tau = [0, 1]^n$  and

$$h_\tau(t) = (1 + t)^n, \quad C_\tau = \text{conv}\{\pm \mathbf{e}_1, \dots, \pm \mathbf{e}_n\}.$$

The gauge is  $\sum_i |y_i|$ . The smaller model is already the simplicial cross polytope, so no boundary refinement is needed.

For the indiscrete preorder, in which every two elements are equivalent,  $\mathcal{Q}_\tau = \{x \geq 0 : x(E) \leq n\}$ . The same stars-and-bars argument used above shows

$$h_\tau(t) = \sum_{k=0}^n \binom{n}{k}^2 t^k.$$

Here  $B_\tau = \Delta_n \times \Delta_n$ , and the lower-dimensional polytope contains all roots  $\pm \mathbf{e}_i$  and  $\mathbf{e}_i - \mathbf{e}_j$ . Its gauge is  $\max(p(y), q(y))$ , since its only ideals are  $\emptyset, E$ . These two extreme examples check both the coefficient bounds and the change of Ehrhart denominator.

### 8.2 A non-self-dual four-element example

Let  $\tau$  have one minimum below three incomparable maximal elements. Its polynomial, obtained directly from (1.1), is

$$h_\tau(t) = 1 + 7t + 12t^2 + 7t^3 + t^4.$$

For example, for supports of size two not containing the minimum, the three pairs of maximal elements each allow the values  $(1, 1), (1, 2), (2, 1)$ , giving nine points; the three supports containing the minimum each allow only  $(1, 1)$ , giving three more. The dual order is not isomorphic to the original, but its directed root polytope is the negative of the original one and its support polynomial agrees. This is a useful test that duality is not merely invariance under relabeling.

### 8.3 A preorder with a nontrivial equivalence class

Use labels  $a, b, c, d, e$ , with  $a \sim c, a < e, c < e, b < e$ , and  $b < d$ . No other comparisons hold beyond those forced by reflexivity and transitivity. The principal ideals are

$$\{a, c\}, \quad \{b\}, \quad \{a, c\}, \quad \{b, d\}, \quad \{a, b, c, e\}.$$

Direct enumeration gives

$$h_\tau(t) = 1 + 11t + 34t^2 + 34t^3 + 11t^4 + t^5. \tag{8.1}$$

This agrees with the running example of [1]. There are 92 points in  $P_\tau$ , 22 vertices of  $B_\tau$ , and 16 vertices of  $C_\tau$ . The number 92 is also the normalized volume of either root model.

For  $x = (0, 0, 1, 2, 2)$ , the matching construction in Appendix A can assign the five demands to the five right vertices as follows: the demand at  $c$  uses  $c_R$ , the two demands at  $d$  use  $b_R, d_R$ , and the two demands at  $e$  use  $a_R, e_R$ . Attach every left vertex to  $0_R$ , then attach each right vertex to its assigned left vertex. This gives the spanning-tree certificate included in the JSON data.

## 8.4 Selected coefficient vectors

| Preorder                  | $n$ | $(h_0, \dots, h_n)$        | $ P_\tau $ |
|---------------------------|-----|----------------------------|------------|
| Antichain                 | 4   | (1, 4, 6, 4, 1)            | 16         |
| Chain                     | 4   | (1, 10, 20, 10, 1)         | 42         |
| Single equivalence class  | 4   | (1, 16, 36, 16, 1)         | 70         |
| One minimum, three maxima | 4   | (1, 7, 12, 7, 1)           | 28         |
| Diamond                   | 4   | (1, 9, 18, 9, 1)           | 38         |
| Example (8.1)             | 5   | (1, 11, 34, 34, 11, 1)     | 92         |
| Six-element crown         | 6   | (1, 12, 48, 75, 48, 12, 1) | 197        |

The crown has minimal elements  $a, b, c$  and maximal elements  $A, B, C$ , with  $a, b < A, b, c < B$ , and  $c, a < C$ . Every row of the table is recomputed by the supplied verifier; no floating-point root finder is involved. [Added 28 September 2026, batch 39: the crown is the case  $r = 3$  of Part II's crown family. Its support polynomial has the closed form (20.2) and only simple negative zeros (Proposition 20.1), yet no lattice triangulation of its  $C_\tau$ , pulling or not, is flag (Theorem 13.6).]

## 8.5 Why arbitrary reflexive relations are not enough

Transitivity cannot simply be dropped from the fiber theorem. Consider the neighborhoods

$$D_1 = \{1\}, \quad D_2 = \{1, 2\}, \quad D_3 = \{2, 3\}.$$

They define a reflexive but nontransitive relation. Its draconian inequalities reduce to

$$x_i \geq 0, \quad x_1 \leq 1, \quad x_1 + x_2 \leq 2, \quad x_3 \leq 2, \quad x_1 + x_2 + x_3 \leq 3.$$

The support counts are (1, 5, 6, 1): one-coordinate supports contribute  $1 + 2 + 2 = 5$ , two-coordinate supports contribute  $1 + 2 + 3 = 6$ , and the full support contributes just (1, 1, 1). This is the nonpalindromic example also recorded in [1, Remark 5.5].

The general support-enumerator/root-polytope theorem remains valid for this graph. What fails is the reduction of all weighted Hall inequalities to matching ideal differences  $u(I) - v(I)$ . In particular, subtracting a common coordinate from the two margins can damage an inequality involving different row and column subsets. Thus the argument does not inadvertently imply palindromicity for all graph support-enumerators. [Added 29 September 2026, batch 44: Part IV explains this example exactly. For every reflexive relation  $R$  the coefficients of degrees 1 and  $n - 1$  are  $|R|$  and the size of its transitive closure, and palindromicity holds if and only if  $R$  is transitive (Theorem 48.3); here  $(1, 5, 6, 1) = (1, |R|, |\bar{R}|, 1)$  with one missing transitive pair (Example 53.3).]

## 9 Exact computational verification

The program code/verify.py uses only Python's standard library and integer arithmetic. It stores a preorder by bitmasks

$$D_i = \{j : j \leq_\tau i\}.$$

It enumerates every reflexive relation on  $[n]$  and retains precisely those for which  $j \in D_i$  implies  $D_j \subseteq D_i$ . This is a direct test of transitivity. All weak compositions with total at most  $n$  are then checked against the ideal inequalities to obtain  $h_\tau$ .

| Ground-set size | Labeled preorders checked | Result |
|-----------------|---------------------------|--------|
| 0               | 1                         | Pass   |
| 1               | 1                         | Pass   |
| 2               | 4                         | Pass   |
| 3               | 29                        | Pass   |
| 4               | 355                       | Pass   |
| 5               | 6,942                     | Pass   |
| Total           | 7,332                     | Pass   |

For each of these preorders, the code checks duality, palindromicity, unimodality, the endpoint coefficients, the relation-count formula for  $h_1$ , and the special-simplex slack identities. These are finite consistency checks, not the reason the theorem holds for arbitrary  $n$ .

Stronger, independent comparisons are performed for all 35 preorders on at most three elements. The program enumerates spanning trees directly, deduplicates their left degree-minus-one vectors, and compares them with  $\{(n - |x|, x) : x \in P_\tau\}$ . It computes internal inactivity by testing all allowed transfers to smaller coordinates, rather than replacing inactivity by support size in the code. Across these instances it enumerates 18,009 spanning trees and checks 399 explicit matching-tree certificates.

For the same cases it counts sums of edge-root vertices directly at every dilation  $0 \leq m \leq 2n$ . Those counts are compared both with Hall-margin enumeration and with (7.3). Finite differences recover all  $2n + 1$  coefficients of the root-polytope Ehrhart numerator, including the predicted zero coefficients above degree  $n$ . Independently, integer points of  $C_\tau$  are counted using the ternary polar description, compared with the gauge formula, and checked through  $m = n$  against (7.2). Full-margin fiber identities and the interior translation are also checked exactly.

Five selected four-element preorders undergo full root Ehrhart checks through  $m = 8$ , quotient checks through  $m = 4$ , and direct edge-sum checks through  $m = 3$ . Further examples include the weighted five-element preorder and the six-element crown. The JSON file records all individual results, and the CSV file contains every preorder and support vector through size five.

The original conjecture paper reports tests through larger sizes for some coefficient properties [1]. The present computations are not offered as an improved size record. Their purpose is to compare independent realizations of the newly proposed construction and detect normalization, orientation, or lattice-index errors.

## 10 What remains unresolved and how to audit the claim

The proof gives an ordinary simplicial polytope, not a flag one. A flag complex is determined by its edges: every collection of pairwise adjacent vertices must form a face. Neither a pulling triangulation nor a general simplicial polytope automatically has that property. Therefore Conjecture 5.1(d) is not settled by the argument. [Added 28 September 2026, batch 39: Part II shows more. If  $\tau$  contains an induced crown on  $2r \geq 6$  elements, every lattice triangulation of  $C_\tau$  or of  $\partial C_\tau$  has a minimal nonface of size  $r$  (Theorem 13.6); so the polytope  $S_\tau$  of Theorem 1.1 is then never flag. Conjecture 5.1(d), which asks for *some* flag polytope, remains open; see Section 11.3.]

Likewise, palindromic unimodality and the  $g$ -theorem do not establish  $\gamma$ -positivity or



real-rootedness. The manuscript proves neither the nonnegativity of all coefficients in

$$h_\tau(t) = \sum_{j=0}^{\lfloor n/2 \rfloor} \gamma_j t^j (1+t)^{n-2j}$$

nor the assertion that every zero of  $h_\tau$  is real. Accordingly it does not claim to settle Conjectures 5.2 or 5.3 of [1]. [Added 28 September 2026, batch 39: the two preceding sentences describe Part I at its state of 20 September 2026. Conjecture 5.3 is false: the first counterexample known here is an eight-element height-two poset posted by Shivam Patel [11], which Part II checks and extends to the family of Theorem 13.3. Conjecture 5.2 holds for posets of height at most two, with  $\gamma_k$  the number of perfectly matchable vertex sets of size  $2k$  in the comparability graph (Theorem 13.2), and is open in general. See Section 11.3.] [Added 29 September 2026, batch 44: Conjecture 5.2 is no longer open. Part IV proves it for every finite preorder, with  $\gamma_k$  the number of ordered pairs of disjoint  $k$ -element sets joined by a  $\tau$ -increasing bijection (Theorem 48.2); see Section 46.3. Its proof does not use Part I's geometric argument. Conjecture 5.3 stays false.]

For independent mathematical review, the shortest audit is to verify four links: the identification of the draconian vectors with  $P_\tau$ ; Proposition 3.2; the unique integer fibers in Theorem 4.1; and the determinant argument for cones over a polytopal pulling boundary. Once these are accepted, the cited support-enumerator theorem and the pulling lemma give the asserted simplicial-polytopal realization. Section 6 is a separate geometric cross-check, not a circular use of the desired symmetry.

The result should be understood as a proposed research proof with reproducible support, not as a claim of community acceptance. The distinction between the known August 2026 result and the additional construction developed here is part of the mathematical claim, not a bibliographic afterthought.



## Part II. Height-two preorders: matching supports, nonreal roots, crown obstructions

Sections 1–10 and the appendices form the original report (Part I) of 20 September 2026, unchanged except for dated pointers to this addition. Part II was added on 28 September 2026 in batch 39 of ProveIt’s incoming-report intake, from one manuscript written against Part I as printed above. It addresses the three questions Part I left open in Section 10: Conjectures 5.3, 5.2 and 5.1(d) of Athanasiadis and Chapoton [1]. Its headline counterexample to Conjecture 5.3 is **not new**: it was posted earlier by Shivam Patel on MathDB [11], and the manuscript says so. Section 11 records the provenance, fixes the notation and states what is answered.

### 11 Provenance, conventions and what Part II answers

#### 11.1 One manuscript, one addition

Part II is batch-39 manuscript 02, *Height-Two Preorder Polynomials: Matching supports, nonreal-root families, and crown obstructions*. With a spectral treatment of bipartite cactus graphs, a research manuscript developed with ChatGPT for Vladimir Reshetnikov and dated 28 September 2026 (26 pages as delivered). It arrived as the archive `ProveIt_Height_Two_Preorder_Research.zip` (inner directory `Preorder_Matching_Research/`) in ProveIt commit `74f7f5bdb` and was placed in `e2b1f016a`. It names no ProveIt commit; it records the Git blob `4c350f5652096b7db91d4a10075eb7905910f647` of Part I’s `article.tex` as the file it consulted. That blob is Part I exactly as printed above (it is unchanged at the placement commit), so every statement the manuscript makes about “the ProveIt report” refers to Part I. The manuscript’s code and data are shipped beside Part I’s files with the prefix `02-height-two-` (Part I’s unprefixes files count as 01), and its status and source notes as `02-height-two-STATUS.md` and `02-height-two-sources.md`. Its manuscript, delivery README and PDF are not shipped and survive in the arrival commit; its checksum list was verified at placement and retired.

The manuscript is printed here as Sections 12 to 25: its Section  $k$  is Section  $k + 11$  here, and its theorem and equation numbers shift accordingly (its Theorem 2.2 is Theorem 13.2); its two figures and ten research questions keep their numbers. Every result, proof, example, remark, limitation, question and non-claim of the manuscript is printed. For the reader’s convenience, its abstract is reproduced here as delivered.

**Abstract of manuscript 02 (as delivered).** For a finite poset  $P$ , let  $h_P(t)$  enumerate the integer points of its preorder polytope by support size. If  $P$  has height at most two and  $G$  is its bipartite comparability graph, we prove

$$h_P(t) = (1+t)^{|P|} p_G\left(\frac{t}{(1+t)^2}\right), \quad p_G(u) = \sum_{\substack{W \subseteq V(G) \\ G[W] \text{ has a perfect matching}}} u^{|W|/2}.$$

Thus the  $\gamma$ -coefficients count distinct matchable vertex sets, not matchings, proving the height-two case of Athanasiadis–Chapoton Conjecture 5.2. For the graph  $\Theta_m$  formed by  $m$  internally disjoint three-edge paths with common endpoints, we obtain an explicit polynomial and a complete real-root count. For  $m \geq 3$ , the associated height-two support polynomial has exactly two real roots when  $m$  is even and four when  $m$  is odd. Consequently its proportion of nonreal roots tends to one, even though the graph has treewidth two. The eight-element case was previously posted by Shivam Patel; we verify it and extend it rather than claiming its discovery. A separate 17-element construction transfers the classical Stembridge–Ohsugi–Tsuchiya example.

An orthant argument proves the height-two instance of the repository’s root-polytope identity. For bipartite cactus graphs, a sixth-root-of-unity matrix gives a determinantal formula and a multivariate stability refinement, with univariate real-rootedness credited to prior work. Induced crowns force arbitrarily large minimal nonfaces in every lattice triangulation of the canonical root polytope and indispensable toric relations of arbitrarily large degree. Exact Python and C++ certificates accompany the proofs.

**Status and scope (manuscript 02).** The proposed contributions are the general height-two connection, the complete root-distribution theorem for  $\Theta_m$ , and the geometric and spectral refinements. The first counterexample to Conjecture 5.3 is *not* claimed as new: a smaller posted example was found during the source audit. Two classical root-polytope results are imported explicitly. The manuscript has not been independently refereed or checked in Lean. General flag realizability and  $\gamma$ -positivity beyond height two remain unsettled here; absolute publication priority is not certified. [Added 29 September 2026, batch 44:  $\gamma$ -positivity beyond height two is proved in Part IV (Theorem 48.2).]

**Priority of the counterexample.** The first known counterexample to Conjecture 5.3 of [1] printed in this report is the eight-element height-two poset  $P_{\Theta_3}$  of Section 16. It is due to Shivam Patel, whose posted solution on MathDB [11] specifies the graph  $\Theta_3$  and both polynomials. The manuscript found that posting during its own source audit, after its 17-element example had been developed, and revised its claims accordingly; it reproduces and independently checks Patel’s example and claims only the all- $m$  extension. The exact calendar date of the posting was not established: the page showed a relative date (about one month before 28 September 2026) and labelled the claim unverified. A withdrawn SSRN listing (abstract 7385158) for an eight-element counterexample was also found; its full text was unavailable, and it is not used as evidence for any mathematical or priority claim. The 17-element counterexample of Section 17 is a transfer of the classical Stembridge–Ohsugi–Tsuchiya graph polynomial [10, 8], neither a first disproof nor a minimum-size example. Univariate real-rootedness for bipartite cacti is Ohsugi–Tsuchiya’s [9, Theorem 7.2], and the even-cycle (crown) formula is Davis–Kohl’s [7]. The source audit behind these statements does not prove the absence of other publications, preprints or postings containing any of Part II’s proposed extensions.

**Shortest audit (manuscript 02).** Read the counting lemma in Section 14, the Boolean summation in Section 15, and the explicit graph and exact certificate in Section 17. Together with Section 16, these establish the root-distribution results without assuming any unrefereed theorem in ProveIt (in particular, without Part I's general theorem). Sections 18–21 develop its geometric and spectral context. The companion programs provide independent finite checks rather than substitutes for the infinite-family proofs.

## 11.2 Notation of Part II

Part II works with *posets* of height at most two, which are preorders; for  $\tau = P$  its  $\mathcal{J}(P)$ ,  $Q(P)$ ,  $h_P$  and  $C_P$  are exactly Part I's  $\mathcal{J}(\tau)$ ,  $Q_\tau$ ,  $h_\tau$  and  $C_\tau$  (compare (13.1) and (13.2) with the two displays opening Section 1, and Theorem 13.6 with (1.2)). Its  $\gamma$ -coefficients are taken in the basis  $t^k(1+t)^{n-2k}$  of Section 10, and  $\text{Vol}_{\mathbb{Z}}$  is the normalized volume of (7.4). Several letters mean different things in the two parts; Table 1 fixes them. No symbol of the manuscript was renamed and no normalization was changed.

**Remark 11.1** (Merge note: Part I's input and Part II's input agree). This remark is the merge's own and is not in manuscript 02. Part I's augmented polytope  $B_\tau$  of (2.1) is the edge polytope  $B_{\widehat{H}}$  of Section 14 for  $H = G_\tau$ , with the same lattice. Proposition 2.2 and the imported identity (14.4) therefore give, for every finite preorder,

$$h_\tau(t) = h^*(B_\tau; t) = p_{G_\tau}(t) :$$

the support polynomial is itself the perfectly-matchable-set polynomial of the doubled graph  $G_\tau$ . This is a consequence of two cited theorems, not a new result, and it is a different statement from Theorem 13.2, in which  $p_G$  of the  $n$ -vertex comparability graph is the  $\gamma$ -polynomial of  $h_{P_G}$ . As a normalization check, both sides were compared by direct enumeration (ideal inequalities against induced perfect matchings of  $G_\tau$ ) for all 390 labeled preorders with  $n \leq 4$ ; they agree. That check is a scratch computation of the write phase and is not shipped.

## 11.3 What Part II answers in Section 10

Section 10 of Part I leaves three statements of [1] unsettled. Their status after Part II, as of 28 September 2026:

- **Conjecture 5.3 (real-rootedness of  $h_\tau$ ) is false.** The first counterexample known to this report is Patel's eight-element poset  $P_{\Theta_3}$  [11], *not* a discovery of Part II. Part II proves that  $h_{P_{\Theta_m}}$  has exactly  $2m$  ( $m$  even) or  $2m - 2$  ( $m$  odd) nonreal zeros for every  $m \geq 3$  (Theorem 13.3, Corollary 16.2), and transfers the classical 17-element example (Theorem 13.4). Part I never asserted real-rootedness, so nothing in it is retracted.
- **Conjecture 5.2 ( $\gamma$ -positivity) holds for posets of height at most two** (Theorem 13.2:  $\gamma_k = |\mathcal{M}_k(G)|$ ). It remains open for general preorders, including height-two posets whose elements are blown up into nontrivial equivalence classes (Research question 1). [Added 29 September 2026, batch 44: it holds in height two and, by Part IV, in general: Theorem 48.2 proves it for every finite preorder, blowups included (Example 54.2), and answers Research question 1; see Section 46.3.]
- **Conjecture 5.1(d) (a flag simplicial polytope) remains open.** Part II shows that no lattice triangulation of  $C_\tau$  or of its boundary is flag when  $\tau$  contains an induced crown on  $2r \geq 6$  elements (Theorem 13.6). A pulling triangulation of  $\partial C_\tau$  uses only vertices of  $C_\tau$ , so for such  $\tau$  the simplicial polytope  $S_\tau$  of Theorem 1.1 is never flag; the smallest case is Part I's

Table 1: Symbols of Part II. The last column prints the tempting false reading, usually a Part I symbol with the same letter.

| Symbol                                                             | Meaning in Part II                                                                                                                | Not to be confused with                                                                                                                                                                                            |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $P, P_G, P_0, P_{\Theta_m}, P_r$                                   | finite posets: $P_G$ has $a < b$ exactly for the edges $ab$ of $G$ ; $P_0 = P_{G_0}$ ; $P_r$ is the crown                         | Part I's lattice-point set $P_\tau = Q_\tau \cap \mathbb{Z}^n$ , which Part II writes $Q(P) \cap \mathbb{Z}^V$ . $P_r$ is not " $P_\tau$ at $\tau = r$ ".                                                          |
| $V, n =  V $                                                       | ground set of $P$                                                                                                                 | Part I's ground set $E = [n]$ .                                                                                                                                                                                    |
| $G = (A \sqcup B, E)$                                              | bipartite comparability graph of a height-two poset on its $n$ elements; $E$ its edge set                                         | Part I's doubled graph $G_\tau$ on $2n$ vertices $i_L, j_R$ (edge iff $j \leq_\tau i$ , loops $i_L i_R$ included) and its augmentation $\widehat{G}_\tau$ ;<br>Part I's ground set $E$ .                           |
| $p =  A , q =  B $                                                 | shore sizes (Section 13)                                                                                                          | Part I's functions $p(y), q(y)$ of (4.3); the polynomial $q_G(x)$ of (19.5).                                                                                                                                       |
| $p_G(u), \mathcal{M}_k(G)$                                         | perfectly matchable <i>vertex sets</i> of $G$ ; $p_G$ is the $\gamma$ -polynomial of $h_{P_G}$                                    | the matching polynomial $M_G(u) = \sum_k m_k(G)u^k$ ; also $p_{G_\tau}$ of Part I's doubled graph, which is $h_\tau$ itself (Remark 11.1).                                                                         |
| $H = (X \sqcup Y, E), \widehat{H}, B_{\widehat{H}}, \mathcal{R}_H$ | any bipartite graph, its augmentation by $x_0, y_0$ , its edge polytope, its demand vectors                                       | Part I's $B_\tau$ , which is $B_{\widehat{H}}$ for $H = G_\tau$ (same augmentation, same lattice);<br>Part I's hyperplanes $H_1, H_m$ .<br>Hypertrees here are <i>right</i> hypertrees; Appendix A uses left ones. |
| $\Theta_m, p_m, R_m(y), k = m - 2$                                 | $m$ parallel three-edge paths, its matchable-set polynomial                                                                       | Part I's dilation $m$ . The vertex $d$ of $\Theta_m$ is not a diagonal vertex $d_i$ of (6.2).                                                                                                                      |
| $G_0, p_0, F_{i,j}(u)$                                             | the 17-element example, its $\gamma$ -polynomial, the board recurrence                                                            | —                                                                                                                                                                                                                  |
| $D_G, T_G, g_T, g_D$                                               | one-sided root polytope lattice-equivalent to $C_{P_G}$ , its positive part, their gauges                                         | Part I's gauge $g_\tau$ on $C_\tau$ ; $g_D$ is $g_\tau$ after negating the $A$ -coordinates. Part I's neighbourhoods $D_i$ .                                                                                       |
| $I(G), I$ in Section 18                                            | independent vertex sets                                                                                                           | order ideals $I \in \mathcal{J}(\tau)$ (also written $I$ in Section 15); the toric ideal $I_{\mathcal{A}}$ .<br>Part I's triangulation $K$ in (5.2).                                                               |
| $K, \zeta, H_G, \Phi_G, q_G, \lambda_j$                            | cactus matrix, sixth root of unity, Hermitian block matrix, stable polynomial, its diagonal specialization, eigenvalues of $KK^*$ |                                                                                                                                                                                                                    |
| $S$                                                                | a subset of $B$ or $Y$ ; in (19.7) the support size of a uniformly random lattice point                                           | Part I's simplicial polytope $S_\tau$ .                                                                                                                                                                            |
| $F, U, W, v_i, w_i, \mathcal{A}$                                   | the exposed circuit face of a crown and its Radon halves                                                                          | $W$ also denotes a perfectly matchable set elsewhere in Part II; $F_{i,j}$ is the board recurrence.                                                                                                                |
| $\nu(G)$                                                           | maximum matching size                                                                                                             | —                                                                                                                                                                                                                  |

six-element crown (Section 8). This does not disprove Conjecture 5.1(d), which asks for *some* flag polytope with  $h$ -polynomial  $h_\tau$ .

Part II also reproves, for height two, three statements of Part I by different routes: the identity  $h^*(C_{P_G}) = h_{P_G}$  and reflexivity by an orthant decomposition (Theorem 18.2, compare Corollary 4.2 and Proposition 5.2), invariance under reversal (Corollary 15.2, compare Section 7), and  $h_1 = n + |E|$  (compare (1.4)). Its hypertree identification in Section 14 is the argument of Lemma A.1 for a general bipartite graph and right hypertrees. These are printed as the manuscript gives them, as second routes.

## 11.4 Where the merge had to choose

- *Placement.* Part II follows Section 10 and precedes Part I's appendices, whose letters and numbers are unchanged.
- *Self-citation.* The manuscript cites Part I as an external repository report (its reference "RepoRoot", with the blob above). Here those citations are replaced by references to Part I's sections, and the phrases "the ProvelIt report" and "the repository's construction" by "Part I".
- *File names.* The manuscript's delivered names (`code/verify.py`, `data/board_certificate.csv`, ...) are replaced by the shipped names in Section 22, because `code/verify.py` in this directory is Part I's different verifier. Its build commands are replaced by commands that run on a copy.
- *Bibliography.* The manuscript's entries for Athanasiadis–Chapoton and Kálmán–Postnikov are Part I's references [1] and [3] (the manuscript adds the DOI 10.1112/plms.12015 for the latter); its other five entries are appended.
- *Labels and front matter.* Labels carry the prefix `htwo:`. Two unlabelled corollaries, the conclusion and the build subsection received labels so that this section can cite them. The manuscript's title page, abstract, status box and audit box are printed in this section.

## 12 The research target and the distinction from prior work

Part I, the ProvelIt report *A Reflexive Root-Polytope Model for Preorder  $h$ -Polynomials*, constructs a canonical root polytope for every finite preorder and gives a proposed proof of simplicial-polytopal realization of its support polynomial (Theorem 1.1). Its final section, Section 10, explicitly leaves flag realizability,  $\gamma$ -positivity, and real-rootedness unresolved. These are different properties, and separating them is essential here.

The underlying questions are Conjectures 5.1(d), 5.2, and 5.3 of Athanasiadis and Chapoton [1]. The first asks for *some* flag simplicial polytope with the prescribed  $h$ -vector. The second asks for a nonnegative expansion in the basis  $t^k(1+t)^{n-2k}$ . The third asserts that all zeros of the support polynomial are real. We focus on posets of height at most two: they are preorders, but unlike arbitrary preorders they admit a direct bipartite-graph description.

The graph polynomial needed for this description is classical. The perfectly matchable set polynomial is studied by Davis and Kohl [7]; its root-polytope realization originates in Ohsugi and Tsuchiya [8]. The nonreal-rooted example used below comes from Stembridge's naturally labeled, width-two counterexample to the Neggers conjecture [10], as translated into a graph polynomial by Ohsugi and Tsuchiya [8, Example 3.7]. It is important not to confuse that older *width-two* poset with the new *height-two* poset obtained by orienting its bipartite incomparability graph.

## 12.1 What the article proves

The height-two matching-support identity is the central theorem. It proves the relevant  $\gamma$ -positivity conjecture for an entire class and converts real-root questions into graph enumeration. The principal negative-family result concerns  $\Theta_m$ : a complete real-root count shows that almost all zeros become nonreal while graph treewidth stays two.

The source audit found an earlier eight-element counterexample posted by Shivam Patel on MathDB [11]. It is exactly  $P_{\Theta_3}$ . The source labels the posting unverified; our independent finite calculation agrees with it, and the general proof below includes it. Consequently a claim to have first disproved Conjecture 5.3 would be inappropriate. The site's exact posting date was not available, and its broad historical metadata is not used as evidence. The 17-element construction is retained as an independent bridge to classical literature, not as the smallest or first counterexample.

A second theme concerns a limitation of the canonical root-polytope construction. Every induced crown of size  $2r$ ,  $r \geq 3$ , forces a minimal nonface of size  $r$  in any lattice triangulation. This rules out obtaining a flag complex by choosing a more clever lattice triangulation of that particular polytope. It does *not* rule out another flag polytope with the same  $h$ -vector.

## 12.2 A provenance ledger

| Statement                                                    | Provenance and status                                                                             |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| $h^*(T_G; u) = p_G(u)$ for bipartite $G$                     | Established input; Ohsugi–Tsuchiya, reformulated by Davis–Kohl.                                   |
| Hypertrees count normalized root-polytope volume             | Established input from Kálmán–Postnikov; used at $u = 1$ .                                        |
| Height-two formula $h_P = (1 + t)^n p_G(t/(1+t)^2)$          | Proved here by a support partition and a counting lemma.                                          |
| Nonreal-rooted degree-eight graph polynomial                 | Classical Stembridge–Ohsugi–Tsuchiya example, independently recomputed here.                      |
| Eight-element counterexample to AC Conjecture 5.3            | Previously posted by Patel; independently checked here, not claimed as new.                       |
| Complete root count for $P_{\Theta_m}$                       | Proved here for every $m$ , extending the eight-element case to a connected treewidth-two family. |
| 17-element counterexample                                    | Independent transfer of the classical graph example; no claim of first disproof or minimality.    |
| Cactus $p_G$ is real-rooted                                  | Already follows from Ohsugi–Tsuchiya [9, Theorem 7.2]; not claimed anew.                          |
| Sixth-root matrix and multivariate refinement                | Explicit proof supplied here; no absolute novelty claim for the matrix technique.                 |
| Induced-crown obstruction in the Provel root model of Part I | Proved here, including a nonregular-triangulation obstruction and a toric degree lower bound.     |

## 13 Definitions and main statements

For a finite poset  $P$  on a ground set  $V$  of size  $n$ , let  $\mathcal{J}(P)$  be its order ideals, including  $\emptyset$  and  $V$ . Write  $x(S) = \sum_{v \in S} x_v$ . Its preorder polytope and support polynomial are

$$\mathcal{Q}(P) = \{x \in \mathbb{R}_{\geq 0}^V : x(I) \leq |I| \text{ for all } I \in \mathcal{J}(P)\}, \quad (13.1)$$

$$h_P(t) = \sum_{x \in \mathcal{Q}(P) \cap \mathbb{Z}^V} t^{|\text{supp}(x)|}. \quad (13.2)$$



The full-set inequality makes the sum finite. Its constant and leading coefficients are both one: the all-zero and all-one vectors are the unique points with support sizes zero and  $n$ , respectively.

Here height at most two means that every chain has at most two elements; we consider ordinary posets, not arbitrary equivalence-class blowups of them. Such a poset can be written  $P = P_G$  for a bipartite graph  $G = (A \sqcup B, E)$ : the only strict comparisons are  $a < b$  for  $ab \in E$ . Isolated elements may be assigned to either shore. Put  $p = |A|$ ,  $q = |B|$ ,  $n = p + q$ , and denote by  $N_G(S) \subseteq A$  the neighborhood of  $S \subseteq B$ .

**Definition 13.1.** A subset  $W \subseteq V(G)$  is *perfectly matchable* if the induced graph  $G[W]$  has a perfect matching. The empty set is perfectly matchable. Let

$$\mathcal{M}_k(G) = \{W \subseteq V(G) : |W| = 2k, G[W] \text{ has a perfect matching}\}, \quad p_G(u) = \sum_{k \geq 0} |\mathcal{M}_k(G)| u^k.$$

For comparison,  $M_G(u) = \sum_k m_k(G) u^k$  counts individual  $k$ -edge matchings. These two polynomials are generally different.

**Theorem 13.2** (Height-two matching-support formula). *For every finite bipartite graph  $G$  on  $n$  vertices,*

$$h_{P_G}(t) = \sum_{k \geq 0} |\mathcal{M}_k(G)| t^k (1+t)^{n-2k} = (1+t)^n p_G\left(\frac{t}{(1+t)^2}\right). \quad (13.3)$$

*In particular, every height-at-most-two poset has a  $\gamma$ -positive support polynomial, with  $\gamma_k(P_G) = |\mathcal{M}_k(G)|$ .*

The rational substitution in (13.3) denotes its polynomial expansion; it is not an instruction to evaluate an undefined expression at  $t = -1$ .

**Theorem 13.3** (A complete root-distribution family). *Let  $\Theta_m$  have lower elements  $d, a_1, \dots, a_m$ , upper elements  $b_0, b_1, \dots, b_m$ , and edges  $a_i b_0, a_i b_i, d b_i$ . For  $m \geq 2$ ,*

$$p_{\Theta_m}(u) = (1+u)^{m-2} (1 + (2m+3)u + (m^2+m+3)u^2 + u^3) - u. \quad (13.4)$$

*The support polynomial of  $P_{\Theta_m}$  is real-rooted exactly when  $m \leq 2$  (for  $m \geq 1$ ). For  $m \geq 3$ , it has precisely two real roots if  $m$  is even, and four if  $m$  is odd; these real roots are simple and negative. Its graph is connected and has treewidth at most two.*

**Theorem 13.4** (An explicit negative answer to real-rootedness). *Let  $G_0$  have shores  $A = \{a_1, \dots, a_9\}$  and  $B = \{b_1, \dots, b_8\}$ , with neighborhoods*

|          |        |        |        |        |        |        |        |        |
|----------|--------|--------|--------|--------|--------|--------|--------|--------|
| $j$      | 1      | 2      | 3      | 4      | 5      | 6      | 7      | 8      |
| $N(b_j)$ | [1, 1] | [1, 2] | [1, 3] | [1, 4] | [1, 5] | [2, 6] | [3, 8] | [4, 9] |

(13.5)

*where  $[l, r]$  means  $\{a_l, a_{l+1}, \dots, a_r\}$ . Then the height-two poset  $P_{G_0}$  is a counterexample to Conjecture 5.3 of [1]. Its  $\gamma$ -polynomial is*

$$p_0(u) = 1 + 32u + 336u^2 + 1420u^3 + 2534u^4 + 1946u^5 + 658u^6 + 86u^7 + 3u^8. \quad (13.6)$$

*Its support polynomial is  $\gamma$ -positive despite having nonreal zeros.*

**Theorem 13.5** (Cactus matrix representation). *For every bipartite cactus graph  $G = (A \sqcup B, E)$  there is an explicitly constructible complex matrix  $K$ , supported exactly on  $E$ , such that*

$$p_G(u) = \det(I_{|A|} + uKK^*). \quad (13.7)$$

*The nonzero entries of  $K$  are sixth roots of unity, and each square minor has squared absolute value either zero or one, according as its support has no perfect matching or has one. Consequently  $h_{P_G}$  is real-rooted. A multivariate stable determinant and a vertex-deletion interlacing theorem are given in Section 19.*

**Theorem 13.6** (Induced-crown obstruction). *Let  $P$  be a finite preorder containing an induced height-two crown on  $2r$  distinct elements,  $r \geq 3$ . The canonical root polytope*

$$C_P = \text{conv}(\{\pm \mathbf{e}_v : v \in P\} \cup \{\mathbf{e}_b - \mathbf{e}_a : a \neq b, a \leq_P b\})$$

*uses all nonzero roots for distinct comparable elements; for equivalent elements both directions are included. Every lattice triangulation of  $C_P$  or of its boundary has a minimal nonface of cardinality  $r$ . The toric ideal on  $C_P \cap \mathbb{Z}^n$  has an indispensable binomial of degree  $r$ , and every initial ideal has a minimal generator of degree  $r$ .*

For preorders, all distinct pairs  $a \leq_P b$  are used, including both directions within an equivalence class.

## 14 A matching–polymatroid counting lemma

For a bipartite graph  $H = (X \sqcup Y, E)$ , define the finite set

$$\mathcal{R}_H = \{c \in \mathbb{Z}_{\geq 0}^Y : c(S) \leq |N_H(S)| \text{ for every } S \subseteq Y\}. \quad (14.1)$$

This set describes nonnegative integer demands that can be supplied by unit-capacity vertices in  $X$ , with adjacency specifying allowable supply.

**Lemma 14.1** (Counting demands by matching supports). *For every finite bipartite graph  $H$ ,*

$$|\mathcal{R}_H| = p_H(1) = \#\{W \subseteq V(H) : H[W] \text{ has a perfect matching}\}. \quad (14.2)$$

We give the reduction to two established root-polytope statements in full. This avoids treating (14.2) as an experimental identity or as an unexplained application of a lattice-point package.

### 14.1 The two classical inputs

Adjoin  $x_0$  and  $y_0$  to the shores of  $H$ , joining  $x_0$  to every old right vertex,  $y_0$  to every old left vertex, and  $x_0$  to  $y_0$ . Write  $\widehat{H}$  for this connected augmented bipartite graph. Its edge polytope is

$$B_{\widehat{H}} = \text{conv}\{\mathbf{e}_x + \mathbf{f}_y : xy \in E(\widehat{H})\}.$$

A *right hypertree* is a vector  $(\deg_T(y) - 1)_{y \in Y \cup \{y_0\}}$  realized by a spanning tree  $T$  of  $\widehat{H}$ . Multiple spanning trees with the same vector count only once.

We use the following published facts, with the full affine integer lattice on the edge polytope:

$$\text{Vol}_{\mathbb{Z}}(B_{\widehat{H}}) = \#\{\text{right hypertrees of } \widehat{H}\}, \quad (14.3)$$

$$h^*(B_{\widehat{H}}; u) = p_H(u). \quad (14.4)$$

The first is the evaluation at  $u = 1$  of the interior-polynomial/root-polytope theorem of Kálmán and Postnikov [3]. The second is Ohsugi–Tsuchiya’s matching-support interpretation, stated explicitly in Davis–Kohl [7, Theorem 3.10]. Recall that  $h^*(B; 1) = \text{Vol}_{\mathbb{Z}}(B)$ . No result about preorder real-rootedness is an input to either statement.



## 14.2 Explicit identification of the hypertrees

Let  $s = |X|$ ,  $r = |Y|$ . A spanning tree of  $\widehat{H}$  has  $s + r + 1$  edges. Its right degree-minus-one vector has total  $s$ . Write it as  $(c_0, (c_y)_{y \in Y})$ . For nonempty  $S \subseteq Y$ , the tree edges incident with  $S$  form a forest on  $S \cup N_{\widehat{H}}(S)$ . Their number is  $|S| + c(S)$ , so

$$|S| + c(S) \leq |S| + |N_{\widehat{H}}(S)| - 1 = |S| + |N_H(S)|.$$

Thus  $c \in \mathcal{R}_H$ , and  $c_0 = s - c(Y)$ .

Conversely, take  $c \in \mathcal{R}_H$  and replace each  $y \in Y$  by  $c_y$  clones, each with neighborhood  $N_H(y) \subseteq X$ . A set of clones with labels in  $S$  has at most  $c(S) \leq |N_H(S)|$  members. Hall's criterion therefore matches all clones to distinct vertices of  $X$ . One can prove this form of Hall's criterion directly by augmenting paths: failure to match a clone produces an alternating-reachable set of clones with fewer neighbors than members, contradicting the stated inequality.

Start a spanning tree with the star joining  $x_0$  to every vertex of  $Y \cup \{y_0\}$ . Attach each old  $x \in X$  as a leaf to the label of its matched clone, or to  $y_0$  if no clone was assigned to it. The resulting graph is a tree and has right degree-minus-one vector

$$(s - c(Y), (c_y)_{y \in Y}).$$

Hence hypertree vectors correspond bijectively to  $\mathcal{R}_H$ . Combining this bijection with (14.3) and (14.4) proves Lemma 14.1. This argument also handles an empty shore:  $\mathcal{R}_H$  then contains only the zero demand and  $p_H = 1$ .

**Remark 14.2.** The lemma equates two cardinalities; it does not provide a canonical bijection between demand vectors and perfectly matchable vertex sets. The tree construction is a bijection with *hypertree vectors*, not with their realizing trees. Counting matchings or spanning trees with multiplicity at this step would invalidate the main formula.

[Added 30 September 2026, batch 64: Part IX constructs an explicit, support-preserving bijection between demand vectors and matchable support pairs (Theorem 135.2), and its disjoint union proves the lemma without the two root-polytope inputs (Corollary 139.2). The bijection depends on a choice of edge costs and is not canonical (Proposition 144.1).]

## 15 The height-two formula and its consequences

### 15.1 Partitioning lattice points by zeros and support

Write a lattice point of  $\mathcal{Q}(P_G)$  as  $(a, b)$  on the two shores. Each singleton in  $A$  is an ideal, so  $a_i \in \{0, 1\}$ . Let

$$Z = \{i \in A : a_i = 0\}, \quad S = \{j \in B : b_j > 0\}.$$

An ideal is  $U \sqcup T$  with  $T \subseteq B$  and  $N_G(T) \subseteq U \subseteq A$ . Its inequality is

$$b(T) \leq |T| + |U \cap Z|.$$

For fixed  $T$ , the strongest choice is  $U = N_G(T)$ . Thus all ideal inequalities are equivalent to

$$b(T) \leq |T| + |N_G(T) \cap Z| \quad (T \subseteq B). \quad (15.1)$$

On  $S$  put  $b_j = 1 + c_j$ , with  $c_j \geq 0$ , and put  $b_j = 0$  elsewhere. For  $T \subseteq S$ , equation (15.1) becomes

$$c(T) \leq |N_G(T) \cap Z|.$$

These inequalities also imply (15.1) for arbitrary  $T \subseteq B$ , by applying them to  $T \cap S$ . Therefore the possible  $c$  are exactly  $\mathcal{R}_{G[Z \sqcup S]}$ .

Every point in this part has support size  $p - |Z| + |S|$ . Lemma 14.1 gives

$$h_{p_G}(t) = \sum_{Z \subseteq A} \sum_{S \subseteq B} t^{p-|Z|+|S|} \sum_{W \in \mathcal{M}(G[Z \sqcup S])} 1, \quad (15.2)$$

where  $\mathcal{M}(H)$  denotes all perfectly matchable vertex sets of  $H$ .

## 15.2 The Boolean summation

Fix a perfectly matchable set  $W = X \sqcup Y$  of  $G$ , with  $X \subseteq A$ ,  $Y \subseteq B$ , and  $|X| = |Y| = k$ . It is counted in (15.2) exactly when  $X \subseteq Z$  and  $Y \subseteq S$ . Its total contribution is

$$\begin{aligned} \left( \sum_{Z \supseteq X} t^{p-|Z|} \right) \left( \sum_{S \supseteq Y} t^{|S|} \right) &= (1+t)^{p-k} t^k (1+t)^{q-k} \\ &= t^k (1+t)^{n-2k}. \end{aligned}$$

Summing over  $W$  proves Theorem 13.2. In particular, the  $\gamma$ -positivity conclusion is not deduced from an assumed flag complex; it follows from an explicit nonnegative counting interpretation.

## 15.3 Roots, matching number, and small coefficients

Let  $\nu(G)$  be the maximum matching size. Then  $\deg p_G = \nu(G)$ , and all its coefficients from degree zero through  $\nu(G)$  are positive: a maximum matching has submatchings of every smaller size.

**Proposition 15.1** (Exact transfer of real-rootedness). *For a bipartite graph  $G$ , the polynomial  $h_{p_G}$  is real-rooted if and only if  $p_G$  is real-rooted. If  $p_G$  has only real roots, they are strictly negative. In all cases, the root  $-1$  of  $h_{p_G}$  has multiplicity  $n - 2\nu(G)$ . In the real-rooted case, its other roots occur in reciprocal negative pairs.*

*Proof.* Write  $p_G(u) = \prod_{j=1}^{\nu} (1 + \lambda_j u)$  over  $\mathbb{C}$ , with  $\lambda_j \neq 0$ . Formula (13.3) gives

$$h_{p_G}(t) = (1+t)^{n-2\nu} \prod_{j=1}^{\nu} ((1+t)^2 + \lambda_j t). \quad (15.3)$$

If  $p_G$  is real-rooted, its positive coefficients and constant term one exclude nonnegative roots. Hence every  $\lambda_j > 0$ , and each quadratic  $t^2 + (2 + \lambda_j)t + 1$  has two distinct negative reciprocal roots. Its value at  $-1$  is  $-\lambda_j \neq 0$ , proving the stated multiplicity.

Conversely, if  $u_0$  is a nonreal zero of  $p_G$ , then  $u_0(1+t)^2 - t = 0$  has two roots different from  $-1$ . Neither is real, since  $t/(1+t)^2$  is real for real  $t \neq -1$ . Both are zeros of  $h_{p_G}$ . Thus real-rootedness of  $h_{p_G}$  forces that of  $p_G$ .  $\square$

**Corollary 15.2.** *The height-two polynomial is unchanged by reversing the poset, and it is multiplicative under disjoint unions. Its first coefficients satisfy*

$$\gamma_0 = 1, \quad \gamma_1 = |E|, \quad \gamma_2 = \binom{|E|}{2} - \sum_{v \in V} \binom{\deg(v)}{2} - c_4(G), \quad h_1 = n + |E|,$$

where  $c_4(G)$  is the number of four-cycles, each counted once.

*Proof.* Reversing the order exchanges the shores of  $G$ , leaving  $p_G$  unchanged. Perfectly matchable subsets split independently across components. For  $\gamma_2$ , subtract adjacent pairs of edges from all edge pairs to obtain  $m_2(G)$ . A four-vertex bipartite subgraph has two perfect matchings exactly when it is a four-cycle; otherwise it has at most one. Thus  $\gamma_2 = m_2 - c_4$ . Finally, the coefficient of  $t$  in (13.3) is  $n + \gamma_1$ .  $\square$

For a forest, every perfectly matchable induced subgraph has a unique perfect matching: the symmetric difference of two perfect matchings would contain a cycle. Hence  $p_G = M_G$ . More generally,  $|\mathcal{M}_k(G)| \leq m_k(G)$ , and equality for all  $k$  holds for a bipartite graph exactly when it is a forest. An even cycle supplies two distinct matchings with the same covered set, proving the converse.

## 16 A complete nonreal-root family at treewidth two

This section strengthens a single counterexample to a uniform theorem with an exact count of real roots. The mechanism is distinct from taking disjoint unions: every graph in the family is connected, and its treewidth is bounded.

### 16.1 Parallel paths and the matching-support polynomial

Let  $\Theta_m$  be the union of the  $m$  paths

$$d - b_i - a_i - b_0 \quad (1 \leq i \leq m),$$

whose only common vertices are the endpoints  $d, b_0$ . Its bipartition is  $A = \{d, a_1, \dots, a_m\}$  and  $B = \{b_0, b_1, \dots, b_m\}$ . For  $m = 1$  this is a four-vertex path; for  $m = 2$  it is a six-cycle. For  $m = 3$  the corresponding height-two poset is the previously posted counterexample of Patel [11].

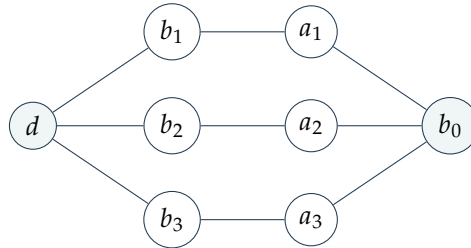


Figure 1: The graph  $\Theta_3$ . To obtain its height-two poset, orient every edge from the shore  $\{d, a_1, a_2, a_3\}$  to the other shore. The horizontal layout shows paths, not increasing order in the poset.

**Proposition 16.1.** *For  $m \geq 2$ , the perfectly matchable set polynomial of  $\Theta_m$  is (13.4). For  $m = 1$ , it is  $1 + 3u + u^2$ .*

*Proof.* Partition matchable vertex sets according to the presence of  $d, b_0$ . If neither endpoint is present, each branch contributes either its two internal vertices or neither; the contribution is  $(1 + u)^m$ .

If  $d$  alone is present, it must be matched to a unique included vertex  $b_i$  whose partner  $a_i$  is absent. Every other included branch is a complete internal pair. There are  $m$  choices of  $i$ , giving  $mu(1 + u)^{m-1}$ . The case with  $b_0$  alone has the same contribution.

Suppose both endpoints are present. If some branch contributes just  $b_i$ , then  $d$  must match it. There must be exactly one different branch contributing just  $a_j$ , to be matched to  $b_0$ ; all

other branches are empty or complete internal pairs. This gives  $m(m-1)u^2(1+u)^{m-2}$ . In the remaining case all selected branches are complete pairs. At least one such branch is needed to match the endpoints, and the selected vertex set is counted once, regardless of how many of its branches could be used for this purpose. This gives  $u((1+u)^m - 1)$ .

Thus the sum is

$$\begin{aligned} & (1+u)^m + 2mu(1+u)^{m-1} + m(m-1)u^2(1+u)^{m-2} + u((1+u)^m - 1) \\ &= (1+u)^{m-2}(1 + (2m+3)u + (m^2+m+3)u^2 + u^3) - u. \end{aligned}$$

The four-vertex path is counted directly. Notice that replacing matchable sets by matchings would overcount the case where both endpoints and several complete branches are present; that distinction is essential to the formula.  $\square$

## 16.2 A positive interval with no roots

Write  $p_m = p_{\Theta_m}$ . For  $m \geq 3$ , put  $k = m - 2 \geq 1$ . A useful rearrangement, valid for  $0 < x < 1$ , is

$$\begin{aligned} p_m(-x) &= (1-x)^k(1-(m+1)x)^2 \\ &\quad + x[1-(1-x)^k(1+kx+x^2)]. \end{aligned} \tag{16.1}$$

The second line is strictly positive. Indeed, the binomial series gives

$$(1-x)^{-k} = 1 + kx + \frac{k(k+1)}{2}x^2 + \sum_{j \geq 3} \binom{k+j-1}{j} x^j > 1 + kx + x^2.$$

All terms are real and positive on this interval; the inequality holds also for  $k = 1$  because the tail is then nonzero. Moreover,  $p_m(0) = p_m(-1) = 1$ . We have proved the particularly strong exclusion

$$p_m(u) > 0 \quad (-1 \leq u \leq 0, \quad m \geq 3). \tag{16.2}$$

The coefficients of  $p_m$  are nonnegative by their interpretation as counts, so there are no nonnegative real roots either.

Already this disproves real-rootedness:  $p_m$  is monic with constant term one. If all its roots were real, (16.2) would put them strictly below  $-1$ , making the absolute value of their product larger than one. We next obtain the exact count, rather than just this contradiction.

## 16.3 Exact root counts by two sign changes

Set  $u = -1 - y$  with  $y > 0$  and define

$$R_m(y) = m(m-1) + 2m^2y + m(m+1)y^2 - y^3.$$

An exact change of variables gives

$$p_m(-1-y) = 1 + y + (-1)^{m-2}y^{m-2}R_m(y). \tag{16.3}$$

We use Descartes' elementary rule of signs: the number of positive roots, counted with multiplicity, is at most the number of changes in the sequence of nonzero coefficients. One way to prove the needed bound is to factor out positive linear factors successively; multiplying a real polynomial by  $y - a$ ,  $a > 0$ , increases the number of sign changes by a positive odd integer. Factoring over  $\mathbb{R}$  then yields the bound.

If  $m$  is even,  $m \geq 4$ , the polynomial on the right of (16.3) has positive coefficients except for its leading coefficient, which is negative. It has one sign change, is positive at  $y = 0$ , and tends to  $-\infty$ . It consequently has exactly one positive root, and that root is simple.

If  $m$  is odd,  $m \geq 3$ , its coefficients have signs

$$+, \underbrace{-, \dots, -}_{\text{nonzero middle block}}, +$$

when zero coefficients are omitted; for  $m > 3$  there is an additional initial positive coefficient, which does not change the count of two sign changes. For  $m = 3$ , the merged coefficient of  $y$  is  $1 - m(m - 1) < 0$ . At  $y = 0$  the value is one, at  $y = 1$  it is

$$2 - (4m^2 - 1) = 3 - 4m^2 < 0,$$

and it tends to  $+\infty$  as  $y \rightarrow \infty$ . There is a root in  $(0, 1)$  and another in  $(1, \infty)$ . The sign bound makes these the only positive roots, both simple.

By (16.2), these exhaust the real roots of  $p_m$ . It has degree  $m + 1$ , so it has  $m$  nonreal roots when  $m$  is even and  $m - 1$  when  $m$  is odd, counted with multiplicity. The substitution in Proposition 15.1 sends each negative root to two distinct negative reciprocal roots of  $h_{P_{\Theta_m}}$  and each nonreal root to two nonreal roots. There is no root at  $-1$ , because  $n = 2m + 2 = 2 \deg p_m$ . This proves the root count in Theorem 13.3. The cases  $m = 1, 2$  are a path and a cycle, respectively, and are real-rooted by the independent cactus proof in Section 19.

For completeness, a width-two tree decomposition is explicit. Use bags  $\{d, b_0, a_i\}$  and  $\{d, a_i, b_i\}$  for each  $i$ , connect the first bags in a path, and attach each second bag to its corresponding first bag. Every edge is covered; bags containing any fixed vertex form a connected subtree. All bags have size three. Thus the asserted treewidth bound requires no general graph-minor theorem.

**Corollary 16.2** (Most roots nonreal in a bounded-treewidth family). *The connected height-two posets  $P_{\Theta_m}$  satisfy*

$$\#\{\text{nonreal zeros of } h_{P_{\Theta_m}}\} = \begin{cases} 2m, & m \text{ even}, \\ 2m - 2, & m \text{ odd}, \end{cases} \quad (m \geq 3),$$

*with multiplicities. Their proportion among all  $2m + 2$  zeros tends to one. Thus bounded treewidth, even together with height two and connectedness, does not force real-rootedness or even a positive limiting proportion of real roots.*

## 16.4 The eight-element specialization and its status

The specialization  $m = 3$  gives

$$p_3(u) = 1 + 9u + 24u^2 + 16u^3 + u^4,$$

$$h_{P_{\Theta_3}}(t) = 1 + 17t + 106t^2 + 303t^3 + 427t^4 + 303t^5 + 106t^6 + 17t^7 + t^8.$$

These are the polynomials in Patel's posting [11], not newly claimed examples. Our direct enumeration from the ideal inequalities finds 1281 lattice points and reproduces every coefficient. Exact arithmetic also verifies the posted Laguerre obstruction

$$(p'_3(-1/4))^2 - p_3(-1/4)p''_3(-1/4) = -95/1024 < 0.$$

The additional result here is the all- $m$  formula and complete root count, not a priority claim for this specialization.

| $m$ | Poset size | Real roots of $p_m$ | Real roots of $h$ | Nonreal roots of $h$ |
|-----|------------|---------------------|-------------------|----------------------|
| 3   | 8          | 2                   | 4                 | 4                    |
| 4   | 10         | 1                   | 2                 | 8                    |
| 5   | 12         | 2                   | 4                 | 8                    |
| 6   | 14         | 1                   | 2                 | 12                   |
| 7   | 16         | 2                   | 4                 | 12                   |
| 8   | 18         | 1                   | 2                 | 16                   |

## 17 The 17-element counterexample in exact arithmetic

### 17.1 A complete graph specification

Use the graph  $G_0$  of (13.5); there are no other edges. Orient each edge from  $a_i$  to  $b_j$  to define  $P_0 = P_{G_0}$ . Since all strict comparisons go from  $A$  to  $B$ , the relation is automatically transitive and antisymmetric, and has height two. There are  $1 + 2 + 3 + 4 + 5 + 5 + 6 + 6 = 32$  strict comparisons.

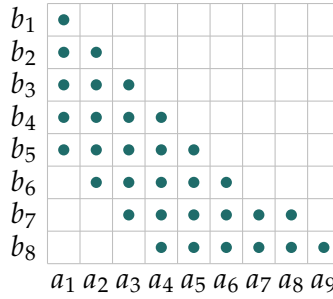


Figure 2: The complete  $8 \times 9$  incidence board of the counterexample. A dot means  $a_i < b_j$ . Increasing pairs of dots give the canonical matching of any matchable support. The diagram is drawn from the exact interval list, not inferred from a numerical polynomial.

### 17.2 Why an increasing matching suffices

**Lemma 17.1** (Uncrossing on a monotone interval board). *Suppose  $N(b_j) = [\ell_j, r_j]$  with both  $(\ell_j)$  and  $(r_j)$  nondecreasing. For sorted subsets  $X = \{a_{i_1}, \dots, a_{i_k}\}$  and  $Y = \{b_{j_1}, \dots, b_{j_k}\}$ , the induced graph on  $X \sqcup Y$  has a perfect matching if and only if every  $a_{i_s} b_{j_s}$  is an edge.*

*Proof.* The backward implication is immediate. In a matching with an inversion, suppose  $j < j'$  and  $i < i'$  but the matching uses  $a_i b_{j'}$  and  $a_{i'} b_j$ . Then

$$\ell_j \leq \ell_{j'} \leq i < i' \leq r_j \leq r_{j'}.$$

Therefore  $a_i b_j$  and  $a_{i'} b_{j'}$  are also edges. Swap the inverted pair. Choosing an adjacent inversion in the permutation of matched row indices reduces the inversion number; iteration gives the unique sorted matching. Thus existence is equivalent to the sorted-edge test.  $\square$

Let  $F_{i,j}(u)$  count increasing matched pairs in the first  $i$  vertices of  $A$  and first  $j$  vertices of  $B$ , by their number of pairs. With  $\varepsilon_{i,j} = 1$  for an edge and zero otherwise,

$$\begin{aligned} F_{i,0}(u) &= F_{0,j}(u) = 1, \\ F_{i,j}(u) &= F_{i-1,j}(u) + F_{i,j-1}(u) + (u\varepsilon_{i,j} - 1)F_{i-1,j-1}(u). \end{aligned} \tag{17.1}$$

Indeed, those objects omitting the last row or the last column are counted by inclusion–exclusion. An object using both must pair them with each other and leave an arbitrary increasing matching on the smaller board. Lemma 17.1 implies  $F_{9,8} = p_{G_0}$ .

For a convenient hand-checkable summary, the last row of this polynomial recurrence is as follows. The complete 90-cell table, including boundary cells, is in `data/02-height-two-board_certificate.csv`.

| $j$ | $F_{9,j}(u)$                                                             |
|-----|--------------------------------------------------------------------------|
| 0   | 1                                                                        |
| 1   | $1 + u$                                                                  |
| 2   | $1 + 3u + u^2$                                                           |
| 3   | $1 + 6u + 6u^2 + u^3$                                                    |
| 4   | $1 + 10u + 20u^2 + 10u^3 + u^4$                                          |
| 5   | $1 + 15u + 50u^2 + 50u^3 + 15u^4 + u^5$                                  |
| 6   | $1 + 20u + 105u^2 + 175u^3 + 105u^4 + 21u^5 + u^6$                       |
| 7   | $1 + 26u + 205u^2 + 595u^3 + 665u^4 + 301u^5 + 49u^6 + 2u^7$             |
| 8   | $1 + 32u + 336u^2 + 1420u^3 + 2534u^4 + 1946u^5 + 658u^6 + 86u^7 + 3u^8$ |

Thus the coefficient formula (13.6) is an exact finite recurrence calculation, not a fit to sampled values. There are  $p_0(1) = 7016$  perfectly matchable sets.

### 17.3 An elementary obstruction to real-rootedness

**Lemma 17.2** (Laguerre inequality). *If a nonzero real polynomial  $f$  has only real zeros, then*

$$(f'(x))^2 - f(x)f''(x) \geq 0 \quad (x \in \mathbb{R}). \quad (17.2)$$

*Proof.* Write  $f(x) = c \prod_i (x - r_i)$  with  $r_i \in \mathbb{R}$ , repeated according to multiplicity. Away from the roots,

$$\frac{(f')^2 - ff''}{f^2} = -\left(\frac{f'}{f}\right)' = \sum_i \frac{1}{(x - r_i)^2} \geq 0.$$

Continuity extends the inequality to the roots. □

Direct substitution into (13.6) gives

$$p_0(-2) = 65, \quad p'_0(-2) = -560, \quad p''_0(-2) = 4912. \quad (17.3)$$

Consequently

$$(p'_0(-2))^2 - p_0(-2)p''_0(-2) = 313600 - 319280 = -5680 < 0. \quad (17.4)$$

By Lemma 17.2,  $p_0$  has nonreal roots. Proposition 15.1 proves Theorem 13.4. This proof requires no isolation of a complex root and no trust in a floating-point solver.

Expanding the height-two formula gives

$$\begin{aligned} h_{P_0}(t) = & 1 + 49t + 952t^2 + 9828t^3 + 61302t^4 + 248816t^5 \\ & + 688516t^6 + 1337738t^7 + 1857081t^8 + 1857081t^9 \\ & + 1337738t^{10} + 688516t^{11} + 248816t^{12} + 61302t^{13} \\ & + 9828t^{14} + 952t^{15} + 49t^{16} + t^{17}. \end{aligned} \quad (17.5)$$



Its coefficient sum is 8 408 566. An independent C++ enumeration of *all* these lattice points, based only on integer capacity allocations, reproduces (17.5); see Section 22.

For orientation only, a nonreal pair of  $p_0$  is approximately  $-1.8588440 \pm 0.1497675 i$ . Its images yield two nonreal reciprocal pairs of  $h_{p_0}$ , approximately

$$-2.0461493 \pm 0.0565649 i, \quad -0.4883497 \pm 0.0135002 i.$$

The exact negative Laguerre value, rather than these approximations, is the certificate used in the proof. The accompanying exact computation does not need numerical roots.

## 17.4 What is old, what is transferred, and what is not settled

Polynomial (13.6) is the normalized descent polynomial in the older Stembridge example, and its interpretation as a bipartite matching-support polynomial is recorded in [8, Example 3.7]. We independently recompute it from the interval board so that the counterexample is specified without relying on a figure or a prior decimal root calculation.

For an explicit comparison, the older poset has chains  $1 < 3 < \dots < 17$  and  $2 < 4 < \dots < 16$ , together with the relations

$$2 < 3, 4 < 5, 6 < 7, 8 < 9, 10 < 11, 12 < 13, 14 < 17, \quad 1 < 12, 3 < 14, 5 < 16.$$

Its cross-chain incomparable pairs have precisely the neighborhoods in (13.5), under  $a_i = 2i - 1$ ,  $b_j = 2j$ . In the present counterexample these incomparable pairs are made into *new* comparabilities, all oriented from  $A$  to  $B$ ; the old chain comparisons are discarded. The two posets are therefore not the same object.

This transfer supplies an independent negative answer to the assertion about the support polynomial of *every preorder*; the existence of such counterexamples was already claimed in the smaller example [11]. It does not contradict  $\gamma$ -positivity, palindromic unimodality, or simplicial-polytopal realizability. The 17-element construction is not smallest: the preceding family already gives an eight-element example. Adding isolated elements multiplies the support polynomial by  $(1 + t)$ , so counterexamples exist in every size  $n \geq 17$ ; this operation alone says nothing about connected counterexamples of every size.

## 18 The canonical root polytope by orthant decomposition

Part I's construction (1.2) has a particularly concrete form in height two. Negate the coordinates in the lower shore  $A$ . This unimodular map sends the roots  $\mathbf{e}_b - \mathbf{e}_a$  to  $\mathbf{e}_b + \mathbf{e}_a$  and preserves the collection of signed coordinate vectors. Thus  $C_{P_G}$  is lattice equivalent to

$$D_G = \text{conv}(\{\pm \mathbf{e}_v : v \in V\} \cup \{\mathbf{e}_a + \mathbf{e}_b : ab \in E\}). \quad (18.1)$$

Only the positive sums are added; the polytope is not generally centrally symmetric. Also put

$$T_G = \text{conv}(\{0\} \cup \{\mathbf{e}_v : v \in V\} \cup \{\mathbf{e}_a + \mathbf{e}_b : ab \in E\}). \quad (18.2)$$

The full coordinate simplex is present, so both polytopes have dimension  $n = |V|$ , with the zero-dimensional empty case understood separately.

### 18.1 An integral gauge from a capacitated matching problem

For  $y \in \mathbb{R}_{\geq 0}^V$ , let  $g_T(y)$  be the least  $s \geq 0$  for which  $y \in sT_G$ . Write  $I(G)$  for the independent vertex sets of  $G$ .



**Lemma 18.1.** *For every bipartite graph  $G$  and nonnegative  $y$ ,*

$$g_T(y) = \max_{I \in I(G)} y(I). \quad (18.3)$$

*Consequently  $g_T$  is coordinatewise monotone and takes integer values on nonnegative integer vectors.*

*Proof.* Represent  $y$  as a nonnegative combination of coordinate vectors and edge-sum vectors. If an edge receives weight  $w_{ab}$ , it supplies that amount at each endpoint, reducing the total coefficient mass by  $w_{ab}$  compared with using only coordinate vectors. The minimum total mass is therefore

$$|y| - \max \left\{ \sum_{ab \in E} w_{ab} : w_{ab} \geq 0, \sum_{e \ni v} w_e \leq y_v \right\}.$$

The maximum is the value of a flow from a source through  $A$ , then  $B$ , to a sink, with vertex capacities  $y_a, y_b$  and across-edge capacities larger than  $|y|$ . A minimum cut never cuts an across-edge arc. Such cuts correspond exactly to vertex covers  $C$ , with capacity  $y(C)$ : the cover consists of lower-shore vertices on the sink side and upper-shore vertices on the source side. Max-flow/min-cut gives the maximum as  $\min_C y(C)$ . Complements of vertex covers are independent sets, proving (18.3). This argument is valid for real capacities; compactness and the usual residual-path proof give max-flow/min-cut in this finite network. The two final conclusions follow directly from the displayed maximum over independent sets.  $\square$

For  $x \in \mathbb{R}^V$ , let  $x_v^+ = \max(x_v, 0)$  and  $x_v^- = \max(-x_v, 0)$ . Since  $D_G = \text{conv}(T_G, \{-e_v\})$ , its gauge is

$$g_D(x) = g_T(x^+) + |x^-|. \quad (18.4)$$

Indeed, any representation  $x = y - z$  with  $y, z \geq 0$  has coefficient mass at least  $g_T(y) + |z|$ . Necessarily  $y \geq x^+$  and  $z \geq x^-$ , so the lower bound in (18.4) follows from monotonicity. Taking  $y = x^+$  and  $z = x^-$  achieves it.

This also proves reflexivity without a triangulation. The gauge can be written as the finite maximum of integer linear forms

$$g_D(x) = \max_{I \in I(G), R \subseteq V \setminus I} \left( \sum_{v \in I} x_v - \sum_{v \in R} x_v \right). \quad (18.5)$$

Every form is bounded by the right side of (18.4); equality is obtained by taking a maximizing independent subset of the positive coordinates and all negative coordinates in  $R$ . Thus  $D_G$  is cut out by integer inequalities with right side one. The origin is interior because  $D_G$  contains the crosspolytope. Its polar is the convex hull of the integer normals in (18.5), hence a lattice polytope.

## 18.2 Counting the lattice points in every orthant

The augmented edge polytope  $B_{\widehat{G}}$  from Section 14 is lattice equivalent to  $T_G$ . Project away the two added coordinates. The inverse recovers them as

$$x_0 = 1 - \sum_{a \in A} x_a, \quad y_0 = 1 - \sum_{b \in B} y_b$$

on the affine span of  $B_{\widehat{G}}$ . These formulas preserve the full affine integer lattice. The projected vertices are exactly 0, the coordinate vectors, and the edge sums. Therefore the established identity (14.4) gives

$$h^*(T_G; t) = p_G(t). \quad (18.6)$$

For an integral gauge  $g$  of a bounded polytope containing the origin,

$$\text{Ehr}(t) = \frac{1}{1-t} \sum_{x \in \mathbb{Z}^n} t^{g(x)};$$

a lattice point of least dilation  $g(x)$  occurs in every subsequent integer dilation. For  $T_G$ , the sum is over nonnegative lattice points. All series here are formal series with finitely many contributions to each coefficient.

Partition the lattice points  $x$  of  $\mathbb{R}^V$  by the set  $R$  of strictly negative coordinates, and put  $S = V \setminus R$ . The negative coordinates contribute  $(t/(1-t))^{|R|}$ . The nonnegative coordinates contribute  $(1-t) \text{Ehr}_{T_{G[S]}}(t)$ , since independent sets restricted to  $S$  give precisely the gauge of the induced graph. Using (18.4),

$$\begin{aligned} \text{Ehr}_{D_G}(t) &= \frac{1}{1-t} \sum_{R \subseteq V} \left( \frac{t}{1-t} \right)^{|R|} (1-t) \text{Ehr}_{T_{G[V \setminus R]}}(t) \\ &= \frac{\sum_{R \subseteq V} t^{|R|} p_{G[V \setminus R]}(t)}{(1-t)^{n+1}}. \end{aligned} \quad (18.7)$$

In the numerator, a fixed matchable set  $W$  of size  $2k$  occurs once for each  $R \subseteq V \setminus W$ . Its contribution is  $t^k(1+t)^{n-2k}$ . Hence:

**Theorem 18.2** (Height-two root identity). *For every bipartite graph  $G$ ,  $D_G$  is reflexive and*

$$h^*(C_{P_G}; t) = h^*(D_G; t) = \sum_{k \geq 0} |\mathcal{M}_k(G)| t^k (1+t)^{n-2k} = h_{P_G}(t). \quad (18.8)$$

*This is a direct height-two proof of the corresponding identity in Part I (Corollary 4.2), independent of its general preorder fiber argument.*

**Remark 18.3** (A normalization trap). The symmetric edge polytope of type B in Ohsugi–Tsuchiya includes all signed edge sums. Its  $\gamma$ -polynomial is  $p_G(4u)$  [8, 9]. Our  $D_G$  has only the positive edge sums, and its  $\gamma$ -polynomial is  $p_G(u)$ . These are different polytopes. The quick check is  $h_1(P_G) = n + |E|$ , not  $n + 4|E|$ .

## 19 Cactus graphs: matrices, stability, and interlacing

A cactus graph is one in which every edge belongs to at most one simple cycle. Distinct cycles may meet in a vertex, but not in an edge; isolated vertices and disconnected components are allowed. In the bipartite case every cycle has even length. Univariate real-rootedness of  $p_G$  for these graphs is already contained in Ohsugi–Tsuchiya [9, Theorem 7.2]. The purpose here is a concrete matrix proof that retains a multivariate and weighted structure.

### 19.1 A unit-modulus certificate for every matchable minor

Let  $\zeta = e^{\pi i/3}$ , so  $\zeta^2 - \zeta + 1 = 0$  and  $|1 - \zeta| = |1 - \bar{\zeta}| = 1$ . Form an  $A$ -by- $B$  matrix  $K$  supported on the edges of  $G$ . Initially every edge entry is one. For each cycle of length  $2r$ , choose one of its edges and replace that entry by  $(-1)^r \zeta$ . These choices do not conflict because cycles are edge-disjoint. All nonzero entries are sixth roots of unity.

**Lemma 19.1** (The minor identity). *For  $X \subseteq A$  and  $Y \subseteq B$  with  $|X| = |Y|$ ,*

$$|\det K_{X,Y}|^2 = \begin{cases} 1, & G[X \cup Y] \text{ has a perfect matching,} \\ 0, & \text{otherwise.} \end{cases} \quad (19.1)$$

*Proof.* If there is no perfect matching, every term in the determinant is zero. Otherwise fix a perfect matching  $M_0$  of the induced subgraph. The symmetric difference of  $M_0$  with another perfect matching is a disjoint union of  $M_0$ -alternating cycles. In a cactus, all  $M_0$ -alternating cycles are vertex-disjoint: if two shared a vertex, the unique matching edge there would have to belong to both cycles, contrary to edge-disjointness. Conversely, flipping any chosen subcollection of these alternating cycles gives a perfect matching. Thus all determinant terms are indexed by independent choices of cycle flips.

For an alternating cycle of length  $2r$ , the ratio of the products of edge entries in its two matching halves is  $\Gamma = (-1)^r \zeta$  or  $(-1)^r \bar{\zeta}$ . There is exactly one modified entry on that cycle; modifications on other cycles are not on its edges. The permutation sign changes by  $(-1)^{r-1}$  when the cycle is flipped. Factoring out the unit-modulus term for  $M_0$ , the determinant therefore factors as

$$\text{a complex unit} \prod_{\text{alternating cycles}} (1 + (-1)^{r-1} \Gamma).$$

Each factor is  $1 - \zeta$  or  $1 - \bar{\zeta}$ , hence has absolute value one. The determinant has absolute value one, proving the claim.  $\square$

**Theorem 19.2** (Determinant and spectral factorization). *Let  $\nu$  be the maximum matching size of a bipartite cactus graph, and let  $\lambda_1, \dots, \lambda_\nu$  be the positive eigenvalues of  $KK^*$ , with multiplicities. Then*

$$p_G(u) = \det(I + uKK^*) = \prod_{j=1}^{\nu} (1 + \lambda_j u), \quad (19.2)$$

$$h_{P_G}(t) = (1 + t)^{n-2\nu} \prod_{j=1}^{\nu} ((1 + t)^2 + \lambda_j t). \quad (19.3)$$

*In particular, all roots of  $p_G$  and  $h_{P_G}$  are real and negative.*

*Proof.* The principal-minor expansion and Cauchy–Binet give

$$\det(I + uKK^*) = \sum_{k \geq 0} u^k \sum_{\substack{X \subseteq A, Y \subseteq B \\ |X|=|Y|=k}} |\det K_{X,Y}|^2.$$

Lemma 19.1 turns its coefficients into  $|\mathcal{M}_k(G)|$ . The largest nonzero minor has size  $\nu$ , so  $\text{rank } K = \nu$ . The Hermitian positive-semidefinite matrix  $KK^*$  has exactly  $\nu$  positive eigenvalues, proving (19.2). Substitute in Theorem 13.2 to get (19.3). Each quadratic factor has discriminant  $\lambda_j(\lambda_j + 4) > 0$  and two negative roots.  $\square$

The distinction between a cactus and a general series-parallel graph is now sharp in a useful sense: the connected graphs  $\Theta_m$  still have treewidth two, but for  $m \geq 3$  their cycles overlap along paths and their support polynomials fail real-rootedness dramatically.

## 19.2 A stable multivariate refinement

Let

$$H_G = \begin{pmatrix} 0 & K \\ K^* & 0 \end{pmatrix}.$$

This is an  $n$ -by- $n$  Hermitian matrix indexed by all vertices. For a vertex set  $W$ , its principal minor is zero if the two shores have unequal sizes. When  $|W \cap A| = |W \cap B| = k$ , it equals  $(-1)^k |\det K_{W \cap A, W \cap B}|^2$ . Expanding a determinant along its diagonal entries therefore proves

$$\Phi_G(\mathbf{z}) := \det(\text{diag}(z_v) - H_G) = \sum_{W \text{ perfectly matchable}} (-1)^{|W|/2} \prod_{v \notin W} z_v. \quad (19.4)$$

In particular, the right side has real coefficients and does not depend on the particular chosen edges or phases in the construction.

**Theorem 19.3** (Stability and deletion interlacing). *For a bipartite cactus graph,  $\Phi_G$  is real stable: it is nonzero whenever  $\text{Im } z_v > 0$  for every  $v$ . Its diagonal specialization*

$$q_G(x) := x^n p_G(-x^{-2}) = \det(xI - H_G) \quad (19.5)$$

*is real-rooted, and  $q_{G-v}$  interlaces  $q_G$  for every vertex  $v$ .*

*Proof.* If  $(\text{diag}(\mathbf{z}) - H_G)w = 0$  for  $w \neq 0$ , taking the imaginary part of its scalar product with  $w$  would give

$$0 = \sum_v \text{Im}(z_v) |w_v|^2 > 0,$$

because  $w^* H_G w$  is real. This proves stability. The diagonal identity follows from (19.4); it denotes a polynomial even at  $x = 0$ . Deleting  $v$  gives the principal submatrix of  $H_G$ . Its characteristic polynomial is  $q_{G-v}$  by the same minor identity, regardless of whether a previously modified edge survives as a bridge. The eigenvalues of a Hermitian principal submatrix interlace those of the whole matrix: the variational characterization of the  $j$ th eigenvalue, restricted to the coordinate hyperplane omitting  $v$ , gives the two inequalities between consecutive eigenvalues. This proves the stated polynomial interlacing.  $\square$

The signed complementary-support polynomial in (19.4) is the stable object. We are not claiming that the unsigned polynomial  $\sum_W \prod_{v \in W} z_v$  is stable; already  $1 + z_a z_b$  for a single edge vanishes at  $z_a = z_b = i$ .

[Added 29 September 2026, batch 42: Part III shows that this construction is exhaustive. Only cacti carry a matrix satisfying (19.1), and on a cactus every such matrix is gauge-equivalent to one of  $2^{\beta(G)}$  sixth-root matrices, the choice  $(-1)^r \zeta$  being allowed on each cycle as well (Theorems 29.1 and 30.3). Real stability of the signed polynomial extends beyond cacti (Theorem 37.1).]

## 19.3 Positive weights and a probabilistic consequence

Given positive vertex weights  $w_v$ , replace  $K$  by  $K_w = \text{diag}(\sqrt{w_a}) K \text{diag}(\sqrt{w_b})$ . The minor identity gives

$$\sum_{W \text{ perfectly matchable}} u^{|W|/2} \prod_{v \in W} w_v = \det(I + u K_w K_w^*). \quad (19.6)$$

Thus the weighted polynomial is real-rooted as well, and its  $\gamma$ -transform has the same negative-root factorization. This is a weighted graph refinement; no weighted preorder-polytope interpretation is presumed.

For the original unweighted preorder polytope, choose a lattice point uniformly and let  $S$  be its support size. Its probability generating function is  $h_{P_G}(t)/h_{P_G}(1)$ . Equation (19.3) identifies its distribution with a sum of  $n - 2v$  independent Bernoulli variables of parameter  $1/2$  and  $v$  independent variables  $Y_j \in \{0, 1, 2\}$  with

$$\Pr(Y_j = 0) = \Pr(Y_j = 2) = \frac{1}{4 + \lambda_j}, \quad \Pr(Y_j = 1) = \frac{2 + \lambda_j}{4 + \lambda_j}.$$

Hence

$$\mathbb{E}S = \frac{n}{2}, \quad \text{Var}(S) = \frac{n - 2v}{4} + \sum_{j=1}^v \frac{2}{4 + \lambda_j}. \quad (19.7)$$

This is an equality in distribution, not a claimed bijection assigning independent coordinates to individual lattice points.

## 20 Crowns: explicit formulas with simple negative roots

Let the crown  $P_r$  have lower elements  $a_1, \dots, a_r$  and upper elements  $b_1, \dots, b_r$ , with comparisons  $a_i < b_i$  and  $a_{i+1} < b_i$ , where indices are cyclic. For  $r \geq 2$ , its comparability graph is  $C_{2r}$ . The graph polynomial formula below is known [7, Corollaries 3.4 and 3.14]; we derive it to identify precisely the polynomials in the geometric obstruction that follows.

A proper induced subgraph of a cycle is a forest, so each of its matchable vertex sets has a unique perfect matching. The whole even cycle has two perfect matchings, but contributes only one vertex set. Therefore

$$p_{C_{2r}}(u) = M_{C_{2r}}(u) - u^r = \sum_{k=0}^r \frac{2r}{2r - k} \binom{2r - k}{k} u^k - u^r. \quad (20.1)$$

To check the matching coefficient, distinguish one of the  $k$  selected edges. There are  $2r$  choices of its position and  $\binom{2r - k - 1}{k - 1}$  choices of positive intervening gap lengths; divide by  $k$  to forget the distinction. The  $k = 0$  case contributes one. The top matching coefficient is two and becomes one after the correction.

Put

$$L_{\pm}(t) = \frac{1 + t \pm \sqrt{1 + 6t + t^2}}{2}.$$

The cycle matching recurrence, or the two characteristic roots of the path recurrence, gives

$$h_{P_r}(t) = L_+(t)^{2r} + L_-(t)^{2r} - t^r. \quad (20.2)$$

The radical expression is a polynomial: its conjugate terms cancel all odd powers of the square root. In particular,

$$|Q(P_r) \cap \mathbb{Z}^{2r}| = (1 + \sqrt{2})^{2r} + (1 - \sqrt{2})^{2r} - 1. \quad (20.3)$$

For  $r = 3$ , this gives the six-element crown vector  $(1, 12, 48, 75, 48, 12, 1)$  and total 197 of Part I's table in Section 8.

**Proposition 20.1** (Explicit location of every crown root). *For  $r \geq 2$ , all  $r$  roots of  $p_{C_{2r}}$  are simple and less than  $-1/4$ . They are*

$$u = -\frac{1}{4 \cos^2 \theta}, \quad 0 < \theta < \frac{\pi}{2}, \quad \cos(2r\theta) = \frac{(-1)^r}{2}. \quad (20.4)$$

All  $2r$  roots of  $h_{P_r}$  are simple and belong to

$$(-3 - 2\sqrt{2}, -1) \cup (-1, -3 + 2\sqrt{2}).$$

*Proof.* The roots of  $z^2 - z - u = 0$  are  $z_{\pm} = (1 \pm \sqrt{1 + 4u})/2$ , and  $M_{C_{2r}}(u) = z_+^{2r} + z_-^{2r}$ . Under the substitution in (20.4), these become  $z_{\pm} = e^{\pm i\theta}/(2 \cos \theta)$ . Hence

$$M_{C_{2r}}(u)/u^r = 2(-1)^r \cos(2r\theta).$$

The correction in (20.1) gives exactly the angle equation. As  $2r\theta$  traverses  $(0, r\pi)$ , each interval  $(j\pi, (j+1)\pi)$  contains one solution. Thus there are exactly  $r$  distinct roots, all strictly below  $-1/4$ . Each gives two negative reciprocal roots through

$$t = \frac{1 - 2u \pm \sqrt{1 - 4u}}{2u}.$$

This map has no double root for  $u < 0$ . Its endpoint behavior as  $u \rightarrow -1/4$  and  $u \rightarrow -\infty$  gives the two asserted open intervals. No roots from different  $u$  can coincide, because  $u = t/(1+t)^2$  recovers the original value.  $\square$

## 21 Induced crowns obstruct every lattice triangulation

A simplicial complex is *flag* if every set of pairwise adjacent vertices is a face. A *minimal nonface* is a set that is not a face although all its proper subsets are faces. Flagness is equivalent to all minimal nonfaces having size two. We show that the canonical root model has forced minimal nonfaces of unbounded size, even when its  $h^*$ -polynomial has only simple negative roots.

### 21.1 An exposed circuit face

Let a finite preorder  $P$  contain an induced crown on distinct elements  $a_i, b_i$  ( $1 \leq i \leq r$ ),  $r \geq 3$ . Thus the only distinct-element comparisons among these  $2r$  points are the crown comparisons. Consider  $C_P$  as in Theorem 13.6. Give a linear functional the coordinate values  $-1/2$  at the  $a_i$ ,  $1/2$  at the  $b_i$ , and zero on all other elements. Every coordinate root has value at most  $1/2$ , and every difference root has value at most one. Equality occurs precisely at the crown roots

$$v_i = \mathbf{e}_{b_i} - \mathbf{e}_{a_i}, \quad w_i = \mathbf{e}_{b_i} - \mathbf{e}_{a_{i+1}}. \quad (21.1)$$

They therefore span an exposed face  $F$  of  $C_P$ .

The coordinate equations show that the unique affine dependence, up to scale, is

$$\sum_{i=1}^r v_i = \sum_{i=1}^r w_i. \quad (21.2)$$

Indeed, a dependence among the cycle edge vectors forces consecutive coefficients alternately equal and opposite. There is one free scalar, and the coefficients sum to zero. Hence  $F$  has dimension  $2r - 2$ , its  $2r$  vertices form an affine circuit, and its Radon partition consists of  $U = \{v_1, \dots, v_r\}$  and  $W = \{w_1, \dots, w_r\}$ .

After negating the  $a_i$  coordinates, the vertices of  $F$  are 0–1 edge-incidence vectors. A convex combination of 0–1 vectors can be integral only if all vectors with positive coefficient agree in every coordinate: an integer coordinate in  $[0, 1]$  is an endpoint, forcing agreement. Thus

$$F \cap \mathbb{Z}^n = \{v_1, \dots, v_r, w_1, \dots, w_r\}. \quad (21.3)$$

There are no additional lattice points with which to subdivide this face.

## 21.2 The nonflag obstruction does not assume regularity

**Lemma 21.1.** *Every triangulation of  $F$  using its vertices has a minimal nonface of size  $r$ .*

*Proof.* Every proper subset  $T$  of  $U$  spans a simplex face of  $F$ . To see this, assign height zero to  $T$ , height  $-r$  to  $U \setminus T$ , and height  $-(r - |T|)$  to  $W$ . The sums of the assigned heights on the two halves are equal, so the unique affine relation (21.2) is respected. There is consequently an affine functional with exactly these values on the vertices. Its maximum face is precisely  $T$ . The vertices in  $T$  are affinely independent because a proper subset of an affine circuit is independent. The same argument works for proper subsets of  $W$ .

On the other hand,  $\text{conv}(U)$  and  $\text{conv}(W)$  meet in their relative interiors at the common average in (21.2). They cannot both be simplices of one triangulation: simplices with disjoint vertex sets may not intersect in a point which is not a common face. At least one of  $U, W$  is therefore missing. All of its proper subsets are forced simplex faces, so it is a minimal nonface of size  $r$ .  $\square$

A lattice triangulation of  $C_P$ , or of its boundary, restricts to a triangulation of its exposed face  $F$ . By (21.3) the restriction has no extra vertices. Every proper subset used in the lemma is already a simplex face of  $C_P$  and cannot disappear under such a triangulation. A missing half cannot occur elsewhere as a simplex, since its geometric simplex lies in  $F$ . This proves the triangulation part of Theorem 13.6 for *all* lattice triangulations, not merely coherent, pulling, or regular ones. Since  $r \geq 3$ , the missing half is a clique in the one-skeleton, so the triangulation is not flag.

For a height-two graph, the hypothesis is exactly an induced cycle of length at least six in the bipartite comparability graph. Thus chordal bipartiteness (absence of such induced cycles) is a necessary condition for a flag lattice triangulation of  $D_G$ . Sufficiency for this particular root model is not proved here. Results about a fully signed symmetric edge polytope cannot be substituted without checking the different configuration.

## 21.3 An indispensable degree- $r$ toric relation

Fix any field  $\mathbb{k}$ . Let  $\mathcal{A} = C_P \cap \mathbb{Z}^n$ , and consider the toric homomorphism

$$\mathbb{k}[X_a : a \in \mathcal{A}] \longrightarrow \mathbb{k}[s, z_1^{\pm 1}, \dots, z_n^{\pm 1}], \quad X_a \longmapsto sz^a.$$

Its kernel  $I_{\mathcal{A}}$  is homogeneous in total degree. Define

$$M_U = \prod_{i=1}^r X_{v_i}, \quad M_W = \prod_{i=1}^r X_{w_i}.$$

Equation (21.2) gives  $M_U - M_W \in I_{\mathcal{A}}$ .

**Proposition 21.2.** *The binomial  $M_U - M_W$  occurs, up to a nonzero scalar, in every binomial generating set of  $I_{\mathcal{A}}$ . For every monomial order, its leading monomial is a minimal generator of the corresponding initial ideal. Consequently neither a quadratic binomial generating set nor a quadratic Gröbner basis is possible when  $r \geq 3$ .*

*Proof.* Consider the monomial fiber with total degree  $r$  and vector sum  $\sum_i v_i$ . The exposing functional of  $F$  has value  $r$  on this sum, and every lattice point of  $C_P$  has value at most one. Every factor of any monomial in the fiber must therefore be a point of  $F$ . By (21.3), only its cycle vertices may occur. After the coordinate sign change, the required sum has a one at every cycle



vertex. Its decompositions into edges are exactly the two perfect matchings of the cycle. Thus the fiber contains only  $M_U$  and  $M_W$ .

Moreover, every proper divisor of either monomial has a singleton fiber. The same face-functional argument restricts any competing factorization to cycle edges, while its endpoint set is a proper subset of the cycle. The induced subgraph is a forest and has at most one perfect matching. No proper divisor can therefore be a term of a nonzero toric binomial.

A binomial generating set connects each monomial fiber by multiplying its generators by monomials. In this two-element fiber, a connection cannot come from a generator of smaller degree, because that would require a proper divisor with another monomial in its fiber. The binomial itself is therefore indispensable. For an initial ideal, the leading monomial of this binomial has no proper divisor in the initial ideal by the same singleton-fiber argument. It is a minimal generator of degree  $r$ .  $\square$

**Corollary 21.3** (A spectral–triangulation separation). *For every  $r \geq 3$  there is a height-two poset on  $2r$  elements whose canonical root polytope has a real-rooted  $h^*$ -polynomial with simple zeros, but every lattice triangulation has a minimal nonface of size  $r$  and every toric initial ideal has a minimal generator of degree  $r$ .*

*Proof.* Take the crown  $P_r$  and combine Theorem 18.2, Proposition 20.1, and the preceding results.  $\square$

**The existential flag question is different.** These obstructions apply to triangulations and the toric configuration of the canonical  $C_P$ . Athanasiadis–Chapoton Conjecture 5.1(d) asks whether *some* flag simplicial polytope has  $h$ -polynomial  $h_P$ . A different polytope might do so even when every lattice triangulation of  $C_P$  fails. The present result does not disprove that conjecture.

## 22 Exact verification and reproducibility

Part II ships three source programs and their recorded outputs, under the names given in Section 22.4; below they are called by their delivered names `verify.py`, `verify_theta.py` and `direct_counterexample.cpp`. (Part I’s own verifier, `code/verify.py` in this directory, is a different program; see Section 9.) No numerical root finder is used by the certificate programs. Python computations use integers, rational numbers, or pairs of integers in  $\mathbb{Z}[\zeta]$ ; the C++ program uses exact finite state sets and integer counts. The proofs establish the infinite-family results. The programs check finite instances and offer independent routes to the explicit counterexamples.

### 22.1 Independent representations, not repeated substitutions

The program `verify.py` represents a bipartite graph by the bitmask neighborhood of each upper vertex. It computes  $p_G$  by adding upper vertices one at a time and keeping a set of pairs of covered-shore masks. Different matchings with the same covered vertices are merged. For the 17-element graph it independently runs the ordered-board recurrence from Section 17; the entire 10-by-9 polynomial table is saved in `data/02-height-two-board_certificate.csv`.

For every one of the 512 labeled bipartite graphs with three vertices on each shore, the program directly enumerates lattice points from the ideal inequalities and compares their support polynomial with the matching-support transform. It independently compares the cardinality of  $\mathcal{R}_G$  with  $p_G(1)$ . A further 24 graphs with shores of sizes four and three use the fixed seed 20260928.

The cactus tests do not infer determinants from matching counts. They compute determinants in  $\mathbb{Z}[\zeta]$  by subset-memoized Laplace expansion, using  $\zeta^2 = \zeta - 1$ . If  $z = a + b\zeta$ , its exact squared modulus is  $a^2 + ab + b^2$ . The code compares that value with an independently enumerated matching-support indicator for every square minor. The tested graphs are cycles of lengths 4, 6, 8, 10, a square and a hexagon sharing a vertex, and that cactus with a two-edge bridge extension. There are 936 square minors in total, including empty minors.

## 22.2 The theta-family audit

The separate program `verify_theta.py` computes (13.4) by binomial expansion and compares it with the covered-shore enumeration for  $1 \leq m \leq 9$ . For  $1 \leq m \leq 4$ , it also enumerates the original lattice points directly from the ideal inequalities. In particular, the eight-element specialization has exactly 1281 lattice points with the coefficients displayed in Section 16.

A rational-arithmetic Sturm implementation independently checks the number of real roots of  $p_m$  for  $1 \leq m \leq 20$ . Its polynomial remainders use exact fractions, normalized only by positive factors. The variations at  $-\infty$  and  $+\infty$  are stored alongside every polynomial. These checks agree with the theorem, but the theorem does not extrapolate from them: (16.1) and the two sign-change arguments prove the count for all  $m$ . The small example's exact Laguerre defect is also recomputed.

## 22.3 All 8,408,566 points of the larger example

The C++17 program `direct_counterexample.cpp` is a third route, not a translation of the polynomial transform. For each lower-zero set  $Z$ , each upper coordinate has its own optional private unit, and there is one optional unit for each element of  $Z$ , assignable only to an adjacent upper coordinate. The set of resulting upper-load vectors is deduplicated.

This is exact because the upper-demand inequalities

$$b(T) \leq |T| + |N(T) \cap Z|$$

are precisely Hall's conditions for assigning demand clones to the upper private resources and the lower resources. Integral demand vectors are therefore realizable by these allocations if and only if they satisfy the original ideal inequalities. Counting the positive coordinates and the fixed lower support gives  $h_P$  directly.

Each of the eight upper coordinates is encoded in three bits. Its maximum load is at most its degree plus one, at most seven in this graph, so the encoding is injective and never overflows a digit. A generation array deduplicates states without hash-dependent behavior. The resulting coefficient vector agrees exactly with (17.5), and its sum is 8 408 566. The source is included; a precompiled executable is not.

| Certificate               | Recorded verification                                                                                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------|
| All labeled 3-by-3 graphs | 512 direct support-polynomial and demand-count comparisons.                                                              |
| Additional 4-by-3 graphs  | 24 direct comparisons with a recorded seed.                                                                              |
| Crown graphs              | Formula checks for $2 \leq r \leq 8$ ; direct lattice checks for $r \leq 4$ .                                            |
| Cactus matrix certificate | 936 exact square-minor norm comparisons.                                                                                 |
| Theta family              | Support enumeration through $m = 9$ , direct lattice enumeration through $m = 4$ , exact Sturm counts through $m = 20$ . |
| 17-element example        | Two graph-polynomial algorithms and a separate direct enumeration of all 8 408 566 lattice points.                       |

All the listed checks passed in the supplied run. None is a Lean proof or a substitute for independent mathematical review.

## 22.4 Build commands

The manuscript's own build commands, run from its package directory, were `python3` on `code/verify.py` and then on `code/verify_theta.py`, the compiled C++ program, and `latexmk` for the PDF. In this report every delivered file name carries the prefix `02-height-two-`:

- programs: `code/02-height-two-verify.py`, `code/02-height-two-verify_theta.py`, `code/02-height-two-direct_counterexample.cpp` and `code/02-height-two-Makefile`;
- recorded outputs: `data/02-height-two-verification.json`, `data/02-height-two-theta_verification.json`, `data/02-height-two-direct_counterexample.json` and `data/02-height-two-board_certificate.csv`.

The programs still use the delivered names internally: `verify_theta.py` imports a module `verify`, which in this directory would be Part I's verifier, and the `Makefile`'s `verify` target would run Part I's verifier and overwrite Part I's data. Therefore rerun them only in a scratch copy with the delivered names restored, from this report's directory:

```
W=/path/to/scratch; mkdir -p "$W/02/code" "$W/02/data"
for f in code/02-height-two-*; do cp "$f" "$W/02/code/${f#code/02-height-two-}";
done
for f in data/02-height-two-*; do cp "$f" "$W/02/data/${f#data/02-height-two-}";
done
mv "$W/02/code/Makefile" "$W/02/Makefile"; cd "$W/02"
python3 code/verify.py
python3 code/verify_theta.py
c++ -O3 -std=c++17 -Wall -Wextra -pedantic code/direct_counterexample.cpp \
-o direct_counterexample
./direct_counterexample > data/direct_counterexample.json
```

The Python programs require Python 3.10 or later and only its standard library. The C++ program needs a C++17 compiler. This article is rebuilt as described in the report's README; its preamble adds TikZ to Part I's packages. No external image or font files are required.

## 23 Proof dependencies and a formalization plan

The main route has a short dependency structure:

$$\begin{aligned} &\text{two published root-polytope inputs} \implies \text{counting lemma} \\ &\implies \text{height-two formula} \implies \text{theta root distribution.} \end{aligned}$$

The canonical root identity is proved separately by orthants. The cactus minor proof and the crown circuit proof are independent of one another. The crown toric result depends only on an explicit exposed lattice face, not on the general root-polytope theorem of Part I.

The following checks are particularly important in an independent audit. First, all matching counts concern *covered vertex sets*; confusing these with individual matchings destroys both the theta and crown formulas. Second, a height-two poset here has singleton elements, not

nontrivial preorder-equivalence blocks. Third,  $h_P$  counts support sizes of lattice points; it is not initially the Ehrhart numerator of  $Q(P)$ . The identity with an Ehrhart numerator concerns the different polytope  $C_P$ . Fourth, a nonflag triangulation obstruction for  $C_P$  does not preclude an unrelated flag polytope having the same  $h$ -vector.

A staged formalization could begin with finite graphs on two finite types, covered-shore pairs, and the elementary theta-support classification. The polynomial identity (13.4), the positive-interval rearrangement, and the Descartes/interval root count then form a compact algebraic project once the required real-polynomial lemmas are available. This would verify the entire graph-family theorem without any polytope formalization.

Transferring that theorem to preorder support polynomials requires either formalizing Lemma 14.1 and its classical inputs, or finding a direct combinatorial bijection. A finite-only route to the eight-element counterexample is different: certify its bounded ideal-inequality enumeration, then verify its polynomial and negative Laguerre value. That route need not formalize the general graph-to-polytope bridge, but it also does not establish the all-height-two theorem.

For the cactus result, exact sixth-root arithmetic can be implemented as a quadratic extension with relation  $z^2 - z + 1 = 0$ . The combinatorial symmetric-difference lemma is a natural intermediate target. The analytic stability proof can then be isolated as a general Hermitian determinant lemma. For the crown obstruction, the finite affine dependence and the two-monomial fiber give a clean algebraic certificate before tackling full triangulation definitions. No uncompiled Lean source is presented as if it had been checked.

## 24 Further research questions and topics

The results replace an undifferentiated real-rootedness question with several sharper problems. The following questions are proposed directions, not assertions that they have been settled by this article. Except for explicitly attributed conjectures, they are directions arising from these proofs, not claims that no answer exists anywhere in the literature.

**Research question 1** (Extend the matching-support interpretation). Does the  $\gamma$ -vector of a larger class of posets admit a similarly concrete nonnegative interpretation? Height three and preorders with nontrivial equivalence classes are natural next cases. Determine exactly which step of the lower-zero-set argument survives weighted blocks.

A successful extension must account for the fact that a coordinate in a nontrivial equivalence block need not be individually bounded by one. Simply replacing the preorder by its quotient poset loses information. [Added 29 September 2026, batch 44: answered in Part IV. For every finite preorder,  $\gamma_k$  counts ordered pairs of disjoint  $k$ -element sets joined by a  $\tau$ -increasing bijection (Theorem 48.2); in height two these are Part II's matchable vertex sets (Section 55.3). The step that survives is the zero-set partition, taken over all zero coordinates rather than a minimal shore (Lemma 50.1), and it allows coordinates larger than one. Positive block formulas are Theorems 54.1 and 54.3. See Section 46.3.]

**Research question 2** (A purely combinatorial bridge). Construct an explicit bijection between  $\mathcal{R}_G$  and perfectly matchable vertex subsets, or an equally elementary proof of their equal cardinality. Can the choice be made compatible with induced subgraphs, vertex deletion, or positive weights?

Such a construction would remove the two root-polytope inputs from the height-two theorem and substantially reduce its formalization burden. The hypertree-vector bijection in Section 14

does not itself solve this problem. [Added 29 September 2026, batch 44: still open. Part IV's general theorem uses the same counting lemma and constructs no such bijection; its Research question 20 asks for one that also preserves the receiver support.] [Added 30 September 2026, batch 64: the bijection is constructed in Part IX (Theorem 135.2), with an elementary proof of equal cardinality (Corollary 139.2). Compatibility with positive weights holds for support weights (Theorem 145.1); compatibility with vertex deletion and induced subgraphs is proved only for receivers outside the support and suppliers outside the selected set (Corollaries 140.3 and 140.2), and no deterministic rule is equivariant under all automorphisms (Proposition 144.1). See Section 132.3.]

**Research question 3** (Which graph structures preserve real-rootedness?). Characterize useful hereditary classes of bipartite graphs for which  $p_G$  is real-rooted. Cacti are a positive class, while the treewidth-two family  $\Theta_m$  shows that series-parallel structure alone is inadequate. What happens under vertex sums, edge subdivisions, or controlled cycle intersections?

The precise boundary between one shared vertex and a shared path is worth studying. The present examples do not imply that every noncactus graph fails real-rootedness. [Added 29 September 2026, batch 42: indeed not every noncactus fails. Part III shows that  $K_{2,3}$  and every graph obtained from it by attaching trees have a real stable  $\Phi_G$ , hence a real-rooted  $p_G$  (Corollary 37.4, Section 39). The question itself remains open; see Section 26.3.] [Added 29 September 2026, batch 55: vertex sums preserve and reflect *stability* (Part VI, Theorem 93.2). Edge subdivision can restore real-rootedness without stability:  $p_{\mathfrak{s}(3,3,3)}$  has nonreal zeros,  $p_{\mathfrak{s}(3,3,5)}$  is real-rooted and  $\Phi_{\mathfrak{s}(3,3,5)}$  is unstable (equation (92.5)). Real-rootedness alone under vertex sums is not treated.]

**Research question 4** (Beyond the sixth-root cactus matrices). Which bipartite graphs admit a complex matrix supported on their edges whose square-minor norms are exactly the matchability indicators? Is this condition equivalent to a useful multivariate stability property, or strictly stronger?

A positive answer for a class immediately gives positive-weight real-rootedness through Cauchy–Binet. The theta counterexamples show that such representations cannot exist for all bipartite graphs. [Added 29 September 2026, batch 42: answered in Part III. Exactly the cacti admit such a matrix (Theorem 29.1), and the condition is strictly stronger than real stability of  $\Phi_G$ :  $K_{2,3}$  and all its tree attachments are real stable without it (Theorem 37.1, Corollary 37.4). See Section 26.3.]

**Research question 5** (Asymptotic zero geometry for  $\Theta_m$ ). The real-root count is exact, but how are the nonreal roots distributed as  $m \rightarrow \infty$ ? Determine their limiting curves and scaling regimes, including the roots near zero and the large negative real root. What is the corresponding distribution after  $u = t/(1+t)^2$ ?

Formula (13.4) gives a concrete exponential-versus-polynomial equation from which such asymptotics may be developed; the present paper does not assert a limiting measure.

**Research question 6** (Flag triangulations of the canonical model). For height-two posets, does chordal bipartiteness suffice for  $D_G$  to admit a flag unimodular triangulation? More generally, characterize which induced configurations obstruct flagness beyond the crown circuits.

The necessity of chordal bipartiteness follows here. Sufficiency is a separate problem for  $D_G$ , even if related statements hold for other signed root configurations.

**Research question 7** (Existential flag realizations). Can the crown polynomial  $h_{P_r}$  be realized as the  $h$ -polynomial of a flag simplicial polytope by a construction unrelated to lattice triangulations of  $C_{P_r}$ ? Can one do this for all height-two posets?

This is the distinction needed to make progress on AC Conjecture 5.1(d), rather than only testing one canonical construction.

**Research question 8** (Exact toric degree bounds). How much of the largest indispensable degree of  $I_{C_P \cap \mathbb{Z}^n}$  is controlled by induced crowns? Is there an upper bound in terms of the largest such crown, or do other faces produce independent unbounded obstructions?

The theorem supplies a lower bound, not a characterization. Additional coordinate roots may change global toric relations while leaving the exposed-face obstruction intact.

**Research question 9** (Minimum counterexamples with checkable completeness). Give a fully reproducible exhaustive certificate for the smallest counterexample to preorder support real-rootedness, including nontrivial preorder blocks. The eight-element example is explicit, while Athanasiadis–Chapoton report successful tests through size seven [1]; a complete independent certificate for that lower bound is not supplied here.

A smaller search space restricted to height-two posets may be practical for identifying every minimal forbidden example and testing structural conjectures about their bipartite graphs.

**Research question 10** (Efficient exact enumeration and formal certificates). Develop a tree-decomposition algorithm for matchable *vertex-set* counts that avoids counting matching multiplicities. Can it output a small certificate that a proof assistant can check independently of the search procedure?

The theta family gives a useful stress test: treewidth is small, yet root behavior is not controlled by real-rootedness. Enumeration and spectral regularity should therefore be treated as distinct algorithmic issues.

## 25 Conclusion

The height-two identity identifies the  $\gamma$ -coefficients of preorder support polynomials with a classical graph invariant. Its value is not merely that it reproduces an isolated counterexample: it exposes a complete root-distribution family, a determinantal positive class, and a sharp distinction between root behavior and canonical triangulations.

The negative examples do not destroy  $\gamma$ -positivity in height two; that property is guaranteed by the matching-support formula. Likewise, real-rooted crown polynomials do not rescue flag lattice triangulations of the canonical root polytope. The remaining questions therefore have different mathematical mechanisms and require different constructions. The accompanying proofs and exact certificates are intended to make those mechanisms, and the limits of the novelty claims, available for direct independent review.



## Part III. Cactus rigidity: exact square-minor models and stability

*Part III was added on 29 September 2026 in batch 42 of ProveIt's incoming-report intake, from one manuscript written against Parts I and II as printed above. It answers Part II's Research question 4, "Beyond the sixth-root cactus matrices": the bipartite graphs that admit a complex matrix on their edges whose square-minor norms are exactly the matchability indicators are precisely the cacti, and this property is strictly stronger than real stability of the signed matching-support polynomial. Parts I and II are unchanged except for dated pointers to this addition. Section 26 records the provenance, fixes the notation and states what is answered.*

### 26 Provenance, conventions and what Part III answers

#### 26.1 One manuscript, one addition

Part III is batch-42 manuscript 05, *Cactus Rigidity for Matching-Support Determinants: Exact classification, a sharp approximation gap, and stability beyond vertex determinants*, an AI-assisted research manuscript prepared for Vladimir Reshetnikov and dated 29 September 2026 (24 US-Letter pages as delivered). It arrived as the archive `ProveIt_Cactus_Rigidity.zip` (inner directory `ProveIt_Cactus_Rigidity/`) in ProveIt commit 8315d24e3 and was placed in 3609d0473. It is pinned to ProveIt commit 9754e8360 (the full hash is printed in Section 27) and records the Git blob 8739a224fa8956adab5d8df9f2d5012a6a220959 of this report's `article.tex`. That blob is Parts I and II exactly as printed above: the file was last changed by the batch-39 write 6a354f52f and is unchanged at the pin and at the placement commit. The manuscript therefore cites Part II in its printed numbering; its "Section 19", "Section 24" and "Theorem 13.2" are Section 19, Section 24 and Theorem 13.2, and are replaced here by cross-references. The manuscript's program, recorded outputs, build file and notes are shipped beside Part I's files with the prefix 03-cactus- (Part I's unprefixed files count as 01, Part II's are 02-); Section 40.3 lists them. Its manuscript, delivery README and PDF are not shipped and survive in the arrival commit; its checksum list `SHA256SUMS.txt` was verified (10/10) at placement and retired.

The manuscript is printed here as Sections 27 to 45: its Section  $k$  is Section  $k + 26$  here, its Appendices A and B are Sections 44 and 45, and its theorem numbers shift accordingly (its Theorem 3.1 is Theorem 29.1); its equations, numbered (1)–(26) consecutively there, are numbered by section here, as in the rest of this report. Its nine research questions continue Part II's numbering as Research questions 11–19. Every result, proof, remark, limitation, question and non-claim of the manuscript is printed; three displays that repeat Part II's formulas are printed once, in Part II (Section 26.4). Its title-page abstract and status note are reproduced as delivered.



**Abstract of manuscript 05 (as delivered).** For a finite bipartite graph, can one assign unit complex numbers to its edges so that every square minor has squared absolute value equal to the indicator that its vertex support is perfectly matchable? The ProveIt preorder-polytope report constructs such assignments for cactus graphs and explicitly asks for the exact boundary of the construction. We prove that the boundary is precisely the cactus class. Equivalently, the matching-support polynomial has a Hermitian vertex-diagonal determinant representation exactly for cacti; equivalently, the graph is the support pattern of a matrix over  $\mathbb{F}_3$  with no vanishing structurally nonzero minor. A cactus of cycle rank  $\beta$  has exactly  $2^\beta$  phase-gauge classes of complex certificates.

The obstruction is quantitative. Every unit-phase matrix on every noncactus has a matchable minor whose squared absolute value differs from one by at least

$$\varepsilon_* = \frac{\sqrt{17} - 3}{2},$$

and this constant is attained on  $K_{2,3}$ . We obtain the classification and the uniform gap by reducing induced theta graphs to three explicit matrices. In contrast, the five-variable matching-support polynomial of  $K_{2,3}$  is real stable, as certified by three exact sums-of-squares identities. Attaching arbitrary trees produces infinitely many connected stable examples without a Hermitian vertex determinant. We also prove that bounded-order minor tests cannot recognize exact representability, even on subcubic graphs of treewidth two.

**Status (manuscript 05, as delivered).** Complete ordinary mathematical proofs are supplied for the stated results. This AI-assisted manuscript is unrefereed and has not been verified in Lean or another proof assistant. The cactus construction and its established real-rootedness consequences are prior work, not claimed here. The classification answers a specified repository question; worldwide priority for its formulations and refinements is not asserted.

**Priority and prior work (from the manuscript’s status and source notes).** The cactus sixth-root construction, its determinant factorization and cactus stability are Part II’s (Section 19); univariate cactus real-rootedness is Ohsugi–Tsuchiya’s [9, Theorem 7.2]; the height-two matching-support transform is Part II’s Theorem 13.2. The binary (forest) observation and the general phenomenon of stability without a determinant representation are not claimed as new; the latter is established elsewhere, for example for Vámos-type polynomials [15]. A related earlier characterization of binary fundamental transversal matroids [16] was located (publisher abstract only; its full text was not available); possible connections to older fundamental-transversal-matroid characterizations deserve further literature review. The source audit was targeted, not exhaustive, and does not establish the absence of an older equivalent theorem; no worldwide publication priority is claimed for the classification, the finite-field reformulations, the gauge count, the quantitative refinements or the stability formulations. The only specialized external theorem used is Brändén’s multiaffine Rayleigh criterion [12, Theorem 5.6], for the stability of  $K_{2,3}$ .

## 26.2 Notation of Part III

Part III uses Part II’s graph conventions:  $G = (A \sqcup B, E)$  is a finite simple bipartite graph with  $p = |A|$ ,  $q = |B|$ ,  $n = p + q$ ; a cactus is a graph in which every edge lies on at most one simple cycle;  $p_G$  and  $\mathcal{M}_k(G)$  count perfectly matchable *vertex sets*;  $\zeta = e^{\pi i/3}$ ;  $\nu$  is the matching number and  $\lambda_j$  are the positive eigenvalues of  $KK^*$ . Table 2 lists the symbols Part III adds or uses differently. One symbol of the manuscript was renamed: its theta graph  $\Theta(a, b, c)$  with three paths of lengths  $a, b, c$  is written  $\vartheta(a, b, c)$  here, because Part II’s  $\Theta_m$  is a different family with  $m$  paths. No normalization was changed.

Table 2: Symbols of Part III. The last column prints the tempting false reading, usually a Part I or Part II symbol with the same letter.

| Symbol                                           | Meaning in Part III                                                                                                                                                | Not to be confused with                                                                                                                                                                                                                                 |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\vartheta(a, b, c)$                             | a theta: three internally disjoint paths of lengths $a, b, c$ between two common endpoints; one length may be 1 (renamed from the manuscript's $\Theta(a, b, c)$ ) | Part II's $\Theta_m$ , which has $m$ paths, all of length three. Only $\Theta_3 = \vartheta(3, 3, 3) = \mathsf{T}$ is both; $\Theta_m$ for $m \neq 3$ is no $\vartheta(a, b, c)$ . Part II's "theta family" means $\Theta_m$ .                          |
| $E, D, \mathsf{T}$                               | the three cores $\vartheta(2, 2, 2) = K_{2,3}$ , $\vartheta(1, 3, 3)$ , $\vartheta(3, 3, 3)$                                                                       | the edge set $E$ ; Part II's polytope $D_G$ and its positive part $T_G$ .                                                                                                                                                                               |
| $\Phi_G(\mathbf{z})$                             | the signed complementary-support polynomial (28.2), defined for <i>every</i> bipartite $G$                                                                         | Part II defines $\Phi_G := \det(\text{diag}(\mathbf{z}) - H_G)$ for cacti; the two agree there (19.4). For a noncactus, $\Phi_G$ is <i>no</i> such determinant (Theorem 29.1).                                                                          |
| $\Phi_G(s, \dots, s)$                            | $s^n p_G(-s^{-2})$ , for every $G$                                                                                                                                 | Part II's $q_G(s)$ of (19.5), stated there for cacti only.                                                                                                                                                                                              |
| $K$ , faithful, $H$ , $H_K$                      | a phase matrix; faithful if (28.3) holds; a Hermitian vertex-indexed matrix; the bipartite Hermitian matrix of $K$                                                 | Part II's $K$ is <i>one</i> faithful certificate (choice $(-1)^r \zeta$ on every cycle) and its $H_G$ is $H_K$ for it; Part II's bipartite graph $H = (X \sqcup Y, E)$ ; Part I's triangulation $K$ . Part II's $D_G$ ; Part I's neighbourhoods $D_i$ . |
| $D_A, D_B$ , gauge                               | diagonal unitary row and column scalings                                                                                                                           | Part II's $D_G$ ; Part I's $\tau$ .                                                                                                                                                                                                                     |
| $T$                                              | a spanning forest                                                                                                                                                  | —                                                                                                                                                                                                                                                       |
| $\beta(G), c(G)$                                 | cycle rank $ E  -  A  -  B  + c(G)$ ; number of components                                                                                                         | —                                                                                                                                                                                                                                                       |
| $\delta_G(K), d(G), \varepsilon_*, d_0$          | minor defect of $K$ , its minimum over phases, the constant $(\sqrt{17} - 3)/2$ , the constant $(\sqrt{5} - 1)/2$                                                  | the vertex $d$ of $\Theta_m$ ; Part I's diagonal vertices $d_i$ .                                                                                                                                                                                       |
| $p, q, r$ in Lemma 36.6; $q$ in Proposition 35.3 | negated real inner products of unit vectors; a field size ( $\mathbb{F}_q$ )                                                                                       | the shore sizes $p =  A $ , $q =  B $ .                                                                                                                                                                                                                 |
| $I, J$                                           | a row set and a column set of a square minor                                                                                                                       | Part I's order ideals $I \in \mathcal{J}(\tau)$ ; Part II's independent sets.                                                                                                                                                                           |
| $U, V, W$ in (37.2)                              | the monomials $xy, xz, yz$                                                                                                                                         | $W$ as a perfectly matchable set elsewhere.                                                                                                                                                                                                             |
| $L$ in Theorem 38.1                              | an odd path length                                                                                                                                                 | Part I's shore index $i_L$ .                                                                                                                                                                                                                            |

## 26.3 What Part III answers in Part II

Status after Part III, as of 29 September 2026:

- **Research question 4 (“Beyond the sixth-root cactus matrices”) is answered.** A bipartite graph has a complex matrix on its edges whose square-minor norms are exactly the matchability indicators if and only if it is a cactus; equivalently,  $\Phi_G$  is a Hermitian vertex-diagonal determinant, or  $G$  carries a support-exact matrix over  $\mathbb{F}_3$  (Theorem 29.1). Part II’s sixth-root matrices are therefore all there is: every such matrix is gauge-equivalent to one of  $2^{\beta(G)}$  sixth-root matrices (Theorem 30.3). The condition is **strictly stronger** than real stability of  $\Phi_G$ , the multivariate property Part II proves for cacti (Theorem 19.3):  $\Phi_{K_{2,3}}$  and  $\Phi_G$  for every tree attachment to  $K_{2,3}$  are real stable with no such matrix (Theorem 37.1, Corollary 37.4). The obstruction is quantitative, with the sharp constant  $\varepsilon_*$  (Theorem 36.3). Graphs with real stable  $\Phi_G$  are not classified (Research question 11 below).
- **Part II’s remark after that question** (“the theta counterexamples show that such representations cannot exist for all bipartite graphs”) is consistent with Part III and gives a second, non-quantitative route for one core. *Merge note, not in manuscript 05:* a faithful matrix gives  $p_G(u) = \det(I + uKK^*)$  (Section 39), which is real-rooted, whereas  $h_{p_{\Theta_3}}$ , and hence  $p_{\Theta_3}$  (Proposition 15.1), is not real-rooted (Theorem 13.3); so  $T = \Theta_3$  has no faithful matrix. Every  $\Theta_m$  with  $m \geq 3$  contains  $\Theta_3$  as an induced subgraph. Part III’s Lemma 36.6 gives the quantitative form, and Lemma 31.1 explains why some core occurs in every noncactus.
- **Research question 3 (which structures preserve real-rootedness) is re-scoped, not answered.** Its remark notes that the examples of Part II do not imply that every noncactus fails real-rootedness. Part III supplies such noncacti:  $K_{2,3}$ , with  $p_{K_{2,3}}(u) = 1 + 6u + 3u^2$ , and every connected graph obtained from it by attaching trees has a real stable  $\Phi_G$  and hence a real-rooted  $p_G$  (Section 39). The question of useful hereditary classes stays open.
- Research question 1 (a  $\gamma$ -interpretation beyond height two) remains open; Research question 18 below proposes stable signed-support methods as one route. [Added 29 September 2026, batch 44: Research question 1 is answered in Part IV, by transport pairs rather than stability (Theorem 48.2); see Section 46.3.]

Part III refutes no statement of Parts I and II, and nothing in them is retracted.

## 26.4 Where the merge had to choose

- *Placement.* Part III follows Part II and precedes Part I’s appendices, whose letters and numbers are unchanged. The manuscript’s own Appendices A and B are printed as the ordinary Sections 44 and 45 of Part III.
- *Self-citation.* The manuscript cites this report as an external repository report (its reference “Repo”, with the pinned commit, Sections 19 and 24 and Theorem 13.2). Here those citations are replaced by references to Part II, and “the repository(’s)” by “Part II(’s)” where it means this report in running text; two headings, in Sections 27 and 39, keep the manuscript’s wording. The pin itself is kept in Section 27.
- *Duplicates.* The manuscript’s cactus factorization (Lemma 30.1) is Part II’s Lemma 19.1, by the same route; it is printed as the manuscript states it because it displays both admissible phases on each cycle, which Theorem 30.3 needs. The manuscript’s height-two formula, Cauchy–Binet identity and two spectral factorizations (its displays (23)–(26), in its Section 13) are Theorem 13.2 and Theorem 19.2; they are printed once, in Part II, and cited in Section 39. The

manuscript’s recall of the cactus stability argument (Section 37) is the proof of Theorem 19.3 and is kept as a short recall.

- *Notation.*  $\Theta(a, b, c)$  is renamed  $\vartheta(a, b, c)$  (Table 2); nothing else is renamed.
- *File names.* The delivered names (`code/verify.py`, `data/verification.json`, the `Makefile`) are replaced by the shipped names in Section 40, because `code/verify.py` in this directory is Part I’s verifier; the build commands are replaced by commands that run on a copy.
- *Bibliography.* The manuscript’s entries for Ohsugi–Tsuchiya’s matching-generating-polynomial paper and for Davis–Kohl are Part II’s references [9] and [7]; its five other external entries are appended.
- *Labels and front matter.* Labels carry the prefix `cac:`. The conclusion and the dependency ledger received labels, and Part II’s Research question 4 received the label `cac:q:beyond` so that Part III can cite it. The manuscript’s title page, abstract and status note are printed in this section; its table of contents is dropped.

## 27 The question, its provenance, and the answer

### 27.1 The repository question being resolved

The starting point is Part II of this report, *A Reflexive Root-Polytope Model for Preorder  $h$ -Polynomials*, whose height-two extension is entitled *Height-Two Preorder Polynomials: Matching supports, nonreal-root families, and crown obstructions*. Section 19 of Part II constructs a sixth-root-of-unity matrix for each bipartite cactus. Its Section 24 asks, under the heading “Beyond the sixth-root cactus matrices” (Research question 4), which bipartite graphs admit a matrix whose square-minor norms reproduce the matchability indicators, and whether this requirement is equivalent to multivariate stability.

This article answers both parts, with the precise meaning of the representation fixed below:

$$\boxed{\text{Exact minor representation} \iff \text{cactus graph} \implies \text{real stability},}$$

where the last implication is strict. The necessity direction is the main extension of Part II’s result. In particular, adding arbitrary complex phases does not extend the exact model to a larger graph class than the existing sixth-root construction.

The source is pinned to commit

9754e83603e223811b7515eb892f22c847144593.

The relevant `article.tex` has Git blob

8739a224fa8956adab5d8df9f2d5012a6a220959.

The main branch may subsequently change. All references to the source’s sections concern this pinned version, not an unspecified current checkout. [Merge note: that blob is this report’s `article.tex` before Part III was added, so the cited sections are those of Part II above; see Section 26.1.]

## 27.2 Why exact support matters

For an ordinary matrix, a determinant is a signed sum over perfect matchings. The invariant studied here counts a vertex set once whenever at least one perfect matching exists. It must not count the number of such matchings. These two tasks agree on forests but diverge on cycles. Phases can correct the overcount on isolated cycles; the question is whether they can do so consistently when cycles overlap.

This is a concrete obstruction problem with implications for determinant proofs of stability, weighted real-rootedness, and exact certificate design. It does not settle general  $\gamma$ -positivity or flag realization questions for preorders. Nor does it characterize all graphs with real-rooted univariate support polynomials.

## 27.3 A map of the contributions

The results are organized around four statements. Theorem 29.1 gives the exact graph classification, including its Hermitian and ternary versions. Theorem 30.3 classifies all complex certificates up to diagonal phase changes. Theorem 36.3 gives the sharp universal separation from noncacti. Theorem 37.1 and Corollary 37.4 separate real stability from the exact vertex determinant property. Theorem 38.1 shows why testing a fixed collection of minor orders cannot substitute for the structural theorem.

The proofs of the classification and approximation gap use only finite graph arguments, determinants, and elementary geometry of unit complex numbers. The stability example uses Brändén's established multiaffine Rayleigh criterion [12, Theorem 5.6]. This is the only specialized external theorem needed for the new stability conclusion.

## 28 Definitions and distinctions

Throughout,  $G = (A \sqcup B, E)$  is a finite simple bipartite graph with a specified bipartition. Write  $p = |A|$ ,  $q = |B|$ ,  $n = p + q$ , and let  $c(G)$  count connected components, including isolated vertices. Its cycle rank is

$$\beta(G) = |E| - |A| - |B| + c(G).$$

Empty shores and isolated vertices cause no difficulty. A cactus is a graph in which every edge belongs to at most one simple cycle. Equivalently, every nontrivial block is an edge or a cycle. We allow disconnected cacti.

**Definition 28.1** (Matching supports). A vertex set  $W \subseteq A \sqcup B$  is *perfectly matchable* when the induced graph  $G[W]$  has a perfect matching. The empty set is perfectly matchable. Let  $\mathcal{M}_k(G)$  be the set of such  $W$  with  $|W| = 2k$ . Define

$$p_G(u) = \sum_{k \geq 0} |\mathcal{M}_k(G)| u^k, \quad (28.1)$$

$$\Phi_G(\mathbf{z}) = \sum_{k \geq 0} \sum_{W \in \mathcal{M}_k(G)} (-1)^k \prod_{v \notin W} z_v. \quad (28.2)$$

The polynomial  $p_G$  is a classical perfectly matchable set polynomial; relevant prior work includes Ohsugi–Tsuchiya and Davis–Kohl [9, 7]. Our symbol  $\Phi_G$  agrees with the signed multivariate refinement used in Section 19 of Part II, equation (19.4), on the cacti treated there; here it is defined for every bipartite graph. Every monomial in (28.2) has coefficient 1 or  $-1$ , regardless of the number of matchings on its omitted vertex set.

**Definition 28.2** (Faithful phase certificate). A matrix  $K \in \mathbb{C}^{A \times B}$  is a *phase matrix* on  $G$  if

$$K_{ab} = 0 \quad (ab \notin E), \quad |K_{ab}| = 1 \quad (ab \in E).$$

It is *faithful* if, for every  $I \subseteq A, J \subseteq B$  with  $|I| = |J|$ ,

$$|\det K_{I,J}|^2 = \mathbf{1}\{G[I \sqcup J] \text{ has a perfect matching}\}. \quad (28.3)$$

The empty determinant is one.

There is no extra restriction in calling the edge entries phases: the one-by-one instances of (28.3) force precisely these moduli and this zero pattern. A nonmatchable square submatrix has determinant zero for every choice of entries on its pattern, since every permutation product contains a zero.

**Definition 28.3** (Support-exact matrices over a field). For a field  $F$ , a matrix  $L \in F^{A \times B}$  is *support-exact* for  $G$  if its nonzero entries occur precisely on  $E$  and every square submatrix whose support has a perfect matching is nonsingular.

This is the pattern-sensitive notion often called superregularity: structurally zero minors are permitted, but no other minor may vanish [13]. We use “support-exact” to avoid confusion with the different convention that a superregular matrix has *every* square minor nonzero.

**Remark 28.4** (Two weaker conditions are not the same problem). Requiring merely that every nonzero complex minor have modulus one permits matchable minors to vanish. That weaker requirement belongs to the theory of complex unimodular and sixth-root-of-unity matrices [14]; it is not (28.3). Likewise, an univariate identity  $p_G(u) = \det(I + uM)$  alone forgets the individual vertex supports and does not imply faithfulness.

## 29 The classification theorem

**Theorem 29.1** (Exact boundary of the determinant model). *For a finite simple bipartite graph  $G$ , the following are equivalent:*

- (i)  $G$  is a cactus.
- (ii)  $G$  has a faithful complex phase certificate.
- (iii)  $G$  has a faithful certificate all of whose nonzero entries are sixth roots of unity.
- (iv) There is an  $n \times n$  Hermitian matrix  $H$ , indexed by  $V(G)$ , for which

$$\Phi_G(\mathbf{z}) = \det(\text{diag}(z_v : v \in V(G)) - H). \quad (29.1)$$

- (v)  $G$  has a support-exact matrix over  $\mathbb{F}_3$ .

No zero pattern for  $H$  is assumed in (iv); the polynomial identity forces the required bipartite pattern.

The implication (i) $\Rightarrow$ (iii) is the construction already present in Part II (Theorem 13.5 and Lemma 19.1). We reproduce its proof in Section 30 to keep the argument self-contained and to prepare the classification of all certificates. Sections 31–33 prove the converse. Section 34 proves the precise Hermitian equivalence, and Section 35 proves the finite-field characterization.

**Remark 29.2** (The strength and the limit of the answer). The theorem rules out every Hermitian matrix of the form (29.1), not only the matrix chosen by Part II’s construction. It does *not* rule out arbitrary larger determinantal pencils, substitutions involving auxiliary variables, sums of determinants, or representations of powers of a polynomial. Those are different questions.



### 30 Cactus certificates and their complete phase freedom

Set  $\zeta = e^{\pi i/3}$ . Thus  $\zeta^2 - \zeta + 1 = 0$  and

$$|1 - \zeta| = |1 - \bar{\zeta}| = 1.$$

Choose a spanning forest  $T$  of a cactus. Each cycle has exactly one edge outside  $T$ , and no edge belongs to two cycles. Put  $K_{ab} = 1$  on the forest edges. On the remaining edge of each cycle of length  $2r$ , put

$$K_{ab} = (-1)^r \zeta \quad \text{or} \quad (-1)^r \bar{\zeta}, \quad (30.1)$$

independently from cycle to cycle. All nonedges have entry zero.

**Lemma 30.1** (The cactus factorization; prior construction). *Every matrix constructed by (30.1) is faithful.*

*Proof.* Fix a square induced support admitting a perfect matching  $M$ . The symmetric difference of  $M$  with any other perfect matching consists of vertex-disjoint  $M$ -alternating cycles. In a cactus, any two  $M$ -alternating cycles are themselves vertex-disjoint: if they met at a vertex, its unique edge of  $M$  would have to belong to both cycles, contradicting the cactus condition.

Consequently all perfect matchings on the support are obtained by independently flipping a subset of its  $M$ -alternating cycles. A flip on a cycle of length  $2r$  changes the permutation sign by  $(-1)^{r-1}$ . The ratio of the products of edge entries in the two alternating halves is  $(-1)^r \zeta$  or  $(-1)^r \bar{\zeta}$ , depending on the orientation of the comparison. Therefore, after factoring out the unit contribution of  $M$ , the determinant is a product of factors  $1 - \zeta$  and  $1 - \bar{\zeta}$ . Its modulus is one. If the support has no perfect matching, every determinant term is zero.  $\square$

This is the content of Part II's cactus minor lemma (Lemma 19.1, which uses the choice  $(-1)^r \zeta$  on every cycle), now with both allowed choices on each cycle displayed. The proof does not invoke any general matrix representability theorem.

**Definition 30.2** (Phase gauge). Two phase matrices on the same labeled bipartite graph are gauge-equivalent if

$$K' = D_A K D_B$$

for diagonal matrices  $D_A, D_B$  whose diagonal entries have modulus one. No row permutation, column permutation, pivot, or complex conjugation is included in this equivalence.

Such changes preserve all square-minor moduli. They are also precisely the relevant diagonal unitary changes of the corresponding bipartite Hermitian matrix, after conjugating the column-side factor as appropriate.

**Theorem 30.3** (Rigidity and moduli of faithful certificates). *For a cactus  $G$ , every faithful certificate is gauge-equivalent to exactly one of the matrices (30.1), relative to a fixed spanning forest. There are exactly  $2^{\beta(G)}$  gauge classes. In particular, arbitrary unit complex phases produce no additional classes beyond sixth-root phases.*

*Proof.* On each tree component, select a root. Choose row and column phases successively away from the root so that every tree edge becomes one. A residual gauge preserving all forest entries uses a common phase on the row vertices of a component and its reciprocal on the column vertices; it changes no edge entry anywhere in that component. Thus normalization gives a unique representative of each gauge class.



A simple cycle in a cactus is induced. In its normalized square submatrix, all entries are one except for the single non-tree entry  $t$ , and there are exactly two perfect matchings. Up to a unit sign, its determinant is

$$1 + (-1)^{r-1}t.$$

Faithfulness and  $|t| = 1$  imply

$$\operatorname{Re}((-1)^{r-1}t) = -\frac{1}{2}.$$

The only possibilities are  $t = (-1)^r \zeta$  and  $t = (-1)^r \bar{\zeta}$ . This proves necessity of every listed cycle choice. Lemma 30.1 shows that the choices are independent and sufficient. A cactus has  $\beta(G)$  cycles, proving the count.  $\square$

**Corollary 30.4** (When real phases suffice). *A bipartite graph has a faithful real matrix if and only if it is a forest. For a forest, every assignment of signs to the edges is faithful and all assignments are gauge-equivalent.*

*Proof.* A forest has at most one perfect matching on each vertex support, because the symmetric difference of two would contain a cycle. Hence every nonzero determinant is a signed product of edge entries. Conversely, a faithful real matrix would be a faithful complex matrix and hence have cactus support. On any cycle the required normalized entry (30.1) is nonreal. One can also argue directly on a shortest induced cycle: its two determinant terms are  $\pm 1$ , so the determinant is 0 or  $\pm 2$ , never of modulus one.  $\square$

### 31 Why every noncactus contains the right obstruction

A *theta* consists of three internally vertex-disjoint paths with common distinct endpoints. Our convention permits one of the paths to have length one; at most one can do so in a simple graph. An *induced theta* has no edges besides those in the three paths. This convention matters: excluding the length-one case would miss two cycles sharing an edge.

**Lemma 31.1** (Induced-theta lemma for bipartite graphs). *Every finite noncactus bipartite graph contains an induced theta. The three path lengths have the same parity.*

*Proof.* A noncactus has a two-connected block  $B$  that is not a cycle or an edge. To see the relevant block fact, two cycles sharing an edge lie in a common two-connected subgraph, and a maximal such block is not a single cycle. Choose a shortest cycle  $C$  of  $B$ . It is induced, and  $B$  has vertices outside  $C$ ; otherwise an extra edge would be a chord contradicting shortestness.

First suppose that a vertex  $v \notin C$  has at least two neighbors on  $C$ . If it has exactly two,  $C \cup \{v\}$  is an induced theta. If it has at least three, take three consecutive neighbors  $x, y, z$  in their cyclic order. Retain  $v$  and the arc of  $C$  from  $x$  through  $y$  to  $z$  having no further neighbors of  $v$  in its interior. The three paths from  $v$  to  $y$  are the edge  $vy$  and the two paths through  $x$  and  $z$ . The arc has no chord, and its endpoints  $x, z$  are in the same bipartition class, so there is no extra edge  $xz$ . This is again an induced theta.

We may therefore assume that each vertex outside  $C$  has at most one neighbor on  $C$ . Some component of  $B - V(C)$  has at least two distinct attachment vertices on  $C$ : if it had only one, that vertex would disconnect  $B$ . Choose a shortest path with distinct endpoints on  $C$  and all internal vertices outside  $C$ . It has no chord. It also has no additional edge from an internal vertex to  $C$ . Such an edge would produce a shorter path of the same kind to one of the two endpoints; at the first or last internal vertex, the assumed uniqueness of its neighbor on  $C$  excludes the only exceptional adjacency. The union of this path with  $C$  is therefore an induced theta.

In a bipartite graph, every path between two fixed vertices has parity determined by the shores of its endpoints. Hence all three lengths have the same parity.  $\square$

The lemma is tailored to induced subgraphs because deleting arbitrary edges from a phase matrix need not preserve any minor identity. Our proof never makes that invalid inference. The only graph reductions used below are restriction to an induced subgraph and a precisely controlled path replacement.

## 32 A path-compression identity that preserves all supports

Consider a path

$$a - v - u - b, \quad a, u \in A, \quad v, b \in B,$$

whose internal vertices  $v, u$  have degree two. Assume that  $a$  and  $b$  are not adjacent. Delete  $u, v$  and replace the three-edge path by the edge  $ab$ . For a matrix supported on the original graph, define the new entry by

$$K'_{ab} = -\frac{K_{av}K_{ub}}{K_{uv}}, \quad (32.1)$$

and leave the other surviving entries unchanged. This operation shortens one path by two while preserving bipartiteness.

**Lemma 32.1** (Compression preserves support and determinant magnitude). *Suppose  $K_{uv} \neq 0$ . For every equally sized pair  $I, J$  of surviving row and column sets,*

$$\det K_{I \cup \{u\}, J \cup \{v\}} = \pm K_{uv} \det K'_{I, J}. \quad (32.2)$$

*Moreover, the two induced supports in (32.2) are perfectly matchable simultaneously. If all edge entries of  $K$  have modulus one, the same is true for  $K'$ , and the squared determinant magnitudes in (32.2) are equal.*

*Proof.* Put row  $u$  and column  $v$  last. The block determinant identity gives the factor  $K_{uv}$  times the determinant of the Schur complement. The column at  $v$  has only one other possible nonzero entry, in row  $a$ , and the row at  $u$  has only one other possible nonzero entry, in column  $b$ . Thus the only altered surviving entry is (32.1). Restoring the original order contributes only a sign.

For matchings there is an explicit bijection. A matching on the compressed support not using  $ab$  lifts by adding  $uv$ . A matching using  $ab$  lifts by replacing it with  $av$  and  $ub$ . Conversely, a matching of the augmented original support either uses  $uv$ , or must match  $v$  to  $a$  and  $u$  to  $b$ . These alternatives also cover the cases in which  $I$  or  $J$  omits one of the endpoints. Finally, (32.1) is a quotient of unit complex numbers with a minus sign, hence has modulus one.  $\square$

**Corollary 32.2** (Preservation of exactness). *Faithful complex certificates and support-exact matrices over any field descend through this compression. Both properties also descend to induced subgraphs.*

Repeated compression of an induced theta produces one of the following three graphs:

| Parity and short path          | Final path lengths | Name used here |
|--------------------------------|--------------------|----------------|
| Even                           | (2, 2, 2)          | $E = K_{2,3}$  |
| Odd, a length-one path present | (1, 3, 3)          | D              |
| Odd, no length-one path        | (3, 3, 3)          | T              |

In the even case shorten only paths of length at least four. In the odd case shorten only paths of length at least five. Consequently no parallel edge is created. At each step the internal vertices have degree two, and the replacement edge does not already exist. These details are why the Schur-complement identity applies throughout.

### 33 Three finite phase obstructions

We now exclude a faithful matrix for each core. Row and column phase changes normalize a spanning tree to ones. Every remaining parameter below has modulus one.

#### 33.1 The even core $E = K_{2,3}$

A normalized matrix is

$$K_E = \begin{pmatrix} 1 & 1 & 1 \\ 1 & z & w \end{pmatrix}. \quad (33.1)$$

Its three two-by-two supports are all matchable, so faithfulness requires

$$|1 - z| = |1 - w| = |z - w| = 1.$$

The first two equalities imply  $z, w \in \{\zeta, \bar{\zeta}\}$ . Their distance is then either zero or  $\sqrt{3}$ , contradicting the last equality. Geometrically, three points on the unit circle cannot have all three chord lengths equal to one.

#### 33.2 Two squares sharing an edge: $D$

A normalized matrix is

$$K_D = \begin{pmatrix} 1 & 1 & 1 \\ x & 1 & 0 \\ y & 0 & 1 \end{pmatrix}. \quad (33.2)$$

The two cycle minors have determinants  $1 - x$  and  $1 - y$ . They force  $x, y \in \{\zeta, \bar{\zeta}\}$ . The full determinant is  $1 - x - y$ , with squared modulus zero if the phases are conjugate and three if they are equal. The full support is matchable, so both possibilities are forbidden.

#### 33.3 Three odd paths: $T$

A normalized matrix is

$$K_T = \begin{pmatrix} 0 & 1 & 1 & 1 \\ x_1 & 1 & 0 & 0 \\ x_2 & 0 & 1 & 0 \\ x_3 & 0 & 0 & 1 \end{pmatrix}. \quad (33.3)$$

Every pair of arms gives a three-by-three cycle minor with determinant  $-(x_i + x_j)$ . Faithfulness would imply

$$\operatorname{Re}(x_i \bar{x}_j) = -\frac{1}{2} \quad (i \neq j).$$

But the full determinant is  $-(x_1 + x_2 + x_3)$ , and these equalities give

$$|x_1 + x_2 + x_3|^2 = 3 + 2 \sum_{i < j} \operatorname{Re}(x_i \bar{x}_j) = 0.$$

Its support has a perfect matching, a contradiction.

*Proof of (ii)  $\Rightarrow$  (i) in Theorem 29.1.* If  $G$  were a noncactus with a faithful matrix, Lemma 31.1 would give an induced theta. Restrict the matrix to it, and apply Lemma 32.1 until one of the three cores remains. The resulting faithful matrix contradicts the corresponding calculation above.  $\square$

### 34 The Hermitian representation has no hidden extra freedom

For a phase matrix  $K$ , put

$$H_K = \begin{pmatrix} 0 & K \\ K^* & 0 \end{pmatrix}.$$

On a balanced vertex set  $W = I \sqcup J$ , with  $|I| = |J| = k$ , its principal determinant is

$$\det(H_K)_W = (-1)^k |\det K_{I,J}|^2. \quad (34.1)$$

On an unbalanced vertex set the principal determinant vanishes: one of the two off-diagonal rectangular blocks has more rows than columns and hence produces a kernel vector.

**Proposition 34.1** (Coefficient recovery). *Identity (29.1) holds for a Hermitian  $H$  if and only if  $H = H_K$  for a faithful phase matrix  $K$ , with vertices ordered by the two shores.*

*Proof.* The diagonal expansion is

$$\det(\text{diag } \mathbf{z} - H) = \sum_{W \subseteq V} (-1)^{|W|} \det H_W \prod_{v \notin W} z_v. \quad (34.2)$$

If  $K$  is faithful, (34.1) and the vanishing of unbalanced minors make this exactly (28.2).

Conversely, compare coefficients for omitted singleton sets. They give  $H_{vv} = 0$ . For an omitted pair  $\{v, w\}$ , the coefficient is

$$\det H_{\{v,w\}} = -|H_{vw}|^2.$$

The coefficient in  $\Phi_G$  is  $-1$  for an edge and zero otherwise. Therefore  $H_{vw} = 0$  on nonedges and  $|H_{vw}| = 1$  on edges. In particular its same-shore blocks vanish, so  $H = H_K$ . Comparing all remaining balanced coefficients using (34.1) yields faithfulness.  $\square$

This proves (ii)  $\Leftrightarrow$  (iv). It also explains exactly which information disappears under diagonal specialization. For example,  $K_{2,3}$  has

$$p_{K_{2,3}}(u) = 1 + 6u + 3u^2 = (1 + (3 + \sqrt{6})u)(1 + (3 - \sqrt{6})u).$$

Thus its univariate polynomial has a positive-semidefinite two-by-two determinant representation using a diagonal matrix. Nevertheless, there is no representation (29.1) for its five-variable polynomial. A one-variable factorization is not a counterexample to the classification.

### 35 The same boundary over the smallest fields

**Theorem 35.1** (Binary and ternary support patterns). *A finite bipartite graph admits a support-exact matrix over  $\mathbb{F}_2$  exactly when it is a forest, and over  $\mathbb{F}_3$  exactly when it is a cactus. A cactus has a unique support-exact  $\mathbb{F}_3$  gauge class under independent nonzero row and column scalings.*

*Proof.* Over  $\mathbb{F}_2$ , all edge entries are one. A shortest cycle is induced, and its two perfect-matching terms sum to zero. Conversely, a forest has at most one perfect matching on any support, so every structurally nonzero determinant is one.

For  $\mathbb{F}_3$ , first consider a cactus. Normalize a spanning forest to ones. On the remaining edge of a cycle of length  $2r$ , choose

$$t = (-1)^{r+1} \in \mathbb{F}_3.$$

The flip ratio including its determinant sign is  $(-1)^{r-1}t = 1$ . The same independent-cycle factorization as in Lemma 30.1 expresses every structurally nonzero determinant as a nonzero matching term times a product of factors  $1 + 1 = 2$ . It cannot vanish in  $\mathbb{F}_3$ .

For necessity, induced restriction and path compression are valid over every field. Normalize each of the three cores as above, now with all parameters in  $\mathbb{F}_3^* = \{1, -1\}$ . In (33.1), three column ratios cannot all be distinct, so some two-by-two minor vanishes. In (33.2), nonvanishing of  $1 - x$  and  $1 - y$  forces  $x = y = -1$ , but  $1 - x - y = 3 = 0$ . In (33.3), nonvanishing of every  $x_i + x_j$  forces the three parameters to be equal, but their sum is again zero. No core is support-exact, so the graph must be a cactus.

Finally, on a normalized cactus cycle,  $1 + (-1)^{r-1}t$  is nonzero for precisely the choice  $t = (-1)^{r+1}$ . The spanning-forest gauge is unique as in Theorem 30.3, proving uniqueness.  $\square$

Together with the preceding sections, this completes every equivalence in Theorem 29.1.

**Remark 35.2** (Reduction of the complex certificate). The ring homomorphism  $\mathbb{Z}[\zeta] \rightarrow \mathbb{F}_3$  sending  $\zeta$  to  $-1$  is well-defined because  $(-1)^2 - (-1) + 1 = 3$ . It sends both complex cycle choices to the unique ternary choice. This provides another explanation of why the  $2^\beta$  complex classes collapse to one ternary class. The determinant factorization, rather than just entrywise reduction, ensures that no required minor is lost.

**Proposition 35.3** (A first larger-field contrast). *For  $m \geq 1$  and a finite field  $\mathbb{F}_q$ , the complete bipartite graph  $K_{2,m}$  admits a support-exact matrix over  $\mathbb{F}_q$  if and only if  $m \leq q - 1$ .*

*Proof.* Scale the columns so that the first row is all ones. The second row consists of nonzero elements  $t_1, \dots, t_m$ . The two-by-two determinants are  $t_j - t_i$ , so support-exactness is equivalent to the  $t_i$  being pairwise distinct. There are exactly  $q - 1$  available elements.  $\square$

This elementary observation shows that the cactus classification is genuinely tied to the ternary field in its finite-field formulation; it is not a field-independent prohibition on overlapping cycles. Broader finite-field classifications are not asserted.

## 36 A sharp universal approximation gap

Exact equality can be fragile in numerical experiments. Here the obstruction is not merely that some equations cannot hold simultaneously: they cannot all be approximately correct beyond an explicit dimension-independent threshold.

**Definition 36.1** (Minor defect). For a phase matrix  $K$  on  $G$ , let

$$\delta_G(K) = \max_{\substack{I \subseteq A, J \subseteq B, |I|=|J| \\ G[I \sqcup J] \text{ perfectly matchable}}} \left| |\det K_{I,J}|^2 - 1 \right|, \quad d(G) = \min_K \delta_G(K). \quad (36.1)$$

The minimum is over unit phases on all edges. It exists because the phase torus is compact and the maximum is over finitely many continuous functions. The empty minor guarantees a nonempty indexing set, even for an edgeless graph.

Nonmatchable minors always have their exact target value zero and need not be included. The normalization  $|K_{ab}| = 1$  is essential: allowing unrestricted edge magnitudes would define a different optimization problem.

**Lemma 36.2** (Defect is monotone under the reductions). *Induced restriction and the path compression of Lemma 32.1 cannot increase the defect of a phase matrix.*

*Proof.* Restriction only removes minors from the maximum. Every minor of a compressed matrix is, by (32.2), equal in squared modulus to an augmented minor of the original matrix, and the two supports are matchable simultaneously. Thus every quantity tested after compression already occurs before it.  $\square$

**Theorem 36.3** (Sharp universal noncactus gap). *Put*

$$\varepsilon_* = \frac{\sqrt{17} - 3}{2} = 0.5615528128 \dots$$

*For every finite bipartite noncactus  $G$  and every phase matrix  $K$  on it,*

$$\delta_G(K) \geq \varepsilon_*. \quad (36.2)$$

*The constant is optimal over all noncactus graphs:  $d(K_{2,3}) = \varepsilon_*$ . In particular,*

$$G \text{ cactus} \iff d(G) = 0, \quad \inf_{G \text{ noncactus}} d(G) = \varepsilon_*.$$

The proof occupies the next two subsections. It yields more information than the exact classification: a phase fit with all matchable minor squared moduli in  $(1 - \varepsilon_*, 1 + \varepsilon_*)$  certifies that the graph is a cactus. This statement concerns all minors, not a sampled subset.

### 36.1 The exact optimization on $K_{2,3}$

**Proposition 36.4** (Three-chord minimax problem). *For the even core,  $d(E) = \varepsilon_*$ . A minimizing normalized matrix is*

$$K_* = \begin{pmatrix} 1 & 1 & 1 \\ e^{-i\varphi} & 1 & e^{i\varphi} \end{pmatrix}, \quad \cos \varphi = \frac{\sqrt{17} - 1}{4}. \quad (36.3)$$

*Its three nontrivial minor squared moduli are  $1 - \varepsilon_*$ ,  $1 - \varepsilon_*$ , and  $1 + \varepsilon_*$ .*

*Proof.* After column normalization, the second-row entries are three unit complex numbers. Every nontrivial determinant is their pairwise difference, so the problem is to approximate one by all three squared chord lengths.

Suppose first that the defect  $\delta$  is less than one; otherwise the desired lower bound is immediate. All squared chord lengths are then less than two, so all pairwise real inner products are positive. The three points lie in an arc of angular length  $\theta < \pi/2$ . To justify this, rotate one point to angle zero and represent the other two angles in  $(-\pi/2, \pi/2)$ . Their difference is in  $(-\pi, \pi)$ , and positivity of their mutual inner product forces its absolute value below  $\pi/2$ . Thus the interval between the extreme angles has the asserted length.

Order the points on that arc. At least one of the two successive gaps is at most  $\theta/2$ . Write  $c = \cos(\theta/2)$ , so  $\sqrt{2}/2 < c \leq 1$ . The smallest squared chord length is at most  $2 - 2c$ , whereas the largest is exactly  $2 - 2\cos \theta = 4(1 - c^2)$ . It follows that

$$\delta \geq \max\{2c - 1, 3 - 4c^2\}. \quad (36.4)$$

The first expression increases with  $c$  and the second decreases. Their unique intersection in the indicated interval occurs at

$$c_* = (\sqrt{17} - 1)/4, \quad 2c_* - 1 = 3 - 4c_*^2 = \varepsilon_*.$$

Therefore (36.4) gives  $\delta \geq \varepsilon_*$ .

For (36.3), the consecutive gaps are equal to  $\varphi$ . The corresponding squared chord lengths are  $2 - 2c_* = 1 - \varepsilon_*$  and  $4(1 - c_*^2) = 1 + \varepsilon_*$ . Thus the lower bound is attained. The argument also shows that any minimizing triple has equal consecutive gaps, up to rotation, reflection, and permutation of its points.  $\square$

### 36.2 The other two cores are separated by at least $3/5$

**Lemma 36.5** (A stronger-than-needed bound for D). *Every phase matrix on D has defect at least  $3/5$ .*

*Proof.* Use (33.2) and suppose the defect is less than  $3/5$ . The two cycle minors imply

$$a = \operatorname{Re} x, \quad b = \operatorname{Re} y \in (1/5, 4/5).$$

Depending on whether the imaginary parts have the same or opposite signs, the squared modulus of the full determinant is

$$g_{\pm}(a, b) = 3 - 2a - 2b + 2ab \pm 2\sqrt{1 - a^2}\sqrt{1 - b^2}.$$

On the closed rectangle  $[1/5, 4/5]^2$ ,  $g_-$  is convex in each variable separately and  $g_+$  is concave in each variable separately. Indeed the second derivative with respect to  $a$  is respectively

$$\frac{2\sqrt{1 - b^2}}{(1 - a^2)^{3/2}} \quad \text{or its negative,}$$

and the same holds with the variables exchanged. Hence the maximum of  $g_-$  and the minimum of  $g_+$  occur at corners. The corner values give

$$g_- \leq \frac{9}{25}, \quad g_+ \geq \frac{9}{5}.$$

For the off-diagonal corners the respective values are  $(33 - 12\sqrt{6})/25$  and  $(33 + 12\sqrt{6})/25$ , so these bounds follow by direct comparison. Both alternatives are outside the interval  $(2/5, 8/5)$  required for a full minor of defect less than  $3/5$ . This is a contradiction.  $\square$

**Lemma 36.6** (A planar Gram obstruction for T). *Every phase matrix on T has defect at least  $3/5$ .*

*Proof.* Use (33.3). Suppose its defect is less than  $3/5$  and put

$$p = -\operatorname{Re}(x_1\bar{x}_2), \quad q = -\operatorname{Re}(x_1\bar{x}_3), \quad r = -\operatorname{Re}(x_2\bar{x}_3).$$

The three pair-cycle minors imply  $p, q, r \in (1/5, 4/5)$ . As three unit vectors in the real plane, the  $x_i$  have a singular real Gram matrix. Expanding its determinant yields

$$p^2 + q^2 + r^2 + 2pqr = 1. \tag{36.5}$$

Write  $s = p + q + r$ . The full minor has squared modulus  $3 - 2s$ , so its lower bound  $3 - 2s > 2/5$  implies  $s < 13/10$ .

For each  $t \in [1/5, 4/5]$ , the inequality  $(t - 1/5)(t - 4/5) \leq 0$  gives  $t^2 \leq t - 4/25$ . Also  $pqr \leq (s/3)^3$  by the arithmetic–geometric mean inequality. Therefore

$$1 = p^2 + q^2 + r^2 + 2pqr \leq s - \frac{12}{25} + 2(s/3)^3 < \frac{13}{10} - \frac{12}{25} + 2(13/30)^3 = \frac{13267}{13500} < 1,$$

a contradiction.  $\square$

*Proof of Theorem 36.3.* Take an induced theta in  $G$  and compress it to a core. The defect cannot increase by Lemma 36.2. On E it is at least  $\varepsilon_*$  by Proposition 36.4; on the other cores it is at least  $3/5 > \varepsilon_*$  by Lemmas 36.5 and 36.6. Formula (36.3) attains  $\varepsilon_*$  on the noncactus  $K_{2,3}$ , proving optimality. Cacti have defect zero by Lemma 30.1.  $\square$

**Remark 36.7** (Why a weaker four-minor bound is not sharp). On T, let  $a_{ij} = |x_i + x_j|^2 - 1$  and  $b = |x_1 + x_2 + x_3|^2 - 1$ . Then  $b = a_{12} + a_{13} + a_{23} - 1$ , so the triangle inequality alone gives  $\delta \geq 1/4$ . It loses the fact that the three complex phases live in a real plane. The Gram identity (36.5) restores that constraint and makes the sharp universal theorem possible.



### 37 Real stability is strictly more general

A nonzero polynomial with real coefficients is *real stable* if it does not vanish when every variable has strictly positive imaginary part. For a cactus certificate, (29.1) immediately gives stability: if  $(\text{diag } \mathbf{z} - H)u = 0$  for a nonzero vector  $u$ , then

$$0 = \text{Im}(u^*(\text{diag } \mathbf{z} - H)u) = \sum_v \text{Im}(z_v)|u_v|^2 > 0,$$

a contradiction. This determinant argument is already used in Part II, as the proof of Theorem 19.3.

The converse fails on only five vertices. Denote the variables on the two shores of  $K_{2,3}$  by  $a, b$  and  $x, y, z$ . Direct support enumeration gives

$$f = \Phi_{K_{2,3}} = abxyz - (a+b)(xy+xz+yz) + (x+y+z). \quad (37.1)$$

The size-four supports are the three choices of two vertices on the larger shore together with both vertices on the smaller shore, explaining the last three terms.

**Theorem 37.1** (A stable polynomial without a Hermitian vertex determinant). *Polynomial (37.1) is real stable, but there is no Hermitian  $5 \times 5$  matrix  $H$  with  $f = \det(\text{diag}(a, b, x, y, z) - H)$ .*

*Proof.* Nonrepresentability follows from Theorem 29.1, or directly from the three-chord obstruction (33.1). For stability, use Brändén's criterion: a nonzero real multiaffine polynomial  $f$  is real stable if and only if

$$\Delta_{ij}f = (\partial_i f)(\partial_j f) - f \partial_i \partial_j f \geq 0$$

for all real choices of all variables and all distinct pairs [12, Theorem 5.6]. Symmetry leaves only three types of pairs.

For the pair on the smaller shore, put  $U = xy$ ,  $V = xz$ ,  $W = yz$ . Direct differentiation gives

$$\begin{aligned} \Delta_{a,b}f &= U^2 + V^2 + W^2 + UV + UW + VW \\ &= \frac{1}{2}((U+V+W)^2 + U^2 + V^2 + W^2) \geq 0. \end{aligned} \quad (37.2)$$

For two variables on the larger shore,

$$\begin{aligned} \Delta_{x,y}f &= (a^2 + ab + b^2)z^2 - (a+b)z + 1 \\ &= \left(1 - \frac{a+b}{2}z\right)^2 + \frac{3a^2 + 2ab + 3b^2}{4}z^2 \geq 0. \end{aligned} \quad (37.3)$$

For a cross-shore pair,

$$\begin{aligned} \Delta_{a,x}f &= (byz)^2 - byz(y+z) + y^2 + yz + z^2 \\ &= \left(byz - \frac{y+z}{2}\right)^2 + \frac{3y^2 + 2yz + 3z^2}{4} \geq 0. \end{aligned} \quad (37.4)$$

The remaining quadratic forms are nonnegative because  $3u^2 + 2uv + 3v^2 = 2(u^2 + v^2) + (u+v)^2$ . The three identities cover all ten pairs by shore symmetries. The criterion proves stability.  $\square$

This is not a claim that stability without determinant representations is new as a general phenomenon. Such separations are established in other settings, including work on Vámos-type polynomials [15]. The result here answers Part II's specific question (Research question 4) for its particular matching-support refinement, with a small explicit example and elementary certificates.

### 37.1 Leaf extension and an infinite connected family

**Lemma 37.2** (Exact leaf recurrence). *Let  $G + \ell$  be obtained by adding a new leaf  $\ell$  adjacent only to a vertex  $v$ . Then*

$$\Phi_{G+\ell} = z_\ell \Phi_G - \Phi_{G-v} = z_\ell \Phi_G - \partial_{z_v} \Phi_G. \quad (37.5)$$

*Proof.* A matchable support omitting  $\ell$  gives the first term. A support containing  $\ell$  must also contain  $v$ , and its perfect matching must use  $\ell v$ . Removing these two vertices gives a unique perfectly matchable support in  $G - v$ , with the opposite sign in (28.2). Finally, differentiation with respect to  $z_v$  retains exactly the supports omitting  $v$  and removes that variable from their monomials.  $\square$

**Lemma 37.3** (Stability of the leaf operation). *If a real multiaffine polynomial  $f$  is stable, then  $wf - \partial_{z_v} f$ , where  $w$  is a new variable, is stable.*

*Proof.* Fix all variables other than  $z_v$  in the upper half-plane. Write  $f = Az_v + B$  at these fixed values. It is not the zero polynomial, since stability would otherwise fail. If  $A = 0$ , then  $f = B \neq 0$  and the claimed expression is  $wB \neq 0$ . If  $A \neq 0$ , its root  $\rho = -B/A$  has imaginary part at most zero. For  $z_v$  in the upper half-plane,

$$\frac{\partial_{z_v} f}{f} = \frac{1}{z_v - \rho}$$

has strictly negative imaginary part. It cannot equal  $w$  in the upper half-plane. Multiplying by the nonzero  $f$  proves the claim.  $\square$

**Corollary 37.4** (Infinitely many connected examples). *Every graph obtained from  $K_{2,3}$  by attaching finite trees at its vertices has a real stable polynomial  $\Phi_G$  and has no Hermitian vertex determinant representation (29.1). These examples can be arbitrarily large and connected.*

*Proof.* Construct the attached trees by successive leaf additions and apply Theorem 37.1 and Lemmas 37.2–37.3. The original  $K_{2,3}$  remains an induced subgraph, so the result is still a noncactus. The classification excludes the representation.  $\square$

[Added 29 September 2026, batch 55: leaf attachment is the edge-block case of Part VI's one-vertex sum theorem (Theorem 93.2), which also extends the stable block  $K_{2,3}$  to every  $K_{\ell,r}$  (Theorem 94.1).]

## 38 No fixed minor order can test exact representability

The classification has a finite structural obstruction mechanism, but not a bounded matrix-size obstruction. Compression may require adjoining many eliminated rows and columns before a core minor becomes a minor of the original matrix.

**Theorem 38.1** (Unbounded order of minor witnesses). *For every positive integer  $r$ , there is a connected bipartite noncactus  $G$  of maximum degree three and treewidth at most two such that every phase matrix on  $G$  satisfies (28.3) for every square minor of order at most  $r$ . Nevertheless no phase matrix on  $G$  satisfies all orders.*

*Proof.* Choose an odd integer  $L > r$ ,  $L \geq 3$ , and take three internally disjoint paths of length  $L$  between two endpoints. This theta is bipartite, subcubic, connected, and not a cactus. Every simple cycle consists of two of the three paths, so its girth is  $2L > 2r$ .

A square minor of order at most  $r$  uses at most  $2r$  vertices. Its induced graph therefore has no cycle. Such a forest has at most one perfect matching; its determinant is zero when there is none and a unit product when there is one, for every phase assignment. Thus all the small-minor tests pass. Theorem 29.1 rules out a faithful matrix globally.

For completeness, a width-two tree decomposition is explicit. Let the common endpoints be  $s, t$  and write the internal vertices of path  $i$  as  $v_{i,1}, \dots, v_{i,L-1}$ . Use a central bag  $\{s, t\}$ . Along path  $i$  use the chain of bags

$$\{s, v_{i,1}, v_{i,2}\}, \dots, \{s, v_{i,L-2}, v_{i,L-1}\}, \{s, v_{i,L-1}, t\},$$

attaching its last bag to the central bag. Every edge is covered and the bags containing any given vertex are connected. Each bag has at most three vertices.  $\square$

**Corollary 38.2** (Failure of uniform low-order numerical testing). *For every  $r$ , unit-phase matrices can pass all minor tests through order  $r$  exactly while some higher-order matchable minor has defect at least  $\varepsilon_*$ .*

*Proof.* Combine Theorem 38.1 with Theorem 36.3.  $\square$

Thus the approximation gap does not justify replacing the complete minor maximum by a fixed-order sample. Even zero observed error on all two-by-two, three-by-three, or any other bounded range of minors can be misleading.

## 39 Consequences for the repository's preorder program

For a height-two poset  $P_G$  with comparability graph  $G$  on  $n$  vertices, Part II establishes the height-two formula  $h_{P_G}(t) = (1+t)^n p_G(t/(1+t)^2)$  with a precise lattice-support interpretation (Theorem 13.2). We use this as a prior bridge rather than reprove it or claim its  $\gamma$ -positivity conclusion.

For a faithful matrix, the principal-minor expansion and Cauchy–Binet give  $p_G(u) = \det(I + uKK^*)$ . Indeed, the coefficient of  $u^k$  is the sum of  $|\det K_{I,J}|^2$  over all  $k$ -element row and column sets, hence counts  $\mathcal{M}_k(G)$ . If  $\lambda_1, \dots, \lambda_\nu$  are the positive eigenvalues of  $KK^*$ , where  $\nu$  is the matching number, the factorizations (19.2) and (19.3) of Part II's Theorem 19.2 follow; that proof uses only faithfulness, so it applies to every certificate of Theorem 30.3. [Merge note: manuscript 05 printed these four formulas again as displays; they are printed once, in Part II.] These cactus formulas and their real-rootedness consequences are old relative to this article; univariate cactus real-rootedness is also contained in Ohsugi–Tsuchiya [9, Theorem 7.2]. Our contribution identifies the exact boundary of this *support-by-support* determinant route.

For the stable noncactus  $K_{2,3}$ , the diagonal specialization of (37.1) is

$$\Phi_{K_{2,3}}(s, s, s, s, s) = s^5 - 6s^3 + 3s = s(s^2 - (3 + \sqrt{6}))(s^2 - (3 - \sqrt{6})),$$

and Theorem 13.2 gives

$$h_{P_{K_{2,3}}}(t) = 1 + 11t + 31t^2 + 31t^3 + 11t^4 + t^5.$$

The graph is not excluded from real-rootedness; only its exact vertex determinant is excluded. More generally, stable  $\Phi_G$  implies real-rooted  $p_G$ : for  $s \neq 0$ ,

$$\Phi_G(s, \dots, s) = s^n p_G(-s^{-2}).$$

The left-hand side has only real zeros. Any nonzero zero of  $p_G$  would otherwise give a nonreal zero on the left after solving  $u = -s^{-2}$ . Since  $p_G$  has positive coefficients and constant term one, its zeros must consequently be negative. This applies to all the tree-attached examples of Corollary 37.4.

There are three logically different classes: faithful support determinants, stable signed support polynomials, and real-rooted univariate support polynomials. We prove the first is properly contained in the second. The second is contained in the third by specialization. We do not assert a classification of the latter two classes or strictness of that second containment. [Added 29 September 2026, batch 55: the containment is strict. The book  $B_m$  has real-rooted  $p_{B_m}$  for every  $m$  but unstable  $\Phi_{B_m}$  for  $m \geq 5$ , and  $\vartheta(3, 3, 5)$  is a second witness (Part VI, Theorems 84.2 and 90.2, equation (92.5)).]

## 40 Exact verification and reproducibility

The included verifier, shipped as `code/03-cactus-verify.py` (delivered as `code/verify.py`), uses only the Python standard library and exact arithmetic. Its checks were executed with Python 3.13.5. The full output is saved in `data/03-cactus-verification.json` (delivered as `data/verification.json`). The tests supplement the proofs; in particular, checking sixth-root assignments does not by itself exclude arbitrary complex phases.

### 40.1 Independent combinatorial and algebraic representations

Graphs with fixed labeled shores are encoded by binary edge masks. Cactus recognition enumerates simple cycles and checks that no edge belongs to two of them. This recognition routine does not use determinants. A separate spanning-forest procedure identifies the free normalized edge entries.

For each support pattern, the verifier enumerates every normalized assignment over  $\mathbb{F}_3^*$ , or over the six complex sixth roots, as appropriate. Structurally nonzero determinant terms are obtained by permutation expansion. Minors with one matching term automatically have unit magnitude; ambiguous minors are evaluated exactly. Complex values are pairs of integers  $a + b\zeta$ , with

$$(a + b\zeta)(c + d\zeta) = (ac - bd) + (ad + bc + bd)\zeta, \quad |a + b\zeta|^2 = a^2 + ab + b^2.$$

The canonical cactus certificates are checked again by evaluating *all* square minors, including empty minors. Their expected zero-or-one values come from a separate covered-shore dynamic program that merges different matchings with the same covered vertices. Thus the determinant checker is compared with a support-set enumeration, not with its own permutation count.

| Check                                                 | Recorded count |
|-------------------------------------------------------|----------------|
| All labeled 3-by-3 bipartite graphs                   | 512            |
| Cacti among those graphs                              | 460            |
| Normalized sixth-root assignments tested              | 5,872          |
| Ambiguous minors evaluated in that search             | 7,792          |
| All labeled 3-by-4 bipartite graphs                   | 4,096          |
| Cacti among those graphs                              | 3,122          |
| Normalized ternary assignments tested                 | 10,576         |
| Ambiguous minors evaluated in that search             | 14,717         |
| Canonical cactus minor checks, both graph families    | 118,470        |
| Compression: augmented-minor norm/support comparisons | 100            |
| Rayleigh identity classes checked symbolically        | 3              |
| Independent leaf-attachment support enumerations      | 5              |

All checks passed. In a separate check, each of the three small cores has zero valid assignments among its 36 normalized sixth-root assignments and four normalized ternary assignments. The 3-by-3 cacti consist of 328 forests and 132 one-cycle graphs. The 3-by-4 cacti consist of 1,856 forests, 1,248 one-cycle graphs, and 18 two-cycle graphs. The exhaustive normalized counts agree with  $2^\beta$  complex classes and one ternary class on every tested cactus, and zero on every tested noncactus.

## 40.2 Symbolic and compression checks

A small rational multivariate polynomial implementation expands and compares the three Rayleigh identities (37.2)–(37.4); by symmetry they cover all ten pairs. The polynomial (37.1) is independently reconstructed from matchable supports. Each of the five possible leaf attachments is enumerated directly and compared with (37.5).

The compression checker uses theta path lengths (2, 2, 4), (1, 3, 5), and (3, 3, 5) with deterministic, nonconstant sixth-root phases. For all compressed square minors it compares the augmented original determinant norm and matching-support indicator. Exact arithmetic in  $\mathbb{Q}(\sqrt{17})$  also verifies the two algebraic equalities defining the extremal constant. None of these finite checks proves the compactness, geometric minimax, or all-graph statements by extrapolation; those are proved in the text.

## 40.3 Build and rerun

The manuscript’s own commands, run from its package root, were `python3 code/verify.py` followed by two `pdflatex` runs on its `article.tex`; its Makefile offered the same. In this report every delivered file name carries the prefix `03-cactus-`:

- program and build file: `code/03-cactus-verify.py` and `code/03-cactus-Makefile`;
- recorded outputs: `data/03-cactus-verification.json` (the recorded run),  
`data/03-cactus-verification_log.txt` (its console output, byte-identical to the JSON)  
and `data/03-cactus-build_status.json` (the manuscript’s record of its own PDF build);
- notes: `03-cactus-STATUS.md` and `03-cactus-SOURCES.md`.

The program still uses the delivered names. Run in this directory, it would write a new unprefixed `data/verification.json` beside Part I’s files; the Makefile’s `verify` target would run Part I’s `code/verify.py`, which without an output option rewrites Part I’s shipped data; and its default target would rebuild this whole article in place. Rerun the program only in a scratch copy with the delivered names restored, from this report’s directory:

```
W=/path/to/scratch; mkdir -p "$W/03/code" "$W/03/data"
for f in code/03-cactus-*; do cp "$f" "$W/03/code/${f#code/03-cactus-}"; done
for f in data/03-cactus-*; do cp "$f" "$W/03/data/${f#data/03-cactus-}"; done
mv "$W/03/code/Makefile" "$W/03/Makefile"; cd "$W/03"
python3 code/verify.py # rewrites data/verification.json of the copy
```

The program needs Python 3.10 or later and only its standard library; no network access or repository checkout is needed, and its elapsed-time field differs from run to run. This article is rebuilt as described in the report's README; Part III needs no package beyond those of Parts I and II.

## 41 Proof audit, priority, and formalization boundary

### 41.1 The parts most worth independent scrutiny

The decisive combinatorial step is the induced-theta lemma, especially the length-one path case. The decisive algebraic step is (32.2), together with its exact matching bijection. Deleting arbitrary edges or suppressing arbitrary degree-two vertices would not be a valid substitute. In the Hermitian equivalence, the singleton and pair coefficients must be compared before concluding that the matrix has the graph's support. In the quantitative theorem, unit moduli of the pivot entries are essential for preserving the defect.

The sharp constant is not inferred from optimization software. Proposition 36.4 proves the lower bound and constructs an extremizer. The other cores are bounded away by  $3/5$ , a strictly larger rational number. The final step uses induced restriction and compression, not a supposition that every noncactus literally contains an induced copy of one of the three *small* cores.

For stability, all ten Rayleigh inequalities are covered by symmetry, and the input theorem applies to nonhomogeneous multiaffine polynomials as well as homogeneous ones. The leaf-preservation proof is independent of a general classification of stability-preserving operators.

### 41.2 What is and is not claimed to be new

The following separation of status is part of the mathematical result, not a bibliographic afterthought.

| Statement                                                                         | Status in this report                                                                                  |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Cactus sixth-root construction and its determinant factorization                  | Prior result of Part II; reproduced and credited.                                                      |
| Cactus univariate real-rootedness                                                 | Prior literature and Part II result; not claimed.                                                      |
| Height-two preorder matching-support transform                                    | Prior result of Part II; used only for consequences.                                                   |
| Cactus necessity, full gauge classification, and ternary support characterization | Proved here as answers/extensions to Part II's Research question 4; absolute priority not established. |
| Sharp universal defect $\varepsilon_*$ and the extremal $K_{2,3}$ phases          | Proved here; no global publication-priority claim.                                                     |
| Explicit stable $K_{2,3}$ separation and its tree extensions                      | Proved here in Part II's signed-support setting; general stability/determinant separation is prior.    |
| Failure of all fixed minor-order tests on subcubic width-two graphs               | Proved here as a consequence of high-girth theta supports.                                             |



The targeted source audit inspected the pinned repository discussion (Part II), the perfectly matchable set literature [9, 7], Brändén’s stability criterion [12], pattern-sensitive superregularity [13], and sixth-root-of-unity matrix terminology [14]. Related fundamental-transversal-matroid literature was also located, including an earlier binary characterization [16]; the present report does not claim novelty for the elementary binary corollary. This audit is not a complete survey of representable transversal matroids or every equivalent formulation of the support-pattern problem. A literature-priority determination would require that broader comparison. The proofs and the resolution of the identified repository question do not depend on a claim that no prior equivalent theorem exists.

### 41.3 A feasible formalization decomposition

A formal development should begin with finite bipartite graphs on two finite types and supports encoded by pairs of finite subsets. One layer proves that a forest support has at most one perfect matching and that symmetric differences decompose into alternating cycles. A second layer proves the exact compression bijection. A third proves the block determinant identity and the three core obstructions.

The graph-theoretic necessity can then be isolated from analysis. The ternary version needs only finite graph arguments and algebra over  $\mathbb{F}_3$ . The complex classification additionally needs elementary norm identities and the two allowed unit-circle solutions. The sharp gap adds real inequalities, the compact phase torus, and the planar Gram identity; the existence of the minimizer can be omitted if one first states the theorem for every phase assignment and gives the explicit extremizer.

Finally, the stability example can be represented as exact polynomial identities plus a separately imported or formalized Rayleigh criterion. This separation prevents the external analytic theorem from entering the trust boundary of the cactus classification. No uncompiled Lean files are supplied or described as checked. The Python verifier is executable evidence for finite cases, not a substitute for these formal layers.

## 42 Further research questions and topics

The original determinant-classification question is answered under the precise hypotheses stated here. The following directions remain outside the proved results. Except where attributed, they are questions generated by this investigation, not assertions that no answer is known anywhere in the literature.

**Research question 11** (Classify real-stable signed support polynomials). Which bipartite graphs have real stable  $\Phi_G$ ? The class contains all cacti and all tree attachments to  $K_{2,3}$ , but the determinant classification shows that no purely faithful-phase characterization can cover it. Determine a useful structural description, at least for graphs of treewidth two.

The three Rayleigh identities suggest searching for graph decompositions that preserve nonnegative Rayleigh differences without preserving a vertex determinant. A first target is a stable gluing theorem more general than leaf attachment. [Added 29 September 2026, batch 55: partly answered in Part VI. Signed-support stability is block-local: a one-vertex sum is stable exactly when both pieces are (Theorem 93.2, Corollary 93.3), which reaches the gluing target above; every complete bipartite graph is a stable block (Theorem 94.1). The tree incidence–apex graphs are stable exactly for the finite and affine ADE trees (Theorem 84.1), and all simple bipartite multi-theta graphs are classified (Theorem 84.3); both families have treewidth at most



two. Infinitely many induced-minimal unstable graphs of treewidth two exist (Section 92.6). The general classification remains open.]

**Research question 12** (Determine the remaining core optimization exactly). Compute  $d(T)$  exactly and classify its minimizing phase configurations. The present bound  $d(T) \geq 3/5$  suffices for the sharp universal theorem but is not claimed optimal. Determine how the value changes when individual minors receive prescribed positive penalties.

The constraints reduce to three unit vectors and their rank-two real Gram matrix. This gives a small semialgebraic problem, but a numerical stationary point alone is not an exact certificate.

**Research question 13** (Describe graphwise sharp defects). Which noncactus graphs satisfy  $d(G) = \varepsilon_*$ ? Having an induced or compressed  $K_{2,3}$  gives a lower bound, not an extension of its optimal phases to every other edge and minor. Characterize when the three-point extremizer can be extended without introducing a larger defect.

This asks for more than the universal infimum, which Theorem 36.3 already determines. Additional cycles can impose new norm constraints not visible in one chosen theta.

**Research question 14** (Larger finite fields and prescribed support). For  $q \geq 4$ , characterize useful classes of bipartite support patterns admitting a support-exact  $\mathbb{F}_q$  matrix. Proposition 35.3 gives a first field-size obstruction. Can the general question be expressed by a manageable family of induced-and-compressed obstructions?

Any proposed characterization must distinguish a fixed zero pattern from unrestricted representability of an associated abstract matroid. Changing the presentation can change the support question.

**Research question 15** (Larger determinantal models). For the stable polynomial  $\Phi_{K_{2,3}}$ , determine the smallest useful Hermitian pencil representation allowing auxiliary dimensions or structured substitutions. How does that size grow under the tree attachments of Corollary 37.4?

Theorem 29.1 addresses only the vertex-indexed diagonal pencil. A no-go result for that form should not be used as an obstruction to every possible determinant construction. [Added 29 September 2026, batch 55: Part VI writes  $F_{K_{2,3}}$ , the basis polynomial behind  $\Phi_{K_{2,3}} = \Phi_{\mathfrak{g}(2,2,2)}$ , as the expected determinant of a pencil with independent random columns (Theorems 91.2 and 91.4). That is a mixture of determinants, which this question does not exclude; it is not an answer.]

**Research question 16** (Quantitative obstructions with variable edge magnitudes). Replace the unit-phase constraint by  $1 - \eta \leq |K_{ab}|^2 \leq 1 + \eta$  on edges. Find an explicit lower bound on the largest support-minor defect, in terms of  $\eta$ , for noncacti. Does a useful dimension-independent gap persist for all sufficiently small  $\eta$ ?

Path compression now multiplies and divides magnitudes, so the proof of Lemma 36.2 no longer gives an exact contraction. Long subdivisions are the natural stress test.

**Research question 17** (Short checkable obstruction certificates). Design an algorithm that, given a noncactus graph and edge phases in an exact number field, outputs a matchable minor with defect at least  $\varepsilon_*$ , together with a compact certificate. Optimize the number of rows and columns of the output witness.

The constructive proof finds a theta and traces a failing core minor back through augmented minors. Theorem 38.1 proves that no constant bound on witness order is possible; it does not determine the best bound in terms of girth, cycle overlap, or input size.

**Research question 18** (Transfer beyond height-two posets). Can stable signed-support methods, rather than faithful matrices, extend Part II’s  $\gamma$ -interpretation (Theorem 13.2; compare Research question 1) to height-three posets or nontrivial preorder-equivalence blocks? Identify an explicit intermediate class where the combinatorial support model remains valid.

Neither the cactus theorem nor the  $K_{2,3}$  example answers general  $\gamma$ -positivity or existential flag-polytope realization. Those questions require an additional bridge beyond Theorem 13.2. [Added 29 September 2026, batch 44: the aim of this question is reached in Part IV by a different route. The combinatorial support model is valid for every finite preorder, with ordered transport pairs in place of matchable sets (Theorem 48.2, Section 55.3); no stable signed-support method is used, and whether one gives a second route is not addressed. Existential flag realization remains open (Research question 22).]

**Research question 19** (Kernel-checked certificate theory). Formalize the ternary support-pattern classification first, then extend it to the complex phase and sharp-gap theorems. Can the induced-theta reduction and the augmented-minor map be packaged as a reusable verified obstruction framework?

A successful implementation would convert a structural nonexistence proof into independently checkable finite witnesses while leaving the harder stability theory in a separate module.

## 43 Conclusion

The phase construction for cactus matching supports is not merely one successful family among many exact vertex determinant models: it exhausts them. Cycle overlaps produce induced theta obstructions, and a support-preserving Schur compression reduces every such obstruction to one of three small matrices. The same reduction determines the ternary support boundary and yields an optimal, dimension-independent approximation threshold.

At the same time, real stability survives beyond that boundary. The explicit  $K_{2,3}$  sums of squares and their leaf extensions show why searching only for a Hermitian vertex determinant would miss valid stability phenomena. The next stage of this report’s program should therefore distinguish structural determinant certificates from more general stability certificates, and should distinguish both from one-variable real-rootedness tests.

## 44 A sharper calculation for the shared-edge core

(Appendix A of manuscript 05.) Although the bound  $3/5$  is enough for the universal theorem, the other small core D also has an exact elementary minimax value.

**Proposition 44.1** (The exact defect of two squares sharing an edge). For  $D = \vartheta(1, 3, 3)$ ,

$$d(D) = d_0 := \frac{\sqrt{5} - 1}{2}.$$

*Proof.* In (33.2), every ambiguous proper square minor is one of the two cycle minors  $1 - x$ ,  $1 - y$ ; the only remaining ambiguous minor is the full determinant  $1 - x - y$ . Thus these three norms determine the defect.

Assume a defect  $\delta < d_0$ . The real parts  $a, b$  lie strictly inside the rectangle  $[l, h]^2$ , where

$$l = (1 - d_0)/2, \quad h = (1 + d_0)/2, \quad l + h = 1.$$

Use the separately convex/concave functions  $g_-$ ,  $g_+$  from Lemma 36.5. The two equal-coordinate corners of  $g_-$  have value  $d_0^2 = 1 - d_0$ . At an off-diagonal corner,

$$g_-(l, h) = (\sqrt{1 - l^2} - \sqrt{1 - h^2})^2 = \frac{(h - l)^2}{(\sqrt{1 - l^2} + \sqrt{1 - h^2})^2} \leq d_0^2.$$

The denominator is at least one:  $l + h = 1$  gives  $(1 - l^2) + (1 - h^2) = 1 + 2lh \geq 1$ . Therefore opposite imaginary signs force  $|1 - x - y|^2 \leq 1 - d_0 < 1 - \delta$ , impossible.

For the same-sign case, the equal-coordinate corners of  $g_+$  have values  $3 - 2d_0$  and  $3 + 2d_0$ . Both exceed  $1 + d_0$ , since  $d_0 < 2/3$ . For the off-diagonal corner,  $lh = d_0/4$  and

$$g_+(l, h) = 1 + \frac{d_0}{2} + 2\sqrt{\frac{d_0}{2} + \frac{d_0^2}{16}} > 1 + d_0.$$

The last strict inequality follows from  $\sqrt{d_0/2 + d_0^2/16} > d_0/4$ . Separate concavity therefore forces  $|1 - x - y|^2 > 1 + d_0 > 1 + \delta$ , another contradiction. Hence  $d(D) \geq d_0$ .

For attainment, take  $x = e^{i\alpha}$  and  $y = e^{-i\alpha}$  with  $\cos \alpha = (1 + d_0)/2$ . The two cycle squared norms are  $1 - d_0$ , while the full squared norm is  $d_0^2 = 1 - d_0$ . Each has defect  $d_0$ , and all other nonzero minors have norm one. This proves equality.  $\square$

This additional calculation is not needed for the main classification or for the sharp universal constant. It leaves the three-odd-path core as the remaining small exact optimization target.

## 45 A compact dependency ledger

(Appendix B of manuscript 05.)

| Result                      | Inputs                                                                                    |
|-----------------------------|-------------------------------------------------------------------------------------------|
| Cactus sufficiency          | Symmetric differences of matchings; independent cycle flips; sixth-root norm identity.    |
| Cactus necessity            | Induced-theta lemma; support-preserving Schur compression; three core calculations.       |
| Hermitian equivalence       | Diagonal coefficient expansion; singleton and pair recovery; bipartite block determinant. |
| Ternary classification      | Same graph reduction; three computations in $\mathbb{F}_3$ ; independent cycle factors.   |
| Gauge count                 | Spanning-forest normalization; two possible normalized phases per cycle.                  |
| Sharp universal gap         | Defect monotonicity; exact three-chord minimax; rational bounds on $D$ , $T$ .            |
| Stable noncactus family     | Three Rayleigh identities; Brändén's criterion; direct leaf-preservation argument.        |
| No bounded minor-order test | High-girth odd theta; uniqueness of a forest perfect matching; classification.            |

## Part IV. Gamma-positivity for every finite preorder

Part IV was added on 29 September 2026 in batch 44 of ProveIt's incoming-report intake, from two manuscripts that prove the same theorem: for every finite preorder  $\tau$ , the  $k$ th  $\gamma$ -coefficient of  $h_\tau$  is the number of ordered pairs of disjoint  $k$ -element sets joined by a  $\tau$ -increasing bijection. This proves Conjecture 5.2 of Athanasiadis and Chapoton [1] in general and answers Part II's Research question 1, "Extend the matching-support interpretation", which Part III's Research question 18 had re-posed. Part II's height-two theorem (Theorem 13.2) is the special case of height at most two. Parts I–III are unchanged except for dated pointers to this addition. Section 46 records the provenance, fixes the notation and states what is answered.

### 46 Provenance, conventions and what Part IV answers

#### 46.1 Two manuscripts, one addition

Part IV merges two batch-44 manuscripts, both AI-assisted research manuscripts prepared with ChatGPT for Vladimir Reshetnikov and dated 29 September 2026, both 22 A4 pages as delivered. They arrived in ProveIt commit ae28ea2db and were placed in 203016015. Both archives wrap an inner directory of the same name, ProveIt\_Preorder\_Gamma/, but they are different packages: no file of one coincides with a file of the other.

- **Manuscript 06 (the base):** *Gamma-Positivity for Every Finite Preorder: Disjoint transport supports, Boolean expansions, block formulas, and a transitivity criterion for palindromicity*, archive ProveIt\_Preorder\_Gamma\_Research.zip. It is pinned to ProveIt commit 9b24a3a8d (full hash in Section 47), at which this report's article.tex was the Git blob 8739a224fa8956adab5d8df9f2d5012a6a220959: Parts I and II exactly as printed above, before Part III. Its statements about "the repository" therefore concern Parts I and II. Its program, recorded outputs, build script and two notes are shipped with the prefix 05-gamma-positivity-.
- **Manuscript 01:** *The Gamma-Vector of Every Finite Preorder: Disjoint matching supports, Boolean expansions, and sharp degree laws*, archive ProveIt\_Preorder\_Gamma.zip. It is pinned to ProveIt commit 077672c84 (full hash in Section 47), at which this report's article.tex was the blob cd80088f04daeabdf6e320e08bdd1b61a5c9cf4c, Parts I–III exactly as printed above; it cites Part III once. Its program, recorded outputs and build script are shipped with the prefix 04-gamma-vector-; it delivered no audit notes.

The two prefixes continue this report's local sequence (Part I's unprefix files count as 01, Part II's are 02-, Part III's 03-) in arrival order; the base took the later number, as in earlier merges of the collection. The manuscripts, delivery READMEs and PDFs of both are not shipped and survive in the arrival commit. Manuscript 01's checksum list SHA256SUMS was verified (7/7) at placement and retired; manuscript 06 shipped none. Section 59 lists the shipped files.

*Why 06 is the base.* The two manuscripts prove the same main theorem by the same route and share most corollaries. Manuscript 06 is the more general: its second theorem (Theorem 48.3) concerns every *reflexive* relation, shows that palindromicity,  $\gamma$ -positivity and transitivity are equivalent there, and has an elementary proof independent of the imported counting identity. Manuscript 01 treats preorders only. Proofs taken from manuscript 01 were re-read for hidden uses of antisymmetry or bounded height; there are none (its own words: its reduction "applies

unchanged to arbitrary height and to supports which split equivalence classes"). What only manuscript 01 contains is printed under its own headings: the Boolean orbits of commuting toggles (Section 52), multivariate complement-duality and the low coefficients (Section 55), the leading Taylor coefficient at  $-1$  (Section 56), the induced-deck identity with odd-cardinality reconstruction (Section 57), an Ehrhart reinterpretation, three false variants, a dependency graph, and a compact proof certificate (Section 64).

**Abstract of manuscript 06 (as delivered).** Let  $h_\tau(t)$  count the lattice points of the preorder polytope of a finite preorder  $\tau$  by support size. We prove

$$h_\tau(t) = \sum_{k=0}^{\lfloor n/2 \rfloor} |\mathcal{T}_k(\tau)| t^k (1+t)^{n-2k}, \quad n = |\tau|,$$

where  $\mathcal{T}_k(\tau)$  consists of ordered pairs of disjoint  $k$ -element sets  $(U, V)$  for which a bijection  $f : U \rightarrow V$  with  $u \leq_\tau f(u)$  exists. The objects counted are support pairs, not individual bijections. This establishes Conjecture 5.2 of Athanasiadis and Chapoton for all finite preorders, without a height or antisymmetry assumption. A support-resolved multivariate identity is proved at the same time. The essential new step is that, after fixing the zero coordinates, every preorder support fibre is a bipartite demand polymatroid. An established matching-support counting identity then turns the fibre sum into Boolean intervals.

We also prove that, for any reflexive relation  $R$  on  $n$  elements, the coefficients of degrees 1 and  $n-1$  of its doubled matching-support polynomial are  $|R|$  and  $|R^*|$ , respectively, where  $R^*$  is transitive closure. Consequently that polynomial is palindromic if and only if  $R$  is transitive, and in that case it is gamma-positive. This gives an exact classification for balanced bipartite graphs admitting a perfect matching. Further consequences include positive quotient-block formulas, coefficientwise monotonicity, sharp universal bounds, an exact multiplicity formula at  $-1$ , and a centered-binomial mixture description of support size. Reproducible integer computations check all 7,332 labeled preorders through size five, 140 larger examples, and all reflexive relations through size four.

**Abstract of manuscript 01 (as delivered).** For a finite preorder  $\tau$  on  $n$  elements, let  $h_\tau(t)$  count the lattice points of its preorder polytope by the number of nonzero coordinates. We prove that its  $k$ th gamma coefficient counts ordered pairs of disjoint  $k$ -element sets  $U, V$  admitting a bijection  $f : U \rightarrow V$  with  $u \leq_\tau f(u)$ . Thus

$$h_\tau(t) = \sum_{\substack{U \cap V = \emptyset \\ U \rightsquigarrow_\tau V}} t^{|V|} (1+t)^{n-2|V|}.$$

This establishes gamma-positivity for every finite preorder, with no restriction on height or on the sizes of its equivalence classes. It extends the height-two matching-support theorem in Provel and proves the full Athanasiadis–Chapoton conjecture in the cited version.

The proof partitions a lattice point by its entire zero set, converts its remaining excess coordinates into a bipartite demand vector, and applies a classical demand–matching-support counting identity. A second reduction uses transitivity to cancel the overlap of two matched sets; it realizes the gamma expansion as Boolean orbits in a support-pair model. We obtain a multivariate refinement, complement-duality, coefficientwise gamma-monotonicity under relation extension, and the exact degree law  $\deg \Gamma_\tau = \nu(C_\tau)$ , where  $C_\tau$  is the undirected comparability graph. Consequently the multiplicity of  $-1$  is exactly the matching deficiency  $n - 2\nu(C_\tau)$ . Sharp extremal bounds, an explicit equivalence-block formula, recoveries of known special cases, exhaustive finite checks, and further research directions are included.

The two abstracts are reproduced as delivered, in their own notation: manuscript 06's  $R^*$  (transitive closure) is written  $\bar{R}$  below, and manuscript 01's  $C_\tau$  (comparability graph) is written  $\text{Comp}(\tau)$ , because  $C_\tau$  is Part I's polytope (Table 3).

**Status and dependence (manuscript 06, as delivered).** Full mathematical proofs are supplied. The general preorder theorem uses a classical bipartite counting identity, whose reduction to explicitly cited root-polytope theorems is included. The transitivity/palindromicity classification has a separate elementary proof. This manuscript is AI-assisted and unrefereed, and is not verified in Lean or Rocq. Computation supports the arguments but does not replace them. The source audit establishes the relation to the inspected sources; it does not certify absolute publication priority.

**Status (manuscript 01, as delivered).** The article supplies ordinary mathematical proofs with explicitly cited classical inputs. It has not been independently refereed or verified in Lean or another proof assistant. The literature and repository comparison is bounded; worldwide priority is not certified. Computations are independent finite checks, not substitutes for the proofs.

**Priority and prior work (from both manuscripts and their notes).** The matching-support interpretation of the augmented bipartite root polytope is Ohsugi–Tsuchiya’s [8, Propositions 3.3–3.4], restated by Davis–Kohl [7, Theorem 3.10]; the hypertree count of root-polytope volume is Kálmán–Postnikov’s [3]; the demand/support counting identity these give is Part II’s Lemma 14.1. None of these is claimed as new. The support-enumerator/root-polytope identity and preorder duality  $h_\tau = h_{\tau^*}$  are Dai, Hou, Liu, Thawinrak and Wang’s [2, Theorems 1.2–1.3]; Part IV recovers duality and refines it, and claims no first proof. Menon [6] gives a bijective proof of  $\gamma$ -positivity for octopuses and lopsided octopuses; the octopus formula (attributed there to Athanasiadis, Xiao and Yan) is recovered in Section 55, not discovered. The height-two case is Part II’s Theorem 13.2. The nonreal-root counterexamples to Conjecture 5.3, with priority credited to Shivam Patel [11], are Part II’s and are not re-claimed. The classical chain (Narayana), binomial and universal-equivalence formulas are checks, not novelty claims; the matching-theoretic ingredients of the reflexive-relation criterion are standard, and its relation to older matching and Gorenstein characterizations deserves further historical review. Targeted searches on 29 September 2026 (manuscript 06: “preorders”, “gamma-positivity”, “Athanasiadis”, “Chapoton”, “Conjecture 5.2”, “perfectly matchable set polynomial”, “transitive”) located no earlier general proof of the main formula, but some search responses were poor or unrelated; this is a bounded search, not a proof of the absence of a preprint, an unpublished argument or parallel work. No worldwide publication priority is claimed for any result of Part IV.

## 46.2 Notation of Part IV

Part IV works with an arbitrary finite preorder  $\tau$  on a ground set  $E$  of size  $n$ , as Part I does, and uses Part I’s  $\mathcal{J}(\tau)$  (all order ideals),  $Q_\tau$ ,  $h_\tau$  and  $\downarrow$ ; its  $\gamma$ -coefficients are those of Section 10, in the basis  $t^k(1+t)^{n-2k}$ , and  $\text{Vol}_{\mathbb{Z}}$  is the normalized volume of (7.4). For a poset of height at most two,  $\tau = P_G$  and Part II’s  $\mathcal{M}_k(G)$ ,  $p_G$  and  $\nu(G)$  are recovered (Section 55.3). Table 3 lists what Part IV adds or uses differently. Four symbols of the manuscripts were renamed and one was retypeset; no normalization was changed:

- manuscript 06’s transitive closure  $R^*$  is written  $\overline{R}$ , because  $\tau^*$  is the *opposite* preorder in Part I and in manuscript 01;
- manuscript 06’s opposite preorder  $\tau^{\text{op}}$  is written  $\tau^*$ , as in Part I;
- manuscript 01’s comparability graph  $C_\tau$  is written  $\text{Comp}(\tau)$ , manuscript 06’s notation, because  $C_\tau$  is Part I’s polytope (manuscript 01 itself warns of this);
- manuscript 01’s  $g_k(\tau)$ ,  $\mathcal{D}_k(\tau)$  and variables  $\mathbf{t} = (t_v)$  are written  $\gamma_k(\tau)$ ,  $\mathcal{T}_k(\tau)$  and  $\mathbf{z} = (z_v)$ , manuscript 06’s notation for the same objects;
- the preorder polytope,  $Q_\tau$  in both manuscripts, is printed  $Q_\tau = Q_\tau$ , as in Part I.



Table 3: Symbols of Part IV. The last column prints the tempting false reading, usually a symbol of Parts I–III with the same letter.

| Symbol                                                            | Meaning in Part IV                                                                                                                                                                                                                                   | Not to be confused with                                                                                                                                                                                                                 |
|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathcal{T}_k(\tau), \mathcal{T}(\tau)$                          | ordered pairs $(U, V)$ of disjoint $k$ -subsets of $E$ joined by a bijection $f : U \rightarrow V$ with $u \leq_\tau f(u)$ , counted once however many bijections exist ( <i>transport pairs</i> ); all $k$ (manuscript 01's $\mathcal{D}_k(\tau)$ ) | Part II's $\mathcal{M}_k(G)$ , a set of <i>unordered</i> vertex sets of size $2k$ (Section 55.3 relates them); the number of such bijections.                                                                                           |
| $\gamma_k(\tau), \Gamma_\tau(u)$                                  | $ \mathcal{T}_k(\tau) $ (manuscript 01's $g_k(\tau)$ ); $\Gamma_\tau(u) = \sum_k \gamma_k(\tau) u^k$                                                                                                                                                 | $\Gamma(1/4)$ in Section 58 is $\Gamma_\tau$ at $1/4$ , <i>not</i> Euler's Gamma function; Part III's cycle phase $\Gamma$ .                                                                                                            |
| $A \rightsquigarrow_\tau B$                                       | $ A  =  B $ and a bijection $f : A \rightarrow B$ with $a \leq_\tau f(a)$ exists; $A, B$ may overlap                                                                                                                                                 | a relation between elements.                                                                                                                                                                                                            |
| $\mathbf{z} = (z_i), z_A, H_\tau(\mathbf{z})$                     | support variables, $z_A = \prod_{i \in A} z_i$ , the multivariate support enumerator $\sum_{x \in Q_\tau \cap \mathbb{Z}^E} z_{\text{supp}(x)}$                                                                                                      | Part III's $\mathbf{z}$ in $\Phi_G(\mathbf{z})$ ; Part II's graph $H$ and matrix $H_G$ .                                                                                                                                                |
| $G_\tau, G_R$                                                     | doubled graph on $E_L \sqcup E_R$ with edge $a_L b_R$ iff $a \leq_\tau b$ (iff $aRb$ )                                                                                                                                                               | Part I's $G_\tau$ , which has $i_L j_R$ iff $j \leq_\tau i$ : the same graph with its two shores exchanged. Matchable vertex sets correspond, so $p_{G_\tau}$ is the same polynomial; the receiver side $B$ here is Part I's left side. |
| $H = (X \sqcup Y, F), \mathcal{R}_H, \mathcal{M}(H), H_{Z,S}$     | a bipartite graph with edge set $F$ ; its demand vectors (Part II's $\mathcal{R}_H$ ); its matchable pairs $(U, V)$ , $U \subseteq X, V \subseteq Y$ ; the zero/support graph of Lemma 50.1 (manuscript 01's $H_S$ )                                 | $\mathcal{M} = \mathcal{M}$ in Part II indexes vertex sets $W = U \sqcup V$ by size; $F$ is not a face or $F_H$ .                                                                                                                       |
| $p_H, p_R, F_H(\mathbf{z})$                                       | perfectly-matchable-set polynomial (Part II's); $p_{G_R}$ ; the demand support enumerator                                                                                                                                                            | Part I's $p(y)$ ; $p =  A $ .                                                                                                                                                                                                           |
| $R, \bar{R},  R $                                                 | a reflexive relation, its transitive closure (manuscript 06's $R^*$ ), number of related ordered pairs including the diagonal                                                                                                                        | $\tau^*$ , the <i>opposite</i> preorder (Part I, manuscript 01, Theorem 55.2); for a preorder $\bar{\tau} = \tau$ .                                                                                                                     |
| $\tau^*$                                                          | opposite preorder (manuscript 06's $\tau^{\text{op}}$ )                                                                                                                                                                                              | transitive closure.                                                                                                                                                                                                                     |
| $\text{Comp}(\tau), \nu$                                          | simple undirected comparability graph on $E$ (manuscript 01's $C_\tau$ ), its matching number                                                                                                                                                        | Part I's polytope $C_\tau$ ; a maximum matching of $G_\tau$ , which is perfect (size $n$ ).                                                                                                                                             |
| $P, E_v, s_v, \mathcal{D}(P, \mathbf{s}), D(d), B_{s,r}, A_{s,r}$ | quotient poset of $\tau$ , blocks, block sizes, feasible signed block vectors and $\sum_v \max(d_v, 0)$ , local polynomials (manuscript 06)                                                                                                          | Part I's $P_\tau$ and $B_\tau$ ; Part II's $P_G, D_G$ .                                                                                                                                                                                 |
| $E_i, m_i, b, (\alpha, \beta), \mathcal{A}(P, \mathbf{m})$        | the same blocks, sizes and number of blocks, source/receiver profiles, feasible profiles (manuscript 01)                                                                                                                                             | Part II's $m$ indexing $\Theta_m$ ; a dilation $m$ in (52.3).                                                                                                                                                                           |
| $d$                                                               | a signed vector in $\{-1, 0, 1\}^E$ or $\mathbb{Z}^P$ ; in Corollary 56.2 the deficiency $n - 2\nu$                                                                                                                                                  | Part I's diagonal vertices $d_i$ ; Part III's defect $d(G)$ .                                                                                                                                                                           |
| $D_\tau(t), D_2(\tau), R_{2,2}(\tau)$                             | the one-element deletion deck polynomial; two counts in Proposition 55.5                                                                                                                                                                             | Part I's neighbourhoods $D_i$ ; $\mathcal{D}(P, \mathbf{s})$ .                                                                                                                                                                          |
| $K, L$                                                            | mixing variable, uniform support size (Section 58)                                                                                                                                                                                                   | Part II's cactus matrix $K$ and its support size $S$ in (19.7); Part I's triangulation $K$ .                                                                                                                                            |
| $L_n, \mathbf{E}_n, C_k$                                          | the $n$ -chain, the universal equivalence preorder, the Catalan number                                                                                                                                                                               | Part I's $C_\tau$ ; the ground set $E$ .                                                                                                                                                                                                |
| $T, Z, S, W$                                                      | a subset; the zero set and the support of a lattice point; a common subset                                                                                                                                                                           | Part I's $S_\tau$ ; Part II's $T_G$ and matchable sets $W$ .                                                                                                                                                                            |



### 46.3 What Part IV answers

Status after Part IV, as of 29 September 2026:

- **Conjecture 5.2 of [1] ( $\gamma$ -positivity) holds for every finite preorder** (Theorem 48.2), with no height or antisymmetry assumption and an explicit count:  $\gamma_k(\tau) = |\mathcal{T}_k(\tau)|$ . This settles what Part I's Section 10 left open for Conjecture 5.2 and what Part II's Section 11.3 left open beyond height two, including height-two posets blown up into nontrivial equivalence classes (Example 54.2). Both manuscripts prove it, by the same route. Its proof does not use Part I's unrefereed polytope theorem.
- **Part II's Research question 1 ("Extend the matching-support interpretation") is answered.** The interpretation extends to every finite preorder with ordered transport pairs in place of matchable vertex sets; for posets of height at most two the two coincide (Section 55.3), so Part II's Theorem 13.2 is the special case. The "step of the lower-zero-set argument" that survives weighted blocks is the zero-set partition itself, taken over *all* zero coordinates rather than a minimal shore (Lemma 50.1); coordinates in nontrivial blocks may exceed one, as the question's remark warns, and the reduction allows that.
- **Part III's Research question 18 ("Transfer beyond height-two posets") is answered in its aim, by a different route:** the combinatorial support model is valid for every preorder, and no stable signed-support method is needed. Whether such methods give a second route is not addressed.
- **Part II's Research question 2 ("A purely combinatorial bridge") remains open:** both manuscripts use the same imported counting identity (Lemma 14.1) and construct no bijection; it is re-posed in support-resolved form as Research question 20.
- **Part I's nonpalindromic reflexive example** (Section 8, subsection 8.5) is an instance of Theorem 48.3: its support counts  $(1, 5, 6, 1)$  are  $(1, |R|, |\overline{R}|, 1)$  with one missing transitive pair (Example 53.3).
- **Conjecture 5.1(d) (flag realization) and Conjecture 5.3 remain as Parts II and III left them:** Part IV proves no flag realization, and  $\gamma$ -positivity does not imply real-rootedness (Section 60). Palindromicity and unimodality, Conjecture 5.1(a)–(b), follow again, by a route independent of Part I's geometry (Corollary 50.3).

Part IV refutes no statement of Parts I–III, and nothing in them is retracted. Parts I–III gain dated pointers: the title block, abstract, scope box and reading route; Section 10; Part I's reflexive example; Part II's status box, Section 11.3 and Research question 1; Part III's Section 26.3 and Research question 18.

### 46.4 Where the merge had to choose

- *Placement.* Part IV follows Part III and precedes Part I's appendices, whose letters and numbers are unchanged. The manuscripts' appendices (06: normalization checklist and source ledger; 01: proof certificate and source ledger) are printed as the ordinary Sections 63–65.
- *Order and base.* Manuscript 06's order of sections is kept; manuscript 01's sections are placed beside the matching ones of 06, and its sections without a counterpart are printed as separate sections or subsections marked "(manuscript 01)". Every result of both is printed; a result proved in both is printed once, with both sources named.

- *Duplicated results.* The main theorem, its support-fibre (06) or all-zero-set (01) lemma and its Boolean summation are the same argument in both manuscripts and are printed once, from 06, with manuscript 01's remarks and example. The same holds for the doubled-graph identity and the second derivation through cancellation, the support-resolved counting identity,  $\gamma$ -monotonicity, disjoint unions,  $\gamma_0$  and  $\gamma_1$ , the universal bounds, the chain and universal-equivalence formulas, the degree and  $-1$ -multiplicity theorem, and the algorithmic remark on block formulas. Where one statement is sharper (manuscript 01's strict unimodality, strict increase of  $\gamma_1$ , bounds on  $h$  and equality cases), the sharper statement is printed and attributed.
- *Second routes.* Genuinely different proofs are kept: manuscript 01's path-decomposition proof of overlap cancellation (after Corollary 51.2) beside 06's ideal-Hall proof; manuscript 01's profile form of the block formula (Theorem 54.3) beside 06's signed form; manuscript 01's direct initial-segment proof of the chain count and its reflection proof of the Narayana formula.
- *The counting identity.* Both manuscripts prove the demand/support identity by the reduction of Part II's Section 14, step for step (augmentation, the two published inputs at  $u = 1$ , the clone-and-Hall hypertree bijection), and say so. It is printed once, in Part II; Section 49 states it in Part IV's notation and records the manuscripts' additional remarks.
- *Research questions.* The two manuscripts' twenty questions continue Part III's numbering as Research questions 20–36. Three pairs ask the same question (flag realization; a support-preserving bijection; formal verification) and are printed once each with both formulations, so there are seventeen.
- *Self-citation.* Both manuscripts cite this report as an external repository report (reference "Repo", with their pins). Here those citations become references to Parts I–III, and "the repository('s)" becomes "Part II('s)" or "this report" where it means this report; the pins are kept in Section 47.
- *File names.* The delivered names, such as `code/verify.py`, `results/verification.json` and `data/verification.json`, are replaced by the shipped names in Section 59, because `code/verify.py` in this directory is Part I's verifier; the run commands are replaced by commands that run on a copy or with explicit output paths. Package-content lists that name unshipped files are kept and annotated.
- *Bibliography.* Every external reference of both manuscripts was already in this report's bibliography: 06's and 01's OT is [8], 01's Dai is [2]; nothing is appended.
- *Labels and front matter.* Labels carry the prefix `gam:`, and `gam:gv:` for material only manuscript 01 contains. Part II's Research questions 1 and 2 and Part I's subsection on reflexive relations received the labels `gam:q:extend`, `gam:q:bridge` and `gam:sec:reflexiveex` so that Part IV can cite them. The two title pages, abstracts and status boxes are printed in this section, and manuscript 01's "shortest proof audit" box in Section 47; both tables of contents and manuscript 06's reading-route box, which only lists its own sections, are dropped.

## 47 The question, its provenance, and the advance

Athanasiadis and Chapoton associate a support polynomial to every finite preorder and ask whether it has nonnegative coefficients in the gamma basis  $t^k(1+t)^{n-2k}$  [1, Conjecture 5.2]. The question is stronger than palindromicity and unimodality: positive combinations of these

centered binomial polynomials have both properties, but a symmetric unimodal polynomial need not be gamma-positive. It is weaker than real-rootedness. Their separate Conjecture 5.1(d) asks for a flag simplicial-polytope realization; that is a geometric statement, not a reformulation of gamma-positivity (manuscript 01).

Part II of this report provides the immediate research context. At the snapshot manuscript 06 inspected, Part II proves gamma-positivity for posets of height at most two. Its explicit open boundary includes arbitrary height and height-two quotient posets whose elements have been replaced by nontrivial equivalence classes. The main result below removes both restrictions. It also identifies every gamma coefficient by a finite matching condition. The answer to Conjecture 5.2 is affirmative. The word “supports” is essential: a pair of vertex sets is counted once whenever at least one suitable matching exists, regardless of how many such matchings it has. Likewise, “disjoint” refers to the underlying elements of the preorder, not merely to left and right copies of a bipartite graph (manuscript 01).

#### 47.1 What is inherited, and what is proved here

The matching-support interpretation for augmented bipartite root polytopes is prior work of Ohsugi and Tsuchiya [8, Propositions 3.3–3.4], restated by Davis and Kohl [7, Theorem 3.10]. The needed total-count identity follows with the hypertree interpretation of root-polytope volume [3]; Part II already uses this identity in its height-two argument (Lemma 14.1). None of these facts is claimed as new.

Dai, Hou, Liu, Thawinrak, and Wang identify bipartite demand support polynomials with the corresponding root-polytope numerators and prove preorder duality [2, Theorems 1.2–1.3]. Menon gives a bijective proof of gamma-positivity for octopuses and an extension to lopsided octopuses [6]. These are relevant antecedents, not results that this addition claims to discover. The theorem below includes Menon’s families and all other finite preorders; their explicit special-case formulas are recoveries (Section 55), and univariate duality is recovered and refined rather than claimed (Theorem 55.2).

The additional argument is the universal support-fibre reduction in Lemma 50.1, together with its Boolean summation. It applies to *every* finite preorder. The cancellation theorem gives another explanation, and the endpoint-defect theorem identifies exactly when palindromicity occurs in the balanced perfect-matching setting. The block, extremal, matching-deficiency, and probabilistic formulas make the resulting structure usable beyond the original conjecture.

#### 47.2 Precise scope of the conclusions

The theorem settles the statement of Conjecture 5.2 in the version of [1] consulted here; it also proves its asserted palindromicity and unimodality consequences. Part I’s ordinary simplicial-polytopal realization is not an input. In particular, no unrefereed geometric assertion in ProveIt is needed to complete the proof.

The stronger flag-polytopal realization question is not settled. Nor does gamma-positivity imply real-rootedness: Part II records and extends previously posted counterexamples to the latter conjecture, with priority credited there to Shivam Patel [11]. We do not re-claim those counterexamples. We also do not identify  $h_\tau$  with the Ehrhart numerator of the *original* preorder polytope; those are different invariants.

The ProveIt tree inspected by manuscript 06 is pinned to commit

9b24a3a8d545af9624f6ac455f5b548be62818b6.

[Merge note: at that commit this report consisted of Parts I and II as printed above; see

Section 46.1.] Its relevant report path and the primary-source details appear in the bibliography and in the accompanying source audit (05-gamma-positivity-SOURCE\_AUDIT.md). A targeted search on 29 September 2026 did not locate an earlier general proof of the main formula, but search results were not uniformly informative. No exhaustive priority claim follows from that search.

### 47.3 The repository frontier seen by manuscript 01

The ProveIt snapshot inspected by manuscript 01 is commit

077672c84f743ff5de82ce6becc3ad6b2ff8787a.

Its report *A Reflexive Root-Polytope Model for Preorder  $h$ -Polynomials*, in

SetTheory/Cardinals/docs/reports/enumerative-combinatorics/  
preorder-root-polytopes/

contains a height-two theorem in Part II and later cactus results in Part III. [Merge note: that is this report, Parts I–III as printed above.] Part II expresses the gamma coefficients of a height-at-most-two *poset* through perfectly matchable subsets of its bipartite comparability graph. Its further-research section explicitly asks for an extension to greater height and to nontrivial preorder-equivalence blocks (Research question 1). The README at that snapshot still distinguishes this restricted theorem from the unsettled general gamma-positivity question.

The new observation is that one should partition by *all* zero coordinates, not only by zeros on a minimal shore. For a support  $S$  with complement  $Z$ , excess demand at any element of  $S$  can be supplied from comparable elements of  $Z$ . Transitivity turns the ideal inequalities into the appropriate Hall inequalities. This construction applies unchanged to arbitrary height and to supports which split equivalence classes.

Part II supplies the useful demand-counting lemma by a reduction to classical root-polytope results. Manuscript 01 reproduces the reduction with its published dependencies identified. Therefore the general theorem does not require accepting Part I’s unrefereed geometric claims.

| Item                             | Status (manuscript 01’s table, as delivered)                                                                                                                    |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General gamma-positivity         | Proved for every finite preorder, with an explicit nonnegative formula for each coefficient.                                                                    |
| Multivariate and structural laws | Proved: support refinement, complement-duality, Boolean support-pair action, relation-monotonicity, exact degree and $-1$ -multiplicity, and block compression. |
| Classical input                  | The demand-matching-support count, derived from the Kálmán-Postnikov and Ohsugi-Tsuchiya root-polytope identities.                                              |
| Previously settled statements    | Height-two gamma-positivity, special arbor formulas, and univariate duality are credited and recovered.                                                         |
| Not established                  | Flag simplicial-polytope realization, a natural lattice-point bijection implementing the Boolean action, or a general real-rootedness statement.                |
| Verification                     | Exhaustive integer tests and explicit proofs; no new proof-assistant verification.                                                                              |

The priority assessment concerns the cited primary papers and the pinned repository report. A search that finds no matching general theorem does not establish that no such theorem exists elsewhere.

**Shortest proof audit (manuscript 01).** Read the main statement in Section 48, the counting lemma in Section 49 (with its proof in Part II, Section 14), and the zero-set reduction in Section 50. The latter reduction is the step that removes Part II's height-two restriction. Section 51 gives a second reduction sharing the same counting lemma. No unrefereed root-polytope theorem from this report is assumed.

## 48 Definitions and main theorems

Let  $E$  be a finite set of size  $n$ . A preorder  $\tau$  is a reflexive, transitive relation  $\leq_\tau$  on  $E$ . We do not assume antisymmetry. An *ideal* is a subset  $I$  such that  $v \in I$  and  $u \leq_\tau v$  imply  $u \in I$ . Write  $\mathcal{J}(\tau)$  for all ideals, including  $\emptyset$  and  $E$ . For  $T \subseteq E$ , put

$$\downarrow T = \{u \in E : u \leq_\tau v \text{ for some } v \in T\}.$$

Reflexivity gives  $T \subseteq \downarrow T$ , and transitivity makes  $\downarrow T$  an ideal. These are exactly the two properties used in the main reduction.

For  $x \in \mathbb{R}^E$  and  $I \subseteq E$ , write  $x(I) = \sum_{i \in I} x_i$ . The preorder polytope and its support polynomials are, as in Part I,

$$Q_\tau = \{x \in \mathbb{R}_{\geq 0}^E : x(I) \leq |I| \text{ for every } I \in \mathcal{J}(\tau)\}, \quad (48.1)$$

$$h_\tau(t) = \sum_{x \in Q_\tau \cap \mathbb{Z}^E} t^{|\text{supp}(x)|}, \quad \text{supp}(x) = \{i : x_i > 0\}, \quad (48.2)$$

$$H_\tau(\mathbf{z}) = \sum_{x \in Q_\tau \cap \mathbb{Z}^E} z_{\text{supp}(x)}, \quad z_A = \prod_{i \in A} z_i, \quad h_\tau(t) = H_\tau(t, \dots, t). \quad (48.3)$$

The full-set inequality makes the lattice-point set finite. The all-zero point is the only point of empty support, and the all-one point is the only point of full support, since its  $n$  positive integer coordinates already sum to at least  $n$ . Thus  $h_\tau$  has degree  $n$  and both endpoint coefficients are one.

The polynomial  $h_\tau$  is initially a *support enumerator*. It must not be confused with the Ehrhart numerator  $h^*(Q_\tau; t)$ . An Ehrhart numerator of a different root polytope appears in Section 52 (manuscript 01).

**Definition 48.1** (Disjoint transport supports). Let  $\mathcal{T}_k(\tau)$  be the set of ordered pairs  $(U, V)$  of subsets of  $E$  such that

$$U \cap V = \emptyset, \quad |U| = |V| = k,$$

and there exists a bijection  $f : U \rightarrow V$  satisfying  $u \leq_\tau f(u)$  for every  $u \in U$ . Set  $\mathcal{T}(\tau) = \bigcup_k \mathcal{T}_k(\tau)$ . More generally, for subsets  $A, B \subseteq E$ , write  $A \rightsquigarrow_\tau B$  if  $|A| = |B|$  and there exists a bijection  $f : A \rightarrow B$  satisfying  $a \leq_\tau f(a)$  for every  $a \in A$  (manuscript 01's "compatible support pairs";  $A$  and  $B$  may overlap).

The empty pair is included. A support pair with several possible bijections is counted once. The order of the pair matters, especially inside a nontrivial equivalence class.

**Theorem 48.2** (General preorder gamma formula; manuscripts 06 and 01). *For every finite preorder  $\tau$  on  $E$ ,*

$$h_\tau(t) = \sum_{k=0}^{\lfloor n/2 \rfloor} |\mathcal{T}_k(\tau)| t^k (1+t)^{n-2k}. \quad (48.4)$$

More strongly, for commuting variables  $\mathbf{z} = (z_i)_{i \in E}$ ,

$$H_\tau(\mathbf{z}) = \sum_{x \in \mathcal{Q}_\tau \cap \mathbb{Z}^E} z_{\text{supp}(x)} = \sum_{(U,V) \in \mathcal{T}(\tau)} z_V \prod_{i \in E \setminus (U \cup V)} (1 + z_i). \quad (48.5)$$

Consequently  $\gamma_k(\tau) = |\mathcal{T}_k(\tau)|$  are nonnegative integers, proving Athanasiadis–Chapoton Conjecture 5.2 for arbitrary finite preorders.

Here and below a gamma expansion is a polynomial identity; no value is assigned to a rational substitution at its pole  $t = -1$ . The expansion is unique:  $t^k(1+t)^{n-2k}$  has lowest degree  $k$  with coefficient one, so successive coefficients in increasing degree determine  $\gamma_0, \gamma_1, \dots$  (manuscript 01). Define the gamma polynomial

$$\Gamma_\tau(u) = \sum_k \gamma_k(\tau) u^k. \quad (48.6)$$

Then (48.4) can be abbreviated as  $h_\tau(t) = (1+t)^n \Gamma_\tau(t/(1+t)^2)$ , with the right side interpreted by its polynomial expansion, not by substitution at the undefined quotient  $t = -1$ . The degree of  $\Gamma_\tau$  and the multiplicity of  $-1$  as a root of  $h_\tau$  are determined exactly in Theorem 56.1, which both manuscripts state (manuscript 01 already in its introduction): with  $\text{Comp}(\tau)$  the simple undirected graph on  $E$  in which distinct vertices are adjacent precisely when they are comparable in at least one direction,  $\deg \Gamma_\tau = \nu(\text{Comp}(\tau))$  and  $\gamma_k(\tau) > 0$  exactly for  $0 \leq k \leq \nu(\text{Comp}(\tau))$ .

There is a second result about a relation which need not be transitive. For a reflexive relation  $R$  on  $E$ , form the bipartite graph  $G_R$  on two copies of  $E$ , with edge  $i_L j_R$  exactly when  $i R j$ ; for a preorder,  $G_\tau$  has edge  $a_L b_R$  exactly when  $a \leq_\tau b$ . Define

$$p_R(t) = \sum_{\substack{A, B \subseteq E, |A|=|B| \\ G_R[A_L, B_R] \text{ has a perfect matching}}} t^{|A|}. \quad (48.7)$$

Its degree is  $n$ , since all diagonal edges are present. The relation  $\bar{R}$  denotes reflexive transitive closure, and  $|R|$  counts ordered related pairs, including the  $n$  diagonal pairs. [Merge note:  $G_\tau$  is Part I's  $G_\tau$  with its shores exchanged, and  $p_\tau = p_{G_\tau}$  is the polynomial of Part II's Remark 11.1; see Table 3.]

**Theorem 48.3** (Transitivity is the palindromicity criterion; manuscript 06). *For  $n \geq 1$  and every reflexive relation  $R$ ,*

$$[t]p_R(t) = |R|, \quad [t^{n-1}]p_R(t) = |\bar{R}|. \quad (48.8)$$

The following conditions are equivalent:

- (i)  $R$  is transitive;
- (ii) every matchable pair  $(A, B)$  remains matchable after removing  $A \cap B$ ;
- (iii)  $p_R(t)$  is palindromic of degree  $n$ ;
- (iv)  $p_R(t)$  is gamma-positive of degree  $n$ ;
- (v)  $[t^{n-1}]p_R(t) = [t]p_R(t)$ .

In particular, the asymmetry between these two coefficients counts exactly the ordered pairs missing from transitive closure:

$$[t^{n-1}]p_R - [t]p_R = |\bar{R} \setminus R|. \quad (48.9)$$

For  $n = 0$ , all statements except the displayed coefficient indices have the obvious trivial interpretation. The theorem's proof in Sections 51 and 53 is elementary and does not use root polytopes or Theorem 48.2.



## 49 The established bipartite counting identity

We state the non-elementary enumerative input precisely and record its reduction to the cited classical theorems. This also fixes the distinction between support counts, matching counts, and hypertree counts.

Let  $H = (X \sqcup Y, F)$  be a finite bipartite graph with specified shores, and let  $N_H(T) \subseteq X$  be the neighborhood of  $T \subseteq Y$ . Its integral demand set is Part II's

$$\mathcal{R}_H = \{c \in \mathbb{Z}_{\geq 0}^Y : c(T) \leq |N_H(T)| \text{ for all } T \subseteq Y\}. \quad (49.1)$$

A vector in  $\mathcal{R}_H$  describes integer requests at right vertices that can be served by distinct unit-capacity left vertices; we count demand *vectors*, not allocations serving them. Write  $\mathcal{M}(H)$  for the pairs  $(U, V)$  with  $U \subseteq X$ ,  $V \subseteq Y$ ,  $|U| = |V|$ , such that  $H[U, V]$  has a perfect matching, and  $p_H(u) = \sum_{(U, V) \in \mathcal{M}(H)} u^{|V|}$ , the perfectly matchable set polynomial; the specified shores allow us to write a matchable vertex set uniquely as  $U \sqcup V$ .

**Lemma 49.1** (Demand/support equinumeracy, inherited). *For every finite bipartite graph  $H$ ,*

$$|\mathcal{R}_H| = |\mathcal{M}(H)| = p_H(1). \quad (49.2)$$

This is Lemma 14.1 of Part II. Both manuscripts prove it by the reduction printed there: adjoin  $x_0, y_0$  with all edges  $x_0y, xy_0, x_0y_0$  to obtain the connected graph  $\widehat{H}$ ; combine (14.3) (Kálmán–Postnikov [3], evaluated at one; manuscript 01 cites its Theorems 1.1 and 3.10) with (14.4) (Ohsugi–Tsuchiya [8, Propositions 3.3–3.4], in the augmented form of Davis–Kohl [7, Theorem 3.10]) through  $h^*(B; 1) = \text{Vol}_{\mathbb{Z}}(B)$ , with the full affine integer lattice; and identify the right hypertree vectors with  $\mathcal{R}_H$  by the forest count and the clone-and-Hall construction of Section 14. The manuscripts' versions differ from Part II's only in presentation: manuscript 06 bounds the forest on  $T \cup N_H(T) \cup \{x_0\}$  directly, and manuscript 01 spells out the augmenting-path proof of the required direction of Hall's criterion (choose a largest matching; absence of an augmenting path from an unmatched clone yields a reached clone set larger than its neighborhood). Both note that empty shores give the single empty matching support, and manuscript 01 that the construction need not choose a unique tree, because the resulting *vector* is unique. Manuscript 01 adds: the augmentation matters; identity (14.4) is about the edge polytope of  $\widehat{H}$ , not the unaugmented edge polytope of  $H$  and not a polytope whose vertices are individual matchings.

**Remark 49.2** (The input is not a matching enumeration; manuscript 06). For  $H = K_{2,2}$ , there are six demand vectors: the nonnegative pairs of total at most two. There are also six matchable support pairs: the empty pair, four single-edge supports, and one full support. There are two perfect matchings on the full support, but it contributes only one. Replacing  $|\mathcal{M}(H)|$  by the number of individual matchings would already fail in this example.

### 49.1 A support-resolved version by Boolean inversion

For later use, the total identity also determines every prescribed right support. This refinement recovers the known support-enumerator identity; it is not a new priority claim for that identity. It requires no additional geometric theorem.

**Proposition 49.3** (Prescribed demand support; manuscripts 06 and 01). *For every  $S \subseteq Y$ ,*

$$\#\{c \in \mathcal{R}_H : \text{supp}(c) = S\} = \#\{U \subseteq X : (U, S) \in \mathcal{M}(H)\}. \quad (49.3)$$

Consequently

$$F_H(\mathbf{z}) := \sum_{c \in \mathcal{R}_H} z_{\text{supp}(c)} = \sum_{(U, V) \in \mathcal{M}(H)} z_V. \quad (49.4)$$



*Proof.* For  $T \subseteq Y$ , the demands supported inside  $T$  are exactly those for  $H[X, T]$ : inequalities involving outside vertices follow from the ones on  $T$ , since outside coordinates are zero and neighborhoods only enlarge. By Lemma 49.1, their number is

$$a(T) = \#\{(U, V) \in \mathcal{M}(H) : V \subseteq T\}.$$

Both sides of (49.3), as functions of  $S$ , therefore have the same sums over all  $S \subseteq T$ . Boolean inversion gives their equality. Explicitly, a fixed  $V \subseteq S$  contributes

$$\sum_{V \subseteq T \subseteq S} (-1)^{|S|-|T|} = \begin{cases} 1 & V = S, \\ 0 & V \neq S. \end{cases}$$

Summing (49.3) against  $z_S$  proves (49.4).  $\square$

Manuscript 01 proves the same refinement by induction on  $|T|$  and records the explicit value: both sides of (49.3) equal  $\sum_{V \subseteq S} (-1)^{|S|-|V|} p_{H[X \sqcup V]}(1)$ .

No explicit canonical bijection between the two sets in (49.3) has been constructed in this argument. The equality is enough for all the polynomial identities below.

**Remark 49.4** (What is not constructed; manuscript 01). The lemmas give equality of cardinalities, refined by the receiver support. They do not specify a natural bijection from a demand vector to a matchable support pair. The explicit tree construction is a bijection with hypertree *vectors*, not with matchings or with the pairs counted by  $p_H$ . This distinction remains important when we introduce Boolean actions (Section 52).

[Added 30 September 2026, batch 64: Part IX gives such a map, support by support (Theorem 135.2); it depends on a cost system and is not natural under automorphisms (Proposition 144.1).]

## 50 Proof of the general preorder theorem

The decisive step is to apply the bipartite identity to the zeros and the positive coordinates of an arbitrary preorder point, rather than to fixed minimal and maximal levels of a height-two poset.

**Lemma 50.1** (Every support fibre is a bipartite demand set; manuscripts 06 and 01). *Fix  $S \subseteq E$  and set  $Z = E \setminus S$ . Let  $H_{Z,S}$  be the bipartite graph with shores  $Z, S$  and edges  $zs$  exactly when  $z \leq_\tau s$ . The map*

$$c \mapsto x, \quad x_z = 0 \ (z \in Z), \quad x_s = 1 + c_s \ (s \in S) \quad (50.1)$$

*is a bijection from  $\mathcal{R}_{H_{Z,S}}$  onto the lattice points of  $\mathcal{Q}_\tau$  with support exactly  $S$ .*

*Proof.* The map is plainly a bijection between  $\mathbb{Z}_{\geq 0}^S$  and all nonnegative integral vectors of support  $S$ . We prove the equivalence of its feasibility conditions. For an ideal  $I$ , the preorder inequality is

$$c(I \cap S) \leq |I \cap Z|. \quad (50.2)$$

Suppose these inequalities hold. Given  $T \subseteq S$ , take the ideal  $I = \downarrow T$ . Then  $T \subseteq I \cap S$ , and  $c$  is nonnegative, so

$$c(T) \leq c((\downarrow T) \cap S) \leq |(\downarrow T) \cap Z| = |N_{H_{Z,S}}(T)|.$$

Thus  $c \in \mathcal{R}_{H_{Z,S}}$ .

Conversely, suppose all the displayed neighborhood bounds hold. For an arbitrary ideal  $I$ , put  $T = I \cap S$ . Downward closure gives  $\downarrow T \subseteq I$ , whence

$$c(I \cap S) \leq |(\downarrow(I \cap S)) \cap Z| \leq |I \cap Z|.$$

This is (50.2), so  $x \in Q_\tau$ . Both implications use only reflexivity, transitivity, and nonnegativity. They do not use a height bound, or require equivalence classes to be singletons.  $\square$

Manuscript 01 calls this the all-zero-set reduction and adds: no property of a minimal or maximal layer is used. In particular, the set  $Z$  need not be an ideal, the set  $S$  need not be a filter, and neither set need be a union of equivalence classes. The only order-theoretic facts used are that  $\downarrow T$  is an ideal and that an ideal contains the lower closure of each of its subsets.

*Proof of Theorem 48.2.* For each  $S \subseteq E$ , Lemmas 50.1 and 49.1 give

$$\#\{x \in Q_\tau \cap \mathbb{Z}^E : \text{supp}(x) = S\} = \#\{(U, V) \in \mathcal{T}(\tau) : U \subseteq E \setminus S, V \subseteq S\}. \quad (50.3)$$

Indeed, the pairs on the right are exactly the matchable support pairs in  $H_{E \setminus S, S}$ . Multiply by  $z_S$  and sum over  $S$ ; all sums are finite, so we may sum first over the pairs. For a fixed  $(U, V) \in \mathcal{T}(\tau)$ , the permissible supports are precisely

$$V \subseteq S \subseteq E \setminus U.$$

Every such  $S$  is uniquely  $V \cup W$  with  $W \subseteq E \setminus (U \cup V)$ . Its total contribution is

$$\sum_{W \subseteq E \setminus (U \cup V)} z_V z_W = z_V \prod_{i \in E \setminus (U \cup V)} (1 + z_i).$$

This proves (48.5). Specializing all  $z_i$  to  $t$  gives (48.4), because  $|U| = |V| = k$  leaves exactly  $n - 2k$  free labels.  $\square$

Equation (50.3) is manuscript 01's Corollary "Support-by-support counting formula": the general theorem is not merely an equality after forgetting which coordinates are nonzero.

**Example 50.2** (Why equivalence classes cause no exception; manuscript 01). If  $E = \{1, 2, 3\}$  is a single equivalence class, the only ideals are  $\emptyset$  and  $E$ . The lattice point  $(3, 0, 0)$  is allowed. Its support  $S = \{1\}$  is not a union of equivalence classes, but the reduction still works: the two zero coordinates supply the excess demand  $c_1 = 2$  through the two edges of  $H_{Z, S}$ . Nothing requires an individual coordinate to be bounded by one. In this example  $h_\tau = 1 + 9t + 9t^2 + t^3$  and  $\Gamma_\tau = 1 + 6u$ .

## 50.1 What the Boolean expansion does and does not provide

Equation (50.3) says that each lattice-point support count is the number of abstract intervals  $[V, E \setminus U]$  containing that support, with  $(U, V)$  a transport pair. An interval has bottom rank  $k$ , top rank  $n - k$ , and rank enumerator  $t^k(1 + t)^{n-2k}$ . This is the source of gamma-positivity.

The intervals are an *enumerative model*; they are not asserted to be actual coordinatewise subposets of the lattice-point set. Manuscript 06 constructs no commuting involutions on those lattice points and no canonical partition of them into geometric cubes. Those stronger structures would require a compatible, explicit realization of the counting identity in Lemma 49.1. (Manuscript 01's commuting toggles, Theorem 52.1, act on a support-pair model, not on the lattice points.)

**Corollary 50.3** (Palindromicity and strict unimodality; manuscripts 06 and 01). *For every finite preorder,  $h_\tau(t) = t^n h_\tau(t^{-1})$ , that is,  $h_j(\tau) = h_{n-j}(\tau)$ , and  $h_0(\tau) \leq h_1(\tau) \leq \dots \leq h_{\lfloor n/2 \rfloor}(\tau)$  (manuscript 06). In fact the coefficients are strictly increasing up to the middle (manuscript 01):*

$$[t^j]h_\tau < [t^{j+1}]h_\tau \quad (0 \leq j < \lfloor n/2 \rfloor).$$

When  $n$  is odd, the two middle coefficients are equal. The inequalities on the other side follow by symmetry.

*Proof.* Each polynomial  $t^k(1+t)^{n-2k}$  has symmetric coefficients centered at  $n/2$ , nondecreasing up to that center. A nonnegative sum preserves both properties. There is no appeal to a geometric realization. Moreover  $\gamma_0 = 1$ , and  $(1+t)^n$  has strictly increasing coefficients for the indicated indices. Adding the remaining nonnegative symmetric-unimodal summands preserves these strict inequalities.  $\square$

Thus the theorem proves the palindromicity and unimodality assertions of Athanasiadis–Chapoton Conjecture 5.1(a)–(b) as well as Conjecture 5.2. It makes no implication in the reverse direction: a palindromic unimodal polynomial need not be gamma-positive. [Merge note: Part I proposes these properties, and more, through its simplicial polytope  $S_\tau$  (Theorem 1.1); the proof here is independent of Part I’s unrefereed geometric argument.] [Added 29 September 2026, batch 55: the coefficients are ultra-log-concave of order  $n$  (Part V, Corollary 67.2), which  $\gamma$ -positivity does not imply.]

## 51 Ideal inequalities, cancellation, and a second proof

The gamma coefficients can be recognized without searching through bijections. This also explains why transitivity is the exact structural hypothesis behind the Boolean intervals.

**Lemma 51.1** (The ideal Hall criterion; manuscript 06). *For subsets  $A, B \subseteq E$  of equal size, there is a bijection  $f : A \rightarrow B$  with  $a \leq_\tau f(a)$  if and only if*

$$|A \cap I| \geq |B \cap I| \quad \text{for every } I \in \mathcal{J}(\tau). \quad (51.1)$$

*Proof.* If  $b \in B \cap I$ , its matched predecessor belongs to  $A \cap I$ . Distinct  $b$  have distinct predecessors, proving necessity. Conversely, let  $T \subseteq B$ . The predecessors of  $T$  inside  $A$  are  $A \cap \downarrow T$ . Since  $\downarrow T$  is an ideal and  $T \subseteq B \cap \downarrow T$ ,

$$|T| \leq |B \cap \downarrow T| \leq |A \cap \downarrow T|.$$

Hall’s criterion gives a matching saturating  $B$ ; equality of the shore sizes makes it a bijection.  $\square$

**Corollary 51.2** (Common-support cancellation; manuscripts 06 and 01). *For any  $A, B \subseteq E$  with  $|A| = |B|$ , the pair  $(A, B)$  is matchable in  $G_\tau$  if and only if  $(A \setminus B, B \setminus A)$  is matchable; that is,*

$$A \rightsquigarrow_\tau B \iff A \setminus B \rightsquigarrow_\tau B \setminus A. \quad (51.2)$$

*Proof.* The common intersection contributes the same quantity to both sides of (51.1), so it cancels. Alternatively, the reverse implication extends a matching by the diagonal edges on  $A \cap B$ .  $\square$

Manuscript 01 states (51.2) as its Lemma “Canceling common support” and proves it without the ideal Hall criterion.

*Second proof of Corollary 51.2, by path decomposition (manuscript 01).* Suppose first that  $f : A \rightarrow B$  is a compatible bijection. On the original ground set draw the directed edges  $a \rightarrow f(a)$ , retaining possible loops. Every vertex has indegree and outdegree at most one. A vertex in  $U = A \setminus B$  has outdegree one and indegree zero; one in  $V = B \setminus A$  has indegree one and outdegree zero; one in  $A \cap B$  has both degrees one. The nontrivial components are therefore directed paths from  $U$  to  $V$  and directed cycles lying entirely in  $A \cap B$ .

To justify that a path beginning in  $U$  cannot enter a cycle, the first entry vertex of that cycle would receive two incoming edges. Since the graph is finite, the path must instead terminate, and it can terminate only in  $V$ . Different starting vertices yield disjoint paths. Along each path, transitivity implies that its starting vertex is below its ending vertex. Replacing every path by this direct relation gives a bijection from  $U$  to  $V$ . Cycles and loops are discarded.

Conversely, a compatible bijection from  $U$  to  $V$  extends to one from  $A$  to  $B$  by fixing every element of  $A \cap B$ . The additional relations are valid by reflexivity.  $\square$

In particular, the truth of  $A \rightsquigarrow_\tau B$  depends on the disjoint core  $(A \setminus B, B \setminus A)$ , not on the choice of common elements (manuscript 01).

### 51.1 Signed vectors and transport feasibility

For a disjoint pair  $(U, V)$ , define  $d_i = 1$  on  $U$ ,  $d_i = -1$  on  $V$ , and  $d_i = 0$  elsewhere. Lemma 51.1 gives a useful equivalent formula:

$$\Gamma_\tau(u) = \sum_k \gamma_k(\tau) u^k = \sum_{\substack{d \in \{-1, 0, 1\}^E \\ d(E)=0 \\ d(I) \geq 0 \ (I \in \mathcal{J}(\tau))}} u^{| \{i: d_i=1\} |}. \quad (51.3)$$

Thus the coefficients count balanced signed labelings whose cumulative sum on every ideal is nonnegative. This is not a count of integer flows with multiplicity: distinct realizations of the same signed labeling still contribute once.

For a chain, these are Dyck words with zero symbols inserted. For a general preorder, ideals replace initial intervals. Equivalent elements are tested only together in the ideal inequalities, but their signs remain individually visible in (51.3). This distinction will produce the block multiplicities in Section 54.

### 51.2 Recovering the doubled-graph identity

**Lemma 51.3** (The doubled graph's demand body; manuscripts 06 and 01). *After identifying right coordinates with  $E$ ,  $\mathcal{Q}_\tau \cap \mathbb{Z}^E = \mathcal{R}_{G_\tau}$ . The analogous equality holds for real nonnegative vectors satisfying the same inequalities.*

*Proof.* The neighborhood of  $T_R$  in  $G_\tau$  is  $(\downarrow T)_L$ . If  $x \in \mathcal{Q}_\tau$ , nonnegativity and the ideal inequality for  $\downarrow T$  give  $x(T) \leq x(\downarrow T) \leq |\downarrow T|$ . Conversely, when the demand inequalities hold, take  $T = I$  for an ideal; then  $\downarrow I = I$ , so  $x(I) \leq |I|$ .  $\square$

[Merge note: this is Part I's Lemma 2.1 with the shores of  $G_\tau$  exchanged.] Proposition 49.3 therefore yields

$$h_\tau(t) = p_\tau(t), \quad H_\tau(\mathbf{z}) = \sum_{x \in \mathcal{Q}_\tau \cap \mathbb{Z}^E} z_{\text{supp}(x)} = \sum_{A \rightsquigarrow_\tau B} z_B, \quad (51.4)$$

where  $p_\tau = p_{G_\tau}$  and the sets  $A$  and  $B$  need not be disjoint as subsets of  $E$  (manuscript 01's Proposition "Relation-graph realization"). The univariate identity is already contained in the

cited support-enumerator and root-polytope results; it is Part II's Remark 11.1. Here it follows from the weaker total-count input by Boolean inversion.

For a matchable pair, set

$$U = A \setminus B, \quad V = B \setminus A, \quad W = A \cap B.$$

Cancellation says exactly that  $(U, V) \in \mathcal{T}(\tau)$ , while  $W$  is an arbitrary subset of  $E \setminus (U \cup V)$ . Conversely, every such triple gives  $A = U \cup W$ ,  $B = V \cup W$  and a matchable pair. Hence

$$\sum_{A \rightsquigarrow_{\tau} B} z_B = \sum_{(U, V) \in \mathcal{T}(\tau)} z_U \sum_{W \subseteq E \setminus (U \cup V)} z_W,$$

which is a second derivation of Theorem 48.2. This proof and the support-fibre proof share the classical counting input but not the same preorder reduction; manuscript 01 stresses that the second route is not an independent replacement for the classical input.

### 51.3 Cancellation characterizes transitivity

**Proposition 51.4** (Manuscript 06). *Let  $R$  be reflexive on a finite set. Common-support cancellation holds for all equally sized pairs if and only if  $R$  is transitive.*

*Proof.* The transitive case is Corollary 51.2. If  $R$  is not transitive, choose  $aRb$  and  $bRc$  with  $aRc$  false. Reflexivity ensures that  $a, b, c$  are distinct. The pair

$$A = \{a, b\}, \quad B = \{b, c\}$$

is matched by  $a \mapsto b$  and  $b \mapsto c$ . After deleting the common vertex  $b$ , the pair  $(\{a\}, \{c\})$  is not matchable. Thus cancellation fails already on a two-edge path.  $\square$

[Added 29 September 2026, batch 55: in matroid terms, cancellation for all pairs says that the two copies of each label are clones in the transversal matroid  $M_{\tau}$  (Part VII, Theorem 103.2), and  $M_{\tau}$  determines  $\tau$  up to isomorphism (Theorem 105.2).]

## 52 Boolean orbits and a recovered Ehrhart interpretation

(Manuscript 01 only.)

**Theorem 52.1** (Boolean orbits). *Let  $\mathcal{M}_{\tau}$  be the set of compatible ordered pairs  $(A, B)$ ,  $A \rightsquigarrow_{\tau} B$ , graded by  $|B|$ . For each  $v \in E$ , define a map  $T_v$  as follows: if  $v$  belongs to exactly one of  $A, B$ , leave the pair unchanged; otherwise toggle its membership in both sets. Then each  $T_v$  is a well-defined involution of  $\mathcal{M}_{\tau}$ , the involutions commute, and their orbits are indexed by the disjoint compatible pairs  $(U, V)$ . The orbit of a core of size  $|U| = |V| = k$  is a Boolean family of dimension  $n - 2k$ , with rank polynomial  $t^k(1+t)^{n-2k}$ .*

*Proof.* Each toggle preserves the disjoint core, so Corollary 51.2 proves that it preserves compatibility. Toggling twice returns the original pair. Different toggles act on different element memberships and commute. For fixed core  $(U, V)$ , the common subset  $C$  can be chosen arbitrarily in the complement of  $U \cup V$ ; the toggles there add or remove its elements. The resulting orbit consists of all  $2^{n-2k}$  choices, and  $|B| = |V| + |C|$ .  $\square$

[Added 29 September 2026, batch 55: the toggles  $T_v$  are the clone transpositions ( $v^+ v^-$ ) of the transversal matroid  $M_\tau$ , and their orbits give Part VII's multivariate orbit polynomial (Theorem 107.1).] This is an explicit combinatorial orbit interpretation of the gamma coefficients. It is an action on the *support-pair model*, not an explicit action on  $Q_\tau \cap \mathbb{Z}^E$ . The counting bridge supplies equal support-fiber cardinalities, but transporting the toggles to lattice points naturally would require an additional bijection. No such bijection is claimed here (compare Research question 21).

## 52.1 A recovered Ehrhart interpretation

Let  $B_{\widehat{G}_\tau}$  be the augmented edge polytope of the relation graph. Its dimension is  $2n$ : its connected graph has  $2n + 2$  vertices, and the sums on the two shores each equal one. Combining (14.4) and (51.4) gives

$$h^*(B_{\widehat{G}_\tau}; t) = h_\tau(t). \quad (52.1)$$

This is a geometric consequence of the counting proof, not an identification of  $h_\tau$  with  $h^*(Q_\tau; t)$ . Root-polytope interpretations of support enumerators are already part of the primary literature [2]; the importance here is the additional explicit gamma interpretation. [Merge note:  $B_{\widehat{G}_\tau}$  is Part I's  $B_\tau$  of (2.1) with its two coordinate blocks exchanged, a lattice isomorphism, so (52.1) is Part I's Proposition 2.2.]

In particular,

$$\text{Vol}_{\mathbb{Z}}(B_{\widehat{G}_\tau}) = |Q_\tau \cap \mathbb{Z}^E| = \sum_k \gamma_k(\tau) 2^{n-2k}, \quad (52.2)$$

$$|mB_{\widehat{G}_\tau} \cap \mathbb{Z}^{2n+2}| = \sum_{j=0}^n [t^j] h_\tau(t) \binom{m+2n-j}{2n} \quad (m \in \mathbb{Z}_{\geq 0}). \quad (52.3)$$

The lattice in the second line is restricted automatically to the affine span of the dilation. These are standard consequences of the Ehrhart numerator identity. A  $2n$ -dimensional root-polytope realization does not, by itself, settle the flag  $n$ -dimensional simplicial-polytope question. (Manuscript 01 writes  $\text{NVol}$  for the normalized volume, which is Part I's  $\text{Vol}_{\mathbb{Z}}$  of (7.4).)

## 53 An exact classification for balanced bipartite graphs

For reflexive relations, failure of transitivity is visible in two coefficients of the matching-support polynomial. This is stronger than merely exhibiting a nonpalindromic counterexample. This section is manuscript 06's.

**Lemma 53.1** (Near-perfect supports detect reachability). *Let  $R$  be reflexive on  $E$ . For  $a, b \in E$ , there is a perfect matching from  $E \setminus \{a\}$  to  $E \setminus \{b\}$  using  $R$  if and only if  $b \bar{R} a$ .*

*Proof.* When  $a = b$ , diagonal edges give the matching and reachability is reflexive. Assume  $a \neq b$ . A matching is a bijection  $f : E \setminus \{a\} \rightarrow E \setminus \{b\}$  with  $iRf(i)$ . Draw its edges as arrows on  $E$ . The vertex  $b$  has one outgoing arrow and no incoming arrow;  $a$  has one incoming arrow and no outgoing arrow; every other vertex has one of each. Starting at  $b$  and following arrows must reach  $a$ . It cannot enter a directed cycle: the first cycle vertex entered would have two incoming arrows, unless the path started on the cycle, which is impossible at  $b$ . Thus there is an  $R$ -path from  $b$  to  $a$ .

Conversely, choose a simple directed path  $b = x_0 R x_1 R \cdots R x_\ell = a$ . Match  $x_i$  to  $x_{i+1}$  along the path, and match every vertex outside the path to itself. These edges match precisely  $E \setminus \{a\}$  onto  $E \setminus \{b\}$ .  $\square$



*Proof of Theorem 48.3.* Matchable singleton pairs are the edges of  $G_R$ , so  $[t]p_R = |R|$ . Pairs of size  $n - 1$  are indexed uniquely by their omitted labels  $(a, b)$ . Lemma 53.1 makes their number  $|\bar{R}|$ , proving (48.8) and (48.9).

Transitivity is equivalent to cancellation by Proposition 51.4. Under transitivity, the decomposition  $(A, B) \leftrightarrow (U, V, W)$  in Section 51, used directly on (48.7), gives

$$p_R(t) = \sum_{(U,V) \in \mathcal{T}(R)} t^{|U|} (1+t)^{n-2|U|}.$$

This implication uses no demand or root-polytope identity. Thus transitivity implies gamma-positivity, which implies palindromicity and hence equality of the coefficients of degrees 1 and  $n - 1$ . If those coefficients are equal, then  $|\bar{R}| = |R|$ . Since  $R \subseteq \bar{R}$  and both sets are finite,  $R = \bar{R}$ ; that is,  $R$  is transitive. All five conditions are equivalent.  $\square$

**Corollary 53.2** (Balanced graphs with a perfect matching). *Let  $H = (X \sqcup Y, F)$  be a bipartite graph with  $|X| = |Y| = n$ , admitting a perfect matching. Label the shores  $x_1, \dots, x_n$  and  $y_1, \dots, y_n$  so that  $x_i y_i$  are the edges of a chosen perfect matching, and define  $iRj$  by  $x_i y_j \in F$ . The following are equivalent:*

- (i) *the perfectly matchable set polynomial  $p_H(t)$  is palindromic;*
- (ii)  *$p_H(t)$  is gamma-positive;*
- (iii) *the relation  $R$  is a preorder.*

*These conditions do not depend on the chosen perfect matching. They are also equivalent to palindromicity, or to gamma-positivity, of the demand support polynomial  $F_H(t)$ .*

*Proof.* The chosen matching supplies every diagonal relation, so Theorem 48.3 applies. The polynomial  $p_H$  is unchanged by relabeling or by a different choice of distinguished matching. Thus the criterion must hold for every such choice whenever it holds for one. Finally, Proposition 49.3 gives  $F_H = p_H$ .  $\square$

This answers the palindromicity/gamma-positivity part of the support-enumerator classification question in [2, Problem 5.3] for the balanced perfect-matching class. It is not a classification for unbalanced graphs or for balanced graphs without perfect matchings, nor is it a real-rootedness criterion. The matching-theoretic ingredients are standard; the explicit endpoint-defect interpretation and its application here are supplied with proofs, without a separate claim of absolute historical priority. [Added 29 September 2026, batch 55: Part V extends this to every bipartite graph:  $p_H$  is palindromic in its actual degree exactly when, after isolated vertices are deleted, the graph is balanced, has a perfect matching and the matched relation is a preorder (Theorem 67.4); unimodality holds for every bipartite graph, in ultra-log-concave form (Corollary 70.2).]

**Example 53.3** (A missing transitive edge). Let  $E = \{a, b, c\}$ , with all loops and just the additional relations  $aRb$  and  $bRc$ . There are five related pairs, but its transitive closure has six. Directly,

$$p_R(t) = 1 + 5t + 6t^2 + t^3.$$

The defect  $6 - 5 = 1$  detects the single missing pair  $(a, c)$ . Adding that pair produces the three-element chain, with

$$p_{\bar{R}}(t) = 1 + 6t + 6t^2 + t^3 = (1+t)^3 + 3t(1+t).$$

This example illustrates why one cannot simply drop transitivity in the preorder theorem. It also shows why using a Hasse diagram instead of the full order relation changes the polynomial.



[Merge note, not in manuscript 06: this is Part I's nonpalindromic example of subsection 8.5. There  $D_i$  is the set of  $j$  with  $jRi$ , so  $D_1 = \{1\}$ ,  $D_2 = \{1, 2\}$ ,  $D_3 = \{2, 3\}$  is the relation above with  $(a, b, c) = (1, 2, 3)$ , and by Proposition 49.3 its draconian support counts  $(1, 5, 6, 1)$  are the coefficients of  $p_R$ :  $(1, |R|, |\bar{R}|, 1)$ . Part I's explanation of what fails (the reduction to ideal differences) is Proposition 51.4 in another form. The endpoint formula (48.8) and the equivalence of palindromicity with transitivity were also checked by the batch-44 dossier on all 4,165 reflexive relations with  $1 \leq n \leq 4$ ; that check is not shipped.]

## 54 Positive formulas for equivalence-class blowups

A preorder is a poset of equivalence classes, each carrying a positive integer size. The signed-vector formula compresses the gamma polynomial to this quotient without sacrificing positivity.

Let  $P$  be the quotient poset of  $\tau$ , with blocks  $E_v$  of sizes  $s_v \geq 1$ , and let  $N = \sum_{v \in P} s_v$ . Ideals of  $\tau$  are exactly unions of blocks whose indices form ideals of  $P$ . Define

$$\mathcal{D}(P, \mathbf{s}) = \left\{ d \in \mathbb{Z}^P : \begin{array}{l} |d_v| \leq s_v \text{ for all } v, \quad \sum_v d_v = 0, \\ \sum_{v \in I} d_v \geq 0 \text{ for every } I \in \mathcal{J}(P) \end{array} \right\}, \quad D(d) = \sum_v \max(d_v, 0). \quad (54.1)$$

For  $0 \leq r \leq s$ , put

$$B_{s,r}(u) = \sum_{j=0}^{\lfloor (s-r)/2 \rfloor} \frac{s!}{j!(j+r)!(s-r-2j)!} u^j. \quad (54.2)$$

All these coefficients are multinomial integers.

**Theorem 54.1** (Quotient-block formula; manuscript 06). *For every finite preorder with quotient data  $(P, \mathbf{s})$ ,*

$$\Gamma_\tau(u) = \sum_{d \in \mathcal{D}(P, \mathbf{s})} u^{D(d)} \prod_{v \in P} B_{s_v, |d_v|}(u). \quad (54.3)$$

*This applies to arbitrary quotient height and arbitrary nontrivial block sizes.*

*Proof.* For a transport pair  $(U, V)$ , put  $a_v = |U \cap E_v|$ ,  $b_v = |V \cap E_v|$ , and  $d_v = a_v - b_v$ . Lemma 51.1 says that the pair is matchable exactly when its block differences lie in (54.1): all labels in a block have the same comparisons to every other block, and all ideals are block unions.

For fixed  $d_v$ , let  $j_v = \min(a_v, b_v)$ . The counts in the block are then  $j_v + |d_v|$  of one sign,  $j_v$  of the other, and  $s_v - |d_v| - 2j_v$  zeros. The number of assignments of these signs to the labeled elements is

$$\frac{s_v!}{j_v!(j_v + |d_v|)!(s_v - |d_v| - 2j_v)!}.$$

Every assignment has the same feasibility status, because the ideal conditions depend only on  $d$ . The common cardinality of the two supports is

$$|U| = \sum_v (\max(d_v, 0) + j_v) = D(d) + \sum_v j_v.$$

Multiplying the local counts and summing first over  $j_v$ , then over  $d$ , proves the formula.  $\square$

### 54.1 A companion formula in the ordinary basis

For  $0 \leq r \leq s$ , define

$$A_{s,r}(t) = \sum_{j=0}^{s-r} \binom{s}{j+r} \binom{s}{j} t^j. \quad (54.4)$$

Extracting the coefficient of  $z^r$  in the Laurent identity

$$(1+z)^s(1+t/z)^s = (1+t+z+t/z)^s$$

gives the positive gamma expansion

$$A_{s,r}(t) = \sum_{j=0}^{\lfloor (s-r)/2 \rfloor} \frac{s!}{j!(j+r)!(s-r-2j)!} t^j (1+t)^{s-r-2j}. \quad (54.5)$$

To see the right-hand count explicitly, choose  $j+r$  copies of  $z$ ,  $j$  copies of  $t/z$ , and  $s-r-2j$  copies of  $1+t$  in the second expression. The first expression yields (54.4).

Since  $\sum_v |d_v| = 2D(d)$ , substitution into Theorem 54.1 gives

$$h_\tau(t) = \sum_{d \in \mathcal{D}(P,s)} t^{D(d)} \prod_{v \in P} A_{s_v, |d_v|}(t). \quad (54.6)$$

No rational functions remain in this identity. Each summand has center  $N/2$  because its degree after the shift is  $N - D(d)$  and its smallest possible exponent is  $D(d)$ .

**Example 54.2** (A nontrivial height-two blowup). Take two equivalence classes of size two, with every member of the first below every member of the second. The feasible differences are  $(r, -r)$ ,  $r = 0, 1, 2$ . Since

$$B_{2,0} = 1 + 2u, \quad B_{2,1} = 2, \quad B_{2,2} = 1,$$

we get

$$\Gamma_\tau(u) = (1 + 2u)^2 + 4u + u^2 = 1 + 8u + 5u^2,$$

and therefore

$$h_\tau(t) = 1 + 12t + 27t^2 + 12t^3 + t^4.$$

The quotient has height two, but the preorder is not a height-two poset. It lies beyond the scope of Part II's height-two theorem. The example also exposes the importance of counting ordered support pairs inside an equivalence class.

### 54.2 Algorithmic consequence and its encoding qualification

The empty quotient gives  $\Gamma_\tau = 1$ . With  $r = |P| \geq 1$  fixed, (54.3) computes  $\Gamma_\tau$  in a number of integer arithmetic operations polynomial in the expanded size  $N$ . Enumerate the ideals of the  $r$ -element quotient once. Enumerate  $d_1, \dots, d_{r-1}$  and determine  $d_r$  from the zero-sum equation, rejecting out-of-range values and failed ideal inequalities. There are at most  $(2N+1)^{r-1}$  candidates. For each accepted candidate multiply the local polynomials, using ordinary polynomial multiplication.

For example, a coarse bound after local precomputation is

$$O(2^r r^2 + (2N+1)^{r-1}(r2^r + rN^2))$$

arithmetic operations. This is an arithmetic, not bit-operation, estimate. It is polynomial for fixed quotient size when  $N$  measures the expanded input. If the block sizes are encoded in binary, the bound is only pseudopolynomial in that compressed input; no stronger complexity assertion is intended.

### 54.3 The profile form of the block formula (manuscript 01)

Manuscript 01 proves an equivalent block formula with a different parametrization and proof. Let  $P = \tau/\sim$  be the quotient poset, where  $u \sim v$  means both  $u \leq_\tau v$  and  $v \leq_\tau u$ . Index its equivalence classes by  $1, \dots, b$ , with positive sizes  $m_1, \dots, m_b$ . Relations between two classes are uniform: every element of class  $i$  lies below every element of class  $j$  precisely when  $i \leq_P j$ . Set  $n = \sum_i m_i$ .

For a disjoint pair  $(U, V)$ , let  $\alpha_i = |U \cap E_i|$  and  $\beta_i = |V \cap E_i|$ . These profiles satisfy

$$\alpha_i, \beta_i \in \mathbb{Z}_{\geq 0}, \quad \alpha_i + \beta_i \leq m_i, \quad \sum_i \alpha_i = \sum_i \beta_i.$$

For  $T \subseteq [b]$ , set  $\downarrow_P T = \{i : i \leq_P j \text{ for some } j \in T\}$ .

**Theorem 54.3** (Exact block formula; manuscript 01). *Let  $\mathcal{A}(P, \mathbf{m})$  consist of the profiles above which also satisfy*

$$\sum_{j \in T} \beta_j \leq \sum_{i \in \downarrow_P T} \alpha_i \quad (T \subseteq [b]). \quad (54.7)$$

Then

$$\Gamma_\tau(u) = \sum_{(\alpha, \beta) \in \mathcal{A}(P, \mathbf{m})} \left( \prod_{i=1}^b \frac{m_i!}{\alpha_i! \beta_i! (m_i - \alpha_i - \beta_i)!} \right) u^{\sum_i \beta_i}. \quad (54.8)$$

*Proof.* A matching covering the chosen receivers can use exactly the selected sources in predecessor classes. For the set of all selected receivers in classes  $T$ , Hall's inequality is (54.7). These inequalities are also sufficient: any subset of selected receivers uses some class set  $T$  and has at most  $\sum_{j \in T} \beta_j$  members; its neighbors are all selected sources in  $\downarrow_P T$ . Thus Hall's criterion applies to the individual selected vertices. Equality of the total source and receiver counts makes a receiver-covering matching a bijection.

For a fixed feasible profile, the choices of disjoint sources and receivers in class  $i$  number  $\binom{m_i}{\alpha_i} \binom{m_i - \alpha_i}{\beta_i}$ . Compatibility depends only on the profiles, not on the chosen labels, because class-to-class relations are uniform. Multiplying these counts and summing the feasible profiles gives (54.8).  $\square$

This is a closed finite formula, not a claim that the quotient poset alone determines  $h_\tau$ : the multiplicities  $m_i$  are indispensable. Replacing the classes by singletons loses both the profile range and the multinomial weights. [Merge note: grouping the profiles of (54.8) by  $d_i = \alpha_i - \beta_i$  and  $j_i = \min(\alpha_i, \beta_i)$  gives (54.3); (54.7) is Hall's condition on receivers, whereas (54.1) is the ideal form of Lemma 51.1.]

*Algorithmic consequence (manuscript 01).* For fixed  $b$ , there are at most

$$\prod_{i=1}^b \binom{m_i + 2}{2} = O(n^{2b}) \quad (54.9)$$

profiles before imposing equal totals. Testing a profile by (54.7) takes at most  $2^b$  subset inequalities, each involving at most  $b$  additions after the lower closures are precomputed. Therefore (54.8) computes the entire gamma polynomial in polynomial time in the *expanded cardinality*  $n$  for a fixed number of classes. This is not a bound polynomial in the bit length of binary-encoded  $m_i$ , and it is not a polynomial bound when  $b$  varies with  $n$ . The weights and their sums use exact integers. The universal gamma bound is at most  $3^n$ , since a disjoint pair assigns each element to source, receiver, or neither, so the output coefficients have  $O(n)$  bits.

Large block sizes therefore enlarge the enumerated state space, but do not create uncontrolled coefficient bit lengths relative to expanded  $n$ .

**Example 54.4** (Examples beyond Part II's domain; manuscript 01). For a chain of three quotient classes of sizes  $(2, 1, 2)$ , the formula gives

$$\Gamma_\tau(u) = 1 + 12u + 17u^2, \quad h_\tau(t) = 1 + 17t + 63t^2 + 63t^3 + 17t^4 + t^5.$$

The quotient has height three and two classes are nontrivial. Both restrictions excluded by Part II's height-two-poset proof fail here.

For two incomparable classes of sizes 2 and 1 below a class of size 3,

$$\Gamma_\tau(u) = 1 + 17u + 36u^2 + 7u^3,$$

$$h_\tau(t) = 1 + 23t + 119t^2 + 201t^3 + 119t^4 + 23t^5 + t^6.$$

Its comparability graph has a perfect matching. Hence  $h_\tau(-1) = -7$ , agreeing with Corollary 56.2. The companion program checks these values both by direct ideal-inequality lattice enumeration and by the block formula.

## 55 Monotonicity, duality, extremal coefficients, and basic families

The transport interpretation gives comparisons between different preorders without comparing their lattice points individually.

**Theorem 55.1** (Monotonicity and products; manuscripts 06 and 01). *Suppose  $\tau$  and  $\sigma$  are preorders on the same set and  $\leq_\tau \subseteq \leq_\sigma$ . Then*

$$\gamma_k(\tau) \leq \gamma_k(\sigma) \quad \text{for every } k. \quad (55.1)$$

*More precisely  $h_\sigma - h_\tau$  has a nonnegative expansion in the  $n$ -centered gamma basis. If the relations differ, then  $\gamma_1(\tau) < \gamma_1(\sigma)$ , so the two gamma vectors and the two support polynomials differ (manuscript 01). Moreover, for disjoint unions and opposites,*

$$\Gamma_{\tau \sqcup \sigma}(u) = \Gamma_\tau(u) \Gamma_\sigma(u), \quad \Gamma_{\tau^*}(u) = \Gamma_\tau(u). \quad (55.2)$$

*Proof.* Every matching allowed by  $\tau$  is allowed by  $\sigma$ , so  $\mathcal{T}_k(\tau) \subseteq \mathcal{T}_k(\sigma)$ . For  $k = 1$ , these pairs are exactly the distinct-element relations; since both preorders contain all diagonal relations, a proper inclusion adds at least one such pair. Subtract the two expansions in (48.4). In a disjoint union, matchings cannot cross components. Thus each transport support pair splits uniquely into a transport pair in each component, and cardinalities add. Finally, duality sends  $(U, V)$  to  $(V, U)$  and reverses a witnessing bijection.  $\square$

The corresponding duality for  $h$  was already proved in [2]; this is a transport-support explanation of it, not a first proof claim. The monotonicity also implies coefficientwise monotonicity of  $h$ , since every gamma-basis polynomial has nonnegative coefficients; that weaker statement also follows from  $\mathcal{Q}_\tau \subseteq \mathcal{Q}_\sigma$ , because every  $\sigma$ -ideal is a  $\tau$ -ideal. Manuscript 01 notes that the theorem is stronger: the difference remains nonnegative in the more restrictive gamma basis.

**Theorem 55.2** (Multivariate complement-duality; manuscript 01). *Let  $\tau^*$  be the opposite preorder. Then*

$$H_\tau(\mathbf{z}) = z_E H_{\tau^*}(\mathbf{z}^{-1}). \quad (55.3)$$

*In particular  $\gamma_k(\tau) = \gamma_k(\tau^*)$  and  $h_\tau = h_{\tau^*}$ .*

*Proof.* The map  $(U, V) \mapsto (V, U)$  is a bijection from  $\mathcal{T}_k(\tau)$  to  $\mathcal{T}_k(\tau^*)$ , since a compatible bijection can be inverted. In the term of (48.5) indexed by  $(V, U)$  for  $\tau^*$ , invert all variables and multiply by  $z_E$ . With  $F = E \setminus (U \cup V)$ , the term becomes

$$z_E z_U^{-1} \prod_{w \in F} (1 + z_w^{-1}) = z_V \prod_{w \in F} (1 + z_w),$$

which is exactly the corresponding term for  $\tau$ . The Laurent expression simplifies to a polynomial. The univariate conclusions also follow directly by swapping the disjoint pairs.  $\square$

Equivalently, the number of lattice points of exact support  $S$  for  $\tau$  equals the number of exact support  $E \setminus S$  for  $\tau^*$ . This does *not* assert that  $H_\tau$  itself is invariant under complementing the support. For the chain  $1 < 2$ ,  $H_\tau = 1 + z_1 + 2z_2 + z_1z_2$ , which illustrates the distinction. [Merge note: the univariate identity  $h_\tau = h_{\tau^*}$  is Conjecture 5.4 of [1], settled by [2, Theorem 1.3]; see Part I, Section 7.]

**Proposition 55.3** (Disjoint unions; manuscript 01). *For preorders on disjoint sets with no cross-relations,*

$$H_{\tau \sqcup \sigma} = H_\tau H_\sigma, \quad h_{\tau \sqcup \sigma} = h_\tau h_\sigma, \quad \Gamma_{\tau \sqcup \sigma} = \Gamma_\tau \Gamma_\sigma.$$

*Adjoining an isolated element multiplies  $h$  by  $1 + t$  and leaves  $\Gamma$  unchanged.*

*Proof.* The defining ideal inequalities separate, giving a product of polytopes. Alternatively a compatible pair and its matching separate into the two ground sets, with equal source and receiver cardinalities within each. Disjoint compatible cores therefore multiply as well.  $\square$

**Corollary 55.4** (The first gamma coefficient; manuscripts 06 and 01). *For every preorder,*

$$\gamma_0(\tau) = 1, \quad \gamma_1(\tau) = \#\{(u, v) : u \neq v, u \leq_\tau v\} = |\tau| - n. \quad (55.4)$$

*In particular,  $h_1(\tau) = |\tau| = \sum_{v \in E} |\downarrow\{v\}|$ . Equivalent distinct elements contribute in both directions to  $\gamma_1$ .*

*Proof.* The  $k = 1$  transport pairs are exactly the non-diagonal related pairs. The coefficient of  $t$  in (48.4) is  $n + \gamma_1$ . Directly (manuscript 01): for a fixed single-element support  $\{v\}$ , the admissible positive coordinate values are  $1, \dots, |\downarrow\{v\}|$ , because every ideal containing  $v$  contains its principal ideal. [Merge note: this is Part I's formula (1.4).]  $\square$

**Proposition 55.5** (The second gamma coefficient; manuscript 01). *Let  $D_2(\tau)$  count unordered pairs of directed distinct-element relations with four distinct endpoints. Let  $R_{2,2}(\tau)$  count ordered disjoint pairs  $(U, V)$  with  $|U| = |V| = 2$  and every relation  $u \leq_\tau v, u \in U, v \in V$ , present. Then*

$$\gamma_2(\tau) = D_2(\tau) - R_{2,2}(\tau). \quad (55.5)$$

*Proof.* Every pair counted by  $D_2$  is a matching witnessing some disjoint  $(U, V)$  of size two. Such a support pair has two witnessing bijections exactly when all four possible source-to-receiver relations exist; otherwise it has one. Subtracting the number of doubled pairs gives (55.5).  $\square$

[Merge note: for a height-two poset  $P_G$  this is Part II's  $\gamma_2 = m_2(G) - c_4(G)$  of Corollary 15.2:  $D_2 = m_2(G)$ , and  $R_{2,2}$  counts the four-cycles.]

**Theorem 55.6** (Sharp coefficientwise bounds; manuscripts 06 and 01). *For every preorder on  $n$  elements and  $0 \leq k \leq \lfloor n/2 \rfloor$ ,*

$$0 \leq \gamma_k(\tau) \leq \frac{n!}{k!^2(n-2k)!} = \binom{n}{2k} \binom{2k}{k}, \quad \binom{n}{j} \leq [t^j]h_\tau(t) \leq \binom{n}{j}^2 \quad (0 \leq j \leq n). \quad (55.6)$$

*For an ordinary poset the sharper bound is*

$$\gamma_k(\tau) \leq \binom{n}{2k} C_k, \quad C_k = \frac{1}{k+1} \binom{2k}{k}. \quad (55.7)$$

*The upper bounds are attained simultaneously for all  $k$  by, respectively, the universal equivalence preorder  $\mathbf{E}_n$  and a total order  $L_n$ . The antichain attains  $\gamma_k = 0$  for all  $k \geq 1$ , and the lower bounds on  $h$ . For  $n \geq 2$ , equality of the complete gamma vector with the upper or lower extremal vector characterizes the corresponding extremal preorder, and equality with the chain vector characterizes a chain among posets (manuscript 01).*

*Proof.* There are  $\binom{n}{k} \binom{n-k}{k} = n!/(k!^2(n-2k)!)$  ordered pairs of disjoint  $k$ -element sets, which gives the first upper bound. Every such pair is matchable in the universal equivalence preorder. The antichain relation is contained in every preorder, and every preorder is contained in  $\mathbf{E}_n$ ; Theorem 55.1 gives the bounds on  $h$  from (55.8) and (55.11) below. Alternatively, the unit cube is contained in  $\mathcal{Q}_\tau$ , while  $\mathcal{Q}_\tau$  is contained in the full simplex  $\{x \geq 0 : x(E) \leq n\}$  (manuscript 01).

Every finite poset has a linear extension, obtained recursively by removing a minimal element. Its relation is a subset of that total order. It is therefore enough, by Theorem 55.1, to compute the chain case. Choose the  $2k$  nonzero positions in the signed-vector formula (51.3). In their increasing order, the signs must have total zero and nonnegative partial sums. There are  $C_k$  such words: among the  $\binom{2k}{k}$  balanced sign words, reflection up to the first negative partial sum bijects the bad ones with words having  $k+1$  plus signs and  $k-1$  minus signs. Hence the good count is  $\binom{2k}{k} - \binom{2k}{k-1} = C_k$ . This proves (55.7) and sharpness. In an antichain a disjoint nonempty transport pair cannot be matched, since the only relation edges are loops.

The characterizations follow from  $\gamma_1$ : zero means that no off-diagonal relations exist,  $n(n-1)$  means that all of them exist, and  $\binom{n}{2}$  for a poset forces total comparability.  $\square$

*Second proof of the chain count, by initial segments (manuscript 01).* Manuscript 01 states  $\gamma_k(L_n) = \binom{n}{2k} C_k$  as its Proposition “The chain gamma vector”. Fix the  $2k$  elements in  $U \cup V$ , and list them in increasing order. A compatible matching exists exactly when every initial segment contains at least as many elements of  $U$  as of  $V$ . Necessity holds because a target must be matched to an earlier source. For sufficiency, pair the  $i$ th source with the  $i$ th target; the initial-segment inequalities force the source to precede the target. Among the  $\binom{2k}{k}$  words with  $k$  source letters and  $k$  target letters, the words whose running source-minus-target balance never becomes negative are counted by  $\binom{2k}{k} - \binom{2k}{k-1} = C_k$ ; the subtraction follows by reflecting the prefix ending at the first negative balance. Multiply by the choice of  $U \cup V$ . Relation-monotonicity under a linear extension gives the bound.  $\square$

## 55.1 Explicit consistency checks from classical families

For an antichain,  $\mathcal{Q}_\tau = [0, 1]^n$ , so

$$h_\tau(t) = (1+t)^n, \quad \Gamma_\tau(u) = 1. \quad (55.8)$$

For a total order, the preceding proof gives

$$\Gamma_\tau(u) = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} C_k u^k. \quad (55.9)$$

This agrees with the classical Narayana specialization for a chain,

$$[t^j]h_{L_n}(t) = \binom{n}{j}^2 - \binom{n}{j-1} \binom{n}{j+1} = \frac{1}{n+1} \binom{n+1}{j} \binom{n+1}{j+1}, \quad (55.10)$$

recorded among the elementary families of [1]; the identity also follows by expanding (55.9). Here a binomial coefficient with lower index outside its usual range is zero. Manuscript 01 gives a direct proof using (51.4): if the two  $j$ -subsets are  $A = \{a_1 < \dots < a_j\}$  and  $B = \{b_1 < \dots < b_j\}$ , compatibility is equivalent to  $a_i \leq b_i$  for all  $i$ , or to  $|A \cap [r]| \geq |B \cap [r]|$  for every  $r$ . There are  $\binom{n}{j}^2$  pairs in total. For a violating pair, exchange the two memberships on the first prefix for which the target count exceeds the source count by one. This is a bijection to a pair of subsets of sizes  $j+1$  and  $j-1$ ; the inverse uses the first prefix with source surplus one. This proves the first equality, and factorial simplification proves the second. Thus (55.10) gives sharp ordinary coefficient bounds for posets as well. These classical formulas are checks, not novelty claims.

For the universal equivalence preorder  $\mathbf{E}_n$ , the only ideals are  $\emptyset$  and  $E$ , so  $\mathcal{Q}_\tau = \{x \geq 0 : x(E) \leq n\}$ . Choosing a support of size  $j$  and then its positive coordinates gives

$$h_{\mathbf{E}_n}(t) = \sum_{j=0}^n \binom{n}{j}^2 t^j, \quad \Gamma_{\mathbf{E}_n}(u) = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{k!^2(n-2k)!} u^k. \quad (55.11)$$

For the first formula, subtract one on the chosen support; the number of resulting nonnegative  $j$ -tuples of total at most  $n-j$  is  $\binom{n}{j}$ . The second follows directly from unrestricted disjoint transport pairs. These are familiar special-case polynomials (Part I's indiscrete preorder, Section 8); the point is their role as universal coefficientwise bounds.

Table 4: Four-element examples (manuscript 06). Each  $h$  is  $1 + h_1 t + h_2 t^2 + h_1 t^3 + t^4$ , and each  $\Gamma$  is  $1 + \gamma_1 u + \gamma_2 u^2$ .

| Preorder                       | $h_1$ | $h_2$ | $\gamma_1$ | $\gamma_2$ |
|--------------------------------|-------|-------|------------|------------|
| Antichain                      | 4     | 6     | 0          | 0          |
| Diamond poset                  | 9     | 18    | 5          | 2          |
| Total order                    | 10    | 20    | 6          | 2          |
| Two ordered blocks of size two | 12    | 27    | 8          | 5          |
| Universal equivalence          | 16    | 36    | 12         | 6          |

## 55.2 The octopus formula as an immediate specialization (manuscript 01)

Consider  $\ell$  pairwise incomparable singleton leaves below a root equivalence class of size  $n - \ell$ . In a nonempty disjoint compatible pair, a receiver cannot be a leaf, because its only predecessor outside the root order is itself and sources and receivers are disjoint. Conversely, once  $V$  is any  $k$ -subset of the root, every  $k$ -subset  $U \subseteq E \setminus V$  is compatible with it. Therefore

$$\gamma_k = \binom{n-\ell}{k} \binom{n-k}{k}. \quad (55.12)$$



This recovers the octopus formula attributed to Athanasiadis, Xiao, and Yan and reproved bijectively by Menon [6, Proposition 1.1]. It is not a new formula claimed here. The general theorem also applies when leaf blocks have arbitrary positive size and when the quotient is not a star; Section 54 gives the exact count without any unit-block assumption.

### 55.3 Recovery of the height-two matching-support formula

If  $\tau$  is a poset of height at most two, write its strict relations as edges of a bipartite graph  $G = (A \sqcup B, F)$  with  $a < b$  on each edge. Every nonempty transport pair has  $U \subseteq A$  and  $V \subseteq B$ . Therefore its union  $U \cup V$  determines the ordered pair uniquely, and

$$\Gamma_\tau(u) = \sum_{\substack{W \subseteq A \sqcup B \\ G[W] \text{ has a perfect matching}}} u^{|W|/2} = p_G(u).$$

This recovers Part II's height-two result (Theorem 13.2) exactly. Beyond height two, the two signs of a transport pair need not occupy fixed shores of the original poset, which is why a fixed comparability-graph support count is no longer the right general definition. In a preorder with equivalence classes, the union may also admit several orientations; those are genuinely distinct ordered pairs and must not be collapsed. [Merge note: Part II's proof partitions a point by its zeros on the minimal shore  $A$  and its support on  $B$  (Section 15); Lemma 50.1 partitions by the whole zero set, which is what removes the height restriction.]

## 56 Matching deficiency and the exact root at minus one

Let  $\text{Comp}(\tau)$  denote the *simple comparability graph*: its vertices are  $E$ , and distinct  $u, v$  are adjacent when  $u \leq_\tau v$  or  $v \leq_\tau u$ . Write  $\nu(\text{Comp}(\tau))$  for its maximum matching size.

**Theorem 56.1** (Degree and root multiplicity; manuscripts 06 and 01). *For every finite preorder,*

$$\deg \Gamma_\tau = \nu(\text{Comp}(\tau)), \quad \text{ord}_{t=-1} h_\tau(t) = n - 2\nu(\text{Comp}(\tau)). \quad (56.1)$$

Furthermore  $\gamma_k(\tau) > 0$  exactly for  $0 \leq k \leq \nu(\text{Comp}(\tau))$ .

*Proof.* A transport pair of size  $k$  has a witnessing directed matching on  $2k$  distinct vertices. Forgetting orientations gives a  $k$ -matching in  $\text{Comp}(\tau)$ , so  $\deg \Gamma_\tau \leq \nu(\text{Comp}(\tau))$ . Conversely, orient each edge of an undirected matching in a permitted preorder direction. Its tails and heads form a disjoint transport pair. Submatchings supply every smaller size. Hence the degree is exactly  $\nu = \nu(\text{Comp}(\tau))$ .

Factor the smallest power of  $1 + t$  from (48.4):

$$h_\tau(t) = (1 + t)^{n-2\nu} \left( \gamma_\nu t^\nu + \sum_{k < \nu} \gamma_k t^k (1 + t)^{2(\nu-k)} \right). \quad (56.2)$$

The expression in parentheses has value  $(-1)^\nu \gamma_\nu \neq 0$  at  $-1$ . Thus the displayed exponent is the exact multiplicity, not just a lower bound.  $\square$

This reduces the gamma degree and the entire  $-1$  multiplicity question to a single ordinary maximum-matching problem, even when the remaining coefficients are expensive to count. It uses the full comparability graph, not merely the cover graph. The maximum matching here is an ordinary matching in the *undirected* comparability graph. It is not a maximum matching in

$G_\tau$ , whose reflexive diagonal already supplies a perfect matching of size  $n$ . Confusing these two graphs would destroy the degree formula (manuscript 01). [Merge note: in height two,  $\text{Comp}(P_G) = G$ , and this is the multiplicity  $n - 2\nu(G)$  of Part II's Proposition 15.1.]

For example, a height-two star with  $m$  edges has  $n = m + 1$  and maximum matching size one. Its polynomial is

$$h_\tau(t) = (1+t)^{m-1}((1+t)^2 + mt),$$

so its exact multiplicity at  $-1$  is  $m - 1$ . A chain has matching number  $\lfloor n/2 \rfloor$ , giving multiplicity zero for even  $n$  and one for odd  $n$ . Equivalent vertices cause no exception: one may choose either allowed direction when both are present.

**Corollary 56.2** (The leading Taylor coefficient at  $-1$ ; manuscript 01). *Put  $d = n - 2\nu(\text{Comp}(\tau))$ . Then*

$$h_\tau^{(j)}(-1) = 0 \quad (0 \leq j < d), \quad h_\tau^{(d)}(-1) = d!(-1)^{\nu(\text{Comp}(\tau))} \gamma_{\nu(\text{Comp}(\tau))}(\tau). \quad (56.3)$$

*For even  $n$ , the signed value  $(-1)^{n/2} h_\tau(-1)$  is positive exactly when  $\text{Comp}(\tau)$  has a perfect matching, and is zero otherwise.*

*Proof.* Equation (56.2) identifies the first nonzero coefficient of the Taylor expansion at  $-1$ . If  $n$  is even,  $d = 0$  is equivalent to  $\nu = n/2$ .  $\square$

## 57 Induced restrictions and a deletion–differentiation law

(Manuscript 01 only.) For  $W \subseteq E$ , let  $\tau|W$  be the induced preorder, retaining every relation of  $\tau$  whose endpoints lie in  $W$ . A matching between specified subsets of  $W$  is compatible in  $\tau|W$  exactly when it is compatible in  $\tau$ . In particular, no choice of an intermediate element outside  $W$  is needed: the relation itself is already transitive.

**Theorem 57.1** (Induced-deck identities). *For  $0 \leq m \leq n$  and  $k \geq 0$ ,*

$$\sum_{\substack{W \subseteq E \\ |W|=m}} \gamma_k(\tau|W) = \binom{n-2k}{m-2k} \gamma_k(\tau), \quad (57.1)$$

*where the right side is zero when  $2k > m$  or  $2k > n$ . In particular,  $\gamma_k(\tau)$  is the sum of the top gamma coefficients of its induced  $2k$ -element subpreorders. If*

$$D_\tau(t) = \sum_{v \in E} h_{\tau|(E \setminus \{v\})}(t),$$

*then*

$$(1-t)D_\tau(t) = nh_\tau(t) - 2th'_\tau(t). \quad (57.2)$$

*Proof.* A fixed disjoint compatible pair of size  $k$  is counted in the induced preorder on  $W$  exactly when its  $2k$  endpoints lie in  $W$ . There are  $\binom{n-2k}{m-2k}$  such  $W$ , proving (57.1). For one-element deletions this gives

$$D_\tau(t) = \sum_{2k \leq n-1} (n-2k) \gamma_k(\tau) t^k (1+t)^{n-1-2k}.$$

On each gamma-basis term, direct differentiation yields

$$n t^k (1+t)^{n-2k} - 2t \frac{d}{dt} (t^k (1+t)^{n-2k}) = (n-2k)(1-t)t^k (1+t)^{n-1-2k}.$$

When  $2k = n$ , the left side is zero and that term is omitted on the right; there is no evaluation of a negative exponent. Sum the identities.  $\square$

**Corollary 57.2** (Odd-cardinality reconstruction). *If  $n$  is odd, the multiset of polynomials of the one-element deletions determines  $h_\tau$ . Explicitly, writing  $D_\tau(t) = \sum_j d_j t^j$  with  $d_{-1} = d_n = 0$ , one has*

$$[t^j]h_\tau(t) = \frac{d_j - d_{j-1}}{n - 2j} \quad (0 \leq j \leq n). \quad (57.3)$$

*For even  $n$ , the same formulas determine every coefficient except the middle coefficient; the differential identity alone leaves one scalar free.*

*Proof.* Compare coefficients in (57.2). For odd  $n$ , none of the integers  $n - 2j$  is zero. For even  $n$ , exactly the middle one vanishes.  $\square$

The final statement is a limitation of this identity, not a claim that the full combinatorial deletion deck never determines the middle coefficient. This separates polynomial reconstruction from the harder reconstruction of the preorder itself. (Here  $d_j$  are coefficients of  $D_\tau$ , not the deficiency  $d$  of Corollary 56.2.)

## 58 Support size as a mixture of centered binomials

The Boolean expansion provides a probability law without assuming a canonical bijection on the original lattice points. Choose a lattice point uniformly from  $\mathcal{Q}_\tau \cap \mathbb{Z}^E$ , and let  $L$  be its support size. All probabilities below refer to this uniform distribution. This section is manuscript 06's.

**Theorem 58.1** (Exact mixture and concentration). *Let  $\Gamma = \Gamma_\tau$  and  $v = \deg \Gamma$ . There is a mixing variable  $K \in \{0, \dots, v\}$  with*

$$\mathbb{P}(K = k) = \frac{\gamma_k 4^{-k}}{\Gamma(1/4)} \quad (58.1)$$

*such that the law of  $L$  is the mixture*

$$L \mid K = k \stackrel{\text{law}}{=} k + \text{Bin}(n - 2k, 1/2). \quad (58.2)$$

*Consequently*

$$|\mathcal{Q}_\tau \cap \mathbb{Z}^E| = 2^n \Gamma(1/4), \quad (58.3)$$

$$\mathbb{E}L = \frac{n}{2}, \quad \text{Var } L = \frac{n}{4} - \frac{\Gamma'(1/4)}{8\Gamma(1/4)}, \quad (58.4)$$

$$\frac{n - 2v}{4} \leq \text{Var } L \leq \frac{n}{4}. \quad (58.5)$$

*For  $n \geq 1$  and  $a \geq 0$ ,*

$$\mathbb{P}(|L - n/2| \geq a) \leq 2 \exp(-2a^2/n). \quad (58.6)$$

*The upper variance bound is an equality exactly for antichains.*

*Proof.* Evaluation of (48.4) at one gives (58.3). Divide (48.4) by that value. The  $k$ th summand has total mass  $\gamma_k 2^{n-2k} / h_\tau(1)$ , which is (58.1), and its normalized coefficients are those in (58.2). Every component has mean  $n/2$  and variance  $(n - 2k)/4$ . Thus the total variance is  $n/4 - \mathbb{E}K/2$ . Since

$$\mathbb{E}K = \frac{\Gamma'(1/4)}{4\Gamma(1/4)},$$

we obtain (58.4) and (58.5). Equality at the upper bound means all  $\gamma_k$  with  $k > 0$  vanish. By Corollary 55.4, this is equivalent to having no non-diagonal comparabilities.

For the tail bound, the centered moment generating function of the  $k$ th component is  $\cosh(s/2)^{n-2k}$ . Since  $\cosh x \leq \exp(x^2/2)$ , this is at most  $\exp(ns^2/8)$  for every real  $s$ . The same bound holds after mixing. For  $s > 0$ , Markov's inequality gives

$$\mathbb{P}(L - n/2 \geq a) \leq \exp(-sa + ns^2/8).$$

Choosing  $s = 4a/n$  gives  $\exp(-2a^2/n)$ , and the lower tail has the same bound. Their sum proves (58.6). The elementary hyperbolic-cosine inequality follows, for example, by integrating  $\tanh x \leq x$  for  $x \geq 0$  and using evenness.  $\square$

[Added 29 September 2026, batch 55: the variance and tail bounds hold under every positive fugacity and external field with the actual mean, and for every bipartite graph, where equality in the variance bound characterizes disjoint equal-strength stars (Part V, Theorems 74.5 and 74.8, Corollary 74.9).] The mixture law is not a representation of  $L$  as a sum of a fixed family of independent Bernoulli variables. In particular it does not imply real-rootedness. Its value is that it gives exact central statistics and a uniform tail bound directly from the proven gamma expansion. [Merge note, not in manuscript 06: for a bipartite cactus  $G$ , Part II's (19.7) gives  $\text{Var } S = (n - 2\nu)/4 + \sum_j 2/(4 + \lambda_j)$  for the same statistic ( $S = L$ ), from  $p_G(u) = \prod_j (1 + \lambda_j u)$ . Since then  $\Gamma'(1/4)/\Gamma(1/4) = \sum_j 4\lambda_j/(4 + \lambda_j)$ , formula (58.4) gives  $n/4 - \sum_j \lambda_j/(2(4 + \lambda_j))$ , the same number, because  $\frac{1}{2} - \lambda_j/(2(4 + \lambda_j)) = 2/(4 + \lambda_j)$ . Here  $\Gamma(1/4)$  is  $\Gamma_\tau$  evaluated at  $1/4$ , not Euler's Gamma function.]

## 59 Exact computations and reproducibility

Both manuscripts supply a standard-library Python program that uses exact integer arithmetic throughout its mathematical checks. Each is designed as a cross-check of several logically different descriptions, not as an unbounded theorem prover. The shipped files are:

| Shipped path (delivered path)                                                                                                                | Content                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| code/05-gamma-positivity-verify.py<br>(code/verify.py)                                                                                       | manuscript 06's verifier                                                                                        |
| data/05-gamma-positivity-preorders.csv<br>(data/preorders.csv)                                                                               | every labeled preorder through size five with its $h$ - and $\gamma$ -vectors (CRLF line endings, as delivered) |
| data/05-gamma-positivity-verification.json, data/05-gamma-positivity-verification.log                                                        | the recorded run: detailed results with all larger samples; its console transcript                              |
| 05-gamma-positivity-PROOF_STATUS.md,<br>05-gamma-positivity-SOURCE_AUDIT.md                                                                  | manuscript 06's theorem-dependency and source-audit notes                                                       |
| code/05-gamma-positivity-build.py<br>(build.py)                                                                                              | manuscript 06's three-pass build script (for its own article; do not run here)                                  |
| code/04-gamma-vector-verify.py<br>(code/verify.py)                                                                                           | manuscript 01's verifier                                                                                        |
| data/04-gamma-vector-verification.json,<br>data/04-gamma-vector-verification.txt<br>(results/verification.json,<br>results/verification.txt) | manuscript 01's recorded run and its console output                                                             |
| code/04-gamma-vector-build.sh (build.sh)                                                                                                     | manuscript 01's two-pass build script (for its own article; do not run here)                                    |

### 59.1 Manuscript 06: exhaustive preorder tests

The program enumerates every reflexive relation on a labeled  $n$ -element set and retains exactly those that are transitive. For each resulting preorder it independently enumerates

all nonnegative integer vectors of total at most  $n$  and tests every ideal inequality. It also enumerates ordered disjoint support pairs and decides matchability by recursive bipartite matching, without using the ideal Hall criterion in that matching routine. The resulting counts are compared in the following ways:

1. Each individual support fibre is compared with the Boolean-interval count in (50.3); this tests the multivariate identity, not only a specialization.
2. Gamma expansion, the quotient-block formula, and both sharp upper bounds are checked. Matchability is also compared with the ideal Hall test.
3. All equal-size support pairs in the doubled graph are tested for common-support cancellation, and its polynomial is compared with  $h_\tau$ . An independent undirected matching recursion checks  $\deg \Gamma_\tau = v(\text{Comp}(\tau))$ .

Table 5: Manuscript 06’s recorded exhaustive run. A support fibre is specified by a preorder and a subset of its labeled ground set.

| $n$   | Preorders | Support fibres | Equal-size doubled pairs |
|-------|-----------|----------------|--------------------------|
| 0     | 1         | 1              | 1                        |
| 1     | 1         | 2              | 2                        |
| 2     | 4         | 16             | 24                       |
| 3     | 29        | 232            | 580                      |
| 4     | 355       | 5,680          | 24,850                   |
| 5     | 6,942     | 222,144        | 1,749,384                |
| Total | 7,332     | 228,075        | 1,774,841                |

All comparisons passed. These are labeled preorders, including nontrivial equivalence classes; the totals do not count only unlabeled posets. The accompanying CSV contains every tested preorder and both coefficient vectors, not merely summary totals.

*Additional independent scopes.* The demand/support identity, including its prescribed-support refinement, was tested on all 689 bipartite graphs with each labeled shore of size at most three, including empty shores. The run counts 5,880 feasible demand vectors across those graphs.

The full transitivity/cancellation/palindromicity characterization and both endpoint coefficient formulas were tested on all 4,166 reflexive relations through size four. This includes nontransitive relations, so it tests the negative direction rather than assuming the input is a preorder. [Merge note: 4,166 counts the empty relation at  $n = 0$ ; there are 4,165 with  $1 \leq n \leq 4$ .]

A further 140 distinct deterministic samples have sizes six, seven, and eight: respectively 46, 47, and 47 examples. They were generated with seed 20260929 by taking transitive closures of both arbitrary directed relations and acyclic ones. This is not uniform sampling from preorders. Every sampled relation is listed in the JSON output. The direct support counts, gamma formula, block formula, ideal Hall criterion, extremal bounds, and matching-number identity were checked on these samples. Exhaustive all-pairs doubled cancellation was reserved for the small-size run.

Finally, the binomial and Narayana specializations were compared with their gamma expansions through  $n = 20$ . These last tests are identities between explicit formulas, not exhaustive lattice-point enumerations in dimension 20. No root approximations, floating-point certificates, or randomized claims of proof are used.

*Files and execution.* The delivered command, run from the package root, was `python code/verify.py -max-n 5 -out data`. In this directory, pass a new output directory, so that

the shipped record is preserved (the default `-out` data is relative to the working directory and would write unprefixed files):

```
py code/05-gamma-positivity-verify.py -max-n 5 -out /path/to/scratch
```

Python 3.10 or later is required. No external Python packages are needed. The program refuses to run under python `-O`, because it uses assertions. It deliberately rejects exhaustive sizes above five, whose relation search grows too rapidly for its intended role. The article’s production algorithm can instead enumerate signed pairs and use an augmenting-path matching test, requiring at most  $O(3^n n^3)$  elementary operations and polynomial workspace; the verifier uses simple recursive matching because its small-instance independence is more important than asymptotic speed.

The package also contains `README.md`, `SOURCE_AUDIT.md`, `PROOF_STATUS.md`, a build script, and the compiled PDF. The theorem proofs are in this article, not in the test log. [Merge note: the two notes and the build script are shipped under the prefixed names above; the delivery README and PDF are not shipped.]

## 59.2 Manuscript 01: independent representations and completed suites

Manuscript 01’s program performs several logically distinct finite checks rather than evaluating the same formula twice. The direct enumerator lists all nonnegative  $n$ -vectors with total at most  $n$ , computes every lower ideal from the input relation, and tests the original inequalities (48.1). It records the number of lattice points *for each exact support*. This part does not use Hall’s condition, matching supports, or a gamma expansion.

A separate recursive matching checker tests bijections between selected source and receiver masks. It does not use the ideal inequalities or the Hall inequalities. Disjoint compatible pairs are accumulated to predict both the gamma coefficients and every multivariate coefficient in (48.5). Independently, compatible pairs with overlapping sets are accumulated to check (51.4). The program also checks cancellation of overlaps, complement-duality, the maximum-matching degree law, and the deletion–differentiation identity using directly enumerated deleted preorders.

Finally, for fixed bipartite shores, the program enumerates integer demands by their defining subset inequalities and compares their exact support counts against matching-support counts. This directly tests the classical bridge and its support refinement on thousands of separate graphs.

Every reflexive relation on a fixed labeled  $n$ -element set is generated, and exactly the transitive ones are retained. Thus the following is an exhaustive test of *labeled preorders*, not merely of naturally labeled posets and not merely of random closures:

| $n$               | 0 | 1 | 2 | 3  | 4   | 5     |
|-------------------|---|---|---|----|-----|-------|
| Preorders checked | 1 | 1 | 4 | 29 | 355 | 6,942 |

All 7,332 cases passed. Additional completed checks comprise all 512 bipartite graphs on fixed shores of sizes (3,3), all 4,096 on shores (3,4), all 64 on shores (2,3), and the empty-shore boundary cases. A seeded generation of 120 preorder closures on six or seven elements produced 119 distinct relations, all of which passed. These latter cases are explicitly *not* exhaustive.

The block formula was checked on 87 cases: every preorder on three indexed blocks, with block-size vectors (1,1,1), (2,1,2), and (2,2,2). Allowing equivalent block indices is harmless



for the capacity formula, although the canonical equivalence-class quotient in Theorem 54.3 is a poset. The five named examples below are also included in the saved output.

| Preorder                             | Gamma vector   | Lattice points | $\text{ord}_{-1} h$ |
|--------------------------------------|----------------|----------------|---------------------|
| Three-element chain                  | (1, 3)         | 14             | 1                   |
| One equivalence class of size 3      | (1, 6)         | 20             | 1                   |
| Diamond poset, size 4                | (1, 5, 2)      | 38             | 0                   |
| Block chain (2, 1, 2)                | (1, 12, 17)    | 162            | 1                   |
| Two lower blocks (2, 1) below size 3 | (1, 17, 36, 7) | 487            | 0                   |

To reproduce the full finite run, manuscript 01 gave, from the extracted package directory, `python3 code/verify.py -max-n 5 -output results/verification.json`. In this directory pass an explicit output path (the default is relative to the working directory):

```
py code/04-gamma-vector-verify.py -max-n 5 \
  -output /path/to/scratch/verification.json
```

Python 3.10 or later is required. No external Python packages, network access, computer algebra system, or proof-assistant installation is used. The JSON records the interpreter version, random seed, case counts, examples, and elapsed times. The recorded times are properties of this run, not performance guarantees.

*What the checks establish (manuscript 01).* Exhaustive agreement detects normalization mistakes, reversed relations, incorrect treatment of equivalence classes, confusion between matchings and supports, and implementation errors in small cases. It does not prove an all-cardinality assertion. The infinite statements are justified by the arguments in Sections 49–58, with the classical inputs (14.3)–(14.4) explicitly imported.

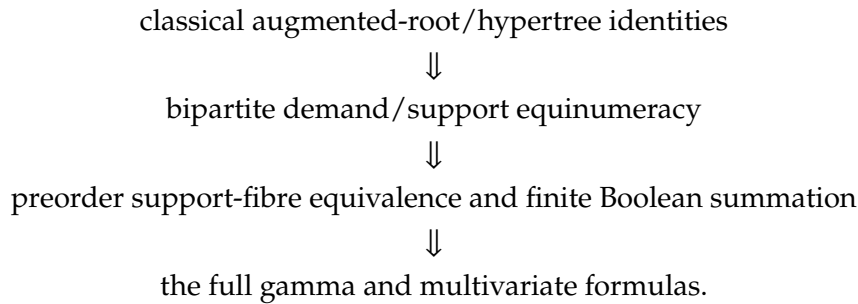
### 59.3 Reruns and cross-checks in the write phase

Both programs were rerun on 29 September 2026 (Python 3.14.4, Windows), from a scratch copy with the shipped names and explicit output paths as above. Both passed. Manuscript 06’s regenerated CSV is byte-identical to `data/05-gamma-positivity-preorders.csv`, and both JSON records differ from the shipped ones only in timings, the Python version (the recorded runs used 3.13.5) and, for 06, the platform string. Independently, the batch-44 placement dossier compared, for all 390 labeled preorders with  $n \leq 4$ , a direct count of lattice points by support size with  $\sum_k \gamma_k t^k (1+t)^{n-2k}$ , and checked the endpoint formula and the transitivity criterion of Theorem 48.3 on all 4,165 reflexive relations with  $1 \leq n \leq 4$  (Part I’s reflexive example gives (1, 5, 6, 1)); there were no mismatches. The write phase recomputed the numerical examples of Sections 50–56 (Examples 50.2, 54.2 and 54.4, Table 4, the star with three edges and an octopus) by direct lattice-point and transport-pair enumeration, and checked the chain and  $\Theta_3$  identities by hand; all agree. These scratch checks are not shipped. None of the programs is a proof.

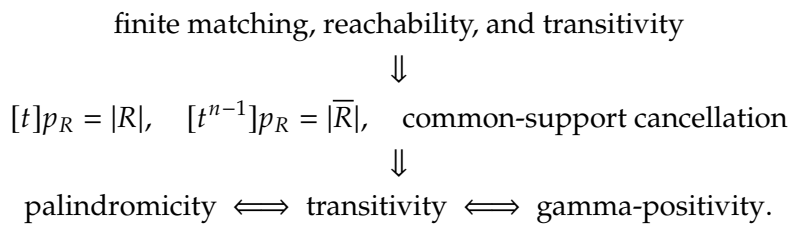


## 60 Proof dependencies, audit, and a formalization route

The mathematical dependency chain for the main theorem is short (manuscript 06):

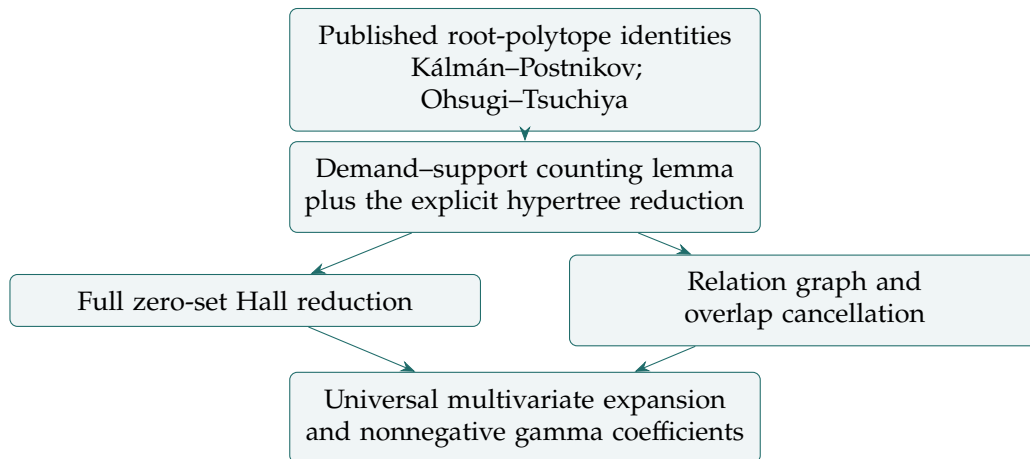


The other principal chain is independent:



A verification of the second chain would not by itself formalize the preorder-polytope theorem: the bridge from lattice points to matching supports must also be checked.

Manuscript 01 displays the central dependencies without any hidden appeal to this report's unrefereed results:



The two lower branches are different reductions but share the same counting bridge. The degree, deletion, monotonicity, and block laws then use only finite combinatorics and polynomial algebra.

### 60.1 Three easy ways to state a false variant (manuscript 01)

**Using only covering relations.** For the chain  $1 < 2 < 3$ , the relation  $1 < 3$  must appear in the matching graph. Using just the two Hasse edges would give  $\gamma_1 = 2$ , whereas the correct value is 3 and  $h = 1 + 6t + 6t^2 + t^3$ . The lower ideals are unchanged by taking transitive closure, but the compatible-pair formula is not unchanged unless the closure is taken as well. The second proof of Corollary 51.2 explains why: paths must be compressible to edges of the relation.

**Testing only principal ideals.** For the poset  $a < b$  and  $a < c$ , the vector  $(x_a, x_b, x_c) = (0, 2, 2)$  satisfies the inequalities for the three principal ideals. It violates the inequality for the full ideal, because its total is four while the ground set has size three. The polytope in (48.1) uses *all* lower ideals.

**Counting witnesses rather than supports.** With two sources and two targets and every source below every target, there are two compatible bijections but only one ordered support pair with those specified sources and targets. That pair contributes one, not two. This distinction is the reason the gamma polynomial is not in general the ordinary matching generating polynomial.

## 60.2 Real-rootedness and flagness remain separate

The universal theorem does not imply real-rootedness. Part II's height-two analysis already contains counterexamples, with the earlier small example credited to Shivam Patel [11]. To see the distinction in an exact instance (manuscript 01), let  $\Theta_3$  be the bipartite graph consisting of three internally disjoint paths of length three with common endpoints. Its height-two poset has gamma polynomial

$$p(u) = 1 + 9u + 24u^2 + 16u^3 + u^4,$$

as counted in Part II (the case  $m = 3$  of (13.4)). The identity

$$p(-x) = (1 - x)(1 - 4x)^2 + x^4$$

shows that  $p$  is positive on  $[-1, 0]$ . Its nonnegative coefficients exclude nonnegative roots. If all its roots were real, all four would therefore be strictly below  $-1$ , contradicting the fact that their product is one. A nonreal zero  $u_0$  of  $p$  lifts through  $t = u_0(1 + t)^2$  to nonreal zeros of  $h_\tau(t) = (1 + t)^8 p(t/(1 + t)^2)$ : the quotient is real for real  $t \neq -1$ , and  $t = -1$  cannot solve this equation. Thus  $h_\tau$  is not real-rooted. This example and its discovery are not claimed anew. [Merge note: this is a short second proof that  $P_{\Theta_3}$  is not real-rooted; Part II's Theorem 13.3 counts its real roots exactly, and Proposition 15.1 is the lifting argument.]

Similarly, Boolean orbits of an auxiliary finite set are not a flag simplicial sphere. Theorem 52.1 supplies a nonnegative gamma expansion, not the face-incidence structure demanded by Conjecture 5.1(d). The root-polytope formula in (52.1) does not remove that distinction.

## 60.3 Suggested Lean development order (manuscript 06)

A useful first module would treat finite relations, disjoint pairs of finite sets, and relation-respecting bijections. It should prove the near-perfect support/reachability lemma and the coefficient defect before introducing any geometry. A second module would establish the ideal Hall criterion and common-support cancellation, then perform the finite-sum reindexing that gives the gamma expansion of  $p_\tau$ .

A third module can express the demand/support identity as an explicit hypothesis and prove the support-fibre reduction and Theorem 48.2 conditional on that hypothesis. Such a theorem must be labeled conditional; it would not be a completed formal proof of the conjecture. Discharging the hypothesis requires formalizing the classical root-polytope route or developing a direct finite combinatorial proof. The quotient-block and deficiency formulas can then be proved by finite counting and elementary polynomial algebra.

The elementary graph classification is therefore a realistic independent formalization target, while the main theorem has one substantial library boundary. No Lean source or successful Lean compilation is claimed in this package. Likewise, the external mathematical theorems

used in the paper are ordinary cited inputs, not assertions that they already exist in the repository’s trusted kernel boundary.

#### 60.4 A staged proof-assistant project (manuscript 01)

A practical formalization can separate the new order-theoretic work from the existing geometric input. First define finite preorders, lower ideals, and the support-fiber excess map. The equivalence in Lemma 50.1 is a finite-sum inequality argument and should be a small, independent target. Next prove the Boolean summation: this gives a formal implication from the explicit counting-bridge statement to the universal gamma theorem.

A second finite target is overlap cancellation. One may formalize the partial-permutation graph and its path decomposition, or iterate a bijection from a source until it reaches a target. Finiteness and injectivity exclude entry into a cycle. The resulting compatible core supports the commuting toggles and every restriction-counting identity.

To obtain a fully closed formal proof, the counting bridge must itself be proved rather than silently postulated. The supplied ordinary proof reduces it to Hall’s theorem and the two published root-polytope results; a natural bijective proof of the bridge could yield a substantially smaller formal library dependency. A Lean theorem with an unproved bridge hypothesis is a valuable intermediate artifact, but is not a complete formal verification of Conjecture 5.2. No uncompiled Lean file or such conditional proof is presented as verified in this package.

[Merge note: neither plan has been started. No Lean or Rocq declaration in ProveIt formalizes any statement of Part IV.]

### 61 Further research questions and topics

The following questions are proposed research directions. Except where an existing conjecture is named, no claim is made that the formulation is new or that the literature has never addressed a related version (manuscript 06). Manuscript 01 adds: the flag question is the remaining named conjecture in the cited source; the others are proposed directions arising from the proof, whose worldwide open status has not been exhaustively surveyed, and they should be viewed as specific research tasks, not as assertions that the literature contains no related results. Research questions 20–29 are manuscript 06’s, in its order, with manuscript 01’s formulation added where it asks the same question; Research questions 30–36 are the rest of manuscript 01’s.

**Research question 20** (A direct demand/support bijection; manuscripts 06 and 01). Construct an explicit bijection in Lemma 49.1 that is compatible with right-support restriction as in Proposition 49.3. A mere lexicographic matching of two equally sized lists is not the target. Can an augmenting-path or activity construction explain the correspondence without importing Ehrhart theory? This would remove the largest formalization dependency of the main theorem. (Manuscript 01: construct an explicit bijection from  $\mathcal{R}_H$  to perfectly matchable support pairs that preserves the exact receiver support. Can it respect induced subgraphs or a controlled choice of vertex ordering?)

Proposition 49.3 proves the required cardinality identities, but the geometric proof does not identify a natural map. Such a construction would make the Boolean action on actual preorder lattice points explicit and would simplify a complete formalization (manuscript 01). [Merge note: this is Part II’s Research question 2, asked again with the receiver support preserved.] [Added 30 September 2026, batch 64: answered in Part IX. The two-flow maps of Theorem 135.2 are explicit, polynomial-time and support-preserving, and their proof imports no Ehrhart theory (Corollary 139.2 and the merge note after it). The ordering sub-question is answered in

part: the maps depend only on the chamber of a cost order (Proposition 141.2) and obey deletion laws outside the selected sets; full automorphism-equivariance is impossible (Proposition 144.1), and the rest is Research question 71.]

**Research question 21** (Boolean orbits on actual lattice points). Can one realize the terms in (48.4) as orbits of a natural system of involutions on  $Q_\tau \cap \mathbb{Z}^E$ , with a  $k$ th orbit containing one point for every subset of  $n - 2k$  free labels? Compatibility with relabeling and restriction is desirable. The present support-count identity does not supply such an action.

[Merge note: manuscript 01's Theorem 52.1 realizes such orbits on the support-pair model  $\mathcal{M}_\tau$ , not on  $Q_\tau \cap \mathbb{Z}^E$ ; the question stays open as stated.]

**Research question 22** (Flag-polytopal realization; manuscripts 06 and 01). Does every  $h_\tau$  occur as the  $h$ -polynomial of an  $n$ -dimensional flag simplicial polytope, as asked in [1, Conjecture 5.1(d)]? The gamma formula gives nonnegative coefficients but no flag face complex. A useful intermediate objective is a simplicial model whose face numbers are the transport-support counts; it must account for the fact that deleting an arbitrary single signed endpoint need not preserve balance. (Manuscript 01: can the disjoint-support model be used to construct a flag simplicial  $n$ -polytope whose  $h$ -polynomial is  $h_\tau$ ?)

The existing Boolean families provide ranks but no face-gluing rules. A successful construction must address intersections and sphere topology, not merely reproduce the gamma coefficients. It need not use the same canonical lattice triangulation considered in Part I (manuscript 01). [Merge note: compare Part II's Research questions 6 and 7 and its crown obstruction, Theorem 13.6.]

**Research question 23** (Classification beyond a perfect matching). Extend Corollary 53.2 to arbitrary bipartite graphs, including unequal shores and graphs whose maximum matching does not saturate either shore. Is there a useful decomposition of the palindromic cases into balanced preorder blocks and residual components? Palindromicity is to be measured in the actual polynomial degree. The endpoint argument here does not directly apply when that degree is smaller than a shore size.

[Added 29 September 2026, batch 55: answered in Part V. Apart from isolated vertices there are no residual components: the palindromic cases are exactly the doubled preorders, and they are recognized by a matching and a transitive closure (Theorem 67.4, Section 72).]

**Research question 24** (Coefficient shape beyond gamma-positivity). Which preorders have log-concave, unimodal, or real-rooted gamma polynomials? The general theorem answers none of these stronger questions for  $\Gamma_\tau$ . For real-rootedness, the already documented height-two counterexamples rule out an unrestricted assertion. Relations to quotient width and induced comparability configurations are concrete targets.

[Added 29 September 2026, batch 55: for every poset of height two the log-concavity and unimodality parts are answered:  $\gamma_k / \binom{\min(m,n)}{k}$  is log-concave with no internal zeros (Part V, Corollary 71.6). For arbitrary preorders Part V proves only that  $h_\tau$  is ultra-log-concave; log-concavity of  $\Gamma_\tau$  remains open.]

**Research question 25** (Extremal profiles with prescribed structure). Determine the maximal and minimal  $\gamma_k$  when block sizes, quotient width, or matching number are fixed. Theorem 55.6 gives sharp unrestricted bounds, but fixing these structural parameters may make the extremizers quite different from chains or universal equivalence classes. The positive block formula offers a way to separate local multiplicities from quotient feasibility.

**Research question 26** (Parameterized exact algorithms). Can (51.3) be evaluated efficiently for bounded-width quotient posets, or when the *comparability graph* has bounded treewidth? A cover-graph parameter is a separate question because transitive closure can enlarge neighborhoods and graph width. Establish precise arithmetic and bit-complexity bounds, and determine the counting complexity without parameter restrictions.

[Added 29 September 2026, batch 55: without parameter restrictions,  $|Q_\tau \cap \mathbb{Z}^n| = h_\tau(1)$  is #P-complete under Turing reductions, already for posets of height two, and it has an FPRAS, as has every coefficient of  $h_\tau$  (Part V, Theorems 76.3, 75.2 and 75.5). The parameterized question remains open.]

**Research question 27** (A geometric interpretation of the gamma coefficients). The signed-vector formula counts balanced points of the cube  $\{-1, 0, 1\}^E$  satisfying ideal inequalities. Can these points be realized as a natural collection of faces, lattice points in a canonical section, or standard monomials of a graded algebra? Such an interpretation might connect positivity to Part I's root-polytope model, but would require more than the enumerative identity established here.

**Research question 28** (Limit laws for support size). For growing families of preorders, what conditions on  $\Gamma_\tau$  or the quotient structure imply a central limit theorem for  $L$ ? The mixture in Theorem 58.1 reduces the question to the fluctuations of its mixing variable and the conditional binomial size. Can the variance and tail bounds be sharpened using comparability density or matching structure, rather than  $n$  alone?

[Added 29 September 2026, batch 55: Part V proves same-mean binomial convex-order domination of  $L$  under every positive external field, with equality exactly for antichains (at every fugacity; Theorem 58.1 has it at uniform weight), and a covariance-matrix bound (Corollary 74.9, Theorem 74.3); no central limit theorem.]

**Research question 29** (Independent formal and mathematical review; manuscripts 06 and 01). Formalize first the elementary endpoint-defect and cancellation theorems, and independently review the support-fibre equivalence. Then discharge the bipartite counting bridge in a proof assistant. In parallel, compare the balanced-graph characterization with existing Gorenstein and matching-decomposition criteria to determine which parts are already implicit in the literature and sharpen the priority assessment. (Manuscript 01: formalize the universal theorem with no unproved assumptions, and develop an independent proof of the counting bridge that avoids the root-polytope machinery.)

The finite-order part is short enough to audit separately. Closing the bridge dependency, while retaining the multivariate support interpretation, would turn the current mathematical proof into a compact and reusable formal combinatorial library result (manuscript 01).

**Research question 30** (Which gamma vectors are face vectors?; manuscript 01). Do the numbers  $\gamma_k(\tau)$  arise as the face numbers of a useful simplicial or balanced complex, with a construction natural under induced restriction?

Simply treating compatible pairs as faces is inadequate: removing one source without a receiver destroys equal cardinality. The locality formula (57.1) suggests studying endpoint-supported structures or quotients of matching complexes with carefully controlled identifications.

**Research question 31** (The even-cardinality reconstruction defect; manuscript 01). What additional deletion, marking, or contraction data determines the top gamma coefficient when  $n$  is even? Can one reconstruct it from a suitably enriched polynomial deck without reconstructing the entire preorder?



Equation (57.2) reconstructs every other coefficient but annihilates the middle monomial. Corollary 56.2 identifies the missing scalar as a signed value at  $-1$ , counted by full endpoint cores. This gives a precise target for richer reconstruction data.

**Research question 32** (Exact counting complexity; manuscript 01). For which classes of quotient posets can the profile formula be evaluated more efficiently than  $O(n^{2b}2^b)$ ? What is the complexity of computing a specified  $\gamma_k$ , or the whole gamma polynomial, for unrestricted preorders?

A single pair's feasibility is a matching problem; counting distinct pairs is a different task. The fixed-block algorithm gives one tractable regime, but does not supply a general efficient counting algorithm or a hardness classification. Tree decompositions of the quotient and repeated neighborhood classes are natural structural parameters. [Added 29 September 2026, batch 55: a specified  $\gamma_k$  is #P-hard to compute for unrestricted preorders, already in height two, where  $\Gamma_{p_H} = p_H$  (Part V, Corollary 76.4); it has an FPRAS in height two (Theorem 75.5). Efficient classes remain open.]

**Research question 33** (Local changes of the relation; manuscript 01). Find effective formulas for the gamma increment when one adds a relation and then takes transitive closure. Can the newly compatible cores be partitioned without overcounting multiple new witnessing matchings?

Theorem 55.1 already proves nonnegativity of every increment. An exact local rule would be stronger and could lead to constructive recurrences, particularly for modular or series-parallel quotient operations. The closure operation must be included in the input model.

**Research question 34** (Root geometry and negative dependence; manuscript 01). Identify structural classes of preorders for which  $\Gamma_\tau$  is real-rooted or admits a useful multivariate stability refinement. Which operations preserve these properties in the presence of nontrivial blocks?

General positivity is now separated from these stronger analytic properties. The arbitrary height-two bipartite subclass already contains nonreal-root examples, so bounded height alone is insufficient. Part III's cactus results offer positive comparison cases, not a classification of all preorders. [Merge note: compare Part II's Research question 3 and Part III's Research question 11.] [Added 29 September 2026, batch 55: Part V proves that the multivariate support polynomial is completely log-concave, which does not upgrade to real stability ( $\Theta_3$ , Section 77); Part VI classifies signed-support stability for tree incidence-apex graphs and multi-theta graphs and shows that it is block-local.  $\Gamma_\tau$  itself is not treated.]

**Research question 35** (Refining the lattice-point statistic; manuscript 01). What counterpart of the Boolean expansion survives if one records both support and coordinate sum, or weights the individual excess coordinates?

The present proof deliberately forgets the total demand in the classical bridge. A two-parameter identity would require extra information about its hypertree or matching-support correspondence. It cannot be obtained by inserting arbitrary weights into an equality of unweighted cardinalities.

**Research question 36** (Beyond unit ideal budgets; manuscript 01). Which integral budget functions  $b(I)$  admit an analogue for  $\{x \geq 0 : x(I) \leq b(I)\}$ ? What replaces the centered gamma basis when the all-one vector is no longer the unique full-support lattice point?

The excess substitution changes the right-hand side to  $b(I) - |I \cap S|$ . The special cancellation into the number of zero coordinates is lost. Any extension must identify new supply capacities or a different symmetry center, rather than assuming the same formula persists.

## 62 Conclusion

The main formula treats the gamma coefficients of a preorder not as mysterious alternating combinations of lattice-point counts, but as counts of disjoint feasible transport supports. Fixing a lattice point's zeros exposes a bipartite demand problem; summing those problems exposes Boolean intervals. This proves gamma-positivity for every finite preorder, including all equivalence-class blowups and all heights.

The matching viewpoint also gives a structural converse. In a balanced bipartite graph with a perfect matching, a palindromic support polynomial forces the relation obtained from that matching to be transitive. The failure is measured exactly by missing reachability pairs. Positive block formulas, coefficientwise monotonicity, sharp bounds, and the exact root multiplicity at  $-1$  then follow from the same finite support model.

The unresolved step toward a more conceptual and more readily formalized proof is not another positivity inequality. It is a natural combinatorial realization of the inherited demand/support equinumeracy. The proofs and checks supplied here isolate that issue while giving an explicit answer to the stated gamma-positivity conjecture.

Manuscript 01 concludes in the same sense. The general preorder problem admits a uniform solution once zero coordinates are allowed on the entire ground set. The ideal inequalities then become a bipartite supply–demand system, and the classical counting bridge turns each support fiber into a family of matchable pairs. Summing over the free coordinates produces the gamma basis with no cancellation of signs. The resulting coefficients count ordered disjoint compatible source and receiver sets. This proves the full gamma-positivity conjecture, explains why arbitrary height and arbitrary equivalence blocks cause no exception, and supplies an exact matching-theoretic meaning for the vanishing at  $-1$ . The same finite model gives relation-monotonicity, complement-duality, induced-deck identities, and a block-capacity formula. The remaining work is not to infer positivity from more examples, but to understand natural bijections, geometric realizations, stronger root properties, and complete formal verification of the classical bridge and its new consequences.

## 63 Normalization and independent-review checklist

(Appendix A of manuscript 06.)

### 63.1 The three different cardinalities

A perfect matching is a set of edges. A matchable support pair  $(U, V)$  is a pair of vertex sets admitting at least one perfect matching. A hypertree is a degree-minus-one vector admitting at least one realizing spanning tree. The proofs count the latter two without realization multiplicity. The  $K_{2,2}$  example in Remark 49.2 is a minimal check against conflating these notions.

### 63.2 Three polynomial models, with different dimensions

The original preorder polytope  $Q_\tau$  lies in  $\mathbb{R}^n$ , and  $h_\tau$  counts its points by support. The doubled graph  $G_\tau$  has  $2n$  vertices and  $p_{G_\tau}$  has degree  $n$ . The augmented edge polytope  $B_{\widehat{G_\tau}}$  has dimension  $2n$  and its Ehrhart numerator is the same polynomial, with higher coefficients zero. These dimension statements are consistent because the degree of an Ehrhart numerator can be strictly smaller than the polytope dimension.



For a general graph  $H$  with shores of sizes  $p, q$ , its augmented graph has  $p + q + 2$  vertices, is connected and bipartite, and its edge polytope has dimension  $p + q$ . The two affine equations say that the coordinate sum on each augmented shore is one. The natural affine lattice is the intersection of that affine space with the ambient integer lattice. No factor of a lattice index is inserted in (14.3).

### 63.3 Orientation and equality cases

A relation edge is always directed from a source in the lower direction to a target in the upper direction. Hence a matching must satisfy  $|A \cap I| \geq |B \cap I|$  for ideals  $I$ , not the reverse inequality. In the near-perfect-support lemma, omitting  $a$  on the source side and  $b$  on the target side detects a path from  $b$  to  $a$ . The count  $|\bar{R}|$  is invariant under swapping their dummy names, but the proof is not.

The signed-vector formula uses signs on individual elements, not just blocks. Once block differences are fixed, the multinomial coefficients in (54.2) recover precisely the lost labeling information. The formula would be wrong without these factors.

### 63.4 A short audit path

An independent reviewer can check the main theorem by verifying the two implications of Lemma 50.1, the normalization in the established input Lemma 49.1, and the finite sum in (50.3). The second principal theorem can be checked separately from the directed-path construction in Lemma 53.1 and the cancellation counterexample in Proposition 51.4. None of those steps depends on the finite verification data.

## 64 A compact proof certificate

(Appendix A of manuscript 01.) This section records the complete logical skeleton for a reader checking the generalization rather than its consequences.

Let  $S \subseteq E$ ,  $Z = E \setminus S$ , and  $H_{Z,S}$  have edge  $zs$  when  $z \leq_\tau s$ . Write  $x = \mathbf{1}_S + c$  on  $S$ , with zero coordinates on  $Z$ . For  $T \subseteq S$ ,

$$x \in Q_\tau \implies c(T) \leq c(S \cap \downarrow T) \leq |Z \cap \downarrow T| = |N_{H_{Z,S}}(T)|.$$

Conversely, for a lower ideal  $I$ ,

$$c(I \cap S) \leq |Z \cap \downarrow(I \cap S)| \leq |Z \cap I| \implies x(I) \leq |I|.$$

Thus the exact-support fiber has size  $|\mathcal{R}_{H_{Z,S}}|$ . The classical bridge says that this is

$$\#\{(U, V) : U \subseteq Z, V \subseteq S, U \rightsquigarrow_\tau V\}.$$

For any fixed disjoint compatible  $(U, V)$ , its allowed supports are exactly  $S = V \sqcup C$ ,  $C \subseteq E \setminus (U \cup V)$ . Hence

$$H_\tau(\mathbf{z}) = \sum_{(U,V)} \sum_C z_{V \cup C} = \sum_{(U,V)} z_V \prod_{w \notin U \cup V} (1 + z_w).$$

This proves the theorem. The only imported count is justified by the explicit demand-to-hypertree-vector correspondence and the published identities (14.3)–(14.4). No statement about preorder gamma-positivity is used in those inputs.

## 65 Provenance and source-use ledgers

(Appendix B of manuscript 06 and Appendix B of manuscript 01.)

| Source                              | Role in manuscript 06                                                                                                          |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Athanasiadis–Chapoton [1]           | Definitions and the precise target, Conjecture 5.2; flag realization is not settled here.                                      |
| Kálmán–Postnikov [3]                | Root-polytope volume equals the number of hypertree vectors, with ordinary lattice normalization.                              |
| Ohsugi–Tsuchiya [8]; Davis–Kohl [7] | The matching-support interpretation of the augmented root-polytope numerator.                                                  |
| Dai et al. [2]                      | Prior support-enumerator identity and preorder duality; Problem 5.3 motivates the balanced-graph classification.               |
| Menon [6]                           | A prior special-family gamma-positivity result.                                                                                |
| ProveIt (Parts I–II)                | The height-two theorem, inherited counting lemma, explicit unresolved boundary, and existing nonreal-root examples.            |
| Manuscript 06                       | The arbitrary-preorder fibre reduction, general gamma formula, graph classification proof, and the stated derived refinements. |

The ledger is a dependency and scope statement, not a certification that every derived refinement is unprecedented. The root-polytope identities are standard mathematical inputs; this report’s unrefereed geometric results are not used as axioms. The delivered proof-status file (`05-gamma-positivity-PROOF-STATUS.md`) repeats this boundary so that computational output is not mistaken for kernel checking.

| Source                    | Use and boundary in manuscript 01                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------|
| Athanasiadis–Chapoton, v1 | Definition of $h_\tau$ and precise target Conjecture 5.2; flag realization is kept separate.                      |
| ProveIt, pinned commit    | Identifies the height-two predecessor and its explicit extension question. Not used as an unproved axiom.         |
| Kálmán–Postnikov          | Published normalized-volume/hypertree consequence, with its original $h$ -polynomial convention evaluated at one. |
| Ohsugi–Tsuchiya, v4       | Published augmented-graph matching-support identity; counts covered vertex sets rather than individual matchings. |
| Davis–Kohl, v1            | Theorem 3.10 states the precise augmentation and Ehrhart identity used in the bridge.                             |
| Dai et al., v2            | Prior univariate duality and root-polytope context; no new claim of having first proved duality.                  |
| Menon, v1                 | Prior octopus and lopsided-octopus results; the special formula is recovered and credited.                        |

All cited online versions were inspected for manuscript 01 on 29 September 2026. The citation labels and conjecture numbers refer to these versions, not to an assumed future revision. The supplied package does not redistribute the source papers or any font files. [Merge note: manuscript 01 cites Davis–Kohl as [arXiv:2207.14759v1](#), and manuscript 06 cites the inspected versions v1 of [1] and [6], v4 of [8] and v2 of [2], as this report’s bibliography does.]

# Part V. Ultra-log-concave support polynomials: palindromic cores, approximate and exact counting

Part V was added on 29 September 2026 in batch 55 of ProveIt’s incoming-report intake, from two manuscripts that prove the same main theorem independently: for every finite bipartite graph, the demand-support enumerator divided by  $\binom{r}{k}$ ,  $r$  the smaller shore, is log-concave with no internal zeros, so every preorder support polynomial  $h_\tau$  is ultra-log-concave of order  $n$ . The unweighted inequality is the case of Example 2.4 of Theorem A of Röhrle and Ulirsch [17], composed with the published support-counting bridge; what is new here is that composition and its consequences. Part V also classifies the palindromic support polynomials of all bipartite graphs, answering Research question 23, gives approximation schemes for partition functions and single coefficients, and shows exact counting #P-complete already for posets of height two. Parts I–IV are unchanged except for dated pointers to this addition. Section 66 records the provenance, fixes the notation and states what is answered.

## 66 Provenance, conventions and what Part V answers

### 66.1 Two manuscripts, one addition

Part V merges two batch-55 manuscripts. Both are AI-assisted research manuscripts dated 29 September 2026, both are 23 A4 pages as delivered, and the title page of each names ChatGPT (“ChatGPT — AI-assisted mathematical research”). Neither saw the other. They are called *source 06* and *source 07* below, after the file prefixes that continue this report’s local sequence (Part I’s unprefixes files count as 01, Parts II–IV used 02– to 05–). Part IV’s “manuscript 06” and “manuscript 01” are different documents, batch-44 manuscripts; the source numbers avoid reusing their names.

- **Source 06 (the base):** batch-55 manuscript 01, *Beyond Real-Rootedness: Ultra-Log-Concavity and Approximate Counting for Preorder Support Polynomials* (subtitle *Transversal-matroid lifts, external fields, and coefficient recovery*), archive `ProveIt_Lorentzian_Support.zip`. It arrived in ProveIt commit `08985ea78` and was held over from batch 54 to be merged with source 07. It is pinned to commit `e9a57d735` (full hash in Section 79.4), at which this report’s `article.tex` was the Git blob `d52a2633b58e74aa653e967eb56adc1aa41bc8ae`: Parts I–IV exactly as printed above. Its verifier, sampler, unit tests, recorded outputs, build script and source notes are shipped with the prefix `06-lorentzian-support-`.
- **Source 07:** batch-55 manuscript 02, *Beyond Real Roots: Ultra-Log-Concavity, Palindromic Cores, and Efficient Counting for Preorder Support Polynomials* (subtitle *A bimatroid bridge resolving two components of the bipartite support-enumerator problem*), archive `ProveIt_Support_Polynomials_Research.zip` (inner directory `ProveIt_Support_Polynomials/`, main file `support_polynomials.tex`). It arrived in `23dd71d2d` and is pinned to commit `de2f8fa8c`, with the same article blob `d52a2633` and this report’s README as blob `846d9f19fadac9cd933b74370cdb8fafa2211c5d`. Its three programs, recorded outputs, build script and two notes are shipped with the prefix `07-support-polynomials-`.

Both were placed in commit `26473dfa0`. This report did not change between the two pins and the placement, so every statement of either source about “the repository” or “the report” concerns Parts I–IV exactly as printed. The manuscripts, delivery READMEs and PDFs are not shipped; they survive in the arrival commits. Source 06’s checksum list `SHA256SUMS.txt` was

verified (13/13) at placement and retired; source 07 shipped none. Sections 78 and 78.1 name the shipped files.

*Why 06 is the base.* The two manuscripts prove the same main theorem with the same transversal matroid, the same hypotheses (a finite bipartite graph, positive vertex weights) and essentially the same Lorentzian argument. Source 06 was held first and already carried the prefix 06-; it states the corollaries that bear on this report's open questions (ultra-log-concave height-two  $\gamma$ -vectors, a single-coefficient approximation scheme, a covariance bound), and its probabilistic and algorithmic layers are the larger. Source 07 contributes whole sections source 06 lacks: palindromic cores (Section 72), an intersection realization (Section 73), convex-order domination with an exact equality classification (Section 74), exact-counting hardness (Section 76), several examples, and the correct attribution to Röhrle–Ulirsch. Proofs taken from source 07 were re-read for hidden uses of stronger hypotheses. The only difference of setting is that source 07 deletes isolated vertices before choosing the normalizing shore; this leaves the polynomial unchanged, so source 06's theorem applied to the nonisolated core gives source 07's order  $d$ .

*A third proof.* Batch-55 manuscript 05 (source 09, printed as Part VII, with its stability sections in Part VI) proves the same ultra-log-concavity theorem with the same matroid and the same Lorentzian proof, and credits Röhrle–Ulirsch correctly. Its coefficient sections are not reprinted; Section 66.4 lists what of them is used here.

**Abstract of source 06 (as delivered).** For a bipartite graph  $H = (X \sqcup Y, E)$ , consider the nonnegative integral vectors  $c$  satisfying  $c(S) \leq |N_H(S)|$  for all  $S \subseteq Y$ , and count these vectors by the number of positive coordinates. We prove that the resulting coefficient sequence is ultra-log-concave, with binomial normalization of order  $\min(|X|, |Y|)$ , even after positive vertex weights are introduced. The proof joins a published support-counting identity to the classical transversal-matroid model for perfectly matchable sets and the Lorentzian theory of matroid bases.

The application gives an affirmative answer to the unimodality component of Dai–Hou–Liu–Thawinrak–Wang's Problem 5.3 for *every* bipartite graph. In particular, every finite preorder has an ultra-log-concave support polynomial. For height-two posets, the gamma polynomial is ultra-log-concave as well, answering the coefficient-shape question from Provelt's preorder report on that entire class. These conclusions do not require real roots.

We establish multivariate complete log-concavity, a covariance-matrix bound, and binomial moment-generating-function domination under external fields. A matching oracle and the matroid basis walk yield fully polynomial-time randomized approximation schemes for partition functions, specified support fibres, and individual ordinary coefficients. The latter assertion is justified by an explicit log-concave tilting reduction, not by unstable interpolation. Integer capacities and all positive integral dilates of preorder polytopes are included, with the input-encoding limitation stated. Exact finite verification accompanies the proofs.

**Abstract of source 07 (as delivered).** The failure of real-rootedness does not prevent strong coefficient inequalities for preorder support polynomials. We identify perfectly matchable vertex supports of a bipartite graph with the bases of an explicit fundamental transversal matroid. Together with the existing support-enumerator/root-polytope identities and Röhrle–Ulirsch’s bimatroid theorem, this proves ultra-log-concavity for every bipartite demand-support enumerator, and therefore for every finite preorder, including arbitrary equivalence classes. In particular, the unimodality component of Dai–Hou–Liu–Thawinrak–Wang’s Problem 5.3 needs no graph restriction. We also give a complete palindromicity criterion: after isolated vertices are deleted, the graph must be a doubled preorder, with its two copies identified along a perfect matching. This extends the repository’s balanced criterion to all bipartite graphs. The resulting theory supplies binomial convex-order domination under every positive fugacity, an exact weighted-star classification of equality in the variance bound, a geometric intersection realization, a rapidly mixing support sampler, and an FPRAS for the corresponding partition functions. A leaf-extension interpolation gives exact-counting hardness even for height-two preorder lattice-point counts. Written proofs are accompanied by exact finite checks on 69,632 bipartite graphs and all 390 labelled preorders through size four. The general bimatroid and mixing theorems are credited inputs, not discoveries of this manuscript.

The two abstracts are reproduced as delivered. Their “the repository” and “ProveIt’s preorder report” mean Parts I–IV of this report; source 07’s “the repository’s balanced criterion” is Part IV’s Corollary 53.2.

**Status and attribution (source 06, as delivered).** The matroid construction, the geometric counting bridge, and the general Lorentzian and mixing theorems are *prior results*, credited at their points of use. The contribution is an explicit application and extension to the current support-polynomial questions, with complete reductions and additional consequences. This is an unrefereed research article, not a Lean-verified development. No claim of worldwide priority, or of solving flag realizability or general gamma log-concavity, is made.

**Status and scope (source 07, as delivered).** This manuscript is not refereed or formalized in Lean. It contains conventional proofs and executable finite regression tests. Global priority of the applications is not established. It does not solve flag realizability, classify real-rooted graphs, or construct a direct demand-to-support bijection. These boundaries are kept explicit throughout.

**Priority and prior work (merge).** The unweighted inequality of Theorem 67.1 is *not new*: it is Theorem A of F. Röhrle and M. Ulirsch, *Logarithmic Concavity of Bimatroids* [17], for the bimatroid of a relation (their Example 2.4, whose regular minors are the pairs  $(I, J)$  joined by a matching), with the same normalization  $\binom{\min(m,n)}{k}$ ; their Proposition 2.2 is the extended-matroid construction of Section 69, and their Theorem E, for realizable bimatroids, contains the intersection realization of Proposition 73.1. Source 07 credits this; source 06 does not cite Röhrle–Ulirsch, and its Section 67 is printed with a merge note to that effect. What Part V adds is the composition with the published support-counting bridge (Ohsugi–Tsuchiya and Oh [8, 22]; Dai–Hou–Liu–Thawinrak–Wang and Davis–Kohl [2, 7]), hence the transfer to demand enumerators and to every preorder, the weighted two-shore form, and the consequences. The transversal-matroid model of perfectly matchable sets is classical (Balas–Pulleyblank [20], as identified by Mori [21]); the Lorentzian theory is Brändén–Huh’s [18]; the basis-walk mixing and approximate-counting machinery is Anari–Liu–Oveis Gharan–Vinzant’s [19]; the #P-completeness input is Colbourn–Provan–Vertigan’s [23]. The palindromicity criterion extends and re-proves Part IV’s Theorem 48.3 and Corollary 53.2; the height-two identity is Part II’s Theorem 13.2; the nonreal-rooted example  $\Theta_3$  is Part II’s, with priority credited to Shivam Patel [11]. Three manuscripts of batch 55 (sources 06, 07 and 09) proved the ultra-log-concavity transfer independently on the same day. The Röhrle–Ulirsch record was checked for this merge on 29 September 2026 against the arXiv abstract page and the HTML of arXiv:2402.15317v3: Theorem A states  $R_k^2 / \binom{m \wedge n}{k}^2 \geq R_{k+1} R_{k-1} / (\binom{m \wedge n}{k+1} \binom{m \wedge n}{k-1})$  for the numbers  $R_k$  of regular  $k \times k$  minors, and Example 2.4 is the bimatroid of a relation. No worldwide publication priority is claimed for any result of Part V.

## 66.2 Notation of Part V

Part V works with a finite bipartite graph  $H = (X \sqcup Y, E)$ , with  $|X| = m$  and  $|Y| = n$ , and with a finite preorder  $\tau$  on  $[n]$ , using Part I’s  $\mathcal{Q}_\tau$ ,  $\mathcal{J}(\tau)$  and  $h_\tau$  and Part IV’s  $\gamma_k(\tau)$  and  $\Gamma_\tau$ . Table 6 lists what Part V adds or uses differently. Three symbols were changed from the sources; no normalization was changed:

- the transversal matroid of  $H$ ,  $M_H$  in both sources, is printed  $M_H$ , because  $M_G(u)$  is Part II’s matching polynomial, and because Part VII’s  $M_G$  is the *dual* convention (Section 102);
- the random support size,  $K$  in both sources, is printed  $L$ , as in Part IV’s Theorem 58.1, where  $K$  is the mixing variable; source 07’s bit length  $L$  is printed  $L_{\text{bit}}$ ;
- the preorder polytope,  $Q_\tau$  in both sources, is printed  $\mathcal{Q}_\tau = Q_\tau$ , and source 06’s  $\mathcal{J}(\tau)$  is printed  $\mathcal{J}(\tau)$ , as in Part I.

Source 07’s macros for  $\mathbb{E}$ ,  $\mathbb{P}$ ,  $\mathcal{B}$ ,  $\mathcal{D}$  are typeset as the host’s. The letters  $r$  and  $d$  are *not* renamed; their meaning depends on the source and is fixed in the table.

## 66.3 What Part V answers

Status after Part V, as of 29 September 2026:

- **Every preorder support polynomial is ultra-log-concave** (Corollary 67.2), with the sharp constant, and positive integral dilations are included (Corollary 71.3). This is stronger than the unimodality of Conjecture 5.1(b) of [1], proved in Part IV (and by Part I’s proposed geometric route), and it does not follow from  $\gamma$ -positivity.
- **The unimodality part of Problem 5.3 of [2] holds** for every bipartite graph (Corollary 70.2), and its **palindromicity part** is classified (Theorem 67.4): after isolated vertices are deleted, the graph must be a doubled preorder.



Table 6: Symbols of Part V. The last column prints the tempting false reading.

| Symbol                               | Meaning in Part V                                                                                                                                                                     | Not to be confused with                                                                                               |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| $H = (X \sqcup Y, E), m, n$          | finite bipartite graph, donors (suppliers) $X$ , receivers $Y$ ; $m =  X , n =  Y $                                                                                                   | Part II's $H$ in Section 14; $n$ is also the size of a preorder.                                                      |
| $\mathcal{D}_H; F_H(z) = D_H(z)$     | demand vectors $c \in \mathbb{Z}_{\geq 0}^Y$ with $c(S) \leq  N_H(S) $ ; their support enumerator, with $z_y$ marking $c_y > 0$ (source 06's $F_H$ is source 07's $D_H$ )             | an enumerator weighted by the values $c_y$ ; Part IV's $F_H$ is the same polynomial.                                  |
| $U \sim_H S; \mathcal{M}(H), (I, J)$ | a matchable support pair (source 06); the set of them, $I \subseteq X, J \subseteq Y$ (source 07)                                                                                     | individual matchings: a pair is counted once.                                                                         |
| $p_H, a_k, p_k, c_k$                 | $\sum_{U \sim S} t^{ S }$ and its coefficients; $a_k(u, v)$ and $c_k$ with vertex weights                                                                                             | Part II's $p_G$ (the same polynomial); Part I's $p(y)$ .                                                              |
| $\mathcal{M}_H, \mathcal{B}_H$       | the transversal matroid with private copies of $X$ , bases $(X \setminus U) \sqcup S$ ; its basis polynomial (source 06's $\mathcal{B}_H(a, z)$ , source 07's $\mathcal{B}_H(x, y)$ ) | Part II's matching polynomial $M_G(u)$ ; Part VII's $M_G$ , the dual matroid; Part IV's pair set $\mathcal{M}_\tau$ . |
| $u, v; \alpha, \beta$                | positive donor and receiver weights (source 06); the same (source 07)                                                                                                                 | Part II's variable $u$ in $p_G(u)$ .                                                                                  |
| $r$                                  | in sources 06's sections, $\min(m, n)$ with isolated vertices kept; in source 07's sections, $v(H)$ , the actual degree                                                               | a rank; Part III's $r$ .                                                                                              |
| $d$                                  | in source 07's sections, the smaller shore after deleting isolated vertices; in Theorem 75.5, $v(H)$ ; in Section 75, a matroid rank                                                  | Part IV's signed vectors $d$ .                                                                                        |
| $v(H)$                               | maximum matching size                                                                                                                                                                 | Part IV's $v(\text{Comp}(\tau))$ .                                                                                    |
| $L$                                  | random support size (sources' $K$ )                                                                                                                                                   | Part IV's $K$ (mixing variable); source 07's divisor classes $L_v$ (Section 73).                                      |
| $\mu, \mathbb{P}_z, \mathbb{E}$      | mean and law under fugacity or external field $z$                                                                                                                                     | Part II's $\mu$ .                                                                                                     |
| $H_\tau, G_\tau$                     | the doubled preorder graph, edge $i \rightarrow j$ iff $i \leq_\tau j$ (source 06 writes $H_\tau$ , source 07 $G_\tau$ )                                                              | Part I's $G_\tau$ with its shores exchanged; the same $p$ .                                                           |
| $P_H$                                | height-two poset with $x < y$ for $xy \in E$                                                                                                                                          | Part I's $P_\tau$ .                                                                                                   |
| $\mathcal{L}, \mathcal{R}$           | left and right $r$ -sets occurring in maximum matchings (Proposition 72.1)                                                                                                            | Part IV's $\mathcal{R}_H$ .                                                                                           |
| $R, \bar{R}$                         | the reflexive relation of a balanced core, its transitive closure                                                                                                                     | Part IV's same notation (agrees).                                                                                     |
| $Z_H, L_v$                           | the variety and nef classes of Proposition 73.1                                                                                                                                       | the support size $L$ .                                                                                                |
| $H^{(b)}, H^{(\ell)}$                | donor cloning by capacities $b$ ; attaching $\ell$ leaves at every vertex                                                                                                             | Part III's $H$ .                                                                                                      |
| $\Theta_3$                           | Part II's $\Theta_3$ , three paths of length three (= $\vartheta(3, 3, 3)$ of Parts III and VI)                                                                                       | Euler's $\Theta$ ; Part III's $T$ core.                                                                               |



- **Part IV's Research question 23 ("Classification beyond a perfect matching") is answered:** there are no residual components beyond isolated vertices, and recognition takes a matching and a transitive closure.
- **Research question 24 ("Coefficient shape beyond gamma-positivity") is answered in its log-concavity and unimodality parts for every poset of height two (Corollary 71.6); for arbitrary preorders log-concavity of  $\Gamma_\tau$  remains open, and real-rootedness of  $\Gamma_\tau$  is untouched.**
- **Research question 26 ("Parameterized exact algorithms"):** its last sentence is answered. Without parameter restrictions,  $|\mathcal{Q}_\tau \cap \mathbb{Z}^n|$  is #P-complete under Turing reductions, already in height two (Theorem 76.3), and it has an FPRAS (Theorem 75.2). The parameterized question remains open.
- **Research question 32 ("Exact counting complexity"):** computing a specified  $\gamma_k$  for unrestricted preorders is #P-hard, already in height two (merge consequence after Theorem 76.3); in height two each  $\gamma_k$  has an FPRAS (Theorem 75.5). Efficient classes remain open.
- **Research question 28 ("Limit laws for support size")** gains same-mean binomial convex-order domination under every positive external field, with equality exactly for antichains (Corollary 74.9), and a covariance-matrix bound (Theorem 74.3); no central limit theorem is proved.
- **Conjecture 5.3 and flag realization** are unchanged: Part V proves no real-rootedness and constructs no flag polytope.  $\Theta_3$  shows that the Lorentzian conclusions do not upgrade to real stability (Section 77).

Part V refutes no statement of Parts I–IV, and nothing in them is retracted. Parts I–IV gain dated pointers: the title block, abstract, scope box and reading route; Section 1; Part IV's remark after Corollary 53.2, Theorem 58.1, Corollary 50.3 and Research questions 23, 24, 26, 28, 32 and 34, which received labels for this purpose.

## 66.4 Where the merge had to choose

- *Placement and order.* Part V follows Part IV. Source 06's order of sections is kept (Sections 67–82); source 07's sections are placed beside the matching ones, and those without a counterpart are printed as sections of their own, marked "(source 07)". Both sources' appendices are printed as ordinary sections.
- *Duplicated results* are printed once, with both sources named: the support-counting bridge (Proposition 68.3; source 07's Lemma 2.1), the basis correspondence (Proposition 69.1; source 07's Proposition 3.1, "Complement one shore"), the weighted inequality (Theorem 70.1), the preorder corollary (Corollary 67.2; source 07's Corollary 1.3), Problem 5.3's unimodality (Corollary 70.2; source 07's Corollary 4.1) and the partition-function FPRAS (Theorem 75.2; source 07's Corollary 8.2). Source 07's first headline theorem is kept as Theorem 67.3 because it states the order on the nonisolated core.
- *Second routes.* Genuinely different proofs are kept and marked: source 07's import of the bridge through [2, 7] with subset inversion, beside source 06's Hall-model import through [8, 22]; source 07's explicit factorial calculation of the Lorentzian criterion; source 06's logistic ODE proof of the binomial moment bound, beside source 07's convex-order proof, which is the main route because it gives the equality classification; source 07's Hessian derivation of the variance bound; source 07's explicit error allocation for the self-reduction.

- *Source 06's palindromicity paragraph* (the ideal Hall criterion and the complement involution) re-proves known facts of Part IV; it is kept, with a merge note, and superseded by Section 72.
- *Source 09* (Part VII) is used in Part V only as follows: its Theorem 4.2 (every preorder ultra-log-concave, with the simultaneous-equality clause) is Corollary 67.2 with the proposition after it, credited there; its binomial convex-order theorem is Lemma 74.4 with Theorem 74.5, and its Hoeffding form of the tail bound is printed as Remark 74.7; its  $\Theta_3$  ultra-log-concavity display duplicates Section 77 and is omitted. Its transport-weighted  $\gamma$ -identity is Part VII's Theorem 107.2.
- *Research questions.* Part V's questions continue the report's numbering as Research questions 37–43. Three pairs ask the same question and are printed once with both formulations; the questions that repeat questions of Parts II–IV, and one answered inside Part V, are printed without numbers in Section 80.1, next to the question they repeat.
- *Self-citation.* Both sources cite this report as an external repository report (reference “Repo”, with their pins). Those citations became references to Parts I–IV; “the repository(’s)” became “Part IV(’s)” or “this report” where it means this report, and was left where it means ProveIt as a whole. The pins are kept in Section 79.4.
- *File names.* The delivered names `code/verify.py`, `results/verification.json`, `data/verification.json` and so on are replaced in the text by merge notes giving the shipped names, because `code/verify.py` in this directory is Part I's verifier. The delivered run commands are kept, marked as such, and the README gives commands that work with the shipped names.
- *Corrections.* Source 06 cites Theorem 4.2 of [2] for the augmented-root identity; the inspected arXiv version 2 has it as Theorem 1.2 (its Section 4 has a Lemma 4.2 on dimensions), as source 07 and Part I cite it. The shipped note `06-lorentzian-support-SOURCES.md` repeats “Theorem 4.2” and is left byte-identical. Source 06's bibliography item is printed as this report's [2].
- *Bibliography.* Source 06's OT is [8]; AC, DHLTW, DK, Patel were already cited. New items: [17, 18, 19, 20, 21, 22, 23].
- *Labels and front matter.* Labels carry the prefix `lor:` (source 06) and `lor:sp:` (source 07). Part IV's Research questions 23, 24, 26, 28, 32 and 34 and Part II's and Part III's Research questions 3, 11 and 15 received new labels (`lor:hq:` and `ade:hq:`) so that Parts V and VI can cite them. Both title pages, abstracts and status boxes are printed in this section; both tables of contents are dropped. Source 07 used `cleveref`; its references are printed with the host's `\ref` and `\eqref`.

## 67 The questions and the answer

A failed real-rootedness conjecture need not mark the end of a coefficient inequality program. Real-rootedness is a sufficient mechanism for many inequalities, but it is stronger than the inequalities themselves. This article isolates a different mechanism for the preorder support polynomials studied in ProveIt: a homogeneous polynomial on a larger ground set whose coefficients are those of a matroid basis enumerator.

The relevant repository is Vladimir Reshetnikov's *ProveIt*, specifically this report, Parts I–IV above. Its four parts develop a root-polytope realization, height-two matching-support formulas and nonreal-rooted examples, cactus determinant models, and gamma-positivity for every

finite preorder. Its current research questions include the shape of the gamma coefficients and the complexity of computing them. Those earlier results are not claimed again here.

The external research target is also precise. Dai, Hou, Liu, Thawinrak and Wang ask in Problem 5.3 of [2] which bipartite support enumerators have the properties conjectured for preorders by Athanasiadis and Chapoton [1]. We settle the *unimodality part* uniformly, and strengthen it to ultra-log-concavity. This does not classify palindromicity or real-rootedness; the three demands have different answers.

### 67.1 Main theorem in finite combinatorial form

Let  $H = (X \sqcup Y, E)$  be a finite bipartite graph, with  $|X| = m$  and  $|Y| = n$ . Parallel edges have no effect and are discarded. Write  $N_H(S) \subseteq X$  for the neighbors of  $S \subseteq Y$ , and put

$$\mathcal{D}_H = \left\{ c \in \mathbb{Z}_{\geq 0}^Y : \sum_{y \in S} c_y \leq |N_H(S)| \quad (S \subseteq Y) \right\}. \quad (67.1)$$

The support polynomial is

$$F_H(z) = \sum_{c \in \mathcal{D}_H} \prod_{y: c_y > 0} z_y, \quad F_H(t, \dots, t) = \sum_{k \geq 0} a_k(H) t^k. \quad (67.2)$$

The exponent of a coordinate in this polynomial is its *presence indicator*, not its value  $c_y$ . Thus  $F_H$  is multiaffine even though  $c_y$  may exceed one.

**Theorem 67.1** (Uniform coefficient theorem; sources 06 and 07). *Set  $r = \min(m, n)$ . The support coefficients satisfy*

$$\frac{a_k(H)^2}{\binom{r}{k}^2} \geq \frac{a_{k-1}(H)}{\binom{r}{k-1}} \frac{a_{k+1}(H)}{\binom{r}{k+1}} \quad (1 \leq k < r). \quad (67.3)$$

*Their nonzero indices are exactly  $0, 1, \dots, v(H)$ , where  $v(H)$  is the maximum matching size. In particular they are log-concave and unimodal. The polynomial  $F_H(z)$  is completely log-concave on the positive orthant. The same coefficient inequality holds after positive product weights are placed on the two vertex supports of each perfectly matchable set.*

The precise meaning of the weighted statement appears in Section 70. If  $r \leq 1$ , the inequalities are vacuous but the conclusions about support and unimodality retain their usual meaning. Zero coefficients beyond the actual degree are included in the normalization.

For a preorder  $\tau$  on  $[n]$ , let  $\mathcal{J}(\tau)$  denote its lower ideals and define

$$\mathcal{Q}_\tau = \{x \in \mathbb{R}_{\geq 0}^n : x(I) \leq |I| \quad (I \in \mathcal{J}(\tau))\}, \quad h_\tau(t) = \sum_{x \in \mathcal{Q}_\tau \cap \mathbb{Z}^n} t^{|\text{supp}(x)|}. \quad (67.4)$$

Reflexivity and transitivity, but not antisymmetry, are required.

**Corollary 67.2** (All finite preorders; sources 06, 07 and 09). *For every finite preorder on  $n$  labels,  $h_\tau(t) = \sum_{k=0}^n h_k t^k$  satisfies*

$$h_k^2 \geq \frac{(k+1)(n-k+1)}{k(n-k)} h_{k-1} h_{k+1} \quad (1 \leq k < n). \quad (67.5)$$

*Every coefficient is positive. Consequently the ordinary coefficients are strictly log-concave at every interior index. The binomial normalization is universally sharp: the antichain has  $h_\tau = (1+t)^n$ .*

This is a statement about  $h_\tau$ , not about its gamma transform. The latter distinction is essential throughout the article.

[Merge note: the unweighted inequality (67.3) is Theorem A of Röhrle and Ulirsch [17] for the relation bimatroid of their Example 2.4, whose regular  $k \times k$  minors are exactly the pairs  $U \sim_H S$  with  $|S| = k$ ; their normalization is the same  $\binom{\min(m,n)}{k}$ . Source 06 does not cite them; the statements of this section that are new relative to [17] are the transfer through Proposition 68.3 to demand vectors and preorders, the weighted two-shore form, and the consequences.]

## 67.2 The same theorems as stated by source 07

The relevant part of ProveIt is the four-part report *A Reflexive Root-Polytope Model for Preorder  $h$ -Polynomials* (this report, Parts I–IV). Its later parts distinguish three questions which must not be conflated: real-rootedness,  $\gamma$ -positivity, and flag-polytopal realization. The report proves a transport-support formula for  $\gamma$ -positivity, records nonreal-root examples, and leaves flag realization open. Its determinant construction is deliberately restricted: a support-exact unit-modulus matrix exists precisely for bipartite cacti. None of those facts implies that the coefficient sequence loses all strong concavity outside the cactus class.

There is a second, broader target. Dai, Hou, Liu, Thawinrak, and Wang [2, Problem 5.3] ask which bipartite graphs have demand-support enumerators satisfying the unimodality, palindromicity, and real-rootedness properties suggested by preorder polynomials. We separate these properties. The answer for *unimodality* is *every bipartite graph*. The answer for *palindromicity* is *an explicit, polynomial-time recognizable class*. Real-rootedness is strictly stronger and is not classified here.

Let  $H = (X \sqcup Y, E)$  be a finite simple bipartite graph. A pair  $(I, J)$ , with  $I \subseteq X, J \subseteq Y$ , is a *matchable support* if  $H[I \sqcup J]$  has a perfect matching; the empty pair is allowed. Define

$$p_H(t) = \sum_{(I,J) \in \mathcal{M}(H)} t^{|I|} = \sum_{k=0}^r p_k t^k, \quad r = \nu(H).$$

Here  $\nu(H)$  is the maximum matching size. Each support is counted once, not once per perfect matching. Delete isolated vertices and let  $a, b$  be the remaining shore sizes. Set  $d = \min(a, b)$ ; an edgeless graph has  $d = r = 0$ .

**Theorem 67.3** (Universal coefficient inequality; Theorem 70.1 on the nonisolated core). *For every bipartite graph, the sequence*

$$\left( \frac{p_k}{\binom{d}{k}} \right)_{k=0}^d, \quad p_k = 0 \quad (k > r),$$

*is log-concave and has no internal zeros. Thus  $p_H$  is unimodal and its positive coefficients are strictly log-concave at all internal indices. The same assertions hold after each support  $(I, J)$  receives any positive vertex-product weight  $\prod_{x \in I} \alpha_x \prod_{y \in J} \beta_y$ .*

The unweighted graph inequality is a direct specialization of the established bimatroid inequality of Röhrle and Ulirsch [17, Theorem A], not a new bimatroid theorem. The important transfer is to the newer support-enumerator and preorder questions. The weighted formulation follows from positive specialization of the same matroid basis polynomial.

**Theorem 67.4** (Complete palindromicity criterion). *For every bipartite graph  $H$ , the following are equivalent.*

- (i)  $p_H(t)$  is palindromic with respect to its actual degree  $r$ .
- (ii) Its nonisolated core has two shores of size  $r$ , has a perfect matching, and, after identifying matched vertices, its edge relation is a preorder.
- (iii)  $p_H(t)$  is  $\gamma$ -positive with respect to degree  $r$ .

If the criterion holds for one perfect matching, it holds for every perfect matching of the core. It can be decided without computing  $p_H$ .

The balanced, perfect-matching case and its transport-pair proof already occur in Part IV (Theorem 48.3, Corollary 53.2). The passage to arbitrary graphs is supplied here by an extremal-support argument, with a quantitative obstruction in Proposition 72.1. A complete proof is included so that the classification does not depend on accepting an unrefereed repository proof.

For a preorder  $\tau$  on  $[n]$ , let  $h_\tau(t) = \sum h_k t^k$  count lattice points of its preorder polytope by the number of positive coordinates. The precise definition is in Section 68. Source 07's Corollary 1.3 ("Every finite preorder") is Corollary 67.2, with the added remark that the normalized inequality is strictly stronger than  $h_k^2 > h_{k-1}h_{k+1}$  and that all equivalence-class sizes and heights are allowed; it is not printed again.

### 67.3 What is new here, and what is imported

There are three substantial external ingredients. First, the number of vectors in (67.1) with a prescribed support is the number of opposite-shore subsets matchable to that support. We use the established geometric counting theorem in the form appearing in the proof of Ohsugi–Tsuchiya's Proposition 3.4 [8], based on Oh's work [22]. Second, every matroid basis polynomial is Lorentzian, and nonnegative linear substitutions preserve that property [18]. Third, the weighted matroid basis down-up walk has a rank-controlled mixing bound [19]. These are cited theorems, not discoveries or formalizations in this article.

The transversal-matroid representation of bipartite perfectly matchable sets is also classical. Balas and Pulleyblank's work [20] is explicitly identified as the source of this observation by Mori [21]. We give the short construction in full because it makes the coefficient, probabilistic and algorithmic consequences transparent.

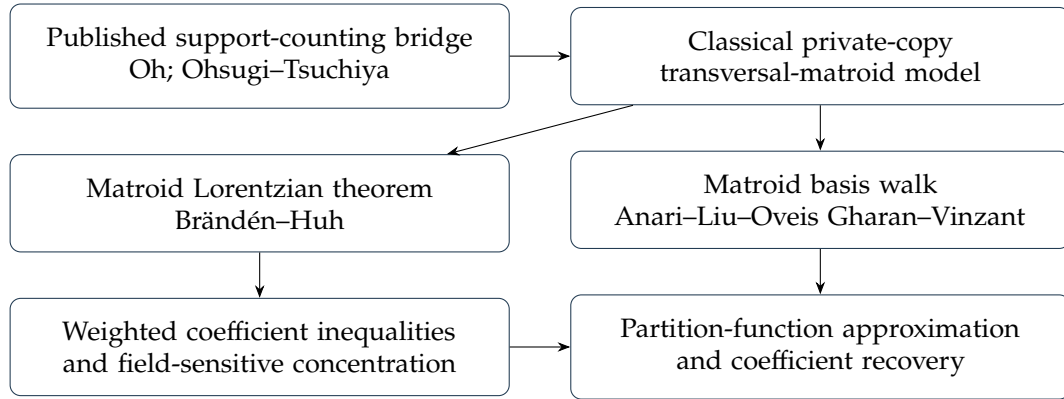
The principal contribution is the resulting package of statements for the current preorder and bipartite support problems: both-shore weighted normalizations, conditional support laws, external-field covariance and tail bounds, and a rigorous reduction for approximating individual coefficients. The inspected sources do not establish a worldwide priority claim for these consequences; some may be implicit in earlier work. The article asserts the proved implications, not the absence of all precedents.

[Merge note: the last paragraph is source 06's own priority statement. The unweighted inequality is in fact published (Theorem A of [17]); see the priority box of Section 66.1. Source 07's ledger of what is imported, derived and left open follows, with source 06's imported items added in the last three rows.]

| Statement                                                                                       | Attribution or boundary                                                                                                   |
|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Support enumerator equals augmented-root $h^*$                                                  | Dai–Hou–Liu–Thawinrak–Wang, Theorem 1.2 [2].                                                                              |
| Augmented-root $h^*$ equals $p_H$                                                               | Ohsugi–Tsuchiya, as stated by Davis–Kohl, Theorem 3.10 [7].                                                               |
| Regular-minor ultra-log-concavity                                                               | Röhrle–Ulirsch, Theorem A [17]; matroid Lorentzian theory [18].                                                           |
| Balanced transitivity criterion and transport $\gamma$ -formula                                 | Part IV (Theorems 48.2 and 48.3); reproved here.                                                                          |
| Universal unimodality of demand supports; all-preorder ULC                                      | Consequences established here by connecting the preceding theories.                                                       |
| Palindromic-core recognition, weighted variance equality, explicit sampling/counting reductions | Proved here; no claim that every component is globally new.                                                               |
| General matroid rapid mixing and FPRAS                                                          | Anari–Liu–Oveis Gharan–Vinzant [19].                                                                                      |
| Direct demand/support bijection; flag realization; full real-root classification                | Not resolved.                                                                                                             |
| Fixed-support positive-coordinate count (source 06’s import route)                              | Ohsugi–Tsuchiya, proof of Proposition 3.4 [8], based on Oh [22].                                                          |
| Private-copy transversal-matroid model                                                          | Balas–Pulleyblank [20], as identified by Mori [21]; extended matroid of a bimatroid, Röhrle–Ulirsch Proposition 2.2 [17]. |
| Weighted basis-walk mixing and approximate counting                                             | Anari–Liu–Oveis Gharan–Vinzant, Theorem 1.1 [19].                                                                         |

[Added 30 September 2026, batch 64: the row “Direct demand/support bijection” is resolved by Part IX (Theorem 135.2); flag realization and the real-root classification are not.]

## 67.4 A map of dependencies



No unrestricted theorem below is inferred from the finite verification program. The computations check independent finite descriptions of the objects entering this diagram.

## 68 The support-counting bridge

For  $U \subseteq X$  and  $S \subseteq Y$ , write  $U \sim_H S$  when  $|U| = |S|$  and  $H[U \sqcup S]$  has a perfect matching. Equivalently, there exists a bijection  $f : S \rightarrow U$  with  $f(y)y \in E$ . Each pair  $(U, S)$  is counted once, not once per matching.



### 68.1 The geometric theorem in the needed form

For  $S \subseteq Y$ , define

$$P_{H,S} = \sum_{x \in X} \text{conv}(\{0\} \cup \{e_y : y \in S, xy \in E\}) \subseteq \mathbb{R}^S. \quad (68.1)$$

An empty neighborhood contributes the single point 0.

**Lemma 68.1** (Published positive-coordinate count). *The number of integer points of  $P_{H,S}$  whose coordinates indexed by  $S$  are all strictly positive is*

$$\#\{U \subseteq X : U \sim_H S\}. \quad (68.2)$$

For  $S = \emptyset$ , both sides are one.

*Source and normalization.* This is the fixed-receiver-support count in the proof of [8, Proposition 3.4]. In that proof the summands are  $\Delta_{\{0\} \cup N(x)}$  in one extra coordinate, and their total coordinate sum is  $m$ . Eliminating the zeroth coordinate gives (68.1). The relevant positive-coordinate lattice-point identity is attributed there to [22, Lemma 22 and Proposition 26]. We import this counting theorem; in particular, we do not replace its geometric proof by an unproved claim of a canonical bijection.

**Lemma 68.2** (The integral feasibility model). *The integer points of  $P_{H,Y}$  are precisely  $\mathcal{D}_H$ . For a prescribed support  $S$ , the points of  $\mathcal{D}_H$  supported exactly on  $S$  are precisely the positive-coordinate integer points of  $P_{H,S}$ , extended by zero outside  $S$ .*

*Proof.* Every point of the Minkowski sum is obtained by distributing at most one unit from each  $x \in X$  among its neighboring receiver coordinates. Therefore its sum on  $T \subseteq Y$  is at most  $|N_H(T)|$ .

Conversely, let  $c$  be an integer vector satisfying those inequalities. Replace each receiver  $y$  by  $c_y$  labeled copies, each with the same neighbors as  $y$ . For any set of these copies, let  $T$  be the original receivers it uses. The number of selected copies is at most  $c(T) \leq |N_H(T)|$ . Hall's theorem therefore matches every copy to a distinct donor. Each donor contributes either 0 or one coordinate vector to (68.1), and their sum is  $c$ .

If  $c$  is zero outside  $S$ , the matching uses only the receiver copies in  $S$ . The same construction is thus valid in  $P_{H,S}$ . Conversely any point of that smaller sum extends by zero to the larger sum. Requiring strict positivity on  $S$  gives exactly the claimed support fibre.  $\square$

This proof needs only the integer version of the polytope correspondence. It does not assert that an integer vector has a unique donor allocation. Such allocations usually have multiplicity, which is why counting all allocations would give the wrong polynomial.

**Proposition 68.3** (Exact multivariate bridge; sources 06 and 07). *For every  $S \subseteq Y$ ,*

$$[z_S]F_H(z) = \#\{U \subseteq X : U \sim_H S\}, \quad z_S := \prod_{y \in S} z_y. \quad (68.3)$$

Consequently

$$F_H(z) = \sum_{U \sim_H S} z_S, \quad F_H(t, \dots, t) = p_H(t) := \sum_{U \sim_H S} t^{|S|}. \quad (68.4)$$

*Proof.* Lemma 68.2 identifies the coefficient with the left-hand side of Lemma 68.1. Apply that lemma separately to each  $S$ , then sum the resulting finite identities.  $\square$

The last polynomial is the perfectly matchable set polynomial of a bipartite graph, with the exponent equal to half the number of selected vertices; this terminology is used in [7]. For example,  $K_{2,2}$  has  $p_H(t) = 1 + 4t + t^2$ , not  $1 + 4t + 2t^2$ .



## 68.2 The bridge in source 07's formulation, and a second proof

For  $S \subseteq Y$ , write  $N_H(S) \subseteq X$  for its neighborhood. Define

$$\mathcal{D}_H = \{c \in \mathbb{Z}_{\geq 0}^Y : c(S) \leq |N_H(S)| \text{ for every } S \subseteq Y\}, \quad c(S) = \sum_{y \in S} c_y. \quad (68.5)$$

The bound  $c(Y) \leq |X|$  makes this set finite. Its support enumerator is

$$D_H(t) = \sum_{c \in \mathcal{D}_H} t^{|\text{supp}(c)|}, \quad D_H(\mathbf{z}) = \sum_{c \in \mathcal{D}_H} \prod_{y: c_y > 0} z_y.$$

The variables mark whether a coordinate is positive, not its numerical value. For example, one receiver with  $s$  suppliers has demands  $0, 1, \dots, s$ ; its polynomial is  $1 + st$ , not  $1 + t + \dots + t^s$ .

Source 07's Lemma 2.1 ("Established counting bridge and its refinement") states  $D_H(\mathbf{z}) = \sum_{(I,J) \in \mathcal{M}(H)} \prod_{y \in J} z_y$ , hence  $D_H(t) = p_H(t)$ : this is Proposition 68.3 in source 07's notation ( $D_H = F_H$ ). Its proof is a different import route.

*Second proof of Proposition 68.3 (source 07).* We first state exactly the imported enumerative input. Augment  $H$  by  $x_0, y_0$ , join  $x_0$  to all of  $Y \cup \{y_0\}$  and  $y_0$  to all of  $X$ , and form

$$B_{\widehat{H}} = \text{conv}\{e_x + f_y : xy \in E(\widehat{H})\}.$$

Use the full affine integer lattice. The two root-polytope results [2, Theorem 1.2] and [7, Theorem 3.10] identify its  $h^*$ -polynomial with  $D_H(t)$  and  $p_H(t)$ , respectively. The first source puts its demand coordinates on the left shore; apply its theorem to the shore-swapped graph. Swapping shores preserves the augmented root polytope up to an integral coordinate permutation. In particular  $|\mathcal{D}_H| = p_H(1)$ . These equalities, not a conjectural positivity assertion, are the imported boundary.

For the refinement, let  $u_J$  count demands of exact support  $J$  and let  $v_J$  count pairs  $(I, J)$  which are matchable. Applying the total-count identity to  $H[X \sqcup T]$ , for every  $T \subseteq Y$ , gives

$$\sum_{J \subseteq T} u_J = \sum_{J \subseteq T} v_J.$$

Induction on  $|T|$ , subtracting all proper subsets, gives  $u_T = v_T$ . This is precisely the multivariate identity. Isolated vertices and empty shores cause no exception.  $\square$

**Remark 68.4** (What the bridge does not give). A polynomial identity is not a direct algorithmic bijection. The proof does not tell us which demand vector should correspond to a given matchable pair. Consequently a uniform support sampler is not, by itself, a uniform sampler of the actual demand vectors. It does sample their support set  $J$  with the correct multiplicity distribution, by Proposition 68.3.

[Added 30 September 2026, batch 64: Part IX supplies the missing algorithmic bijection (Theorem 135.2) and a second proof of Proposition 68.3 that imports no lattice-point theorem (Corollary 139.2).]

## 68.3 A consistency check with the recent geometric formulation

Augment  $H$  by adding one vertex on each shore, adjacent to every vertex of the opposite shore, including each other. Denote the result by  $\widehat{H}$ . Its bipartite root polytope can be written

$$R_{\widehat{H}} = \text{conv}\{e_x - e_y : xy \in E(\widehat{H})\}.$$

The equivalent plus-sign convention differs by a lattice coordinate reflection. The identities in [8, 2] give

$$h^*(R_{\widehat{H}}, t) = p_H(t) = F_H(t, \dots, t). \quad (68.6)$$

Here  $h^*$  is the numerator of the Ehrhart series with its dimensionally correct denominator. Thus our inequalities also apply to this entire class of *augmented* bipartite root polytopes. Nothing here asserts ultra-log-concavity for every lattice polytope or every unaugmented root polytope.

## 69 The classical matroid lift

We now explain why counting matchable *supports* rather than matching *witnesses* is exactly the right convention.

Make a disjoint private copy  $X^\circ = \{x^\circ : x \in X\}$  of  $X$ . The ground set of a transversal matroid  $M_H$  is

$$\Omega = X^\circ \sqcup Y,$$

and its matching slots are the original elements of  $X$ . The element  $x^\circ$  is adjacent only to slot  $x$ , whereas  $y \in Y$  is adjacent to the slots in  $N_H(y)$ . A subset of  $\Omega$  is independent when it can be matched injectively into the slots.

This family is hereditary. Its augmentation property is the usual alternating-path augmentation between matchings of two independent sets of unequal size. Thus it is a transversal matroid. The private elements form a basis, so its rank is exactly  $m$ , even when  $H$  has isolated vertices. This is the concrete version of the classical connection credited in Section 67.

**Proposition 69.1** (Basis correspondence; sources 06 and 07). *The bases of  $M_H$  are exactly*

$$B(U, S) = (X^\circ \setminus U^\circ) \sqcup S, \quad U \sim_H S. \quad (69.1)$$

*In particular this is a bijection between support pairs and bases.*

*Proof.* Every basis has size  $m$ . If its receiver part is  $S$ , write its private part as  $X^\circ \setminus U^\circ$ . The size equation gives  $|U| = |S|$ . The selected private elements occupy precisely their own slots in  $X \setminus U$ . Therefore the remaining elements  $S$  must be matched to  $U$ , and a basis is possible exactly when  $U \sim_H S$ .

Conversely, choose one matching between  $S$  and  $U$ . Together with the forced private-slot edges it matches all of  $B(U, S)$  into  $X$ . The resulting basis set does not depend on which matching was chosen. Recovering  $S$  as  $B \cap Y$  and  $U$  as the missing private elements gives the inverse.  $\square$

Source 07 proves the same bijection (its Proposition 3.1, “Complement one shore”, with the private elements identified with  $X$  itself) by the same forced-slot argument, and adds: This is the fundamental-transversal/extended-matroid construction, not a new kind of matroid. Its role here is to make the support model compatible with existing matroid geometry and algorithms.

### 69.1 Generic minors, without a unit-determinant requirement

Let  $A_H$  be the  $m \times n$  matrix whose entry on edge  $xy$  is an independent indeterminate  $q_{xy}$ , with zero elsewhere. Work over  $\mathbb{Q}(q_{xy} : xy \in E)$ . The determinant of  $A_H[I, J]$  is a signed sum

indexed by perfect matchings of  $H[I \sqcup J]$ . Different matchings use different edge monomials, so no two terms cancel. Hence

$$\det A_H[I, J] \neq 0 \iff (I, J) \in \mathcal{M}(H). \quad (69.2)$$

Expanding along identity columns shows that  $[I_m \mid A_H]$  represents  $\mathbf{M}_H$ . Thus matchable supports are precisely the regular minors of a realizable bimatroid, in the terminology of [17].

The entries need not have modulus one, and the nonzero determinants need not have equal magnitudes. This is why the construction works beyond the cactus class in the repository. Substituting an all-ones adjacency matrix would be wrong: a  $2 \times 2$  all-ones minor vanishes although its support is matchable. Generic *nonvanishing* and faithful *unit modulus* are different requirements.

Write  $a_A = \prod_{x \in A} a_x$ . The basis generating polynomial is

$$\mathcal{B}_H(a, z) = \sum_{B \text{ basis of } \mathbf{M}_H} \prod_{e \in B} w_e = \sum_{U \sim_H S} a_{X \setminus U} z_S, \quad (69.3)$$

where  $w_{x^\circ} = a_x$  and  $w_y = z_y$ . It is multiaffine and homogeneous of degree  $m$ . Proposition 68.3 now gives the key equality

$$\boxed{F_H(z) = \mathcal{B}_H(\mathbf{1}, z)}. \quad (69.4)$$

Thus a nonhomogeneous support polynomial is an affine slice of a homogeneous matroid polynomial. No determinant representation is required.

## 69.2 The homogeneous lift and conditioning in source 07's notation

The basis polynomial is

$$\mathcal{B}_H(\mathbf{x}, \mathbf{y}) = \sum_{(I, J) \in \mathcal{M}(H)} \prod_{x \in X \setminus I} x_x \prod_{y \in J} y_y. \quad (69.5)$$

It is homogeneous of degree  $m$  and multiaffine. The original occupied-vertex polynomial, whose terms have varying degree  $2|I|$ , is not this polynomial. Complementing one shore is essential.

For positive activities  $\alpha_x, \beta_y$ , write

$$p_H^{\alpha, \beta}(t) = \sum_{(I, J) \in \mathcal{M}(H)} \alpha_I \beta_J t^{|I|} = \sum_k c_k t^k.$$

Then

$$\left( \prod_{x \in X} \alpha_x \right) \mathcal{B}_H((s/\alpha_x)_{x \in X}, (t\beta_y)_{y \in Y}) = \sum_k c_k s^{m-k} t^k. \quad (69.6)$$

Supplier activities on the right side are inverse activities for private matroid elements on the left. Confusing these two conventions would sample the wrong probability distribution.

**Proposition 69.2** (Conditioning). *Fix any collection of vertices required to be covered or uncovered. If at least one support satisfies the requirements, the corresponding conditional weighted basis distribution is the positive-activity basis distribution of a matroid minor of  $\mathbf{M}_H$ .*

*Proof.* For  $x \in X$ , being covered means the private element  $x$  is absent from the basis; being uncovered means it is present. For  $y \in Y$  these meanings are reversed. Required present elements are contracted and required absent elements are deleted. A feasible set of required present elements is independent. The bases of the minor correspond bijectively to the surviving bases after removing the required elements. Their common weight factor cancels on normalization.  $\square$

This fact does not assert that every minor has a presentation by a graph of exactly the same form. The ordinary minor oracle is enough. [Merge note: Proposition 69.2 conditions vertices of both shores; source 06's Proposition 74.1 below conditions receivers and adds complete log-concavity of the conditioned polynomial.]

### 69.3 Exactly which analytic theorem is used

A polynomial with nonnegative coefficients is *completely log-concave* if every iterated directional derivative in nonnegative directions is either zero or log-concave on the positive orthant. For a homogeneous polynomial, this agrees with being Lorentzian.

We use the following parts of [18]: matroid basis polynomials are Lorentzian (Theorem 3.10); nonnegative linear substitutions preserve Lorentzian polynomials (Theorem 2.10); and Lorentzian and complete log-concavity are equivalent in the homogeneous case (Theorem 2.30). A bivariate degree- $d$  homogeneous polynomial  $\sum b_k s^{d-k} t^k$  is Lorentzian exactly when its nonnegative coefficient sequence has no internal zeros and  $b_k / \binom{d}{k}$  is log-concave (Example 2.26).

**Theorem 69.3** (Multivariate completion). *For  $m \geq 1$ ,*

$$\tilde{F}_H(s, z) := s^m F_H(z/s) = \mathcal{B}_H(s\mathbf{1}, z) \quad (69.7)$$

*is Lorentzian. The polynomial  $F_H(z)$  is completely log-concave.*

*Proof.* The first assertion follows from the nonnegative linear substitution  $a_x = s$  in (69.3). For the second, extend each nonnegative direction on the  $z$  coordinates by zeros on the  $a$  coordinates. Differentiation in those directions commutes with restriction to  $a = \mathbf{1}$ . Each resulting derivative of  $\mathcal{B}_H$  is log-concave or zero. A log-concave function restricted to an affine convex slice remains log-concave. This proves the required assertion for every sequence of directions. If  $m = 0$ ,  $F_H = 1$  and the conclusion is immediate.  $\square$

The homogenizing degree in (69.7) is  $m$ , not necessarily the actual degree  $\nu(H)$ . Dividing a Lorentzian polynomial by a power of a homogenizing variable does not automatically preserve the property. We therefore do not silently replace  $m$  by  $\nu(H)$ .

## 70 Weighted inequalities on both shores

For positive real weights  $u = (u_x)_{x \in X}$  and  $v = (v_y)_{y \in Y}$ , define

$$A_H(t; u, v) = \sum_{U \sim_H S} u_U v_S t^{|S|} = \sum_{k=0}^{\nu(H)} a_k(u, v) t^k. \quad (70.1)$$

When  $u = \mathbf{1}$ , this is the original support enumerator with the external field  $z_y = t v_y$ . General  $u$  weights a donor *support*; it is not an unspecified weighting of individual demand vectors.

**Theorem 70.1** (Two-shore ultra-log-concavity; sources 06, 07 and 09). *For  $r = \min(m, n)$ , the sequence*

$$\left( \frac{a_k(u, v)}{\binom{r}{k}} \right)_{k=0}^r$$

*is log-concave and has no internal zeros, with zero padding after  $\nu(H)$ . In particular Theorem 67.1 holds.*

*Proof.* First group all private variables and all receiver variables in the basis polynomial, but use the inverse donor weights:

$$\begin{aligned} \left( \prod_{x \in X} u_x \right) \mathcal{B}_H((s/u_x)_{x \in X}, (tv_y)_{y \in Y}) &= \sum_{U \sim_H S} u_U v_S s^{m-|S|} t^{|S|} \\ &= \sum_k a_k(u, v) s^{m-k} t^k. \end{aligned} \quad (70.2)$$

This is a nonnegative linear substitution in a Lorentzian polynomial, followed by positive scalar multiplication. The bivariate criterion gives ultra-log-concavity with normalization  $\binom{m}{k}$ .

Now interchange  $X$  and  $Y$ . A support pair  $(U, S)$  in  $H$  becomes  $(S, U)$  in the transposed graph, with the weights interchanged. It has the same weight  $u_U v_S$  and the same cardinality. The identical construction therefore gives normalization  $\binom{n}{k}$ . Choose the smaller shore to obtain the displayed conclusion.

Every matching of size  $k$  supplies a support pair, and every such pair has a witnessing matching of size  $k$ . Submatchings show that all indices from 0 to  $v(H)$  occur. All weights are positive, so no cancellation or internal zero is possible. Finally, for  $0 < k < r$ , the binomial factor in (67.3) is greater than one; this implies ordinary log-concavity, and a nonnegative log-concave sequence without internal zeros is unimodal.  $\square$

**Corollary 70.2** (The published unimodality question; sources 06 and 07). *Every bipartite graph in [2, Problem 5.3] has a unimodal support-enumerator. More strongly, it satisfies (67.3), and so does the corresponding augmented-root-polytope  $h^*$ -polynomial in (68.6).*

*Proof.* Up to interchanging the shore names, their support-enumerator is (67.2); the neighborhood inequalities specify the same integer vectors. Apply Theorem 70.1 with unit weights and use (68.6) for the Ehrhart formulation.  $\square$

This is an answer to one component of that problem. For example,  $K_{1,2}$  gives  $1 + 2t$ , which is not palindromic in its actual degree. Section 77 gives an example with nonreal roots.

Source 07 proves the same corollary (its Corollary 4.1, “The unimodality component of Problem 5.3”) by Proposition 68.3 and, for preorders, by (71.3), noting that for the doubled preorder graph both active shores have size  $n$  and all coefficients are positive.

## 70.1 Sharpness and elementary extremal checks

If  $H$  consists of  $r$  disjoint edges, with any number of isolated vertices added, then

$$p_H(t) = (1 + t)^r.$$

When the smaller shore has size  $r$ , equality holds in every normalized inequality. Thus the normalization cannot be universally strengthened beyond the binomial one at the stated shore sizes.

If  $H = K_{m,n}$ , every equal-size pair is matchable, so

$$p_H(t) = \sum_{k=0}^{\min(m,n)} \binom{m}{k} \binom{n}{k} t^k. \quad (70.3)$$

For an arbitrary  $H$ ,  $0 \leq a_k(H) \leq \binom{m}{k} \binom{n}{k}$ . The upper bound is a count of possible pairs, not of edge matchings.

## 70.2 The Lorentzian step made explicit (source 07)

We use the following established facts from [18, Theorems 2.10, 2.30, 3.10]. Every matroid basis polynomial is Lorentzian. Nonnegative linear substitutions and nonnegative directional derivatives preserve this property. A nonzero homogeneous Lorentzian polynomial, and each of its nonzero partial derivatives, is log-concave on the positive orthant. For a quadratic polynomial, the relevant Hessian has at most one positive eigenvalue. Röhrle–Ulirsch [17, Theorem A] already apply this mechanism to regular-minor counts. We give the graph specialization explicitly, including the normalization constant.

*Proof of Theorem 67.3, a second route to Theorem 70.1 (source 07).* Delete isolated vertices and orient the shores so that  $|X| = d \leq |Y|$ . They do not change  $p_H^{\alpha,\beta}$ . By Proposition 69.1 and the imported matroid theorem, equation (69.6) is a Lorentzian polynomial

$$F(s, t) = \sum_{j=0}^d c_j s^{d-j} t^j.$$

Fix  $1 \leq k < d$ . The derivative  $\partial_s^{d-k-1} \partial_t^{k-1} F$  is  $As^2 + Bst + Ct^2$ , where

$$\begin{aligned} A &= \frac{1}{2} c_{k-1} (d-k+1)! (k-1)!, \\ B &= c_k (d-k)! k!, \\ C &= \frac{1}{2} c_{k+1} (d-k-1)! (k+1)!. \end{aligned}$$

Its Hessian has at most one positive eigenvalue, so  $B^2 \geq 4AC$ . Dividing factorials gives

$$c_k^2 \geq \frac{(k+1)(d-k+1)}{k(d-k)} c_{k-1} c_{k+1}. \quad (70.4)$$

This is exactly log-concavity of  $c_k / \binom{d}{k}$ . A maximum matching supplies a support of each size  $0, \dots, r$  by taking submatchings; all its weights are positive. Hence there are no internal zeros. Since the multiplier in equation (70.4) is greater than one, ordinary log-concavity is strict whenever both neighboring coefficients are positive. The usual decreasing-ratio criterion then gives unimodality.  $\square$

## 70.3 Bounds from the first and last coefficients

**Proposition 70.3.** *For a positive vertex-weighted support polynomial  $p(t) = \sum c_k t^k$ , with  $c_0 = 1$  and active-shore parameter  $d \geq 1$ ,*

$$c_k \leq \binom{d}{k} (c_1/d)^k, \quad p(z) \leq (1 + c_1 z/d)^d \quad (z \geq 0). \quad (70.5)$$

*If  $c_d > 0$ , then also*

$$c_k \geq \binom{d}{k} c_d^{k/d}, \quad p(z) \geq (1 + c_d^{1/d} z)^d. \quad (70.6)$$

*Proof.* Set  $b_k = c_k / \binom{d}{k}$ . On the positive interval, the ratios  $b_k / b_{k-1}$  are nonincreasing, with  $b_0 = 1$  and  $b_1 = c_1/d$ . Their product gives the upper bound, including any trailing zeros. When  $b_d > 0$ , concavity of  $\log b_k$  puts it above the chord between  $(0, 0)$  and  $(d, \log c_d)$ , which gives the lower bound. Summing the coefficient bounds proves the polynomial inequalities.  $\square$



For unweighted graphs,  $c_1 = |E|$ . For a preorder,  $c_d = 1$  and  $c_1 = |\tau|$ . Consequently

$$(1 + z)^n \leq h_\tau(z) \leq (1 + |\tau|z/n)^n.$$

The right bound is often weaker than bounds using the entire relation; it is useful because it uses only one coefficient. We do not call it an optimal fixed-size preorder bound. For graphs with  $d$  disjoint equal-degree stars, equation (70.5) is attained exactly.

#### 70.4 Why the order is the smaller active shore

The theorem permits removal of isolated vertices before normalization. It does not silently replace  $d$  by the maximum matching size  $r$ . That would be a stronger assertion when  $r < d$ ; no such strengthening is proved here. Nor may one divide by both shore binomial coefficients. The latter plausible variant is false even for forests; see Example 70.4.

The coefficient bound is sharp uniformly in  $d$  and  $k$ : a graph consisting of  $d$  independent edges has  $p_H(t) = (1 + t)^d$ . For preorders, the same example is the antichain. Sharpness concerns the displayed universal constant, not the classification of equality at an individual index in an arbitrary graph.

#### 70.5 Two-sided normalization fails even for a forest

**Example 70.4.** Take three suppliers with neighborhoods

$$N(x_1) = \{y_1, y_2, y_4\}, \quad N(x_2) = \{y_3\}, \quad N(x_3) = \{y_1\}.$$

This graph is a forest, and

$$p_H(t) = 1 + 5t + 6t^2 + 2t^3 = (1 + t)(1 + 4t + 2t^2).$$

Its zeros are real, but the sequence obtained by division by  $\binom{3}{k} \binom{4}{k}$  is

$$1, \quad \frac{5}{12}, \quad \frac{1}{3}, \quad \frac{1}{2}.$$

At index two,  $1/9 < 5/24$ . Thus real-rootedness, let alone our theorem, does not justify simultaneous division by both shore binomial coefficients.

#### 70.6 Integer capacities and dilations

Let each donor  $x$  have capacity  $b_x \in \mathbb{Z}_{\geq 0}$  and replace the right-hand side of (67.1) by  $\sum_{x \in N_H(S)} b_x$ . Put  $B = \sum_x b_x$ . Construct  $H^{(b)}$  by replacing  $x$  with  $b_x$  distinct copies of the same neighborhood.

**Corollary 70.5** (Capacity extension). *The capacity support polynomial is  $F_{H^{(b)}}$ . Its ordinary coefficients are ultra-log-concave with normalization  $\binom{\min(B, n)}{k}$ , including arbitrary positive receiver external fields. Its multivariate polynomial is completely log-concave, with the Lorentzian completion of degree  $B$ .*

*Proof.* The neighborhood size in the cloned graph is exactly the capacity sum in the original inequalities. Thus the feasible integer vectors, not merely their counts, coincide. Apply Theorems 69.3 and 70.1 to the cloned graph.  $\square$

The distinction between actual degree, receiver count, and homogenizing degree matters here. A capacity of one million is not a single unit donor. Algorithmic consequences of cloning are polynomial in  $B$ , hence only *pseudopolynomial* when capacities are encoded in binary.

## 71 Consequences for preorders and gamma coefficients

### 71.1 The direct reduction uses the full relation

Let  $H_\tau$  have a donor copy and a receiver copy of  $[n]$ , with an edge  $i \rightarrow j$  exactly when  $i \leq_\tau j$ . Diagonal edges are included.

**Lemma 71.1.**  $\mathcal{Q}_\tau \cap \mathbb{Z}^n = \mathcal{D}_{H_\tau}$ , and therefore  $h_\tau(t) = F_{H_\tau}(t, \dots, t)$ .

*Proof.* The predecessor set of  $S$  is  $\downarrow S$ , which is a lower ideal. The ideal inequalities and nonnegativity give  $x(S) \leq x(\downarrow S) \leq |\downarrow S| = |N_{H_\tau}(S)|$ . Conversely, if  $I$  is a lower ideal, reflexivity and downward closure give  $N_{H_\tau}(I) = I$ . The neighborhood inequality for  $I$  is exactly the required ideal inequality. The argument works for real nonnegative vectors as well as for integral ones.  $\square$

Using only the Hasse edges would invalidate this identification. Likewise, checking only principal ideals is not sufficient in general. Neither shortcut is used here.

Corollary 67.2 follows immediately from this lemma and Theorem 70.1. Positivity follows because  $\{0, 1\}^n$  is contained in  $\mathcal{Q}_\tau$ . We also obtain, for all  $k$ ,

$$\binom{n}{k} \leq h_k \leq \binom{n}{k}^2. \quad (71.1)$$

The first bound counts binary vectors; the second counts all possible matchable support pairs in the doubled graph. [Merge note: both bounds are also in Part IV's Theorem 55.6.]

**Remark 71.2** (Source 07's form of the reduction). Source 07 proves Lemma 71.1 by the same ideal-to-neighbourhood argument and records the following facts, printed as delivered.

A preorder is a reflexive, transitive relation, not necessarily antisymmetric. An ideal  $A \subseteq [n]$  satisfies  $j \in A, i \leq_\tau j \Rightarrow i \in A$ . Following Athanasiadis and Chapoton [1], define

$$\mathcal{Q}_\tau = \{c \in \mathbb{R}_{\geq 0}^n : c(A) \leq |A| \text{ for every ideal } A\}, \quad h_\tau(t) = \sum_{c \in \mathcal{Q}_\tau \cap \mathbb{Z}^n} t^{|\text{supp}(c)|}. \quad (71.2)$$

This support polynomial is *not, by definition, the Ehrhart numerator of  $\mathcal{Q}_\tau$* . The augmented root polytope is a different polytope.

Let  $G_\tau$  have suppliers  $x_1, \dots, x_n$  and receivers  $y_1, \dots, y_n$ , with  $x_i y_j$  an edge exactly when  $i \leq_\tau j$ . For  $S \subseteq [n]$ , its supplier neighborhood is the ideal  $\downarrow S$ . If  $c \in \mathcal{Q}_\tau$ , then  $c(S) \leq c(\downarrow S) \leq |\downarrow S|$ . Conversely, the Hall inequality for an ideal  $A$  is exactly  $c(A) \leq |A|$ , because  $N(A) = A$ . Therefore

$$\mathcal{Q}_\tau \cap \mathbb{Z}^n = \mathcal{D}_{G_\tau}, \quad h_\tau(t) = p_{G_\tau}(t). \quad (71.3)$$

The first equality is elementary; the second uses Proposition 68.3. The diagonal matching is perfect. Thus  $h_0 = h_n = 1$ ,  $h_k > 0$  for all  $0 \leq k \leq n$ , and  $h_1 = |\tau|$ , where  $|\tau|$  counts ordered related pairs, including all  $n$  reflexive pairs.

**Corollary 71.3** (Every positive integral dilation). *For an integer  $q \geq 1$ , let*

$$h_{\tau,q}(t) = \sum_{x \in q\mathcal{Q}_\tau \cap \mathbb{Z}^n} t^{|\text{supp}(x)|}.$$

*Its coefficients are ultra-log-concave with normalization  $\binom{n}{k}$ , also under arbitrary positive support weights. Its multivariate version is completely log-concave.*

*Proof.* The scaled inequalities are  $x(I) \leq q|I|$ . The proof of Lemma 71.1 identifies these with capacity  $q$  on each of the  $n$  donors. In Corollary 70.5,  $B = qn$ , so  $\min(B, n) = n$ .  $\square$

Palindromicity is special to the original normalization and is not asserted for  $q > 1$ . For instance, one isolated label has support polynomial  $1 + qt$  in the  $q$ th dilation.

## 71.2 Palindromicity by an elementary support-pair involution

The following known preorder facts are included to specify how our stronger inequality fits the earlier theory, not as additional novelty claims.

[Merge note: Lemma 71.4 and the involution below re-prove Part IV's ideal Hall criterion (Lemma 51.1) and complement-duality (Theorem 55.2); the palindromicity it gives is part of Part IV's Theorem 48.3 for preorders. Source 07's Section 72 extends the palindromicity classification to every bipartite graph.]

**Lemma 71.4** (Ideal Hall criterion). *For  $A, B \subseteq [n]$  of equal size, there is a bijection  $f : A \rightarrow B$  satisfying  $a \leq_\tau f(a)$  exactly when*

$$|A \cap I| \geq |B \cap I| \quad (I \in \mathcal{J}(\tau)). \quad (71.4)$$

*Proof.* Necessity follows because every predecessor matched to a point of an ideal lies in that ideal. Conversely, for  $T \subseteq B$ , use the ideal  $\downarrow T$ . Its inequality gives  $|A \cap \downarrow T| \geq |B \cap \downarrow T| \geq |T|$ , which is Hall's condition for a matching of  $B$  into  $A$ .  $\square$

The transformation

$$(A, B) \mapsto ([n] \setminus B, [n] \setminus A) \quad (71.5)$$

preserves (71.4), exchanges size  $k$  with size  $n - k$ , and is its own inverse. By Proposition 68.3, it proves  $h_k = h_{n-k}$ . Interchanging the two supports also proves invariance under reversal of the preorder.

**Proposition 71.5** (Equality throughout the normalized inequalities; sources 06 and 09). *For  $n \geq 2$ , equality holds in all  $n - 1$  inequalities in (67.5) if and only if  $\tau$  is the antichain preorder.*

*Proof.* Put  $b_k = h_k / \binom{n}{k} > 0$ . Equality throughout forces consecutive ratios  $b_k / b_{k-1}$  to be constant. Since  $h_0 = h_n = 1$ , all  $b_k$  equal one. In particular  $h_1 = n$ . But the support fibre on a single receiver  $j$  contains exactly the values  $1, \dots, \downarrow j$ , so  $h_1 = \sum_j \downarrow j$ . Reflexivity contributes  $n$ , and any non-diagonal relation contributes a further term. Hence  $h_1 = n$  precisely for the antichain. The converse follows from  $h = (1 + t)^n$ .  $\square$

This classifies simultaneous equality, not equality in one specified inequality for a general graph or a weighted model.

[Merge note: source 09 proves the same equality clause, as part of its Theorem 4.2, by the same argument ( $q_k = h_k / \binom{n}{k}$ ) with constant ratios and  $h_1$  counting related ordered pairs). Source 07's Theorem 74.8 classifies equality in the variance bound for weighted graphs, which contains the simultaneous case for preorders (Corollary 74.9).]

## 71.3 The height-two gamma question has a uniform answer

Write the palindromic support polynomial as

$$h_\tau(t) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_k(\tau) t^k (1+t)^{n-2k}, \quad \Gamma_\tau(u) = \sum_k \gamma_k(\tau) u^k. \quad (71.6)$$

Part IV already establishes the disjoint transport-support interpretation of these coefficients (Theorem 48.2). In height two, its specialization is particularly simple.

Let  $P_H$  be the height-two poset on  $X \sqcup Y$  whose strict relations are  $x < y$  for the edges  $xy$  of  $H$ . Then

$$h_{P_H}(t) = (1+t)^{m+n} p_H\left(\frac{t}{(1+t)^2}\right), \quad \Gamma_{P_H}(u) = p_H(u). \quad (71.7)$$

For completeness, the support-pair proof takes any compatible pair in the doubled preorder graph and cancels its common intersection. Cancellation preserves all the differences in (71.4); conversely one can restore the intersection by diagonal edges. A disjoint nonempty pair in height two has its sources in  $X$  and receivers in  $Y$ . A pair of size  $k$  leaves  $m+n-2k$  labels that may be placed independently in the common intersection. Summing  $t^{k+|W|}$  over that intersection  $W$  gives  $t^k(1+t)^{m+n-2k}$ , proving (71.7).

**Corollary 71.6** (All height-two gamma sequences). *For every height-two poset  $P_H$ , the sequence  $\gamma_k(P_H)/\binom{r}{k}$ ,  $r = \min(m, n)$ , is log-concave with no internal zeros. Thus every such gamma polynomial has log-concave, unimodal coefficients, even when it is not real-rooted.*

This answers Part IV's "Coefficient shape beyond gamma-positivity" question (Research question 24) for all height-two posets. It does not answer it for arbitrary preorders or even for all posets of height three. In particular, applying (67.5) to  $h_\tau$  is not a proof of a corresponding inequality for  $\Gamma_\tau$ .

[Merge note: source 07 states the height-two identity as its equation (76.3) and remarks that it identifies the height-two  $\gamma$ -polynomial with  $p_H$ , but does not state Corollary 71.6; source 09 remarks the same identity.]

## 72 Palindromic cores: an exact structural classification

All polynomials in this section are unweighted. Palindromicity refers to actual degree:  $p(t) = t^r p(t^{-1})$ , where  $r = \deg p$ . Arbitrary external fields need not preserve it.

### 72.1 A maximum-support factorization

**Proposition 72.1** (Top coefficient and a defect bound). *Let the nonisolated shores have sizes  $a, b$ , and let  $r = v(H)$ . Let  $\mathcal{L}$  be the collection of left  $r$ -sets which occur in a maximum matching, and let  $\mathcal{R}$  be the analogous right collection. Then*

$$\mathcal{M}_r(H) = \mathcal{L} \times \mathcal{R}, \quad p_r = |\mathcal{L}| |\mathcal{R}| \geq (a-r+1)(b-r+1). \quad (72.1)$$

*In particular,  $p_r = 1$  precisely when  $a = b = r$ .*

*Proof.* The generic matrix  $A_H$  has rank  $r$  by equation (69.2). The row bases are exactly  $\mathcal{L}$  and the column bases exactly  $\mathcal{R}$ . If  $I$  is a row basis, restriction to coordinates  $I$  is an isomorphism from the column space to the coordinate space of dimension  $r$ : a vector vanishing on  $I$  vanishes on all rows because the rows in  $I$  span. Applying this isomorphism to any column basis  $J$  shows that  $A_H[I, J]$  is invertible. This proves the Cartesian-product statement.

Fix one row basis. Every other active row is nonzero and can replace at least one of its members, yielding a different basis containing that row. Hence there are at least  $1 + (a-r)$  row bases. The column argument is identical. If  $p_r = 1$ , the bound forces  $a = b = r$ . Conversely, when  $a = b = r$ , the full pair of shores is the sole support of size  $r$ .  $\square$

There is also a graph-only proof of the qualitative conclusion. Every nonisolated vertex is covered by some maximum matching: if a maximum matching leaves it uncovered, swap in an incident edge, removing the edge currently covering its neighbor. A unique maximum support must therefore contain every nonisolated vertex. The generic-matrix proof additionally supplies the product formula and the quantitative lower bound.

## 72.2 The balanced reachability calculation

Suppose now that the active core has  $r$  vertices on each shore and has a perfect matching. Label the matched pairs  $(x_i, y_i)$ , and define  $iRj$  when  $x_i y_j$  is an edge. The relation  $R$  is reflexive. Let  $\overline{R}$  be its reflexive transitive closure.

[Merge note: Lemmas 72.2 and 72.3 re-prove Part IV's endpoint theorem (Theorem 48.3) and overlap cancellation (Proposition 51.4) for a balanced core; source 07 credits this. The new content of this section is Proposition 72.1 and the extension of the classification to all bipartite graphs, including unequal shores and deficient matchings.]

**Lemma 72.2** (First and penultimate coefficients). *For  $r \geq 1$ ,*

$$p_1 = |R|, \quad p_{r-1} = |\overline{R}|. \quad (72.2)$$

*Thus  $p_{r-1} - p_1$  is exactly the number of missing transitive pairs.*

*Proof.* The first equality counts edges. A support of size  $r - 1$  deletes one supplier  $x_a$  and one receiver  $y_b$ . If the remaining graph has a perfect matching, its directed assignment on  $[r]$  has one vertex without an outgoing edge ( $a$ ) and one without an incoming edge ( $b$ ). For  $a \neq b$ , following its edges from  $b$  produces a directed path ending at  $a$ : it cannot enter a cycle, since that would give a vertex two incoming edges. Therefore  $b \overline{R} a$ . For  $a = b$ , use the reflexive path.

Conversely, choose a simple directed path from  $b$  to  $a$ . Match along this path and use diagonal edges at all vertices off the path. This covers every supplier except  $x_a$  and every receiver except  $y_b$ . Thus deletion pairs are in bijection with ordered reachability pairs. The assertion remains valid when  $r = 1$ ; then both sides of the second equality equal one.  $\square$

**Lemma 72.3** (Transitive complement cancellation). *Let  $R$  be reflexive and transitive on  $[r]$ . A pair  $(I, J)$  of equal-size subsets is matchable exactly when  $(I \setminus J, J \setminus I)$  is matchable. Moreover,*

$$(I, J) \mapsto ([r] \setminus J, [r] \setminus I) \quad (72.3)$$

*is an involution on matchable pairs, reversing their size from  $k$  to  $r - k$ .*

*Proof.* Represent a matching  $I \rightarrow J$  as a directed graph on  $[r]$ . Each vertex has indegree and outdegree at most one, so its nontrivial components are directed paths or cycles. A path starts in  $I \setminus J$  and ends in  $J \setminus I$ . Transitivity shortcuts each path to a permitted edge from its start to its end. The cycles require no surviving edge. This proves the forward cancellation. Conversely, a matching between the differences extends by diagonal edges on  $I \cap J$ .

For the complement pair, use the shortcut edges just constructed and add identity edges at vertices outside  $I \cup J$ . Its source is  $[r] \setminus J$  and its target  $[r] \setminus I$ . Applying the same set transformation twice returns  $(I, J)$ , proving the involution.  $\square$

*Proof of Theorem 67.4.* If  $p_H$  is palindromic then  $p_r = p_0 = 1$ , so Proposition 72.1 forces a balanced core with a perfect matching. By Lemma 72.2, palindromicity also gives  $|R| = |\overline{R}|$ . Since  $R \subseteq \overline{R}$ , the relation is transitive. This proves (i) $\Rightarrow$ (ii), including the edgeless case separately.

Assume (ii). For each matchable pair  $(I, J)$ , write  $U = I \setminus J$ ,  $V = J \setminus I$ , and  $C = I \cap J$ . By Lemma 72.3,  $U, V$  are disjoint and matchable, and  $C$  may be *any* subset of  $[r] \setminus (U \cup V)$ . Consequently

$$p_H(t) = \sum_{k=0}^{\lfloor r/2 \rfloor} \gamma_k t^k (1+t)^{r-2k}, \quad \gamma_k = \#\{(U, V) : U \cap V = \emptyset, |U| = |V| = k, U \rightarrow_R V\}. \quad (72.4)$$

This proves (ii) $\Rightarrow$ (iii), with the already-known transport-pair interpretation from Part IV (Theorem 48.2). Each basis polynomial  $t^k(1+t)^{r-2k}$  is palindromic of degree  $r$ , proving (iii) $\Rightarrow$ (i). Finally, starting with any perfect matching in the same graph, the necessity argument applies again and proves transitivity of its associated relation.  $\square$

### 72.3 Recognition without enumerating the polynomial

Delete isolated vertices. If the remaining shore sizes differ, return “not palindromic.” Otherwise find a maximum matching. If it is not perfect, return “not palindromic.” Use a perfect matching to label both shores and test transitivity of  $R$ , for example by computing its Boolean transitive closure. Return “palindromic” exactly when the closure adds no pair. Matching and closure are polynomial-time operations; a straightforward closure takes  $O(r^3)$  Boolean operations. In particular this is not an algorithm which first computes exponentially many coefficients and then compares them.

This proves the palindromicity component of the bipartite support-enumerator problem as well, using Proposition 68.3. For a preorder, the criterion always holds. Combining symmetry with strict log-concavity yields strict increase toward the middle, except for the unavoidable two equal middle coefficients when the degree is odd.

## 73 An intersection-theoretic realization

The general geometric statement below is a special case of the realizable bimatroid theory in [17, Theorem E]. We include the construction to fix the factorials and explain what it does *not* imply about flag polytopes.

**Proposition 73.1** (Normalized intersection realization). *For a graph with  $m$  suppliers, there is a projective variety  $Z_H$  of dimension  $m$  and nef divisor classes  $L_v$ , indexed by  $X \sqcup Y$ , such that*

$$\mathcal{B}_H(\mathbf{w}) = \frac{1}{m!} \int_{Z_H} \left( \sum_v w_v L_v \right)^m. \quad (73.1)$$

In particular, with  $U = \sum_{x \in X} L_x$  and  $V = \sum_{y \in Y} L_y$ ,

$$\sum_k p_k s^{m-k} t^k = \frac{1}{m!} \int_{Z_H} (sU + tV)^m, \quad \frac{p_k}{\binom{m}{k}} = \frac{1}{m!} \int_{Z_H} U^{m-k} V^k. \quad (73.2)$$

The same construction applies to  $s^n h_\tau(t/s)$ .

*Proof.* There is a complex specialization of the edge variables for which exactly the structurally nonzero minors stay nonzero: their finite product is a nonzero polynomial, so its nonvanishing set is nonempty. Let  $C = [I_m \mid A_H]$  be such a specialization. Its columns give linear coordinate functions on  $\mathbb{C}^m$ . Take the closure of the resulting affine linear space in  $(\mathbb{P}^1)^{X \sqcup Y}$ ; normalizing, or passing to a smooth projective resolution, does not change the intersection calculation below.



Let  $L_v$  be the pullback of the point/hyperplane class of the  $v$ th projective line. These classes are nef and satisfy  $L_v^2 = 0$ .

For a set  $B$  of  $m$  coordinates, the associated coordinate projection has generic degree one if those columns form a basis, because its restriction to the affine linear space is an invertible linear map. If the columns are dependent, its image has dimension less than  $m$ , so its top intersection is zero. The boundary of the affine space has smaller dimension and does not contribute over a general target point. Thus

$$\int_{Z_H} \prod_{v \in B} L_v = \begin{cases} 1, & B \text{ a basis of } M_H, \\ 0, & \text{otherwise.} \end{cases}$$

Expanding the  $m$ th power, square terms vanish and every basis monomial occurs  $m!$  times. This proves equation (73.1); grouping the two shores gives equation (73.2).  $\square$

This construction is compatible with the Hodge-theoretic source of ultra-log-concavity. It is an intersection realization, not a construction of a simplicial flag polytope with the prescribed  $h$ -vector. Nor does it require the repository's proposed equality with the Ehrhart numerator of its lower-dimensional directed root polytope. The normalizing factor  $1/m!$  is explicit to avoid different conventions for "volume polynomial."

## 74 Conditional laws, covariance, and concentration

### 74.1 Conditioning on present and absent coordinates

For a positive external field  $z \in \mathbb{R}_{>0}^Y$ , give  $c \in \mathcal{D}_H$  probability proportional to  $z_{\text{supp}(c)}$ . Equivalently, its support law is

$$\mathbb{P}_z(S) = \frac{[z_S] F_H z_S}{F_H(z)}. \quad (74.1)$$

Fix disjoint  $T, A \subseteq Y$  and condition on  $T \subseteq S$  and  $S \cap A = \emptyset$ , assuming this event has positive probability. Once the forced variables are removed, its generating polynomial is

$$G_{T,A}(z) = \left( \prod_{y \in T} \partial_{z_y} \right) F_H(z) \Big|_{z_a=0 \ (a \in A)}. \quad (74.2)$$

The external-field factor on  $T$  is constant and disappears upon normalization.

**Proposition 74.1** (Hereditary complete log-concavity). *The nonzero polynomial  $G_{T,A}$  is completely log-concave. It is an affine slice of the basis polynomial of a minor of  $M_H$  of rank  $m - |T|$ . Consequently its extra-support-size coefficients, under positive external fields, are ultra-log-concave with normalization  $\binom{m-|T|}{k}$ .*

*Proof.* Differentiating a multiaffine basis polynomial in the variables of an independent set  $T$  gives the basis polynomial of the contraction  $M_H/T$ . Setting the variables of  $A$  to zero then retains exactly the bases avoiding  $A$ . Since the conditioning event is nonempty, deleting  $A$  does not lower the rank of that contracted matroid. Its rank is  $m - |T|$ . Setting the surviving private variables to one gives (74.2). Theorem 69.3's argument and the bivariate Lorentzian criterion apply without alteration. A rank-zero case is a nonzero constant.  $\square$

The specified normalization is the rank of the displayed minor. No unjustified replacement by the number of unforced receiver coordinates is made.

## 74.2 A homogeneous concavity inequality

We record the standard homogeneous concavity equivalence [18, Proposition 2.33], with a proof fixing the covariance constant.

**Lemma 74.2.** *If  $B$  is a positive log-concave homogeneous polynomial of degree  $d \geq 1$  on the positive orthant, then  $B^{1/d}$  is concave there. Equivalently,*

$$\nabla^2 \log B \leq -\frac{1}{d}(\nabla \log B)(\nabla \log B)^\top. \quad (74.3)$$

*The inequality remains valid on any affine slice, with the restricted gradient and Hessian.*

*Proof.* For  $d = 1$  the polynomial is linear and direct differentiation proves the claim. For general  $d$ , let  $x$  be a positive point and write any direction  $v$  as  $v = w + \alpha x$ , where  $\alpha = (\nabla B \cdot v)/(dB)$  and  $\nabla B \cdot w = 0$ . Log-concavity implies  $w^\top (\nabla^2 B) w \leq 0$ . Euler's identities give  $\nabla^2 B x = (d-1)\nabla B$  and  $x^\top \nabla^2 B x = d(d-1)B$ . The cross term vanishes, and hence

$$v^\top \nabla^2 B v \leq \frac{d-1}{dB}(\nabla B \cdot v)^2.$$

Subtract the gradient-square term in the Hessian of  $\log B$  to obtain (74.3). The Hessian of  $B^{1/d}$  has the same required sign. Restriction amounts to applying the inequality only to directions tangent to the affine slice.  $\square$

**Theorem 74.3** (External-field covariance bound). *For  $m \geq 1$ , let  $I_y = \mathbf{1}_{\{y \in S\}}$  under (74.1), let  $p_y = \mathbb{E}I_y$ , and write  $p = (p_y)_{y \in Y}$ . Then*

$$\boxed{\text{Cov}(I) \leq \text{diag}(p) - \frac{1}{m}pp^\top.} \quad (74.4)$$

*In particular, for every real vector  $b$ ,*

$$\text{Var}\left(\sum_y b_y I_y\right) \leq \sum_y p_y b_y^2 - \frac{1}{m} \left(\sum_y p_y b_y\right)^2. \quad (74.5)$$

*The same statement holds for the remaining indicators after the conditioning above, with  $m$  replaced by  $m - |T|$  when that rank is positive.*

*Proof.* Apply Lemma 74.2 to  $\mathcal{B}_H$  of degree  $m$  and restrict its private variables to one. Let  $D = \text{diag}(z_y)$ . Logarithmic differentiation of the probability generating function gives

$$D \nabla \log F_H = p, \quad D(\nabla^2 \log F_H)D = \text{Cov}(I) - \text{diag}(p). \quad (74.6)$$

For distinct indices this is the usual covariance identity. On the diagonal, multiaffineness gives  $z_y^2 \partial_{yy} \log F_H = -p_y^2$ , which is exactly  $\text{Var}(I_y) - p_y$ . Conjugate (74.3) by  $D$  and rearrange. The conditioned claim uses the minor polynomial from Proposition 74.1.  $\square$

This is a positive-semidefinite matrix comparison, not a claim that each off-diagonal covariance is nonpositive. Complete log-concavity is weaker than real stability and does not justify importing every negative dependence property of determinantal models.

### 74.3 Binomial convex-order domination (source 07)

Let  $p(t) = \sum_{k=0}^d c_k t^k$  be a positive vertex-weighted support polynomial, where  $d$  is the smaller active shore. For fugacity  $z > 0$ , define

$$\mathbb{P}_z(L = k) = \frac{c_k z^k}{p(z)}, \quad \mu = \mathbb{E}_z L = \frac{z p'(z)}{p(z)}. \quad (74.7)$$

The random variable counts matched pairs, or equivalently half the number of covered vertices. Under the demand interpretation it counts positive coordinates when the activities are attached to receiver support. If both shores carry activities, the graph measure remains well defined, but no unspecified weight on individual demand vectors is being asserted.

### 74.4 A finite convex-order lemma

**Lemma 74.4** (Binomial domination from ultra-log-concavity). *Suppose a distribution  $\pi$  on  $\{0, \dots, d\}$  is ultra-log-concave of order  $d$  and has mean  $\mu$ . If  $B \sim \text{Bin}(d, \mu/d)$ , then*

$$\mathbb{E}_\pi \phi(L) \leq \mathbb{E} \phi(B) \quad (74.8)$$

for every convex function  $\phi$  on  $[0, d]$ . Equality for  $\phi(k) = k^2$  holds if and only if the two distributions are identical.

*Proof.* The endpoint means  $0, d$  are deterministic. Otherwise let  $q_k$  be the binomial mass function, which is positive everywhere. The ratios  $\pi_k/q_k$  form a log-concave sequence: division by  $q_k$  differs from division by  $\binom{d}{k}$  only by a positive geometric factor and a constant. Therefore the set where  $\pi_k - q_k$  is positive is an interval, allowing zeros between its endpoints.

If the two laws differ, their equal mass and equal mean force negative differences on both sides of the positive interval. Otherwise one distribution would have strictly larger mean. Choose real cutpoints  $u < v$ , between the sign regions, so that differences are nonnegative inside  $(u, v)$  and nonpositive outside it. Let  $\ell$  be the affine line joining  $(u, \phi(u))$  to  $(v, \phi(v))$ . Convexity gives  $\phi \leq \ell$  between the cutpoints and  $\phi \geq \ell$  outside. Hence

$$\sum_k (\pi_k - q_k)(\phi(k) - \ell(k)) \leq 0.$$

The affine part sums to zero by equal mass and mean, proving equation (74.8). For  $\phi(k) = k^2$ , choose the cutpoints between integer sign regions. The quadratic-minus-secant is strictly negative inside and strictly positive outside, so some nonzero difference gives a strict inequality. The converse when the laws coincide is immediate.  $\square$

This lemma is a general consequence of ultra-log-concavity, not a new theorem specific to preorders. Its short proof is included because it exposes exactly the equality condition used below.

**Theorem 74.5** (All-fugacity binomial bounds). *For every bipartite support model with at least one edge, positive vertex activities, and every  $z > 0$ ,*

$$\text{Var}_z L \leq \mu(1 - \mu/d), \quad (74.9)$$

$$\mathbb{E}_z e^{\lambda(L-\mu)} \leq e^{-\lambda\mu}(1 - \mu/d + (\mu/d)e^\lambda)^d \quad (\lambda \in \mathbb{R}). \quad (74.10)$$

For  $\mu/d < q \leq 1$ ,

$$\mathbb{P}_z(L \geq dq) \leq \exp\{-d \text{KL}(q\|\mu/d)\}, \quad (74.11)$$

where  $\text{KL}(q\|p) = q \log(q/p) + (1-q) \log((1-q)/(1-p))$ . There is an analogous lower-tail inequality for  $0 \leq q < \mu/d$ .

*Proof.* Exponential tilting preserves ultra-log-concavity. Apply Lemma 74.4 to  $k^2$  and  $e^{\lambda k}$ . The usual exponential Markov bound, minimized at  $e^\lambda = q(1-p)/(p(1-q))$ , gives equation (74.11). Endpoint probabilities are interpreted by limits. An edgeless graph has  $L = 0$  and is treated separately.  $\square$

One can also derive the variance bound directly from the two-variable Hessian. For  $F(s, t) = s^d p(t/s)$ , nonpositivity of its Hessian determinant gives

$$dp(z)p''(z) \leq (d-1)p'(z)^2.$$

Substitution in  $\text{Var } L = \mu + z^2 p''(z)/p(z) - \mu^2$  recovers equation (74.9). The convex-order argument is stronger and is needed for the clean equality statement.

#### 74.5 The same bound at the actual weighted mean (source 06)

Let  $L = |S|$  in the positive weighted-pair distribution associated to (70.1), and let  $\mu = \mathbb{E}L$ . Let  $r = \min(m, n) > 0$ .

**Theorem 74.6** (Binomial moment bound; sources 06, 07 and 09). *For every real  $s$ ,*

$$\mathbb{E}e^{sK} \leq \left(1 - \frac{\mu}{r} + \frac{\mu}{r}e^s\right)^r, \quad \text{Var } L \leq \mu \left(1 - \frac{\mu}{r}\right). \quad (74.12)$$

*Thus the moment-generating function is dominated by that of a binomial random variable with  $r$  trials and the same mean.*

This is Theorem 74.5 for the weighted pair law at unit fugacity, and Lemma 74.4 proves it, with all convex test functions in place of  $e^{sL}$ . Source 06's own proof is a different, analytic route.

*Second proof (source 06).* Write  $A(t) = A_H(t; u, v)$ , absorbing the initial activity into  $v$  if necessary. Choose the smaller-shore lift in the proof of Theorem 70.1. Lemma 74.2 gives

$$AA'' \leq (1 - 1/r)(A')^2.$$

Under a scalar tilt  $t > 0$ , the mean is  $\mu(t) = tA'(t)/A(t)$ . Therefore

$$\text{Var}_t K = t\mu'(t) \leq \mu(t) - \frac{\mu(t)^2}{r}. \quad (74.13)$$

Put  $\psi(s) = \log(A(e^s)/A(1))$  and  $v(s) = \psi'(s)$ . Then  $v'(s) \leq v(s)(1 - v(s)/r)$ , and  $v(0) = \mu$ . The zero-mean case is constant; otherwise  $0 < v(s) < r$  at every finite  $s$  unless the law is deterministic, which can be handled directly. The derivative of  $\log(v/(r-v))$  is at most one. Hence for  $s \geq 0$ ,  $v(s)$  is at most the logistic solution with initial value  $\mu$ ; for  $s \leq 0$  it is at least that solution. Integrating from zero in the appropriate direction gives, for both signs of  $s$ ,

$$\psi(s) \leq r \log(1 - p + pe^s), \quad p = \mu/r.$$

Exponentiating proves the first assertion. Equation (74.13) at  $t = 1$  proves the second.  $\square$

For  $0 < p < 1$  and  $q \in [0, 1]$ , put

$$D(q||p) = q \log \frac{q}{p} + (1-q) \log \frac{1-q}{1-p},$$

with the continuous endpoint conventions. Chernoff's argument applied to (74.12) gives

$$\begin{aligned}\mathbb{P}(L \geq rq) &\leq e^{-rD(q\|p)} \quad (q \geq p), \\ \mathbb{P}(L \leq rq) &\leq e^{-rD(q\|p)} \quad (q \leq p).\end{aligned}\tag{74.14}$$

For a uniformly chosen lattice point of  $\mathcal{Q}_\tau$ , symmetry gives  $\mu = n/2$ , recovering  $\text{Var } L \leq n/4$  and  $\mathbb{P}(|L - n/2| \geq a) \leq 2e^{-2a^2/n}$ . Part IV already obtains these unweighted estimates by its gamma-binomial mixture (Theorem 58.1). The additional content here is the arbitrary external field, the actual weighted mean, and the multivariate conditional covariance bound.

**Remark 74.7** (The Hoeffding form for every ultra-log-concave law; source 09). Source 09 proves Lemma 74.4 and the bounds of Theorem 74.5 again, for any nonzero ultra-log-concave sequence of order  $d$  tilted by  $t > 0$ , by the same single-crossing and chord argument, and adds the two-sided bound

$$\mathbb{P}(|L - \mu| \geq s) \leq 2e^{-2s^2/d} \quad (s \geq 0).$$

Its proof, as delivered: the logarithm of the centered Bernoulli moment function has value and first derivative zero at  $\theta = 0$ , and its second derivative is the variance of a tilted Bernoulli variable, at most  $1/4$ . Integrating twice gives the upper bound  $\theta^2/8$ . The moment-generating-function bound (74.10) consequently implies  $\mathbb{E}e^{\theta(L-\mu)} \leq e^{d\theta^2/8}$ . Markov's inequality, optimized at  $\theta = 4s/d$ , bounds the upper tail by  $e^{-2s^2/d}$ ; using negative  $\theta$  bounds the lower tail. Add the two bounds. Source 09 adds: for  $h_{\tau,w}$  at  $t = 1$ , palindromicity gives  $\mu = n/2$ , so  $\text{Var}(L) \leq n/4$ ; the argument works for all positive transport activities (Part VII's Theorem 107.2), all size tilts and all bipartite support models, and it is a consequence of ultra-log-concavity, not a new general theorem about log-concave measures.

## 74.6 Equality is exactly an equal-strength star forest

Orient the active core so that  $|X| = d \leq |Y|$ . Define the weighted strength of a supplier by

$$s_x = \alpha_x \sum_{y \in N_H(x)} \beta_y.$$

**Theorem 74.8** (Sharp equality classification). *Assume  $H$  has at least one edge and all vertex activities are positive. The following are equivalent.*

- (i) *Equality holds in equation (74.9) for one  $z > 0$ .*
- (ii)  *$p_H^{\alpha,\beta}(t) = (1 + at)^d$  for some  $a > 0$ .*
- (iii) *The neighborhoods of the smaller-shore vertices are pairwise disjoint, and all strengths  $s_x$  are equal.*

*When these conditions hold, equality holds for every  $z > 0$ . In the unweighted case, the active graph is a disjoint union of  $d$  stars of the same degree, centered on the smaller shore.*

*Proof.* By the equality clause of Lemma 74.4, (i) makes the tilted coefficient law exactly binomial. Dividing its masses by  $z^k$  and using  $c_0 = 1$  gives  $c_k = \binom{d}{k} a^k$  for a positive  $a$ . Thus (i) implies (ii).

Suppose (ii). Then  $c_1 = da = \sum_x s_x$  and  $c_d = a^d > 0$ . The total weight of supports covering every supplier is at most the total weight of matchings covering every supplier, which is at most the weight of all independent choices of one neighboring receiver for each supplier. Hence

$$c_d \leq \sum_{\text{matchings covering } X} \text{wt}(M) \leq \prod_{x \in X} s_x \leq \left( \frac{1}{d} \sum_{x \in X} s_x \right)^d = a^d = c_d.\tag{74.15}$$

All inequalities are equalities. The arithmetic–geometric mean equality forces equal strengths. If two suppliers share a receiver, there is a positive-weight choice assigning that receiver to both, and arbitrary neighbors to the other suppliers. This is counted in the product but is not a matching, contradicting equality in the second inequality. Thus neighborhoods are disjoint.

Conversely, under (iii) every support chooses independently whether to use each star and, when it does, which receiver to use. Therefore  $p(t) = \prod_x (1 + s_x t) = (1 + at)^d$ . The tilted law is binomial, completing all implications. Positivity is essential for the strict collision argument; zero activities should first be handled by deleting their forbidden vertices.  $\square$

**Corollary 74.9** (Preorder support fluctuations). *For  $n \geq 1$ , choose a lattice point of  $\mathcal{Q}_\tau$  with probability proportional to  $z^{|\text{supp}(c)|}$ . Its support size obeys equations (74.9), (74.10) and (74.11) with  $d = n$ . Equality in the variance bound at one  $z > 0$  holds precisely when  $\tau$  is an antichain. In particular, for the uniform law,*

$$\mathbb{E}L = n/2, \quad \text{Var } L \leq n/4, \quad \mathbb{P}(|L - n/2| \geq s) \leq 2e^{-2s^2/n},$$

with strict variance inequality for every non-antichain preorder.

*Proof.* Use equation (71.3). In the balanced graph  $G_\tau$ , pairwise disjoint nonempty neighborhoods force exactly one receiver per supplier. Since all diagonal edges are present, these are precisely the diagonal edges; there are no nontrivial comparisons. Palindromicity gives the mean at  $z = 1$ . The tail follows from equation (74.11) and the elementary bound  $\text{KL}(1/2 + u || 1/2) \geq 2u^2$ , obtained by differentiating twice.  $\square$

Part IV already obtains the uniform mean, variance formula, and a Hoeffding-type bound from its centered-binomial mixture (Theorem 58.1). The additional content here is domination by the same-mean binomial for every positive fugacity, the graph-wide weighted theorem, and its exact equality classification.

## 75 Polynomial-time approximation and support sampling

We distinguish exact counting, approximate counting, sampling supports, and sampling actual integer vectors. They are not interchangeable tasks. The input graph is given explicitly. A positive rational weight has a standard binary encoding. An approximation scheme is fully polynomial if its running time is polynomial in the input bit length,  $\varepsilon^{-1}$ , and  $\log(\delta^{-1})$ , and its relative error is at most  $\varepsilon$  with probability at least  $1 - \delta$ .

### 75.1 A matching oracle and the basis walk

Independence in  $M_H$  is decided by a bipartite matching from the selected ground elements to the  $m$  slots. The presentation has  $m + n$  ground elements and  $m + |E|$  incidence edges. A basic augmenting-path algorithm already gives a polynomial-time oracle. A starting basis is  $X^\circ$ .

At a basis  $B$  of rank  $d$ , the down-up step first removes a uniformly chosen element  $e \in B$ . It then chooses a feasible extension  $B - e + f$  with probability proportional to its product weight. Since the weight of the retained set is common, the extension probabilities are proportional simply to the weight of  $f$ . There are at most  $m + n$  candidates, each testable by the matching oracle.

Theorem 1.1 of [19] gives a spectral gap at least  $1/d$  for this walk and the mixing estimate

$$t_{B_0}(\eta) \leq d \log \frac{1}{\eta \pi(B_0)}, \quad (75.1)$$



where  $\pi$  is the normalized weighted basis measure. Our product weights preserve the required log-concavity by a positive variable scaling. The same theorem applies after contractions and deletions.

For clarity about bit complexity, suppose there are  $N$  ground elements, rank  $d$ , and all positive element weights lie between  $2^{-b}$  and  $2^b$ . Every basis weight lies between  $2^{-db}$  and  $2^{db}$ , and there are at most  $2^N$  bases. Consequently

$$\pi(B_0) \geq 2^{-N-2db}. \quad (75.2)$$

Thus the logarithm in (75.1) is polynomially bounded in the input bit length. Exact rational extension probabilities are computable with polynomial bit length; no exponentially small probability has to be estimated directly just to run a step.

### 75.2 The chain in explicit form (source 07)

For  $S \subseteq X \sqcup Y$ , let  $S_X = S \cap X$  and  $S_Y = S \cap Y$ . The private elements  $S_X$  reserve their own slots. Thus

$$S \text{ is independent in } M_H \iff S_Y \text{ can be matched into } X \setminus S_X. \quad (75.3)$$

This is a bipartite maximum-matching query; no symbolic determinant is computed. For a size- $m$  set it is a basis query. The supplied implementation uses augmenting paths and exact integer bitmasks. It is a reference implementation, not a claim of optimal matching complexity.

### 75.3 The correct local chain

Give matroid elements positive activities

$$\lambda_x = \alpha_x^{-1} \quad (x \in X), \quad \lambda_y = z\beta_y \quad (y \in Y).$$

The product weight of  $B_{I,J}$  differs from  $\alpha_I \beta_J z^{|J|}$  by the constant  $\prod_x \alpha_x^{-1}$ . Starting at a basis  $B$ , perform one step:

1. Remove a uniformly chosen element  $b \in B$  and set  $S = B \setminus \{b\}$ .
2. Enumerate  $C(S) = \{e \notin S : S \cup \{e\} \text{ is a basis}\}$ .
3. Choose  $e \in C(S)$  with probability  $\lambda_e / \sum_{f \in C(S)} \lambda_f$ , and move to  $S \cup \{e\}$ .

The old element is an allowed completion, so the chain has self-loops. Detailed balance follows because for adjacent bases  $B = S + b$ ,  $B' = S + e$ ,

$$\pi(B)P(B, B') = \frac{\lambda_S \lambda_b \lambda_e}{Z_M m \sum_{f \in C(S)} \lambda_f} = \pi(B')P(B', B).$$

Basis exchange ensures irreducibility.

In support coordinates, an exchange can replace a supplier, replace a receiver, add a supplier–receiver pair, or remove such a pair. An added pair need not be an edge in a fixed previously chosen matching: global rematching may be needed. The chain acts on supports, not on chosen perfect matchings.

**Theorem 75.1** (Rapid mixing, by the general matroid theorem). *For a nonzero rank- $m$  model, this chain has spectral gap at least  $1/m$ . If  $B_0$  is its starting basis, the conservative bound*

$$t \geq m \log \frac{1}{\varepsilon \pi(B_0)} \quad (75.4)$$

is sufficient for total variation error at most  $\varepsilon$ . In the uniform case, starting at  $B_0 = X$ , it is enough to take

$$t \geq m \left( \log \binom{m+n}{m} + \log \frac{1}{\varepsilon} \right). \quad (75.5)$$

The shores can first be exchanged and isolated vertices removed to lower the rank used in this bound.

*Proof.* The positively rescaled basis polynomial is strongly log-concave by the Lorentzian theorem. The chain is exactly the down-up chain in [19, Theorem 1.1], which supplies the gap and equation (75.4). In the uniform case  $\pi(X) = 1/p_H(1)$  and the number of bases is at most  $\binom{m+n}{m}$ . The rank-zero model is deterministic.  $\square$

The bound is not advertised as the latest or sharpest possible mixing bound. Its advantage is that the exact chain, oracle, and imported theorem match without an unproved transfer. For rational positive activities of bit length  $L_{\text{bit}}$ , a crude starting-mass bound is

$$\log \pi(B_0)^{-1} \leq \log \binom{m+n}{m} + m \log(\lambda_{\max}/\lambda_{\min}),$$

which is polynomial in the input bit size. Conditional models use Proposition 69.2 and the same theorem on a matroid minor.

**Theorem 75.2** (Partition functions and support laws; sources 06 and 07). *For every explicit bipartite graph,  $F_H(z)$  at positive rational  $z$  has an FPRAS. So does  $A_H(1; u, v)$  at positive rational vertex weights. There is a polynomial-time sampler arbitrarily close in total variation to the support law of the corresponding weighted integer vectors when  $u = \mathbf{1}$ . In particular,  $|Q_\tau \cap \mathbb{Z}^n|$  has an FPRAS for every explicit finite preorder.*

*Proof.* The polynomial identities reduce each partition function to a positive known scalar times a weighted matroid basis partition function. Sampling bases using (75.1) and retaining  $B \cap Y$  gives the support law, because Proposition 69.1 supplies exactly the multiplicity in (68.3). Taking an image cannot increase total variation distance.

Here is the counting self-reduction, to make the algorithmic implication explicit. In any weighted rank- $d$  matroid on at most  $N$  elements, the sum of element inclusion probabilities is  $d$ . Some element therefore has probability at least  $1/N$ . Approximate all inclusion probabilities to additive error  $1/(4N)$ , using independent sufficiently mixed samples, and choose a largest empirical marginal. On the simultaneous success event, the chosen element has true inclusion probability at least  $1/(2N)$ . Estimate that probability  $p_e$  to relative accuracy  $\varepsilon/(10N)$  and contract  $e$ . For the partition functions,

$$Z(M) = \frac{w_e Z(M/e)}{p_e}. \quad (75.6)$$

Repeat until rank zero, where  $Z = 1$ . There are at most  $N$  stages. Allocate failure probability over all stages and estimate all marginals and chosen probabilities with standard bounded-variable concentration. The lower bound  $p_e \geq 1/(2N)$  makes all sample counts polynomial; the product of the relative errors in (75.6), including their reciprocals, is at most  $\varepsilon$  with the displayed conservative accuracy.

Every contracted set is independent, so its oracle can be implemented by adjoining that set before querying the original matching presentation. A greedy extension supplies a starting basis in each minor. The weight bound (75.2) remains valid. Fresh randomness and mixing accuracy small enough for a union bound complete the claimed FPRAS. Finally apply Lemma 71.1 for  $Q_\tau$ .  $\square$

The general basis mixing theorem and its use in approximate counting are established results [19]; the proof above spells out their concrete application rather than presenting a new universal matroid-counting algorithm.

Source 07 states the same result as its Corollary 8.2 (“Counting consequences”): given a bipartite graph and positive rational vertex activities and fugacity,  $p_H^{\alpha,\beta}(z)$  has an FPRAS, and consequently so do  $|\mathcal{D}_H|$ ,  $h_\tau(1)$  and  $h_\tau(z)$  for positive rational  $z$ , by the positive-activity matroid-base counting theorem [19] applied through (75.3) and (69.6), whose oracle calls are polynomial-time matching calls. It then makes the self-reduction concrete with explicit constants:

For clarity, the self-reduction underlying the imported algorithm can be made concrete. In a rank- $q$  conditional minor on at most  $N$  elements,  $\sum_e \pi(e \in B) = q$ . Therefore some element has marginal at least  $q/N \geq 1/N$ . Estimate all marginals by independent, sufficiently mixed samples, choose an element whose estimate is largest, and contract it. Repeat until the rank is zero. If  $p_i$  is the successive inclusion marginal and  $e_i$  the chosen element, then

$$Z_M = \prod_{i=1}^m \frac{\lambda_{e_i}}{p_i}. \quad (75.7)$$

The conditional rank-zero partition function is one.

Here is a deliberately conservative error allocation. Set  $\rho = \varepsilon/(4m)$  and  $\eta = \rho/(4N)$  for  $m > 0$ . At each stage use

$$T \geq \frac{1}{2\eta^2} \log \frac{4Nm}{\delta}$$

independent runs, each within  $\delta/(4mT)$  of its exact conditional law. Hoeffding’s inequality and a union bound ensure that all empirical marginals are within  $\eta$  of their true values, except with probability at most  $\delta$  after also accounting for sampling error. The selected element has true marginal at least  $1/N - 2\eta \geq 1/(2N)$ , and its relative estimation error is at most  $2N\eta \leq \rho$ . The product in equation (75.7) then has multiplicative error bounded by  $(1 \pm \rho)^{\mp m}$ , contained in  $[1 - \varepsilon, 1 + \varepsilon]$  for  $0 < \varepsilon < 1$  with these constants. This proves polynomial sample complexity; it is not a practical optimization.

The shipped program implements the oracle, exact enumeration, the rational weighted down-up sampler, and the finite verifier. It does *not* claim to be a production implementation of this full FPRAS wrapper. Sampling and counting individual coefficients at a prescribed internal index are also not silently included in Theorem 75.2.

## 75.4 Precisely what is sampled for a preorder

A sample from  $M_{G_\tau}$  gives a pair  $(I, J)$  with the desired support-model law. Its  $J$ -marginal equals the distribution of the positive-coordinate set of a correspondingly weighted lattice point, by Proposition 68.3. Its cardinality has exactly the  $h_\tau$  distribution. The algorithm does not output the positive *values* of that lattice point. A direct or efficiently invertible demand/support bijection would close this remaining algorithmic gap. [Added 30 September 2026, batch 64: Part IX closes it; see Corollary 146.2.]

## 75.5 A specified exact support fibre

**Corollary 75.3.** *For a specified  $S \subseteq Y$ , the number of points of  $\mathcal{D}_H$  with support exactly  $S$  is either detectably zero or has an FPRAS.*

*Proof.* Its count is the number of  $|S|$ -subsets  $U \subseteq X$  matchable to  $S$ . Use the transversal matroid on ground set  $X$  with slots  $S$ . If its rank is smaller than  $|S|$ , the count is zero, which a maximum matching detects. Otherwise the desired sets are exactly its bases. Apply the same basis counting algorithm.  $\square$

This is a count of integer vectors in a fibre. The proof does not produce a vector from a sampled opposite-shore basis.

## 75.6 Why coefficient approximation needs another argument

A partition-function approximation does not justify numerical interpolation of the coefficient vector. Interpolation can amplify relative errors severely. Nor should one simply restrict the basis walk to bases with exactly  $k$  receivers and assume its mixing theorem still applies: that slice need not be the bases of a matroid.

Instead we use log-concavity to tilt the full distribution until the requested coefficient has polynomially large probability.

**Lemma 75.4** (Mass at an index between substantial tails). *Let  $(p_0, \dots, p_d)$  be a probability sequence, log-concave with no internal zeros. If*

$$\sum_{i < k} p_i \geq \frac{1}{4}, \quad \sum_{i \geq k} p_i \geq \frac{1}{4},$$

*then  $p_k \geq 1/(4(d+1))$ .*

*Proof.* There is an  $i < k$  with  $p_i \geq 1/(4(d+1))$ , and a  $j \geq k$  with the same bound. Log-concavity and absence of internal zeros imply that every entry between  $i$  and  $j$  is at least the smaller endpoint entry: interpolate linearly in the concave sequence  $\log p_\ell$ . Apply this at  $k$ .  $\square$

**Theorem 75.5** (Individual coefficient FPRAS). *For an explicit bipartite graph  $H$  and an index  $k$ ,  $a_k(H)$  is either detectably zero or has an FPRAS. Consequently every ordinary coefficient  $[t^k]h_\tau(t)$  and every height-two gamma coefficient  $\gamma_k(P_H)$  has an FPRAS. The entire vector can be approximated to simultaneous relative accuracy in fully polynomial time.*

*Proof.* Set  $N = m + n$  and  $A(t) = p_H(t) = \sum_{j=0}^d a_j t^j$ , where  $d = v(H)$  is computable by maximum matching. Then  $a_0 = 1$ , every  $a_j$  for  $0 \leq j \leq d$  is a positive integer, and

$$\sum_{j=0}^d a_j \leq 2^N \tag{75.8}$$

by the basis correspondence. Indices outside this interval have zero coefficient, and  $a_0$  needs no approximation. Assume  $1 \leq k \leq d$ .

For an activity  $t > 0$ , define  $p_j(t) = a_j t^j / A(t)$  and  $q(t) = \mathbb{P}_t(L < k)$ . The sequence  $p_j(t)$  is log-concave. Differentiating with respect to  $\log t$  gives

$$\frac{d}{d \log t} q(t) = \text{Cov}_t(\mathbf{1}_{\{L < k\}}, L) \leq 0, \quad \left| \frac{d}{d \log t} q(t) \right| \leq d. \tag{75.9}$$

For the sign, values of  $L$  in the indicated event are smaller than values in its complement; condition on the event and its complement. The absolute bound follows since both conditional expectations lie in  $[0, d]$ .

Put  $T = 2^{N+5}$ . By (75.8),

$$q(T^{-1}) \geq \mathbb{P}_{T^{-1}}(L = 0) \geq 31/32, \quad q(T) \leq \mathbb{P}_T(L < d) \leq 1/32.$$

The first estimate bounds the total positive-degree weight by  $2^N/T$ ; the second compares the lower-degree weight with  $a_d T^d$ , using  $a_d \geq 1$ .

Consider the rational grid

$$t_j = \left(1 + \frac{1}{32d}\right)^j, \quad -J \leq j \leq J, \quad J = 64d(N+5). \quad (75.10)$$

Since  $\log(1 + 1/(32d)) \geq 1/(64d)$ , the endpoints bracket  $T^{-1}$  and  $T$ . The change of  $q$  between neighboring grid points is at most  $1/32$  by (75.9). Each grid activity is a rational with  $O(dN \log(d+1))$  bits.

Use binary search on this grid, estimating  $q(t_j)$  to additive error  $1/64$  by the weighted basis sampler. Accept a point when its estimate belongs to  $[3/8, 5/8]$ . Otherwise move the lower endpoint to the point if the estimate exceeds  $5/8$ , and move the upper endpoint to it if the estimate is below  $3/8$ . Allocate the failure probabilities over  $O(\log(J+1))$  tests. On their simultaneous success event, an unaccepted lower endpoint has  $q > 39/64$  and an unaccepted upper endpoint has  $q < 25/64$ . They cannot become adjacent, because their difference would exceed the adjacent-grid bound  $1/32$ . Thus the search accepts a point. At that point  $q \in [23/64, 41/64] \subseteq [1/4, 3/4]$ . Lemma 75.4 now gives

$$p_k(t) \geq \frac{1}{4(d+1)}. \quad (75.11)$$

Estimate  $A(t)$  using Theorem 75.2 and estimate  $p_k(t)$  by independent samples. The latter relative estimate takes  $O(d\varepsilon^{-2} \log(1/\delta))$  ideal samples by (75.11). Choose each approximate sampler's total-variation error small enough that a coupling and union bound cover the entire sample batch. Equation (75.1) depends only logarithmically on this extra accuracy. Finally use the exact identity

$$a_k = \frac{A(t) p_k(t)}{t^k}. \quad (75.12)$$

Relative accuracy  $\varepsilon/4$  in both factors is more than sufficient for relative accuracy  $\varepsilon$  in the product. All activity bit lengths, sample counts, oracle costs and mixing times are polynomial.

The preorder case uses  $H_\tau$ , and the height-two gamma case uses (71.7). Run the scheme for all at most  $N+1$  coefficients and divide the permitted failure probability among them to get the final assertion.  $\square$

The same argument applies to positive rational product weights after replacing (75.8) and the lower coefficient bound by the bounds obtained from their input bit lengths. The unweighted theorem already covers the integer counts asked about above.

## 75.7 Sampling after fixing the support size

At the accepted activity in the preceding proof, rejection until  $L = k$  takes at most  $4(d+1)$  trials in expectation for the ideal sampler. Conditional on acceptance, the factor  $t^k$  is constant. Thus the accepted bases are uniform among bases with  $k$  receivers in the unit-weight case. Taking their receiver supports gives the support distribution of a uniform integer vector conditioned to have support size  $k$ .

To obtain a high-probability running-time bound, truncate after  $O(d \log(1/\delta))$  trials. For approximate samples, choose the unconditioned total-variation accuracy smaller than the desired conditional accuracy times the lower bound in (75.11). Conditioning can amplify errors by the reciprocal of the acceptance probability, but that reciprocal is polynomially bounded here.

**Sampling boundary.** All samplers above return a support set, or an auxiliary matchable support pair. They do *not* yet return a uniform integer vector in  $\mathcal{D}_H$  or  $Q_\tau$ . Equal fibre cardinalities do not by themselves provide an efficient fibrewise bijection. The direct, restriction-compatible demand/support bijection remains a separate research task.

[Added 30 September 2026, batch 64: Part IX answers this. Theorems 135.2 and 145.2 return actual demand vectors, and Corollary 146.2 actual lattice points of  $Q_\tau$ , with the total-variation error transferred exactly (Theorem 145.1); the bijection is restriction-compatible for receivers outside the support and suppliers outside the selected set, not for arbitrary minors. See Section 132.3.] Similarly, these are randomized approximation results, not polynomial-time exact counting algorithms. No #P-hardness classification is asserted. [Merge note: source 07 supplies one; see Section 76.]

## 76 Exact counting remains hard

Fast randomized approximation is compatible with exact counting being #P-complete. We give an explicit reduction rather than infer hardness from a superficial resemblance to counting matchings.

### 76.1 The leaf-extension identity

For an integer  $\ell \geq 0$ , form  $H^{(\ell)}$  by attaching  $\ell$  private leaves to every original vertex of  $H$ . Leaves are placed on the opposite shore. Let  $N = |X| + |Y|$  count the original vertices.

**Lemma 76.1** (Leaf transform).

$$p_{H^{(\ell)}}(u) = (1 + \ell u)^N p_H \left( \frac{u}{(1 + \ell u)^2} \right). \quad (76.1)$$

*The right side is a polynomial after cancellation.*

*Proof.* In a matchable support, an included leaf forces its original neighbor to be included and matched to it. No original vertex can have two included leaves. Delete these forced pairs. The remaining support  $W$  lies in  $H$  and is perfectly matchable there. Conversely, choose any such  $W$ , and at each original vertex outside  $W$  either add nothing or add that vertex together with one of its  $\ell$  leaves. These choices are uniquely recoverable from the resulting support.

If  $|W| = 2k$ , its contribution is  $u^k (1 + \ell u)^{N-2k}$ . Summing over  $W$  gives equation (76.1). This proof counts supports rather than matchings, so it survives multiple matchings on  $W$ .  $\square$

### 76.2 Hardness for graph supports

The standard input is that counting bases of a transversal matroid, presented by a bipartite graph, is #P-complete [23]. One can assume a presentation with  $r$  slots, all saturable, where  $r$  is the rank. To see the reduction to such a presentation, choose  $r$  independent rows of the generic



presentation matrix using a maximum matching. They span its row space, so restriction to them preserves independence of every set of columns. This is again a graph presentation and can be obtained in polynomial time.

For that graph, with exactly  $r$  vertices on one shore, the top coefficient  $p_r$  is the number of transversal bases on the other shore. By equation (76.1) at  $u = 1$ ,

$$p_{H^{(\ell)}}(1) = (1 + \ell)^N p_H((1 + \ell)^{-2}). \quad (76.2)$$

The  $r + 1$  values  $\ell = 0, \dots, r$  give distinct rational arguments. Exact interpolation recovers  $p_r$ . Each enlarged graph has at most  $N(r + 1)$  vertices, and the interpolation data have polynomial bit length.

**Theorem 76.2** (Exact graph-counting complexity). *Computing  $p_H(1) = |\mathcal{D}_H|$  is #P-complete under polynomial-time Turing reductions.*

*Proof.* The reduction above proves hardness using [23]. Membership in #P is immediate by guessing a vertex subset and testing, in polynomial time, whether its induced bipartite graph has a perfect matching. Each subset is one witness regardless of how many matchings it has.  $\square$

### 76.3 Hardness even for height-two preorder lattice counts

Orient the edges of an arbitrary bipartite graph  $H$  from its lower shore to its upper shore, obtaining a height-two poset  $P_H$  on  $N$  elements. The height-two identity of Part II (Theorem 13.2) is

$$h_{P_H}(t) = (1 + t)^N p_H\left(\frac{t}{(1 + t)^2}\right). \quad (76.3)$$

For completeness, partition lattice points by the zero set  $Z$  on the lower shore and the positive-coordinate set  $S$  on the upper shore. The excess above one on  $S$  is a demand vector of  $H[Z \sqcup S]$ . The total-count bridge replaces it by a matchable pair  $(I, J)$  with  $I \subseteq Z, J \subseteq S$ . For a fixed size- $k$  pair, summing all allowed  $Z, S$  contributes  $t^k(1 + t)^{N-2k}$ , which proves equation (76.3). This also identifies the height-two  $\gamma$ -polynomial with  $p_H$ .

Apply equation (76.3) to  $H^{(\ell)}$ , which has  $N_\ell = N(1 + \ell)$  vertices. At  $t = 1$ , the leaf transform gives

$$h_{P_{H^{(\ell)}}}(1) = 2^{N_\ell} (1 + \ell/4)^N p_H\left(\frac{4}{(\ell + 4)^2}\right). \quad (76.4)$$

Once again  $r + 1$  oracle values interpolate  $p_H$  and recover its hard top coefficient.

**Theorem 76.3** (Exact preorder-counting complexity). *The number  $|\mathcal{Q}_\tau \cap \mathbb{Z}^n| = h_\tau(1)$  is #P-complete under polynomial-time Turing reductions, even when  $\tau$  is an ordinary poset of height at most two. Nevertheless it has an FPRAS.*

*Proof.* Equation (76.4) and the preceding transversal-base reduction prove hardness. Membership in #P follows from  $h_\tau(1) = p_{G_\tau}(1)$ , with a guessed support and a deterministic matching test. Alternatively, one may guess the bounded integer coordinates and verify their Hall feasibility by a flow computation. The FPRAS is Theorem 75.2. This is a Turing reduction, not a parsimonious reduction.  $\square$

The hardness input and general approximation machinery are established results. The displayed graph transforms make their application to these particular support and preorder quantities explicit. No deterministic polynomial-time exact counter is claimed.

**Corollary 76.4** (A specified height-two  $\gamma_k$ ; merge consequence). *Computing a specified coefficient  $\gamma_k(\tau)$  of an unrestricted finite preorder is #P-hard under polynomial-time Turing reductions, already for posets of height two. For posets of height two every  $\gamma_k(P_H)$  has an FPRAS.*

*Proof.* By (76.3),  $\Gamma_{P_H} = p_H$ , so  $\gamma_k(P_H) = p_k(H)$ . In the reduction above, the graph presentation has exactly  $r$  vertices on one shore, and  $p_r$  is the number of transversal bases, which is #P-hard to compute [23]; an oracle for  $\gamma_r(P_H)$  therefore returns this number. The FPRAS is Theorem 75.5.  $\square$

[Merge note: this corollary is stated by neither source; it combines source 07's reduction with the height-two identity, which both sources state. It answers the "specified  $\gamma_k$ " part of Research question 32.]

## 77 Examples that separate the properties

(Source 06's Section 8, with source 07's examples.)

### 77.1 A four-cycle and the danger of multiplicities

Let  $H = K_{2,2}$  with receivers  $y_1, y_2$ . The demand set consists of the nonnegative pairs with  $c_1 + c_2 \leq 2$ . Its multivariate polynomial is

$$F_H(z_1, z_2) = 1 + 2z_1 + 2z_2 + z_1z_2.$$

The corresponding matroid on  $x_1^\circ, x_2^\circ, y_1, y_2$  is uniform of rank two: every two-element subset is a basis. Setting the private variables to one yields exactly the displayed polynomial. There are six lattice points and six bases, but seven edge matchings if both perfect matching witnesses are counted separately.

At unit fields, the mean receiver support size is one and its variance is  $1/3$ . This is below the binomial upper bound  $2(1/2)(1 - 1/2) = 1/2$ . The model is a simple check that bases must be counted as sets.

### 77.2 The eight-element nonreal example

Let  $\Theta_3$  be the graph with three internally disjoint paths of length three joining common endpoints. Choose shores

$$X = (a, c_1, c_2, c_3), \quad Y = (b, d_1, d_2, d_3)$$

and edges  $ad_i, d_ic_i, c_ib$  for  $i = 1, 2, 3$ . Its perfectly matchable set polynomial is

$$p_{\Theta_3}(u) = 1 + 9u + 24u^2 + 16u^3 + u^4. \tag{77.1}$$

This is the previously documented example in Parts II and IV (Theorem 13.3), where the earlier small preorder counterexample is credited to Shivam Patel. We use the example and independently check its coefficients; we do not claim to have discovered it.

Its order-four normalized sequence is

$$1, \quad \frac{9}{4}, \quad 4, \quad 4, \quad 1,$$

which is log-concave. Yet the polynomial does not have all real roots. Indeed,

$$p_{\Theta_3}(-x) = (1-x)(1-4x)^2 + x^4 > 0 \quad (0 \leq x \leq 1).$$

Positive coefficients exclude nonnegative roots. If all four roots were real, all would be less than  $-1$ , contradicting their product being one. Thus ultra-log-concavity is genuinely weaker than real-rootedness in the class covered by the theorem.

The height-two preorder on these eight vertices has

$$\begin{aligned} h_{p_{\Theta_3}}(t) = & 1 + 17t + 106t^2 + 303t^3 + 427t^4 \\ & + 303t^5 + 106t^6 + 17t^7 + t^8. \end{aligned} \quad (77.2)$$

It has 1281 lattice points. Both its ordinary coefficient sequence, normalized by  $\binom{8}{k}$ , and its gamma sequence, normalized by  $\binom{4}{k}$ , are log-concave. Its ordinary polynomial is nevertheless not real-rooted: a nonreal root  $u$  of (77.1) lifts through  $t = u(1+t)^2$ , and real  $t \neq -1$  would give real  $u$ .

The completion  $s^4 p_{\Theta_3}(t/s)$  is Lorentzian but not real stable. If it were stable, setting  $s = 1$  would give a univariate stable polynomial with real coefficients, hence a real-rooted one. This example prevents an incorrect upgrade from the Lorentzian conclusion to real stability.

### 77.3 A numerical view of the improvement

For (77.2), the unweighted support law is symmetric about four. Its variance, computed exactly from the coefficients, is

$$\frac{2(16 + 17 \cdot 9 + 106 \cdot 4 + 303)}{1281} = \frac{1792}{1281} < 2.$$

The bound 2 is  $n/4$ . More importantly, after changing the positive coordinate fields, symmetry disappears but the covariance estimate and the same-mean binomial comparison remain valid. They require neither a symmetric law nor a real-root factorization.

### 77.4 The same example as delivered by source 07

Let  $\Theta_3$  have suppliers  $u, b_1, b_2, b_3$  and receivers  $v, a_1, a_2, a_3$ , with edges  $ua_i, b_i a_i$ , and  $b_i v$  for  $i = 1, 2, 3$ . It consists of three internally disjoint length-three paths from  $u$  to  $v$ . Its polynomial is

$$p_{\Theta_3}(t) = 1 + 9t + 24t^2 + 16t^3 + t^4. \quad (77.3)$$

The height-two preorder on these eight vertices has

$$\begin{aligned} h(t) = & 1 + 17t + 106t^2 + 303t^3 + 427t^4 \\ & + 303t^5 + 106t^6 + 17t^7 + t^8. \end{aligned} \quad (77.4)$$

This is the existing counterexample credited to Shivam Patel by Part II and [11]; neither its discovery nor its minimality is claimed here. It has  $h(1) = 1281$ .

The quartic in equation (77.3) has discriminant  $-5243 < 0$ , hence exactly two real roots and one nonreal conjugate pair. Indeed a real square-free quartic has negative discriminant exactly when it has one conjugate pair. The quadratic substitution in equation (76.3) then gives nonreal roots of  $h$ : a nonreal value of  $t/(1+t)^2$  cannot arise from real  $t$ . Nevertheless both polynomials satisfy the appropriate ultra-log-concavity inequalities. A real-stable homogeneous lift would force its univariate specialization to be real-rooted, so the Lorentzian lift here is genuinely weaker than stability.

### 77.5 Complete bipartite graphs and antichains

For  $K_{m,n}$  with  $m, n \geq 1$ , every equally sized pair is matchable, so

$$p_{K_{m,n}}(t) = \sum_{k=0}^{\min(m,n)} \binom{m}{k} \binom{n}{k} t^k.$$

When  $m = n$ , the active graph is the doubled indiscrete preorder and is palindromic. When  $m \neq n$ , its top coefficient exceeds one, so it is not palindromic. Both cases satisfy ultra-log-concavity. For  $d$  disjoint edges,  $p = (1 + t)^d$ , corresponding to the antichain and equality in the sharp universal coefficient bound.

### 77.6 A reflexive but nontransitive core

Take the relation on  $[3]$  with all diagonal pairs and the additional pairs  $(1, 2), (2, 3)$ , but not  $(1, 3)$ . Direct counting gives

$$p(t) = 1 + 5t + 6t^2 + t^3.$$

Its penultimate coefficient exceeds its first coefficient by one, exactly the one missing transitive pair. It is not palindromic, although its normalized coefficients  $1, 5/3, 2, 1$  are log-concave. This separates the two components of the support-enumerator problem.

### 77.7 Cardinality conditioning is not a matroid minor

Even for two disjoint edges, restricting to supports of size exactly one selects two basis monomials  $x_2 y_1 + x_1 y_2$ . Their supports do not satisfy basis exchange: replacing  $x_2$  from  $\{x_2, y_1\}$  by either element of  $\{x_1, y_2\}$  gives neither allowed basis. Thus fixed-cardinality support sampling does not follow by pretending that the selected layer is a matroid minor. Conditioning individual vertices, as in Proposition 69.2, is a different operation.

## 78 Finite verification and reproducibility

The accompanying standard-library Python verifier checks distinct representations, using exact integer and rational arithmetic. The proofs remain those in the preceding sections.

For a bipartite graph it enumerates demand vectors by all neighborhood inequalities, counts support pairs by a recursive perfect-matching test, and independently enumerates bases using an augmenting-path oracle on the private-copy presentation. It compares the resulting coefficient for *every receiver support*, not merely the diagonal specialization. It then checks the binomially normalized inequalities by integer cross-multiplication. A second pass assigns positive integer weights to both shores and checks the weighted inequalities.

On every bipartite graph with both shores of size at most three, the program also checks every nonempty conditioning event that specifies each receiver as forced present, forced absent, or free. For covariance, it computes the matrix on the right of (74.4) minus the actual covariance using rational arithmetic and checks all principal minors. This is an exact positive-semidefiniteness test in those finite instances.

For preorders, a separate enumerator tests *all lower-ideal inequalities* on nonnegative integer vectors of total at most  $n$ , then compares with the doubled-graph support counts. Every reflexive labeled relation is considered and exactly the transitive ones are retained.

Table 7: Completed exact checks. The graph families in the first two rows are disjoint; seeded examples are a separate, nonexhaustive suite.

| Scope                                               | Cases |
|-----------------------------------------------------|-------|
| All bipartite graphs with $0 \leq m, n \leq 3$      | 689   |
| All bipartite graphs on fixed shores (3, 4)         | 4096  |
| Additional seeded graphs on (4, 4), (5, 4), (4, 5)  | 75    |
| Nonempty conditioning events in the first family    | 14021 |
| Graphs in the exact covariance suite ( $m \geq 1$ ) | 685   |
| All labeled preorders with $0 \leq n \leq 5$        | 7332  |
| Integer-capacity examples on shores (2, 2)          | 64    |

The preorder counts by cardinality are

| $n$       | 0 | 1 | 2 | 3  | 4   | 5     |
|-----------|---|---|---|----|-----|-------|
| preorders | 1 | 1 | 4 | 29 | 355 | 6942. |

All comparisons passed. The eight-element  $\Theta_3$  preorder was checked separately by direct ideal enumeration, including the 1281 point count. The capacity suite compares the capacity inequalities with donor cloning. The complete list of seeded adjacency rows and the capacity examples is recorded in `results/verification.json` [shipped as `data/06-lorentzian-support-verification.json`].

The full exact run used Python 3.13.5. Reproduce it from the package root with

```
python code/verify.py -max-preorder 5 -output results/rerun.json
```

[Merge note: this is the delivered command. Here the program is `code/06-lorentzian-support-verify.py`; the sampler and its tests import it under its delivered name `verify`, which in this directory would load Part I's `code/verify.py`. Run source 06's programs on a copy with the delivered names, as the README describes.] The default size is four for a shorter check; `-skip-3x4` further reduces the finite graph suite. The program raises explicit exceptions on failure and does not rely on removable Python assertions.

The separate `code/support_sampler.py` [shipped as `code/06-lorentzian-support-support_sampler.py`] implements the weighted private-copy basis walk and returns receiver supports. It uses an explicit conservative mixing budget from (75.1), rational extension probabilities, and independent restarts for the requested samples. It is a reference sampler, not an optimized implementation of the entire coefficient FPRAS. A 100-sample seeded smoke run and separate unit tests are included; a finite-seed experiment is not a proof of a mixing guarantee. The article proves the FPRAS reductions; the package does not claim a benchmarked production coefficient-counting system.

The shipped files of source 06 are its verifier, sampler and unit tests (`code/06-lorentzian-support-verify.py`, `-support_sampler.py`, `-test_sampler.py`), its build script (`code/06-lorentzian-support-build.sh`, which builds the delivered article and cannot build this one), the recorded outputs `data/06-lorentzian-support-verification.json` and `.log`, `-sampler_smoke.json` and `.log`, `-sampler_tests.log`, and the note `06-lorentzian-support-SOURCES.md`.

## 78.1 Source 07's exact verification

The package contains `code/support_tools.py` and `code/verify.py` [shipped as `code/07-support-polynomials-support_tools.py` and `code/07-support-polynomials-verify.py`].

They use Python’s standard library, integer bitmasks, and exact rational arithmetic. No floating-point root finder is used for the universal coefficient or structural checks. The separate example certificate also uses only the standard library: fraction-free integer elimination computes the discriminant, and exact polynomial division implements Sturm’s rule.

| Test family                                      | Instances | Exact checks                                                                                         |
|--------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------|
| All labelled $4 \times 4$ bipartite graphs       | 65,536    | Basis/support bijection, ULC, palindromic-core test, top factorization.                              |
| All labelled $3 \times 4$ bipartite graphs       | 4,096     | Same, plus receiver-refined Hall-demand identity.                                                    |
| All labelled preorders, $0 \leq n \leq 4$        | 390       | Direct ideal-inequality counts versus doubled-graph supports; ULC and symmetry.                      |
| Seeded positive rational vertex weights          | 300       | ULC, variance, all finite stop-loss convex-order tests, equality classification at three fugacities. |
| All $2 \times 2$ graphs, two leaf multiplicities | 32        | Full coefficient identity for the leaf transform.                                                    |
| Seeded sampler runs on $\Theta_3$                | 30        | Feasibility of every returned support; not an empirical proof of mixing.                             |

The  $4 \times 4$  family contains 2,165,695 supports in total and 9,268 graphs with palindromic support polynomials. The  $3 \times 4$  family contains 68,832 supports and 707 palindromic graphs. The verifier obtains the coefficient sequence from a dynamic enumeration of partial matchings, merging equal support pairs. Independently, it tests all candidate matroid bases with an augmenting-path oracle and compares the resulting support sets.

For demand vectors, the second route enumerates all nonnegative coordinate vectors of total at most  $|X|$  and tests every Hall inequality directly. For preorders, it enumerates reflexive relations and retains precisely the transitive ones. It counts lattice points using the order-ideal inequalities, not the matching identity under test. The numbers of labelled preorders found are 1, 1, 4, 29, 355 for sizes 0, 1, 2, 3, 4.

The weighted test uses seed 20260929 and fugacities  $1/3, 1, 7/2$ . Convex-order checking is exact because on a finite integer interval it is enough to test all stop-loss functions  $(k - c)_+$ , together with equal mass and mean. The sampler tests only confirm that the implementation stays in the stated state space. The general mixing theorem remains an imported mathematical theorem.

## 78.2 Rebuild and rerun

From the package directory:

```
python code/verify.py --output rerun/verification.json
python code/example_certificate.py --output rerun/example_certificate.json
python code/support_tools.py
pdflatex -interaction=nonstopmode -halt-on-error support_polynomials.tex
pdflatex -interaction=nonstopmode -halt-on-error support_polynomials.tex
pdflatex -interaction=nonstopmode -halt-on-error support_polynomials.tex
```

Use `-quick` for the smaller  $3 \times 3$  regression run. The default recorded output is in `data/verification.json`; use a different output name to preserve it. Elapsed time is environment-dependent and is not a mathematical assertion. The delivered PDF was compiled and visually inspected. [Merge note: these are the delivered commands, for



the package layout. Here the programs are `code/07-support-polynomials-verify.py`, `-example_certificate.py` and `-support_tools.py`; the first two import `support_tools` under its delivered name and fail with `ModuleNotFoundError` if run under the shipped names, and the verifier’s default output `../data/verification.json` is resolved from the working directory. The recorded outputs are `data/07-support-polynomials-verification.json`, `-verification.log`, `-example_certificate.json`, `-example_certificate.log` (byte-identical to the JSON) and `-build_validation.json` (a record of the delivered PDF’s build). The delivered build script `code/07-support-polynomials-build.sh` builds `support_polynomials.tex`, which is not shipped. The README gives commands that work on a copy.]

## 79 Proof audit and limits of the conclusions

### 79.1 The imported boundary is small and explicit

The general graph conclusions rely on the positive-coordinate count in Lemma 68.1 and the matroid Lorentzian theorem. The counting and sampling complexity statements additionally rely on the basis-walk theorem. These are ordinary mathematical theorem citations; none is being claimed as a checked declaration in ProveIt.

The ordinary preorder inequalities have a route independent of the repository’s unrefereed gamma and simplicial-polytope arguments:

$$\begin{aligned} \text{ideal inequalities} &\implies \text{bipartite demand vectors} \\ &\implies \text{matroid bases} \implies \text{Lorentzian inequalities.} \end{aligned}$$

The height-two gamma corollary additionally uses the finite cancellation identity, whose proof was supplied above. It therefore does not require assuming the full unrefereed geometric realization of the preorder polynomial.

### 79.2 Claims deliberately not made

The article does not solve flag-polytopal realization, classify all palindromic bipartite support polynomials, or restore real-rootedness. It does not prove log-concavity of the gamma polynomial for arbitrary preorders. It does not assert pairwise negative correlation merely from complete log-concavity. It does not count matchings with their witness multiplicities. It does not give a polynomial-time exact counting algorithm or a polynomial-in-binary-capacity algorithm by expanding exponentially many donor copies. Finally, support sampling is not silently promoted to sampling actual lattice points.

### 79.3 Proof dependencies and a formalization route (source 07)

The logical structure is short enough to audit without reading every application. The support-counting identity is the only geometric enumeration input needed to move from graph supports to demand vectors. The matroid basis bijection is elementary. The coefficient inequality then uses the established Lorentzian theorem, while the complete palindromic-core characterization has an independent finite graph proof. Probability and algorithmic consequences branch from those two mechanisms rather than depend on the repository’s unrefereed flag or root-polytope claims.

A realistic formalization should be divided into the following layers.

1. **Finite combinatorics.** Formalize matchable support pairs, the private-slot matroid, the complement-one-shore equivalence, path cancellation, and reachability of deletion pairs. These statements do not need analysis.
2. **Coefficient algebra.** Formalize the bivariate derivative calculation and the factorial/binomial conversion. It can initially be stated with the relevant quadratic Hessian inequality as an explicit hypothesis.
3. **Imported mathematics.** Prove or instantiate the general Lorentzian basis theorem and the two root-polytope counting identities. Until these are formalized, they remain visible theorem dependencies, not hidden axioms labelled as completed proofs.
4. **Probability and algorithms.** Formalize the finite convex-order lemma, the star equality chain, and detailed balance. The full spectral mixing theorem is a separate major dependency; implementation tests do not discharge it.

No Lean file is shipped as if it had been checked. The value of the present separation is that the finite structural classification can be formalized independently of the much larger Hodge-theoretic and mixing libraries.

## 79.4 Pins and review scope

The repository tree was inspected at commit

e9a57d735db2177a8cca6aaecd95a3d53a0ba678

and the preorder article had blob

d52a2633b58e74aa653e967eb56adc1aa41bc8ae.

The current README, report index, relevant article sections, and final research questions were consulted. This was a targeted review, not a proof audit of every report in the repository. The source log records the primary papers and the points of dependence. Independent mathematical review and formalization are still appropriate before treating the unrefereed application results as established repository infrastructure.

Source 07 records (as delivered, its Section 1.3):

The inspection is pinned to commit `de2f8fa8c6f0107b1a0f1f77e2e25c343e562180`. The report's `article.tex` has Git blob `d52a2633b58e74aa653e967eb56adc1aa41bc8ae`; its README has blob `846d9f19fadac9cd933b74370cdb8fafa2211c5d`. The relevant article contains Parts I–IV. The surrounding repository was still changing during inspection, which is why the file identities, rather than a vague reference to “the latest report,” are recorded. No repository files were modified.

Source 09 (Part VII) records commit `e1afd75e35a4de734d5aa47aec5cdb917b82ed3e` with the same two blobs. All three pins resolve, and this report was unchanged from the earliest of them to the placement commit `26473dfa0`.

## 80 Further research questions and topics

The following are specific research targets, not assertions that their worldwide status has been exhaustively surveyed. They separate statements settled here from stronger statements that would require new arguments.

**Research question 37** (Normalization by the matching number; sources 06 and 07). Can  $\binom{v(H)}{k}$  replace  $\binom{\min(m,n)}{k}$  for all bipartite graphs, or under a useful structural hypothesis? The present proof supplies the shore-size normalization only. The question is nontrivial when no maximum matching saturates either shore. (Source 07: can  $\binom{d}{k}$  in equation (70.4) be replaced by  $\binom{r}{k}$  for  $r = v(H) < d$ , either for all graphs or for a meaningful subclass? Prove the stronger claim or find a smallest counterexample, including positive vertex weights as a separate version.)

Deleting isolated vertices can sharpen a displayed bound, but does not remove a general Hall deficiency. The obstruction to dividing a Lorentzian completion by a homogenizing monomial must be addressed rather than ignored.

The present proof cannot simply cancel the factor  $s^{d-r}$  from a Lorentzian homogenization: division by a monomial is not, in general, a Lorentzian preserving operation. For example  $s(s^2 + 4st + 5t^2)$  satisfies the degree-three bivariate criterion, while  $s^2 + 4st + 5t^2$  fails the degree-two criterion.

[Merge note: the batch-55 placement dossier searched for a counterexample (evidence only): among all 4,160 bipartite graphs on shores (4, 4) with  $2 \leq v(H) < d$ , and 264 seeded random graphs with shores up to (6, 6), none violates log-concavity of  $a_k / \binom{v(H)}{k}$ . The question stays open.]

**Research question 38** (Equality, rigidity, and sharper covariance bounds; sources 06 and 07). Characterize equality at a single ultra-log-concavity index under positive weights, and characterize the null directions of  $\text{diag}(p) - pp^\top/m - \text{Cov}(I)$ . Can neighborhood redundancy or a canonical decomposition replace the ambient rank in part of this bound? (Source 07: classify equality at one internal coefficient in equation (70.4), not merely simultaneous binomial equality. Quantify how a shared neighbor or unequal weighted star strengths forces a positive gap in equation (74.9).)

The antichain classification above concerns simultaneous equality only. A complete answer is likely to require equality information in the Lorentzian Hessian inequalities, not just the coefficient count.

The exact equality theorem suggests using collision mass in equation (74.15). Converting that endpoint deficit into a uniform positive-fugacity variance gap is the missing step; it is not supplied by qualitative compactness when activities approach zero.

[Merge note: simultaneous equality in the variance bound, for every positive weighting, is settled by source 07's Theorem 74.8; source 06's antichain proposition is its preorder case.]

**Research question 39** (Binary capacities and approximation complexity). Can partition functions or individual support coefficients for large binary-encoded capacities be approximated in time polynomial in the encoding length, without explicitly cloning donors? A compressed transversal-polymatroid walk, or another representation of the same support counts, would be needed.

This is the correct next step after Corollary 70.5; the pseudo-polynomial cloned presentation is not already an answer. [Added 30 September 2026, batch 64: still open. Part IX's bijection and sampler also reach dilations only by cloning, and its Research question 68 asks the same for them.]

**Research question 40** (Deterministic and practical coefficient algorithms). Develop a deterministic approximation method for these special transversal-matroid partition functions, or a practically competitive implementation of the tilting FPRAS. Analyze dependence on graph width, repeated neighborhoods, and sparse presentations. Compare with exact algorithms for the same classes.

The rational-grid construction is designed to prove polynomial complexity, not to be the fastest numerical tuning rule. Its probability lower bound provides a useful correctness target for faster implementations.

**Research question 41** (Algorithmic use of the preorder quotient). Exploit equivalence classes and the quotient poset to accelerate the independence oracle or aggregate the Markov chain without losing its stationary law. Can the running time depend primarily on the number of equivalence classes or on a structural width parameter?

Any aggregation must count supports, not matchings or orbit representatives with uniform weight. The general rank- $n$  sampler gives a correctness baseline against which compressed implementations can be tested.

**Research question 42** (Distributional comparison under relation extension). When a preorder is enlarged by adding comparisons, does its support-size law become smaller in convex order after the appropriate mean normalization? Determine which order extensions preserve such a comparison.

Coefficientwise growth of transport counts is not enough by itself: the partition function also changes. A proof would need a likelihood-ratio or coupling argument, and could yield sharper variance bounds than the universal binomial comparison.

**Research question 43** (Formalization in layers; sources 06 and 07). Formalize the private-copy basis correspondence and preorder reduction first, then the finite support-fibre count, and finally the analytic and probabilistic implications. State imported results explicitly until they are discharged. Can the elementary coefficient-recovery reduction be verified independently of the implementation of the matching oracle? (Source 07, “A small independently checked formal core”: formalize the complete palindromic-core theorem, including isolated vertices and the actual-degree convention, and verify the recognition algorithm against that theorem. Then formalize the weighted-star equality theorem assuming the ULC inequality.)

A conditional Lean theorem about a supplied support-counting identity is a useful intermediate object, but should not be labeled an unconditional formal proof of the present theorem. [Added 30 September 2026, batch 64: the finite support-fibre count now has an elementary proof (Part IX, Corollary 139.2), a natural target for this layer; Research question 73 asks for a certified checker of its certificates.]

These two projects isolate finite and elementary arguments from the large imported theories. They would substantially increase confidence in the new applications without pretending that a few computational examples formalize Lorentzian Hodge theory.

## 80.1 Questions of sources 06 and 07 already posed in Parts II–IV

These questions of the sources repeat earlier questions of this report, or are answered inside Part V. They are printed as delivered, without new numbers, each after a pointer to the question it repeats.

*Repeats Research question 24* (coefficient shape beyond gamma-positivity):

**Gamma log-concavity beyond height two (source 06).** Is  $\Gamma_\tau$  log-concave for every finite preorder? If not, determine a minimal counterexample and structural restrictions that recover it. Height-three posets and chains of nontrivial equivalence classes are useful first test families. Corollary 71.6 settles height two, but the ordinary  $h$ -inequalities do not settle this question. A

productive approach would identify a homogeneous completion of the *disjoint* transport-support polynomial, rather than reusing the completion for all doubled support pairs.

*Repeat Research questions 20 and 2 (a bijective demand/support bridge):*

**An effective demand/support bijection (source 06).** Construct a support-preserving bijection that is compatible with a controlled ordering or restriction of the receiver set and can be evaluated and inverted in polynomial time. Such a construction would upgrade the support samplers to actual integer-vector samplers and might substantially reduce the proof-assistant dependency on geometry. The existing geometric counting theorem is the starting point, and Oh's bijections must be compared carefully before claiming a genuinely new construction. Compatibility and computational complexity are part of the target, not consequences of a cardinality identity.

**A direct demand/support bijection (source 07).** Construct a natural, efficiently invertible map from  $\mathcal{D}_H$  to matchable supports, preferably preserving the exact receiver support. The equality of these refined counts is proved above, but an arbitrary ranking-and-unranking bijection would not explain the combinatorics. Such a map would turn the support sampler into an actual demand-vector sampler. A first test should reject any rule depending on a noncanonical chosen matching. Lexicographically selecting allocations and retaining one extreme supplier from each allocation can merge different demand vectors. A successful construction must control this loss of information. [Added 30 September 2026, batch 64: both paragraphs are answered by Part IX. Its maps are evaluated and inverted in polynomial time, preserve the exact receiver support, and read the demand vector off a spanning tree extracted from an optimum, with no chosen matching; they turn the support sampler into an actual demand-vector sampler (Theorem 145.2). Section 133 compares them with Oh's bijection, whose existence they realize.]

*Repeat Research questions 3, 11 and 34, and bear on Part VI (real stability and real roots):*

**Real stability of the multivariate support polynomial (source 06).** Which bipartite graphs have a real-stable  $F_H(z)$ , or a real-stable private-copy basis polynomial? The two formulations should not be assumed equivalent. The repository's cactus models give a natural comparison class, while  $\Theta_3$  shows that the unrestricted real-stability claim fails. A stable affine slice may arise from a nonstable completion. Conversely, a stable completion gives a stable slice. This directionality should be retained in any classification.

**Real-rootedness inside the Lorentzian class (source 07).** Classify graph operations which preserve real-rootedness of  $p_H$ , and seek a structural recognition theorem broader than cacti. The nonreal-root family of Part II (Theorem 13.3) shows that low treewidth alone is insufficient. The aim should be a criterion for the univariate polynomial, not an unnecessary unit-modulus determinant representation. The latter is a strictly stronger requirement in the existing repository results.

[Merge note: Part VI classifies stability for two treewidth-two families and proves that stability of  $\Phi_G$  is strictly stronger than real-rootedness of  $p_G$ .]

*Answered in Part V:*

**Efficient sampling of a fixed coefficient layer (source 07).** For prescribed  $k$ , construct a polynomial-time approximate sampler of  $\mathcal{M}_k(H)$ , or determine the approximation complexity of  $p_k$ . Separate  $k = v(H)$ , where Proposition 72.1 factors the problem into two transversal-base counts, from internal layers. The example in Section 77.7 blocks the most direct matroid-minor

argument. A fugacity method would need a proved lower bound on the mass of the desired layer and an effective way to choose a suitable fugacity.

[Merge note: answered by source 06's Theorem 75.5 and its fixed-size sampler (Section 75): every  $a_k$  has an FPRAS, and at the tilted activity found there the layer  $k$  has mass at least  $1/(4(d+1))$ , so rejection sampling gives the layer. The fugacity method the delivered text asks for is exactly that argument.]

*Repeat Research questions 22 and 30 (flags and face numbers):*

**Flags and the preorder  $\gamma$ -vector (source 07).** Does every preorder support polynomial occur as the  $h$ -polynomial of a flag simplicial polytope? Does the transport-pair  $\gamma$ -vector satisfy stronger face-number inequalities or admit a canonical simplicial-complex model? Neither the intersection realization nor ultra-log-concavity supplies a flag triangulation. The repository's induced-crown obstruction concerns its canonical lattice model and must not be promoted into a disproof of all flag realization.

*Repeats Research question 28 (limit laws):*

**Limit laws without real-rootedness (source 06).** Which growing graph or preorder families have asymptotically normal support size? The same-mean binomial upper bound gives concentration but not a central limit theorem. Seek hypotheses phrased in terms of component structure, correlation decay, or the growing matroid lift, including sequences whose polynomials have many nonreal zeros.

## 81 Conclusion

The failure of real-rootedness leaves a substantial uniform theorem intact. Every bipartite demand support enumerator admits a classical transversal-matroid completion, and that completion imports the full Lorentzian coefficient mechanism. The resulting theorem is stronger than unimodality, applies to all finite preorders and their positive integral dilations, and settles the gamma coefficient-shape question throughout height two.

The same structure has algorithmic content. Matching feasibility supplies a basis oracle; the matroid walk supplies approximate sampling; and log-concave tilting makes a requested coefficient visible with polynomially large probability. This gives approximation schemes without confusing support sets with their matching witnesses or with actual integer vectors. The remaining questions concern genuinely stronger structures: general gamma inequalities, effective lattice-point bijections, sharper normalizations, and compressed-capacity algorithms.

Source 07 concludes in the same sense:

The useful replacement for universal real-rootedness is not merely unimodality. Bipartite matchable supports have a homogeneous matroid lift, which imports the full binomially normalized log-concavity inequality. The existing demand/root-polytope identities then transfer this structure to every preorder support polynomial and to the general bipartite support-enumerator problem. Palindromicity has a different explanation: uniqueness of the largest support forces a balanced core, and its first/penultimate coefficient defect measures failure of transitivity exactly.

Together these mechanisms answer the unimodality and palindromicity components of the stated research target, while preserving the distinction from the still unclassified real-rooted case. They also expose a useful computational dichotomy: exact counts are hard even in height two, yet uniform or positively weighted support measures admit rapid sampling and



randomized approximation. The mathematical advance claimed here is this explicit transfer, classification, and set of consequences relative to the inspected repository, not a rediscovery of general matroid concavity or a claim to have solved every remaining preorder conjecture.

## 82 Compact proof certificates

(Appendix A of source 06, and Appendix A of source 07 as Section 82.3.)

### 82.1 The coefficient inequality in four steps

For an ordinary unweighted graph, a reader can check the central argument through the following equalities and one imported theorem:

$$\begin{aligned}
 F_H(z) &= \sum_{U \sim_H S} z_S && \text{(published support-counting bridge),} \\
 \mathcal{B}_H(a, z) &= \sum_{U \sim_H S} a_{X \setminus U} z_S && \text{(private-slot basis correspondence),} \\
 \mathcal{B}_H(s\mathbf{1}, t\mathbf{1}) &= \sum_k a_k s^{m-k} t^k && \text{(homogeneous grouping),} \\
 \frac{a_k^2}{\binom{m}{k}^2} &\geq \frac{a_{k-1}}{\binom{m}{k-1}} \frac{a_{k+1}}{\binom{m}{k+1}} && \text{(Lorentzian bivariate criterion).}
 \end{aligned}$$

Transpose the graph to obtain the other shore normalization. For preorders, neighborhood sets are lower closures, giving the direct identification of the integer inequalities.

### 82.2 The coefficient-recovery algorithm

The theoretical algorithm of Theorem 75.5 is summarized below. Its probability budgets are specified in the proof.

```

Input: graph H, coefficient index k, relative error eps, failure delta.
Compute d = maximum-matching size.
Return 0 for k outside [0, d]; return 1 for k = 0.
Form the rational activity grid of equation (75.10).
Binary-search using samples to estimate Pr(L < k).
Stop when both sides of the threshold have substantial mass.
Approximate Z = p_H(t) by weighted basis self-reduction.
Estimate q = Pr(L = k) by repeated weighted basis samples.
Return Z * q / t^k.

```

The lower bound  $q \geq 1/(4(d+1))$  is the reason the final empirical step is fully polynomial. This bound is proved; it is not an assumption about the shape of a numerically observed histogram.

### 82.3 The example discriminant and exact normalization checks (source 07)

For  $q(t) = t^4 + 16t^3 + 24t^2 + 9t + 1$ , direct integer resultant computation gives

$$\text{disc}(q) = \text{Res}(q, q') = -5243.$$

The monic quartic sign factor is  $(-1)^{4 \cdot 3/2} = 1$ . The supplementary script code/example\_certificate.py [shipped as code/07-support-polynomials-example\_certificate.py]

computes this resultant using fraction-free integer elimination, independently counts two real roots by Sturm's rule, and constructs the height-two transform. The existence of nonreal roots does not depend on a numerical root plot.

For the eight-element support polynomial, the binomial-normalized sequence is

$$1, \quad \frac{17}{8}, \quad \frac{53}{14}, \quad \frac{303}{56}, \quad \frac{61}{10}, \quad \frac{303}{56}, \quad \frac{53}{14}, \quad \frac{17}{8}, \quad 1.$$

Each internal log-concavity inequality holds exactly. These numbers are an illustration of the theorem, not a substitute for its all-graph proof.

# Part VI. Stability of signed matching supports: block locality, ADE trees, books and multi-theta graphs

Part VI was added on 29 September 2026 in batch 55 of ProveIt's incoming-report intake. It continues Part III's question which bipartite graphs have a real-stable signed support polynomial  $\Phi_G$  (Research question 11). Its base manuscript proves that for a tree  $T$  the incidence–apex graph  $\mathcal{A}(T)$  is stable exactly when  $\rho(A_T) \leq 2$ , that is, for the finite and affine ADE trees, locates a four-page threshold for books whose univariate diagonals stay real-rooted, and classifies all simple bipartite multi-theta graphs. A second manuscript contributes the one-vertex sum theorem for all simple graphs, so that stability is block-local, and the stability of every complete bipartite block. Together they settle the point Part III left open: stability of  $\Phi_G$  is strictly stronger than real-rootedness of  $p_G$ . Parts I–V are unchanged except for dated pointers. Section 83 records the provenance, fixes the notation and states what is answered.

## 83 Provenance, conventions and what Part VI answers

### 83.1 One manuscript and part of another

- **Source 08 (the base):** batch-55 manuscript 04, *An ADE Threshold for Matching-Support Stability: Tree incidence–apex graphs, sharp book transitions, and a classification of bipartite multi-theta graphs*, dated 29 September 2026, 25 A4 pages as delivered, archive `ProveIt_ADE_Stability.zip`, arrived in ProveIt commit 2d4919838. Its byline reads “Research draft prepared for Vladimir Reshetnikov / Mathematical development and exposition with AI assistance”; it names no assistant, and its PDF metadata says “AI-assisted research draft”. It is pinned to commit e1afd75e3 (full hash in Section 98) and records the Git blob 846d9f19fadac9cd933b74370cdb8fafa2211c5d of this report's README, not of its article: it read the summary of Parts II–IV. The report did not change between that pin and the placement, so its statements about “the repository” concern Parts I–IV as printed. Its verifier, its recorded output, its requirements file and its source audit are shipped with the prefix 08-ade-stability-; its checksum list `SHA256SUMS.txt` was verified (7/7) at placement and retired.
- **Source 09, Sections 11–14 (stability):** batch-55 manuscript 05, *Preorders as Transversal Matroids: Reconstruction, Ultra-Log-Concavity, and Stability of Matching Supports*, 24 A4 pages, archive `ProveIt_Matroid_Lifts.zip`, arrived in the same commit and pinned to the same commit, with the blobs of both this report's article and README. Its byline credits ChatGPT. Its matroid sections are Part VII, which records its provenance in full (Section 100.1); its ultra-log-concavity sections are credited in Part V.

Both were placed in commit 26473dfa0. Neither saw the other. The manuscripts, delivery READMEs and PDFs are not shipped; they survive in the arrival commit.

*Why 08 is the base.* Source 08 treats stability for whole infinite families of 2-connected graphs (ADE incidence–apex graphs, books, multi-theta graphs) and is the only source for them; source 09's stability sections are shorter and overlap source 08 in the bridge to matroids, the inheritance by induced subgraphs, the one-vertex gluing theorem and even cycles. Source 09 is the more general on gluing (any simple graphs, with block localization) and alone proves every  $K_{\ell,r}$  stable, so its gluing theorem is printed in place of source 08's (Section 83.4).

**Abstract of source 08 (as delivered).** A matching-support polynomial counts an endpoint set once whenever it admits at least one perfect matching. Replacing matching multiplicities by this Boolean existence condition destroys the usual determinantal construction. We study the resulting multivariate stability problem, left open beyond special graph classes in the inspected ProveIt research corpus. For a tree  $T$ , form a bipartite graph whose left vertices are  $V(T)$ , whose right vertices are  $E(T)$  together with one apex, and whose apex is adjacent to every left vertex. We prove that its signed matching-support polynomial is real stable if and only if  $C_T = I - \frac{1}{2}A_T$  is positive semidefinite, equivalently  $\rho(A_T) \leq 2$ . Thus its stable members are exactly the trees of finite  $A, D, E$  and affine  $D, E$  type. The mechanism is the exact identity  $F_T = \det L + s \operatorname{tr}(\operatorname{adj}(L)C_T)$ ; failure of the spectral condition produces an explicit rational negative Rayleigh certificate. The criterion survives arbitrary subdivisions between the original apex neighbors. For quadrilateral books, the threshold is exactly four pages, although the univariate support polynomial is real-rooted for every number of pages. At five pages, two elements of the associated uniformly random matroid basis have covariance  $1/576$ . An independent expected-determinant argument proves stability for every even-path multi-theta graph. Combined with the tree criterion and a credited earlier nonreal-root example, this gives a complete stability classification of simple bipartite multi-theta graphs. Proofs, explicit examples, exact finite verifiers, provenance boundaries, and a further research program are included.

The abstract is reproduced as delivered; its  $\Theta(3, 3, 3)$  is printed  $\vartheta(3, 3, 3)$  below (Section 83.2), and its “inspected ProveIt research corpus” is Parts I–IV of this report.

**Status and scope (source 08, as delivered).** This is an unrefereed, AI-assisted mathematical research draft. Its general claims rest on the written proofs, not on the finite tests. No Lean or Rocq verification is claimed. The results are presented as developments beyond the inspected repository material; worldwide priority has not been established. The classical ADE classification and the earlier eight-vertex nonreal-root example are explicitly credited, not claimed as discoveries of this article.

**Priority and prior work (merge).** The ADE list of trees with spectral radius at most two is classical (McKee–Smyth [25] recall the context); the half-plane-property framework is Choe–Oxley–Sokal–Wagner’s and Brändén’s [24, 12], and Brändén’s Theorem 5.6 is the multiaffine Rayleigh criterion, whose necessary direction is reproved here. The nonreal-root obstruction  $\vartheta(3, 3, 3) = \Theta_3$  is Part II’s, with priority credited to Shivam Patel [11]; even-cycle stability is contained in Part II’s cactus theorem (Theorem 19.3) and in Ohsugi–Tsuchiya’s univariate result [9]. Part III already proved that  $\Phi_{K_{2,3}}$  and its tree attachments are stable without a faithful matrix (Theorem 37.1, Corollary 37.4); source 09’s complete bipartite blocks extend that example, and its vertex-sum theorem generalizes Part III’s leaf lemmas (Lemmas 37.2 and 37.3). Source 09 does not cite these Part III results; merge notes add the credit. The one-vertex gluing theorem was proved independently by both sources on the same day. Source 08’s focused literature search found no earlier statement of the tree incidence–apex equivalence; that is not a priority certificate, and no worldwide priority is claimed for any result of Part VI.

## 83.2 Notation of Part VI

Part VI uses Part III’s signed support polynomial  $\Phi_G$  (equation (28.2)) and Part II’s  $p_G$ ; source 08’s shores are written  $U \sqcup V$  where Parts II–III write  $A \sqcup B$ . Table 8 lists what Part VI adds or uses differently. Two symbols of the sources were changed; no normalization was changed:

- source 08’s theta graphs  $\Theta(\ell_1, \dots, \ell_q)$  are printed  $\vartheta(\ell_1, \dots, \ell_q)$ , extending Part III’s rule (Section 26), because Part II’s  $\Theta_m$  has  $m$  paths of length three; only  $\vartheta(3, 3, 3) = \Theta_3$  is both;
- source 08’s canonical transversal matroid  $M_G$  is printed  $\mathbf{M}_G$ , because  $M_G(u)$  is Part II’s matching polynomial and because source 09’s matroid, printed  $M_G$  as delivered, is its dual.

Table 8: Symbols of Part VI. The last column prints the tempting false reading.

| Symbol                                 | Meaning in Part VI                                                                                                              | Not to be confused with                                          |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| $G = (U \sqcup V, E)$                  | bipartite graph, left shore $U$ , right shore $V$ (source 08); $A \sqcup B$ in source 09's sections                             | Part I's ground set $E$ and $U, V$ as vertex sets in Part IV.    |
| $\mathcal{M}(G), (I, J)$               | matchable pairs, $I \subseteq U, J \subseteq V$                                                                                 | Part II's $\mathcal{M}_k(G)$ , sets of $2k$ vertices.            |
| $\Phi_G, p_G$                          | Part III's signed support polynomial (source 09 defines it for every simple graph); Part II's support polynomial                | a matching polynomial.                                           |
| $M_G, F_G(x; y)$                       | source 08's matroid, bases $(U \setminus I) \sqcup J$ , and its basis polynomial $\sum x^{U \setminus I} y^J$                   | Part II's $M_G(u)$ ; source 09's $M_G$ and $F_G$ .               |
| $M_G, F_G(x, y)$                       | source 09's matroid, bases $I \sqcup (B \setminus J)$ , and its basis polynomial $\sum x^I y^{B \setminus J}$ ; $M_G = (M_G)^*$ | source 08's $M_G, F_G$ .                                         |
| $T, r, A_T, C_T$                       | a tree on $r$ vertices, its adjacency matrix, $C_T = I - \frac{1}{2}A_T$                                                        | Part III's spanning forest $T$ and core $T$ ;                    |
| $\mathcal{A}(T), \star, s$             | incidence–apex graph, its apex, the apex variable                                                                               | Part I's cones $C_P$ and polytope $C_T$ .                        |
| $L, B, b_e, w$                         | the pencil $\text{diag}(x) + \sum y_e b_e b_e^T$ , the incidence matrix, its columns, an auxiliary vector                       | Part V's support size $L$ ; Part III's matrix $L$ .              |
| $B_m$                                  | the quadrilateral book with $m$ pages; $B_m = \Theta_m$ plus the edge $d b_0$ (Section 90)                                      | a shore $B$ ; Part I's $B_T$ .                                   |
| $\mathfrak{g}(\ell_1, \dots, \ell_q)$  | multi-theta graph with paths of lengths $\ell_i$ (source 08's $\Theta(\dots)$ )                                                 | Part II's $\Theta_m$ (only $\mathfrak{g}(3, 3, 3) = \Theta_3$ ). |
| $\Delta_{ij} f$                        | Rayleigh difference $(\partial_i f)(\partial_j f) - f \partial_i \partial_j f$                                                  | a Laplacian.                                                     |
| $\text{Lap}(T), \text{adj}, \text{tr}$ | tree Laplacian, adjugate, trace                                                                                                 | Part IV's deck operator.                                         |
| $\mathbb{H}, \text{Im}$                | open upper half-plane, imaginary part                                                                                           | Part III's $H, H_K$ .                                            |
| $A_i, B_i$                             | in the vertex-sum theorem, $\Phi_{G_i} = z_v A_i + B_i$                                                                         | shores $A, B$ .                                                  |
| $K$                                    | random incidence matrix in Section 91; $K_{\ell, r}$ complete bipartite                                                         | Part III's phase matrix $K$ ; Part IV's mixing variable.         |

Source 08's set of matchable pairs  $\mathcal{M}(G)$  is typeset with the host's macro and is Part IV's  $\mathcal{M}(H)$ . *The two matroid conventions.* With the same shores  $A = U$  (left) and  $B = V$  (right), source 08's  $M_G$  has one slot per left vertex and the bases  $(U \setminus I) \sqcup J$ , rank  $|U|$ ; source 09's  $M_G$  has one slot per right vertex and the bases  $I \sqcup (V \setminus J)$ , rank  $|V|$ . Each basis of one is the complement of a basis of the other, so  $M_G = (M_G)^*$  on the same ground set, and the two basis polynomials are  $F_G$  of source 08 and  $F_G$  of source 09, different polynomials under one letter (Table 8). Stability of a basis polynomial is invariant under duality, so the two bridges (Proposition 85.2 and Theorem 85.3) say the same thing. Part V's  $M_H$  is source 08's convention, with the private copies relabelled.

### 83.3 What Part VI answers

Status after Part VI, as of 29 September 2026:

- **Stability of  $\Phi_G$  is strictly stronger than real-rootedness of  $p_G$ .** Part III proved the containment and did not assert strictness (Section 39). The book  $B_m$  has real-rooted  $p_{B_m}$  for every  $m$  (Theorem 90.2) and unstable  $\Phi_{B_m}$  for  $m \geq 5$  (Theorem 84.2);  $\mathfrak{g}(3, 3, 5)$  is a second witness (equation (92.5)).
- **Part III's Research question 11** is partly answered: stability is block-local (Corollary 93.3), which reaches that question's first target, a gluing theorem more general than leaf attachment; every complete bipartite block is stable (Theorem 94.1); the tree incidence–apex graphs are classified (Theorem 84.1, also after terminal subdivision, Theorem 89.1); and all simple bipartite multi-theta graphs are classified (Theorem 84.3). Both families have treewidth at most two. The general question remains open.

- **Part II's Research question 3** gains two facts: vertex sums preserve and reflect stability, and edge subdivision can restore real-rootedness of  $p_G$  without restoring stability. Real-rootedness alone under vertex sums is not treated.
- **Part III's Research question 15** is touched, not answered:  $F_{K_{2,3}}$  is an expected determinant with independent random columns (Theorem 91.4), a mixture of determinants that the question does not exclude.
- **Source 09's own question on minimal induced obstructions** is answered negatively inside the merge (Section 92): there is no finite obstruction set even for treewidth two and maximum degree three.

Part VI refutes no statement of Parts I–V, and nothing is retracted. Parts II and III gain dated pointers at Research questions 3, 11 and 15, after Corollary 37.4, and at Part III's statement of the containment in Section 39.

### 83.4 Where the merge had to choose

- *Order.* Source 08's order of sections is kept. Source 09's Section 11 (the bridge and induced subgraphs) follows source 08's bridge, its Section 12 (vertex sums and blocks) replaces source 08's gluing proposition, and its Section 13 (stable blocks), its  $K_{3,3} \vee C_6$  example and its  $\Theta_3$  certificates follow source 08's closure section and its  $\mathfrak{I}(3, 3, 3)$  subsection. Both sources' appendices are printed as ordinary sections.
- *The gluing theorem is printed once, in source 09's form* (Lemma 93.1, Theorem 93.2, Corollary 93.3): arbitrary simple graphs, with block localization. Source 08's Proposition 10.1 ("One-vertex gluing") is its case of two bipartite graphs with aligned bipartitions, with the same formula (93.1) and the same parity and upper-half-plane argument; its closing paragraph is kept. Inheritance by induced subgraphs is proved in both (Lemma 85.5 and Lemma 85.6); both are printed, because source 08's lemma also treats matroid minors and source 09's covers non-bipartite graphs.
- *Second routes.* Even cycles: source 08's expected-determinant proof (Lemma 92.1) and source 09's Hermitian phase determinant (Proposition 94.2). The  $\mathfrak{I}(3, 3, 3)$  obstruction: source 08's discriminant, source 09's Sturm chain and its explicit Rayleigh certificate. The dual bridges (Proposition 85.2 and Theorem 85.3).
- *Merge observations* (in neither source; checked for this merge): the answer to source 09's minimal-obstruction question and the relation of source 08's books to Part II's  $\Theta_m$  (Section 90); an exact probe of source 08's Research question on tree diagonals (Section 96).
- *Research questions.* Source 08's nine questions and source 09's two stability questions continue the numbering as Research questions 44–53; source 08's first question and source 09's first ask the same and are printed once with both formulations.
- *Self-citation.* Source 08's reference "ProveIt" and source 09's became references to Parts II–IV; the pins are kept in Section 98. "The repository" is left where it means this report as the sources read it.
- *File names.* Source 08's `code/verify.py`, `data/verification_results.json` and `requirements.txt` are shipped as `code/08-ade-stability-verify.py`, `data/08-ade-stability-verification_results.json` and `data/08-ade-stability-requirements.txt`; merge notes give the shipped names, and the README gives a rerun recipe that cannot overwrite Part I's `data/verification_results.json`.



- *Bibliography and layout.* Source 08's OT is [9] and its DaiEtAl is [2]; Branden was already cited; new items [24, 25]. Its longtable ledger is set as a tabularx table and its scope box in the host's status-box style. Labels carry ade: (source 08) and ade:mat: (source 09). Part II's and Part III's Research questions 3, 11 and 15 received the labels ade:hq:preserve, ade:hq:classify and ade:hq:detmodels. Source 08's abstract and status box are printed in this section; its title page, table of contents and "Reading routes" section (which lists only its own sections) are dropped.

## 84 Research target and main conclusions

### 84.1 What is counted, and why stability is difficult

Let  $G = (U \sqcup V, E)$  be a finite simple bipartite graph. A pair  $(I, J)$ , with  $I \subseteq U$ ,  $J \subseteq V$  and  $|I| = |J|$ , is *matchable* when the induced graph on  $I \sqcup J$  has a perfect matching. The empty pair is matchable. Write  $\mathcal{M}(G)$  for these pairs. The univariate polynomial is

$$p_G(t) = \sum_{(I,J) \in \mathcal{M}(G)} t^{|I|}. \quad (84.1)$$

A pair with seven different perfect matchings still contributes only one monomial. This distinction from the ordinary matching-generating polynomial is essential throughout.

The multivariate polynomial naturally used in the repository is

$$\Phi_G(z) = \sum_{(I,J) \in \mathcal{M}(G)} (-1)^{|I|} \prod_{u \in U \setminus I} z_u \prod_{v \in V \setminus J} z_v. \quad (84.2)$$

A real polynomial is *real stable* if it is nonzero whenever every variable lies in the open upper half-plane  $\mathbb{H} = \{z \in \mathbb{C} : \text{Im } z > 0\}$ . Thus stability is a zero-exclusion statement in many complex variables, not just a statement about one diagonal.

For a matrix  $K$  supported on the edges of  $G$ , Cauchy–Binet assigns the weight  $|\det K_{I,J}|^2$  to a matchable endpoint pair. Usually that weight is neither zero nor one. The issue is to recover the Boolean support coefficients without losing the analytic zero-exclusion property. The principal construction here accomplishes this by a second-moment identity, rather than by a single faithful edge-phase matrix.

### 84.2 The exact tree criterion

For a nonempty tree  $T$ , let  $\mathcal{A}(T)$  have left vertex set  $V(T)$  and right vertex set  $E(T) \sqcup \{\star\}$ . A right vertex  $e = \{i, j\}$  is adjacent to  $i, j$ , and  $\star$  is adjacent to every vertex of  $T$ . There are  $2r$  vertices when  $r = |V(T)|$ .

**Theorem 84.1** (Main criterion). *For every finite nonempty tree  $T$ ,*

$$\Phi_{\mathcal{A}(T)} \text{ is real stable} \iff I - \frac{1}{2}A_T \geq 0 \iff \rho(A_T) \leq 2. \quad (84.3)$$

Here  $A_T$  is the adjacency matrix. The same criterion holds after subdividing the edges of  $T$  arbitrarily before forming incidence vertices, provided that the apex remains adjacent only to the original vertices.

[Added 30 September 2026, batch 63: Part VIII removes the terminal convention for a single apex over a tree: for an arbitrary apex neighbourhood  $S \subseteq V(T)$ , stability holds exactly when

every branching vertex of the subtree spanned by  $S$  is marked and the suppressed skeleton is an ADE tree (Theorem 118.2).]

The list of trees with adjacency spectral radius at most two is classical: the finite simply-laced Dynkin trees  $A, D, E$  and the affine simply-laced Dynkin trees  $\tilde{D}, \tilde{E}$ ; affine  $A$  diagrams are cycles, not trees. We give an elementary proof of the tree list in Section 88. The contribution at issue is the equivalence with matching-support stability, not the spectral classification by itself.

The proof supplies more information than a yes/no test. Let

$$L = \text{diag}(x_i) + \sum_{\{i,j\} \in E(T)} y_{ij}(e_i - e_j)(e_i - e_j)^T, \quad C_T = I - \frac{1}{2}A_T. \quad (84.4)$$

The homogeneous support polynomial  $F_T$ , defined precisely in Section 85, satisfies

$$F_T(x; y, s) = \det L + s \operatorname{tr}(\operatorname{adj}(L)C_T). \quad (84.5)$$

This is a polynomial identity for *every* tree, including those for which  $C_T$  is indefinite. Positivity of  $C_T$  is exactly what turns the identity into a stability proof. When  $C_T$  is not positive semidefinite, Section 87 gives real rational variable values at which a Rayleigh difference is negative.

### 84.3 Two sharp consequences

Let  $B_m$  be the quadrilateral book with a common edge and  $m$  internally disjoint three-edge paths between its endpoints. Then  $B_m = \mathcal{A}(K_{1,m})$ .

**Theorem 84.2** (The book transition). *The polynomial  $\Phi_{B_m}$  is real stable exactly when  $m \leq 4$ . Nevertheless,  $p_{B_m}$  has only negative real zeros for every  $m \geq 0$ . For  $m \geq 2$ ,*

$$p_{B_m}(t) = (1+t)^{m-2}(1 + (2m+3)t + (m^2 + m + 3)t^2 + t^3). \quad (84.6)$$

At  $m = 5$ , the associated basis polynomial has a Rayleigh difference  $-256$  at the all-ones point.

The threshold persists if the  $m$  paths are lengthened by even numbers of edges while the common edge is retained. The last assertion is stronger than failure of stability: it violates the ordinary positive-weight Rayleigh inequality and yields a positively correlated pair under the uniform basis measure.

A *multi-theta graph* consists of  $q \geq 2$  internally vertex-disjoint paths with the same two endpoints. Simplicity permits at most one one-edge path. Bipartiteness says that all path lengths have the same parity.

**Theorem 84.3** (Multi-theta classification). *For a finite simple bipartite multi-theta graph  $G$ :*

- (i) *if all path lengths are even,  $\Phi_G$  is real stable;*
- (ii) *if all path lengths are odd and no path has length one,  $\Phi_G$  is real stable exactly when there are two paths;*
- (iii) *if one path has length one and there are  $m$  other paths,  $\Phi_G$  is real stable exactly when  $m \leq 4$ .*

Part (i) uses a separate independent-column expected-determinant method. Part (ii) uses a matroid-minor reduction to the earlier  $\mathfrak{g}(3,3,3)$  obstruction, not a claim of a new first counterexample to real-rootedness.

### 84.4 Source 09's stability statement

Source 09 states, in its introduction (as delivered): For a finite simple graph  $G$ , let  $\Phi_G$  be its signed matching-support polynomial, defined in Section 85.2. We also prove

$$\Phi_G \text{ is real stable} \iff \Phi_B \text{ is real stable for every block } B \text{ of } G. \quad (84.7)$$

For bipartite  $G$ , this connects directly to the half-plane property of the matching-support matroid. Complete bipartite graphs and even cycles supply stable blocks of arbitrary size.

### 84.5 Relation to ProveIt and the literature

The inspected ProveIt report (this report's Parts II–IV) gives a faithful-matrix classification for bipartite cacti, a stable noncactus example  $K_{2,3}$  and its tree attachments, and further questions about stability and real-rootedness. It also develops the height-two preorder identity

$$h_{P_G}(t) = (1+t)^{|U|+|V|} p_G\left(\frac{t}{(1+t)^2}\right). \quad (84.8)$$

Its discussion credits Shivam Patel for the eight-element nonreal-root example corresponding to  $\mathfrak{J}(3, 3, 3)$ . We preserve that attribution. The general gamma-positivity developments in that report are not targets of the present article.

The half-plane-property theory of matroid basis polynomials is classical [24, 12]. We use its language, while proving the special closure and determinant facts needed here. Cactus real-rootedness has prior literature [9]. The ADE spectral classification is likewise classical, and the account of McKee and Smyth [25] provides an inspected primary source for its context. The root-polytope identifications and duality in the surrounding preorder literature also have prior work [2]; none is claimed here.

The scope is specific. The article resolves the *restricted classification problems* in Theorems 84.1 and 84.3, gives sharp new consequences within that development, and offers proofs for checking. It does not classify stability for all bipartite graphs; it does not solve the flag-polytope question; and it does not establish that no earlier matroid paper contains an equivalent formulation of these restricted results.

## 85 A homogeneous bridge to matroids and stability

### 85.1 The canonical basis polynomial

Put  $r = |U|$ . For a subset  $S$ , use  $x^S = \prod_{i \in S} x_i$ . Define

$$F_G(x; y) = \sum_{(I, J) \in \mathcal{M}(G)} x^{U \setminus I} y^J. \quad (85.1)$$

This is homogeneous of degree  $r$ , multiaffine, and has coefficients in  $\{0, 1\}$ . The coefficient of  $x^U$  is one.

**Lemma 85.1** (Canonical transversal matroid). *The sets  $(U \setminus I) \sqcup J$ , for  $(I, J) \in \mathcal{M}(G)$ , are the bases of a rank- $r$  transversal matroid  $\mathbf{M}_G$ . They are also the bases of  $[I, K]$ , where  $K$  has an independent indeterminate in every edge position and zero elsewhere, over the field of rational functions in those indeterminates.*

*Proof.* Present a transversal matroid by one set  $\{u\} \cup N_G(u)$  for each  $u \in U$ , on the ground set  $U \sqcup V$ . A chosen left element  $u$  can represent only its own row. After choosing the left elements  $U \setminus I$ , representing the remaining rows is exactly a perfect matching from  $I$  to  $J$ . This proves the first claim. For the matrix claim, expand the minor on columns  $(U \setminus I) \sqcup J$  along the identity columns. The remaining minor is  $\det K_{I,J}$ . In its Leibniz expansion, different perfect matchings produce different monomials in independent indeterminates, so cancellation cannot make a structurally nonzero minor vanish.  $\square$

**Proposition 85.2** (Inversion bridge). *The polynomials  $F_G$  and  $\Phi_G$  are real stable simultaneously. Moreover, stability implies that  $p_G$  has only negative real zeros.*

*Proof.* Directly from the definitions,

$$\Phi_G(x, z_V) = z^V F_G(x; -z_V^{-1}). \quad (85.2)$$

The transformation  $z \mapsto -1/z$  maps  $\mathbb{H}$  bijectively onto itself. The factor  $z^V$  does not vanish there, giving equivalence. For the last assertion,  $p_G(t) = F_G(\mathbf{1}; t\mathbf{1})$ . Specialization of the  $x$  variables to the real number one preserves stability or gives the zero polynomial; the latter is excluded by the constant coefficient one. Diagonalization of the  $y$  variables preserves stability. A univariate real stable real polynomial has no nonreal zeros, since a nonreal zero and its conjugate would put one zero in  $\mathbb{H}$ . Finally, the nonnegative coefficients and constant coefficient one exclude nonnegative real zeros.  $\square$

## 85.2 The dual bridge for all simple graphs (source 09)

For a finite simple graph  $G$  on  $V$ , a subset  $S$  is perfectly matchable when the induced graph  $G[S]$  has a perfect matching. Define

$$\Phi_G(z) = \sum_{S \text{ perfectly matchable}} (-1)^{|S|/2} z^{V \setminus S}. \quad (85.3)$$

This is multi-affine, real, and has leading monomial  $z^V$ . A real polynomial is *real stable* if it is nonzero whenever all variables lie in  $\mathbb{H} = \{z \in \mathbb{C} : \text{Im } z > 0\}$ . For a homogeneous matroid basis polynomial this is the usual half-plane property, after a common rotation of variables if another half-plane convention is used [12].

**Theorem 85.3** (Exact stability bridge). *For bipartite  $G = (A \sqcup B, \mathcal{E})$ ,*

$$\Phi_G(z_A, z_B) = \left( \prod_{a \in A} z_a \right) F_G((-1/z_a)_{a \in A}, (z_b)_{b \in B}). \quad (85.4)$$

*Consequently  $\Phi_G$  is real stable if and only if the basis polynomial of  $M_G$  is real stable.*

*Proof.* The term belonging to  $(I, J)$  on the right is  $(-1)^{|I|} z^{A \setminus I} z^{B \setminus J}$ , exactly the term in (85.3). The map  $z \mapsto -1/z$  is a bijection of  $\mathbb{H}$  to itself, and none of the prefactors vanishes there. Applying the same reciprocal substitution again, or using its inverse, proves both implications.  $\square$

Thus the stability problem is a half-plane-property problem for a concrete class of transversal matroids. Lorentzianity holds for every  $M_G$ , but stability is stronger and can fail. The partial reciprocal substitution is important: the positive support polynomial in the matched variables is not being asserted stable.

[Merge note: source 09's  $M_G$  is the dual of source 08's  $M_G$  (Section 83.2); Theorem 85.3 is Proposition 85.2 read through that duality. Source 09's  $\Phi_G$  agrees with Part III's (28.2) on bipartite graphs and is defined for every simple graph.]

### 85.3 The analytic facts used below

We record details to make the proof boundary transparent. The standard specialization statement follows by substituting  $a + i\varepsilon$  and letting  $\varepsilon \downarrow 0$ , using Hurwitz's theorem on the product of upper half-planes. A limit is zero-free or identically zero. Differentiation also preserves stability or gives zero: for fixed remaining variables, logarithmic derivatives in one variable have nonpositive imaginary part, and the derivative is obtained as a stable limit. These are standard closure facts in [24, 12].

For a real multiaffine polynomial  $f$ , set

$$\Delta_{ij}f = (\partial_i f)(\partial_j f) - f \partial_i \partial_j f. \quad (85.5)$$

The expression does not depend on the two variables  $z_i, z_j$ .

**Lemma 85.4** (Necessary Rayleigh inequality). *If a nonzero real multiaffine polynomial  $f$  is real stable, then  $\Delta_{ij}f(a) \geq 0$  for every real assignment  $a$  and every pair  $i, j$ .*

*Proof.* Specialize all other variables to real values. The resulting polynomial is zero or a stable bivariate polynomial  $g(z, w) = A + Bz + Cw + Dz w$  with real coefficients. If  $AD - BC > 0$ , the zero of  $g$  in the  $z$  variable is

$$z = -\frac{A + Cw}{B + Dw}, \quad \operatorname{Im} z = \frac{(AD - BC) \operatorname{Im} w}{|B + Dw|^2} > 0$$

for  $w \in \mathbb{H}$  away from the possible denominator zero. This contradicts stability. Thus  $BC - AD \geq 0$ , which is the required Rayleigh difference. The identically zero specialization has difference zero.  $\square$

Brändén's Theorem 5.6 [12] gives the converse too. The main necessity proof in this article only needs the elementary direction just proved. In particular, no numerical root-finding is part of that proof.

**Lemma 85.5** (Matroid and induced-graph minors). *Real stability of a matroid basis polynomial is preserved under matroid deletion and contraction, apart from the harmless zero-polynomial case for deleting a coloop or contracting a loop in the direct formulas. Also, if  $\Phi_G$  is stable and  $H$  is an induced subgraph, then  $\Phi_H$  is stable.*

*Proof.* For a noncoloop, deletion is specialization of its variable to zero; for a nonloop, contraction is differentiation. Loop and coloop cases reduce by a constant or linear factor. For induced graph deletion there is an even more direct formula:  $\partial_{z_v} \Phi_G = \Phi_{G-v}$ , because differentiation selects exactly the supports that leave  $v$  unmatched. Its leading monomial is nonzero. Iterate this identity.  $\square$

**Lemma 85.6** (Induced-subgraph inheritance). *For  $W \subseteq V(G)$ ,*

$$\left( \prod_{v \notin W} \partial_{z_v} \right) \Phi_G = \Phi_{G[W]}. \quad (85.6)$$

*Hence signed-support stability passes to induced subgraphs.*

*Proof.* Differentiation retains exactly the terms for which the matching support avoids every deleted vertex, and removes those vertices from the unmatched monomial. Real stability is preserved by differentiation, with zero as the only possible exception; here the full remaining monomial has coefficient one, so the derivative is not zero. The differentiation property follows from univariate Gauss–Lucas and limits, or from the standard stable polynomial closure properties [12].  $\square$

[Merge note: Lemma 85.6 is the second half of Lemma 85.5, for every simple graph.]

## 86 Forest cofactors recover Boolean matching supports

### 86.1 Incidence matrices and the single unrooted component

Fix a tree  $T$  on  $U = [r]$ . For every edge  $e = \{i, j\}$ , choose the column  $b_e = e_i - e_j$ ; reversing a column has no effect on  $b_e b_e^T$ . Let  $B$  be the matrix of these columns. The right apex variable will be denoted by  $s$ .

Consider  $I \subseteq U$ ,  $J \subseteq E(T)$  with  $|I| = |J| + 1$  and the rectangular matrix  $B_{I,J}$ . A selected edge with both endpoints in  $I$  is an *internal edge*. A selected edge with exactly one endpoint in  $I$  is a *half-edge*. A selected edge disjoint from  $I$  gives a zero column. Components below mean components of the internal-edge forest on  $I$ , including isolated vertices.

**Lemma 86.1** (Cofactor support lemma). *The pair  $(I, J \sqcup \{\star\})$  is matchable in  $\mathcal{A}(T)$  if and only if there is no zero column, every internal component has at most one half-edge, and exactly one component  $W$  has no half-edge. In this case*

$$\det[B_{I,J} \ w_I] = \pm \sum_{i \in W} w_i. \quad (86.1)$$

*The sign depends on the ordering of rows and columns, not on  $w$ . If the pair is not matchable, the determinant vanishes identically.*

*Proof.* A component with  $k$  rows has  $k - 1$  internal columns, because the underlying graph is a forest. If it has  $h$  half-edges, it has  $k - 1 + h$  selected columns other than the apex. More than  $k$  such columns cannot all be matched to its rows, so  $h \leq 1$  is necessary. The apex can supply a missing column to only one component, so exactly one component must have  $h = 0$  and all others must have  $h = 1$ .

Conversely, a component with one half-edge has a square incidence matrix with determinant  $\pm 1$ . This follows by peeling off a leaf, or by rooting the component at that half-edge and matching edges toward their unique available child rows. For an unrooted component  $W$ , the incidence matrix has rank  $|W| - 1$ . Deleting any one row gives determinant  $\pm 1$ , and its left kernel is spanned by  $\mathbf{1}_W$ . Appending  $w_W$  therefore gives determinant  $\pm \sum_{i \in W} w_i$ . Combining the rooted components with this last component proves (86.1) and also constructs a perfect matching when a chosen apex entry on  $W$  is nonzero. If there is no matching, every term in the Leibniz expansion vanishes, which proves the last assertion.  $\square$

The set  $W$  is a connected vertex set in  $T$ , and the induced graph  $T[W]$  contains exactly  $|W| - 1$  edges. This uses the uniqueness of paths in a tree: an omitted tree edge between two vertices of  $W$  could not be replaced by another path in the selected forest.

### 86.2 The uniquely determined covariance

For every connected nonempty vertex set  $W \subseteq U$ ,

$$\mathbf{1}_W^T C_T \mathbf{1}_W = |W| - |E(T[W])| = 1, \quad C_T = I - \frac{1}{2} A_T. \quad (86.2)$$

This simple Euler characteristic is the source of the spectral threshold.

**Proposition 86.2** (Covariance rigidity). *If a real symmetric matrix  $C$  satisfies  $\mathbf{1}_W^T C \mathbf{1}_W = 1$  for every nonempty connected vertex set  $W$  of  $T$ , then  $C = C_T$ .*



*Proof.* Singletons give  $C_{ii} = 1$ . An adjacent pair gives  $2 + 2C_{ij} = 1$ , hence  $C_{ij} = -1/2$ . Proceed by induction on the distance between two nonadjacent vertices  $i, j$ . Apply the hypothesis to the vertex set of their unique path. All entries between closer nonadjacent pairs are zero by induction, and the diagonal and adjacent-pair contributions already sum to one. The only remaining contribution is  $2C_{ij}$ , so it is zero.  $\square$

Thus the covariance is not selected merely because it is convenient. The coefficients corresponding to connected subtrees force it. In particular, testing positive semidefiniteness of this specific matrix has a structural reason.

### 86.3 The determinant–adjugate identity

**Theorem 86.3** (Exact Boolean coefficient identity). *For every nonempty tree  $T$ , with  $L$  as in (84.4),*

$$F_{\mathcal{A}(T)}(x; y, s) = \det L + s \operatorname{tr}(\operatorname{adj}(L)C_T). \quad (86.3)$$

*No definiteness assumption is required.*

*Proof.* Introduce an auxiliary real vector  $w$  of indeterminates and use Cauchy–Binet:

$$\begin{aligned} \det(L + sww^T) &= \sum_{\substack{I \subseteq U, J \subseteq E(T) \\ |I|=|J|}} x^{U \setminus I} y^J \det(B_{I,J})^2 \\ &\quad + s \sum_{\substack{I \subseteq U, J \subseteq E(T) \\ |I|=|J|+1}} x^{U \setminus I} y^J \det[B_{I,J} \ w_I]^2. \end{aligned} \quad (86.4)$$

The square minors in the first line are 0 or  $\pm 1$  and are nonzero exactly on matchable supports of the subdivided tree. Indeed, a forest has at most one perfect matching on any fixed induced vertex set, so there is at most one nonzero Leibniz term.

Apply to the coefficients in the second line the linear functional on quadratic polynomials in  $w$  defined by  $w_i w_j \mapsto (C_T)_{ij}$ . By Lemma 86.1, a feasible coefficient becomes  $\mathbf{1}_W^T C_T \mathbf{1}_W = 1$ ; an infeasible coefficient stays zero. The result is exactly (85.1) for  $\mathcal{A}(T)$ . On the other hand, the rank-one determinant identity gives

$$\det(L + sww^T) = \det L + s w^T \operatorname{adj}(L)w.$$

Applying the same functional gives (86.3). This argument takes place in a polynomial ring, so it remains valid when  $L$  is singular and when  $C_T$  is indefinite.  $\square$

**Remark 86.4** (What the formula is not). In general (86.3) is *not*  $\det(L + sC_T)$ . The latter would include higher powers of  $s$  and count more than one apex column. Nor does the formula supply a faithful unit-phase matrix on the edges of  $\mathcal{A}(T)$ . It therefore does not contradict the repository’s much more restrictive cactus classification.

## 87 The spectral criterion and explicit failure certificates

### 87.1 Sufficiency, including the affine boundary

**Lemma 87.1** (A positive covariance gives stability). *Let  $L = \operatorname{diag}(x_i) + \sum_e y_e b_e b_e^T$  with real vectors  $b_e$  and let  $0 \neq C \geq 0$  be real symmetric. Then*

$$\det L + s \operatorname{tr}(\operatorname{adj}(L)C) \quad (87.1)$$

*is nonzero when all  $x_i, y_e, s \in \mathbb{H}$ .*

*Proof.* For these assignments,  $\text{Im } L > 0$ , because the diagonal imaginary part is already positive definite and every other summand has nonnegative imaginary part. In particular,  $L$  is invertible. Write

$$q = \text{tr}(CL^{-1}).$$

The exact identity

$$\text{Im } L^{-1} = -L^{-*}(\text{Im } L)L^{-1} < 0 \quad (87.2)$$

implies  $\text{Im } q < 0$ , since  $C$  is nonzero and positive semidefinite. Expression (87.1) equals  $\det L(1 + sq)$ . Its vanishing would give  $s = -1/q$ , whose imaginary part is negative. This contradicts  $s \in \mathbb{H}$ .  $\square$

By Theorem 86.3,  $C_T \geq 0$  implies stability of  $F_{\mathcal{A}(T)}$ , and Proposition 85.2 transfers it to  $\Phi$ . The proof includes singular positive semidefinite matrices. There is no missing limiting argument at the affine Dynkin diagrams, where  $C_T$  has a one-dimensional kernel.

## 87.2 Necessity by a rank-one adjugate

**Theorem 87.2** (Rational negative Rayleigh certificate). *Suppose  $\rho(A_T) > 2$ . There is a vector  $v \in \mathbb{Q}_{>0}^r$  with  $v^T C_T v < 0$ . For any such vector, make the real rational assignments*

$$y_{ij} = \frac{1}{v_i v_j} \quad (ij \in E(T)), \quad (87.3)$$

$$x_i = \frac{\deg_T(i)}{v_i^2} - \sum_{j \sim i} \frac{1}{v_i v_j}, \quad s = 0. \quad (87.4)$$

Put  $c = \prod_i v_i^{-2}$ . Then, for every  $i$ ,

$$\Delta_{s, x_i} F_{\mathcal{A}(T)} = c^2 v_i^2 (v^T C_T v) < 0. \quad (87.5)$$

Consequently neither  $F_{\mathcal{A}(T)}$  nor  $\Phi_{\mathcal{A}(T)}$  is real stable.

*Proof.* A connected tree has a strictly positive Perron eigenvector for  $A_T$ . Its quadratic form under  $C_T$  is negative when the eigenvalue is greater than two. Strict inequalities and density of rational vectors give a positive rational  $v$  with the same property.

At (87.3)–(87.4),

$$L_0 = D_v^{-1} \text{Lap}(T) D_v^{-1}, \quad D_v = \text{diag}(v_i). \quad (87.6)$$

The ordinary tree Laplacian has kernel  $\mathbb{R}\mathbf{1}$ , rank  $r - 1$ , and adjugate  $\mathbf{1}\mathbf{1}^T$ . The last fact follows either from the elementary incidence cofactors of a tree or from the matrix-tree theorem with its single spanning tree. Therefore

$$\det L_0 = 0, \quad \text{adj}(L_0) = c v v^T. \quad (87.7)$$

At  $s = 0$ , Theorem 86.3 now gives

$$F = 0, \quad F_s = c v^T C_T v, \quad F_{x_i} = c v_i^2.$$

In the Rayleigh difference, the term  $FF_{s, x_i}$  vanishes, leaving (87.5). Lemma 85.4 excludes stability.  $\square$

Combining sufficiency and necessity proves the first equivalence in Theorem 84.1. The second follows because a tree is bipartite, so the adjacency spectrum is symmetric about zero, and the largest eigenvalue is the spectral radius. In fact, positive semidefiniteness of  $I - A_T/2$  already says that the largest eigenvalue is at most two.

**Example 87.3** (An entirely rational excluded case). For  $T = K_{1,5}$ , order the center first and take  $v = (2, 1, 1, 1, 1)^T$ . Then  $v^T C_T v = -1$ , all edge variables are  $1/2$ , the center variable is  $-5/4$ , and the five leaf variables are  $1/2$ . With  $s = 0$ ,  $c = 1/4$ , so  $\Delta_{s,x_0} F = -1/4$ . This certificate uses a negative center variable. Section 90 gives a stronger certificate at the all-positive all-ones point for the same graph.

### 87.3 An algorithmic form

The theorem admits exact certificates on both sides. A positive result can be certified by the ADE shape description in the next section or by an exact rational positive-semidefinite decomposition of  $C_T$ . A negative result can be certified by a positive rational vector with  $v^T C_T v < 0$ , followed by the explicit formulas above. Finding the vector numerically and rationalizing it is legitimate only when the final quadratic inequality is checked exactly. No estimated eigenvalue is needed in the final certificate.

The role of positivity is also sharp at the level of the construction: Proposition 86.2 shows that no different real symmetric covariance on the same oriented tree-incidence columns can repair an indefinite  $C_T$  while keeping every connected-support coefficient equal to one. The Rayleigh certificate then proves that the failure is not merely a defect of that particular representation.

## 88 Why the stable trees are exactly of ADE type

We now classify the trees with  $2I - A_T \geq 0$ . This is a classical spectral classification, included here with proof to make the main criterion usable without an external classification theorem. Its historical context and the finite/affine diagrams are discussed in [25].

### 88.1 The explicit list

For a tree with one trivalent vertex, its *arm lengths* are the numbers of edges from that vertex to each of its three leaves. Sort them as  $a \leq b \leq c$ .

**Theorem 88.1** (Classical tree list, in explicit form). *A nonempty finite tree  $T$  satisfies  $\rho(A_T) \leq 2$  exactly in the following cases:*

- (a) a path  $A_r$ ,  $r \geq 1$ ;
- (b) one trivalent vertex with arms  $(1, 1, c)$ ,  $c \geq 1$  (the finite diagram  $D_{c+3}$ );
- (c) one trivalent vertex with arms  $(1, 2, 2)$ ,  $(1, 2, 3)$ ,  $(1, 2, 4)$  (respectively  $E_6, E_7, E_8$ );
- (d) one trivalent vertex with arms  $(2, 2, 2)$ ,  $(1, 3, 3)$ ,  $(1, 2, 5)$  (respectively  $\widetilde{E}_6, \widetilde{E}_7, \widetilde{E}_8$ );
- (e) the four-leaf star  $K_{1,4} = \widetilde{D}_4$ ;
- (f) two trivalent vertices joined by a path of length  $d \geq 1$ , with exactly two additional leaves at each end (the diagram  $\widetilde{D}_{d+4}$ ).

The matrices  $2I - A_T$  are positive definite in (a)–(c) and positive semidefinite of corank one in (d)–(f).

*Proof.* First, a proper connected tree extension of a subtree with a strictly positive adjacency eigenvector of eigenvalue two has spectral radius greater than two. To see this directly, extend the eigenvector by zero outside the smaller graph and then put a small positive value  $\varepsilon$  at one

neighboring new vertex. The quadratic form gains a term linear in  $\varepsilon$ , while the squared norm gains only  $\varepsilon^2$ . Its Rayleigh quotient is therefore greater than two for sufficiently small  $\varepsilon > 0$ .

A vertex of degree at least five contains  $K_{1,5}$  as a subtree, whose spectral radius is  $\sqrt{5} > 2$ . A vertex of degree four contains  $K_{1,4}$ , whose positive eigenvector of eigenvalue two is two at the center and one at each leaf. The extension observation forces the whole tree to equal that star.

Suppose there are at least two trivalent vertices and none of larger degree. Choose two with no other trivalent vertex on their connecting path, and include the two neighbors off the path at each end. This gives a double-fork subtree: the connecting path with four attached leaves. The vector that is two on the path and one on the four leaves is a positive eigenvector of eigenvalue two. Again, the tree must equal that subtree. This proves (f) and excludes further branching.

With no branching the tree is a path, and  $2I - A_T$  is positive definite. One elementary verification uses successive Schur pivots  $2, 3/2, 4/3, \dots, (r+1)/r$ , all positive. With one trivalent vertex, eliminate each of the three path arms. For an arm of length  $a$ , the endpoint entry of the inverse of its path matrix  $2I - A_{A_a}$  is  $a/(a+1)$ ; the same Schur recursion proves this. The remaining center pivot is nonnegative exactly when

$$2 - \frac{a}{a+1} - \frac{b}{b+1} - \frac{c}{c+1} \geq 0, \quad \text{or equivalently} \quad \frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1} \geq 1. \quad (88.1)$$

If  $a \geq 3$ , this is impossible. If  $a = 2$ , it forces  $b = c = 2$ . If  $a = 1$ , then  $b = 1$  permits every  $c \geq 1$ ;  $b = 2$  permits  $c = 2, 3, 4, 5$ ;  $b = 3$  permits only  $c = 3$ ; and  $b \geq 4$  is impossible. These are exactly (b)–(d). The strict and equality cases of the center pivot give the stated ranks. The positive eigenvectors already displayed handle (e)–(f), or one may eliminate their four leaves and use a path Schur calculation.  $\square$

| $r$                         | 1 | 2 | 3 | 4 | 5 | 6 | 7  | 8  | 9  |
|-----------------------------|---|---|---|---|---|---|----|----|----|
| All unlabeled trees         | 1 | 1 | 1 | 2 | 3 | 6 | 11 | 23 | 47 |
| Stable incidence–apex types | 1 | 1 | 1 | 2 | 3 | 4 | 5  | 5  | 4  |

Table 9: Exact finite enumeration checked by the accompanying program. The stable count for every  $r \geq 10$  is three, by the classification:  $A_r, D_r, \bar{D}_{r-1}$ .

## 88.2 Consequences for the graph family

The restriction is on the *auxiliary tree*  $T$ , not on the number of cycles of  $\mathcal{A}(T)$ . Even a path  $T = A_r$  gives arbitrarily large graphs with overlapping cycles and stable support polynomials. Thus the theorem goes well beyond cactus graphs in a controlled family.

On the other hand, the first excluded auxiliary trees occur at six vertices. There are six unlabeled trees on six vertices; only  $A_6, D_6, E_6, \bar{D}_5$  qualify. One excluded type is  $K_{1,5}$ , which produces the five-page book. The other is a proper extension of  $K_{1,4}$ . Their exclusion follows from explicit certificates, not from an extrapolation of Table 9.

The equivalence also distinguishes three logically different statements. A tree may fail the spectral test and hence multivariate stability; its univariate diagonal may nevertheless remain real-rooted. The book family proves this separation exactly. Finally, even on the stable side, the polynomial need not admit the repository’s faithful vertex-indexed determinant. Stability is the broad analytic property; a faithful determinant is a much narrower representation.

## 89 Subdividing between apex neighbors

### 89.1 The terminal-skeleton construction

Let  $T$  be a tree, whose vertices will be called *terminals*. Replace each edge of  $T$  by a path of positive length, using internally disjoint paths, to obtain a tree  $\widehat{T}$ . The terminal set is still  $V(T) \subseteq V(\widehat{T})$ . Define  $\mathcal{A}(\widehat{T}; T)$  to have left vertices  $V(\widehat{T})$ , right vertices  $E(\widehat{T}) \sqcup \{\star\}$ , and the usual incidence edges. The apex is adjacent to terminals only. In particular, inserting a new left vertex does *not* make it a new apex neighbor.

Embed  $C_T$  into a matrix  $\widehat{C}_T$  on  $V(\widehat{T})$  by putting zeros in every row and column belonging to a nonterminal. Use the incidence vectors of  $\widehat{T}$  to define  $\widehat{L}$ .

**Theorem 89.1** (Subdivision invariance of the criterion). *For every terminal-skeleton construction above,*

$$F_{\mathcal{A}(\widehat{T}; T)} = \det \widehat{L} + s \operatorname{tr}(\operatorname{adj}(\widehat{L}) \widehat{C}_T). \quad (89.1)$$

Furthermore,

$$\Phi_{\mathcal{A}(\widehat{T}; T)} \text{ is real stable} \iff \rho(A_T) \leq 2. \quad (89.2)$$

*Proof of the identity and sufficiency.* Use the same forest-cofactor argument as before. A potentially nonzero apex minor has a unique unrooted component  $W$  of selected rows. Because the apex vector is supported only on terminals, its cofactor expression is  $\pm \sum_{i \in W \cap V(T)} w_i$ .

If  $S = W \cap V(T)$  is empty, the endpoint pair is not matchable using the apex and the quadratic functional gives zero. If  $S$  is nonempty, it is a connected vertex set in the skeleton  $T$ . Indeed, a connected set in  $\widehat{T}$  containing two terminals contains their entire unique path, including every intervening terminal. Therefore

$$\mathbf{1}_W^T \widehat{C}_T \mathbf{1}_W = \mathbf{1}_S^T C_T \mathbf{1}_S = 1.$$

This proves (89.1) coefficient by coefficient. If  $C_T \geq 0$ , its zero extension is positive semidefinite and nonzero, so Lemma 87.1 proves stability.  $\square$

For necessity we record the precise minor operation. It is important not to assume that arbitrary graph edge contraction preserves the support polynomial.

**Lemma 89.2** (Generic path shortening). *Suppose a bipartite graph contains a path  $a - c - b - d$ , where  $a, b$  are left vertices and  $c, d$  are right vertices. Assume  $b$  has only neighbors  $c, d$ ,  $c$  has only neighbors  $a, b$ , and there is no edge  $a - d$ . In the canonical matroid, contract the right element  $c$  and delete the left identity element  $b$ . The resulting matroid is the canonical matroid of the graph obtained by removing  $b, c$  and replacing the path by the edge  $a - d$ , leaving all other incidences unchanged.*

*Proof.* Use the generic representation  $[I \ K]$  of Lemma 85.1. The column for  $c$  is  $\alpha e_a + \beta e_b$ , with independent nonzero indeterminates. In the quotient by this column,  $e_b = -(\alpha/\beta)e_a$ . Deleting the identity element  $b$  leaves the other identity elements as a basis of the quotient. The only other column affected is  $d$ , where its  $b$  entry  $\gamma$  becomes the new  $a$  entry  $-\alpha\gamma/\beta$ .

There is no preexisting  $a - d$  entry to combine with it. The new entry is algebraically independent of all remaining edge entries: the variable  $\gamma$  occurred nowhere else and can be solved from the new entry after adjoining the discarded nonzero parameters  $\alpha, \beta$ . Thus the resulting zero pattern has a generic representation of exactly the shortened graph, as claimed.  $\square$

*Completion of Theorem 89.1.* Apply Lemma 89.2 repeatedly to every inserted nonterminal of  $\widehat{T}$ . Such a vertex has degree two and is not adjacent to the apex, and the tree condition ensures there is no preexisting shortcut edge. We obtain  $M_{\mathcal{A}(T)}$  as a matroid minor. If the subdivided polynomial were stable, Lemma 85.5 and the inversion bridge would force stability of  $\Phi_{\mathcal{A}(T)}$ . Theorem 87.2 then forces  $\rho(A_T) \leq 2$ .  $\square$

## 89.2 Why the terminal convention matters

Subdividing and then joining the apex to *all* new left vertices is a different construction: it replaces the skeleton itself by  $\widehat{T}$  and tests  $\rho(A_{\widehat{T}})$ . The theorem does not say that this modified operation is neutral. The invariant is the tree connecting the designated apex neighbors after suppressing the new nonterminals. This distinction is particularly visible for books: the common spine endpoints stay fixed, while a page path can be lengthened without adding new apex edges.

[Added 30 September 2026, batch 63: when the apex neighbourhood is not the terminal set, an unmarked vertex of degree at least three in the subtree spanned by the neighbours forbids stability (Part VIII, Theorem 120.2), and otherwise the suppressed skeleton decides (Theorem 118.2).]

## 90 Quadrilateral books: four stable pages, all-real diagonals

### 90.1 The graph and its spectral threshold

The graph  $B_m$  has left vertices  $u, b_1, \dots, b_m$  and right vertices  $v, a_1, \dots, a_m$ . Its edges are

$$uv, \quad ua_i, \quad a_ib_i, \quad b_iv \quad (1 \leq i \leq m). \quad (90.1)$$

For  $m = 0$  it is a single edge. For  $m = 1$  it is a four-cycle. For  $m \geq 2$  the pages overlap in the edge  $uv$ , so it is not a cactus.

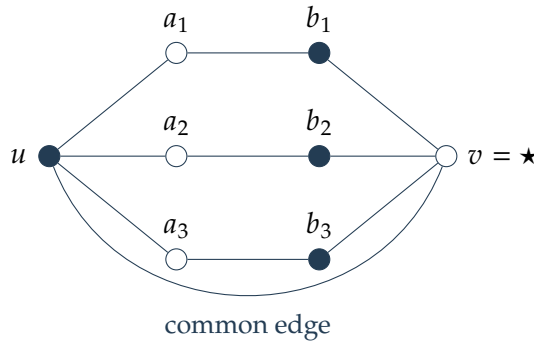


Figure 3: The three-page book  $B_3 = \mathcal{A}(K_{1,3})$ . Filled and open vertices indicate the two bipartition classes. Each displayed three-edge path, together with the common edge, is a page.

The matrix forced by the cofactor identity is

$$C_{K_{1,m}} = \begin{pmatrix} 1 & -\frac{1}{2}\mathbf{1}^T \\ -\frac{1}{2}\mathbf{1} & I_m \end{pmatrix}. \quad (90.2)$$

Its Schur complement is  $1 - m/4$ . Hence the matrix is positive semidefinite exactly when  $m \leq 4$ , proving the stability threshold in Theorem 84.2. By Theorem 89.1, the same threshold holds for a common edge together with  $m$  odd paths of any length at least three.



[Merge note: in Part II's labelling,  $B_m$  is  $\Theta_m$  plus the edge  $d b_0$  joining its endpoints: source 08's  $u, v, a_i, b_i$  are Part II's  $d, b_0, b_i, a_i$ , with the letters  $a_i, b_i$  exchanged (Section 16 draws  $\Theta_m$ ). The formula of Proposition 90.1 is Part II's (13.4) plus  $u$ :  $p_{B_m}(u) = p_{\Theta_m}(u) + u$ , the single extra support being the edge  $d b_0$  itself.]

## 90.2 A closed form for the entire univariate family

**Proposition 90.1** (Book support polynomial). *For  $m \geq 2$ , formula (84.6) holds. The two smaller cases are  $p_{B_0}(t) = 1 + t$  and  $p_{B_1}(t) = 1 + 4t + t^2$ .*

*Proof.* In the identity of Theorem 86.3, put every left variable equal to one and every right variable, including the apex, equal to  $t$ . With the center first, the matrix  $L$  is

$$L = \begin{pmatrix} 1 + mt & -t\mathbf{1}^T \\ -t\mathbf{1} & (1 + t)I_m \end{pmatrix}. \quad (90.3)$$

Let  $d = 1 + t$  and  $S = 1 + mt/d$ . Its determinant is  $d^m S$ , and block inversion gives

$$\text{tr}(C_{K_{1,m}} L^{-1}) = \frac{m}{d} + \frac{1 - mt/d^2}{S}. \quad (90.4)$$

For example, the center diagonal contributes  $1/S$ , the leaf diagonals contribute  $m/d + mt^2/(d^2 S)$ , and the two off-diagonal blocks of  $C$  together subtract  $mt/(dS)$ , which combine to (90.4). Thus

$$\begin{aligned} p_{B_m}(t) &= d^m \left[ S + t \left( \frac{mS}{d} + 1 - \frac{mt}{d^2} \right) \right] \\ &= d^{m-2} (1 + (2m + 3)t + (m^2 + m + 3)t^2 + t^3). \end{aligned}$$

The calculation is an identity of rational functions that simplifies to a polynomial, and therefore holds at the values initially excluded by block inversion. At  $m = 0, 1$  the apparent negative powers of  $d$  cancel, giving the two separately stated polynomials.  $\square$

**Theorem 90.2** (Every book diagonal is real-rooted). *For every  $m \geq 0$ , all zeros of  $p_{B_m}$  are real and negative. For  $m \geq 2$ , its three cubic-factor zeros are distinct, and the multiplicity of  $-1$  is exactly  $m - 2$ .*

*Proof.* Put

$$Q_m(t) = 1 + (2m + 3)t + (m^2 + m + 3)t^2 + t^3.$$

The cubic discriminant is the exact expression

$$\text{disc}_t Q_m = m^2(8m^3 + 13m^2 + 54m - 27) > 0 \quad (m \geq 2). \quad (90.5)$$

For a transparent positivity check, put  $m = k + 2$ ; the factor in parentheses becomes  $8k^3 + 61k^2 + 202k + 197$ . A real cubic with positive discriminant has three distinct real roots. Its positive coefficients and constant coefficient one exclude zero and positive roots, so all three are negative. Since  $Q_m(-1) = m(m - 1) > 0$ , none equals  $-1$ . The cases  $m = 0, 1$  follow directly from their displayed formulas.  $\square$

Every  $B_m$  is chordal bipartite: a cycle using the common edge is a four-cycle, while a cycle using two page paths has the common edge as a chord. For  $m \geq 1$  it has treewidth two, as seen by gluing the width-two decompositions of the four-cycle pages along the common edge. Consequently this is an infinite connected, chordal-bipartite, treewidth-two family in which the diagonal is real-rooted for every size but multivariate stability fails from  $m = 5$  onward. No general first separation between real-rootedness and stability is claimed; the result here is this explicit sharp family and its exact threshold.

### 90.3 A positive-weight Rayleigh failure and its probability law

Let  $a$  be the left center variable and  $b$  the right apex variable, and set all other variables of  $F_{B_m}$  equal to one. The same block calculation, now with edge variables one, gives

$$F_{B_m}(a, b) = 2^m \left[ a + \frac{m}{2} + b \left( \frac{m}{2} \left( a + \frac{m}{2} \right) + 1 - \frac{m}{4} \right) \right]. \quad (90.6)$$

Its Rayleigh difference is independent of  $a, b$ :

$$\Delta_{a,b} F_{B_m} = 4^{m-1}(4 - m). \quad (90.7)$$

This also holds for  $m = 0$ , with the expression interpreted as the rational number  $4^{-1} \cdot 4 = 1$ .

For a homogeneous multiaffine basis polynomial  $F$  with nonnegative coefficients, the uniform basis measure has

$$\text{Cov}(\mathbf{1}_{i \in B}, \mathbf{1}_{j \in B}) = \frac{F(\mathbf{1})F_{ij}(\mathbf{1}) - F_i(\mathbf{1})F_j(\mathbf{1})}{F(\mathbf{1})^2} = -\frac{\Delta_{ij}F(\mathbf{1})}{F(\mathbf{1})^2}. \quad (90.8)$$

For books,

$$Z_m = F_{B_m}(\mathbf{1}) = 2^{m-2}(m^2 + 3m + 8), \quad (90.9)$$

and hence

$$\text{Cov}(\mathbf{1}_{u \in B}, \mathbf{1}_{\star \in B}) = \frac{4(m - 4)}{(m^2 + 3m + 8)^2}. \quad (90.10)$$

**Example 90.3** (Five pages, with all counts exposed). For  $B_5$  there are 384 bases. The numbers containing the center, the apex, and both are respectively 112, 272, 80. Thus

$$\Delta_{u,\star} F = 112 \cdot 272 - 384 \cdot 80 = -256, \quad \text{Cov}(\mathbf{1}_{u \in B}, \mathbf{1}_{\star \in B}) = \frac{1}{576} > 0.$$

The univariate polynomial is nevertheless

$$p_{B_5}(t) = 1 + 16t + 75t^2 + 140t^3 + 115t^4 + 36t^5 + t^6 = (1 + t)^3(1 + 13t + 33t^2 + t^3).$$

The random objects here are bases of  $M_{B_5}$ , not uniformly random perfect matchings. In the support correspondence, the left center is in the basis when it is *not* in the matched left endpoint set, whereas the apex is in the basis when it *is* in the matched right endpoint set. This distinction prevents an incorrect interpretation of the covariance as a matching-multiplicity statement.

## 91 Independent columns and all even-path multi-theta graphs

The tree covariance construction does not include all stable multi-theta graphs. When the two endpoints are in the same bipartition class, a separate expected-determinant argument works for arbitrarily many paths. We give the analytic closure lemma first, because an arbitrary average of stable polynomials need not be stable.

### 91.1 A squarefree expected-determinant lemma

**Lemma 91.1** (A differential stability operation). *If  $f$  is stable, then  $f + y\partial_z f$ , with  $y$  a new variable, is stable.*

*Proof.* Fix all original variables other than  $z$  in  $\mathbb{H}$ . The resulting nonzero polynomial in  $z$  has roots outside  $\mathbb{H}$ . At  $z \in \mathbb{H}$ , its logarithmic derivative is a sum of terms  $1/(z - \alpha)$ , each of negative imaginary part. It is either zero because the polynomial is constant, or has negative imaginary part. In the nonconstant case,  $f + yf_z = f(1 + yf_z/f)$  could vanish only at  $y = -1/(f_z/f)$ , which is in the lower half-plane. The constant case is immediate. This applies at every upper-half-plane assignment.  $\square$

**Theorem 91.2** (Independent-column expectation). *Let  $a_1, \dots, a_c$  be independent real random column vectors in  $\mathbb{R}^r$ , with finite second moments. Then*

$$P(x; y) = \mathbb{E} \det \left( \text{diag}(x_i) + \sum_{j=1}^c y_j a_j a_j^T \right) \quad (91.1)$$

*is real stable. It has coefficient one on  $\prod_i x_i$ .*

*Proof.* Let  $C_j = \mathbb{E}(a_j a_j^T) \geq 0$ . The polynomial

$$H(x; z) = \det \left( \text{diag}(x_i) + \sum_j z_j C_j \right)$$

is stable, because the imaginary part of its matrix is positive definite at upper-half-plane assignments. Apply Lemma 91.1 once for each variable  $z_j$ , then specialize all  $z_j$  to zero. The resulting polynomial is stable or zero, and is

$$\prod_{j=1}^c (1 + y_j \partial_{z_j}) H(x; z) \Big|_{z_1=\dots=z_c=0}. \quad (91.2)$$

Its  $x_1 \cdots x_r$  coefficient is one, so it is not zero.

It remains to identify it with (91.1). A rank-one update  $a_j a_j^T$  appears at most linearly in the determinant as a function of  $y_j$ . For a squarefree set of selected columns, the coefficient is multilinear in their outer products. Independence permits replacing each outer product by its expectation. This is exactly the squarefree coefficient of  $\prod z_j$  selected by (91.2). Equivalently, expand squared minors by Cauchy–Binet and take the expectation of the product of the two entries contributed by each column. Only that column's second moment is used.  $\square$

This lemma is an elementary instance of the established stability machinery around positive-semidefinite determinants and differential operators [24, 12]. Its use here is to realize Boolean support coefficients for a particular graph family. Independence of the columns and the squarefree truncation are essential; the proof is not a claim that all convex combinations preserve stability.

## 91.2 The random incidence construction

Let  $G$  consist of  $q \geq 2$  internally disjoint paths of lengths  $2\ell_1, \dots, 2\ell_q$ , each joining the left endpoint  $u$  to the left endpoint  $v$ . Every right vertex lies between two consecutive left vertices along a path. Orient each path from  $u$  to  $v$ .

Create a random matrix  $K$  with rows indexed by left vertices and columns by right vertices. On an ordinary path column between consecutive left vertices  $a, b$ , use  $e_a - e_b$ . On the *first* right

column of path  $i$ , instead use

$$e_u + \varepsilon_i e_b, \quad \Pr(\varepsilon_i = 1) = \frac{1 - 1/\sqrt{2}}{2}, \quad \Pr(\varepsilon_i = -1) = \frac{1 + 1/\sqrt{2}}{2}. \quad (91.3)$$

The signs are independent, so  $\mathbb{E}\varepsilon_i = -1/\sqrt{2}$  and  $\varepsilon_i^2 = 1$ . When  $\ell_i = 1$ , the first column is also the only column on the path; the construction still applies.

**Lemma 91.3** (Expected minor weights are exactly Boolean). *For every pair of equally sized row and column sets  $I, J$ ,*

$$\mathbb{E} \det(K_{I,J})^2 = \begin{cases} 1, & (I, J) \in \mathcal{M}(G), \\ 0, & (I, J) \notin \mathcal{M}(G). \end{cases} \quad (91.4)$$

*Proof.* An unsupported minor vanishes structurally. Suppose the support is matchable. It cannot contain three complete endpoint-to-endpoint paths. Indeed, for  $k \geq 3$  complete paths, the right vertices on those paths number  $\sum \ell_i$ , while their left neighbors number

$$2 + \sum (\ell_i - 1) = \sum \ell_i + 2 - k < \sum \ell_i.$$

Hall's condition for matching those columns fails. Every cycle of a multi-theta graph uses two complete paths, so a matchable selected support has at most one cycle.

A forest support has at most one perfect matching. The corresponding determinant then has one term of magnitude one, giving expectation one. A support with one cycle has either one perfect matching, covered by the same argument, or two. In the two-matching case, the matchings agree off the cycle and alternate on it. The cycle consists of two complete paths, say  $i, j$ . The squared magnitudes of the two determinant terms are both one. Their product depends on the two random first-column signs  $\varepsilon_i, \varepsilon_j$  and on no other random sign with odd exponent.

If both special columns were ordinary oriented-incidence columns ( $\varepsilon_i = \varepsilon_j = -1$ ), the two terms would cancel: they are the two terms of a cycle-incidence determinant. Therefore their product, as a function of the actual signs, is  $-\varepsilon_i \varepsilon_j$ . Its expectation is  $-1/2$ . The expected squared sum is consequently  $1 + 1 + 2(-1/2) = 1$ .  $\square$

**Theorem 91.4** (Even multi-theta stability). *Every simple multi-theta graph whose paths all have even length has real stable signed matching-support polynomial.*

*Proof.* Cauchy–Binet and Lemma 91.3 give

$$F_G(x; y) = \mathbb{E} \det(\text{diag}(x) + K \text{diag}(y) K^T).$$

The random columns of  $K$  are independent, with the nonspecial columns viewed as constant random variables. Theorem 91.2 proves stability of  $F_G$ , and Proposition 85.2 transfers it to  $\Phi_G$ .  $\square$

**Example 91.5** (Arbitrarily many even paths). The graph  $K_{2,q}$  has  $q$  paths of length two. [Merge note: it is also the case  $\ell = 2$  of source 09's Theorem 94.1, which proves every  $K_{\ell,r}$  stable.] Its support polynomial is

$$p_{K_{2,q}}(t) = 1 + 2qt + \binom{q}{2} t^2.$$

The theorem strengthens real-rootedness of this quadratic to multivariate stability, for every  $q$ . More generally, it permits any positive even length on each path, not only length two. Exact examples are

$$p_{\mathfrak{S}(2,2,4)} = 1 + 8t + 16t^2 + 4t^3, \quad p_{\mathfrak{S}(4,4,4)} = 1 + 12t + 51t^2 + 92t^3 + 63t^4 + 6t^5.$$

The examples illustrate the theorem, but their finite verification is not its proof.

## 92 The full simple bipartite multi-theta classification

### 92.1 The single-cycle case

**Lemma 92.1** (Even cycles are stable). *If  $G$  is a simple even cycle, then  $\Phi_G$  is real stable.*

*Proof.* Index the cycle so that each right column lies between two consecutive left vertices. Use oriented-incidence columns except at one right vertex, where the two entries are  $1, \varepsilon$  with  $\Pr(\varepsilon = 1) = 1/4$  and  $\Pr(\varepsilon = -1) = 3/4$ . Every proper matchable square support is a forest and has a unique matching, hence squared determinant one. The full support has two alternating matchings. Their product is  $\varepsilon$ : at  $\varepsilon = -1$  the oriented incidence determinant cancels. Its expected squared determinant is  $2 + 2\mathbb{E}\varepsilon = 1$ . Thus the expected determinant is  $F_G$ , and Theorem 91.2 proves stability. This is an alternative short proof of a special case already included in the cactus results of the repository and the surrounding literature.  $\square$

**Remark 92.2** (Second proof of Lemma 92.1, source 09). Source 09's Proposition 94.2 (Section 94) proves the same statement with a Hermitian adjacency matrix whose cycle phase has real part  $(-1)^r/2$ , so that  $\Phi_{C_{2r}}$  is a vertex determinant; it is Part II's cactus construction for one cycle, as source 09 says.

### 92.2 The credited three-odd-path obstruction

The graph  $\vartheta(3, 3, 3)$  has left vertices  $u, b_1, b_2, b_3$ , right vertices  $v, a_1, a_2, a_3$ , and edges  $ua_i, a_ib_i, b_iv$  for  $1 \leq i \leq 3$ . It is the three-page book *with its common edge removed*. Its polynomial is

$$p_{\vartheta(3,3,3)}(t) = 1 + 9t + 24t^2 + 16t^3 + t^4. \quad (92.1)$$

For completeness, its coefficients can be checked without numerical search. There are nine edges. For a two-element left set containing  $u$ , there are five matchable two-element right sets, giving  $3 \cdot 5$ ; for a left set  $\{b_i, b_j\}$  there are three, giving  $3 \cdot 3$ . Every three-element left set can be matched to every three-element right set, giving  $4 \cdot 4 = 16$ . The full set is matchable and contributes one.

The exact discriminant is

$$\text{disc}_t(1 + 9t + 24t^2 + 16t^3 + t^4) = -5243 < 0. \quad (92.2)$$

A real polynomial all of whose roots are real has nonnegative discriminant, so this is a nonreal-root obstruction. It follows from Proposition 85.2 that  $\Phi_{\vartheta(3,3,3)}$  is not stable. The repository credits the corresponding earlier preorder counterexample to Shivam Patel. Equation (92.1) is reproduced as an exact input to the classification, not as a new first disproof.

**Source 09's certificates for the same graph.** Source 09 prints the same obstruction in Part II's labelling (Section 94.2), with a Sturm chain for the diagonal and an exact Rayleigh certificate for the multivariate polynomial. As delivered: Let  $\Theta_3$  have endpoints 0, 1 and three internally disjoint length-three paths

$$0 - 2 - 3 - 1, \quad 0 - 4 - 5 - 1, \quad 0 - 6 - 7 - 1.$$

The repository credits the first preorder real-rootedness counterexample based on this graph to Shivam Patel, not to its own later report (Part II, Theorem 13.3; [11]). We do not claim the counterexample as new. Direct support enumeration gives

$$p_{\Theta_3}(u) = 1 + 9u + 24u^2 + 16u^3 + u^4. \quad (92.3)$$

A positively rescaled Sturm chain is

$$\begin{aligned} q_0 &= u^4 + 16u^3 + 24u^2 + 9u + 1, \\ q_1 &= 4u^3 + 48u^2 + 48u + 9, \\ q_2 &= 144u^2 + 165u + 32, \\ q_3 &= 4557u + 1120, \\ q_4 &= -1. \end{aligned}$$

At  $-\infty$  the signs are  $+, -, +, -, -$ , giving three variations; at  $+\infty$  they are  $+, +, +, +, -$ , giving one. Hence the quartic has exactly two real roots. The other two are nonreal.

[Merge note: source 09 then derives  $h_{P_{\Theta_3}}$  and checks its ultra-log-concavity at  $k = 1$ ; that display is Part V's (77.2) and is not repeated.]

### 92.3 A small exact failure of multivariate stability

For a multi-affine polynomial  $f$ , its Rayleigh difference is

$$\Delta_{ij}f = (\partial_i f)(\partial_j f) - f \partial_i \partial_j f.$$

If  $f$  is real stable, this expression is nonnegative at every real assignment of the other variables. One can see the necessity directly: a real bivariate specialization has the form  $axy + bx + cy + d$ , and its potential root  $y = -(bx + d)/(ax + c)$  has imaginary part  $(ad - bc) \operatorname{Im}(x)/|ax + c|^2$ . Stability therefore forces  $bc - ad \geq 0$ . Real specializations follow by a limit; if the specialization is zero the Rayleigh difference is zero. This is the necessary direction of the multi-affine criterion of [12].

For  $f = \Phi_{\Theta_3}$  and the endpoints  $i = 0, j = 1$ , set

$$(z_2, z_3, z_4, z_5, z_6, z_7) = (3, 2, -2, -2, 2, 2). \quad (92.4)$$

The resulting bivariate polynomial is

$$f(z_0, z_1) = 45z_0z_1 - 27z_0 - 18z_1 + 11,$$

so  $\Delta_{01}f = (-27)(-18) - 45 \cdot 11 = -9$ . The included standard-library program recomputes this integer directly from feasible vertex subsets. Thus the corresponding matroid polynomial is Lorentzian but not stable. Negative coordinates in this certificate are legitimate: real stability requires the Rayleigh inequalities at *all* real assignments, not only positive activities.

### 92.4 Proof of the classification

*Proof of Theorem 84.3.* If the common endpoints belong to the same bipartition class, all paths are even, and Theorem 91.4 applies.

Suppose all paths are odd and none has length one. With two paths the graph is an even cycle, so Lemma 92.1 applies. With at least three paths, choose three and delete all internal vertices of the others. This is an induced subgraph. Each remaining odd path has length at least three. While a path has length at least five, apply Lemma 89.2 at its first two internal vertices, starting at the left common endpoint. Those two vertices have degree two and there is no existing shortcut edge. The operation decreases its length by two. Repeating gives the canonical matroid of  $\mathfrak{g}(3, 3, 3)$  as a minor. A stable original basis polynomial would force a stable basis polynomial on that minor, contradicting (92.2).



Finally, suppose there is one one-edge path and  $m$  other paths. Every other path is odd of length at least three. This graph is exactly the terminal-skeleton construction of a star with  $m$  leaves: the star center is the left spine endpoint, its leaves are the last left vertices on the page paths, and the right apex is the other spine endpoint. Lengths greater than three come from nonterminal subdivisions of the star arms. Theorem 89.1 and the Schur complement  $1 - m/4$  prove the criterion  $m \leq 4$ .  $\square$

| Path pattern                            | Exact stability condition               |
|-----------------------------------------|-----------------------------------------|
| All paths even                          | Always stable, with any number of paths |
| All paths odd, no direct edge           | Stable exactly for two paths            |
| One direct edge and $m$ other odd paths | Stable exactly for $m \leq 4$           |

Table 10: Classification of simple bipartite multi-theta graphs. The distinction between a direct edge and a lengthened path is essential.

## 92.5 Subdivision can hide a diagonal obstruction

The minor proof cannot be replaced by a blanket claim that all odd three-path diagonals are nonreal-rooted. For example,

$$p_{\mathfrak{S}(3,3,5)}(t) = 1 + 11t + 41t^2 + 59t^3 + 25t^4 + t^5 \quad (92.5)$$

has five negative real roots, although its multivariate polynomial is not stable by the classification. A completely rational root certificate is obtained from the following five disjoint open intervals:

$$(-23, -22), \quad (-2, -1), \quad (-1/2, -2/5), \quad (-1/3, -1/4), \quad (-1/4, -1/5). \quad (92.6)$$

The signs at their endpoints alternate within each interval. More explicitly, the values at  $-23, -22, -2, -1, -1/2, -2/5, -1/3, -1/4, -1/5$  are respectively

$$-136734, 94139, 39, -4, -3/32, 43/3125, 2/243, -13/1024, 24/3125.$$

The intermediate value theorem gives five distinct roots, exhausting the degree. Thus even an odd-path graph whose obstruction becomes visible through a matroid minor can have an entirely real-rooted diagonal. This is a second reason to formulate the classification genuinely multivariately.

## 92.6 Infinitely many induced-minimal obstructions (merge observation)

Source 09 asks (Research question 53) whether the unstable bipartite graphs have a finite set of induced-minimal obstructions within a fixed treewidth or maximum-degree class. Theorem 84.3(ii) answers this negatively, although neither source says so. Every  $\mathfrak{S}(a, b, c)$  with  $a, b, c$  odd and at least three is bipartite, subcubic, of treewidth two, and unstable. Every proper induced subgraph of it is a cactus: deleting an internal vertex of one path leaves a cycle with a pendant path, deleting an endpoint leaves a tree, and induced subgraphs of cacti are cacti. Bipartite cacti have stable  $\Phi$  by Part II's Theorem 19.3. Hence each such  $\mathfrak{S}(a, b, c)$  is an induced-minimal unstable graph, and there are infinitely many of them in the class of subcubic graphs of treewidth two. Source 09's same question also names subdivisions of theta graphs as “a useful first target” of its first question; Theorem 84.3 is exactly that classification.

## 93 Closure, weights, and preorder consequences

### 93.1 Articulation gluing and tree attachments

*Source 08 proves the one-vertex gluing theorem for two bipartite graphs with aligned bipartitions (its Proposition 10.1). Source 09 proves the same theorem for all simple graphs, with the same formula and argument, and adds block localization. It is printed once, in source 09's form (as delivered, its Section 12).*

Suppose  $G_1, G_2$  have exactly one common vertex  $v$ , and no edges run between their other vertices. Write  $G = G_1 \vee_v G_2$  and

$$\Phi_{G_i} = z_v A_i + B_i, \quad A_i = \Phi_{G_i - v}.$$

**Lemma 93.1** (Support-exact vertex sum). *For arbitrary finite simple graphs in this configuration,*

$$\begin{aligned} \Phi_G &= z_v A_1 A_2 + B_1 A_2 + A_1 B_2 \\ &= \Phi_{G_1} A_2 + A_1 \Phi_{G_2} - z_v A_1 A_2. \end{aligned} \tag{93.1}$$

*Proof.* A support avoiding  $v$  is matchable exactly when its restrictions to the two pieces are matchable; this gives the first term. A support containing  $v$  must match  $v$  to a vertex in exactly one piece. In that piece the number of selected vertices other than  $v$  is odd; in the other it is even. These parities identify the piece uniquely from the support, irrespective of the matching witness. Therefore the two remaining cases are disjoint and contribute  $B_1 A_2$  and  $A_1 B_2$ . The sign exponents add and the unmatched monomials multiply, establishing the formula without counting any support twice.  $\square$

**Theorem 93.2** (Stability under a one-vertex sum). *The polynomial  $\Phi_{G_1 \vee_v G_2}$  is real stable if and only if both  $\Phi_{G_1}$  and  $\Phi_{G_2}$  are real stable.*

*Proof.* Necessity follows from induced-subgraph inheritance. For sufficiency, fix every variable other than  $z_v$  in  $\mathbb{H}$ . Each  $A_i$  is a nonzero stable polynomial, so  $A_i \neq 0$  at the chosen values. Since  $z_v A_i + B_i$  has no zero with  $z_v \in \mathbb{H}$ , its sole zero  $-B_i/A_i$  lies outside  $\mathbb{H}$ . Thus

$$\operatorname{Im}(B_i/A_i) \geq 0.$$

By (93.1),

$$\frac{\Phi_G}{A_1 A_2} = z_v + B_1/A_1 + B_2/A_2.$$

The right-hand side has strictly positive imaginary part for  $z_v \in \mathbb{H}$ . It cannot vanish. This proves stability.  $\square$

**Corollary 93.3** (Exact block localization). *A finite simple graph has stable signed-support polynomial if and only if every block does. Here bridges are two-vertex blocks and isolated vertices may be treated separately as one-vertex blocks.*

*Proof.* For disjoint unions,  $\Phi$  is a product. A connected graph is assembled from its blocks by successive one-vertex sums, along its block-cut tree. Use Theorem 93.2 repeatedly. Conversely every block is an induced subgraph on its vertex set, so Lemma 85.6 applies.  $\square$

This answers the articulation-gluing problem completely. It reduces, but does not solve, the classification of stable bipartite graphs to their 2-connected blocks. Ordinary univariate real-rootedness alone is not the hypothesis of the gluing theorem.

On the diagonal, (93.1) gives the useful recurrence

$$p_{G_1 \vee_v G_2} = p_{G_1} p_{G_2 - v} + p_{G_1 - v} p_{G_2} - p_{G_1 - v} p_{G_2 - v}. \quad (93.2)$$

Its stability consequence requires the multivariate argument, not merely inspection of this subtraction formula.

In particular, attaching arbitrary trees at any vertices preserves stability, and cannot repair a nonstable core. Thus every stable family in this article yields a larger family by tree attachments. This closure principle is routine stability machinery, included with proof to specify what is and is not preserved. It is not a replacement for the classification of the two-connected core.

### 93.2 Positive vertex weights

For positive real weights  $w_v$ , define the weighted support polynomial

$$p_{G,w}(t) = \sum_{(I,J) \in \mathcal{M}(G)} t^{|I|} \prod_{u \in I} w_u \prod_{v \in J} w_v. \quad (93.3)$$

The inversion bridge gives the exact identity

$$p_{G,w}(t) = \left( \prod_{u \in U} w_u \right) F_G((w_u^{-1})_{u \in U}; (tw_v)_{v \in V}). \quad (93.4)$$

Consequently every stable graph proved here has a real-rooted weighted support polynomial for *every* positive vertex-weight assignment. This is stronger than the unweighted statement. For books with more than four pages we have proved the unweighted real-rootedness theorem, not this full weighted assertion. The negative Rayleigh certificate must not be read as a proof that every weighted diagonal fails, or even as an exhibited nonreal-root weighted diagonal.

Source 09 proves the same consequence in its own normalization (as delivered, its Section 13.1): For positive vertex activities  $w_v$ , put

$$p_{G,w}(u) = \sum_{S \text{ perfectly matchable}} w^S u^{|S|/2}.$$

Positive scaling and diagonalization of a stable  $\Phi_G$  show that

$$\left( \prod_v w_v \right) \Phi_G((x/w_v)_v) = x^{|V|} p_{G,w}(-x^{-2}) \quad (93.5)$$

is a real-rooted polynomial in  $x$ . The expression on the right is understood after expansion, so no value at  $x = 0$  is required. A nonzero root  $u$  of  $p_{G,w}$  would give roots of  $x^2 = -1/u$  on the left; all of those roots are real, forcing  $u < 0$ . Thus all roots of  $p_{G,w}$  are real and negative. This applies to the entire block family above.

### 93.3 Height-two preorder polynomials

Part II's height-two matching-support formula (Theorem 13.2) (84.8) immediately transfers the univariate results. This subsection imports that identity; the preceding graph and matroid theorems do not depend on it.

If  $p_G(t) = \prod_{j=1}^d (1 + \lambda_j t)$  with  $\lambda_j > 0$ , then

$$h_{P_G}(t) = (1+t)^{n-2d} \prod_{j=1}^d ((1+t)^2 + \lambda_j t), \quad n = |U| + |V|. \quad (93.6)$$

Each quadratic is  $t^2 + (2 + \lambda_j)t + 1$  and has two negative real roots. Thus  $h_{P_G}$  is real-rooted. Its reciprocal pairing of roots is visible directly in this factorization.

For  $B_m$ ,  $m \geq 2$ , the formula becomes

$$h_{P_{B_m}}(t) = (1 + 3t + t^2)^{m-2} R_m(t), \quad (93.7)$$

$$R_m(t) = (1 + t)^6 + (2m + 3)t(1 + t)^4 + (m^2 + m + 3)t^2(1 + t)^2 + t^3. \quad (93.8)$$

This proves real-rootedness of the entire associated height-two preorder family, even though the stronger signed support stability fails for all  $m \geq 5$ . The underlying preorder gamma-positivity and its general transport-pair interpretation are prior repository results, not claims of this article.

### 93.4 What these results do not imply

A stable signed support polynomial is not itself a construction of a flag simplicial polytope with a prescribed  $h$ -vector. Conversely, the failure of stability shown here is not a disproof of such a flag realization. The lattice-triangulation obstructions in the repository concern a different geometric requirement.

Similarly, the ADE criterion classifies only incidence–apex graphs over a tree, with the stated terminal convention. It does not classify all fundamental transversal matroids with the half-plane property, all bipartite graphs with stable support polynomial, or all real-rooted support diagonals. These remaining distinctions motivate the research questions below. [Added 30 September 2026, batch 63: Part VIII removes the terminal convention for a single apex over a tree (Theorem 118.2); within one-apex tree hosts the criterion now classifies every apex neighbourhood. The other limitations of this paragraph stand.]

## 94 Stable blocks beyond cactus determinants (source 09)

**Theorem 94.1** (Every complete bipartite block is stable). *For all  $\ell, r \geq 1$ ,  $\Phi_{K_{\ell,r}}$  is real stable and*

$$p_{K_{\ell,r}}(t) = \sum_{k=0}^{\min(\ell,r)} \binom{\ell}{k} \binom{r}{k} t^k. \quad (94.1)$$

*Proof.* Every balanced pair  $(I, J)$  is matchable, proving the coefficient formula. Proposition 102.2 says that every  $r$ -element subset of  $A \sqcup B$  is a basis: the lift is the uniform matroid  $U_{r, \ell+r}$ . Thus  $F_G = e_r$ , the elementary symmetric polynomial in all vertex variables. It is stable, since it is, up to a positive factorial, the specialization at  $s = 0$  of

$$\partial_s^\ell \prod_{v \in A \sqcup B} (s + z_v).$$

The product is stable, differentiation preserves stability, and real specialization preserves stability or gives zero; the resulting elementary symmetric polynomial is not zero. The reciprocal bridge proves the claim.  $\square$

**Proposition 94.2** (Even cycles, with an explicit certificate). *For each  $r \geq 2$ ,  $\Phi_{C_{2r}}$  is real stable.*

*Proof.* Let  $H$  be a Hermitian adjacency matrix of the cycle, with all edge entries of modulus one. Choose one edge phase so that the product around the oriented cycle,  $\eta$ , has real part

$(-1)^r/2$ ; set the other phases to one and reverse phases to their conjugates. Every proper induced subgraph is a forest, so every perfectly matchable proper support has a unique perfect matching. Its contribution to  $\det(\text{diag}(z) - H)$  is therefore exactly  $(-1)^{|S|/2}$ .

For the full vertex set, the two perfect matchings contribute  $2(-1)^r$ . The two oriented cycle permutations contribute  $-2\Re(\eta)$ . Their sum is  $(-1)^r$ , the coefficient required for counting the full support once. Hence this determinant equals  $\Phi_{C_{2r}}$ .

If all  $z_v \in \mathbb{H}$ , then for nonzero  $u \in \mathbb{C}^{2r}$ ,

$$\text{Im}(u^*(\text{diag}(z) - H)u) = \sum_v \text{Im}(z_v)|u_v|^2 > 0.$$

The matrix has no nonzero null vector and its determinant does not vanish.  $\square$

The cycle certificate is a restatement of the determinant method already used in the repository; it is included so the block-family theorem has a complete proof within this article.

**Corollary 94.3** (Biclique–cycle block family). *Every graph whose blocks are complete bipartite graphs or even cycles has a real-stable signed-support polynomial. The conclusion allows arbitrary articulation patterns, arbitrary biclique sizes, bridges, and isolated vertices.*

*Proof.* Apply Theorem 94.1, Proposition 94.2, and Corollary 93.3; an isolated vertex contributes  $z_v$ .  $\square$

## 94.1 Why this is genuinely beyond faithful matrices

A faithful unit-phase matrix for a bipartite graph has zeros off the edges, unit-modulus entries on them, and each matchable square minor has squared modulus one. The inspected report proves that such matrices exist exactly on cacti (Part III, Theorem 29.1). Here is the elementary obstruction already visible inside  $K_{2,3}$ . Normalize its first row to  $(1, 1, 1)$  by column phases. The second row consists of unit complex numbers  $u_1, u_2, u_3$ . Faithfulness would require  $|u_i - u_j| = 1$  for each distinct pair. Three such points form an equilateral triangle of side one, whose circumradius is  $1/\sqrt{3}$ , not one. They cannot all lie on the unit circle.

Yet Theorem 94.1 proves signed stability for every  $K_{\ell,r}$ . In particular, stable blocks with  $\min(\ell, r) \geq 2$  and  $\max(\ell, r) \geq 3$  contain a  $K_{2,3}$  obstruction and admit no faithful matrix. Stability is therefore not merely the old determinant construction with different phases.

[Merge note: Part III already separated stability from faithful matrices with the block  $K_{2,3}$  (Theorem 37.1, Corollary 37.4), including the equilateral-triangle obstruction for faithful phases on  $K_{2,3}$ ; source 09 does not cite it. Theorem 94.1 extends that example to every  $K_{\ell,r}$ , and its abstract’s “strictly extending cactus-based constructions” should be read with that credit.]

## 94.2 A stable noncactus with two different block types

Glue  $K_{3,3}$  and  $C_6$  at one vertex. The resulting graph has eleven vertices and is not a cactus. The relevant polynomials are

$$\begin{aligned} p_{K_{3,3}} &= 1 + 9u + 9u^2 + u^3, & p_{K_{3,3}-v} &= 1 + 6u + 3u^2, \\ p_{C_6} &= 1 + 6u + 9u^2 + u^3, & p_{C_6-v} &= 1 + 4u + 3u^2. \end{aligned}$$

Equation (93.2) gives

$$p_G(u) = 1 + 15u + 66u^2 + 107u^3 + 55u^4 + 6u^5. \quad (94.2)$$

Its roots are all negative by the proved multivariate theorem, not by a numerical root plot. Arbitrary trees or additional stable blocks may be attached while retaining the conclusion.

## 95 Exact verification and reproducibility

### 95.1 The finite checks that were actually run

The included program code/verify.py uses exact integer and rational arithmetic in SymPy, plus NetworkX to generate nonisomorphic trees. Its complete machine-readable output is in data/verification\_results.json [shipped as data/08-ade-stability-verification\_results.json; the program is code/08-ade-stability-verify.py]. The delivered run records Python 3.13.5, SymPy 1.14.0, and NetworkX 3.6.1.

| Check                                                               | Count |
|---------------------------------------------------------------------|-------|
| Unlabeled trees on one through nine vertices                        | 95    |
| Exact weighted determinant-identity evaluations on these trees      | 190   |
| All apex-containing coefficient checks, trees through five vertices | 462   |
| Book sizes checked, $m = 0, \dots, 8$                               | 9     |
| Even-theta expected-minor checks across three examples              | 507   |
| Exact path-contraction/deletion basis-set comparisons               | 4     |
| Terminal-subdivision identity evaluations                           | 6     |

Table 11: Executed finite checks. The all-size statements rely on the proofs in the article, not on extrapolation from this table.

For each of the 95 trees, the program independently compares an exact Schur-complement positive-semidefiniteness test with a combinatorial ADE shape classifier. It then enumerates matchable endpoint pairs and checks Theorem 86.3 at two deterministic positive integral weight assignments. Those evaluations alone are not an identity proof; the coefficient argument in Section 86 supplies that. For every tree through five vertices, every apex-containing coefficient is also checked directly using integer incidence cofactors and the exact quadratic form  $q^T C_T q$ .

For books, the program verifies every coefficient of the displayed univariate formula through eight pages, the all-ones Rayleigh formula, and the uniform covariance counts. It symbolically verifies the cubic discriminant and the quartic obstruction discriminant. It checks the rational rank-one-adjugate certificate for  $K_{1,5}$ . For  $\vartheta(2, 2, 2)$ ,  $\vartheta(2, 2, 4)$  and  $\vartheta(4, 4, 4)$ , it enumerates all sign assignments in (91.3) and every square minor, retaining  $\sqrt{2}$  symbolically. The numbers of checked minors are respectively 10, 35, 462, totaling 507. Four additional checks compare the complete basis sets before and after path contraction/deletion with the predicted shortened graph. Six weighted evaluations test the zero-extended covariance identity for three subdivided stars, including a nonstable five-leaf skeleton.

### 95.2 The support enumerator deliberately removes multiplicities

A dynamic program enumerates pairs  $(I, J)$  as follows. Set  $D(\emptyset) = \{\emptyset\}$ . For a nonempty row set  $I$ , choose its least element  $i$  and put

$$D(I) = \{J \cup \{j\} : J \in D(I \setminus \{i\}), j \in N(i) \setminus J\}. \quad (95.1)$$

The braces denote a *set*; repeated ways to produce the same  $J$  are removed. Induction on  $|I|$  shows that  $D(I)$  consists precisely of the column sets matchable to  $I$ . This avoids the most dangerous computational error in this subject: accidentally checking the ordinary matching polynomial instead of the matching-support polynomial.



The exact PSD checker uses symmetric Schur complements. A negative diagonal excludes positive semidefiniteness. A zero diagonal must have a zero row in a positive semidefinite matrix; otherwise the checker rejects. Positive pivots reduce the dimension by an exact rational Schur complement. Thus the test does not classify borderline affine diagrams using floating-point tolerances.

### 95.3 Rebuilding the package

From the package directory, the following commands reproduce the checks and compile the document:

```
python -m pip install -r requirements.txt
python code/verify.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

[Merge note: these are the delivered commands, for the package layout. Here the requirements file is `data/08-ade-stability-requirements.txt`; without `-output` the program writes `data/verification_results.json` beside its parent directory, which in this directory is *Part I*'s recorded output. The README gives a command with an explicit scratch output.] The verifier has no network calls and contains no downloaded data. The requirements file records the versions used in the delivered run. The source has a self-contained bibliography. No external font files or third-party article PDFs are included.

### 95.4 A proposed formalization boundary

A future Lean or Rocq development could keep the main proof modular. The finite combinatorial layer defines matchability, the support polynomial and the unique-unrooted-component decomposition. A linear algebra layer proves the incidence-cofactor identity and the exact adjugate formula. An analytic layer proves the upper-half-plane resolvent lemma and the necessary Rayleigh inequality. The spectral layer then supplies Perron positivity and the elementary ADE tree list.

The negative certificates are particularly suitable for a small exact checker: the checker need only verify positive rational entries in  $v$  and  $v^T C_T v < 0$ , after a general theorem has connected those data to (87.5). Evaluating a rational inequality without that connecting theorem would not by itself certify the original stability claim. No part of this planned formalization is represented as already completed in the delivered package.

**Source 09's checks of its stability sections.** Source 09's single verifier (Part VII, Section 110) also checks the sixteen exact signed vertex-sum identities, the  $\Theta_3$  support polynomial and the Rayleigh value  $-9$  from vertex-subset feasibility.

## 96 Further research questions and topics

The following questions are proposed directions, not theorems asserted by this article. Their purpose is to separate the next genuinely missing steps from what the present results already settle.

**Research question 44** (Beyond the classified graph families; sources 08 and 09). Characterize the bipartite graphs  $G$  for which the canonical transversal matroid  $M_G$  has stable basis polynomial.

Can a useful class of forbidden *canonical matroid minors*, rather than merely induced graphs, organize the answer? (Source 09, “Which 2-connected bipartite graphs are stable?”: classify the bipartite blocks  $B$  for which  $\Phi_B$  is real stable. Theorem 85.3 reformulates this as the half-plane property for the concrete matroid  $M_B$ , and Corollary 93.3 removes all articulation complications. A useful first target is the exact classification for subdivisions of theta graphs. A proof must distinguish univariate real-rootedness from full multivariate stability.)

The path-shortening proof shows why induced-graph obstructions alone can miss a persistent failure. The five-page book and the three-odd-path obstruction give different mechanisms and should be treated separately. A general classification is not supplied by the ADE or multi-theta results.

[Merge note: by Corollary 93.3 the two formulations are equivalent; source 09’s “useful first target”, the subdivisions of theta graphs, is Theorem 84.3. The question overlaps Part III’s Research question 11.]

**Research question 45** (All tree incidence–apex diagonals). For which trees  $T$  is the univariate polynomial

$$p_{\mathcal{A}(T)}(t) = \det(I + t \operatorname{Lap}(T)) + t \operatorname{tr}(\operatorname{adj}(I + t \operatorname{Lap}(T))C_T) \quad (96.1)$$

real-rooted? In particular, is it real-rooted for every tree?

The spectral stability criterion settles the ADE cases, while the book formula settles every star, including non-ADE stars. These facts do not settle the question for arbitrary trees. A plausible route is to analyze the residues of  $\operatorname{tr}(C_T(I + t \operatorname{Lap}(T))^{-1})$  in Laplacian eigenspaces; positivity of those residues would require a separate proof and is not assumed here.

[Merge note: evidence only. For this merge the diagonal was checked exactly for every unlabelled tree with  $2 \leq r \leq 12$  (986 trees;  $r = 1$  is immediate): all are real-rooted.] [Added 30 September 2026, batch 63: source 10 poses the same question for arbitrary markings (Part VIII, Research question 58).]

**Research question 46** (Positive-weight Rayleigh versus full stability). Within the incidence–apex tree family, characterize the trees for which all Rayleigh differences are nonnegative on the *positive* orthant. Does this condition have a spectral description different from ADE?

The universal failure certificate uses arbitrary real left variables. The book certificate is stronger and already fails at positive weights. Equating these two levels without proof would be a mistake.

**Research question 47** (Several apex columns). Replace one apex by two or more independently selectable apex vertices, with prescribed terminal neighborhoods. Is there an exact Boolean coefficient formula involving mixed covariances or higher exterior moments that yields a tractable stability criterion?

The single-apex case only sees one unrooted component. Several apex columns must resolve simultaneous components and remove the multiplicities of their assignments. A product of single-apex formulas would generally overcount. [Added 30 September 2026, batch 63: source 10 poses the same question (Part VIII, Research question 59).]

**Research question 48** (Base graphs with cycles). Which parts of the covariance method survive if the auxiliary incidence graph is not a tree? Can cycle-space corrections replace the identity  $|W| - |E(T[W])| = 1$  without destroying positive semidefiniteness?

The tree proof exploits both unique paths and unimodular incidence cofactors. The even-theta construction suggests that independent random columns can sometimes correct cycle

multiplicities, but it does not yet provide a general replacement for the connected-subtree energy law. [Added 30 September 2026, batch 63: source 10 poses the same question for unicyclic, cactus and bounded-cycle-rank hosts (Part VIII, Research question 60).]

**Research question 49** (Boundary Rayleigh certificates). Find compact sums-of-squares expressions for the Rayleigh differences of the affine  $\widetilde{D}, \widetilde{E}$  incidence–apex polynomials. Can their one-dimensional covariance kernel yield a uniform formula?

The analytic proof already establishes nonnegativity. A transparent algebraic certificate would be valuable for formalization, exact optimization and a comparison with the repository’s explicit  $K_{2,3}$  certificates.

**Research question 50** (Weighted and perturbed covariance models). For which deformations of the support coefficients can the forced matrix  $C_T$  be replaced by a weighted adjacency expression while preserving nonnegative coefficients and a useful combinatorial meaning?

Merely replacing  $A_T$  by a weighted matrix no longer leaves every connected-support coefficient equal to one. A successful generalization must state exactly what the new coefficients count.

**Research question 51** (Sampling and computational certificates). Can the determinant–adjugate and independent-column representations be turned into exact samplers for the Boolean endpoint-support measure, or into polynomial-time symbolic stability certificates for the classified families?

The formulas provide compact partition functions on the stable side, but sampling a determinant mixture and forgetting its auxiliary vector need not automatically give a simple uniform-support sampler. The conditional distributions and normalization must be analyzed.

**Research question 52** (Formal integration into ProveIt). Formalize the cofactor support lemma, the covariance rigidity theorem, and the rational Rayleigh certificate compiler, then connect them to the repository’s existing matching-support and preorder semantics.

This route would turn the written argument into a machine-checked extension without importing the whole surrounding refereed research corpus. The separate expected-determinant proof could then be added as a second, independently useful stability module.

[Merge note: compare Part III’s Research question 19 (kernel-checked certificate theory).]

**Research question 53** (Minimal induced obstructions; source 09). Describe the induced-minimal bipartite graphs with unstable  $\Phi_G$ . The eight-vertex theta graph gives one obstruction, but the present manuscript does not claim a minimality classification. Is there a finite obstruction set within a fixed treewidth or fixed maximum-degree class? Any proposed list can be tested using exact Rayleigh certificates.

[Merge note: the finite-obstruction part has a negative answer, Section 92.6; describing all induced-minimal unstable graphs remains open.]

## 97 Conclusion

The central obstruction in matching-support enumeration is that several perfect matchings must contribute only once. For tree incidence–apex graphs, an oriented forest cofactor identifies exactly one unrooted component, and the matrix  $I - A_T/2$  assigns that component the required

weight one. This matrix is forced by the coefficients. Its positive semidefiniteness proves stability through a resolvent, while its failure produces a rank-one-adjugate Rayleigh obstruction. The resulting exact criterion is therefore both constructive and sharp.

The classical ADE tree list becomes a stability classification through this mechanism. The book family exhibits a four-page boundary that cannot be detected from its univariate roots, and the independent-column method gives the complementary even-path family. Together with exact minor reduction, these arguments classify the simple bipartite multi- $\theta$  graphs while preserving the distinction between new restricted results, classical inputs, earlier counterexamples, and the still-open general questions.

Source 09 concludes on stability (as delivered):

The stability question is correspondingly sharper. The reciprocal bridge places it inside half-plane-property theory, while the block theorem isolates the genuinely difficult 2-connected cases. Complete bipartite blocks show that stability extends far beyond faithful cactus matrices. Together these results provide a concrete continuation of the repository's program without reviving a false real-rootedness conjecture or relabeling its existing gamma-positivity theorem as a new discovery.

## 98 Provenance, dependency ledger, and claim audit

### 98.1 Repository pin

The repository snapshot used for the final provenance check is

`e1afd75e35a4de734d5aa47aec5cdb917b82ed3e.`

The principal comparison document is

`SetTheory/Cardinals/docs/reports/  
enumerative-combinatorics/preorder-root-polytopes/README.md`

with Git blob

`846d9f19fadac9cd933b74370cdb8fafa2211c5d.`

It was fetched both from the default branch and from the pinned commit; the blob agreed. Its Parts II–IV summary supplies the nonduplication boundary used here. This is a targeted comparison, not a claim to have read, audited, or rebuilt every file in the repository.

## 98.2 What is imported and what is proved here

| Statement or tool                                                            | Status in this article                                                                                             |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Upper-half-plane terminology, standard closure facts, matroid HPP framework  | Classical; sources [24, 12]. Special facts used in the main proof are reproved.                                    |
| ADE spectral classification                                                  | Classical; explicit elementary tree proof in Section 88. Not a novelty claim.                                      |
| Cactus faithful matrices and the stable $K_{2,3}$ example                    | Prior repository results used as context, not as dependencies of the main criterion.                               |
| $\mathfrak{g}(3, 3, 3)$ nonreal-root obstruction                             | Earlier example credited by the repository to Patel. Its support coefficients and discriminant are rechecked here. |
| Height-two preorder transform                                                | Imported from the repository; used only in Section 93.                                                             |
| Tree determinant–adjugate identity and forced covariance                     | Proved coefficientwise here, independently of the repository’s unrefereed classification claims.                   |
| Tree spectral criterion, rational failure certificate, terminal subdivisions | Proved here using the cofactor identity, elementary matrix analysis and generic matroid path shortening.           |
| Sharp book threshold, all-real diagonal formula, positive covariance         | Proved here; exact finite checks included.                                                                         |
| Even multi-theta and full bipartite multi-theta classification               | Proved here by expected minors, closure operations and the credited obstruction.                                   |
| Global classification, worldwide priority, formal verification               | Not established and not claimed.                                                                                   |

## 98.3 Search and priority limitations

The literature review checked the cited primary sources and performed focused searches combining matching-support polynomials, half-plane property, transversal matroids, tree incidence, spectral radius and ADE. No source found in that focused review was identified as stating the precise tree incidence–apex equivalence of this manuscript. This negative search result does *not* prove priority: equivalent results may use different matroid terminology or occur in sources not retrieved. The accompanying `SOURCE_AUDIT.md` records the same limitation.

Independent proof review and a specialized matroid-literature comparison remain necessary before treating this manuscript as a refereed or priority-established contribution.

Source 09’s pins (its Appendix B) are recorded in Part VII (Section 115).

## 99 Compact certificates for readers and proof checkers

### 99.1 The main theorem in six equations

The mathematical core can be reduced to the following chain, with each step justified in the main text:

$$\begin{array}{ll}
 \det[B_{I,J} \ w_I] = \pm \mathbf{1}_W^T w, & \text{unique unrooted component;} \\
 \mathbf{1}_W^T (I - A_T/2) \mathbf{1}_W = 1, & \text{connected subtree;} \\
 F_T = \det L + s \operatorname{tr}(\operatorname{adj} L C_T), & \text{coefficient identity;} \\
 \operatorname{Im} \operatorname{tr}(C_T L^{-1}) < 0 \quad (C_T \geq 0), & \text{upper-half-plane nonvanishing;} \\
 \operatorname{adj}(D_v^{-1} \operatorname{Lap}(T) D_v^{-1}) = \left( \prod_i v_i^{-2} \right) v v^T, & \text{tree cofactors;} \\
 \Delta_{s,x_i} F_T = c^2 v_i^2 (v^T C_T v) < 0 \quad (v^T C_T v < 0), & \text{explicit obstruction.}
 \end{array}$$

The last line is evaluated at the rational assignment in (87.3)–(87.4); it is not an identity at arbitrary variables. The first line applies to feasible apex minors and is zero on infeasible ones.

### 99.2 A simple exact PSD checker

For a rational symmetric matrix  $M$ , repeat the following operations. A negative diagonal entry rejects  $M$ . A zero diagonal with a nonzero entry in its row also rejects it; a zero row and column can be removed. Otherwise choose a positive pivot  $a$  and write

$$M = \begin{pmatrix} a & b^T \\ b & D \end{pmatrix}.$$

Then

$$M \geq 0 \iff D - b b^T / a \geq 0.$$

The dimension decreases, so the procedure terminates and uses exact rational operations. The implementation in the verifier is independent of its ADE shape classifier. The comparison of these two finite procedures is a regression check on the explicit diagram list, not a



## Part VII. The transversal matroid of a preorder: reconstruction, duality and gammoids

*Part VII was added on 29 September 2026 in batch 55 of ProveIt's incoming-report intake, from one manuscript. It studies the transversal matroid  $M_\tau$  on two copies of the ground set of a finite preorder, whose bases are the matchable support pairs of Part IV:  $M_\tau$  determines  $\tau$  up to isomorphism, its automorphism group is computed, order reversal is matroid duality, its rank function is an upset minimum, and height-two preorder matroids generate exactly the finite gammoids under minors. The manuscript's ultra-log-concavity sections are the third proof of Part V's main theorem and are credited there; its stability sections are in Part VI. Parts I–VI are unchanged except for dated pointers. Section 100 records the provenance, fixes the notation and states what is answered.*

### 100 Provenance, conventions and what Part VII answers

#### 100.1 The manuscript

Part VII is printed from batch-55 manuscript 05, called *source 09* here after its file prefix: *Preorders as Transversal Matroids: Reconstruction, Ultra-Log-Concavity, and Stability of Matching Supports*, dated 29 September 2026, 24 A4 pages as delivered, prepared for Vladimir Reshetnikov as “AI-assisted mathematical research by ChatGPT”. Its archive `ProveIt_Matroid_Lifts.zip` arrived in ProveIt commit `2d4919838` and was placed in `26473dfa0`. It is pinned to commit `e1afd75e3` (full hash in Section 115), with this report's `article.tex` as blob `d52a2633b58e74aa653e967eb56adc1aa41bc8ae` and its `README` as blob `846d9f19fadac9cd933b74370cdb8fafa2211c5d`: Parts I–IV exactly as printed, unchanged from the pin to the placement. Its verifier, its Makefile, its recorded output (two byte-identical files) and its claim ledger are shipped with the prefix `09-matroid-lifts-`; its checksum list `SHA256SUMS.txt` was verified (8/8) at placement and retired. The manuscript, delivery `README` and PDF are not shipped; they survive in the arrival commit.

The manuscript has three strands. *Ultra-log-concavity* (its Sections 2–4, the general part of Section 15) is the same theorem, with the same matroid and proof, as Part V's sources 06 and 07; it is credited in Part V and not reprinted, except that its matroid definition (Section 2) is printed here because every later section uses it. *Stability* (its Sections 11–14.3) is merged into Part VI with source 08. *Matroid structure* (Sections 5–10) is new relative to this report and is Part VII, with the manuscript's verification, formalization, questions, conclusion and appendices.

**Abstract of source 09 (as delivered).** A matching support remembers the vertices covered by a matching, but not the choice of matching. We use its classical transversal-matroid lift to develop new structural consequences for the preorder-polynomial program in ProveIt. For a reflexive relation  $R$ , we prove that transitivity is equivalent to the two copies of each element being clones in the associated matroid. For a preorder, we give an upset formula for every matroid rank, identify all intrinsic clone classes, reconstruct the entire weighted quotient poset from the unlabeled matroid, and determine its full automorphism group. Order reversal becomes matroid duality. Every transversal matroid is an explicit minor of the matroid of a height-two poset; consequently these height-two matroids generate exactly the finite gammoids under minors. Combining established matching-support identities with the Brändén–Huh and Röhrle–Ulirsch theory proves ultra-log-concavity of *every* finite preorder  $h$ -polynomial, including a positive transport-weighted version. This retains Newton-type coefficient inequalities even though real-rootedness fails. We distinguish this application from the already published general bimatroid inequality. Finally, a partial reciprocal substitution identifies stability of a signed matching-support polynomial with the half-plane property of its matroid. We prove that signed-support stability is local to the blocks of a graph, and obtain stable families with arbitrary complete-bipartite blocks, strictly extending cactus-based constructions. All new structural arguments are supplied in full. Exact finite checks and a negative Rayleigh certificate are included as supplementary evidence, not as substitutes for proof.

The abstract describes the whole manuscript. Its stability claims are Part VI’s Theorems 93.2 and 94.1; “strictly extending cactus-based constructions” is to be read with Part III’s  $K_{2,3}$  example (Theorem 37.1), which source 09 does not cite.

**Status and priority (source 09, as delivered).** Unrefereed research manuscript; no Lean or Rocq verification is claimed. Classical inputs are explicitly separated from the contributions. The source audit is targeted, not exhaustive, so absolute publication priority is not asserted. The inspected repository snapshot is e1afd75e35a4de734d5aa47aec5cdb917b82ed3e. Its UTC timestamp is 30 September 2026, still 29 September in America/Los\_Angeles.

**Priority and prior work (merge).** The extended-matroid construction and the ultra-log-concavity of the regular minors of a relation are Röhrle–Ulirsch’s [17], which source 09 credits. Two results that source 09 lists as principal contributions are known in this report in non-matroid form: the clone/transitivity criterion (Theorem 103.2) is Part IV’s Proposition 51.4, with the same counterexample, read through the matroid, and the orbit formula (Theorem 107.1) is Part IV’s Theorem 52.1 with a multivariate orbit polynomial added; the unweighted transport  $\gamma$ -formula it refines is Part IV’s Theorem 48.2. The gammoid characterization is Ingleton–Piff’s [26]. The reconstruction and automorphism theorems, the upset rank formula, the duality and restriction identities and the explicit height-two minor construction are, relative to the sources inspected, new; source 09’s audit was targeted, and no worldwide priority is claimed for any result of Part VII.

## 100.2 Notation of Part VII

Part VII uses Part I’s  $\tau$ ,  $Q_\tau$  and  $h_\tau$ , Part IV’s transport pairs  $\mathcal{T}_k(\tau)$ ,  $\gamma_k(\tau)$  and opposite preorder, and Part V’s support polynomial  $p_G$ . Table 12 lists what Part VII adds or uses differently. Source 09’s matroid is printed  $M_G$ ,  $M_\tau$  as delivered; its bases are  $I \sqcup (B \setminus J)$ , so it is the *dual* of Part V’s and Part VI’s  $M_G$  (Section 83.2). Source 09’s  $\mathcal{T}_k(\tau)$  is printed with the host’s  $\mathcal{T}_k(\tau)$  (the same set), its  $\mathcal{M}(G)$  with the host’s  $\mathcal{M}(G)$ , and its  $\tau^{\text{op}}$  is kept (Part I writes  $\tau^*$ ). No normalization was changed.

Table 12: Symbols of Part VII. The last column prints the tempting false reading.

| Symbol                                                | Meaning in Part VII                                                                                                                                      | Not to be confused with                                                                                                                                             |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $M_G, F_G$                                            | transversal matroid of a bipartite $G = (A \sqcup B, \mathcal{E})$ with one slot per $b \in B$ , bases $I \sqcup (B \setminus J)$ ; its basis polynomial | Part II's matching polynomial $M_G(u)$ ; the dual $M_G$ of Parts V–VI.                                                                                              |
| $M_\tau, M_R$<br>$E^\pm, i^\pm, U^\pm$                | $M_{G_\tau}, M_{G_R}$ on $E^+ \sqcup E^-$<br>the two labelled copies of $E$ ; the copies of $i$ ;<br>$U^+ \sqcup U^-$                                    | Part IV's set $\mathcal{M}_\tau$ of compatible pairs.<br>Part V's private copies $X^\circ$ .                                                                        |
| $C(I, J)$<br>$\bar{\tau}, C, m_C$<br>$m_k(G)$         | the basis $I^+ \sqcup (E \setminus J)^-$<br>quotient poset, an equivalence class, its size<br>$ \{(I, J) \in \mathcal{M}(G) :  I  = k\} $                | Part I's $C_\tau$ , Part III's cores.<br>Part IV's transitive closure $\bar{R}$ ; Part IV's $m_i$ .<br>$m_C$ ; Part II's matching counts $m_k(G)$ (edge matchings). |
| $\ell, r$<br>$\uparrow S$ , upset<br>$\text{rk}, x_i$ | shore sizes $ A ,  B $ in Section 102<br>up-closure, sets closed upward<br>matroid rank, number of selected copies of $i$                                | Part V's $r$ .<br>Part I's $\downarrow$ , ideals.<br>a rank of a matrix.                                                                                            |
| $\text{Aut}(M), \mathfrak{S}_k$<br>$T_G$              | automorphism group, symmetric group<br>transversal matroid on $A$ with slots $B$                                                                         | Part II's $T_G$ ; a tree $T$ of Part VI.                                                                                                                            |
| $h_{\tau, w}, \gamma_k(\tau; w)$                      | transport-weighted polynomial and its $\gamma$ -coefficients                                                                                             | Part V's vertex-weighted $a_k(u, v)$ .                                                                                                                              |
| $\mathbb{Z}_{\geq 0}$                                 | $\{0, 1, 2, \dots\}$ (source 09's $\mathbb{N}$ , the host's $\mathbb{Z}_{\geq 0}$ )                                                                      |                                                                                                                                                                     |

### 100.3 What Part VII answers

Part VII answers no numbered question of Parts I–VI; its matroid strand opens new territory. It gives Part IV's overlap cancellation and Boolean orbits a matroid meaning (clones), shows that the support-pair model loses no information about  $\tau$  (Theorem 105.2), and identifies the minor-closed class generated by height-two preorders (Corollary 108.2). Part IV's Proposition 51.4 and Theorem 52.1 gain dated pointers. Part VII refutes nothing, and nothing is retracted.

### 100.4 Where the merge had to choose

- *The split.* Source 09's Sections 1–2, 5–10, 16–18, the first paragraph of its conclusion, its Appendices A–B and its Section 14.4 (“Claims deliberately not made”) are printed here in its order. Its Sections 11–14.3 and the stability paragraphs of its introduction and conclusion are in Part VI.
- *Not reprinted (duplicates of Part V).* Its Section 3 (Theorem 3.1, weighted matching-support ultra-log-concavity, and Corollary 3.2, strictness) is Theorem 70.1 and Theorem 67.3, with the same Lorentzian substitution; its Section 4 (the counting bridge through [2, 7], Proposition 4.1, and Theorem 4.2, every preorder ultra-log-concave with  $\binom{n}{k} \leq h_k \leq \binom{n}{k}^2$  and the simultaneous-equality clause) is Proposition 68.3, Lemma 71.1, Corollary 67.2, equation (71.1) (Part IV's Theorem 55.6) and the equality proposition after it; its remark that unimodality holds throughout Problem 5.3 of [2] is Corollary 70.2; its Section 15 (binomial convex domination) is Lemma 74.4 with Theorem 74.5, its Hoeffding tail is Remark 74.7; its  $\Theta_3$  ultra-log-concavity display is Part V's (77.2). References to these statements point to Part V.
- *Attribution.* Merge notes credit Part IV at Theorems 103.2 and 107.1 and at the coefficient bounds.
- *Research questions.* Source 09's Questions 5, 6, 8 and 10 continue the numbering as Research questions 54–57. Its Questions 1 and 2 are in Part VI; Questions 3, 4, 7 and 9 repeat questions of Parts IV–V and are printed unnumbered next to a pointer (Section 112.1).

- *Self-citation and file names.* Its reference “ProveIt” became references to Parts II–IV. Its `verify.py`, `verification_results.json`, `verification_run.txt` and `Makefile` are shipped as `code/09-matroid-lifts-verify.py`, `data/09-matroid-lifts-verification_results.json`, `data/09-matroid-lifts-verification_run.txt` and `code/09-matroid-lifts-Makefile`; merge notes give the shipped names.
- *Macros and labels.* Its preamble is dropped: its `\DeclareMathOperator` forms of `supp`, `Var`, `conv` would clash with the host’s commands of the same meaning; its  $\mathbb{P}$ ,  $\mathbb{E}$ ,  $\text{Im}$  are typeset with the host’s macros. Labels carry the prefix `mat:`; the questions, unlabelled in the source, gained labels. Its abstract and status box are printed in this section; its title page and table of contents are dropped.

## 101 The question left after real-rootedness fails

The preorder program of Athanasiadis and Chapoton associates a polynomial  $h_\tau(t)$  with a finite preorder  $\tau$  by counting lattice points according to support size [1]. The inspected ProveIt report (this report, Parts I–IV) develops root-polytope realizations, gamma expansions, counterexamples to real-rootedness, and matrix constructions for matching supports. In particular, its recent Parts II–IV already establish many conclusions that must not be presented again as new: the height-two matching-support formula, gamma-positivity for every preorder, and the classification of faithful unit-phase matrices by bipartite cacti.

The useful next question is therefore not whether every  $h_\tau$  has only real roots. It is what robust structure survives the counterexamples, and whether the support model retains more information than its univariate polynomial. This manuscript gives two answers. The coefficient sequence always satisfies ultra-log-concavity. More surprisingly, an associated unlabeled matroid retains the entire preorder, including the sizes of its equivalence classes.

### 101.1 Principal conclusions

Let  $E$  be an  $n$ -element set. A preorder is a reflexive, transitive relation on  $E$ ; antisymmetry is not assumed. Let  $E^+$  and  $E^-$  be disjoint labeled copies. We construct a rank- $n$  transversal matroid  $M_\tau$  on  $E^+ \sqcup E^-$ . Write  $\bar{\tau}$  for the quotient poset of equivalence classes, and let  $m_C$  be the size of a class  $C$ .

The main structural results are

$$M_\tau \cong M_\sigma \iff \tau \cong \sigma, \quad (101.1)$$

$$\text{Aut}(M_\tau) \cong \left( \prod_{C \in \bar{\tau}} \mathfrak{S}_{2m_C} \right) \rtimes \text{Aut}(\bar{\tau}, m), \quad (101.2)$$

$$M_\tau^* = M_{\tau^{\text{op}}}, \quad (101.3)$$

where the duality identity uses the fixed labels on the two copies. The semidirect-product isomorphism depends on choices of identifications inside the clone classes; its kernel and quotient have intrinsic meanings.

The coefficient conclusion is the following: if  $h_\tau(t) = \sum_{k=0}^n h_k t^k$ , then

$$h_k^2 \geq \frac{(k+1)(n-k+1)}{k(n-k)} h_{k-1} h_{k+1} \quad (1 \leq k < n). \quad (101.4)$$

The same inequalities hold for the transport-weighted coefficients defined in Section 107. In particular, the ordinary coefficient sequence is strictly log-concave at every interior index when  $n \geq 2$ . This statement does not assert strictness after binomial normalization.

[Merge note: the introduction continues with the block-stability statement (84.7), printed in Part VI (Section 84.4). The ultra-log-concavity display (101.4) is Part V's (67.5).]

## 101.2 What is imported, and what is contributed

The extended-matroid construction for a bimatroid is classical. In particular, Röhrle and Ulirsch explicitly treat relations as bimatroids and prove ultra-log-concavity for their regular-minor counts [17, Proposition 2.2, Example 2.4, Theorem A]. The general unweighted matching-support inequality is therefore *not* a new theorem of this manuscript. We give its weighted specialization and use it to connect two previously separate parts of the investigated program.

The principal contributions developed here are the preorder-specific reconstruction and automorphism theorems, the clone/transitivity criterion, the upset rank description, the explicit height-two minor construction, and the signed-support block-stability theorem. The universal preorder ultra-log-concavity statement is a new application relative to the sources inspected, rather than a replacement proof of the deep Lorentzian theorem. All these priority descriptions are bounded by the source audit in Appendix 115.

[Merge note: the clone/transitivity criterion is Part IV's Proposition 51.4 in matroid form, and the block-stability theorem is printed in Part VI; see the priority box in Section 100.1.]

## 102 Matching supports and their classical matroid lift

Throughout,  $\mathbb{Z}_{\geq 0} = \{0, 1, 2, \dots\}$  and all sets and graphs are finite. Let  $G = (A \sqcup B, \mathcal{E})$  be a finite simple bipartite graph, with  $\ell = |A|$  and  $r = |B|$ . Put

$$\mathcal{M}(G) = \{(I, J) : I \subseteq A, J \subseteq B, |I| = |J|, G[I \sqcup J] \text{ has a perfect matching}\}.$$

The empty pair belongs to  $\mathcal{M}(G)$ . We count a pair once even when it has several matching witnesses. Define

$$p_G(t) = \sum_{(I,J) \in \mathcal{M}(G)} t^{|I|} = \sum_{k=0}^{\min(\ell,r)} m_k(G) t^k. \quad (102.1)$$

For instance,  $p_{K_{2,2}}(t) = 1 + 4t + t^2$ , not  $1 + 4t + 2t^2$ .

A transversal matroid is defined by a family of sets, viewed as a collection of slots: a set of ground elements is independent when its elements can be assigned injectively to slots that allow them. The usual augmenting-path argument proves the independence axioms; a proof is recalled in Appendix 114.

**Definition 102.1** (Matching-support matroid). For each  $b \in B$ , take a slot allowing exactly the elements  $\{b\} \cup N_G(b)$  of  $A \sqcup B$ . Let  $M_G$  be the resulting transversal matroid. Its distinguished basis is  $B$ , because every  $b$  can use its own slot.

**Proposition 102.2** (Classical lift in graph form). *The rank of  $M_G$  is  $r$ , and its bases are exactly*

$$\mathcal{B}(M_G) = \{I \sqcup (B \setminus J) : (I, J) \in \mathcal{M}(G)\}. \quad (102.2)$$

Consequently its basis-generating polynomial is

$$F_G(x, y) = \sum_{(I,J) \in \mathcal{M}(G)} x^I y^{B \setminus J}, \quad (102.3)$$

where products over empty sets are one.

*Proof.* There are  $r$  slots and the elements of  $B$  fill all of them, proving the rank statement. In a basis of the displayed form, each selected element  $b \in B \setminus J$  can use only its private slot. The remaining slots are exactly  $J$ , and the remaining elements are exactly  $I$ . Filling those slots is equivalent to a perfect matching between  $I$  and  $J$ . The construction is reversible. Notice that a basis is a set, not a slot assignment, so multiple perfect matchings do not create multiple coefficients.  $\square$

This proposition is the extended-matroid construction for the relation bimatroid, in the opposite shore convention to some sources [17]. It is useful to keep the concrete slot description: it makes the later minor constructions entirely explicit.

### 102.1 Three different polynomials

We will consistently distinguish the support polynomial  $p_G$ , the homogeneous matroid polynomial  $F_G$ , and the signed vertex polynomial  $\Phi_G$ . In particular,

$$F_G(s\mathbf{1}_A, t\mathbf{1}_B) = \sum_k m_k(G) s^k t^{r-k}. \quad (102.4)$$

The degree on the right is  $r$ , even when the maximum matching size is smaller. Interchanging the shores supplies a different lift of rank  $\ell$ . This distinction controls the normalization in all coefficient inequalities below.

*Source 09's Sections 3 and 4 are not reprinted.* They prove, for  $M_G$  and with positive weights, that the coefficients of  $p_G$  are ultra-log-concave of order  $\min(\ell, r)$  with exactly the positive indices  $0, \dots, \nu(G)$ , and deduce ultra-log-concavity of every  $h_\tau$  with  $\binom{n}{k} \leq h_k \leq \binom{n}{k}^2$  and simultaneous equality exactly for the antichain. These are Part V's Theorem 70.1, Corollary 67.2 and the proposition after it, proved by the same Lorentzian substitution; source 09 credits the unweighted graph inequality to Röhrlé–Ulirsch's Theorem A [17]. Source 09 imports the demand/support identity by the route of Dai–Hou–Liu–Thawinrak–Wang (Theorem 1.2) and Davis–Kohl (Theorem 3.10), as Part V's source 07 does.

## 103 Transitivity is exactly paired clonal symmetry

For a reflexive relation  $R$  on  $E$ , write  $M_R = M_{G_R}$ , labeling the left copy by  $E^+$  and the private right copy by  $E^-$ . Its distinguished basis is  $E^-$ . A basis has the unique form

$$C(I, J) = I^+ \sqcup (E \setminus J)^-, \quad (I, J) \in \mathcal{M}(G_R). \quad (103.1)$$

Two ground elements of a matroid are *clones* if their transposition, fixing every other element, is an automorphism. This does not mean that the elements are parallel: clones can form an independent pair.

**Lemma 103.1** (Overlap cancellation). *If  $R$  is a preorder, then  $(I, J)$  is matchable if and only if  $(I \setminus J, J \setminus I)$  is matchable.*

*Proof.* A matching  $f : I \rightarrow J$  gives a directed graph with at most one incoming and one outgoing edge at each vertex. Each component is a directed cycle or a directed path. Every path starts in  $I \setminus J$  and ends in  $J \setminus I$ ; its intermediate vertices lie in  $I \cap J$ . Transitivity replaces that path by one allowed edge between its endpoints. The resulting edges are a matching of the differences. Conversely, extend such a matching by the diagonal edges on  $I \cap J$ , using reflexivity.  $\square$



This cancellation is already present in the repository's gamma argument (Part IV, Corollary 51.2). The next theorem identifies its intrinsic matroid meaning.

**Theorem 103.2** (Clone/transitivity criterion). *For a reflexive relation  $R$ , the following conditions are equivalent:*

- (i)  $R$  is transitive;
- (ii) every transposition  $(i^+ i^-)$  is an automorphism of  $M_R$ ;
- (iii) the whole group generated by these  $n$  commuting transpositions acts on the bases of  $M_R$ .

*Proof.* Assume transitivity. In a basis  $C(I, J)$ , a label in  $I \setminus J$  has both copies selected, and a label in  $J \setminus I$  has neither copy selected. Every other label has exactly one selected copy. Swapping that single copy changes only whether the label belongs to both  $I, J$  or to neither. By Lemma 103.1, this leaves matchability unchanged. The transpositions commute, so (ii) and (iii) are equivalent.

Conversely, suppose  $iRj$  and  $jRk$  but not  $iRk$ . Reflexivity implies that  $i, j, k$  are distinct. The support pair  $I = \{i, j\}, J = \{j, k\}$  is matchable. In its basis,  $j^+$  is selected and  $j^-$  is not. Swapping these two copies produces exactly  $C(\{i\}, \{k\})$ , which is not a basis because  $iRk$  fails. Thus (ii) forces transitivity.  $\square$

[Merge note: Theorem 103.2 is Part IV's Proposition 51.4 (cancellation for all pairs holds exactly for transitive reflexive relations, with the same two-edge counterexample) read in  $M_R$ : by the basis description (103.1), swapping  $i^+$  and  $i^-$  is the toggle of  $i$  in the common part of  $(I, J)$ . Source 09 lists it as a principal contribution without citing Part IV; its new content is the matroid formulation.]

**Remark 103.3.** The theorem is a classification within the specified fundamental transversal presentations. It does not say that every matroid with paired clones comes from a preorder. The distinguished private-slot presentation and reflexivity are essential hypotheses.

## 104 Symmetric presentations and an exact upset rank formula

For a preorder  $\tau$ , write  $\uparrow S = \{j : \exists i \in S, i \leq_\tau j\}$ . An upset is a set  $U$  with  $\uparrow U = U$ . For a set  $X \subseteq E^+ \sqcup E^-$ , put  $U^\pm = U^+ \sqcup U^-$  and let  $x_i = |X \cap \{i^+, i^-\}|$ .

**Theorem 104.1** (Symmetric presentation and rank). *The matroid  $M_\tau$  has a transversal presentation with one slot  $j$  for each  $j \in E$ , in which both copies of  $i$  are allowed in slot  $j$  exactly when  $i \leq_\tau j$ . In particular,*

$$X \text{ is independent} \iff |X \cap U^\pm| \leq |U| \text{ for every upset } U, \quad (104.1)$$

and for every  $X$ ,

$$\text{rk}_{M_\tau}(X) = \min_{U \text{ upset}} (|U| + |X \setminus U^\pm|). \quad (104.2)$$

*Proof.* First consider an  $n$ -element set  $X$  of copies. Let  $P$  be the labels whose two copies are selected and  $N$  those whose two copies are omitted. Then  $|P| = |N|$  and

$$|X \cap U^\pm| = |U| + |P \cap U| - |N \cap U|.$$

The upset inequalities are therefore equivalent to  $|P \cap U| \leq |N \cap U|$  for all upsets  $U$ . If  $P$  can be matched upward into  $N$ , these inequalities hold, since an edge starting in an upset also ends there. Conversely, for any  $S \subseteq P$ , take  $U = \uparrow S$ . Then

$$|S| \leq |P \cap U| \leq |N \cap U| = |N \cap \uparrow S|,$$

which is Hall's criterion for a matching from  $P$  to  $N$ . By overlap cancellation, these are exactly the bases of  $M_\tau$ .

In the symmetric slot presentation, Hall's inequalities for an arbitrary set  $Z$  of selected copies read  $|Z| \leq |\uparrow\pi(Z)|$ , where  $\pi$  forgets signs. The upset inequalities imply these because  $Z \subseteq X \cap (\uparrow\pi(Z))^\pm$ . Conversely, for an upset  $U$ , every neighbor of  $X \cap U^\pm$  lies in  $U$ , so Hall implies the upset inequality. The two presentations consequently have identical bases of rank  $n$ , hence identical independent sets.

For the rank formula, the maximum-matching deficiency formula gives

$$\text{rk}(X) = \min_{Z \subseteq X} (|X \setminus Z| + |N(Z)|).$$

For a given  $Z$ , its neighborhood  $U = N(Z) = \uparrow\pi(Z)$  is an upset, and  $Z \subseteq X \cap U^\pm$ . Hence the expression for  $Z$  is at least  $|X \setminus U^\pm| + |U|$ . Conversely, given an upset  $U$ , take  $Z = X \cap U^\pm$ . Its neighborhood is contained in  $U$ , so the matching minimum is at most  $|X \setminus U^\pm| + |U|$ . Taking minima proves equality.  $\square$

**Example 104.2** (Why transitive edges, not only covers, matter). For the chain  $1 < 2 < 3$ , both copies of 1 may use every slot, both copies of 2 may use slots 2, 3, and both copies of 3 may use only slot 3. Thus the two copies of 3 are parallel, while the two copies of 1 are independent clones. Replacing the order relation by the cover graph would remove the allowed assignment  $1 \mapsto 3$  and change the matroid.

The formula is useful even when listing all upsets is expensive: the symmetric bipartite matching problem is a polynomial-size rank oracle. It does not assert that all coefficients can be counted in polynomial time.

## 105 Intrinsic reconstruction and the full automorphism group

The paired symmetry is not the end of the story. It identifies all clone classes and allows the distinguished labels to be discarded.

**Theorem 105.1** (All clone classes). *Let  $C$  range over the equivalence classes of  $\tau$ , where  $i \sim_\tau j$  means  $i \leq_\tau j$  and  $j \leq_\tau i$ . The clone classes of  $M_\tau$  are exactly the sets  $C^\pm$ . Each has size  $2m_C$ .*

*Proof.* Equivalent labels have the same upward neighborhood. In the symmetric presentation of Theorem 104.1, every permutation of their copies therefore preserves independent sets. So  $C^\pm$  lies in one clone class.

Conversely, suppose first that  $i^+$  and  $j^+$  are clones. The set  $E^- \setminus \{i^-\} \cup \{i^+\}$  is a basis. Swapping  $i^+, j^+$  shows that  $E^- \setminus \{i^-\} \cup \{j^+\}$  is a basis, so  $j \leq_\tau i$ . Interchanging  $i, j$  gives  $i \leq_\tau j$ . If arbitrary copies of  $i, j$  are clones, conjugate their transposition by the paired clone transpositions from Theorem 103.2. This reduces to the plus-plus case. Thus no clone class merges two distinct preorder equivalence classes.  $\square$

Call a set of matroid elements *balanced* if it contains exactly half of every clone class. For  $M_\tau$ , all clone classes have even size and every balanced set is a basis: start from  $E^-$  and permute within each clone class.

**Theorem 105.2** (Reconstruction from the unlabeled matroid). *The isomorphism type of  $M_\tau$  determines the isomorphism type of  $\tau$ . More explicitly, recover its clone classes  $C$ , give  $C$  weight  $|C|/2$ , and choose any balanced basis  $B_0$ . For clone classes  $C, D$ , define*

$$C \leq D \iff B_0 \setminus \{d\} \cup \{c\} \text{ is a basis,} \quad c \in C \setminus B_0, \quad d \in D \cap B_0. \quad (105.1)$$

This is well-defined and recovers the quotient poset of  $\tau$ . Consequently (101.1) holds.

*Proof.* Permutations within clone classes show that the answer in (105.1) does not depend on  $c, d$ . They also carry any balanced basis to any other, preserving clone classes, so the answer does not depend on  $B_0$ . We may therefore take  $B_0 = E^-$ . Then  $c = i^+$  and  $d = j^-$ , and Proposition 102.2 says precisely that the exchanged set is a basis if and only if  $i \leq_\tau j$ . Theorem 105.1 identifies the classes with preorder equivalence classes, so the recovered relation is exactly their quotient order. Their weights recover the number of original elements in each equivalence class. These data determine a finite preorder up to isomorphism. Conversely, any preorder isomorphism lifts to a matroid isomorphism by applying it to both copies.  $\square$

This is stronger than recovering a preorder from a *labeled* fundamental presentation. The distinguished basis and the original pairing are unnecessary for isomorphism reconstruction, although they remain convenient for explicit formulas.

**Corollary 105.3** (Automorphism group). *There is a split exact sequence*

$$1 \longrightarrow \prod_{C \in \bar{\tau}} \mathfrak{S}_{2m_C} \longrightarrow \text{Aut}(M_\tau) \longrightarrow \text{Aut}(\bar{\tau}, m) \longrightarrow 1.$$

In particular, (101.2) holds and

$$|\text{Aut}(M_\tau)| = |\text{Aut}(\bar{\tau}, m)| \prod_C (2m_C)!. \quad (105.2)$$

For a poset, this specializes to  $\text{Aut}(M_\tau) \cong (\mathbb{Z}/2\mathbb{Z})^n \rtimes \text{Aut}(\tau)$ .

*Proof.* An automorphism preserves the intrinsic clone partition and the reconstructed weighted order. Its kernel is exactly the product of all permutation groups inside clone classes; every such permutation is allowed by the symmetric presentation. Any weight-preserving automorphism of the quotient poset lifts by matching copies in classes of equal size; (104.2) is invariant under each such lift. Choose once and for all an ordering of the  $2m_C$  copies in every class. Lifting a quotient automorphism by preserving these local indices supplies a group homomorphism section and proves the split description.  $\square$

**Example 105.4.** For an antichain of  $n$  singleton classes,  $M_\tau$  is the direct sum of  $n$  copies of the two-element rank-one uniform matroid. Its automorphism group is  $(\mathbb{Z}/2\mathbb{Z})^n \rtimes \mathfrak{S}_n$ . For the universal equivalence relation on  $n$  elements,  $M_\tau = U_{n,2n}$  and the automorphism group is  $\mathfrak{S}_{2n}$ . For a chain, the order has no nontrivial automorphisms, so the group has order  $2^n$ .

## 106 Order duality and induced restrictions as matroid operations

**Theorem 106.1** (Duality and restriction). *On the fixed ground set  $E^+ \sqcup E^-$ ,*

$$M_\tau^* = M_{\tau^{\text{op}}}. \quad (106.1)$$

*If  $S \subseteq E$  and  $T = E \setminus S$ , then after retaining the natural labels on  $S^+ \sqcup S^-$ ,*

$$M_{\tau|S} = (M_\tau/T^-) \setminus T^+. \quad (106.2)$$

*In particular,  $M_\tau$  is isomorphic to its dual exactly when  $\tau$  is isomorphic to its opposite preorder.*

*Proof.* Let  $\iota$  exchange the two copies of every label. The complement of  $C(I, J)$  is  $(E \setminus I)^+ \sqcup J^-$ . Applying  $\iota$  gives  $J^+ \sqcup (E \setminus I)^- = C(J, I)$ . A matching  $I \rightarrow J$  in  $\tau$  is a matching  $J \rightarrow I$  in  $\tau^{\text{op}}$ . Thus  $\iota$  identifies  $M_\tau^*$  with  $M_{\tau^{\text{op}}}$ . But  $\iota$  is already an automorphism of  $M_{\tau^{\text{op}}}$  by paired clonal symmetry. This proves the labeled identity.

The set  $T^-$  is independent. A basis contributing to the contraction and deletion in (106.2) contains  $T^-$  and avoids  $T^+$ . In (103.1) this means  $I, J \subseteq S$ . The remaining support condition is precisely matching in the induced preorder  $\tau|_S$ . These remaining bases have size  $|S|$  and include  $S^-$ , so no rank loss is hidden by the deletion. The final assertion follows from Theorem 105.2.  $\square$

For relations  $\tau \subseteq \sigma$  on the same labeled set,  $\mathcal{B}(M_\tau) \subseteq \mathcal{B}(M_\sigma)$ : every old matching remains available. This gives the corresponding rank inequalities and explains the coefficient monotonicity directly at the level of bases.

[Merge note (neighbouring report): the report *preorder-q-zeta-reciprocity* in this collection defines, for a set  $B$ , a transversal matroid  $M_B$  on  $E$  whose independent sets are the  $S$  with  $\mathbf{1}_B + \mathbf{1}_S$  in its lattice-point set, of rank  $n - |B|$  (its Proposition “Tutte evaluation”, label `prop:matroid`). The batch-55 placement dossier compared it with the minor  $(M_{\tau^{\text{op}}}/B^-) \setminus (E \setminus B)^-$  of the present matroid, on  $E^+$  identified with  $E$ , for all 5,930 pairs  $(\tau, B)$  with  $n \leq 4$ , and found them equal. This is a finite check, not a theorem of either report.]

## 107 The complete orbit formula and weighted preorder polynomials

A *transport pair* is an ordered pair  $(U, V)$  of disjoint subsets of  $E$ , of equal size, admitting a bijection  $U \rightarrow V$  along  $\tau$ . Write  $\mathcal{T}_k(\tau)$  for the pairs of size  $k$ . We count the pair once, not once for each bijection.

**Theorem 107.1** (Matroid orbit polynomial; Part IV’s Theorem 52.1 with multivariate weights). *For arbitrary commuting variables  $a_i, b_i$ ,*

$$F_\tau(a, b) = \sum_{(U, V) \in \mathcal{T}(\tau)} \left( \prod_{i \in U} a_i b_i \right) \prod_{i \notin U \cup V} (a_i + b_i). \quad (107.1)$$

*The summands are the generating polynomials of the orbits of the paired clone group on the bases.*

*Proof.* For a basis  $C(I, J)$ , put  $U = I \setminus J$  and  $V = J \setminus I$ . Overlap cancellation says that  $(U, V)$  is a transport pair. Exactly the labels of  $U$  contribute both copies, those of  $V$  contribute neither, and every other label contributes either copy independently. The paired clone group is transitive on these independent choices and changes neither  $U$  nor  $V$ . Conversely, every choice described by a transport pair is a basis. Its monomial is counted in exactly one orbit, proving the identity.  $\square$

For positive numbers  $w = (w_i)_{i \in E}$ , define the *transport-weighted preorder polynomial*

$$h_{\tau, w}(t) = \sum_{(I, J) \in \mathcal{M}(G_\tau)} \frac{w^I}{w^J} t^{|I|}. \quad (107.2)$$

At  $w = \mathbf{1}$ , this is the actual lattice-point support polynomial by Proposition 71.1. For general  $w$ , we do not claim a weight-preserving bijection with the original lattice points.

**Theorem 107.2** (Weighted symmetry, gamma expansion, and inequalities). *Put*

$$\gamma_k(\tau; w) = \sum_{(U, V) \in \mathcal{T}_k(\tau)} \frac{w^U}{w^V}.$$

Then

$$h_{\tau,w}(t) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_k(\tau; w) t^k (1+t)^{n-2k}. \quad (107.3)$$

In particular  $h_{\tau,w}$  is palindromic and gamma-positive. Its coefficients are ultra-log-concave of order  $n$ , and are strictly log-concave without normalization at every interior index. They are at least  $\binom{n}{k}$ .

*Proof.* Set  $W = \prod_i w_i$ ,  $a_i = w_i s$ ,  $b_i = w_i t$  in (107.1). A transport pair of size  $k$  contributes

$$W \frac{w^U}{w^V} (st)^k (s+t)^{n-2k}.$$

On the other hand, the basis definition gives

$$F_{\tau}(ws, wt) = W \sum_{(I,J) \in \mathcal{M}(G_{\tau})} \frac{w^I}{w^J} s^{|I|} t^{n-|I|}.$$

Equating the two expressions and setting  $(s, t) = (t, 1)$  proves the gamma formula. The homogeneous specialization is Lorentzian, so its coefficients satisfy ultra-log-concavity of order  $n$ . Alternatively use Theorem 70.1 with  $\alpha_i = w_i$  and  $\beta_i = w_i^{-1}$ . Diagonal supports have weight one, proving the lower bound and positivity; strict ordinary log-concavity follows as before.  $\square$

The unweighted transport-pair gamma formula is already in Part IV of the repository report (Theorem 48.2); the Boolean orbits of the clone transpositions are Part IV's Theorem 52.1. Here it is placed inside the intrinsic matroid orbit structure and extended to the indicated activities. It does not imply that the gamma sequence itself is ultra-log-concave, nor that  $h_{\tau,w}$  has only real roots.

## 108 Height-two preorders generate all finite gammoids under minors

A height-two poset is already sufficient to encode an arbitrary transversal matroid as a minor. This makes the new class structurally rich rather than merely a narrow collection of special matroids.

**Theorem 108.1** (Explicit height-two lift). *Let  $G = (A \sqcup B, \mathcal{E})$  be any finite bipartite graph, and let  $P_G$  be the poset on  $A \sqcup B$  whose strict relations are exactly  $a < b$  for  $ab \in \mathcal{E}$ . Then*

$$M_G \cong (M_{P_G}/A^-) \setminus B^-. \quad (108.1)$$

*The surviving ground elements are  $A^+ \sqcup B^+$ , identified with  $A \sqcup B$  on the left. If  $T_G$  is the transversal matroid on  $A$  with slots  $B$  and adjacency  $G$ , then*

$$T_G \cong (M_{P_G}/A^-) \setminus (B^- \sqcup B^+). \quad (108.2)$$

*Proof.* Put  $N = M_{P_G}$ , of rank  $|A| + |B|$ . The contracted set  $A^-$  is independent. After contraction, delete  $B^-$ . A remaining basis has  $r = |B|$  elements and can be written  $I^+ \sqcup (B \setminus J)^+$  with  $I \subseteq A$ ,  $J \subseteq B$ ,  $|I| = |J|$ . Together with  $A^-$ , its double-selected labels are  $I$ , its omitted labels are  $J$ , and all other labels are neutral. By overlap cancellation, it is a basis exactly when  $I$  matches upward to  $J$  in  $P_G$ , that is, when  $(I, J) \in \mathcal{M}(G)$ . These are the bases of  $M_G$  from Proposition 102.2.

The minor has rank  $r$ , since  $A^- \sqcup B^+$  is a basis of  $N$ . Restricting  $M_G$  to  $A$  removes the private ground elements  $B$  but not their slots. Its independent sets are precisely the subsets of  $A$  that can be matched into  $B$ , which is  $T_G$ . This proves (108.2), even if that final deletion lowers rank.  $\square$

On basis-generating polynomials, the first identity is the explicit contraction/deletion operation

$$F_G = \left( \prod_{a \in A} \partial_{a^-} \right) F_{P_G} \Big|_{b^- = 0 \ (b \in B)}, \quad (108.3)$$

with the surviving plus variables renamed. Because  $A^-$  is independent and the deletion in (108.1) preserves the contracted rank, there is no missing scalar or lower-degree component.

A finite gammoid can be defined through vertex-disjoint directed paths. We use the classical characterization that the finite gammoids are exactly the minors of transversal matroids, equivalently the contractions of transversal matroids [26].

**Corollary 108.2** (Exact minor closure). *The smallest minor-closed class containing  $M_P$  for all finite height-two posets  $P$  is exactly the class of finite gammoids. The same is true if all finite preorder matroids are used as generators.*

*Proof.* Every  $M_\tau$  is transversal, so every minor of one is a finite gammoid. Conversely, a finite gammoid is a minor of some transversal matroid  $T_G$ . Theorem 108.1 realizes  $T_G$  as a minor of a height-two poset matroid. Transitivity of the minor relation completes the proof.  $\square$

The gammoid characterization is not new. The explicit two-copy height-two encoding is the contribution used to obtain this particular generating class. No converse is claimed for stability: a stable minor need not have a stable height-two lift.

## 109 Claims deliberately not made

(Source 09’s Section 14.4, as delivered.) These results do not prove real-rootedness for all preorder polynomials, do not classify the stable 2-connected bipartite graphs, and do not construct a flag simplicial polytope for every preorder. They do not provide a canonical bijection from lattice points to matching supports. They also do not show that the gamma sequence of an arbitrary preorder is ultra-log-concave. Distinguishing these statements prevents a valid matroid lift from being used to infer properties it does not possess. [Added 30 September 2026, batch 64: this describes source 09. Part IX constructs a bijection from lattice points to matching supports (Theorem 135.2); it is not canonical (Proposition 144.1).]

## 110 Exact computational verification

The archive includes `verify.py` [shipped as `code/09-matroid-lifts-verify.py`], using Python’s standard library only. It uses integer and rational arithmetic; no floating-point root finder, random sampling, or numerical optimization is involved. Its finite conclusions are supplementary evidence. The all-size statements rest on the proofs above.

The run supplied with this manuscript passed the following checks:



| Check family                                                                                                                      | Instances |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------|
| All bipartite graphs in shore sizes $(1, 1), (2, 2), (2, 3), (3, 3), (3, 4)$ ,<br>with unweighted and fixed positive weighted ULC | 4,690     |
| All reflexive relations on sets of sizes 1, 2, 3, 4                                                                               | 4,165     |
| All preorders among those relations                                                                                               | 389       |
| Subset-rank comparisons with upset and symmetric-slot formulas                                                                    | 92,804    |
| Full automorphism-group orders, all preorders of sizes 1, 2, 3                                                                    | 34        |
| Height-two minor constructions, shore sizes $(2, 2), (2, 3), (3, 3)$                                                              | 592       |
| Exact signed vertex-sum identities                                                                                                | 16        |

For each preorder through size four, the program independently enumerates integer points of (67.4) and compares their support counts to the matching-support coefficients. It checks the intrinsic clone relation, order-dual/matroid-dual identity, unweighted and weighted gamma expansions, and weighted palindromicity. The rank comparison uses three routes: intersection with an enumerated basis, the upset minimum, and an independent augmenting-path matching implementation.

For each tested graph, the support-to-basis bijection is also checked against the separate slot-matching oracle. The theta polynomial and the Rayleigh value  $-9$  are reconstructed from vertex-subset feasibility. The counts, certificate assignment, and execution result are recorded in `verification_results.json`. Running

```
python3 verify.py --output verification_results.json
```

repeats the checks. Execution time is recorded as an observation of the local run, not a portability guarantee. [Merge note: this is the delivered command, run in the package root. The recorded output is shipped as `data/09-matroid-lifts-verification_results.json` and, byte for byte the same, `data/09-matroid-lifts-verification_run.txt`. The program writes `verification_results.json` in the working directory unless `-output` is given, and the delivered Makefile (`code/09-matroid-lifts-Makefile`) runs `pdflatex` on `article.tex` and deletes `article.aux` and similar files in its working directory: run from this report's directory it would act on this article. The README gives a command that runs on a copy. The checks of the stability sections recorded here belong to Part VI.]

## 111 A staged formalization plan

The elementary and deep parts have a clean separation, which is useful for future integration into ProveIt.

**Finite combinatorial core.** Formalize a finite relation, matchable support pairs, and the private-slot transversal presentation. Prove the basis equivalence using matchings as existence witnesses, explicitly quotienting away multiplicity by counting sets. Then prove overlap cancellation and the clone/transitivity criterion. These require finite graph decompositions and finite-set algebra, not real analysis.

**Reconstruction and minors.** The symmetric presentation, upset rank formula, intrinsic clone classes, reconstruction, automorphism kernel and quotient, duality, and the two minor constructions can be developed on top of the finite matroid API. The automorphism splitting should be stated with chosen local identifications, while its exact-sequence version is intrinsic.

The supplied small instances are suitable regression tests but are not proof certificates for arbitrary finite sets.

**Coefficient theorem.** There are two explicit external boundaries: the enumerative bridge (68.4), and the Lorentzian theorem for matroid basis polynomials plus nonnegative substitutions. A formal development may first parameterize the statements by these named theorems, but should not mark such a module as an unconditional end-to-end formal proof until the assumptions are discharged. In particular, `native_decide` checks of finitely many coefficients cannot establish the universal theorem.

**Complex-analytic stability.** The reciprocal bridge is elementary algebra and the fact that  $-1/z$  preserves  $\mathbb{H}$ . The block theorem additionally uses differentiation closure of real stability. The complete-bipartite seed uses elementary symmetric stability; the cycle seed can instead be formalized directly by the Hermitian positive-imaginary-part argument. No claim is made here that these modules have been written or compiled.

## 112 Further research questions and concrete targets

The following questions are organized around the remaining obstructions, not around the already proved gamma expansion or the already disproved universal real-rootedness statement.

**Research question 54** (Characterize the image among matroids). Give an intrinsic characterization of the matroids isomorphic to some  $M_\tau$ , without presupposing a private-slot relation presentation. Necessary conditions include even clone-class sizes, balanced bases, and the reconstructed order. Are those conditions together with the upset rank formula sufficient in a short axiom system? Theorem 105.2 proves uniqueness inside the image, not an intrinsic recognition theorem for arbitrary matroids.

**Research question 55** (Which minors remain preorder matroids?). Determine exactly which deletions and contractions preserve the class. Paired contraction/deletion realizes induced preorders by (106.2), whereas unrestricted minors generate all finite gammoids. This large gap suggests a useful intermediate minor theory that preserves balanced clone classes and explains additional quotient-order operations.

**Research question 56** (Weighted geometric interpretation). Find a natural geometric statistic on  $Q_\tau \cap \mathbb{Z}^E$  whose generating function is  $h_{\tau,w}$ , or prove that a specified natural class of local-coordinate weights cannot realize it. The ratios  $w^U/w^V$  are natural on transport pairs and matroid bases, but that does not identify a canonical lattice-point weight.

**Research question 57** (Proof-assistant completion). Formalize first the clone/transitivity criterion and the explicit height-two minor construction, whose dependencies are finite. Then extend to intrinsic reconstruction and the automorphism exact sequence. These provide substantial unconditional milestones before the larger task of formalizing the Lorentzian and root-polytope input needed for the coefficient theorem.

### 112.1 Questions of source 09 printed elsewhere or already posed

Source 09's first two questions ("Which 2-connected bipartite graphs are stable?" and "Minimal induced obstructions") are Part VI's Research questions 44 and 53. The following repeat earlier questions of this report and are printed as delivered, without new numbers.

*Close to Research question 38* (equality at a single index):

**Strict normalized inequalities (source 09).** Characterize equality in a single interior preorder inequality (67.5), including positive transport activities. Simultaneous equality at every index is settled by Theorem 67.2, but that does not settle individual equality cases. A structural criterion in terms of tight upset constraints or the symmetric matroid presentation would be more informative than a coefficient-only test.

*Repeats Research question 24* (coefficient shape; Part V answers height two):

**Gamma log-concavity beyond height two (source 09).** Are the transport coefficients  $\gamma_k(\tau)$  log-concave for every finite preorder? Which normalization, if any, produces a valid universal ultra-log-concavity statement? The ordinary  $h$ -coefficient theorem does not answer this: the transformation between  $h$  and  $\gamma$  is not a nonnegative linear substitution on the matroid variables. Height two is a controlled starting case because its gamma polynomial is  $p_G$ .

*Repeats Research question 20* (a support-resolved bijection):

**Direct lattice-point bijections (source 09).** Construct a canonical, support-resolved bijection realizing (68.4), ideally compatible with deletion or with the paired clone action in preorder cases. The present proof uses established root-polytope identities; equality of cardinalities alone does not lift the matroid automorphisms to explicit transformations of lattice points. [Added 30 September 2026, batch 64: Part IX constructs a support-resolved bijection realizing (68.4), compatible with deletion of receivers outside the support and of suppliers outside the selected set. It is not canonical: it depends on the chamber of a cost system, no deterministic rule is equivariant under all automorphisms (Proposition 144.1), and compatibility with the paired clone action is not obtained. See Section 132.3.]

*Repeats Research question 28* (limit laws):

**Sharper distributional consequences (source 09).** For sequences of preorders with growing size, determine hypotheses on upset structure that force a central limit theorem for the support size. Binomial convex domination gives concentration but not Gaussian limits; for example, a family may have variance much smaller than its size. A promising target is a quantitative theorem with an explicit variance lower bound and a decomposition condition on the quotient poset.

## 113 Conclusion

The matching-support model has two distinct levels of strength. Its univariate polynomial satisfies universal coefficient inequalities by the classical bimatroid theory. Its full matroid, in the preorder case, remembers enough to reconstruct the entire preorder and compute all its matroid automorphisms. Height-two examples already generate every finite gammoid under minors, so this structural connection is not confined to chains or complete relations.

[Merge note: source 09's conclusion continues with a paragraph on stability, printed in Part VI's conclusion.]

## 114 The matching facts used in the elementary proofs

*(Appendix A of source 09.)*

### 114.1 Why slot-matchable sets form a matroid

Hereditary closure is immediate: restrict an injective slot assignment. For augmentation, let  $I, J$  be matchable ground sets with  $|I| < |J|$ , and choose assignments saturating each. Superimpose the two bipartite matchings on ground elements and slots. Their symmetric difference is a disjoint union of alternating paths and cycles. Since the  $J$ -matching has more edges, one alternating path has one more  $J$ -edge than  $I$ -edge. Its endpoints are a ground element  $j \in J \setminus I$  not saturated by the  $I$ -matching and a slot not saturated by that matching. Flipping along the path augments the  $I$ -matching without losing any of its previously matched ground elements. Thus  $I \cup \{j\}$  is matchable, proving augmentation.

### 114.2 The maximum-matching deficiency formula

For a bipartite graph with left set  $X$ , every matching has size at most  $|X \setminus Z| + |N(Z)|$  for each  $Z \subseteq X$ : it can match at most  $|N(Z)|$  members of  $Z$ . Let a maximum matching leave  $d$  left vertices unmatched. Start alternating searches from all of them and let  $Z$  be the reached left set and  $W$  the reached right set. No right vertex in  $W$  is unmatched, since that would give an augmenting path. Every neighbor of a reached left vertex is reached, and every matched partner of a reached right vertex is reached; hence  $W = N(Z)$ . The matching pairs  $W$  with  $Z$  minus the  $d$  unmatched starting vertices. Therefore  $|Z| - |N(Z)| = d$ , and the upper bound is attained:

$$\nu = |X| - d = |X \setminus Z| + |N(Z)|.$$

This proves the deficiency formula used in Theorem 104.1 and also proves Hall's criterion by taking  $d = 0$ .

## 115 A compact dependency and claim audit

(Appendix B of source 09.)

### 115.1 Repository snapshot and inspected material

The source repository is <https://github.com/VladimirReshetnikov/ProveIt>, pinned to e1afd75e35a4de734d5aa47aec5cdb917b82ed3e. The investigated report is located at

SetTheory/Cardinals/docs/reports/  
enumerative-combinatorics/preorder-root-polytopes/

Its `article.tex` blob at this pin is d52a2633b58e74aa653e967eb56adc1aa41bc8ae; its `README.md` blob is 846d9f19fadac9cd933b74370cdb8fafa2211c5d. The audit used the report's status and result summaries and its statements and proofs of the matching-support, transitivity, transport, and degree identities. It did not audit the entire repository, and it does not endorse unrelated claims in the repository's top-level README.

### 115.2 Dependencies that remain external

The preorder lattice-point coefficient theorem uses the established root-polytope identification from [2, 7], and the Lorentzian matroid/substitution theorems from [18]. The unweighted graph inequality is already Theorem A of [17]. The exact gammoid minor closure uses the classical characterization in [26] after our explicit height-two construction. Stability uses standard differentiation and real-specialization closure; the new graph gluing proof is then

the one-line upper-half-plane quotient argument, with its support-counting identity proved separately.

None of the reconstruction, clone classification, rank, automorphism, duality, orbit, or explicit minor arguments depends on the repository's unrefereed polytope-realization claims. The theta example is a prior counterexample, reproduced only to illustrate the boundary between the properties.

### 115.3 Novelty boundary

Targeted searches covered preorder ultra-log-concavity, relation bimatroids, paired matroid clones, preorder reconstruction from transversal matroids, and signed matching-support stability. The primary bimatroid paper was particularly important: it prevents claiming the general matching-support ULC theorem as new. No matching statement of the preorder reconstruction/automorphism theorem or the exact block-stability theorem was identified in the inspected sources. That negative search result is not an exhaustive priority certificate, and a literature check by specialists remains appropriate before publication.

The manuscript does not claim that all ten further questions are known published open problems in exactly this formulation. They are a mixture of remaining repository questions, natural sharpenings, and new questions suggested by the proved construction. The concrete new claims are the numbered theorems with written proofs, not the speculative research list.

### 115.4 Reproducibility and trust boundary

The delivered TeX compiles using ordinary pdf<sub>l</sub>at<sub>e</sub>x packages. The bibliography is embedded, so no BibTeX database is needed. The exact verification program is independent of the PDF generation and records its own output. Finite enumeration checks cannot certify global stability, universal coefficient inequalities, or all-size reconstruction; those are proved mathematically in the text. No Lean, Rocq, or external theorem prover was run for this manuscript.

## Part VIII. Marked-tree stability: arbitrary apex neighbourhoods, enumeration and repair

*Part VIII was added on 30 September 2026 in batch 63 of ProveIt's incoming-report intake, from one manuscript. It removes the terminal convention of Part VI's tree criterion for a single apex: for a tree  $T$  and an arbitrary set  $S$  of apex neighbours, the signed support polynomial of the incidence graph of  $T$  with one apex adjacent exactly to  $S$  is real stable if and only if  $S$  is empty, or every branching vertex of the subtree spanned by  $S$  is marked and the tree obtained by suppressing its unmarked degree-two vertices is a finite or affine ADE tree. This classification is a corollary of Part VI's theorems. The manuscript's direct host-level proof, the existence dichotomy for the supported covariance, and the consequences—linear-time recognition, a sharp diameter window, an exact endpoint formula for the stable neighbourhoods with  $O(n^5)$  counting, optimization and repair, and a path-only real-rootedness theorem for the stable-neighbourhood count—are new relative to this report. Parts I–VII are unchanged except for dated pointers. Section 116 records the provenance, fixes the notation and states what is answered.*

### 116 Provenance, conventions and what Part VIII answers

#### 116.1 The manuscript

Part VIII is printed from batch-63 manuscript 03, called *source 10* here after its file prefix: *Marked-Tree Stability: Complete Apex-Neighborhood Classification, Exact Enumeration, and Optimal Repair*, dated 30 September 2026, 24 A4 pages as delivered. Its title page reads “Research prepared for Vladimir Reshetnikov / Mathematical development and exposition with ChatGPT”, and its PDF metadata name ChatGPT; that byline is recorded here and not reprinted. Its archive `ProveIt_Marked_Tree_Stability.zip` (inner directory of the same name, main file `article.tex`) arrived in ProveIt commit `a4268e78e` and was placed in `62f1ad07c`. It is pinned to commit `b57c0b5ff` (full hash in Section 117), with this report's `article.tex` as blob `920246d173bfbcb708bcfdd6d67e0a6fc358cb3dc`: Parts I–VII exactly as printed, unchanged from the pin to the placement (the last change before it was `8c6517c0f`). Its library, verifier and build script are shipped in `code/`, its requirements, console log and recorded output in `data/`, and its source audit at the report root, all with the prefix `10-marked-trees-`; its checksum list `SHA256SUMS.txt` was verified (10/10) at placement and retired. The manuscript, delivery README and PDF are not shipped; they survive in the arrival commit.

Source 10 is a continuation of Part VI and credits it exactly: the full-apex criterion, the terminal-subdivision theorem, the canonical matroid and inversion bridge, path shortening, the  $\mathfrak{d}(3, 3, 3)$  obstruction with its credit to Shivam Patel, and stability under tree attachments. Its new material is a completion of Part VI's tree criterion to arbitrary apex neighbourhoods and the enumeration, optimization and root-geometry consequences of that completion.



**Abstract of source 10 (as delivered).** Given a tree  $T$  and a set  $S$  of marked vertices, form the bipartite incidence graph of  $T$  and adjoin one apex adjacent exactly to  $S$ . We classify real stability of its Boolean matching-support polynomial for *every* marking. The answer has two independent parts: every branching vertex of the minimal subtree spanning  $S$  must itself be marked, and the tree obtained by suppressing its unmarked degree-two vertices must be a finite or affine ADE tree. An unmarked branch forces an induced odd-theta obstruction; on the remaining markings a uniquely determined, terminal-supported covariance gives an exact determinant–adjugate identity. Indefiniteness produces a rational negative Rayleigh certificate on the original tree, including subdivisions and unmarked attachments.

The classification has consequences not requiring enumeration of all  $2^{|V(T)|}$  markings. We derive a finite endpoint formula for the complete weighted stable-neighborhood enumerator and obtain exact  $O(n^5)$  arithmetic algorithms for counting, maximum-weight selection, and minimum-cost apex-edge repair. The largest stable marking lies between  $\text{diam}(T) + 1$  and  $\text{diam}(T) + 3$ , with both bounds sharp. The polynomial counting stable markings by size is real-rooted precisely when  $T$  is a path; even log-concavity and unimodality fail on small stars.

The starting ADE theorem, its terminal-subdivision version, and the odd-theta obstruction are credited to the pinned ProveIt report rather than claimed anew. The arbitrary-marking completion, covariance-existence dichotomy, endpoint enumeration, and optimization results are the research continuation developed here. Full proofs, exact finite checks, limitations, and further questions are included.

The abstract is reproduced as delivered. Its “pinned ProveIt report” is this report (Part VI), and its “arbitrary-marking completion” is read with the priority box below: the classification follows from Part VI’s theorems, and what is new is its statement, its direct proof and its consequences.

**Status (source 10, as delivered).** This is an unrefereed, AI-assisted research article. The all-size statements rest on the written proofs; the supplied tests are not a Lean or Rocq formalization. Novelty is assessed relative to the inspected repository and a focused literature review, not certified worldwide.

**Priority and prior work (merge).** Theorem 118.2, the classification of all markings, is new as a statement but is a corollary of Part VI: for a branch-complete marking,  $\mathcal{A}(T; S)$  is the terminal-skeleton graph of Theorem 89.1 with trees attached by one-vertex sums, which Theorem 93.2 controls, and an unmarked hull branch yields an induced multi-theta graph excluded by Theorem 84.3(ii). Source 10 says so itself (Section 130: “a synthesis and extension of specific existing results, not a first discovery of ADE stability”), although its comparison table lists the classification as “continuation developed here”; that row carries a merge note. The theorem is printed with the corollary proof first and source 10’s direct proof second. Theorems 121.1, 121.3 and 122.2 generalize Part VI’s Proposition 86.2, Theorem 86.3 with identity (89.1), and Theorem 87.2 to arbitrary markings and to hosts with unmarked fringe; the nonexistence half of Theorem 121.1 is new. The canonical matroid, the inversion bridge, the Rayleigh lemma, path shortening, the resolvent lemma, the ADE list, the  $\mathfrak{g}(3, 3, 3)$  quartic and the  $K_{1,5}$  certificate are Part VI’s and are printed as pointers (Section 116.4); the quartic’s nonreal-root example is Shivam Patel’s [11]. The ADE list is classical, and Kummer and Sert classify the half-plane-property matroids on at most eight elements [28]. Recognition, the diameter window, the endpoint enumerator, counting, optimization, repair and the neighbourhood polynomial have no counterpart in Parts I–VII. Source 10’s literature review was focused, and no worldwide priority is claimed for any result of Part VIII.

## 116.2 Notation of Part VIII

Part VIII uses Part VI’s  $\Phi_G$ ,  $F_G$ ,  $p_G$ ,  $\mathcal{A}(\cdot)$ ,  $\star$ ,  $L$ ,  $\Delta_{ij}$  and  $\mathfrak{g}(\cdots)$ , and Part VI’s convention that the canonical matroid is  $M_G$  with bases  $(U \setminus I) \sqcup J$  (Section 83.2). Table 13 lists what Part VIII adds or uses differently. Two symbols of source 10 were changed; no normalization was changed:

Table 13: Symbols of Part VIII. The last column prints the tempting false reading.

| Symbol                                                      | Meaning in Part VIII                                                                                         | Not to be confused with                                                                                                                                   |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| $T = (V, E), n$                                             | the host tree, $n =  V $                                                                                     | Part VI's $r$ (the same number); Part I's ground set $E = [n]$ .                                                                                          |
| $S$ , marks                                                 | the apex neighbourhood (a <i>marking</i> )                                                                   | Part VI's $S = W \cap V(T)$ in the proof of Theorem 89.1; Part II's subsets $S$ ; Part I's polytope $S_\tau$ .                                            |
| $\mathcal{A}(T; S), \star, s$                               | incidence graph of $T$ plus one apex $\star$ adjacent exactly to $S$ (source 10's $G(T, S)$ ); apex variable | $\mathcal{A}(T) = \mathcal{A}(T; V(T))$ of Part VI.                                                                                                       |
| $F_{T,S}(x; y, s)$<br>$K = \text{Hull}_T(S)$                | Part VI's basis polynomial $F_G$ of $\mathcal{A}(T; S)$<br>the hull: the smallest subtree containing $S$     | source 09's $F_G$ (the dual convention).<br>the generic matrix $K$ of Lemma 85.1;<br>Part VI's random matrix $K$ ; the stars $K_{1,m}$ and $K_{\ell,r}$ . |
| $H = H(T, S), C_H, C$                                       | terminal skeleton on $S$ ; $I - \frac{1}{2}A_H$ ; its zero extension to $V(T)$                               | the half-plane $\mathbb{H}$ ; Part III's $H$ ; Part VI's $C_T$ (the case $S = V(T)$ ); Part I's $C_\tau$ .                                                |
| $L$                                                         | in Sections 121–122, the pencil $\text{diag}(x) + \sum_e y_e b_e b_e^T$ (Part VI's $L$ )                     | a leaf set $L$ in Sections 124–125.                                                                                                                       |
| $L$ , Leaf, $\mathcal{L}_3, \mathcal{L}_4$                  | a set of hull leaves; the leaf set of a tree; eligible three- and four-leaf sets                             | the pencil $L$ .                                                                                                                                          |
| $o(L), p(L), R_i$                                           | the branch vertices of $\text{Hull}_T(L)$ ; internal vertices of its arms                                    | Part VI's $p_G$ .                                                                                                                                         |
| $\mathcal{S}(T), \mathcal{Z}_T(z), Z_T(t; w)$               | stable markings and their multivariate and univariate enumerators                                            | $p_{\mathcal{A}(T; S)}$ , which counts supports of <i>one</i> marking.                                                                                    |
| $P_{ab}(z), \hat{P}_{ab}$                                   | $\prod (1 + z_v)$ over the internal vertices of the $a$ – $b$ path                                           | the path graph $P_n$ ; Part I's $P_\tau$ ; Part VI's $P_G$ .                                                                                              |
| $E_k(R), Q(R_1, R_2, R_3), \mathcal{E}$                     | elementary symmetric polynomial; three-leaf factor; exceptional arm triples                                  | the edge set $E$ ; Part I's polytope $Q_\tau$ and its equation (1.1).                                                                                     |
| $\alpha_{\text{st}}(T)$ , diam<br>$c_v, d_v, a_v, w_v, p_v$ | largest stable marking size; diameter<br>scores, deletion and addition costs, weights, marking probabilities | $c = \prod_i v_i^{-2}$ in Theorem 122.2; Part VI's auxiliary vector $w$ .                                                                                 |

- source 10's graph  $G(T, S)$  is printed  $\mathcal{A}(T; S)$ , extending Part VI's  $\mathcal{A}(T)$  and  $\mathcal{A}(\hat{T}; T)$ ;
- source 10's  $\Theta_3$  is printed  $\mathfrak{g}(3, 3, 3)$ , following Part VI's rule (Section 83.2); it is also Part II's  $\Theta_3$ .

Source 10's sign  $(-1)^{|U|/2}$  in (118.2) is Part VI's  $(-1)^{|I|}$ , since a matchable support  $U$  has  $|U| = 2|I|$ . Source 10 writes  $n = |V(T)|$  where Part VI writes  $r$ .

### 116.3 What Part VIII answers

Status after Part VIII, as of 30 September 2026:

- **Part VI's terminal convention is removed for one apex over a tree.** Part VI's criterion covers the full apex and, after subdivision, apex neighbourhoods equal to the terminals (Theorems 84.1 and 89.1), and its Section “What these results do not imply” restricts it to that convention. Theorem 118.2 classifies every apex neighbourhood on every tree host. Beyond one apex over a tree, nothing changes.
- **Stability and its certificates are decidable in linear time** (Theorem 123.1), and the stable neighbourhoods of a tree are counted, optimized and repaired exactly in  $O(n^5)$  arithmetic operations (Theorems 124.1–125.1, Corollary 125.2).
- **No numbered research question of Parts I–VII is answered.** Part VI's Research questions 45, 47 and 48 are posed again by source 10, with added detail, as Research questions 58–60;

Research questions 44, 52 and 53 are overlapped by Research questions 64 and 65.

Part VIII refutes no statement of Parts I–VII, and nothing is retracted. Its remark that unmarked fringe is irrelevant agrees with Part VI’s statement after Corollary 93.3 that attaching trees preserves stability. Part VI gains dated pointers after Theorem 84.1, at “Why the terminal convention matters”, at “What these results do not imply”, and after Research questions 45, 47 and 48; those three questions and Research question 52, unlabelled before, received labels.

#### 116.4 Where the merge had to choose

- *Order.* Source 10’s order of sections is kept. Its two appendices are printed as ordinary sections at the end of Part VIII, so that Part I’s appendices keep their letters.
- *The main theorem* is printed once, with a first proof assembled for this merge from Part VI’s theorems (Section 118) and source 10’s direct proof as the second (Section 122).
- *Printed as pointers (duplicates of Part VI).* Source 10’s four bridge lemmas, printed as Lemmas 119.1, 119.2, 119.3 and 119.4, are Lemma 85.1, Proposition 85.2, Lemmas 85.5, 85.6 and 85.4 with the same proofs; its path-shortening lemma (Lemma 120.1) is Lemma 89.2 almost word for word; its three-odd-path paragraph is Theorem 84.3(ii); its resolvent lemma (Lemma 122.1) is Lemma 87.1 with the trivial case  $C = 0$  added; its ADE proposition (Proposition 122.4) is Theorem 88.1, whose proof is not reprinted; Table 14 is kept because later statements cite it. Their statements and labels are kept, their proofs are replaced by pointers.
- *Second routes kept.* Source 10’s coefficient count for (120.1) and its direct proof of Theorem 118.2.
- *Research questions.* Source 10’s eight questions continue the numbering as Research questions 58–65, each with its label; the three that restate Part VI’s questions are numbered, with merge notes, because their formulations are broader, and Part VI’s questions point to them.
- *Self-citation and file names.* Source 10’s reference “ProveIt” (Parts II–VII of this report at the pin, especially Part VI) became references to Part VI, and its macro for the repository name was expanded. Its source-range table, which cites line numbers of the pinned article.tex, became a table of Part VI’s sections (Section 130). Its seven code, data and audit files keep their delivered base names behind the prefix 10-marked-trees-: marked\_trees.py, verify.py and build.sh in code/; requirements.txt and the two files of its results/ directory, run.log and verification.json, in data/; and SOURCE\_AUDIT.md at the report root. Merge notes give the shipped names, and its reproduction commands are paraphrased with delivery names.
- *Cross-references.* Source 10 uses cleveref, which this report does not load (Parts I–VII use plain \ref); its \cref and \Cref references were converted to explicit “Theorem \ref” forms, with the same targets.
- *Macros, labels and layout.* Its preamble is dropped: its \R, \C, \HH, \one, \ind, \diag, \tr, \adj, \Lap, \supp, \im, \del agree with the host’s; its  $\mathbb{N}$  macro (unused in its text, and the host’s  $\mathbb{Z}_{\geq 0}$  is  $\mathbb{Z}_{\geq 0}$ ) and an unused \weight are dropped, and its  $\mathbb{Q}$  is typeset directly. Its \Hull, \Leaf, \diam, \Stab, \Zpoly and \pathin were added to the preamble. Labels carry the prefix mts:; the eight questions and this section’s headings gained labels, as did four questions of Part VI (mts:hq:). Its abstract and status box are printed in this section; its title page, table of contents and running head are dropped.

- *Bibliography.* [24] and [12] were already cited. Source 10's McKee–Smyth reference is a different paper from Part VI's [25] and is [27]; Kummer–Sert is new [28]; the Patel credit uses the host's [11].

## 117 The research problem and its provenance

A matching polynomial ordinarily counts a matching once for each choice of its edges. A *matching-support polynomial* instead counts a set of endpoints once whenever some perfect matching on those endpoints exists. This replacement of multiplicity by existence changes the analytic problem: the familiar determinantal or stable-polynomial constructions need not retain their coefficients after multiplicities are removed.

This report develops this distinction, especially in Parts II, III and VI. Part VI proves a striking criterion (Theorem 84.1): if an apex is joined to every vertex on the left of the incidence graph of a tree  $H$ , the signed support polynomial is real stable exactly when  $I - A_H/2$  is positive semidefinite. It also proves that subdividing edges between the original apex neighbors, without marking the new vertices, preserves the criterion (Theorem 89.1). These are existing results, not the principal novelty claims of this article.

A natural missing step is to permit an *arbitrary* apex neighborhood on a given host tree. The original vertices need not all be marked; an unmarked vertex may have degree three or more in the subtree connecting the marks. Such a configuration is not a terminal subdivision of a tree whose vertices are exactly the marks. The distinction is essential: three marked leaves around an unmarked center already produce a nonstable example, whereas marking the center repairs it.

We resolve this arbitrary-neighborhood classification within the tree incidence–apex family. We then ask a different, algorithmic question: *how many neighborhoods are stable, and which stable neighborhood has maximum weight?* The classification imposes at most four leaves on the marked hull. This yields an endpoint decomposition with polynomially many outer cases, even though a path has exponentially many stable markings. It is this decomposition, rather than generic matroid sampling machinery, that gives exact counting and optimization.

## 117.1 A precise boundary between old and new

| Ingredient or result                                                              | Status in this article                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Half-plane property, stable minors, Rayleigh inequalities                         | Established stable-polynomial theory [24, 12]; the special facts used here are stated explicitly.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Trees with adjacency spectral radius at most two                                  | Classical ADE classification; an elementary proof is included for completeness.                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Full-apex and terminal-subdivision determinant identities and ADE criterion       | Prior: Part VI (Theorems 84.1, 86.3 and 89.1). Their method is extended, with credit, to arbitrary unmarked fringe and branch-complete markings.                                                                                                                                                                                                                                                                                                                                                                      |
| Three odd paths and the quartic obstruction                                       | Prior results of Parts II and VI (Theorem 84.3(ii)); the report credits Shivam Patel for the earlier eight-vertex example.                                                                                                                                                                                                                                                                                                                                                                                            |
| Arbitrary-marking classification and supported-covariance existence               | Continuation developed here, including the unmarked-branch obstruction and exact necessity of the geometric condition. [Merge note: the classification, Theorem 118.2, is a corollary of Part VI's Theorems 89.1, 93.2 and 84.3(ii) (first proof in Section 118), as source 10 itself says in Section 130. New here are the statement for arbitrary markings, the direct host-level identity (Theorem 121.3) and certificate (Theorem 122.2), which give a second proof, and the nonexistence half of Theorem 121.1.] |
| Weighted neighborhood enumeration, selection, and repair                          | Continuation developed here: explicit endpoint formulas and exact polynomial-time algorithms.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Diameter window and the path-only real-rootedness theorem for neighborhood counts | Consequences proved here; the latter follows from a two-coefficient root-rigidity argument, not numerical root plots.                                                                                                                                                                                                                                                                                                                                                                                                 |

The general half-plane-property literature is much broader than this family. Choe–Oxley–Sokal–Wagner develop structural constructions and show that some, but not all, transversal matroids have the property [24]. Brändén gives the all-real-specialization Rayleigh criterion [12, Theorem 5.6]. Kummer–Sert classify the matroids on at most eight elements with the half-plane property [28]; therefore our eight-vertex obstruction is not presented as a new small-matroid discovery. The spectral classification has a classical setting in the work of McKee–Smyth [27].

Our focused review did not establish whether every new formulation below has an equivalent earlier formulation elsewhere. In particular, a specialist comparison with principal extensions and transversal-matroid presentations remains appropriate before a priority claim. What is unambiguous here is the statement, proof, implementation, and boundary relative to the pinned source.

## 117.2 The repository snapshot

The inspected snapshot is

`b57c0b5ff112e1a08af8add12b2d687b8ece91d0`.

The main source is the file

`SetTheory/Cardinals/docs/reports/enumerative-combinatorics/  
preorder-root-polytopes/article.tex`,

whose blob is `920246d173bfb708bcfdd6d67e0a6fc358cb3dc`. The relevant source ranges and headings are recorded in Section 130. We do not rely on unrelated claims elsewhere in the repository, and no repository file was modified. [Merge note: that blob is Parts I–VII of this report exactly as printed before Part VIII was added (Section 116.1).]

## 118 Definitions and the classification theorem

All trees are finite, simple, undirected, and nonempty. Let  $T = (V, E)$  be a tree with  $n = |V|$ . A subset  $S \subseteq V$  is a *marking*. Define  $\mathcal{A}(T; S)$  to be the bipartite graph with left shore  $V$  and right shore  $E \sqcup \{\star\}$ , where an edge-element  $e = uv$  is adjacent to  $u, v$ , and  $\star$  is adjacent exactly to the vertices in  $S$ . Thus  $\mathcal{A}(T; S)$  has  $2n$  vertices, even when  $\star$  is isolated. [Merge note: source 10 writes  $G(T, S)$ . The notation extends Part VI's:  $\mathcal{A}(T; V(T)) = \mathcal{A}(T)$ , and Part VI's  $\mathcal{A}(\widehat{T}; T)$  is  $\mathcal{A}(\widehat{T}; V(T))$  (Section 89).]

For  $I \subseteq V$  and  $J \subseteq E \sqcup \{\star\}$ , write  $I \sim J$  when  $|I| = |J|$  and the induced bipartite graph on  $I \sqcup J$  has a perfect matching. Define

$$F_{T,S}(x; y, s) = \sum_{I \sim J} \left( \prod_{v \in V \setminus I} x_v \right) \left( \prod_{e \in J \cap E} y_e \right) s^{\mathbf{1}_{\{\star \in J\}}}. \quad (118.1)$$

A feasible pair contributes one term, irrespective of the number of witnessing matchings. The polynomial is homogeneous of degree  $n$ , multiaffine, and has coefficients zero or one.

For a finite simple graph  $G$  on a vertex set  $W$ , its signed matching-support polynomial is

$$\Phi_G(z) = \sum_{U \subseteq W: G[U] \text{ has a perfect matching}} (-1)^{|U|/2} \prod_{v \in W \setminus U} z_v. \quad (118.2)$$

The empty support is allowed. A nonzero real polynomial is *real stable* when it does not vanish if every variable belongs to  $\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$ . [Merge note:  $F_{T,S}$  is Part VI's basis polynomial (85.1) of  $\mathcal{A}(T; S)$ , with left variables  $x$  and right variables  $y$  and  $s$ , and (118.2) is source 09's (85.3), which agrees with Part VI's (84.2) on bipartite graphs.]

**Definition 118.1** (Hull and terminal skeleton). For nonempty  $S$ , let  $K = \operatorname{Hull}_T(S)$  be the smallest connected subtree containing  $S$ . For  $|S| \geq 2$ , it is the union of the unique paths between marked vertices. For  $|S| = 1$ , it consists of that vertex. Call  $S$  *branch-complete* if every vertex of degree at least three in  $K$  lies in  $S$ . In this case suppress all vertices of  $K \setminus S$ , which necessarily have degree two, and denote the resulting tree on  $S$  by  $H(T, S)$ .

Every leaf of  $K$  is marked. An edge of  $H(T, S)$  joins two marks exactly when the path between them contains no other mark. Vertices outside  $K$  are necessarily unmarked but may form arbitrarily large attached trees. The degree condition is on  $K$ , not on the host  $T$ .

**Theorem 118.2** (Complete marked-tree classification). *Let  $T$  be a nonempty finite tree and  $S \subseteq V(T)$ . The following conditions are equivalent:*

- (i)  $F_{T,S}$  is real stable;
- (ii)  $\Phi_{\mathcal{A}(T;S)}$  is real stable;
- (iii) either  $S = \emptyset$ , or  $S$  is branch-complete and

$$C_H = I - \frac{1}{2}A_{H(T,S)} \geq 0; \quad (118.3)$$

- (iv) either  $S = \emptyset$ , or  $S$  is branch-complete and its terminal skeleton is one of the finite or affine ADE trees in Table 14.

Equivalently, in the nonempty branch-complete case,  $\rho(A_{H(T,S)}) \leq 2$ . A marking with zero or one mark is always stable.



*First proof: a corollary of Part VI (merge).* The equivalence (i) $\Leftrightarrow$ (ii) is Proposition 85.2 for  $\mathcal{A}(T; S)$ , and (iii) $\Leftrightarrow$ (iv) is Theorem 88.1 for the skeleton; since  $H(T, S)$  is bipartite,  $I - \frac{1}{2}A_{H(T,S)} \geq 0$  is equivalent to  $\rho(A_{H(T,S)}) \leq 2$ , as after Theorem 87.2.

*Unmarked fringe.* Let  $S \neq \emptyset$  and  $K = \text{Hull}_T(S)$ . Every component  $R$  of  $T - V(K)$  is joined to exactly one vertex  $u \in V(K)$ , by one edge  $e$ , and contains no mark. In  $\mathcal{A}(T; S)$  the left vertices of  $R \cup \{u\}$  and the edge-elements of  $E(R) \cup \{e\}$  span the incidence graph of the tree  $R + e$ , which meets the rest of  $\mathcal{A}(T; S)$  only in  $u$  and has no apex edge. Hence  $\mathcal{A}(T; S)$  is obtained from  $\mathcal{A}(K; S)$  by one-vertex sums with incidence graphs of trees. A tree is stable: its blocks are single edges, and  $\Phi_{K_2} = z_1 z_2 - 1$  does not vanish on  $\mathbb{H}^2$ , because a product of two points of  $\mathbb{H}$  is never a positive real number; apply Corollary 93.3. By Theorem 93.2,  $\Phi_{\mathcal{A}(T;S)}$  is stable if and only if  $\Phi_{\mathcal{A}(K;S)}$  is. The same argument with  $\mathcal{A}(T; \emptyset)$ , the incidence graph of  $T$  with an isolated apex, proves that the empty marking is stable.

*Branch-complete markings.* If  $S$  is branch-complete, every vertex of  $K \setminus S$  has degree two in  $K$ , and  $K$  is obtained from the skeleton  $H = H(T, S)$  by replacing each edge by the path of  $K$  between consecutive marks. The apex is adjacent exactly to the terminals  $S = V(H)$ , so  $\mathcal{A}(K; S)$  is the terminal-skeleton graph  $\mathcal{A}(\hat{H}; H)$  with  $\hat{H} = K$ , and Theorem 89.1 gives stability exactly when  $\rho(A_H) \leq 2$ .

*Unmarked hull branches.* If  $v \in V(K) \setminus S$  has degree at least three in  $K$ , choose in three components of  $K - v$  a mark  $a_i$  nearest to  $v$ . As in the proof of Theorem 120.2 below, the incidence vertices of the three paths from  $v$  to  $a_i$ , together with  $\star$ , induce in  $\mathcal{A}(T; S)$  a multi-theta graph  $\vartheta(2d_1 + 1, 2d_2 + 1, 2d_3 + 1)$  with endpoints  $v$  and  $\star$ , where  $d_i = d_T(v, a_i) \geq 1$ . By Theorem 84.3(ii) it is not stable, and by Lemma 85.6 neither is  $\Phi_{\mathcal{A}(T;S)}$ .  $\square$

The first proof is assembled for this merge from Part VI's theorems; source 10 states that its classification is such a synthesis (Section 130). Source 10's own proof, printed as the second proof after Proposition 122.4, occupies Sections 119–122; it proves the identity and the failure certificate directly on the host tree, without the vertex-sum theorem or matroid minors. The full-apex criterion is the special case  $S = V(T)$ ; the repository's terminal-subdivision theorem is the case  $K = T$  with every suppressed vertex of degree two. The unmarked branching obstruction is the additional case needed for arbitrary neighborhoods.

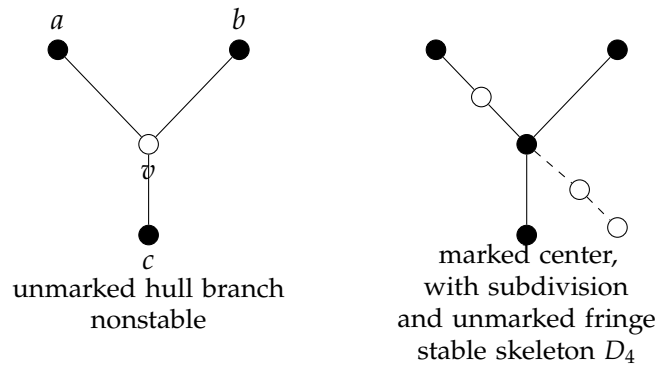


Figure 4: Filled vertices are apex neighbors. The diagrams show the host tree, not its bipartite incidence graph. Unmarked degree-two subdivision vertices and unmarked fringe do not alter the terminal skeleton.

## 119 The support–matroid bridge and elementary analytic facts

*Merge note.* The four lemmas of this section are Part VI's, in source 10's formulation; source 10 proves them by the same arguments as Part VI, and its proofs are replaced by pointers. Its statements are kept because later sections cite them.

**Lemma 119.1** (Canonical transversal matroid). *For a bipartite graph with shores  $U$  and  $B$ , the sets  $(U \setminus I) \sqcup J$ , with  $I \sim J$ , are the bases of a rank- $|U|$  transversal matroid. Its basis polynomial is  $F_G(x; y) = \sum_{I \sim J} x^{U \setminus I} y^J$ .*

*Proof.* This is Lemma 85.1 (the matroid  $M_G$  and its basis polynomial (85.1)), proved there by the same one-slot-per-left-vertex presentation.  $\square$

Source 10 also recalls the generic linear representation  $[I \ K]$  of Lemma 85.1, with an independent indeterminate  $K_{ub}$  on each graph edge; a basis minor is nonzero precisely when  $I \sim J$ .

**Lemma 119.2** (Inversion bridge and induced subgraphs). *For a bipartite graph  $G$  with shores  $U, B$ ,*

$$\Phi_G(x, z_B) = \left( \prod_{b \in B} z_b \right) F_G(x; (-1/z_b)_{b \in B}). \quad (119.1)$$

*Thus  $F_G$  and  $\Phi_G$  are stable simultaneously. Stability of  $\Phi_G$  passes to every induced subgraph.*

*Proof.* Identity (119.1) and the first conclusion are Proposition 85.2, equation (85.2). The second is Lemmas 85.5 and 85.6, by the identity

$$\partial_{z_v} \Phi_G = \Phi_{G-v}, \quad (119.2)$$

iterated; its leading unmatched monomial excludes the zero polynomial.  $\square$

We use the standard closure properties under differentiation and real specialization [24, 12]. Real specialization follows by approaching the real point from  $\mathbb{H}$  and applying Hurwitz's theorem; the limit is zero-free or identically zero. For a matroid basis polynomial, deletion and contraction are respectively specialization at zero and differentiation, with the usual loop/coloop factors treated separately. Hence matroid minors inherit stability.

**Lemma 119.3** (Necessary Rayleigh inequality). *If a real multiaffine polynomial  $f$  is stable, then, for all distinct variables  $z_i, z_j$  and every real assignment  $a$ ,*

$$\Delta_{ij} f(a) := f_i(a) f_j(a) - f(a) f_{ij}(a) \geq 0. \quad (119.3)$$

*Proof.* This is Lemma 85.4, with the same bivariate specialization argument.  $\square$

The converse is Brändén's general theorem [12]; our negative certificates require only the elementary direction just proved.

**Lemma 119.4** (A stable polynomial has real-rooted support diagonal). *If  $F_G$  is stable, then*

$$p_G(t) = F_G(\mathbf{1}; t\mathbf{1}) = \sum_{I \sim J} t^{|I|} \quad (119.4)$$

*has only negative real zeros, or is constant.*

*Proof.* This is the last assertion of Proposition 85.2, with the same specialization and diagonalization argument.  $\square$

The converse is false; the repository's books provide examples (Theorem 84.2). This distinction will also matter when we introduce a second enumerator, counting *which markings* are stable.

## 120 The unmarked-branch obstruction

### 120.1 A small obstruction, with its original credit

Let  $\mathfrak{g}(3, 3, 3)$  consist of three internally disjoint paths of length three between a left endpoint and a right endpoint. It is  $\mathcal{A}(K_{1,3}; S)$  when  $S$  consists of the three leaves and excludes the center. Its support polynomial is

$$p_{\mathfrak{g}(3,3,3)}(t) = 1 + 9t + 24t^2 + 16t^3 + t^4. \quad (120.1)$$

This is an existing obstruction in this report, which attributes the earlier counterexample to Shivam Patel [11]. We reproduce the small certificate rather than claim a new example. [Merge note: (120.1) is Part VI's (92.1) and Part II's  $p_{\Theta_3}$  (Theorem 13.3). The coefficient count below differs from Part VI's (which counts matchable right sets per left set) and is kept as a second route.]

Here is a direct coefficient count (source 10's route). The graph has nine edges and degree sequence  $(3, 3, 2, 2, 2, 2, 2, 2, 2)$ . Its number of two-edge matchings is

$$\binom{9}{2} - 2\binom{3}{2} - 6\binom{2}{2} = 24.$$

There is no four-cycle, so two matchings cannot share a four-vertex support. Each shore has four vertices, and every choice of three vertices from each shore is matchable. To verify this last assertion, label the left vertices  $u, b_1, b_2, b_3$  and the right vertices  $a_1, a_2, a_3, \star$ , with edges  $ua_i, b_ia_i, b_i\star$ . If  $u$  is omitted, match the leaves to their private  $a_i$  columns, using  $\star$  for the unique missing private column when necessary. If  $u$  is retained and  $\star$  is omitted, match the two retained leaves privately and give  $u$  the third column. Finally, if both  $u$  and  $\star$  are retained, either an available private column belongs to the omitted leaf, in which case give it to  $u$ , or both available private columns belong to retained leaves. In the latter case give one to  $u$ , its leaf to  $\star$ , and match the other leaf privately. Thus the size-three coefficient is  $\binom{4}{3}^2 = 16$ . The full graph has a perfect matching, giving final coefficient one.

For an exact nonreal-root certificate,

$$\text{disc}(t^4 + 16t^3 + 24t^2 + 9t + 1) = -5243. \quad (120.2)$$

If all four roots were real and distinct, the discriminant would be positive; repeated roots would give zero. Thus the polynomial has a nonreal root. The included program independently enumerates the supports and computes the discriminant over the integers. By Lemma 119.4,  $F_{\mathfrak{g}(3,3,3)}$  is not stable. [Merge note: (120.2) is Part VI's (92.2); Part VI also prints source 09's Sturm chain and Rayleigh certificate for the same graph. The "included program" is shipped as code/10-marked-trees-verify.py; its recorded output data/10-marked-trees-verification.json lists the coefficients 1, 9, 24, 16, 1 and the discriminant  $-5243$ .]

## 120.2 Shortening odd paths by legitimate matroid minors

It is not correct to assume that arbitrary graph contractions preserve matching-support stability. We use the following specific operation, already present in the repository's terminal-subdivision argument.

**Lemma 120.1** (Path shortening). *Suppose a bipartite graph has a path  $a - c - b - d$ , where  $a, b$  are on the left and  $c, d$  on the right. Suppose  $N(b) = \{c, d\}$ ,  $N(c) = \{a, b\}$ , and  $a$  is not adjacent to  $d$ . In its canonical matroid, contract the right element  $c$  and delete the left private element  $b$ . The result is the canonical matroid of the graph obtained by removing  $b, c$  and replacing this path by the edge  $a - d$ .*

*Proof.* This is Lemma 89.2; source 10's proof, in the generic representation  $[I \ K]$ , is the same argument almost word for word and is not reprinted.  $\square$

Consequently, a graph consisting of three internally disjoint odd paths, each of length at least three, is nonstable. Shorten each path by pairs of edges until all three have length three. The intermediate internal vertices satisfy the degree conditions in Lemma 120.1. Stability of the original canonical matroid would imply stability of the resulting  $\mathfrak{g}(3, 3, 3)$  minor, a contradiction. [Merge note: this is Theorem 84.3(ii) for three paths, with the same proof (Section 92).]

**Theorem 120.2** (An unmarked hull branch forbids stability). *If  $S \neq \emptyset$  and  $K = \text{Hull}_T(S)$  has a vertex  $v \notin S$  of degree at least three, then  $F_{T,S}$  and  $\Phi_{\mathcal{A}(T;S)}$  are not stable.*

*Proof.* Choose three components of  $K - v$ . Each contains a mark. In each choose a mark  $a_i$  at minimum distance from  $v$ , and take the path from  $v$  to  $a_i$ . Its internal vertices are unmarked. The three paths meet only at  $v$ . In  $\mathcal{A}(T; S)$  their incidence subdivisions, followed by the three edges  $a_i \star$ , form three paths from  $v$  to  $\star$  of lengths  $2d_T(v, a_i) + 1 \geq 3$ .

The graph induced on precisely these path vertices is exactly this odd theta graph. There are no cross-edges, because  $T$  is a tree; there are no extra apex edges, because there are no internal marks on the chosen paths. The preceding minor argument proves it nonstable. Induced-subgraph inheritance in Lemma 119.2 excludes stability of the original graph, and the inversion bridge transfers the conclusion to  $F_{T,S}$ .  $\square$

[Merge note: Theorem 120.2 is new as a statement; its proof applies Theorem 84.3(ii) and induced inheritance (Lemma 85.6) to the induced multi-theta graph  $\mathfrak{g}(2d_1 + 1, 2d_2 + 1, 2d_3 + 1)$  between  $v$  and  $\star$ . It is the case that Part VI's terminal convention excludes (Section 89, "Why the terminal convention matters").]

## 121 Exactly when a supported covariance exists

The geometric obstruction also has a precise algebraic interpretation. A matrix is *supported on  $S$*  if all its rows and columns outside  $S$  are zero. Consider the connected-set interpolation conditions

$$\mathbf{1}_W^T C \mathbf{1}_W = \mathbf{1}_{\{W \cap S \neq \emptyset\}} \quad \text{for every nonempty connected vertex set } W \subseteq V(T). \quad (121.1)$$

They ask that the quadratic form turn the number of usable apex entries in a connected component into the Boolean assertion that at least one is present.

**Theorem 121.1** (Existence and uniqueness of the supported covariance). *For  $S \neq \emptyset$ , a real symmetric matrix supported on  $S$  satisfying (121.1) exists if and only if  $S$  is branch-complete. When it exists it is unique and equals the zero extension of*

$$C_H = I - \frac{1}{2} A_{H(T,S)}. \quad (121.2)$$

For  $S = \emptyset$ , the unique supported matrix is zero and it satisfies (121.1).

*Proof.* First suppose the marking is branch-complete. For connected  $W$ , the set  $R = W \cap S$  is either empty or connected in the skeleton  $H$ . Indeed, the unique path in  $T$  between two elements of  $R$  belongs to  $W$ , and contains every intervening mark on their skeleton path. Consequently

$$\mathbf{1}_W^T C \mathbf{1}_W = |R| - |E(H[R])| = \begin{cases} 0, & R = \emptyset, \\ 1, & R \neq \emptyset. \end{cases}$$

This proves existence even when  $C_H$  is indefinite.

For uniqueness, singleton marks give  $C_{aa} = 1$ . For adjacent marks in  $H$ , their connecting path contains exactly those two marks, so its energy is  $2 + 2C_{ab} = 1$  and  $C_{ab} = -1/2$ . Induct on distance in  $H$  for a nonadjacent pair  $a, b$ . On their connecting path, diagonal and adjacent-pair contributions already sum to one. All other pairs of smaller distance have zero entry by induction. The remaining term  $2C_{ab}$  must vanish. This determines every entry supported on  $S$ .

Conversely, suppose an unmarked branch  $v$  exists. Choose three first marks  $a, b, c$  in distinct directions as in Theorem 120.2. The path between any two of them contains no other mark. Singleton and two-mark path energies force

$$C_{aa} = C_{bb} = C_{cc} = 1, \quad C_{ab} = C_{ac} = C_{bc} = -\frac{1}{2}.$$

The connected union of the three paths contains exactly these three marks, so its energy is  $3 + 2(-3/2) = 0$ , contradicting its required energy one. The empty case is immediate from the support requirement.  $\square$

[Merge note: for  $S = V(T)$  the uniqueness half is Part VI's covariance rigidity, Proposition 86.2, with the same induction on distance, and the existence half is (86.2). The support restriction, the branch-complete generality and the nonexistence half at an unmarked branch are new.]

**Remark 121.2** (The support restriction is part of the theorem). We have classified matrices supported on the apex neighborhood and satisfying every connected-set equation. We have not ruled out unrelated determinantal models, indefinite matrices with additional off-neighborhood entries, or higher-moment constructions. On the positive-semidefinite side, zero rows outside  $S$  follow automatically from the zero diagonal entries forced by singleton equations.

### 121.1 The coefficient identity, including unmarked fringe

Choose an orientation of  $T$ . For  $e = uv$ , let  $b_e = e_u - e_v$ , with the sign chosen according to the orientation, and put

$$L(x, y) = \text{diag}(x_v)_{v \in V} + \sum_{e \in E} y_e b_e b_e^T. \quad (121.3)$$

Changing an orientation does not change  $L$ .

**Theorem 121.3** (Exact Boolean determinant–adjugate identity). *Suppose  $S$  is branch-complete, or empty, and let  $C$  be the matrix in Theorem 121.1. Then*

$$F_{T,S}(x; y, s) = \det L + s \, \text{tr}(\text{adj}(L)C). \quad (121.4)$$

*This is a polynomial identity, with no invertibility or definiteness assumption. It holds on the original host tree, including every unmarked subdivision vertex and every unmarked attached subtree.*

*Proof.* Let  $B$  be the oriented incidence matrix. By Cauchy–Binet,

$$\begin{aligned} \det(L + sww^T) &= \sum_{|I|=|J|} x^{V \setminus I} y^J \det(B_{I,J})^2 \\ &\quad + s \sum_{|I|=|J|+1} x^{V \setminus I} y^J \det[B_{I,J} \ w_I]^2, \end{aligned} \quad (121.5)$$

where  $I \subseteq V$ ,  $J \subseteq E$ , and  $w$  is an auxiliary vector. The first-line minor is zero or  $\pm 1$  and is nonzero exactly on a matchable support. One can see this by leaf elimination; alternatively, the underlying incidence graph is a tree, so a fixed support has at most one perfect matching and hence at most one nonzero determinant term.

For the second line, consider the internal-edge forest on the chosen rows  $I$ . A selected edge with one endpoint in  $I$  is a half-edge; an edge with no endpoint in  $I$  gives a zero column. A component with  $k$  vertices and  $h$  half-edges has  $k - 1 + h$  selected columns. If  $h > 1$ , those columns cannot be assigned distinct rows. When there is no zero column and every component has at most one half-edge, the equality  $|I| = |J| + 1$  forces exactly one component  $W$  to have no half-edge; every other component has one.

A component with one half-edge gives a square incidence block of determinant  $\pm 1$ . In the component  $W$ , the internal incidence block has left kernel spanned by  $\mathbf{1}_W$ , and each maximal cofactor has modulus one. Thus

$$\det[B_{I,J} \ w_I] = \pm \sum_{v \in W} w_v. \quad (121.6)$$

In every other component configuration the augmented determinant vanishes identically. If  $w$  is supported on  $S$ , an apex matching exists exactly when this unique unrooted component meets  $S$ .

Apply the linear functional  $w_u w_v \mapsto C_{uv}$  to the quadratic coefficients of (121.5). By (121.1), the square in (121.6) becomes one when  $W \cap S \neq \emptyset$  and zero otherwise. Each feasible apex support is therefore counted exactly once. The first line is unchanged. On the left, the rank-one determinant identity gives

$$\det(L + sww^T) = \det L + s w^T \operatorname{adj}(L) w.$$

The same functional produces (121.4). Everything takes place in a polynomial ring, so singular substitutions are allowed.  $\square$

[Merge note: Theorem 121.3 generalizes Part VI’s Theorem 86.3 ( $S = V(T)$ ) and the terminal-subdivision identity (89.1) ( $K = T$ ) to hosts with unmarked fringe; its cofactor analysis is Lemma 86.1. Part VI reaches fringe only through the vertex-sum theorem, not by an identity.]

**Remark 121.4.** The right side is generally *not*  $\det(L + sC)$ : that determinant would introduce higher powers of  $s$  and represent several apex choices. Nor is the coefficient identity the assertion that every matching has the same weight. The connected-component quadratic functional is precisely what removes the matching multiplicity.

## 122 Stability, rational certificates, and the ADE list

### 122.1 Positive covariance gives stability

The following resolvent argument is the analytic mechanism already used for full-apex trees in Part VI. The extended identity allows the same argument on every branch-complete marking.



**Lemma 122.1** (Resolvent positivity). *Let  $L = \text{diag}(x_i) + \sum_e y_e b_e b_e^T$ , with real vectors  $b_e$ . If  $C \geq 0$ , then  $\det L + s \text{tr}(\text{adj}(L)C)$  is stable.*

*Proof.* For  $C \neq 0$  this is Lemma 87.1, with the same argument through

$$\text{Im } L^{-1} = -L^{-*}(\text{Im } L)L^{-1} < 0, \quad (122.1)$$

which gives  $\text{Im } \text{tr}(CL^{-1}) < 0$ ; source 10 notes that this follows also by factoring  $C = RR^*$  and taking the trace of the negative definite form on the nonzero columns of  $R$ . If  $C = 0$ , the polynomial is  $\det L$ , which does not vanish because  $\text{Im } L > 0$  for  $x_i, y_e \in \mathbb{H}$ .  $\square$

Singular positive-semidefinite matrices cause no difficulty; in particular, the affine ADE boundary is included without an extra approximation step.

## 122.2 A negative certificate directly on the original tree

**Theorem 122.2** (Rational Rayleigh certificate with subdivisions and fringe). *Suppose  $S$  is nonempty and branch-complete, but  $C_H$  is not positive semidefinite. There is a vector  $v \in \mathbb{Q}_{>0}^{V(T)}$  with  $v^T C v < 0$ . For any such vector set*

$$y_{ij} = \frac{1}{v_i v_j} \quad (ij \in E(T)), \quad (122.2)$$

$$x_i = \frac{\deg_T(i)}{v_i^2} - \sum_{j \sim i} \frac{1}{v_i v_j}, \quad s = 0. \quad (122.3)$$

Writing  $c = \prod_i v_i^{-2}$ , the exact Rayleigh values are

$$\Delta_{s,x_i} F_{T,S} = c^2 v_i^2 (v^T C v) < 0 \quad (i \in V(T)). \quad (122.4)$$

*Proof.* For the connected nonnegative adjacency matrix  $A_H$ , failure of  $I - A_H/2 \geq 0$  means its largest eigenvalue exceeds two. A strictly positive Perron eigenvector therefore has negative  $C_H$  energy. Approximate it sufficiently closely by positive rational entries; strict negativity persists. Extend to a positive rational vector on the other vertices of  $T$ . Since  $C$  is supported on  $S$ , the energy is unchanged.

At (122.2)–(122.3), with  $D_v = \text{diag}(v_i)$ ,

$$L_0 = D_v^{-1} \text{Lap}(T) D_v^{-1}. \quad (122.5)$$

The tree Laplacian has kernel  $\mathbb{R}\mathbf{1}$  and adjugate  $\mathbf{1}\mathbf{1}^T$ : any cofactor is one, by elimination of leaves or the incidence determinant of the tree. Adjugate transformation under the invertible diagonal congruence gives

$$\det L_0 = 0, \quad \text{adj}(L_0) = c v v^T.$$

Using Theorem 121.3 at  $s = 0$  yields  $F = 0$ ,  $F_s = c v^T C v$ , and  $F_{x_i} = c v_i^2$ . In the Rayleigh expression  $F_s F_{x_i} - F F_{s,x_i}$ , the second term vanishes, proving (122.4). Lemma 119.3 then excludes stability.  $\square$

This theorem extends the source’s skeleton certificate to the full host without contracting away its extra vertices. It does not assert that all negative certificates occur at positive variable values: the  $x_i$  may be negative, which is entirely legitimate for a test of real stability. [Merge note: “the source” is Part VI; the formulas are those of Theorem 87.2, equations (87.3)–(87.5). On the host tree this is a second route to the necessity half of Theorem 89.1, which Part VI proves by matroid minors (Lemma 89.2).]

**Example 122.3** (A transparent spectral failure). For the fully marked  $K_{1,5}$ , take center coordinate  $v_0 = 2$  and all five leaf coordinates equal to one. Then  $v^T C v = -1$ , all  $y_e = 1/2$ , the center  $x_0 = -5/4$ , and each leaf  $x_i = 1/2$ . Thus  $c = 1/4$  and  $\Delta_{s,x_0} F = -1/4$ . This bare-star certificate is already in the repository (the example after Theorem 87.2). Inserting an unmarked subdivision vertex on an arm, and attaching an unmarked two-edge twig there, preserves the energy: put value one at all new vertices and use (122.3). The delivered output checks this nine-vertex extension exactly as well. [Merge note: both certificates are recorded under `rayleigh_certificates` in `data/10-marked-trees-verification.json`.]

### 122.3 An elementary form of the classical classification

| Name                                    | Shape of the terminal skeleton $H$                                                                        |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------|
| $A_r, r \geq 1$                         | A path on $r$ vertices.                                                                                   |
| $D_{c+3}, c \geq 1$                     | One trivalent vertex with arm lengths $(1, 1, c)$ .                                                       |
| $E_6, E_7, E_8$                         | One trivalent vertex with arm lengths $(1, 2, 2), (1, 2, 3), (1, 2, 4)$ , respectively.                   |
| $\tilde{E}_6, \tilde{E}_7, \tilde{E}_8$ | One trivalent vertex with arm lengths $(2, 2, 2), (1, 3, 3), (1, 2, 5)$ , respectively.                   |
| $\tilde{D}_4$                           | The four-leaf star $K_{1,4}$ .                                                                            |
| $\tilde{D}_{d+4}, d \geq 1$             | Two trivalent vertices joined by a path of length $d$ , with two additional immediate leaves at each end. |

Table 14: All trees with adjacency spectral radius at most two. Arm lengths count edges of the terminal skeleton, not edges of the host tree. The finite types have  $2I - A_H > 0$ ; affine types have corank one.

**Proposition 122.4** (Classical tree criterion). *A nonempty tree  $H$  satisfies  $I - A_H/2 \geq 0$  if and only if it occurs in Table 14.*

*Proof.* This is Theorem 88.1, and Table 14 is its list (a)–(f) in source 10’s layout. Source 10’s proof is the same elementary argument (excluded degrees five and four through  $K_{1,5}$  and the  $K_{1,4}$  eigenvector, the double-fork eigenvector, and Schur pivots along the arms) and is not reprinted. For a skeleton with one trivalent vertex and arm lengths  $a \leq b \leq c$ , the decisive condition is the arm inequality (88.1),

$$\frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1} \geq 1, \quad (122.6)$$

with equality exactly in the affine cases.  $\square$

*Second proof of Theorem 118.2 (source 10, direct).* The inversion bridge gives (i) $\Leftrightarrow$ (ii). An unmarked hull branch is excluded by Theorem 120.2. On every remaining nonempty marking, Theorem 121.3 and Lemma 122.1 give stability when  $C_H \geq 0$ , and Theorem 122.2 excludes it otherwise. The empty case is  $F = \det L$ . Finally Proposition 122.4 identifies the matrix criterion with (iv). For a nonnegative adjacency matrix its largest eigenvalue is its spectral radius, giving the stated equivalent numerical criterion.  $\square$

**Corollary 122.5** (Positive covariance is also necessary). *The graph  $\mathcal{A}(T; S)$  is stable if and only if a positive-semidefinite symmetric matrix satisfying (121.1) exists. Such a matrix is automatically supported on  $S$  and is unique.*

*Proof.* Singleton equations make every diagonal outside  $S$  zero. A positive-semidefinite matrix with  $C_{ii} = 0$  has its entire  $i$ th row zero, as follows from its two-by-two principal minors. Now use Theorems 121.1 and 118.2.  $\square$

## 123 Recognition and geometric consequences

**Theorem 123.1** (Linear-time recognition). *Given a tree by adjacency lists and a marking, stability of  $\mathcal{A}(T; S)$  can be decided in  $O(n)$  combinatorial time, without constructing its support polynomial and without numerical eigenvalue computation.*

*Proof.* Handle  $S = \emptyset$  separately. Repeatedly remove unmarked leaves, updating degrees with a queue. Every vertex and edge is processed a constant number of times, and the remaining graph is  $K = \text{Hull}_T(S)$ . Reject if a remaining unmarked vertex has degree at least three. Suppress the unmarked degree-two vertices, following each path once. This constructs  $H$  in linear time. Its membership in Table 14 is checked from its degrees, the number of branches, and, in the one-branch case, its three arm lengths. These traversals are linear.  $\square$

A geometric rejection can include three explicit first-mark paths, giving the induced theta certificate. A positive result can include the skeleton shape and the subdivision paths. A spectral rejection can be supplemented by the rational vector of Theorem 122.2. The theorem's linear-time claim concerns the structural decision, not an unproved uniform bound on the bit length of every possible rationalized eigenvector.

**Corollary 123.2** (Four-leaf bound). *If  $S$  is stable and  $|S| \geq 2$ , then  $\text{Hull}_T(S)$  has at most four leaves.*

*Proof.* Suppressing degree-two vertices does not change the leaves, and every ADE tree in Table 14 has at most four.  $\square$

The bound is on the number of *hull leaves*, not the number of marks: a path can have all its vertices marked, and a double-fork skeleton can have arbitrarily many marks along its central path.

**Theorem 123.3** (A sharp diameter window). *Let  $\alpha_{\text{st}}(T)$  be the largest cardinality of a stable marking. Then*

$$\text{diam}(T) + 1 \leq \alpha_{\text{st}}(T) \leq \min\{n, \text{diam}(T) + 3\}. \quad (123.1)$$

*The offsets one, two, and three are all attained.*

*Proof.* Mark all vertices on a longest path. Its hull and skeleton are paths, so it is stable and has  $\text{diam}(T) + 1$  vertices. Conversely, every nonempty stable marking has an ADE skeleton  $H$  on  $|S|$  vertices. Inspection of Table 14, or its arm formulas, gives  $|V(H)| \leq \text{diam}(H) + 3$ . Distances between terminals can only increase under subdivision, so  $\text{diam}(H) \leq \text{diam}(T)$ . Also  $|S| \leq n$ . Paths attain offset one,  $K_{1,3}$  attains offset two, and  $K_{1,4}$  attains offset three.  $\square$

**Proposition 123.4** (No monotonicity under changing the neighborhood). *The stable markings are in general neither downward closed nor upward closed under inclusion. Every marking of  $T$  is stable if and only if  $T$  is a path.*

*Proof.* On  $K_{1,3}$  the fully marked set has skeleton  $D_4$  and is stable, but deleting its center mark leaves the unstable three-leaf marking. On  $K_{1,5}$ , the center together with four leaves is stable, whereas adding the fifth leaf gives the excluded five-leaf star. Thus neither monotonicity holds.

If  $T$  is a path, every hull and skeleton are paths. If  $T$  is not a path, it has a vertex with at least three neighbors. Mark precisely three of those neighbors. The hull is a three-leaf star with unmarked center, so this marking is unstable.  $\square$

These examples explain why minimum repair cannot be reduced to a naive procedure which only deletes marks or only adds missing branch marks.

## 124 An exact endpoint formula for all stable neighborhoods

Define the stable-marking family and its enumerators by

$$\begin{aligned} \mathcal{S}(T) &= \{S \subseteq V(T) : F_{T,S} \text{ is real stable}\}, \\ \mathcal{Z}_T(z) &= \sum_{S \in \mathcal{S}(T)} z^S, & Z_T(t; w) &= \mathcal{Z}_T((tw_v)_{v \in V}). \end{aligned} \quad (124.1)$$

Here  $z^S = \prod_{v \in S} z_v$ . Write  $Z_T(t) = Z_T(t; \mathbf{1})$  for unweighted size enumeration. This polynomial counts *apex neighborhoods*; it is not  $p_{\mathcal{A}(T;S)}$ , which counts supports for one fixed neighborhood.

For distinct vertices  $a, b$ , let  $\mathring{P}_{ab}$  be the internal vertices of the unique  $a$ – $b$  path, and put

$$P_{ab}(z) = \prod_{v \in \mathring{P}_{ab}} (1 + z_v). \quad (124.2)$$

For a vertex set  $R$  let

$$E_k(R) = \sum_{A \subseteq R, |A|=k} z^A, \quad E_0(R) = 1,$$

with  $E_k(R) = 0$  for  $k > |R|$ .

For  $\ell = 3, 4$ , define

$$\mathcal{L}_\ell(T) = \{L \subseteq V : |L| = \ell, \text{Leaf}(\text{Hull}_T(L)) = L\}. \quad (124.3)$$

Not every three- or four-element set is eligible: an element lying between two others is not a leaf of their hull and must be excluded. Each pair of distinct vertices is automatically eligible as a two-leaf hull.

### 124.1 The three-leaf factor

For  $L = \{a, b, c\} \in \mathcal{L}_3(T)$ , let  $o = o(L)$  be the unique trivalent vertex of its hull. Its three arms have internal vertex sets  $R_1, R_2, R_3$ , excluding  $o$  and the corresponding leaf. Let  $\mathcal{E}$  be the set of all *distinct permutations* of the six triples

$$(0, 1, 1), (0, 1, 2), (0, 1, 3), (1, 1, 1), (0, 2, 2), (0, 1, 4). \quad (124.4)$$

Define

$$Q(R_1, R_2, R_3) = 1 + \sum_{i=1}^3 \left( \prod_{v \in R_i} (1 + z_v) - 1 \right) + \sum_{(k_1, k_2, k_3) \in \mathcal{E}} \prod_{i=1}^3 E_{k_i}(R_i). \quad (124.5)$$

The first part counts choices of internal marks on at most one arm. The second part contains the six exceptional finite/affine patterns. Repeated coordinates in a triple do not create extra multiplicities.

### 124.2 Four-leaf hulls

Every four-leaf tree, after degree-two vertices are suppressed, has either one degree-four vertex or two degree-three vertices. Partition  $\mathcal{L}_4(T)$  accordingly as  $\mathcal{L}_4^{(4)}(T)$  and  $\mathcal{L}_4^{(3,3)}(T)$ . In the first case denote the branch by  $o(L)$ ; in the second denote the two branches by  $o(L), p(L)$ . These are vertices of the host tree, and may be separated by any positive path length.

**Theorem 124.1** (Weighted endpoint decomposition). *For every nonempty finite tree,*

$$\begin{aligned}
Z_T(z) = & 1 + \sum_{v \in V} z_v + \sum_{\{a,b\} \subseteq V} z_a z_b P_{ab}(z) \\
& + \sum_{L \in \mathcal{L}_3(T)} z^{L \cup \{o(L)\}} Q(R_1, R_2, R_3) \\
& + \sum_{L \in \mathcal{L}_4^{(4)}(T)} z^{L \cup \{o(L)\}} \\
& + \sum_{L \in \mathcal{L}_4^{(3,3)}(T)} z^{L \cup \{o(L), p(L)\}} P_{o(L)p(L)}(z). \tag{124.6}
\end{aligned}$$

Every monomial on the right occurs once. In particular the expression, although (124.5) uses a subtraction shorthand, has nonnegative coefficients after expansion.

*Proof.* The empty marking and all singletons give the first two terms. Every other stable marking has a unique leaf set  $L = \text{Leaf}(\text{Hull}_T(S))$  with  $2 \leq |L| \leq 4$ , by Corollary 123.2. All vertices outside this hull are unmarked. The leaves and all branch vertices of the hull must be marked; only its degree-two vertices can be chosen freely subject to the ADE condition. Moreover  $\text{Hull}_T(S) = \text{Hull}_T(L)$ , so each outer case in (124.6) is recoverable uniquely from the resulting marking.

If  $|L| = 2$ , the hull is a path. Its endpoints must be marked and any subset of internal vertices is allowed, giving  $z_a z_b P_{ab}$ .

If  $|L| = 3$ , its branch  $o$  and three leaves are mandatory. If  $k_i$  internal marks are chosen on arm  $i$ , the terminal skeleton has arm lengths  $k_i + 1$ . By Table 14, it is of  $D$  type when at least two of the  $k_i$  vanish, giving the first part of (124.5). The remaining allowed unordered triples are exactly (124.4). For a specified triple, the weighted choices are  $E_{k_1}(R_1)E_{k_2}(R_2)E_{k_3}(R_3)$ . These cases are disjoint from the  $D$  cases and from one another.

If  $|L| = 4$  and there is one degree-four branch, the only allowed skeleton is  $\widetilde{D}_4$ . Hence none of its four pendant arms may contain an extra mark. If there are two degree-three branches, the only allowed skeleton is an affine double fork. Again the four pendant arms may have no extra marks, but every subset of the internal joining path is allowed. These are precisely the last two lines. No other hull leaf count or branch type can occur, completing the disjoint decomposition.  $\square$

This is a polynomial-size *description*, not a promise to print every monomial of the multivariate expansion in polynomial time. For example,  $Z_{P_n}(z) = \prod_v (1 + z_v)$  has  $2^n$  distinct monomials.

**Theorem 124.2** (Exact counting in polynomial time). *For rational vertex weights  $w_v$ , the complete coefficient vector of  $Z_T(t; w)$  can be computed using  $O(n^5)$  exact arithmetic operations and polynomial space. For unit weights, counting stable neighborhoods and counting those of any prescribed cardinality have polynomial bit complexity.*

*Proof.* Enumerate all subsets of at most four endpoints, of which there are  $O(n^4)$ . Using unique tree paths, construct their hulls, check that the chosen vertices are exactly the leaves, and identify the branches in  $O(n)$  time per candidate. Store path polynomials  $\prod_{v \in \hat{P}_{ab}} (1 + w_v t)$  for endpoint pairs. There are  $O(n^2)$  such products, each of degree at most  $n$ ; even independent multiplication for each pair costs at most  $O(n^4)$  arithmetic operations and uses  $O(n^3)$  coefficient storage.

After substituting  $z_v = tw_v$ , each  $E_k(R_i)$  is a scalar multiple of  $t^k$ , obtained from the appropriate path polynomial. The exceptional list is fixed and has  $k_i \leq 4$ . Thus every endpoint contribution can be added to the answer in  $O(n)$  operations. The total is  $O(n^5)$ .

For unit weights, every output coefficient is at most  $2^n$  and has  $O(n)$  bits. The path products, elementary coefficients, and accumulated nonnegative contributions have polynomial bit length as well, so exact integer arithmetic preserves polynomial bit complexity. With rational weights of total input bit length  $B$ , all squarefree monomial products have a common denominator dividing the product of the input denominators; the coefficient numerators and denominators have  $O(B + n)$  bits up to standard representation constants. The same argument therefore also supplies polynomial bit complexity in  $n + B$ .  $\square$

### 124.3 Independent random markings

For independently marked vertices with probabilities  $0 < p_v < 1$ , the exact probability of obtaining a stable neighborhood is

$$\Pr(S \in \mathcal{S}(T)) = \left( \prod_{v \in V} (1 - p_v) \right) Z_T \left( 1; \left( \frac{p_v}{1 - p_v} \right)_{v \in V} \right). \quad (124.7)$$

This follows by factoring the probability of each marking. Values  $p_v = 0$  or  $1$  are handled by the original finite sum, or by continuity. Thus rational independent marking probabilities also admit exact polynomial-time stability-probability evaluation, despite the lack of monotonicity.

## 125 Maximum-weight choice and minimum-cost repair

**Theorem 125.1** (Exact maximum-weight stable marking). *Given arbitrary rational scores  $c_v$ , one can find a set  $S \in \mathcal{S}(T)$  maximizing  $\sum_{v \in S} c_v$  in  $O(n^5)$  arithmetic operations and comparisons. Negative scores are allowed, and an optimizing set, not just its value, is returned.*

*Proof.* Use the endpoint cases of Theorem 124.1, but maximize rather than sum. Consider the empty marking and every singleton separately. For each other case, include its mandatory leaves and branch vertices.

On a two-leaf path, choose exactly the positive-score internal vertices; zero-score choices can be resolved arbitrarily. For a three-leaf hull, first consider the  $D$  cases: either no internal marks, or positive-score internal marks on one selected arm only. Take the best of these options. For each exceptional triple  $(k_1, k_2, k_3) \in \mathcal{E}$ , choose the  $k_i$  highest-score vertices of arm  $i$ ; reject that triple if an arm is too short. These selections can include negative scores, because the required cardinalities are fixed within the case. Only the four largest entries per arm are needed, and they can be found in a linear scan.

For a four-leaf hull with one degree-four branch, there are no optional marks. With two trivalent branches, choose all positive-score internal vertices on their joining path and none on the pendant arms. Compare all candidate totals and retain their vertex sets. Each endpoint case is processed in  $O(n)$  time and there are  $O(n^4)$  cases. Correctness follows from the exhaustive, disjoint geometric description in Theorem 124.1 and elementary optimization within each free or fixed-cardinality region.  $\square$

This is not a generic theorem about optimizing stable matroids. The host is a tree and only its single apex neighborhood is changed. The small number of hull leaves and fixed list of exceptional arm counts are doing the algorithmic work.



**Corollary 125.2** (Nearest stable neighborhood). *Given an initial neighborhood  $S_0$ , a stable neighborhood minimizing  $|S \Delta S_0|$  is computable within the same bound. More generally, nonnegative rational costs for adding and deleting individual apex edges can be minimized exactly.*

*Proof.* For Hamming distance, assign  $c_v = 2\mathbf{1}_{\{v \in S_0\}} - 1$ . Then

$$|S \Delta S_0| = |S_0| - \sum_{v \in S} c_v.$$

For weighted repair, let  $d_v$  be the deletion cost on  $v \in S_0$  and  $a_v$  the addition cost on  $v \notin S_0$ . The total repair cost is

$$\sum_{v \in S_0} d_v - \sum_{v \in S \cap S_0} d_v + \sum_{v \in S \setminus S_0} a_v.$$

Use score  $d_v$  on the original marks and score  $-a_v$  elsewhere, and apply Theorem 125.1.  $\square$

**Example 125.3** (Why additions can be optimal). On the three-leaf star, start with the three leaves marked. Adding the center costs one and makes the marking stable. Deleting one leaf also costs one, but deletion cost two per leaf and center-addition cost one makes addition uniquely cheaper. Conversely, on a fully marked five-leaf star, deletion of one leaf repairs the spectral obstruction; adding marks cannot help because all vertices are already marked.

The implementation exposes four public entrypoints:

```
recognize          stable_polynomial
maximum_weight_stable nearest_stable
```

All use integer or rational arithmetic, never a floating-point spectral tolerance.

## 126 Explicit families and a second real-rootedness classification

### 126.1 Paths and stars

**Proposition 126.1** (Path and star formulas). *For a path,*

$$\mathcal{Z}_{P_n}(z) = \prod_{v \in V(P_n)} (1 + z_v), \quad Z_{P_n}(t) = (1 + t)^n. \quad (126.1)$$

*For the star  $K_{1,m}$ , with center  $o$  and leaf set  $L$ ,*

$$\mathcal{Z}_{K_{1,m}}(z) = \sum_{k=0}^{\min(2,m)} E_k(L) + z_o \sum_{k=0}^{\min(4,m)} E_k(L), \quad (126.2)$$

$$Z_{K_{1,m}}(t) = \sum_{k=0}^{\min(2,m)} \binom{m}{k} t^k + t \sum_{k=0}^{\min(4,m)} \binom{m}{k} t^k. \quad (126.3)$$

*Proof.* All path markings are stable. For the star with unmarked center, at most two leaves may be marked, since three create an unmarked hull branch. With marked center, the skeleton is a star with the selected leaves, and its covariance is positive semidefinite precisely when at most four leaves are selected. Count these disjoint alternatives.  $\square$

For  $m \geq 4$  the total number of stable star neighborhoods is

$$Z_{K_{1,m}}(1) = 2 + 2m + 2\binom{m}{2} + \binom{m}{3} + \binom{m}{4}. \quad (126.4)$$

Under uniform independent marking its probability is this count divided by  $2^{m+1}$ , asymptotic to  $m^4/(24 \cdot 2^{m+1})$ . Thus paths have stable probability one, while large stars have exponentially small stable probability with a polynomial prefactor.

## 126.2 Two initial coefficients force complete root rigidity

The following elementary equality argument is useful beyond this example. It explains why good root geometry cannot survive the act of counting stable neighborhoods.

**Lemma 126.2** (Two-level root rigidity). *Let  $\mathcal{F} \subseteq 2^{[n]}$  contain every subset of cardinality at most two, and put  $f(t) = \sum_{S \in \mathcal{F}} t^{|S|}$ . Then  $f$  has only real roots if and only if  $\mathcal{F} = 2^{[n]}$ .*

*Proof.* The assertion is immediate for  $n \leq 1$ . Otherwise  $f(0) = 1$ ,  $[t]f = n$ , and  $[t^2]f = \binom{n}{2}$ . If  $f$  has only real roots, its nonnegative coefficients imply all roots are negative. Writing  $d = \deg f \leq n$ , we can therefore factor

$$f(t) = \prod_{i=1}^d (1 + \lambda_i t), \quad \lambda_i > 0.$$

Comparison of the first two coefficients gives

$$\sum_i \lambda_i = n, \quad \sum_i \lambda_i^2 = n^2 - 2\binom{n}{2} = n.$$

Cauchy-Schwarz now gives  $n^2 \leq dn$ , so  $d \geq n$ . Thus  $d = n$  and equality holds; all  $\lambda_i = 1$ . Hence  $f = (1 + t)^n$ . Since each coefficient counts a subcollection of the  $\binom{n}{k}$  available  $k$ -subsets, equality of every coefficient forces every subset to belong to  $\mathcal{F}$ . The reverse implication is immediate.  $\square$

**Theorem 126.3** (Root classification for the stable-neighborhood enumerator). *For every nonempty finite tree  $T$ , the following are equivalent:*

- (i)  $T$  is a path;
- (ii) every marking of  $T$  is stable;
- (iii) the univariate polynomial  $Z_T(t)$  has only real roots;
- (iv) the multivariate polynomial  $Z_T(z)$  is real stable.

*Proof.* The equivalence of (i) and (ii) is Proposition 123.4. A marking of size at most two always has an empty, singleton, or path skeleton, so  $\mathcal{S}(T)$  contains all subsets of those sizes. Lemma 126.2 gives (iii)  $\Leftrightarrow$  (ii). If (iv) holds, diagonalizing all variables gives (iii). If  $T$  is a path, (126.1) is a product of stable linear factors, giving (iv).  $\square$

This theorem concerns  $Z_T$ , which varies the apex neighborhood. It does *not* assert that the support polynomial  $p_{\mathcal{A}(T;S)}$  for an individual nonpath tree is nonreal-rooted. In particular, the repository proves real-rootedness for the full-apex star support polynomial for every  $m$ , even though  $Z_{K_{1,m}}$  is nonreal-rooted for every  $m \geq 3$  (Theorem 84.2: the book  $B_m$  is  $\mathcal{A}(K_{1,m})$ ).

### 126.3 Even weaker coefficient-shape properties fail

Small exact examples from (126.3) are

$$Z_{K_{1,3}}(t) = 1 + 4t + 6t^2 + 3t^3 + t^4 = (1 + t)^4 - t^3, \quad (126.5)$$

$$Z_{K_{1,4}}(t) = 1 + 5t + 10t^2 + 6t^3 + 4t^4 + t^5, \quad (126.6)$$

$$Z_{K_{1,6}}(t) = 1 + 7t + 21t^2 + 15t^3 + 20t^4 + 15t^5. \quad (126.7)$$

The first has no real zeros at all: it is positive for  $t \geq 0$  by its coefficients, and for  $t < 0$  its expression  $(1 + t)^4 - t^3$  is positive. In the second,  $6^2 < 10 \cdot 4$ , so log-concavity fails. In the third,  $21 > 15 < 20$ , so unimodality fails.

More generally, for every  $m \geq 4$ , the coefficients at degrees two, three, and four are respectively  $m(m+1)/2$ ,  $\binom{m}{2}$ , and  $\binom{m}{3}$ . The log-concavity inequality at degree three fails exactly when

$$3(m-1) < (m+1)(m-2),$$

which holds for every integer  $m \geq 4$ . For every  $m \geq 6$ , those same three coefficients have a strict dip, since  $m(m+1)/2 > \binom{m}{2} < \binom{m}{3}$ . Thus the exact algorithms do not rely on unimodality, log-concavity, or a hidden stable-polynomial property of the marking family.

## 127 Exact verification and the computational trust boundary

The supplied checks were executed with Python 3.13.5, NetworkX 3.6.1, and SymPy 1.14.0. All decisions about positive semidefiniteness, coefficients, and optimization use exact integers or rationals. Random test weights and scores are reproducible, with seed 20260930. The complete output is `results/verification.json`. [Merge note: shipped as `data/10-marked-trees-verification.json`, with the console output as `data/10-marked-trees-run.log`; the programs are `code/10-marked-trees-marked_trees.py` and `code/10-marked-trees-verify.py`. The recorded run used `--max-n 9`; the verifier's own docstring and default say 8 (README, "Discrepancies and delivery names").]

### 127.1 Exhaustive marked instances through nine vertices

For each unlabelled tree representative, every subset of its vertex set was tested. The counts below are *not* counts of isomorphism classes of marked trees: automorphism-equivalent subsets of the same representative are deliberately counted separately, because the enumerator counts vertex subsets.

| $n$ | Trees | Markings | Stable | Unmarked branch | Spectral failure |
|-----|-------|----------|--------|-----------------|------------------|
| 1   | 1     | 2        | 2      | 0               | 0                |
| 2   | 1     | 4        | 4      | 0               | 0                |
| 3   | 1     | 8        | 8      | 0               | 0                |
| 4   | 2     | 32       | 31     | 1               | 0                |
| 5   | 3     | 96       | 88     | 8               | 0                |
| 6   | 6     | 384      | 324    | 58              | 2                |
| 7   | 11    | 1408     | 1075   | 306             | 27               |
| 8   | 23    | 5888     | 3942   | 1707            | 239              |
| 9   | 47    | 24064    | 13850  | 8569            | 1645             |

Table 15: 31,886 marked instances on 95 unlabelled tree representatives. The all-size classification is proved in the text; this table tests the implementation of its two obstruction mechanisms.

| Executed check                                                                               | Count         |
|----------------------------------------------------------------------------------------------|---------------|
| Combinatorial ADE decisions compared with exact rational PSD tests                           | 21,237        |
| Explicit three-first-mark-path witnesses at unmarked branches                                | 10,649        |
| Connected-set energy feasibility, marked trees through order six                             | 526 instances |
| Complete determinant coefficient identities, branch-complete marked trees through order five | 133 instances |
| Individual incidence-minor/cofactor coefficients in those identities                         | 24,534        |
| Endpoint enumerators compared with exhaustive stable-subset sums                             | 190           |
| Signed-score optimizations compared with exhaustive maxima                                   | 285           |
| Nearest-neighborhood repairs compared with exhaustive distances                              | 95            |
| Sharp diameter-window checks on tree representatives                                         | 95            |
| Path formulas / star formulas checked                                                        | 5 / 11        |
| Exact rank-one-adjugate Rayleigh certificates                                                | 2             |

The distinction between these checks matters. In particular, the exhaustive counting and optimization comparisons use the structural classification to identify admissible subsets; they are not independent analytic tests of stability in every number of variables. The full coefficient checks *do* independently enumerate matching supports and compare every coefficient of the determinant formula for the stated small orders. Neither kind of finite check replaces the general proofs.

## 127.2 Independent exact routines

The matching-support enumerator uses sets of attainable column masks, not lists of matching witnesses. If  $D(I)$  is the collection of column sets matchable to a row set  $I$ , then for a chosen row  $i \in I$ ,

$$D(I) = \{J \cup \{j\} : J \in D(I \setminus \{i\}), j \in N(i) \setminus J\}.$$

Identical column sets are deduplicated. This is essential: counting matchings would verify the wrong polynomial.

The covariance test is independent of the ADE shape routine. It constructs the matrix forced by singleton and consecutive-mark path equations, then uses exact Schur complements. A negative diagonal rejects positive semidefiniteness; a zero diagonal must have a zero row; a positive pivot is eliminated exactly. In particular affine diagrams are not classified by a floating-point cutoff.

For the connected-set test, every nonempty connected  $W \subseteq V(T)$  is enumerated in the stated small orders. The program checks that all energies (121.1) hold exactly if and only if no unmarked hull branch exists. The determinant tests use integer incidence cofactors and the rational quadratic form, not sampled numerical polynomial evaluations.

## 127.3 Reproduction and formalization boundary

Source 10's instructions, run from its delivered package directory, install its `requirements.txt` with `pip`, run `python code/verify.py --max-n 9 --output results/verification.json`, and build its article twice with `pdflatex`. [Merge note: in this report the verifier imports its library by the delivered name `marked_trees`, so it is rerun on a copy with the delivered layout; the README gives a recipe that cannot overwrite the recorded output. The article is built with this report.] The code has no network calls. Dependencies are ordinary mathematical Python packages; their versions used here are pinned for reproduction. No external fonts or third-party article PDFs are shipped.

A formalization could isolate a small finite combinatorial core: hulls, branch-completeness, the three-mark obstruction, and the endpoint partition. Separately it would prove the forest-cofactor identity, resolvent lemma, and exact Rayleigh certificate. The classical stable minor closure and spectral facts should be explicit dependencies until proved in the target library. No delivered Lean or Rocq theorem, kernel check, or formal correctness claim is made for the Python implementation. [Merge note: nor does this report or ProveIt's Lean and Rocq developments formalize any statement of Parts VI or VIII; compare Part VI's formalization boundary (Section 95).]

## 128 Further research questions and topics

The following questions retain the distinctions exposed by the proofs. They are not claims established by this article. [Merge note: source 10's eight questions continue the numbering as Research questions 58–65. The first three restate Part VI's Research questions 45, 47 and 48 with added detail and are printed with pointers, because source 10's formulations are broader (arbitrary markings; a terminal-geometry criterion; unicyclic and cactus hosts). Research questions 64 and 65 overlap Part VI's Research questions 44, 52 and 53. The other three are new.]

**Research question 58** (The individual support diagonal). Classify real-rootedness of  $p_{\mathcal{A}(T;S)}(t)$  for arbitrary marked trees. In particular, resolve the repository's question whether the full-apex diagonal  $p_{\mathcal{A}(T;V(T))}$  is real-rooted for every tree.

Our criterion classifies multivariate stability, not this weaker property. On branch-complete markings, (121.4) gives a concrete resolvent formula for investigating residues on Laplacian eigenspaces. The theorem that  $Z_T$  is real-rooted only for paths does not answer this question, because  $Z_T$  varies  $S$  rather than counting supports for one  $S$ . [Merge note: the full-apex case is Part VI's Research question 45; its merge note records an exact check of every tree with  $2 \leq r \leq 12$  (all real-rooted), which is evidence, not an answer. For  $S = V(T)$ , the specialization  $x = \mathbf{1}$ ,  $y = t\mathbf{1}$ ,  $s = t$  of (121.4) is Part VI's formula (96.1).]

**Research question 59** (More than one apex). For two or more separately selectable apex elements with specified neighborhoods, characterize when the canonical support polynomial is stable. Is there a finite terminal-geometry criterion analogous to branch-completeness?

One apex leaves precisely one unrooted incidence component. Several apices must coordinate several components while removing multiplicities of their assignments. Simply multiplying the one-apex formulas, or replacing the rank-one term by  $\det(L + sC)$ , counts the wrong objects. [Merge note: this is Part VI's Research question 47, asked there with prescribed terminal neighbourhoods; source 10 adds the request for a criterion analogous to branch-completeness.]

**Research question 60** (Cycle-space corrections). Extend the covariance-existence and stability dichotomies when the host is unicyclic, a cactus, or has bounded cycle rank.

The present proof uses unique paths twice: to create the unmarked-branch obstruction and to make every relevant selected incidence component a tree with one-dimensional cofactor direction. Cycles change both the connected-set energy and the multiplicities of incidence determinants. A useful first goal is an exact analogue of Theorem 121.1 for one cycle, before attempting a general stable representation. [Merge note: this is Part VI's Research question 48, with the covariance-existence dichotomy added as a target.]

**Research question 61** (Faster exact enumeration and repair). Can the  $O(n^5)$  endpoint algorithms be replaced by quadratic or near-linear tree dynamic programs, retaining exact weighted coefficients or optimal signed-score repair?

The finite arm-pattern list suggests a bounded set of states, but the unbounded  $A$ ,  $D$ , and affine- $D$  path lengths must be compressed without counting the same hull repeatedly. The formulas here give an exact baseline for validating such a faster implementation.

**Research question 62** (Coefficient shape after real-rootedness fails). Characterize the trees whose stable-neighborhood size sequence is log-concave or unimodal. Find structural explanations for the number and location of its modes, rather than expecting a universal positive answer.

Stars already disprove universal log-concavity and unimodality. Any useful positive theorem must impose a genuine structural restriction. The path-only real-rootedness theorem settles the strongest version entirely; it should not be reposed as an open question.

**Research question 63** (Random trees and asymptotic stable probabilities). For random labelled trees, or deterministic families with growing branching complexity, determine the asymptotic probability that an independent random marking is stable.

The exact formula (124.7) reduces the problem to counts of small-leaf hulls and their allowable marked paths. Paths and stars show that no universal order of magnitude is possible without hypotheses. A meaningful analysis must specify both the host-tree model and the marking probabilities. [Added 30 September 2026, batch 66: answered in Part X for uniform labelled trees with independent marks of one common probability  $p$ , in every regime of  $p$  (Theorems 156.2, 159.1, 160.1, 161.1 and 162.2): for instance the probability is asymptotic to  $\frac{9}{8}\sqrt{2\pi}p^3n^{5/2}e^{-pn}$  whenever  $p\sqrt{n} \rightarrow \infty$ . For deterministic families Part X gives the probability only within a polynomial factor, from the diameter (Theorem 165.2); an asymptotic equivalent there, other random-tree laws and vertex-dependent markings remain open (Research questions 74, 79 and 80). See Section 152.3.]

**Research question 64** (Restricted obstruction certificates). Within this family, identify economical canonical-matroid certificates for the spectral failures and compare them with the geometric odd-theta certificate. Which different marked trees yield the same excluded matroid minor?

The existing theory of small half-plane-property matroids [28] and cyclotomic spectral obstructions [27] must be consulted before claiming a new minimal-obstruction list. Our criterion gives a recognition theorem, not such a minimality classification. [Merge note: compare Part VI's Research questions 44 (forbidden canonical matroid minors) and 53 (minimal induced obstructions), and Section 92.6: the unstable bipartite graphs have no finite set of induced-minimal obstructions.]

**Research question 65** (A small verified compiler from trees to proofs). Formalize the endpoint partition and construct a checker that accepts an ADE skeleton certificate or a negative obstruction certificate and returns a theorem about the original support polynomial.

The input should include the actual marking and host-tree paths, not just an abstract diagram. Correctness of the compiler must connect the combinatorial certificate to the determinant or minor semantics. Checking an isolated rational inequality is insufficient without that bridge. [Merge note: compare Part VI's Research question 52 (formal integration of the cofactor lemma, covariance rigidity and the Rayleigh certificate compiler).]

## 129 Conclusion

The full-apex ADE theorem leaves a geometric issue when the neighborhood is arbitrary: an unmarked branch connecting several marks is not an innocent subdivision vertex. It



simultaneously destroys the connected-set covariance interpolation and creates a persistent odd- $\theta$  obstruction. Once those branches are marked, the terminal-supported covariance is forced, and its positive semidefiniteness is both sufficient and necessary. This gives a complete classification for all markings of all finite trees.

The classification does more than decide one instance. The marked hull has at most four leaves, producing a disjoint weighted endpoint formula, exact counting, and exact signed-score selection and repair. The sharp diameter window limits the maximum cardinality, while the two-level rigidity argument determines the real-rootedness of the neighborhood counting polynomial completely. These conclusions remain meaningful even though that enumerator can fail both log-concavity and unimodality.

The general bipartite stability problem and the individual tree-support diagonal problem remain distinct further targets. The present contribution is a proved and executable completion of one structural family, with its new enumeration and optimization consequences, rather than an unsupported claim about all half-plane-property matroids.

### 130 Source audit and dependency ledger

(*Appendix A of source 10.*) The repository was read at the fixed commit in Section 117. [Merge note: source 10 lists the material it inspected by line ranges of the pinned `article.tex`; those ranges no longer match this file, which has grown, and are replaced by the corresponding sections of Part VI. The line ranges survive in `10-marked-trees-SOURCE_AUDIT.md`. The last range stopped inside Part VI's research questions, at the first line of Research question 50.]

| Part VI        | Material inspected                                                                                                         |
|----------------|----------------------------------------------------------------------------------------------------------------------------|
| Section 83     | Part VI provenance, status, notation, and previously established scope.                                                    |
| Sections 84–85 | Existing ADE/book/multi- $\theta$ statements, canonical matroid, inversion, minor closure, and Rayleigh inequality.        |
| Sections 86–87 | Forest cofactors, full-apex covariance rigidity, determinant–adjugate identity, and rational negative certificate.         |
| Sections 88–89 | Elementary ADE list and proof; terminal-subdivision theorem and generic path-shortening matroid minor.                     |
| Section 93     | Vertex gluing, unmarked tree attachments, weighted support diagonals, and the distinction from univariate real-rootedness. |
| Sections 95–96 | Prior finite checks and further questions, including full-apex diagonals, several apices, and cyclic hosts.                |

The source's tree and subdivision theorems already prove the positive case when all branches are terminals and there is no outside fringe; its gluing theorem handles fringe abstractly. We give a direct host-level cofactor proof and certificate, and identify the missing unmarked-branch case. The resulting classification is therefore a synthesis and extension of specific existing results, not a first discovery of ADE stability. The exact endpoint enumerator and weighted repair algorithm are not contained in the inspected sections. [Merge note: this paragraph is the basis of the first proof of Theorem 118.2; "the source" is Part VI, its tree and subdivision theorems are Theorems 84.1 and 89.1, and its gluing theorem is Theorem 93.2.]

The public literature review checked the arXiv records and relevant full-text material for [24, 12, 27, 28]. The main external dependencies are standard stable-polynomial closure properties and Perron positivity. All tree-specific combinatorics, the explicit ADE case analysis, the support coefficients, and the endpoint algorithms are proved in the article. No literature search can by itself certify global priority; the package keeps that claim separate from mathematical correctness.

## 131 A compact proof certificate for the main results

(Appendix B of source 10.) The classification can be audited in the following order.

1. If a hull branch is unmarked, select three first-mark paths. Their incidence–apex graph is induced, and legitimate matroid path shortening reduces it to the credited quartic obstruction.
2. Otherwise the terminal skeleton is a tree. Its zero-extended  $C_H = I - A_H/2$  has connected-set energy exactly the indicator of meeting  $S$ , and singleton/path equations force this matrix uniquely.
3. Forest cofactors turn those energies into the Boolean support identity  $F = \det L + s \operatorname{tr}(\operatorname{adj}(L)C)$ .
4. Positive semidefiniteness gives stability through the sign of  $\operatorname{Im} \operatorname{tr}(CL^{-1})$ . Indefiniteness gives a rational negative Rayleigh value through a diagonally rescaled tree Laplacian.
5. The classical tree spectrum calculation identifies the ADE list. Its at-most-four-leaf property partitions every stable marking by a unique endpoint hull. The allowed arm counts yield (124.6); free choices and fixed-cardinality choices yield the optimization algorithm.
6. Every one- and two-element marking is stable. If  $Z_T$  were real-rooted, its first two coefficients would force all its factors to be  $1 + t$ , hence every marking would be stable, forcing  $T$  to be a path.

This certificate lists logical dependencies. It is not machine-checked formal proof syntax, and the accompanying finite tests should not be substituted for any of its all-size steps.

# Part IX. Two-flow bijections: full-coordinate sampling of demand vectors and preorder lattice points

*Part IX was added on 30 September 2026 in batch 64 of ProveIt's incoming-report intake, from one manuscript. For a bipartite graph and a fixed receiver support  $J$ , it gives explicit, mutually inverse, polynomial-time maps between the demand vectors of  $\mathcal{D}_H$  with support  $J$  and the supplier sets matchable to  $J$ : two integer-margin minimum-cost transportation problems under one cycle-generic cost system have the same optimal spanning tree, whose supplier degrees give the supplier set and whose supplier leaves give every coordinate of the vector. The existence of such a correspondence is due to Oh; the realization, its certificates, a core factorization, restriction laws for receivers outside  $J$  and suppliers outside the selected set, and two obstructions are the manuscript's. Composed with the matroid basis walk it samples actual demand vectors, and actual lattice points of every finite preorder polytope, with the total-variation error preserved exactly. This answers the "Sampling boundary" of Part V and the bijection part of Research questions 2 and 20, and the disjoint union of the bijections is an elementary, Ehrhart-free proof of the support-counting identity of Parts II, IV and V. Parts I–VIII are unchanged except for dated pointers. Section 132 records the provenance, fixes the notation and states what is answered.*

## 132 Provenance, conventions and what Part IX answers

### 132.1 The manuscript

Part IX is printed from batch-64 manuscript 01, called *source 11* here after its file prefix: *Two-Flow Bijections for Demand Polytopes: Full-coordinate sampling, exact certificates, restriction laws, and preorder applications*, dated 30 September 2026, 23 A4 pages as delivered. Its title page reads "Prepared for Vladimir Reshetnikov / Mathematical development and implementation: ChatGPT", and its PDF metadata call it an "AI-assisted research manuscript prepared for Vladimir Reshetnikov"; that byline is recorded here and not reprinted. Its archive `ProveIt_Two_Flow_Lattice_Sampling.zip` (inner directory `Two_Flow_Lattice_Sampling/`, main file `article.tex`) arrived in ProveIt commit 7747fcfdd and was placed in 3025c15df. It is pinned to commit 6d04e1e385f2fe7d1cfbdbd8fd8d4e45bd6b6c72, with this report's `article.tex` as blob 920246d173bfb708bcfdd6d67e0a6fc358cb3dc: Parts I–VII exactly as printed, before Part VIII was written (the same blob that Part VIII's source 10 read). Source 11 locates the question it answers as the "Sampling boundary" box in lines 9510–9555 of that blob, immediately after "Sampling after fixing the support size"; the passage is unchanged and is now in Part V's Section 75. Its library, verifier and build script are shipped in `code/`, its recorded outputs, worked certificate, environment record and console log in `data/`, and its proof-status and source-audit notes at the report root, all with the prefix `11-two-flow-`. The delivery had no checksum list. The manuscript, delivery README and PDF are not shipped; they survive in the arrival commit.

Source 11 continues Part V: it answers Part V's sampling boundary and composes its decoder with Part V's tilting method for a prescribed support size, both credited to "the repository". Its demand set, lifted matroid, conditioning minor, preorder reduction and cloning construction are also Part V's, which source 11 does not say for the last four; merge notes record it. It did not read Part VIII, and nothing in Part VIII bears on it.

**Abstract of source 11 (as delivered).** For a bipartite graph  $H = (X \sqcup Y, E)$ , let  $\mathcal{D}_H$  consist of the nonnegative integral demand vectors  $c$  satisfying  $c(A) \leq |N_H(A)|$  for all  $A \subseteq Y$ . A classical correspondence of Suho Oh identifies the vectors with a fixed positive support  $J$  with the supplier sets  $I$  matchable to  $J$ . We give an explicit algorithmic realization using two integer-margin minimum-cost transportation problems, each of total flow  $|X|(|J| + 1)$ . A fixed cycle-generic cost system forces both optimal supports to be the same spanning tree. The supplier degrees recover  $I$ , while the supplier leaves recover every coordinate of  $c$ . We prove the inverse identities directly, without enumerating a mixed subdivision, and obtain polynomial bit complexity and short exact primal–dual certificates. A core factorization reduces the forward flow to  $|J|(|J| + 1)$ , followed by independent supplier attachments.

The construction closes the full-coordinate sampling gap recorded in the pinned ProveIt report on preorder root polytopes. Composing it with the established matroid basis walk gives almost-uniform samples of actual lattice points, not merely their supports; total-variation error is preserved exactly. We also prove support restrictions, unused-supplier deletion laws, perturbation invariance, and an obstruction to fully symmetry-natural deterministic bijections. The results apply to every finite preorder. An executable standard-library implementation is checked exhaustively on 4,785 bipartite graphs and on all 390 labeled preorders of order at most four.

The abstract is reproduced as delivered, with source 11’s  $\mathcal{D}_H$  printed  $\mathcal{D}_H$ . Its “pinned ProveIt report” is this report, and the gap is Part V’s sampling boundary.

**Contribution and status (source 11, as delivered).** The existence of the demand/base correspondence, its counting consequences, and rapid mixing on matroid bases are prior results. The contribution developed here is the explicit two-flow realization, its proofs and certificates, its compatibility laws, and the resulting full-coordinate sampler for the repository’s problem. The manuscript does not assert first priority over all possible algorithmic realizations of Oh’s construction. The proofs below are ordinary mathematical proofs; the implementation has been run and tested, but neither has been checked by Lean or Rocq or independently refereed.

**Priority and prior work (merge).** The correspondence between fixed-support demand vectors and matchable supplier sets, with its leaf-count description, is Oh’s [22] (source 11 cites Section 4, Remark 4.4 and Proposition 4.5 of the arXiv version; this report’s entry cites the journal numbering, Lemma 22 and Proposition 26), and Ohsugi–Tsuchiya use it [8]; Loho and Smith [29] study closely related degree-vector and lattice-point bijections, and source 11’s audit did not read their whole paper. The counting identity (Corollary 139.2) is Part II’s Lemma 14.1, Part IV’s Proposition 49.3 and Part V’s Proposition 68.3; its proof here is a further, elementary route (merge note after the corollary). The lifted matroid, the conditioning minor, the preorder reduction and the cloning construction are Part V’s Proposition 69.1, Proposition 69.2, Lemma 71.1 and Corollary 70.5; the preorder reduction is printed once, as Proposition 146.1, with its proof replaced by a pointer. The basis-walk mixing theorem is Anari–Liu–Oveis Gharan–Vinzant’s [19], imported as in Part V. New relative to Parts I–VIII: the two-flow realization (Theorem 135.2) with its explicit margins, the core factorization and restriction laws (Section 140), the perturbation and chamber laws, the certificate theorem, the two obstructions (Section 144), the exact error transfer and the full-coordinate sampler (Section 145), its preorder corollary, and the inverse prices (Section 150). No worldwide priority is claimed for any result of Part IX.

## 132.2 Notation of Part IX

Part IX uses Part V’s  $\mathcal{D}_H$ ,  $\mathcal{M}(H)$ ,  $\mathcal{M}_H$ ,  $N_H(\cdot)$  and  $\mathcal{Q}_\tau$ , and Part V’s orientation of the preorder graph  $H_\tau$  (an edge  $x \rightarrow y$  exactly when  $x \leq_\tau y$ ). Table 16 lists what Part IX adds or uses differently. Four symbols of source 11 were changed; no normalization was changed:

- source 11’s  $\mathcal{B}_H(J)$  (supplier sets matchable to  $J$ ) and  $\mathcal{B}_H$  (all matchable pairs) are printed

Table 16: Symbols of Part IX. The last column prints the tempting false reading.

| Symbol                                     | Meaning in Part IX                                                                                                     | Not to be confused with                                                                                                         |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| $H = (X \sqcup Y, E), m, n$                | bipartite graph, suppliers $X$ , receivers $Y$ , $m =  X , n =  Y $                                                    | Part I's ground set $E = [n]$ ; Part VIII's tree edges $E$ .                                                                    |
| $\mathcal{D}_H, \mathcal{D}_H(J)$          | demand vectors; those with support exactly $J$                                                                         | Part V's support enumerators $D_H(t)$ , $D_H(z)$ (polynomials); Part II's polytope $D_G$ .                                      |
| $\mathcal{M}(H), \mathcal{M}_J(H)$         | matchable support pairs $(I, J)$ ; the supplier sets matchable to $J$ (source 11's $\mathcal{B}_H, \mathcal{B}_H(J)$ ) | Part V's basis polynomials $\mathcal{B}_H(x, y)$ and $\mathcal{B}_H(a, z)$ .                                                    |
| $I, J, r =  J $                            | selected supplier set, receiver support, its size                                                                      | Part V's $r = \min(m, n)$ (source 06) or $v(H)$ (source 07); Part VI's tree size $r$ ; Part I's order ideals $I$ .              |
| $0, G_J, E_J, c_0$                         | dummy receiver; the augmented graph; its edges; $m - \sum_{y \in J} c_y$                                               | receiver index 0 in the code, where the dummy is $-1$ ; Part I's doubled graph $G_\tau$ .                                       |
| $w, q, q_J, q_I$                           | edge costs (for instance $w(e_i) = 2^i$ ); numbers of augmented edges                                                  | Part VII's weights $w_i$ ; in Section 146 $q$ is a dilation factor, and in $\prod_y q_y^{c_y}$ the $q_y$ are magnitude weights. |
| $\Phi_w, \Psi_w, \widehat{\Phi}_w$         | forward and inverse maps; the lifted bijection (145.2)                                                                 | Part VI's signed support polynomial $\Phi_G$ .                                                                                  |
| $(A), (B), L = r + 1$                      | the forward and inverse margin systems; the inverse scale                                                              | Part V's random support size $L$ ; Part VI's pencil $L$ .                                                                       |
| $T, T_I$                                   | the common optimal spanning tree; the core tree                                                                        | Part VIII's host tree $T$ ; Part V's subsets $T \subseteq Y$ .                                                                  |
| $S_y, a_y, A_y, \ell_y$                    | receivers, suppliers, supplier set and supplier leaves below $y$                                                       | Part V's supports $S$ .                                                                                                         |
| $u, v, s_{xy}, F, K$                       | potentials, reduced costs, flows, core flow                                                                            | Part IV's mixing variable $K$ ; Part VIII's hull $K$ .                                                                          |
| $M_H$                                      | the lifted transversal matroid, bases $(X \setminus I) \cup J$ (source 11's $M_H$ )                                    | Part VII's $M_G$ , its dual; Part II's matching polynomial $M_G(u)$ .                                                           |
| $\theta, \mu_\theta, \pi_\theta, Z_\theta$ | receiver activities, the demand law, the basis law, its partition function                                             | Part VIII's enumerator $Z_T$ ; Part V's activities $\alpha, \beta$ .                                                            |
| $N = m + n, W, t, \eta$                    | ground-set size, largest activity, number of walk steps, target error                                                  | the variable $t$ of $h_\tau(t)$ ; Part V's $N$ .                                                                                |
| $\tau, V, H_\tau, \mathcal{Q}_\tau$        | a preorder on $V$ , its graph ( $x$ supplies $y$ iff $x \leq_\tau y$ ), its polytope                                   | Part I's $G_\tau$ , which has the two shores exchanged (Part IV's convention).                                                  |

$\mathcal{M}_J(H)$  and  $\mathcal{M}(H)$ , because Part V's  $\mathcal{B}_H$  is the basis polynomial (69.5);

- source 11's  $\mathcal{D}_H$  is printed  $\mathcal{D}_H$  (the same set as Part V's);
- source 11's  $M_H$  is printed  $M_H$ , Part V's symbol for the same matroid;
- source 11's  $Q_\tau$  is printed  $\mathcal{Q}_\tau$ , as in every other Part.

Source 11's macros  $\Phi_w, \Psi_w$  are typeset directly, and its TV is the report's \TV.

### 132.3 What Part IX answers

Status after Part IX, as of 30 September 2026:

- **Part V's "Sampling boundary" is answered.** Its samplers return support sets or support pairs; Theorem 145.2 and Corollary 146.2 return almost-uniform actual vectors of  $\mathcal{D}_H$  and of  $\mathcal{Q}_\tau \cap \mathbb{Z}^V$ , with every coordinate value, and Theorem 145.1 transfers the total-variation error exactly. The bijection is restriction-compatible in two stated senses: for receivers outside the support (Corollary 140.3) and for suppliers outside the selected set (Corollary 140.2); arbitrary minors are not covered. The approximate, not exact, sense of "almost-uniform" is

that of Part V's support samplers. The same answers Part V's "Precisely what is sampled for a preorder" and its two unnumbered questions "An effective demand/support bijection (source 06)" and "A direct demand/support bijection (source 07)" (Section 80.1): the tree is extracted from an optimum, so no noncanonical chosen matching enters, as source 07's test demands.

- **The bijection of Research questions 2 and 20 is constructed**, explicitly, in polynomial time, preserving the exact receiver support, and with an elementary proof that imports no Ehrhart theory (Theorem 135.2 and the merge note after Corollary 139.2). Their compatibility clauses are answered only in part: the cost system is a controlled ordering of the augmented edges, positive support weights are preserved (Theorem 145.1), and deletion laws hold outside  $I$  and  $J$ ; Proposition 144.1 shows that no deterministic support-preserving rule is equivariant under all graph automorphisms, so full naturality is impossible. The rest is Research question 71.
- **Part VII's "Direct lattice-point bijections (source 09)"** is answered in its support-resolved part; a *canonical* bijection, and compatibility with the paired clone action, are not obtained (the maps depend on the cost chamber, Proposition 141.2; symmetry is restored only in law, by a random cost order).
- **Not answered:** Part V's Research question 39 (binary capacities), which Research question 68 extends; Part V's Research question 43 (formalization), which Research question 73 overlaps. No other numbered research question of Parts I–VIII is affected, and Conjectures 5.1(d) and 5.3 of [1] are untouched.

Part IX refutes no statement of Parts I–VIII, and nothing is retracted: the passages that said no bijection or no lattice-point sampler had been constructed described the report as it then was, and keep their text with dated pointers here (in Parts II, IV, V and VII). Part V's Research question on binary capacities, unlabelled before, received the label `tf:hq:capacities`.

## 132.4 Where the merge had to choose

- *Order.* Source 11's order of sections is kept; its Section  $k$  is Section  $k + 132$  here, and its Appendices A and B are printed as ordinary Sections 150 and 151 at the end of Part IX, so that Part I's appendices keep their letters.
- *Section 133.* Its first subsection, "A precise gap in the repository", cited the pinned commit, the blob and a line interval of this report; it was rewritten as a pointer to Part V's section, and the pin, blob and line interval are recorded in Section 132.1. Its other subsections, the claim ledger included, are kept.
- *Printed as pointers (duplicates of Parts II–V).* The proof of the preorder reduction (Proposition 146.1), which is Part V's; the statement and its label are kept. The lifted matroid, the conditioning minor, the Hall lemma, the basis walk and the cloning construction keep source 11's short text and gain merge notes naming Part V's statements.
- *Second route kept.* Corollary 139.2, whose proof through Theorem 135.2 is a further, elementary proof of the counting identity of Parts II, IV and V; source 11 itself calls the identity known.
- *Research questions.* Source 11's eight questions continue the numbering as Research questions 66–73, each with a label; the three that extend earlier questions carry merge notes.
- *Self-citation and file names.* Source 11's reference "ProveIt repository" (Part V of this report at the pin) became references to Part V's sections and statements, and its macro for the pinned



commit was expanded. Its code, data and audit files keep their delivered base names behind the prefix `11-two-flow-`: `transport_bijection.py`, `verify.py` and `build.sh` in `code/`; the six suite records `small.json`, `four_by_three.json`, `preorders.json`, `random.json`, `chain.json`, `example.json`, and `environment.json` and `verification_run.txt`, in `data/`; and `PROOF_STATUS.md`, `SOURCE_AUDIT.md` at the report root. Merge notes give the shipped names; its two code listings are kept with delivery names.

- *Cross-references and layout.* Source 11 uses plain `\ref`, like this report; no `cleveref` is loaded. Its preamble is dropped: its `\N`, `\R`, `\Z`, `\supp`, `\one` agree with the host's, its `\cost` and `\NN` were unused, and `\TV` was added to the preamble. Its box environment became the host's statusbox, and its claim-ledger table, placed with the float package's `[H]`, which this report does not load, is placed `[htbp]`. Its one `\choose` is typeset with `\binom`, and its two tagged displays (A) and (B) were given PDF anchors of their own. Labels carry the prefix `tf`; the eight questions, this section and its subsections, the notation table and the rewritten subsection gained labels. Its abstract and status box are printed in this section; its title page, table of contents and running head are dropped.
- *Bibliography.* [22], [8], [19] and [1] were already cited (source 11 records the published version of [19], *Ann. of Math.* **199** (2024), 259–299, and takes its theorem numbering from arXiv:1811.01816v3); Loho–Smith is new [29].

## 133 The question answered and the relation to prior work

### 133.1 The question left by Part V

Parts II, IV and V of this report develop support enumerators, matroid models, and approximate counting for bipartite demand vectors. Part V ends its sampling discussion (Section 75) with an explicit boundary, the “Sampling boundary” box after “Sampling after fixing the support size”: the algorithms produce support sets or matchable support pairs, not the coordinate values of a uniform lattice point (see also “Precisely what is sampled for a preorder” in the same section). It asks for an efficiently invertible, restriction-compatible demand/support correspondence.

[Merge note: source 11 identifies this passage by the pinned commit, the path and blob of this report's `article.tex`, and the line interval 9510–9555 of that blob (Section 132.1); those line numbers refer to Parts I–VII as printed before Part VIII and no longer apply. Source 11 calls it “a dated, identified question, not an inference from a missing theorem in a summary.” Parts II and IV asked the same bijective question earlier, as Research questions 2 and 20; Part V repeats it in Section 80.1, and Part VII in its questions (“Direct lattice-point bijections (source 09)”). What Part IX answers of each is stated in Section 132.3.]

The distinction matters. Suppose a probability measure has the right support-size distribution. This does not specify the individual positive coordinates. Even knowing every fibre cardinality does not provide a polynomial-time decoder from a uniformly chosen object of an equinumerous family. The algorithm below supplies precisely that decoder and its inverse.

### 133.2 What is already known

Oh constructs a correspondence between bases of a transversal matroid and positive-coordinate lattice points of an associated generalized permutohedron [22]. In the arXiv version inspected here, Section 4, Remark 4.4 and Proposition 4.5 describe the relevant zonotopal-cell correspondence and its leaf-count formula. Thus an abstract demand/base bijection is not a new theorem of this manuscript. Ohsugi–Tsuchiya use the same correspondence in the graph-polynomial

setting [8]. Loho–Smith study closely related degree-vector and lattice-point bijections for matching fields and triangulations [29]. Those works are important context for the present use of coherent spanning trees.

The rapid-mixing theorem used later is due to Anari, Liu, Oveis Gharan, and Vinzant [19]. Their theorem gives polynomial-time approximate sampling of matroid bases. We do not prove a new general matroid mixing theorem. Likewise, the definition and lattice-point study of preorder polytopes originate in the work of Athanasiadis–Chapoton [1]. [Merge note: source 11 prints the author order “Chapoton–Athanasiadis” here and in its bibliography; this report’s entry [1] and Part I use the order of the arXiv record, Athanasiadis and Chapoton.]

Our approach locates the relevant coherent tree with one min-cost flow in each direction. The important extra information is the explicit margins, finite perturbation sizes, bit bounds, independently checkable optimality certificate, and exact compatibility statements. No list of cells in a triangulation is an input. No numerical infinitesimal is required.

### 133.3 Claim ledger

| Statement                                                               | Status and role                                                                              |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Existence of a positive-support demand/base correspondence              | Prior work of Oh; independently proved in the special form needed here.                      |
| Two explicit integer-margin optimizations with inverse decoders         | Main constructive result proved in Sections 135–139.                                         |
| Finite perturbation and restriction laws                                | Proved here for the stated cost-labelled setting; no arbitrary-minor naturality is asserted. |
| Polynomial-time weighted basis sampling                                 | Imported from the established matroid-walk theorem.                                          |
| Full-coordinate sampling of $\mathcal{D}_H$ and preorder lattice points | Constructive consequence, closing the identified repository sampling gap.                    |
| Lean or Rocq formalization                                              | Not supplied or claimed; an explicit formalization plan is given in Section 147.             |

Table 17: Logical and novelty boundaries. “Proved here” identifies the manuscript’s proof, not an assertion of worldwide first priority.

## 134 Demand vectors and matchable support pairs

Let  $H = (X \sqcup Y, E)$  be a finite simple bipartite graph, with  $m = |X|$  suppliers and  $n = |Y|$  receivers. For  $A \subseteq Y$  put

$$N_H(A) = \{x \in X : \text{some } y \in A \text{ satisfies } xy \in E\}, \quad c(A) = \sum_{y \in A} c_y.$$

We study the finite set

$$\mathcal{D}_H = \{c \in \mathbb{Z}_{\geq 0}^Y : c(A) \leq |N_H(A)| \text{ for every } A \subseteq Y\}. \quad (134.1)$$

In particular,  $\sum_y c_y \leq m$ . For  $J \subseteq Y$ , write

$$\begin{aligned} \mathcal{D}_H(J) &= \{c \in \mathcal{D}_H : \text{supp}(c) = J\}, \\ \mathcal{M}_J(H) &= \{I \subseteq X : |I| = |J|, H[I, J] \text{ has a perfect matching}\}. \end{aligned} \quad (134.2)$$

A member of  $\mathcal{M}_J(H)$  is a supplier *set*, not a perfect matching. Multiple perfect matchings on the same pair  $(I, J)$  are counted once. We also use the disjoint union

$$\mathcal{M}(H) = \{(I, J) : J \subseteq Y, I \in \mathcal{M}_J(H)\}.$$

There is one empty pair, corresponding to the zero vector. [Merge note:  $\mathcal{D}_H$  is Part V's demand set (68.5) and Parts II and IV's  $\mathcal{R}_H$ ;  $\mathcal{M}(H)$  is Part V's set of matchable support pairs (written  $U \sim_H S$  there, and  $\mathcal{M}(H)$  in Part IV). Source 11 writes  $\mathcal{B}_H(J)$  and  $\mathcal{B}_H$  for  $\mathcal{M}_J(H)$  and  $\mathcal{M}(H)$ ; they are renamed because Part V's  $\mathcal{B}_H$  is the basis polynomial (69.5) (Section 132.2).]

**Lemma 134.1** (Integral allocation). *A vector  $c \in \mathbb{Z}_{\geq 0}^Y$  belongs to  $\mathcal{D}_H$  if and only if there is a partial assignment of suppliers to adjacent receivers in which each supplier is used at most once and receiver  $y$  receives exactly  $c_y$  suppliers.*

*Proof.* An assignment immediately proves all the inequalities in (134.1). Conversely, replace receiver  $y$  by  $c_y$  identical copies. For a set  $U$  of copies, let  $A$  be the original receivers represented in  $U$ . Then  $|U| \leq c(A) \leq |N_H(A)| = |N(U)|$ . These inequalities imply a matching saturating every copy. To recall the augmenting-path argument, take a maximum matching. If a copy is unmatched, follow alternating paths from all unmatched copies. Every reachable supplier must be matched, or an augmenting path exists. Its matched copy is reachable too. The reachable copies therefore outnumber their entire supplier neighbourhood, contradicting the displayed inequality. The saturating matching is the required assignment.  $\square$

[Merge note: this is the clone-and-Hall argument of Part II (Section 14) and Part V's Lemma 68.2, with the augmenting-path step spelled out as in Part IV's manuscript 01 (after Lemma 49.1).]

For  $J \neq \emptyset$  write  $r = |J|$  and add a dummy receiver 0 adjacent to every supplier. We use 0 as a separate symbol, not as a member of  $Y$ . Let

$$G_J = (X \sqcup (\{0\} \cup J), E_J), \quad E_J = \{xy \in E : y \in J\} \cup \{x0 : x \in X\}. \quad (134.3)$$

An assignment may send an unused supplier to the dummy. For a demand vector define

$$c_0 = m - \sum_{y \in J} c_y. \quad (134.4)$$

This number is nonnegative. An empty fibre is detected by asking whether  $J$  can be saturated by a matching; if it cannot, neither side of (134.2) has an element.

## 135 The two-flow theorem

We first state the complete construction. Its proof occupies the next four sections.

**Definition 135.1** (Cycle-generic costs). A real edge-cost assignment  $w$  on the full dummy-augmented graph is *cycle-generic* if the alternating sum of its costs around every simple even cycle is nonzero. All restrictions to  $G_J$  use the same costs.

For an explicit integral choice, order all  $q = |E| + m$  augmented edges as  $e_0, \dots, e_{q-1}$  and set

$$w(e_i) = 2^i. \quad (135.1)$$

The largest power in a cycle sum exceeds the sum of all the smaller powers, so this choice is cycle-generic. Each cost has at most  $q$  bits.

**Theorem 135.2** (Explicit two-flow correspondence). *Fix cycle-generic costs  $w$ . For each nonempty  $J$  with  $r = |J|$ , the following maps are well-defined inverse bijections*

$$\Phi_w : \mathcal{M}_J(H) \longrightarrow \mathcal{D}_H(J), \quad \Psi_w : \mathcal{D}_H(J) \longrightarrow \mathcal{M}_J(H).$$

**Forward map.** *Given  $I \in \mathcal{M}_J(H)$ , minimize  $\sum_{xy \in E_J} w_{xy} F_{xy}$  over nonnegative flows with margins*

$$\sum_y F_{xy} = m \mathbf{1}_{x \in I} + 1 \quad (x \in X), \quad \sum_x F_{xy} = m \quad (y \in \{0\} \cup J). \quad (\text{A})$$

*The positive-flow support is a unique spanning tree  $T$ . A supplier in  $I$  has degree two and any other supplier has degree one. Set*

$$\Phi_w(I)_y = \begin{cases} 1 + \#\{x \in X \setminus I : xy \in T\}, & y \in J, \\ 0, & y \notin J. \end{cases} \quad (135.2)$$

**Inverse map.** *Given  $c \in \mathcal{D}_H(J)$ , put  $L = r + 1$  and use (134.4). Minimize the same objective with margins*

$$\sum_y F_{xy} = L \quad (x \in X), \quad \sum_x F_{x0} = Lc_0 + r, \quad \sum_x F_{xy} = Lc_y - 1 \quad (y \in J). \quad (\text{B})$$

*Again the support is a unique spanning tree  $T$ , every supplier has degree one or two, and*

$$\Psi_w(c) = \{x \in X : \deg_T(x) = 2\}. \quad (135.3)$$

*If  $c = \Phi_w(I)$ , the two optimal trees in (A) and (B) are identical. Each optimization has integral optimal flow and total flow  $m(r + 1)$ . For the costs (135.1), both maps are computable in polynomial bit time and admit exact certificates of polynomial length.*

For  $J = \emptyset$  the maps send  $\emptyset$  to the zero vector and back. This convention covers  $m = 0$ ,  $n = 0$ , and graphs with no ordinary edges. The theorem concerns *one* optimization per direction, not an alternating iterative process between (A) and (B).

## 136 Why generic optimal supports are trees

For a bipartite graph  $G = (X \sqcup C, E_G)$  and positive margins  $a_x, b_y$  with equal sums, a transportation flow is a family  $F_{xy} \geq 0$  on its edges satisfying the row and column sums. A feasible set is compact: every flow coordinate lies between zero and the total margin. Thus a linear cost has a minimum whenever the set is nonempty.

**Lemma 136.1** (Acyclicity, uniqueness, and integrality). *Under cycle-generic costs, every minimum-cost flow has an acyclic positive support, and the minimum-cost flow is unique. If its margins are integral, its positive flow values are integral.*

*Proof.* Suppose an optimal positive support contains a cycle. Add a sufficiently small  $\eta$  to the alternate edges and subtract  $\eta$  from the other edges. Both signs of  $\eta$  preserve nonnegativity for sufficiently small  $|\eta|$ , and all margins are unchanged. The objective varies by a nonzero multiple of  $\eta$ , so one sign improves the cost. This is a contradiction.

If  $F$  and  $F'$  were distinct optima, their average would also be optimal and its positive support would be the union of their supports. That union is therefore a forest. But  $F - F'$  has zero row

and column sums. Repeatedly remove a leaf of the forest: the difference on its sole edge must be zero. This removes every edge and shows  $F = F'$ .

For integral margins, the same leaf elimination proves integrality of the unique feasible flow on its support forest. The sole edge at a leaf has its integral residual margin; subtracting that value leaves integral margins on the remaining forest.  $\square$

**Lemma 136.2** (Strict tree certificates). *Suppose a minimum-cost flow is positive on a spanning tree  $T$ . There are potentials  $u_x, v_y$ , unique after fixing one of them, such that*

$$u_x + v_y = w_{xy} \quad (xy \in T), \quad u_x + v_y < w_{xy} \quad (xy \in E_G \setminus T). \quad (136.1)$$

*Consequently, every other positive flow on  $T$ , for any other margins, is the unique minimum-cost flow for those margins.*

*Proof.* Fix one potential and propagate the equations along the tree. For an off-tree edge  $xy$ , its reduced cost  $s_{xy} = w_{xy} - u_x - v_y$  is the alternating cost of the fundamental cycle formed by adding  $xy$  to  $T$ . If  $s_{xy} < 0$ , add a small positive flow on  $xy$  and alternate along the tree path. All edges from which flow is subtracted are positive. The margins stay fixed and the cost decreases. Thus  $s_{xy} \geq 0$ , and cycle-genericity excludes equality.

For any flow  $G$  of the new margins  $a', b'$ , one has

$$\sum_{xy} w_{xy} G_{xy} = \sum_x u_x a'_x + \sum_y v_y b'_y + \sum_{xy \notin T} s_{xy} G_{xy}. \quad (136.2)$$

A flow on  $T$  attains the first two terms. Strict positivity of the slacks forces every other optimum to be supported on  $T$ ; leaf elimination then forces equality of the flows. This proves the assertion without requiring a separate appeal to a linear-programming duality theorem.  $\square$

These lemmas separate two issues. Genericity gives a forest, but not necessarily a connected support for arbitrary margins. The particular congruences in (A) and (B) supply connectedness. Once a tree is known, (136.2) transfers its optimality between margin systems.

## 137 The forward optimization and the leaf decoder

Assume  $I \in \mathcal{M}_J(H)$ ,  $r = |J| > 0$ . Choose any perfect matching between  $I$  and  $J$ . Send  $m$  units along every matching edge and one unit from every supplier to the dummy. This is a feasible flow for (A). In particular, all margins are strictly positive.

**Lemma 137.1** (Connectedness from the forward margins). *The unique optimizer of (A) has connected support.*

*Proof.* Take a component of its positive-support forest, with supplier set  $A$  and receiver set  $C \subseteq \{0\} \cup J$ . Positive margins imply that every vertex is incident to a positive edge, so the component contains at least one supplier. Balancing its total margins gives

$$m|I \cap A| + |A| = m|C|.$$

Hence  $m$  divides  $|A|$ . Since  $1 \leq |A| \leq m$ , the component contains all suppliers. There can be no second component, again because every component has a supplier.  $\square$

The optimizer is therefore a spanning tree with  $m + r$  edges. Each supplier in  $I$  has row sum  $m + 1$ , larger than the column sum  $m$  of any single receiver. Its degree is at least two. Every other supplier has degree at least one. The sum of these lower bounds is  $2r + (m - r) = m + r$ , exactly the number of tree edges. Thus all the degree bounds are equalities.

Root the tree at the dummy 0. A supplier in  $I$  has one parent receiver and one child receiver; a supplier outside  $I$  is a leaf. Conversely, every nonroot receiver has a parent supplier, and that supplier must have degree two. Consequently the parent–child edges from  $I$  to  $J$  form a perfect matching. This matching is extracted from the tree, and need not be the matching used to establish feasibility of (A).

Assign each supplier in  $I$  to its child receiver, and every supplier outside  $I$  to its sole neighbour. Ignore assignments to 0. Every receiver in  $J$  gets one supplier from  $I$ , together with precisely its supplier leaves. It follows from Lemma 134.1 that the vector in (135.2) lies in  $\mathcal{D}_H(J)$ . This proves well-definedness of the forward map, including all coordinate values, rather than only its support.

### 138 The inverse optimization and its degree bound

Now let  $c \in \mathcal{D}_H(J)$  with  $r > 0$ , and put  $L = r + 1$ . By Lemma 134.1, allocate suppliers to realize  $c$ , sending unused ones to 0. Scale that assignment by  $L$ . For each  $y \in J$ , choose one supplier assigned to  $y$  and divert one unit of its flow from  $y$  to the dummy. These chosen suppliers are distinct. This gives exactly the margins (B). All receiver margins are positive, including the dummy margin  $Lc_0 + r$ .

**Lemma 138.1** (Connectedness from the inverse margins). *The unique optimizer of (B) has connected support.*

*Proof.* If a positive-support component does not contain 0, let its receiver set be  $S \subseteq J$  and its supplier set be  $A$ . Positive margins imply  $1 \leq |S| \leq r$ . The margin equality is

$$L|A| = Lc(S) - |S|,$$

which says that  $L = r + 1$  divides  $|S|$ . This is impossible. Hence every component contains the same vertex 0, and there is only one.  $\square$

Root the resulting spanning tree at 0. For a receiver  $y \in J$ , let  $S_y$  denote the *number* of receiver vertices from  $J$  in the subtree rooted at  $y$ ; let  $A_y$  denote the set of suppliers in that subtree. Its parent supplier  $x$  lies outside this subtree. Summing flow balances over the subtree yields

$$F_{xy} = L \left( \sum_{z \text{ in the receiver subtree of } y} c_z - |A_y| \right) - S_y. \quad (138.1)$$

The left side is positive and at most the whole row supply  $L$  of  $x$ . The quantity in parentheses is an integer, and  $1 \leq S_y \leq r = L - 1$ . The only possible integer is therefore one. We have proved

$$F_{xy} = L - S_y \quad \text{for every supplier-to-child-receiver edge.} \quad (138.2)$$

**Lemma 138.2** (No branching at a supplier). *Every supplier in the optimal tree of (B) has degree at most two.*



*Proof.* If a supplier had two distinct child receivers  $y, z$ , their subtrees would be disjoint, so  $S_y + S_z \leq r$ . By (138.2), the sum of its flows to just those two children would be

$$2L - (S_y + S_z) \geq 2L - r = L + 1,$$

exceeding its entire row supply  $L$ . Thus a supplier has at most one child receiver, in addition to its parent receiver.  $\square$

All supplier degrees are one or two. Since the tree has  $m + r$  edges, exactly  $r$  suppliers have degree two. Denote their set by  $I$ . The parent–child construction from the preceding section again gives a perfect matching between  $I$  and  $J$ . It remains to check that the leaf formula recovers the original vector, not merely some demand vector.

For  $y \in J$ , let  $\ell_y$  be its number of supplier-leaf children. If a nonleaf supplier child of  $y$  has child receiver  $z$ , its flow to  $y$  is  $L - (L - S_z) = S_z$ . The flow into  $y$  from its parent supplier is  $L - S_y$ . Since

$$S_y = 1 + \sum_z S_z,$$

where the sum runs over these grandchildren, its column balance becomes

$$Lc_y - 1 = (L - S_y) + L\ell_y + \sum_z S_z = L(1 + \ell_y) - 1.$$

Thus  $c_y = 1 + \ell_y$ , exactly (135.2). At the dummy, the analogous identity gives  $c_0 = \ell_0$ . This proves the local recovery assertion.

### 139 Why the two trees are identical: the inverse proof

It is not enough to establish that the formulas have the right codomains. We must show that an optimal tree for one flow is optimal for the other. The strict tree certificate makes this possible.

Call a rooted spanning tree of  $G_J$  *admissible* if every supplier has degree one or two. Write  $I(T)$  for its degree-two suppliers and  $c(T)$  for its leaf vector in (135.2). For a receiver  $y \in J$ , write  $S_y$  for its subtree's number of ordinary receivers and  $a_y$  for its subtree's number of suppliers. The parent supplier of  $y$  is outside that subtree. Consequently

$$1 \leq S_y \leq r, \quad 0 \leq a_y \leq m - 1. \quad (139.1)$$

**Lemma 139.1** (Two positive flows on every admissible tree). *Every admissible tree supports a strictly positive flow for (A) with  $I = I(T)$  and a strictly positive flow for (B) with  $c = c(T)$ . The flows are as follows. A supplier leaf sends 1 in (A) and  $L$  in (B) to its parent. If a degree-two supplier  $x$  has child receiver  $y$ , then*

|                 | $x \longrightarrow y$ | $x \longrightarrow \text{parent}(x)$ |         |
|-----------------|-----------------------|--------------------------------------|---------|
| <i>flow (A)</i> | $m - a_y$             | $a_y + 1$                            |         |
| <i>flow (B)</i> | $L - S_y$             | $S_y$                                | (139.2) |

*All entries are strictly positive integers.*

*Proof.* Positivity follows from (139.1). Row sums are immediate. For a nonroot receiver  $y$ , its supplier subtree consists of its supplier leaves and its nonleaf supplier children, together with the subtrees of their child receivers. Thus

$$a_y = \ell_y + \sum_z (1 + a_z).$$

Its column sum in (A) is therefore

$$(m - a_y) + \ell_y + \sum_z (a_z + 1) = m.$$

At 0, every supplier is in one of the attached subtrees or is a leaf there, so the same identity gives column sum  $m$ .

For (B), the column computation in Section 138 gives  $L(1 + \ell_y) - 1 = Lc_y - 1$  for  $y \in J$ . The dummy receives  $L\ell_0 + \sum_z S_z = Lc_0 + r$ . This is the required margin.  $\square$

*Proof of Theorem 135.2.* The preceding sections prove feasibility, uniqueness, connectedness, integrality, the supplier degree pattern, and both decoding formulas. Let  $T$  be the optimizer of (A) for  $I$ , and set  $c = c(T)$ . By Lemma 139.1,  $T$  supports a positive flow with margins (B). Its strict potentials from Lemma 136.2 prove that this is the unique optimizer of (B). Therefore  $\Psi_w(\Phi_w(I)) = I$ .

Conversely, let  $T$  be the optimizer of (B) for  $c$  and let  $I = I(T)$ . Section 138 proves  $c(T) = c$  and  $I \in \mathcal{M}_J(H)$ . Lemma 139.1 puts a positive (A)-flow on  $T$ , and the same strict potentials make it the unique (A)-optimizer. Thus  $\Phi_w(\Psi_w(c)) = c$ . The total margin in each case is  $m(r + 1)$ . The polynomial complexity and certificate claims are proved in Section 142.  $\square$

**Corollary 139.2** (Supportwise counting identity). *For every bipartite graph  $H$ ,*

$$\sum_{c \in \mathcal{D}_H} \prod_{y \in \text{supp}(c)} z_y = \sum_{(I,J) \in \mathcal{M}(H)} \prod_{y \in J} z_y. \quad (139.3)$$

*In particular,  $|\mathcal{D}_H(J)| = |\mathcal{M}_J(H)|$  for every  $J$ .*

*Proof.* Take the disjoint union of the support-preserving bijections, including the empty fibre.  $\square$

The identity is a known consequence of the Oh correspondence, not a new counting formula claimed here. The theorem supplies an explicit decoder for each term on its right-hand side.

[Merge note (second route): Corollary 139.2 is Part V's Proposition 68.3 (identity (68.4), with  $S = J$ ) and Part IV's Proposition 49.3; setting every  $z_y = 1$  gives Part II's Lemma 14.1. Its proof above is therefore a further proof of those three statements, and the only one in this report that uses no Ehrhart theory and no imported lattice-point theorem. Part II and Part IV reduce the count to Kálmán–Postnikov and Ohsugi–Tsuchiya (as stated by Davis–Kohl); Part V's source 06 imports Ohsugi–Tsuchiya's fixed-support count, which rests on Oh (Lemma 68.1); its source 07 imports Dai et al. and Davis–Kohl. Theorem 135.2 uses only Hall's theorem (Lemma 134.1), the existence of a minimum of a linear function on a nonempty compact set, and the forest and potential arguments of Section 136. This is the elementary proof that Research questions 2 and 20 ask for, as a way to remove “the two root-polytope inputs” and “the largest formalization dependency”. It replaces only the counting input: the other imported theorems of Parts II, IV and V (for example Part V's Lorentzian and Röhrle–Ulirsch inputs) are untouched, and, as source 11 says, the identity itself is not new.]

## 140 A smaller core flow and independent supplier attachments

The forward optimization can be made substantially smaller when  $m$  is large and the requested support has size  $r$ . Its essential tree is already determined by the  $r$  selected suppliers.

**Theorem 140.1** (Core-tree factorization). *Fix  $I \in \mathcal{M}_J(H)$  with  $r = |I| = |J| > 0$ . Restrict the augmented graph to  $I \sqcup (\{0\} \cup J)$ , keeping the original costs. Solve its forward problem with  $m$  replaced by  $r$ :*

$$\sum_y K_{xy} = r + 1 \quad (x \in I), \quad \sum_{x \in I} K_{xy} = r \quad (y \in \{0\} \cup J). \quad (140.1)$$

*Let  $T_I$  be its optimal tree and let  $v_y$  be its receiver potentials, normalized by  $v_0 = 0$ . For every unselected supplier define*

$$a_I(x) = \arg \min_{y \in (N_H(x) \cap J) \cup \{0\}} (w_{xy} - v_y), \quad x \in X \setminus I. \quad (140.2)$$

*Each minimizer is unique. The full optimal tree of  $(A)$  is*

$$T = T_I \cup \{xa_I(x) : x \in X \setminus I\}, \quad (140.3)$$

*and its demand vector is*

$$\Phi_w(I)_y = 1 + \#\{x \notin I : a_I(x) = y\} \quad (y \in J). \quad (140.4)$$

*Thus the forward map needs a flow of total value  $r(r + 1)$  on only  $r$  supplier vertices, followed by independent reduced-cost comparisons for the other suppliers.*

*Proof.* The restriction has a perfect matching from  $I$  to  $J$ , so the earlier forward argument applies. All its suppliers have degree two. Its strict potentials satisfy (136.1) on the restricted graph.

If two distinct receivers  $y, z$  tied in (140.2), the path from  $y$  to  $z$  in  $T_I$ , together with  $xy$  and  $xz$ , would form a simple cycle of alternating cost zero. This contradicts cycle-genericity. Define the new supplier potential by  $u_x = w_{x, a_I(x)} - v_{a_I(x)}$ . All its other incident edges then have strictly positive reduced cost.

The graph (140.3) is an admissible spanning tree: it is obtained by attaching supplier leaves to  $T_I$ . The old potentials and the new  $u_x$  satisfy the strict certificate on the entire graph. By Lemma 139.1, the tree supports a positive flow with the full margins (A), and (136.2) proves it uniquely optimal. The leaf decoder is exactly (140.4).  $\square$

This factorization supplies a useful interpretation. The chosen supplier set determines a small price system  $v$ . Every other supplier chooses the unique cheapest receiver after subtracting that receiver's price, or chooses the dummy. Its contribution to the demand vector is independently one unit at the chosen ordinary receiver, or zero at the dummy. The independence here is *deterministic additivity*, not a probabilistic independence assertion about a random supplier set.

**Corollary 140.2** (Unused-supplier deletion and insertion). *Keep  $I, J$  and inherited costs fixed, and replace  $X$  by any subset  $X' \supseteq I$ . Then*

$$\Phi_w^{X'}(I)_y = 1 + \#\{x \in X' \setminus I : a_I(x) = y\} \quad (y \in J). \quad (140.5)$$

*In particular, deleting one supplier outside  $I$  subtracts one unit at its chosen receiver, or subtracts nothing if it chooses the dummy. All such deletions commute. If  $I \subseteq X' \subseteq X''$ , then*

$$\Phi_w^{X'}(I) \leq \Phi_w^{X''}(I) \quad \text{coordinatewise}, \quad \|\Phi_w^{X''}(I) - \Phi_w^{X'}(I)\|_1 \leq |X'' \setminus X'|. \quad (140.6)$$

*Proof.* The core tree, its receiver potentials, and the minimizer for any remaining supplier do not change when another unselected supplier is deleted. Apply Theorem 140.1. Every ordinary coordinate stays positive because of its initial contribution one.  $\square$

**Corollary 140.3** (Support restriction). *Deleting or adding receivers outside the fixed support  $J$ , together with their incident edges, does not change either map when the costs on the retained augmented edges are unchanged.*

*Proof.* Both optimizations are performed on exactly  $G_J$ , and the core construction uses only the same receivers.  $\square$

These are precise senses in which the correspondence is restriction compatible. Deleting a supplier *inside*  $I$ , deleting a positive receiver, or changing the cost choices is a different operation and is not covered by these statements.

## 141 Finite perturbations and stability of the construction

### 141.1 No infinitesimals are needed

The integral margins are normalized versions of two finite perturbations. They can be varied over explicit open ranges without changing the tree.

**Proposition 141.1** (Margin invariance). *Let  $T$  be the common optimal tree corresponding to  $(I, c)$ . Both of the following transportation problems have the same optimal support  $T$ :*

$$a_x = \mathbf{1}_{x \in I} + \delta_x, \quad b_y = 1 \quad (y \in \{0\} \cup J), \quad \delta_x > 0, \quad \sum_x \delta_x = 1; \quad (141.1)$$

$$a_x = 1, \quad b_0 = c_0 + r\varepsilon, \quad b_y = c_y - \varepsilon \quad (y \in J), \quad 0 < \varepsilon < 1/r. \quad (141.2)$$

*Proof.* Use the notation of Lemma 139.1. In (141.1), a supplier leaf sends  $\delta_x$  to its parent. A selected supplier  $x$  with child receiver  $y$  sends

$$1 - \sum_{z \in A_y} \delta_z \quad \text{to } y, \quad \delta_x + \sum_{z \in A_y} \delta_z \quad \text{to its parent,}$$

where  $A_y$  is the supplier subtree of  $y$ . These quantities are strictly positive because  $x \notin A_y$  and all  $\delta$  are positive. The row sums are as required; partitioning supplier subtrees verifies that every column sum is one.

For (141.2), a supplier leaf sends one. A selected supplier with child  $y$  sends  $1 - \varepsilon S_y$  to  $y$  and  $\varepsilon S_y$  to its parent. These values are positive in the stated range. The calculation used for (B) gives the desired column sums. The strict potentials of  $T$  prove optimality for both families by Lemma 136.2.  $\square$

Taking  $\delta_x = 1/m$  and multiplying by  $m$  gives (A). Taking  $\varepsilon = 1/(r+1)$  and multiplying by  $r+1$  gives (B). These choices have small exact denominators. The upper bound on  $\varepsilon$  cannot be replaced uniformly by a non-strict bound: when a receiver subtree contains all  $r$  ordinary receivers, the corresponding edge becomes zero at  $\varepsilon = 1/r$ .

### 141.2 Cost chambers and gauge invariance

**Proposition 141.2** (Dependence only on cycle signs). *Two cycle-generic cost assignments with the same signs on every oriented alternating cycle induce the same maps. In particular, the maps are constant on each chamber of the finite arrangement of cycle-cost hyperplanes. Replacing  $w_{xy}$  by  $w_{xy} + \alpha_x + \beta_y$  leaves the maps unchanged.*

*Proof.* A positive feasible tree is optimal precisely when all its off-tree reduced costs are positive. Each is an oriented fundamental-cycle sum. Its feasibility depends on the margins, not on the costs. Thus the same trees are selected whenever all cycle signs agree. For the gauge change, every fixed-margin objective changes by the same constant  $\sum_x \alpha_x a_x + \sum_y \beta_y b_y$ , or equivalently every alternating cycle sum is unchanged.  $\square$

For the integral costs (135.1), every nonzero cycle sum has absolute value at least one. A simple cycle has at most  $m + n + 1$  edges. Consequently any perturbation satisfying

$$\max_e |\widetilde{w}_e - w_e| < \frac{1}{m + n + 1} \quad (141.3)$$

preserves every cycle sign, hence all maps simultaneously. This is a mathematical robustness estimate, not a recommendation to replace exact arithmetic by unvalidated floating-point optimization.

## 142 Algorithms, bit complexity, and exact certificates

### 142.1 Explicit complexity bounds

Let  $q_J = |E_J| \leq q = |E| + m$ . With costs (135.1), either full flow problem is a network with  $O(m + r)$  vertices,  $O(q_J + m + r)$  arcs, and total integral flow

$$F = m(r + 1) \leq m(m + 1).$$

A successive shortest augmenting-path implementation makes at most  $F$  augmentations because each augmentation increases the sent flow by at least one. Reduced-cost potentials permit Dijkstra's algorithm at each iteration. This gives the conservative arithmetic-operation bound

$$O(m(r + 1)(q_J + m + r) \log(m + r + 2)). \quad (142.1)$$

The bit length of costs, distances, and potentials is  $O(q + \log(m + r + 2))$ : initial costs have  $q$  bits, and the algorithm uses shortest residual paths and tree-potential sums of polynomial length. Flow values have  $O(\log(m + r + 2))$  bits. Thus (142.1) is a polynomial bit-time bound as well, with the usual cost of operations on these integers included. It is not a unit-cost claim for exponentially large numbers. Reading the graph, constructing the globally ordered cost system, and writing the  $n$  output coordinates add input-processing work; a canonical sorted edge order takes  $O(q \log(q + 2) + n)$  elementary operations. These costs are separate from the flow bound.

The core version improves the forward bound. If  $q_I$  is the number of augmented edges incident to  $I$  and ending in  $\{0\} \cup J$ , its flow uses only  $2r + 1$  bipartite vertices, total flow  $r(r + 1)$ , and then a linear scan of the remaining incident edges. A corresponding bound is

$$O(r(r + 1)(q_I + r) \log(r + 2) + q_J + n) \quad (142.2)$$

on integers of the same polynomial bit length. This bound explains why it is useful to separate selected suppliers from unused suppliers. This is the mathematical flow-and-scan bound after the input representation has been prepared. The reference code favors a simple, checkable interface and may rebuild restricted graph data rather than cache it.

The shipped optimizer starts with zero flow and nonnegative integral edge costs. It repeatedly uses a shortest residual augmenting path, increases flow by its bottleneck, and updates vertex potentials. It then derives a new set of row/column potentials directly from the final support tree and checks the certificate below. The correctness theorem does not require trusting the optimizer's internal floating-point tolerances: there are none. A different exact or numerical solver may be substituted, but its output must pass the same exact certificate checks.

### 142.2 A small trusted checker

For a claimed optimum on  $G_J$ , supply the following data: the  $m + r$  positive tree edges with their integral flows,  $m$  row potentials, and  $r + 1$  column potentials. A checker verifies that the

edges form a spanning tree, that all flows are positive, that the required margins hold, that every tree edge has reduced cost zero, and that every other allowed edge has strictly positive reduced cost.

**Theorem 142.1** (Soundness of the certificate). *If these checks pass, the supplied flow is the unique minimum-cost flow for the stated margins. Checking the expected margins (A) or (B) and the appropriate degree/leaf decoder then certifies the corresponding map value. The certificate has  $O(m + r)$  integer entries apart from endpoint labels, and can be verified with  $O(q_I + m + r)$  elementary graph and arithmetic operations, up to the usual near-linear disjoint-set overhead.*

*Proof.* The objective-gap identity (136.2) is valid for every competing feasible flow. All off-tree terms are nonnegative and have strictly positive coefficients. An optimum must therefore be supported on the claimed tree. Zero-margin differences on a tree vanish by leaf elimination. The remaining assertions follow from counting the listed entries and scanning the edges and margins.  $\square$

For the core algorithm, only the  $2r$  core-tree edges and the core potentials need a transportation certificate. For each of the  $m - r$  remaining suppliers, the checker verifies the unique minimum in (140.2), then counts its contribution. This checks the smaller computation directly. It does not have to re-run the full flow.

### 142.3 Reproducible interface

The standard-library module `code/11-two-flow-transport_bijection.py` (delivered as `code/transport_bijection.py`) exposes the full forward map, the faster core forward map, the inverse map, both certificate checkers, and a basis-walk sampler. Suppliers and ordinary receivers use zero-based indices; the dummy is represented by  $-1$ . The mathematical  $0$  denoting the dummy should not be confused with ordinary receiver index  $0$  in the program.

```
from transport_bijection import BipartiteDemand

H = BipartiteDemand(4, 3,
    [(0, 0), (1, 0), (1, 1), (2, 1), (2, 2), (3, 2)])
I, J = (0, 1, 3), (0, 1, 2)

c, cert = H.basis_to_demand_core(I, J, certificate=True)
H.verify_core_certificate(c, cert)
assert c == (1, 1, 2)
assert H.demand_to_basis(c) == (I, J)
assert H.basis_to_demand(I, J) == c
```

[Merge note: the import uses the delivered module name `transport_bijection`; with the shipped file name it works only in a copy that restores the delivered layout, as the report's `README.md` describes.] The inverse rejects nonintegral, negative, overlarge, or infeasible inputs. Matching routines use iterative augmenting paths, so their graph size is not limited by Python's recursion depth. Enumeration routines occur only in the verifier, not in the production maps.

## 143 A complete worked certificate

For readability this section numbers the suppliers 1, 2, 3, 4 and the ordinary receivers 1, 2, 3. Take edges

11, 21, 22, 32, 33, 43



and choose  $I = \{1, 2, 4\}$ ,  $J = \{1, 2, 3\}$ . Add all four dummy edges and assign powers-of-two costs in the following order:

| edge | 10 | 11 | 20 | 21 | 22 | 30 | 32 | 33  | 40  | 43  |
|------|----|----|----|----|----|----|----|-----|-----|-----|
| cost | 1  | 2  | 4  | 8  | 16 | 32 | 64 | 128 | 256 | 512 |

The full forward margins are  $(5, 5, 1, 5)$  on suppliers and  $(4, 4, 4, 4)$  on receivers  $(0, 1, 2, 3)$ . The inverse margins for the decoded vector  $c = (1, 1, 2)$  are  $(4, 4, 4, 4)$  on suppliers and  $(3, 3, 3, 7)$  on receivers. Both positive supports are the tree in Figure 5.

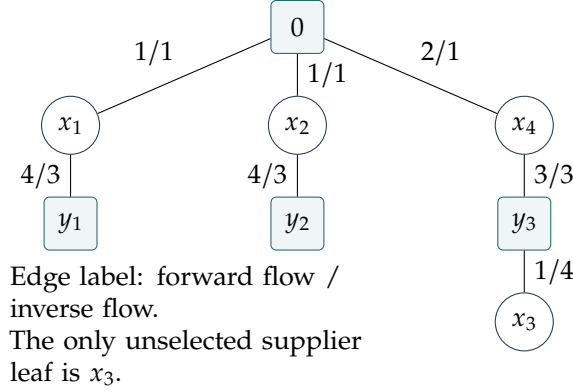


Figure 5: The common tree. Edges are drawn as undirected tree edges; transportation flow always goes from a supplier to a receiver.

The row and column potentials are

$$u = (1, 4, -128, 256), \quad v = (0, 1, 12, 256). \quad (143.1)$$

Every tree edge satisfies  $u_x + v_y = w_{xy}$ . The three off-tree edges 21, 30, 32 have positive slacks 3, 160, 180, respectively. The exact primal costs are 2253 for (A) and 2363 for (B); substituting the respective margins in  $u \cdot a + v \cdot b$  gives those same values. These equalities and strict slacks certify both optima.

Deleting supplier 3 leaves the core tree. Its receiver potentials are still  $v$ . Supplier 3 compares its three available reduced costs:

$$w_{30} - v_0 = 32, \quad w_{32} - v_2 = 52, \quad w_{33} - v_3 = -128.$$

It chooses receiver 3. The core contributes  $(1, 1, 1)$ , and that single leaf contributes  $(0, 0, 1)$ , producing  $(1, 1, 2)$ . The core transport sends only  $r(r + 1) = 12$  units, rather than the full transport's  $m(r + 1) = 16$ . The complete machine-readable certificates appear in `data/11-two-flow-example.json` (delivered as `data/example.json`), with the program's zero-based labels.

## 144 What compatibility can and cannot mean

### 144.1 Equivariance with specified costs

A graph isomorphism which also transports the cost assignment takes feasible flows to feasible flows with the same objective. Uniqueness then transports the selected tree, and the degree and leaf formulas commute with the isomorphism. The correspondence is therefore natural for *cost-labelled* graphs. The auxiliary costs cannot in general be removed while keeping a deterministic equivariant bijection.

**Proposition 144.1** (A symmetry obstruction). *There is no rule assigning to every unlabelled bipartite graph a support-preserving bijection  $\mathcal{M}(H) \rightarrow \mathcal{D}_H$  that is equivariant under all graph automorphisms.*

*Proof.* Take two suppliers  $x_1, x_2$ , one receiver  $y$ , and both ordinary edges. In the positive-support fibre the two supplier sets are  $\{x_1\}$  and  $\{x_2\}$ . The two demand vectors are  $c_y = 1$  and  $c_y = 2$ . The automorphism swapping  $x_1, x_2$  exchanges the supplier sets but fixes each demand vector. An equivariant map would have to give the exchanged sets the same image, contradicting injectivity.  $\square$

There is nevertheless an equivariant *randomized* correspondence: choose a uniformly random ordering of all augmented edges, use (135.1), and apply the deterministic map for that order. A graph isomorphism transports the uniform distribution on orders to the uniform distribution on orders. This construction restores symmetry in law, not as a single deterministic bijection.

## 144.2 Basis locality does not imply coordinate locality

**Proposition 144.2** (Unavoidable large jumps). *There is no constant  $C$ , independent of the graph, such that a bijection from lifted matroid bases to full demand vectors always sends every basis exchange to a coordinate change of  $\ell_1$ -size at most  $C$ .*

*Proof.* Take  $H = K_{m,1}$ . Its demand vectors are the integers  $0, 1, \dots, m$ . The lifted matroid defined in the next section is the uniform matroid of rank  $m$  on  $m + 1$  elements: omitting the receiver gives the empty support, and omitting any supplier gives a matchable singleton support. Its basis exchange graph is complete. Under any bijection, the bases mapped to 0 and  $m$  are adjacent, so an exchange has coordinate displacement  $m$ .  $\square$

Thus the transported rapidly mixing chain is an exact chain on lattice points, but it need not be a nearest-neighbour lattice walk. This prevents an unwarranted local-move interpretation of the sampling theorem. It does not contradict the one-unit supplier-insertion law (140.6), which changes the graph while keeping the selected supplier set fixed.

## 145 From matroid samples to full-coordinate samples

### 145.1 The lifted transversal matroid

Construct a transversal matroid  $M_H$  on the disjoint ground set  $X \sqcup Y$  using a separate set of  $m$  slots indexed by  $X$ . The ground element  $x \in X$  can occupy only its own slot; the ground element  $y \in Y$  can occupy any slot in  $N_H(y)$ . The supplier elements themselves occupy all slots, so the matroid has rank  $m$ .

A size- $m$  ground set  $B$  is a basis exactly when

$$B = (X \setminus I) \cup J \quad \text{for some } (I, J) \in \mathcal{M}(H). \quad (145.1)$$

Indeed, the supplier elements in  $B$  consume their own slots, leaving exactly  $I$  available for the receiver elements  $J$ . These receivers fill those slots exactly when  $I$  is matchable to  $J$ . This is a polynomial-time basis oracle using ordinary bipartite matching. [Merge note: this is Part V's private-copy matroid  $M_H$  and its basis correspondence, Proposition 69.1 (sources 06 and 07, the same forced-slot argument), with the private elements identified with  $X$  as in source 07; its basis polynomial is  $\mathcal{B}_H$  in (69.5). It is the dual of Part VII's matroid (Section 100).]

Taking the disjoint union of Theorem 135.2 gives an explicit bijection

$$\widehat{\Phi}_w : \{\text{bases of } M_H\} \longrightarrow \mathcal{D}_H, \quad (X \setminus I) \cup J \longmapsto \Phi_w(I, J). \quad (145.2)$$

The faster core decoder from Theorem 140.1 may be used on the right. No perfect matching is included in the state, so matching multiplicity does not bias this correspondence.

### 145.2 The exact target measure

Give each receiver a positive rational activity  $\theta_y$  and set the supplier activity to one. For a basis  $B$ , its weight is  $\prod_{y \in B \cap Y} \theta_y$ . The corresponding demand law is

$$\mu_\theta(c) = \frac{1}{Z_\theta} \prod_{y \in \text{supp}(c)} \theta_y, \quad Z_\theta = \sum_{c \in \mathcal{D}_H} \prod_{y \in \text{supp}(c)} \theta_y. \quad (145.3)$$

For  $\theta_y = 1$ , this is the uniform distribution on individual demand vectors. It is not the distribution obtained by assigning suppliers independently and then recording their loads; the latter would weight a vector by the number of assignments realizing it.

**Theorem 145.1** (Exact error transfer). *Let  $\pi_\theta$  be the weighted basis distribution on  $M_H$ , and let  $\nu$  be any probability distribution on its bases. Then*

$$(\widehat{\Phi}_w)_* \pi_\theta = \mu_\theta, \quad \text{TV}((\widehat{\Phi}_w)_* \nu, \mu_\theta) = \text{TV}(\nu, \pi_\theta). \quad (145.4)$$

*In particular, decoding loses no accuracy and requires no fibre-size correction or rejection step.*

*Proof.* The bijection preserves the support, hence the weight of each individual state. For the second assertion, the sum  $\frac{1}{2} \sum_B |\nu(B) - \pi_\theta(B)|$  is merely reindexed by the bijection. This proves equality, rather than only the general non-increase under a pushforward.  $\square$

The exact target law is independent of the cost system used for decoding. A random independent choice of a cycle-generic cost system can therefore also be used: conditioning on that choice gives the same exact target law, and averaging preserves a uniform error bound.

### 145.3 A fully specified almost-uniform sampler

Use the weighted down-up walk on bases. From  $B$ , remove a uniformly chosen element, obtaining  $A$ . Among elements  $e$  with  $A \cup \{e\}$  a basis, choose  $e$  with probability proportional to its activity and add it. The general matroid theorem gives spectral gap at least  $1/m$  and a polynomial mixing bound for this walk [19]. Its applicability to positive product activities follows from the weighted basis-polynomial form of that theorem.

For completeness, the quantities needed to turn the theorem into an explicit input-size bound are elementary. Put

$$N = m + n, \quad W = \max(1, \max_y \theta_y).$$

Start at  $B_0 = X$ , which is always a basis of weight one. Its inverse stationary mass obeys

$$\pi_\theta(B_0)^{-1} = Z_\theta \leq \binom{N}{m} W^m \leq 2^N W^m. \quad (145.5)$$

For  $0 < \eta < 1$  and  $m > 0$ , the conservative budget

$$t = \lceil m (N \log 2 + m \log W + 2 \log(1/\eta) + 2) \rceil \quad (145.6)$$

is sufficient for total-variation error at most  $\eta$ . For example, the usual reversible-chain estimate gives

$$\text{TV}(P^t(B_0, \cdot), \pi_\theta) \leq \frac{1}{2} \pi_\theta(B_0)^{-1/2} e^{-t/m},$$

using the nonnegative spectrum of the down-up operator. Substituting (145.5) proves the claim with room to spare. Equivalently one can use the direct mixing-time bound in [19, Theorem 1.1].

**Theorem 145.2** (Polynomial-time full-vector sampling). *Given an explicit bipartite graph, positive rational support activities, and  $0 < \eta < 1$ , one can sample a full vector whose distribution is within  $\eta$  in total variation of (145.3), in time polynomial in the graph size, the activity bit lengths, and  $\log(1/\eta)$ . In the unit-activity case this is an almost-uniform sampler for  $\mathcal{D}_H$ .*

*Proof.* Run the basis walk for (145.6) steps and decode once at the end. At a step, list at most  $N$  possible additions and test them by the matching oracle from (145.1). Positive rational choices can be made exactly using integer random choices after clearing denominators; the bit lengths remain polynomial. The mixing budget is polynomial in the claimed parameters. Apply the core decoder and Theorem 145.1. The cases  $m = 0$  or empty ground set return the unique zero vector. Exact bounded-integer draws give expected polynomial bit time. A worst-case polynomial-time approximate version runs the chain to error  $\eta/2$  and caps each random-integer rejection routine, allocating total failure probability at most  $\eta/2$  across all draws. Binary rejection has success probability at least  $1/2$  at each attempt, so logarithmically many attempts per draw suffice; on failure one may return a fixed valid vector. This preserves the claimed total error and polynomial bound.  $\square$

The theorem uses ideal independent random bits. The shipped reference sampler uses Python's pseudorandom generator, with an optional seed for reproducibility, and implements positive *integer* receiver activities directly. It uses an integer upper bound on (145.6), replacing natural logarithms by binary bit-length bounds, so a rounded logarithm cannot shorten its budget. The maps and certificate checkers themselves are fully exact and deterministic. An exact rational-activity implementation requires only replacing the weighted discrete choice, not a new mathematical argument. A deliberately shortened user-specified walk does not carry the stated mixing guarantee. [Merge note: the walk is the one of Part V's Theorem 75.2 (both import [19]); there a sample is a support pair, here it is decoded once, at the end, into the full vector.]

#### 145.4 Conditioning on support information

For a fixed exact support  $J$ , the domain is the set of bases of the rank- $r$  transversal matroid on suppliers  $X$  presented by slots  $J$ . A maximum matching either detects an empty fibre or supplies a starting basis. Sampling that matroid and applying  $\Phi_w$  gives an almost-uniform *full vector* conditioned on  $\text{supp}(c) = J$ . Its internal coordinate values, not just the support, are sampled.

More generally, prescribe disjoint sets  $J_+$  and  $J_-$  of receivers which must be positive and zero, respectively. In the lifted matroid, these are the conditions  $J_+ \subseteq B$  and  $B \cap J_- = \emptyset$ . If feasible, contracting  $J_+$  and deleting  $J_-$  gives a matroid whose bases, after adjoining  $J_+$ , are exactly the allowed bases. Feasibility requires that  $J_+$  be independent in the deletion and that the deletion retain rank  $m$ ; both are matching tests. The weighted basis sampler on this minor, followed by the same decoder, gives the corresponding conditioned full-vector law. General minor oracles are implemented by adjoining the contracted set before the original oracle query. [Merge note: the minor is Part V's Proposition 69.2; what is new here is the decoding of its bases into full vectors.]

The repository also provides a log-concave tilting and rejection method for a prescribed support size (Part V, “Sampling after fixing the support size” in Section 75, with Theorem 75.5). Composing its resulting basis distribution with (145.2) supplies full vectors with no further loss of total-variation accuracy. This last assertion is a composition with the report’s stated method; it is not a new proof of its coefficient-approximation theorem.

Activities depending on coordinate magnitudes, such as  $\prod_y q_y^{c_y}$ , are not covered by the support-product theorem. Nor is an exact independent uniform sampler claimed. These are distinct problems.

## 146 Preorder polytopes and integral dilations

Let  $\tau$  be a preorder on a finite set  $V$ : the relation is reflexive and transitive, but need not be antisymmetric. A lower ideal is a set  $A \subseteq V$  such that  $y \in A$  and  $x \leq_\tau y$  imply  $x \in A$ . The associated preorder polytope is

$$\mathcal{Q}_\tau = \{c \in \mathbb{R}_{\geq 0}^V : c(A) \leq |A| \text{ for every lower ideal } A\}. \quad (146.1)$$

This is the family studied in [1] and in this report (Part I’s (1.1); the lower ideals are Part I’s order ideals  $\mathcal{J}(\tau)$ ).

Use one supplier and one receiver for each element of  $V$ , and define

$$x \text{ supplies } y \iff x \leq_\tau y. \quad (146.2)$$

The orientation is explicit because the output coordinates are labelled: reversing the preorder is not silently the same operation on those coordinates.

**Proposition 146.1** (Exact demand realization). *For the bipartite graph  $H_\tau$  in (146.2),*

$$\mathcal{D}_{H_\tau} = \mathcal{Q}_\tau \cap \mathbb{Z}^V. \quad (146.3)$$

*Proof.* This is Part V’s Lemma 71.1, with the same orientation (an edge  $i \rightarrow j$  exactly when  $i \leq_\tau j$ , diagonal edges included) and the same lower-closure proof, which source 11 gives again: the neighbourhood of a receiver set  $S$  is  $\downarrow S$ , so  $c(S) \leq c(\downarrow S) \leq |\downarrow S| = |N_{H_\tau}(S)|$  for  $c \in \mathcal{Q}_\tau \cap \mathbb{Z}^V$ ; conversely  $N_{H_\tau}(A) = A$  for a lower ideal  $A$ , and the demand inequality for  $A$  is the ideal inequality in (146.1).  $\square$

**Corollary 146.2** (Full lattice-point sampling for every finite preorder). *Every finite preorder admits an explicit invertible support-preserving encoding of the integer points of  $\mathcal{Q}_\tau$  by bases of a transversal matroid. The encoding and decoding are polynomial-time and admit the certificates above. In particular, there is a polynomial-time almost-uniform sampler of the actual integer points of  $\mathcal{Q}_\tau$ , including their coordinate values, and corresponding samplers with positive support activities or prescribed zero/nonzero coordinates.*

*Proof.* Combine Proposition 146.1, Theorem 135.2, (145.1), and Theorem 145.2 and its support-conditioning construction.  $\square$

No bound on the height of the preorder is imposed, and equivalence classes can have any size. The result does not rely on any of the repository’s unformalized gamma-positivity or polytopal-realization arguments; its preorder specialization is the direct equality (146.3).

## 146.1 Dilations and the encoding-size boundary

For a positive integer  $q$ , replace each supplier of  $H_\tau$  by  $q$  distinguishable copies with the same neighbours. The neighbourhood of  $S$  then has size  $q|\downarrow S|$ , so its demand set is exactly  $qQ_\tau \cap \mathbb{Z}^V$ . The construction and sampler therefore apply to all positive integral dilations as well.

This is polynomial in the *expanded* input size  $q|V|$ , not a claim of polynomial complexity in  $\log q$ . When capacities are given in binary, explicit cloning may be exponentially expensive. Distinguishable clones also inherit distinct costs; their exchange symmetry is not used to identify bases. The symmetry obstruction in Proposition 144.1 explains why careless quotienting can lose an injective correspondence. [Merge note: the cloned graph and its pseudopolynomial boundary are Part V's capacity extension, Corollary 70.5 and the remark after it; Part V's Research question 39 asks for the binary-capacity algorithm, and Research question 68 below asks it again for the bijection and the sampler.]

## 147 Computational verification and the formal proof boundary

### 147.1 Independent finite checks

The package includes a deterministic exact verifier using independent matching-permutation tests and demand-inequality enumeration. On every exhaustive graph instance it enumerates all matchable support pairs, checks both inverse identities, checks equality of the two optimal tree supports, checks the full and core forward implementations against each other, verifies the exact certificates, and compares the entire image with an independently enumerated demand set.

| Suite                                          | Instances | Coverage                                                                                     |
|------------------------------------------------|-----------|----------------------------------------------------------------------------------------------|
| All bipartite graphs with $0 \leq m, n \leq 3$ | 689       | 5,880 support pairs; all demand vectors.                                                     |
| All bipartite graphs with $(m, n) = (4, 3)$    | 4,096     | 68,832 support pairs; all demand vectors.                                                    |
| All labeled preorders with $0 \leq  V  \leq 4$ | 390       | 12,709 lattice points; ideal inequalities checked against demand inequalities.               |
| Seeded random graph and cost-order trials      | 250       | 4,256 feasible-pair trials, 9,427 restriction/deletion checks, 8,512 chamber/gauge checks.   |
| Exact rational Markov chains                   | 2         | 17 states each; row sums, detailed balance, target point weights, and finite-time distances. |

Table 18: Executed verification. Counts refer to the reported suites; random trials can revisit a pair, and the preorder suite can overlap the small graph suite.

The 4,785 exhaustive bipartite graphs contain 74,712 matchable support pairs in total. The preorder counts by order are 1, 1, 4, 29, 355, including the empty preorder. The random seed is 20260930. The tests also replace powers of two by powers of three, which preserve all alternating-cycle signs, and add nonnegative row and column gauges. Three deliberately corrupted primal/dual certificates are rejected.

For the two exact Markov-chain tests, the target partition functions are 17 in the uniform case and 144 with receiver activities (2, 3, 5). Starting from the supplier basis, the exact total-variation distances after eight steps are approximately 0.00875421 and 0.00833044, respectively. These numbers are exact rational computations rounded for display, not statistical estimates or a proof of the general mixing bound. Their full fractions appear in `data/11-two-flow-chain.json` (delivered as `data/chain.json`).



The verifier may be run, in the delivered layout, with

```
python code/verify.py --suite all
```

It writes fresh JSON reports under `data/`; run it on a copy to preserve the delivered records. [Merge note: the shipped files are `code/11-two-flow-verify.py`, which imports `transport_bijection` by its delivered name, and `data/11-two-flow-*.json`; the report's `README.md` gives a recipe that restores the delivered names in a copy.] No external Python package is required. Elapsed times are environment-dependent and are not mathematical claims.

## 147.2 What has and has not been certified

There are three distinct levels of assurance. First, the article proves the general theorems for arbitrary finite graphs. Second, the program emits exact arithmetic certificates whose mathematical soundness is Theorem 142.1. Third, the exhaustive tests check many finite inputs with independent combinatorial predicates. None of these is a Lean or Rocq kernel check. The implementation and tests could still contain errors not found in the tested cases; the mathematical proofs remain the general justification.

The supplied sampler implements the unconditioned law with integer support activities. The conditional samplers and rational-activity extension are specified and proved above, but are not separate packaged API wrappers. The package does not include an exact uniform sampler, a theorem-prover build, or a new implementation of the general matroid counting FPRAS.

## 147.3 A concrete formalization plan

A useful first milestone for ProvelIt is a small certificate checker rather than a monolithic formalization of an optimization library. Represent a finite graph, flow, and potential by finite arrays of integers. Prove the margin-summation identity (136.2), then prove that a zero-margin flow supported on a finite tree vanishes. These two lemmas suffice for unique-optimality certificates.

Next formalize the two component-balance congruences and the inverse subtree estimate (138.2). These establish connectedness and the degree-two condition. The leaf formula and the two explicit positive flows in (139.2) then prove inverse correctness. The core factorization needs only the unique reduced-cost minimizer and the same certificate identity. The preorder specialization is the elementary lower-closure argument in Proposition 146.1.

An external optimizer can remain outside the trusted boundary for *concrete* checked examples. A universal existence theorem for all inputs requires more: either a verified terminating min-cost-flow algorithm, or a formal proof that a feasible transportation polytope has an optimal vertex followed by the forest argument. Checking certificates alone does not prove that a solver always produces one. The general matroid mixing theorem is a separate, substantial dependency; it should be imported explicitly or formalized separately, not hidden behind a certificate for a single sample.

## 148 Further research questions and an actionable programme

The following questions are proposed continuations of this work. They are not assertions that each formulation is a recognized unsolved problem in the published literature.

**Research question 66** (A direct inverse price algorithm). Can the unique inverse price vector described in Proposition 150.1 below be computed by a substantially faster combinatorial procedure than a full transportation solve? The forward map already isolates a small core. An inverse method that discovers its  $r$  indifferent suppliers without processing  $m(r + 1)$  units of flow would give a symmetric improvement. A useful first target is sparse incidence graphs of bounded treewidth.

**Research question 67** (Cost compression). For which graph classes can a cycle-generic cost system be chosen with polynomially bounded *magnitudes*, rather than merely polynomial bit length? Independently, can the required comparisons be implemented lexicographically without manipulating the powers of two explicitly? The distinction is important: the current algorithm is already polynomial in bit complexity, but its constants and integer arithmetic can still be improved.

**Research question 68** (Binary-encoded capacities). Can cloning be replaced by a compressed correspondence and sampler whose running time is polynomial in capacity bit lengths? One must preserve unit weight per demand vector rather than introduce the binomial multiplicities of identical supplier copies. A precise first case is a fixed small number of receiver types with large supplier capacities.

[Merge note: this extends Part V’s Research question 39 (“Binary capacities and approximation complexity”), which asks for the same compression for partition functions and support coefficients, from counting to the bijection and the sampler.]

**Research question 69** (Updating the selected core). Supplier insertions outside  $I$  are independent leaf attachments. What is the cheapest update when a basis exchange changes  $I$  itself? A dynamic algorithm should reuse old potentials and identify the changed tight edges. Proposition 144.2 rules out uniformly bounded coordinate movement, but not an efficient update of a possibly large movement.

**Research question 70** (Magnitude-sensitive probability measures). Which coordinate-dependent weights on  $\mathcal{D}_H$  become tractable weights on the lifted bases? In particular, determine useful conditions under which  $\prod_y q_y^{c_y}$  or conditioning on a prescribed coordinate sum admits a rapid sampler through this correspondence. Support-product weights are already handled; arbitrary magnitude weights need a new argument.

**Research question 71** (Stronger functoriality). Characterize the receiver deletions or selected-supplier operations for which a meaningful commuting diagram survives, possibly after changing the support fibre. The existing results cover receiver operations outside  $J$  and supplier operations outside  $I$ . Any stronger naturality theorem must respect the explicit symmetry obstruction.

[Merge note: this is what remains of the compatibility clauses of Research questions 2 (induced subgraphs, vertex deletion) and 20 (induced subgraphs, a controlled vertex ordering), and of source 09’s “Direct lattice-point bijections” (deletion, the paired clone action; Part VII’s questions); see Section 132.3.]

**Research question 72** (Geometric cells and chamber crossings). Describe how the decoded demand vector changes when the cost system crosses a single cycle-cost hyperplane. The chamber theorem proves constancy away from the walls but does not classify the corresponding bijection permutation across a wall. Such a description could connect local tree changes, mixed-subdivision flips, and coupling constructions.

**Research question 73** (Certified implementation and formal extraction). Implement the certificate checker in Lean or Rocq and prove its soundness, then formalize the two-flow inverse

theorem. A practical intermediate deliverable is a verified core certificate with an external optimizer: it would certify full coordinate values with a small trusted checker before a full min-cost-flow algorithm or a matroid mixing theorem is formalized.

[Merge note: this overlaps Part V’s Research question 43 (“Formalization in layers”), whose first layer is the private-copy basis correspondence and the preorder reduction; with Corollary 139.2 the finite support-fibre count of that question’s second layer also has an elementary proof to formalize.]

For an immediate continuation, the strongest sequence is: certify the small checker; benchmark and optimize the selected-core routine; then investigate dynamic core updates and binary capacities. This sequence produces useful artifacts at each stage and separates easy verification work from the genuinely new algorithmic questions.

## 149 Conclusion

The repository’s missing step is not another equality of enumerators. It is a way to turn one sampled combinatorial state into a specified integer vector, with the right multiplicity and an efficiently checkable inverse. The two-flow construction supplies that step.

A single generic cost system chooses an admissible tree. Two deliberately chosen margin systems recover that same tree from opposite sides of the correspondence. Supplier degrees encode the selected basis; supplier leaves encode the full coordinates. The smaller core formulation goes further: after one flow on the selected suppliers, every other supplier contributes by a separate reduced-cost minimum. The full-vector sampler then follows by a literal change of coordinates on the known matroid basis walk, preserving total-variation error exactly.

These are constructive refinements of an existing geometric correspondence, not a replacement of the prior literature. They provide a complete answer to the identified full-coordinate sampling gap at the pinned repository snapshot, with precise restriction laws, counterexamples to stronger interpretations, explicit code, and a feasible route to formal verification.

## 150 The unique dual price vector

(Source 11’s Appendix A.) The core construction has a concise dual interpretation which also suggests an inverse algorithm. For  $c \in \mathcal{D}_H(J)$ ,  $r > 0$ , and  $\varepsilon = 1/(r + 1)$ , set  $v_0 = 0$  and consider the concave piecewise-linear function

$$f_c(v) = \sum_{x \in X} \min_{y \in (N_H(x) \cap J) \cup \{0\}} (w_{xy} - v_y) + \sum_{y \in J} (c_y - \varepsilon) v_y. \quad (150.1)$$

**Proposition 150.1** (Unique inverse prices). *Under cycle-generic costs, (150.1) has a unique maximizer. It is the receiver-potential vector of the optimal tree for (B), normalized by  $v_0 = 0$ . At this price vector, precisely the suppliers in  $\Psi_w(c)$  have two minimizing receivers; every other supplier has a unique minimizing receiver. No supplier has three minimizing receivers.*

*Proof.* For any  $v$ , put  $u_x = \min_y (w_{xy} - v_y)$ . Then  $u_x + v_y \leq w_{xy}$  on every edge. Applying the objective-gap identity to the flow (B) divided by  $L = r + 1$  shows that  $f_c(v)$  is bounded above by that flow’s cost. The optimal tree potentials attain this bound, because every positive-flow edge is tight and every row has positive flow. Thus the maximum exists.

If another  $v$  attained the same bound, every positive-flow edge of the optimal tree would also have to be tight, since the objective gap is a sum of nonnegative edge slacks weighted by positive flow. Together with  $v_0 = 0$ , the tree equalities determine every potential uniquely. Finally, the strict off-tree slacks show that the minimizing edges at a supplier are exactly its incident tree edges. Their numbers are one or two, and the degree-two suppliers are precisely  $\Psi_w(c)$ .  $\square$

For the graph  $K_{m,1}$  this interpretation becomes an ordinary ranking. Write  $d_x = w_{xy} - w_{x0}$ . Cycle-genericity makes these numbers distinct. For the singleton selected set  $I = \{i\}$ , the core price is  $v_y = d_i$ , and an unselected supplier chooses  $y$  precisely when  $d_x < d_i$ . Therefore

$$\Phi_w(\{i\})_y = 1 + \#\{x : d_x < d_i\}.$$

The positive demand values  $1, \dots, m$  are exactly the ranks of the supplier differences. The empty support gives zero. This simplest case already exhibits the role of costs, the impossibility of a completely symmetry-natural deterministic rule, and the difference between sampling supplier sets and sampling assignments.

## 151 Algorithm summary and delivered files

(Source 11's Appendix B.)

**Forward, full form.** Validate the matchable pair  $(I, J)$ . Return zero if  $J$  is empty. Otherwise solve (A), verify its strict tree certificate, and output one on every receiver in  $J$  plus its number of unselected supplier leaves.

**Forward, core form.** Solve (140.1) on the selected suppliers only. Verify its certificate and extract  $v$ . For each unselected supplier, find the unique minimum (140.2); add one at its chosen receiver unless it chooses the dummy.

**Inverse.** Set  $J = \text{supp}(c)$  and  $c_0 = m - \sum c_y$ . Return the empty pair for zero. Otherwise solve (B), verify its strict tree certificate, and select exactly the suppliers of degree two.

**Sampling.** Walk on lifted matroid bases for the explicit mixing budget, then run the core decoder once. No transportation optimization is needed during an ordinary basis-walk step.

| File (shipped name)                     | Purpose                                                                                           |
|-----------------------------------------|---------------------------------------------------------------------------------------------------|
| code/11-two-flow-transport_bijection.py | Exact flows, full/core maps, inverse, checkers, matching oracle, reference sampler.               |
| code/11-two-flow-verify.py              | Independent finite enumeration, compatibility checks, exact chain tests, and certificate example. |
| data/11-two-flow-*.json                 | Recorded results and explicit certificates.                                                       |
| data/11-two-flow-verification_run.txt   | Console output of the recorded run.                                                               |
| 11-two-flow-PROOF_STATUS.md             | Claim-by-claim assurance and implementation boundaries.                                           |
| 11-two-flow-SOURCE_AUDIT.md             | Repository provenance and literature/novelty audit.                                               |
| code/11-two-flow-build.sh               | The delivered build script (three pdf <sub>l</sub> atex passes on the delivered article.tex).     |

[Merge note: source 11's table lists its own `article.tex`, `article.pdf` and `README.md`, which are not shipped (they survive in the arrival commit; this report's `article.tex`, `article.pdf`

and `README.md` replace them), and describes `build.sh` as rebuilding “with two LaTeX passes”; the script runs three. The table above gives the shipped names; the report’s `README.md` maps them to the delivered ones.]

## Part X. Random trees: rare stable markings, sparse transitions and annealed–quenched separation

*Part X was added on 30 September 2026 in batch 66 of ProveIt’s incoming-report intake, from one manuscript. It answers Part VIII’s Research question 63 for the uniform labelled (Cayley) tree on  $n$  vertices with every vertex marked independently with probability  $p$ . From Part VIII’s classification it derives an exact joint count of stable markings by number of marks and hull size, an exact two-colour generating function and a Poisson identity with eight finite corrections; the annealed stability probability is asymptotic to  $\frac{9}{8}\sqrt{2\pi} p^3 n^{5/2} e^{-pn}$  whenever  $p\sqrt{n} \rightarrow \infty$ , an explicit entire function governs the window  $p\sqrt{n} \rightarrow \lambda$ , and below it stability is asymptotically the event of at most two marks. At fixed  $p$  it proves a joint Gaussian law for marks and hull size given stability, and a linear conditioned diameter. For every fixed host it determines the stability probability up to a polynomial factor by the diameter, which separates the typical-host rate  $\log(1-p)$  from the annealed rate  $-p$ . Parts I–IX are unchanged except for dated pointers. Section 152 records the provenance, fixes the notation and states what is answered.*

### 152 Provenance, conventions and what Part X answers

#### 152.1 The manuscript

Part X is printed from batch-66 manuscript 06, called *source 12* here after its file prefix: *Rare Stable Markings on Random Trees: Exact enumeration, sparse transitions, conditioned geometry, and annealed–quenched separation*, dated 30 September 2026, 23 A4 pages as delivered. Its title page reads “Prepared for Vladimir Reshetnikov / Mathematical development and computation: ChatGPT” under the kicker “ProveIt research continuation”, and its PDF metadata give the author as “ChatGPT; prepared for Vladimir Reshetnikov”; that byline is recorded here and not reprinted. Its archive `ProveIt_Random_Tree_Stability.zip` (inner directory of the same name, main file `article.tex`) arrived in ProveIt commit 9ad899cbe and was placed in 4fee1cd07. It is pinned to commit b8b0fa2184a044d46ce9ed0f25f88d7bb60fa042, with this report’s `article.tex` as blob ba1cc4a6c99abd5ff808aae97df75efe434833e9: Parts I–IX exactly as printed, unchanged from the pin to the placement (the last change before it was 8a8c07210, the writing of Part IX). Its two programs and its build script are shipped in `code/`; its recorded outputs, two CSV tables, console logs, environment and requirements records and its own PDF-validation record in `data/`; and its proof-status and source-audit notes at the report root, all with the prefix 12-random-trees-. Its checksum list `SHA256SUMS.txt` was verified (19/19) at placement and retired. The manuscript, delivery `README` and PDF are not shipped; they survive in the arrival commit.

Source 12 continues Part VIII only. It imports Part VIII’s classification (Theorem 118.2 with Table 14) as a stated dependency, credits Part VIII’s path-only real-rootedness theorem, endpoint enumerator, repair algorithm and spectral test, and does not re-prove any of them. It did not use Part IX, which is unrelated.



**Abstract of source 12 (as delivered).** A research question in ProveIt’s marked-tree stability report asks for the probability that an independently marked random tree has a real-stable incidence–apex matching–support polynomial. We answer it for uniform labelled trees. Using the report’s credited ADE-skeleton classification, we derive an exact exponential generating function and an exact joint count by the number of marks and the marked-hull size. For Bernoulli marking with probability  $p$ , the annealed stability probability satisfies

$$\pi_n(p) \sim \frac{9}{8} \sqrt{2\pi} p^3 n^{5/2} e^{-pn}$$

whenever  $p\sqrt{n} \rightarrow \infty$ , including every fixed  $p > 0$ . An exact finite-correction Poisson identity makes this statement uniform up to  $p = 1$ . In the critical window  $p\sqrt{n} \rightarrow \lambda$ , an explicit entire function describes the crossover, the limiting number of marks, and a mixture of chi-distributed hull sizes. Below that window, stability is asymptotically equivalent to having at most two marks. For fixed  $0 < p < 1$ , conditioning on stability produces  $np^2 + O_{\mathbb{P}}(\sqrt{n})$  marks and a hull of size  $np + O_{\mathbb{P}}(\sqrt{n})$ , with an explicit joint Gaussian law and constant-order moment corrections. The host diameter differs from the hull size by  $O_{\mathbb{P}}(\log n)$ . In contrast, for every fixed host tree the stability probability is determined, up to a polynomial factor, by its diameter. This yields a strict separation between the typical-host exponential rate  $\log(1 - p)$  and the annealed rate  $-p$ . Exact finite checks and a dependency ledger accompany the proofs.

The abstract is reproduced as delivered. Its “marked-tree stability report” is Part VIII of this report, and “the report’s credited ADE-skeleton classification” is Theorem 118.2.

**Status and scope (source 12, as delivered).** The ADE characterization is an imported repository theorem, not a new result of this article. The enumerative and probabilistic results below are proved here for the precisely specified model. The proofs and programs are not Lean- or Rocq-verified, and have not been independently refereed. Global publication priority is not asserted. No repository file was modified.

**Priority and prior work (merge).** The stability classification is Part VIII’s Theorem 118.2 (itself a corollary of Part VI), imported and not re-proved: every probability in Part X is a probability of Part VIII’s branch-complete ADE markings, and Part X’s finite checks test that combinatorial class, not polynomial stability. The ADE list is classical and is Part VI’s Theorem 88.1; its spectral context is credited to McKee–Smyth [30], a different paper from the two McKee–Smyth papers already cited ([25] and [27]). The half-plane-property framework is Choe–Oxley–Sokal–Wagner’s [24]. The labelled-tree and rooted-forest counts, the matrix-tree expansion, Prüfer codes, Lagrange inversion, Cauchy coefficient bounds, Stirling’s formula, Poisson central limits, Chernoff bounds and the characteristic-function method are classical (Flajolet–Sedgewick [31]); the long-path calculation behind the diameter rate (165.4) is included with its proof and is not claimed as new. The model of Section 154 restates Part VIII’s, and Lemma 165.1 sharpens the inspection behind Part VIII’s Theorem 123.3 (merge note after it). New relative to Parts I–IX: the skeleton series and its eventual constant  $\frac{9}{8}$  (Proposition 155.1, with the automorphism column of Table 20), the joint count, the generating functions and the Poisson identity (Sections 156–159), the fixed- $k$ , supercritical, sparse and critical limits (Sections 158, 160–162), the conditional Gaussian laws and family weights (Section 163), the conditioned diameter (Section 164), the quenched sandwich and the annealed–quenched gap (Section 165), and the universality theorem (Section 166). Source 12’s audit was focused and found no prior publication of these formulas, which is not evidence of first priority; no worldwide priority is claimed for any result of Part X.

## 152.2 Notation of Part X

Part X uses Part VIII’s incidence–apex graph  $\mathcal{A}(T; S)$ , its polynomial  $F_{T,S}$  of (118.1), its hull  $\text{Hull}_T(S)$ , its branch-complete markings and its terminal skeletons, and the report’s macros for



Table 19: Symbols of Part X. The last column prints the tempting false reading.

| Symbol                                                                                    | Meaning in Part X                                                                                                                                                                      | Not to be confused with                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $T_n, n, [n]$                                                                             | the uniform labelled (Cayley) tree on $[n]$ ; a fixed host is $T$                                                                                                                      | Part VIII's fixed host $T$ ; Part I's ground set $E = [n]$ .                                                                                                                             |
| $p, q = 1 - p, r = p/q$                                                                   | common marking probability, its complement, the odds                                                                                                                                   | Part VIII's vertex probabilities $p_v$ ; Part VI's polynomial $p_G$ and tree size $r$ ; Part V's $r$ ; Part IX's $q$ and $r =  J $ .                                                     |
| $S, E_n = E_n$                                                                            | the random marking; the event that it is stable                                                                                                                                        | Part III's $E = E$ (the same glyph, a different object).                                                                                                                                 |
| $K_n, H_n$                                                                                | the <i>numbers</i> of marks and of hull vertices ( $H_n = 0$ without marks)                                                                                                            | Part VIII's hull $K = \text{Hull}_T(S)$ and terminal skeleton $H(T, S)$ , which are trees.                                                                                               |
| $\pi_n(p), \pi_p(T)$                                                                      | annealed and quenched stability probabilities                                                                                                                                          | Part IX's basis law $\pi_\theta$ .                                                                                                                                                       |
| $a_k, c_k, c_*, \delta_k$                                                                 | allowed labelled skeletons on $k$ labels; $a_k/k!$ ; the eventual value $\frac{9}{8}$ ; $c_k - c_*$                                                                                    | Part VIII's scores $c_v$ ; the diameter fraction $\delta$ of (165.4); Part VI's $\Delta$ .                                                                                               |
| $A(y), \mathcal{H}, c_k^{\mathcal{H}}$                                                    | the skeleton series; a class of skeletons and its coefficients                                                                                                                         | the graph $\mathcal{A}(T; S)$ ; adjacency matrices $A_H$ ; Part IV's $\mathcal{M}_\tau$ .                                                                                                |
| $C_{n,k,\ell}, C(u, v)$<br>$f_{n,\ell}, R_{n,k}, b_{n,k}, W_{n,k}$                        | joint counts; the moving-pole amplitude<br>rooted-forest count; attachment factor;<br>stability probability of a fixed $k$ -set; the coefficient-one contribution                      | Part VIII's $C_H$ and $C$ ; Part I's $C_\tau$ .<br>Part VIII's arms $R_i$ .                                                                                                              |
| $U(x), s, t, B_n$                                                                         | rooted-tree series; $s = U(qt)$ ; the series variable; $n!/n^{n-2}$                                                                                                                    | the apex variable $s$ of $F_{T,S}(x; y, s)$ ; the variable $t$ of $h_\tau(t)$ ; Part I's $B_\tau$ and Part V's $\mathcal{B}_H$ .                                                         |
| $\mathcal{F}, \mathcal{F}_p$<br>$\alpha_k, \chi_k$<br>$\lambda, \Psi, \Phi, w_k(\lambda)$ | series of stable marked trees<br>chi constants (158.1) and chi laws<br>window parameter; the crossover function;<br>the standard normal distribution function;<br>the critical weights | the polynomial $F_{T,S}$ .<br>Part VIII's largest stable size $\alpha_{\text{st}}(T)$ .<br>Part IX's maps $\Phi_w, \Psi_w$ ; Part VI's polynomial $\Phi_G$ ; Part VIII's weights $w_v$ . |
| $\tau(u, v), d(u), s_0, \Lambda, \Sigma_p, \theta$                                        | moving pole and its data; log-growth;<br>covariance; a ratio below one                                                                                                                 | a preorder $\tau$ ; the multi-theta graphs $\vartheta(\dots)$ ; Part IX's activities $\theta$ .                                                                                          |
| $I(p), J(\delta), J_0, M_n, X$                                                            | rate gap; diameter rate; a constant; the main term; a Poisson variable                                                                                                                 | Part I's order ideals $I$ and $\mathcal{J}(\tau)$ ; Part IX's receiver support $J$ and suppliers $X$ .                                                                                   |
| $d, D, N_d$                                                                               | a diameter (in edges); the random diameter in the variational formula; the number of length- $d$ paths                                                                                 | Part IX's neighbourhoods $N_H(\cdot)$ ; the $D$ and $\tilde{D}$ skeletons.                                                                                                               |

$\mathbb{P}, \mathbb{E}, \text{Var}, \text{Cov}, \mathbf{1}, \text{Aut}$  and  $\text{diam}$ . Table 19 lists what Part X adds or uses differently. No symbol of source 12 was renamed and no normalization was changed: its  $\mathcal{A}(T; S)$  is printed with the report's `\caLa`, which is the same symbol. The letters  $K, H, \alpha, C, A, s, \tau, \Phi, \Psi$  and the sans-serif  $E$  mean something else in Part X than in earlier Parts; the table prints the false readings.

### 152.3 What Part X answers

Status after Part X, as of 30 September 2026:

- **Part VIII's Research question 63 is answered for random labelled trees** under independent marking with one common probability  $p$ , in every regime of  $p$ : the probability is exact (Theorems 156.2, 157.1 and 159.1) and asymptotically determined below, in and above the window  $p \asymp n^{-1/2}$  (Theorems 161.1, 162.2 and 160.1, Corollary 161.2), and Section 171 summarizes the regimes. The question's second half, *deterministic families with growing branching complexity*, is answered only on the exponential scale: Theorem 165.2 determines the stability probability of every host within a polynomial factor from its diameter, without an asymptotic equivalent (Research question 74). Other random-tree laws and vertex-dependent or correlated markings remain open (Research questions 79 and 80). Part VIII's remark that a meaningful analysis must specify the host model and the marking law is

borne out: the constant  $\frac{9}{8}$  and the rate  $-p$  belong to this model (Research question 79).

- **No other numbered research question of Parts I–IX is answered.** Part VIII’s Research question 59 is posed again, with a random-tree layer, as Research question 81, and Part VIII’s Research question 65 is overlapped by Research question 83.

Part X refutes no statement of Parts I–IX, and nothing is retracted. Part VIII gains a dated pointer after Research question 63; the title block, the abstract, the scope box and the reading route gain dated sentences.

## 152.4 Where the merge had to choose

- *Order.* Source 12’s order of sections is kept: its Section  $k$  is Section  $k + 152$  here. Its two appendices are printed as ordinary sections at the end of Part X (Sections 170 and 171), so that Part I’s appendices keep their letters.
- *Duplicates of Part VIII, kept with merge notes.* Source 12’s model (Section 154) restates Part VIII’s definitions, and its imported Theorem 154.1 is Part VIII’s Theorem 118.2; both are short and cited throughout, so they are printed with pointers rather than replaced. Its Table 20 is Part VIII’s Table 14 in another layout with a new automorphism column. Lemma 165.1 is kept as a second, stronger route to the inspection behind Part VIII’s Theorem 123.3.
- *Self-citation.* Source 12’s reference “ProveIt” (this report at the pin, Part VIII) became references to Part VIII’s statements, and the reference was dropped from the bibliography. Its statement of the pin, blob and inspected line ranges is kept in Section 153.1, with a merge note mapping the ranges to Part VIII’s sections.
- *Research questions.* Source 12’s ten unlabelled questions continue the numbering as Research questions 74–83 and received labels; the two that restate or overlap Part VIII’s questions are numbered, with merge notes, because their formulations are broader.
- *File names.* Source 12’s programs and records keep their delivered base names behind the prefix 12-random-trees- (verify.py, extra\_checks.py and build.sh in code/; its ten data/ files and requirements.txt in data/; PROOF\_STATUS.md and SOURCE\_AUDIT.md at the report root). Merge notes in Section 167 give the shipped names; its reproduction commands are printed with delivery names.
- *Cross-references.* Source 12 does not use cleveref; its references are plain \ref and \eqref, as in the rest of the report, and nothing was converted.
- *Macros, labels and layout.* Its preamble is dropped. Its colour definitions Ink, Accent and Pale use other hex values than the report’s and would have recoloured the whole report; its redefinitions of statusbox, of the section formats and running heads, and of the question environment are dropped in favour of the report’s own. Its \R, \C, \PP, \EE, \Var, \Cov, \ind, \Aut, \Hull and \diam agree with the report’s; its  $\mathbb{N}$  macro (unused in its text, and the report’s  $\mathbb{Z}_{\geq 0}$  is  $\mathbb{Z}_{\geq 0}$ ) and an unused big-O macro are dropped. Its \Pois, \st, \fall, \dto, \pto and \cst were added to the preamble. Its 111 labels carry the prefix rts: (one of them, app:audit, equals a bare Part I label); this section’s headings, its notation table, the subsection of Section 153 that states the pin and the ten questions gained labels. Its abstract and status box are printed in this section; its title page, table of contents and running head are dropped.
- *Bibliography.* [24] was already cited (the report’s entry gives the arXiv record, source 12’s the DOI and the LSU record of the same paper). Source 12’s McKeeSmyth is J. Algebra 317 (2007),

a different paper from the report's [25] (arXiv:2002.06082) and from [27]; it is printed as [30], so that its one citation does not silently point to the wrong paper. Flajolet–Sedgewick [31] and Aldous [32] are new.

## 153 The question answered and the contribution boundary

### 153.1 An explicit repository question

Part VIII of this report asks for asymptotic stable-marking probabilities for random labelled trees, while emphasizing that the tree model and marking law must be specified (Research question 63). The source label is `mts:q:random`. This is a repository research question, not a claim that a named conjecture in the published literature is being settled. We choose the uniform Cayley tree and independent Bernoulli marking, and determine the probability throughout its principal scaling regimes. We also obtain a deterministic estimate that applies to every host tree.

The inspected source was pinned to commit

`b8b0fa2184a044d46ce9ed0f25f88d7bb60fa042`.

The relevant file is

`SetTheory/Cardinals/docs/reports/enumerative-combinatorics/  
preorder-root-polytopes/article.tex`,

with Git blob `ba1cc4a6c99abd5ff808aae97df75efe434833e9`. The source ranges inspected include lines 14560–14710 (definitions and classification), 15210–15385 (ADE list and geometric consequences), and 15900–16200 (the question and the contribution boundary). These refer to the pinned file, not to an indefinitely changing branch. [Merge note: the pinned blob is this report exactly as it stood after Part IX was written, so “the source” in this section is Part VIII. Lines 14560–14710 of that blob are the end of Section 117 and Section 118 into the first proof of Theorem 118.2; lines 15210–15385 are Table 14, the rest of Section 122, Section 123 with Theorem 123.3, and the opening of Section 124; lines 15900–16200 are Research questions 61–65, among them Research question 63 itself, Part VIII's conclusion, Sections 130 and 131, and the opening of Part IX (the last range runs past Part VIII; nothing in Part IX bears on Part X). Part X's additions to the front matter moved these lines by 30 in the present file, and those after Research question 63 by 42, because of the dated pointer added there.]

The source has already classified all markings on all finite trees (Theorem 118.2) and already proved that the stable-neighbourhood counting polynomial is real-rooted exactly when the host is a path (Theorem 126.3). Neither statement is posed as an open question here. Nor do we reclaim the source's deterministic endpoint enumerator (Theorems 124.1 and 124.2), optimal-repair algorithm (Theorem 125.1, Corollary 125.2), or ADE spectral test.

### 153.2 What is proved here

Let  $T_n$  be uniform among the  $n^{n-2}$  trees on  $[n]$ , and mark each vertex independently with probability  $p$ . Write  $E_n$  for stability,  $K_n$  for the number of marks, and  $H_n$  for the number of vertices of their hull, with  $H_n = 0$  for no marks. The main conclusions are:

1. An exact labelled decomposition and a finite joint counting formula for  $(K_n, H_n)$  on  $E_n$ ; an exact Poisson-tail formula with only eight exceptional coefficient corrections.

2. A three-regime probability theory: at most two marks dominate below  $p \asymp n^{-1/2}$ ; an explicit crossover function governs that window; and  $\frac{9}{8}\sqrt{2\pi}p^3n^{5/2}e^{-pn}$  is an equivalent throughout  $p\sqrt{n} \rightarrow \infty$ .
3. At fixed marking density, explicit conditional Gaussian laws and constant-order moment corrections, limiting skeleton-family weights, and a linear-diameter conditioned geometry.
4. A diameter-based, polynomial-factor estimate for every host, and therefore an annealed–quenched exponential-rate separation.

The factor  $9/8$  has a transparent source: the three infinite families of allowed labelled skeletons contribute  $1/2$ ,  $1/2$ , and  $1/8$  to the exponential generating-function coefficients. The probabilistic conclusions are not numerical extrapolations; they follow from exact identities.

### 153.3 Prior methods and limits of the audit

The general relation between stable multiaffine polynomials and matroids belongs to the established half-plane-property literature [24]. The spectral setting of the ADE list is classical [30]. Labelled substitution, Lagrange inversion, and meromorphic coefficient asymptotics are standard methods of analytic combinatorics [31]. We use these methods with their usual credit and supply the needed calculations. The continuum-random-tree literature [32] provides natural context for later quenched fluctuation questions, but no continuum limit theorem is needed for our proofs.

The source audit was focused, not an exhaustive reading of the entire repository or all random-tree literature. It did not establish a prior publication containing these precise stable-marking formulas. That negative search outcome is not evidence of first priority. The defensible claim is an explicit mathematical continuation of the pinned question, with an auditable separation of imported and newly derived statements.

## 154 The model and the imported structural theorem

All trees are finite, simple, undirected, and nonempty. A marking is a subset  $S$  of the vertex set of a tree  $T = (V, E)$ . Form a bipartite graph  $\mathcal{A}(T; S)$  with left shore  $V$  and right shore  $E \sqcup \{\star\}$ . An edge-element  $e = ab$  is adjacent to  $a, b$ ; the apex element  $\star$  is adjacent precisely to  $S$ .

For  $I \subseteq V$  and  $J \subseteq E \sqcup \{\star\}$ , write  $I \sim J$  when  $|I| = |J|$  and there is a perfect matching between them. The canonical basis polynomial is

$$F_{T,S}(x; y, s) = \sum_{I \sim J} \prod_{v \in V \setminus I} x_v \prod_{e \in J \cap E} y_e s^{\mathbf{1}_{\{\star \in J\}}}. \quad (154.1)$$

A feasible support contributes once, even if several matchings realize it. This distinction is essential. A real polynomial is *real stable* when it is nonzero whenever every variable has positive imaginary part.

For  $S \neq \emptyset$ , the marked hull  $\text{Hull}_T(S)$  is the least connected subtree containing  $S$ . Every leaf of this hull belongs to  $S$ . A marking is *branch-complete* when every vertex of degree at least three in its hull is marked. Suppressing the unmarked degree-two hull vertices then produces a tree on  $S$ , the *terminal skeleton*. [Merge note: this is Part VIII’s model (Section 118), restated by source 12:  $\mathcal{A}(T; S)$  is Part VIII’s incidence–apex graph, (154.1) is (118.1), the hull is Part VIII’s  $K = \text{Hull}_T(S)$  and the terminal skeleton is Part VIII’s  $H(T, S)$ . In Part X the letters  $K$  and  $H$  denote the *numbers*  $K_n = |S|$  and  $H_n = |\text{Hull}_T(S)|$  instead (Table 19).]

**Theorem 154.1** (Imported marked-tree classification, Part VIII). *The polynomial (154.1) is real stable if and only if either  $S = \emptyset$ , or  $S$  is branch-complete and the adjacency matrix of its terminal skeleton has spectral radius at most two. Equivalently, the skeleton belongs to Table 20.*

This is the pinned source's theorem `mts:thm:main`, with its spectral list `mts:tab:ADE`: Part VIII's Theorem 118.2, (i) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (iv), with Table 14. It is used as a dependency, not silently reproved or declared formally verified by this article. All subsequent combinatorics can alternatively be read as unconditional statements about the explicitly defined class of ADE-admissible markings.

| Family                                              | Skeleton shape                                                                             | Aut     |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------|---------|
| $A_k, k \geq 2$                                     | Path on $k$ vertices                                                                       | 2       |
| $D_4$                                               | Three-leaf star                                                                            | 6       |
| $D_k, k \geq 5$                                     | Three arms $(1, 1, k-3)$                                                                   | 2       |
| $E_6, E_7, E_8$                                     | Arms $(1, 2, 2), (1, 2, 3), (1, 2, 4)$                                                     | 2, 1, 1 |
| $\widetilde{E}_6, \widetilde{E}_7, \widetilde{E}_8$ | Arms $(2, 2, 2), (1, 3, 3), (1, 2, 5)$                                                     | 6, 2, 1 |
| $\widetilde{D}_4$                                   | Four-leaf star, five vertices                                                              | 24      |
| $\widetilde{D}_{d+4}, d \geq 1$                     | Two trivalent vertices joined by a length- $d$ path, with two immediate leaves at each end | 8       |

Table 20: The allowed nontrivial skeletons. Arm lengths count skeleton edges, not host edges. The singleton  $A_1$  has one automorphism. The spectral classification is imported; the automorphism counts are elementary.

[Merge note: Table 20 is Part VIII's Table 14 (Part VI's list of Theorem 88.1) in source 12's layout:  $D_4$  is listed apart from  $D_k, k \geq 5$  (Part VIII's  $D_{c+3}, c \geq 1$ ),  $A_1$  is in the caption, and the |Aut| column is new. It is kept because later statements of Part X cite it and because the automorphism counts give (155.1).]

**Definition 154.2** (Annealed and quenched probabilities). Fix  $n \geq 1$ . Let  $T_n$  be uniform among labelled trees on  $[n]$ , independently of Bernoulli marks with parameter  $p$ . Put

$$\pi_n(p) = \mathbb{P}(E_n), \quad \pi_p(T) = \mathbb{P}(S \text{ is stable} \mid T).$$

The first averages over both sources of randomness (*annealed*); the second holds the tree fixed (*quenched*). We use  $q = 1 - p$  and, when  $q > 0$ ,  $r = p/q$ . The hull size counts vertices; diameter counts edges.

For  $n = 1$ , the number of trees is one and both markings are stable. For  $p = 0$ ,  $\pi_n(0) = 1$ . For  $p = 1$ , stability means that the entire tree is an ADE tree. Statements involving  $r$  or  $\log q$  exclude  $p = 1$ .

## 155 The skeleton generating function

Let  $a_k$  be the number of allowed skeletons on a prescribed  $k$ -element label set, and put  $c_k = a_k/k!$ . Relabelling a fixed unlabelled tree produces  $k!/|\text{Aut}|$  labelled trees. Consequently Table 20 gives

$$\begin{aligned} A(y) := \sum_{k \geq 1} c_k y^k &= y + \frac{y^2}{2(1-y)} + \frac{y^4}{6} + \frac{y^5}{2(1-y)} + \frac{y^5}{24} + \frac{y^6}{8(1-y)} \\ &\quad + \frac{y^6}{2} + \frac{7y^7}{6} + \frac{3y^8}{2} + y^9. \end{aligned} \tag{155.1}$$

For example, at seven vertices the exceptional contributions are  $E_7$  and  $\widetilde{E}_6$ , of weights 1 and  $1/6$ . At eight vertices they are  $E_8$  and  $\widetilde{E}_7$ , of weights 1 and  $1/2$ .

**Proposition 155.1** (Eventual coefficient constancy). *With  $c_* = 9/8$ , one has*

$$A(y) = y + c_* \frac{y^2}{1-y} + \sum_{k=2}^9 \delta_k y^k, \quad c_k = c_* \quad (k \geq 10), \quad (155.2)$$

where Table 21 gives the complete exceptional prefix.

*Proof.* Expand the three geometric series in (155.1). Their eventual coefficients are  $1/2 + 1/2 + 1/8 = 9/8$ ; every other summand is a polynomial.  $\square$

| $k$        | 1 | 2              | 3              | 4                | 5               | 6              | 7               | 8              | 9              |
|------------|---|----------------|----------------|------------------|-----------------|----------------|-----------------|----------------|----------------|
| $c_k$      | 1 | $\frac{1}{2}$  | $\frac{1}{2}$  | $\frac{2}{3}$    | $\frac{25}{24}$ | $\frac{13}{8}$ | $\frac{55}{24}$ | $\frac{21}{8}$ | $\frac{17}{8}$ |
| $\delta_k$ | — | $-\frac{5}{8}$ | $-\frac{5}{8}$ | $-\frac{11}{24}$ | $-\frac{1}{12}$ | $\frac{1}{2}$  | $\frac{7}{6}$   | $\frac{3}{2}$  | 1              |

Table 21: Skeleton coefficients and finite corrections to  $9/8$ .

The constant coefficients are not a statement about uniform unlabelled random trees. The factorial normalization comes from labelled enumeration, and changing the random-tree model changes the analysis.

## 156 Exact joint enumeration by marks and hull size

### 156.1 A rooted-forest counting lemma

We use the falling factorial  $(n)_k = n(n-1)\cdots(n-k+1)$ .

**Lemma 156.1** (Prescribed roots). *For a prescribed root set of size  $\ell$  in  $[n]$ , the number of forests whose components each contain exactly one root is*

$$f_{n,\ell} = \ell n^{n-\ell-1} \quad (1 \leq \ell \leq n), \quad (156.1)$$

with value one when  $\ell = n$ .

*Proof.* Orient every edge toward its component root. The directed forest matrix-tree expansion reduces the count to the determinant of  $nI_m - J_m$ , where  $m = n - \ell$  indexes the nonroots. To recall the combinatorics of this expansion, choose one outgoing edge from each nonroot; the alternating terms in the determinant cancel exactly the choices containing a directed cycle among nonroots. The remaining functional digraphs all reach a root and are precisely the desired forests. The matrix has eigenvalues  $n$  with multiplicity  $m - 1$  and  $n - m = \ell$  with multiplicity one, giving (156.1). If  $m = 0$ , the empty determinant is one, also the value of  $\ell n^{-1}$ .  $\square$

**Theorem 156.2** (Exact joint count). *Let  $C_{n,k,\ell}$  count pairs  $(T, S)$  with stable marking,  $|S| = k$ , and  $|\text{Hull}_T(S)| = \ell$ . For  $2 \leq k \leq \ell \leq n$ ,*

$$C_{n,k,\ell} = \binom{n}{k} a_k (n-k)_{\ell-k} \binom{\ell-2}{k-2} f_{n,\ell}. \quad (156.2)$$



The remaining nonzero entries are  $C_{n,0,0} = n^{n-2}$  and  $C_{n,1,1} = n^{n-1}$ , with the single-vertex convention already stated. Thus

$$\pi_n(p) = \frac{1}{n^{n-2}} \sum_{k,\ell} C_{n,k,\ell} p^k q^{n-k}. \quad (156.3)$$

*Proof.* Choose the  $k$  marked labels and one of their  $a_k$  allowed skeletons. Set  $j = \ell - k$ . On the  $k - 1$  skeleton edges, insert ordered lists of  $j$  distinct unmarked labels. Orient each skeleton edge by any fixed label-dependent convention; this avoids introducing a spurious factor of two. The list lengths form a weak composition of  $j$  into  $k - 1$  parts, of which there are  $\binom{j+k-2}{k-2}$ . Choosing and ordering the subdivision labels gives  $(n - k)_j$  possibilities.

The resulting subdivided skeleton is exactly the hull. Attach the remaining vertices by a rooted forest whose roots are the  $\ell$  hull vertices; Lemma 156.1 counts these attachments. Every attached vertex is unmarked. Conversely, pruning all unmarked fringe and suppressing the unmarked degree-two hull vertices recovers each choice uniquely. This proves the product formula. Zero and one mark are always stable and give the two remaining entries.  $\square$

**Corollary 156.3** (A fixed marked set). *For a prescribed  $k$ -element marked set,  $k \geq 2$ , write  $b_{n,k} = \mathbb{P}(E_n \mid S = [k])$ . Then*

$$\begin{aligned} b_{n,k} &= a_k R_{n,k}, \\ R_{n,k} &= n^{1-k} \sum_{j=0}^{n-k} \frac{(n-k)_j}{n^j} \binom{k+j-2}{k-2} (k+j). \end{aligned} \quad (156.4)$$

Conditional on stability and this marked set, the labelled terminal skeleton is exactly uniform among its  $a_k$  possibilities, for every finite  $n \geq k$ .

*Proof.* Divide (156.2) by  $\binom{n}{k} n^{n-2}$  and sum over  $\ell$ . The attachment factor does not depend on the shape of the skeleton.  $\square$

## 156.2 A nonnegative one-dimensional kernel

Substituting  $a_k = k!c_k$  into (156.2) and collecting by  $\ell$  gives the useful exact identity

$$\frac{\pi_n(p)}{q^n} = 1 + nr + n \sum_{\ell=2}^n \frac{(n)_\ell}{n^\ell} \ell \sum_{k=2}^{\ell} c_k \binom{\ell-2}{k-2} r^k. \quad (156.5)$$

Every summand here is nonnegative. Since  $c_k = c_*$  after  $k = 9$ , the inner sum has the computationally economical form

$$c_* r^2 (1+r)^{\ell-2} + \sum_{k=2}^{\min(9,\ell)} \delta_k \binom{\ell-2}{k-2} r^k. \quad (156.6)$$

This evaluates the probability in  $O(n)$  arithmetic operations after updating the falling-factorial ratio recursively. It is an arithmetic operation count, not a claim of linear bit complexity at arbitrary precision. The correction form can exhibit cancellation; the nonnegative form remains available for independent exact verification.

## 157 The labelled generating function and its deformations

Let  $U(x)$  be the exponential generating function for rooted labelled trees:

$$U(x) = xe^{U(x)}, \quad U(x) = \sum_{n \geq 1} \frac{n^{n-1}}{n!} x^n. \quad (157.1)$$

Its unrooted counterpart is  $U - U^2/2$ . For completeness,  $x \frac{d}{dx}(U - U^2/2) = U$ , so the latter has coefficient  $n^{n-2}/n!$  and zero constant term. The rooted equation follows by removing the root and obtaining a labelled set of rooted trees, or by Lagrange inversion [31].

**Theorem 157.1** (Exact two-colour generating function). *Let  $x$  count unmarked vertices and  $z$  marked vertices in the labelled exponential convention. Stable marked trees have generating function*

$$\mathcal{F}(x, z) = U(x) - \frac{U(x)^2}{2} + (1 - U(x))A\left(\frac{ze^{U(x)}}{1 - U(x)}\right). \quad (157.2)$$

For  $0 < p < 1$ , put  $s = U(qt)$ . Then

$$\mathcal{F}_p(t) = s - \frac{s^2}{2} + (1 - s)A\left(\frac{ps}{q(1 - s)}\right), \quad (157.3)$$

$$\pi_n(p) = B_n[t^n]\mathcal{F}_p(t), \quad B_n = \frac{n!}{n^{n-2}}. \quad (157.4)$$

More generally, marking fugacity  $u$  and hull-size fugacity  $v$  give

$$\mathcal{F}_p(t; u, v) = s - \frac{s^2}{2} + (1 - vs)A\left(\frac{vpu s}{q(1 - vs)}\right), \quad (157.5)$$

and  $B_n[t^n]\mathcal{F}_p(t; u, v) = \mathbb{E}[u^{K_n} v^{H_n} \mathbf{1}_{E_n}]$ .

*Proof.* At each marked skeleton vertex, attach an arbitrary labelled set of rooted unmarked trees, giving the factor  $ze^U$ . Each unmarked vertex inserted along a skeleton edge, together with its own fringe, has weight  $xe^U = U$ . An ordered sequence of such vertices has weight  $(1 - U)^{-1}$ . Thus a skeleton with  $k$  vertices and  $k - 1$  edges contributes  $c_k(ze^U)^k(1 - U)^{-(k-1)}$ . Summing gives the last term of (157.2); the first two terms count the empty marking. This decomposition is the labelled version of Theorem 156.2.

Substitute  $x = qt$ ,  $z = pt$  and use  $te^s = s/q$ . For the refinement, each marked vertex acquires  $uv$ , every subdivision vertex acquires  $v$ , and no fringe vertex acquires  $v$ . Replacing  $ze^U$  by  $uvpte^s$  and  $U$  along edges by  $vs$  gives (157.5). In particular,  $v$  counts hull vertices, not the total size of the attached rooted trees.  $\square$

The rational skeleton function has its only pole at  $y = 1$ . For fixed  $0 < p < 1$ , this produces a pole at  $s = q$ , hence at

$$t_0 = e^{-q}. \quad (157.6)$$

The rooted-tree branch point is at  $t = 1/(eq)$ , and  $t_0 < 1/(eq)$  because  $qe^{1-q} < 1$  for  $0 < q < 1$ . The collision of these two singularities as  $p \downarrow 0$  explains why a separate critical analysis is needed; the fixed- $p$  pole argument alone is not a uniform proof in that limit.

## 158 A fixed number of marks: chi hulls and uniform subdivisions

Define

$$\alpha_k = \frac{2^{k/2-1}\Gamma(k/2)}{(k-2)!} = \frac{\sqrt{\pi} 2^{1-k/2}}{\Gamma((k-1)/2)} \quad (k \geq 2). \quad (158.1)$$

**Theorem 158.1** (Fixed- $k$  asymptotics). *For every fixed  $k \geq 2$ ,*

$$R_{n,k} \sim \alpha_k n^{1-k/2}, \quad b_{n,k} \sim a_k \alpha_k n^{1-k/2}. \quad (158.2)$$

*There is a constant  $M_k$  with  $R_{n,k} \leq M_k n^{1-k/2}$  for all  $n \geq k$ . Given stability and  $K_n = k$ ,*

$$\frac{H_n}{\sqrt{n}} \xrightarrow{d} \chi_k, \quad (158.3)$$

*where  $\chi_k$  has density  $x^{k-1}e^{-x^2/2}/(2^{k/2-1}\Gamma(k/2))$  on  $x > 0$ .*

*Proof.* For  $j/\sqrt{n}$  in a fixed compact interval,

$$\frac{(n-k)_j}{n^j} = \prod_{i=0}^{j-1} \left(1 - \frac{k+i}{n}\right) = e^{-j^2/(2n)+o(1)},$$

uniformly in that interval. Also  $\binom{k+j-2}{k-2}(k+j) \sim j^{k-1}/(k-2)!$  as  $j \rightarrow \infty$ . The Riemann sum in (156.4) therefore yields

$$n^{k/2-1}R_{n,k} \longrightarrow \frac{1}{(k-2)!} \int_0^\infty x^{k-1}e^{-x^2/2} dx = \alpha_k.$$

To justify tails and obtain the stated upper bound, use

$$\frac{(n-k)_j}{n^j} \leq \exp\left(-\frac{kj + j(j-1)/2}{n}\right)$$

and bound the remaining factor by a constant depending on  $k$  times  $(j+1)^{k-1}$ . Comparison with a Gaussian integral is uniform in  $n$ ; the finitely many small  $n$  can be absorbed into  $M_k$ .

The exact joint count assigns to  $j = H_n - k$  precisely the normalized summands of (156.4). Applying the same Riemann-sum argument to bounded continuous test functions proves the chi limit.  $\square$

For  $k = 2$ ,  $a_2 = \alpha_2 = 1$ , and in fact stability has probability one. For  $k = 3$ ,  $a_3 = 3$  and

$$b_{n,3} \sim 3\sqrt{\frac{\pi}{2}} n^{-1/2}. \quad (158.4)$$

Three marks fail generically because the branching point of their hull need not itself be marked; (158.4) quantifies the exceptional event.

**Corollary 158.2** (Limiting metric skeleton at fixed  $k$ ). *Given stability and  $K_n = k \geq 2$ , the labelled skeleton is exactly uniform among the  $a_k$  possibilities. Conditional on  $H_n = k + j$ , its subdivision counts are uniform among weak compositions of  $j$  into  $k-1$  parts. As  $n \rightarrow \infty$ , the normalized edge-length vector converges to Dirichlet(1, ..., 1), independently of the uniform skeleton and the limiting chi radius  $H_n/\sqrt{n}$ .*

*Proof.* For a fixed composition, the ordered subdivision labels and the attached forests have counts independent of that composition and of the skeleton. Uniform lattice points in the dilated simplex, divided by  $j$ , converge by elementary Riemann sums to uniform simplex measure, namely the stated Dirichlet law. By Theorem 158.1,  $j \rightarrow \infty$  in probability. Conditioning on  $j$  and then using bounded test functions proves joint convergence and independence. Replacing subdivision counts by edge lengths adds one to each coordinate and does not alter the limit.  $\square$

## 159 An exact Poisson identity with finite corrections

For  $2 \leq k \leq n$  put

$$W_{n,k}(p) = (n)_k p^k q^{n-k} R_{n,k}. \quad (159.1)$$

This is the probability contribution that a skeleton coefficient equal to one would give at size  $k$ ; the actual contribution is  $c_k W_{n,k}$ .

**Theorem 159.1** (Finite-correction Poisson formula). *For  $n \geq 2$  and  $0 < p < 1$ , if  $X \sim \text{Pois}(nq)$ , then*

$$\begin{aligned} \pi_n(p) = & q^n + npq^{n-1} + c_* p^3 B_n e^{nq} \mathbb{P}(X \leq n-2) \\ & + c_* p^2 n(n-1)q^{n-1} + \sum_{k=2}^{\min(9,n)} \delta_k W_{n,k}(p). \end{aligned} \quad (159.2)$$

*The formula is exact, not asymptotic. The exponential and the Poisson probability may be replaced together by their finite sum.*

*Proof.* The singleton and empty terms in (155.2) give the first two terms of (159.2); its finite polynomial gives the last sum by Theorem 156.2. For the geometric tail, set  $y = ps/[q(1-s)]$  and calculate

$$(1-s)c_* \frac{y^2}{1-y} = c_* p^2 \left( \frac{q}{q-s} - 1 - \frac{s}{q} \right). \quad (159.3)$$

Lagrange inversion for  $s = qte^s$  gives

$$[t^n] \frac{q}{q-s} = \sum_{i=0}^{n-1} \left( 1 - \frac{i}{n} \right) \frac{(nq)^i}{i!}. \quad (159.4)$$

Indeed, apply  $[t^n]g(s) = q^n[n^{-1}z^{n-1}]g'(z)e^{nz}$ , expand  $g'(z) = q/(q-z)^2$ , and change the summation index. Since  $[t^n]s/q = (nq)^{n-1}/n!$ , subtraction and the identity  $\sum_{i=1}^{n-1} i(nq)^i/i! = nq \sum_{i=0}^{n-2} (nq)^i/i!$  yield

$$[t^n] \left( \frac{q}{q-s} - 1 - \frac{s}{q} \right) = p \sum_{i=0}^{n-2} \frac{(nq)^i}{i!} + \left( 1 - \frac{1}{n} \right) \frac{(nq)^{n-1}}{(n-1)!}.$$

Multiplying by  $c_* p^2 B_n$  gives the middle two terms of (159.2).  $\square$

A useful exact interpretation is that every departure from the universal geometric skeleton tail is confined to the eight sizes  $2, \dots, 9$ . The negative corrections in Table 21 are algebraic correction terms, not probabilities. Their signed appearance must not be used to infer a probabilistic mixture.

## 160 The rare-event equivalent above the critical window

**Theorem 160.1** (Uniform supercritical equivalent). *For any sequence  $0 < p_n \leq 1$  with  $p_n \sqrt{n} \rightarrow \infty$ ,*

$$\pi_n(p_n) \sim c_* p_n^3 B_n e^{n(1-p_n)} \sim \frac{9}{8} \sqrt{2\pi} p_n^3 n^{5/2} e^{-np_n}. \quad (160.1)$$

*In particular, for every fixed  $0 < p \leq 1$ ,  $n^{-1} \log \pi_n(p) \rightarrow -p$ .*

*Proof.* First suppose  $0 < p \leq 1/2$ , set  $x = p\sqrt{n}$ , and assume  $x \geq 2$ . Define the positive rate gap

$$I(p) = -p - \log(1-p) \geq p^2/2. \quad (160.2)$$

Let  $M_n = c_* p^3 B_n e^{nq}$ . Stirling's lower bound gives  $M_n \geq C p^3 n^{5/2} e^{-np}$  with an absolute  $C > 0$ . For each fixed  $2 \leq k \leq 9$ , Theorem 158.1 implies

$$\frac{|\delta_k| W_{n,k}}{M_n} \leq C_k x^{k-3} q^{-k} e^{-nI(p)}. \quad (160.3)$$

The zero- and one-mark terms, and the extra boundary term in (159.2), are bounded similarly; the latter has ratio at most  $C x^{-1} q^{-1} e^{-nI(p)}$ .

For the Poisson tail use the Chernoff bound at  $n-1 > nq$ :

$$\mathbb{P}(X \geq n-1) \leq \exp \left( -(n-1) \log \frac{n-1}{nq} + n-1 - nq \right) \leq C e^{-nI(p)}.$$

The last constant is absolute because  $q \geq 1/2$ ; the difference between the two exponents is  $\log(1/q) - (n-1) \log(1-1/n) - 1$ , which is bounded. Thus, uniformly in this range,

$$\frac{\pi_n(p)}{M_n} = 1 + O \left( (1+x^6) e^{-nI(p)} \right). \quad (160.4)$$

It tends to one whenever  $x \rightarrow \infty$ .

For  $1/2 \leq p < 1$ , set  $J_0 = \log 2 - 1/2 > 0$ . For every fixed  $k \leq 9$  and  $n \geq 18$ , the function  $q^{n-k} e^{n(1-q)}$  is increasing on  $0 < q \leq 1/2$ , so it is at most  $2^k e^{-nJ_0}$ . The same bounds as above, now retaining this expression instead of  $q^{-k} e^{-nI(p)}$ , show that all correction ratios are  $O(n^3 e^{-nJ_0})$ . The Poisson-tail ratio is bounded uniformly by its value at mean  $n/2$ , with the same exponential rate. The remaining terms are smaller. Hence  $\pi_n/M_n \rightarrow 1$  uniformly in this second range.

At  $p = 1$  and  $n \geq 10$ , (156.3) reduces exactly to  $\pi_n(1) = a_n/n^{n-2} = c_* B_n$ . Finally,  $B_n \sim \sqrt{2\pi} n^{5/2} e^{-n}$  proves the second equivalent.  $\square$

**Proposition 160.2** (Fixed-density meromorphic strengthening). *For fixed  $0 < p < 1$ , there is  $0 < \theta_p < 1$  such that*

$$\pi_n(p) = c_* p^3 B_n e^{nq} (1 + O(\theta_p^n)). \quad (160.5)$$

*Consequently the expansion from Stirling, beginning with the factor  $1 + 1/(12n) + O(n^{-2})$ , may be appended to (160.1).*

*Proof.* The same conclusion follows from (159.2), or directly from (157.3). Near  $t_0 = e^{-q}$ ,  $s' = s/[t(1-s)]$  implies

$$\mathcal{F}_p(t) = \frac{c_* p^3}{1 - t/t_0} + \text{an analytic remainder}.$$

Choose  $R$  strictly between  $t_0$  and  $1/(eq)$ . The principal rooted-tree branch is analytic in  $|t| < 1/(eq)$ ; the equation  $s = q$  has the unique solution  $t = t_0$  there. After subtracting the displayed pole, all other terms are analytic on a slightly larger disc than  $|t| \leq R$ . Cauchy's coefficient bound gives an error  $O(R^{-n})$ , proving the result.  $\square$

The exponential factor is  $e^{-pn}$ , not  $(1-p)^n$ . Replacing the former by the latter loses an exponentially significant factor when  $p$  is fixed. Section 165 explains why the latter nevertheless appears naturally when the host tree is held fixed.

## 161 Below the window: at most two marks dominate

**Theorem 161.1** (Uniform sparse comparison). *As  $p\sqrt{n} \rightarrow 0$ , with  $0 \leq p < 1$ ,*

$$\pi_n(p) = q^n \left( 1 + nr + \binom{n}{2} r^2 \right) (1 + O(r\sqrt{n})). \quad (161.1)$$

*At  $p = 0$  interpret the formula exactly. In particular, stability and the event  $K_n \leq 2$  have asymptotically equal probabilities.*

*Proof.* Every zero-, one-, or two-mark set is stable, so the displayed main term is a lower bound. By (156.5) and  $c_k \leq 21/8$ , the contribution of  $k \geq 3$ , divided by  $q^n$ , is at most

$$Cnr^2 \sum_{\ell=2}^n \ell \frac{\binom{n}{\ell}}{n^\ell} ((1+r)^{\ell-2} - 1) \leq Cnr^3 \sum_{\ell=2}^n \ell^2 \exp\left(-\frac{\ell(\ell-1)}{2n} + r\ell\right).$$

When  $r\sqrt{n} \leq 1/4$ , Gaussian integral comparison bounds the sum by  $C'n^{3/2}$ . The remainder is therefore at most  $C''n^{5/2}r^3$ . For  $n \geq 2$ , divide by the two-mark main term  $\binom{n}{2}r^2$  to obtain  $O(r\sqrt{n})$ ; the zero case is immediate. Since  $p\sqrt{n} \rightarrow 0$  also implies  $r\sqrt{n} \rightarrow 0$ , the claim follows.  $\square$

**Corollary 161.2** (Two sparse subregimes). *If  $np \rightarrow \mu \in [0, \infty)$ , then*

$$\pi_n(p) \longrightarrow e^{-\mu}(1 + \mu + \mu^2/2). \quad (161.2)$$

*For  $\mu > 0$ , the conditional mark count tends to a Poisson random variable of mean  $\mu$  conditioned to be at most two; for  $\mu = 0$  it tends to zero. If instead  $np \rightarrow \infty$  and  $p\sqrt{n} \rightarrow 0$ , then*

$$\pi_n(p) \sim \binom{n}{2} p^2 q^{n-2}, \quad \mathbb{P}(K_n = 2 \mid E_n) \longrightarrow 1. \quad (161.3)$$

*Proof.* Use the elementary binomial-to-Poisson limit in the first case. In the second case the two-mark term dominates the zero- and one-mark terms. Theorem 161.1 shows that stable configurations with three or more marks have negligible mass in both cases.  $\square$

In the second regime the original, unconditioned expected number of marks  $np$  diverges, but stability selects just two. This selection changes in the window  $p \asymp n^{-1/2}$ , not when  $np$  first becomes large.

## 162 The critical crossover and its limiting distributions

Define the entire function

$$\Psi(\lambda) = \sum_{k \geq 2} c_k \alpha_k \lambda^k. \quad (162.1)$$

The gamma denominator in (158.1) and boundedness of  $c_k$  give infinite radius of convergence. For real  $\lambda > 0$  it is strictly positive. A form involving just eight correction terms is

$$\Psi(\lambda) = c_* \left[ \lambda^2 + \sqrt{\frac{\pi}{2}} \lambda^3 e^{\lambda^2/2} \left( 1 + \operatorname{erf} \frac{\lambda}{\sqrt{2}} \right) \right] + \sum_{k=2}^9 \delta_k \alpha_k \lambda^k. \quad (162.2)$$



**Lemma 162.1** (Integral form). *For real  $\lambda \geq 0$ ,*

$$\Psi(\lambda) = \int_0^\infty e^{-x^2/2} \sum_{k \geq 2} \frac{c_k \lambda^k x^{k-1}}{(k-2)!} dx. \quad (162.3)$$

*This identity gives (162.2). It also yields*

$$\Psi(\lambda) \sim \frac{1}{2} \lambda^2 \quad (\lambda \downarrow 0), \quad \Psi(\lambda) \sim c_* \sqrt{2\pi} \lambda^3 e^{\lambda^2/2} \quad (\lambda \rightarrow \infty). \quad (162.4)$$

*Proof.* Integrate each monomial using (158.1); nonnegativity justifies interchange for real  $\lambda \geq 0$ . For the constant tail the inner sum is  $c_* \lambda^2 x e^{\lambda x}$ . Completing the square and integrating by parts gives

$$\int_0^\infty x e^{-x^2/2 + \lambda x} dx = 1 + \lambda \sqrt{\frac{\pi}{2}} e^{\lambda^2/2} \left( 1 + \operatorname{erf} \frac{\lambda}{\sqrt{2}} \right).$$

The finite corrections are immediate. The first nonzero series coefficient is  $c_2 \alpha_2 = 1/2$ ; at infinity the finite polynomial is negligible beside the positive exponential term.  $\square$

**Theorem 162.2** (Critical probability). *If  $p_n \sqrt{n} \rightarrow \lambda \in (0, \infty)$ , then*

$$\frac{\pi_n(p_n)}{n q_n^n} \rightarrow \Psi(\lambda). \quad (162.5)$$

*The normalization uses the actual  $q_n^n$ ; replacing it merely by  $e^{-\lambda \sqrt{n}}$  can lose a constant or a larger factor for general sequences approaching the same  $\lambda$ .*

*Proof.* Use the exact formula (159.2). For  $X \sim \text{Pois}(n q_n)$ , the centered standardized threshold tends to  $\lambda$ , so  $\mathbb{P}(X \leq n - 2) \rightarrow \Phi(\lambda)$ , where  $\Phi$  is the standard normal CDF. This Poisson central limit follows, for example, by expanding its characteristic function  $\exp(m(e^{it/\sqrt{m}} - 1 - it/\sqrt{m}))$  as  $m \rightarrow \infty$ . Also  $n I(p_n) \rightarrow \lambda^2/2$ . Thus the main Poisson term, divided by  $n q_n^n$ , tends to  $c_* \sqrt{2\pi} \lambda^3 e^{\lambda^2/2} \Phi(\lambda)$ . The boundary term tends to  $c_* \lambda^2$ . For each fixed  $k$ , Theorem 158.1 gives

$$\frac{W_{n,k}(p_n)}{n q_n^n} = \frac{(n)_k r_n^k R_{n,k}}{n} \rightarrow \alpha_k \lambda^k.$$

The zero- and one-mark terms vanish after normalization. Since  $2\Phi(\lambda) = 1 + \operatorname{erf}(\lambda/\sqrt{2})$ , these limits are exactly (162.2).  $\square$

**Theorem 162.3** (Critical mark count and hull mixture). *Under the hypotheses of Theorem 162.2, conditional on stability,*

$$\mathbb{P}(K_n = k \mid E_n) \rightarrow w_k(\lambda) := \frac{c_k \alpha_k \lambda^k}{\Psi(\lambda)} \quad (k \geq 2). \quad (162.6)$$

*These limits sum to one. The limiting probability generating function is  $\Psi(\lambda u)/\Psi(\lambda)$ , with convergence of every fixed moment. Jointly with this discrete limit,  $H_n/\sqrt{n}$  tends to the mixture that, given  $K = k$ , has law  $\chi_k$ . Equivalently, the limiting pair  $(K_\lambda, X_\lambda)$  has joint mass-density*

$$\mathbb{P}(K_\lambda = k, X_\lambda \in dx) = \frac{c_k \lambda^k}{(k-2)! \Psi(\lambda)} x^{k-1} e^{-x^2/2} dx, \quad x > 0. \quad (162.7)$$

*Proof.* The exact probability contribution for  $K_n = k$  is  $c_k W_{n,k}$ . The calculation in the preceding proof gives (162.6), and (162.1) shows that no probability escapes to infinity. The chi mixture follows from Theorem 158.1 and truncating the mark count at a large fixed value.

For moment convergence, let  $u > 0$  be fixed and put  $d = q + pu$ ,  $p' = pu/d$ ,  $q' = q/d$ . Tilting the independent mark law gives exactly

$$\mathbb{E}[u^{K_n} \mathbf{1}_{E_n}] = d^n \pi_n(p'). \quad (162.8)$$

Now  $p' \sqrt{n} \rightarrow \lambda u$  and  $d^n (q')^n = q^n$ . The critical probability theorem yields the asserted generating-function limit for every fixed positive  $u$ . Taking some  $u > 1$  bounds exponential moments and gives uniform integrability of every fixed power of  $K_n$ .  $\square$

Conditional on the limiting value  $K = k$ , the skeleton and edge proportions retain the independent uniform and Dirichlet laws of Corollary 158.2. Formula (162.7) describes the marked hull, not the full host diameter: in this sparse window the unmarked fringe need not have logarithmic height.

The large- $\lambda$  asymptotic in (162.4) agrees with Theorem 160.1, because  $q^n = e^{-np - np^2/2 + O(np^3)}$  when the expansion is applicable. This is a consistency check of the overlap of the two descriptions; the uniform supercritical theorem itself was proved without extrapolating a fixed- $\lambda$  limit.

## 163 Fixed-density conditional Gaussian laws

Throughout this section  $0 < p < 1$  is fixed. Introduce complex fugacities  $u, v$  near one, and define

$$\begin{aligned} d(u) &= q + pu, \\ s_0(u, v) &= \frac{q}{vd(u)}, \\ \tau(u, v) &= \frac{1}{vd(u)} \exp\left(-\frac{q}{vd(u)}\right), \end{aligned} \quad (163.1)$$

$$C(u, v) = c_* \left(\frac{pu}{d(u)}\right)^2 \left(1 - \frac{q}{vd(u)}\right). \quad (163.2)$$

At  $u = v = 1$ ,  $\tau = e^{-q}$  and  $C = c_* p^3$ .

**Lemma 163.1** (Uniform moving-pole estimate). *In a complex neighbourhood of  $(1, 1)$ ,*

$$\mathbb{E}[u^{K_n} v^{H_n} \mid E_n] = \frac{C(u, v)}{c_* p^3} \left(\frac{e^{-q}}{\tau(u, v)}\right)^n (1 + O(\theta^n)), \quad (163.3)$$

for some  $\theta < 1$ , uniformly in that neighbourhood. Derivatives of any fixed order satisfy the corresponding exponentially small error.

*Proof.* In (157.5), the equation making the skeleton argument equal to one is  $s = s_0(u, v)$ . Invert  $s = qte^s$  to obtain (163.1). The singular summand of  $A$  is  $c_*/(1 - y)$ ; consequently the singular part of the last term of (157.5) is

$$\frac{c_* q (1 - vs)^2}{q - vd(u)s}.$$

Since  $ts' = s/(1 - s)$ , its coefficient in front of  $(1 - t/\tau)^{-1}$  is  $c_*(1 - vs_0)^2(1 - s_0) = C(u, v)$ . Choose a radius strictly between  $e^{-q}$  and  $1/(eq)$ . For  $u, v$  sufficiently close to one, this remains a disc

with exactly that pole; potential poles at  $s = 1/v$  and the tree branch point stay outside a slightly larger circle. Subtract the moving pole and apply Cauchy's bound uniformly on the circle. Divide by the estimate at  $(1, 1)$ . Cauchy's formula in the parameter variables controls the derivatives.  $\square$

Put  $u = e^a, v = e^b$ . The logarithm of the exponential factor is

$$\Lambda(a, b) = b + \log(q + pe^a) + \frac{qe^{-b}}{q + pe^a} - q. \quad (163.4)$$

Its gradient and Hessian at zero are

$$\nabla \Lambda(0, 0) = (p^2, p), \quad \Sigma_p = \begin{pmatrix} 2p^2q & pq \\ pq & q \end{pmatrix}. \quad (163.5)$$

The determinant is  $p^2q^2 > 0$ .

**Theorem 163.2** (Joint central limit and moment corrections). *Conditional on  $E_n$ ,*

$$\frac{1}{\sqrt{n}} \begin{pmatrix} K_n - np^2 \\ H_n - np \end{pmatrix} \xrightarrow{d} N(0, \Sigma_p). \quad (163.6)$$

For a suitable  $\theta_p < 1$ , the conditional moments satisfy

$$\mathbb{E}K_n = np^2 + 3q + O(\theta_p^n), \quad \mathbb{E}H_n = np + q/p + O(\theta_p^n), \quad (163.7)$$

$$\text{Var } K_n = 2np^2q - 3pq + O(\theta_p^n), \quad \text{Var } H_n = nq - q/p^2 + O(\theta_p^n), \quad (163.8)$$

$$\text{Cov}(K_n, H_n) = npq - q/p + O(\theta_p^n). \quad (163.9)$$

All expectations in these displays are conditional on stability.

*Proof.* Insert  $a = i\xi/\sqrt{n}$ ,  $b = i\eta/\sqrt{n}$  into (163.3) and expand (163.4) to second order. After centering, the logarithm of the characteristic function tends to  $-\frac{1}{2}(\xi, \eta)\Sigma_p(\xi, \eta)^T$ . The amplitude tends to one. The continuity theorem proves (163.6).

For the refined moments, differentiate the logarithm of (163.3). The first derivatives of  $\log C(e^a, e^b)$  at zero are  $3q$  and  $q/p$ ; its second derivatives in  $(a, a), (b, b), (a, b)$  are respectively  $-3pq, -q/p^2, -q/p$ . Combining these with (163.5) proves the formulas. The uniform analytic estimate justifies the exponentially small differentiated errors.  $\square$

**Corollary 163.3** (Three limiting skeleton families). *At fixed  $0 < p < 1$ , conditional on stability, the skeleton lies asymptotically in the path, long-D, and double-fork affine-D families with probabilities*

$$\frac{4}{9}, \quad \frac{4}{9}, \quad \frac{1}{9}, \quad (163.10)$$

*respectively. Every other family has exponentially small conditional probability. The limiting family is independent of the Gaussian vector in (163.6).*

*Proof.* The three family generating functions have residues  $1/2, 1/2$ , and  $1/8$  at  $y = 1$ ; their finite exceptional terms have no such pole. Repeat Lemma 163.1 separately for each family. The moving pole and its exponential factor are unchanged; the amplitude is merely multiplied by its residue divided by  $9/8$ . This proves both the family probabilities and independence from the limiting Gaussian fluctuations.  $\square$

These statements require fixed  $p$  for their present uniformity. They do not assert a triangular-array central limit theorem throughout the full supercritical range of Theorem 160.1.

## 164 The conditioned host has linear diameter

The hull-size law alone does not determine the host diameter: fringe might be tall, or a hull might branch too much. The next two estimates control these issues in the fixed-density conditional model.

**Lemma 164.1** (Height of a uniform rooted forest). *Let a forest on  $[n]$  be uniform among forests rooted at a prescribed set of  $\ell$  vertices, and write  $m = n - \ell$ . For  $1 \leq h \leq m$ , the expected number of vertices at distance exactly  $h$  from the root set is*

$$(\ell + h) \frac{\binom{m}{h}}{n^h}. \quad (164.1)$$

If  $\ell \geq \varepsilon n$  for a fixed  $\varepsilon > 0$ , the maximum root distance is  $O_{\mathbb{P}}(\log n)$ , uniformly in the root set and  $\ell$ .

*Proof.* Choose a root and an ordered list of  $h$  distinct nonroots, prescribing a directed path to that root. There are  $\ell(m)_h$  choices. The probability that a fixed such path occurs is

$$\frac{(\ell + h)n^{m-h-1}}{\ell n^{m-1}}. \quad (164.2)$$

To see the numerator, contract the fixed path to its root. An edge from an outside nonroot into the contracted component has  $h + 1$  possible original targets; edges to each other root have one. The directed forest matrix on the remaining  $m - h$  nonroots is again  $nI - J$ , whose determinant is  $(\ell + h)n^{m-h-1}$ . When  $m = h$ , it is the empty determinant, and the same formula is one. The denominator is Lemma 156.1. Each vertex at distance  $h$  has exactly one path to its root, so multiplying (164.2) by the number of choices proves (164.1).

A forest reaching depth at least  $h$  has a vertex at depth  $h$ . Markov's inequality gives probability at most  $(\ell + h)(m/n)^h \leq 2n(1 - \varepsilon)^h$ . Take a sufficiently large constant times  $\log n$ .  $\square$

**Lemma 164.2** (Longest subdivision chain). *If  $(J_1, \dots, J_d)$  is uniform among weak compositions of  $j$  into  $d \geq 2$  parts, then*

$$\mathbb{P}(J_i \geq a) = \frac{\binom{j-a+d-1}{d-1}}{\binom{j+d-1}{d-1}} \leq \left( \frac{j}{j+d-1} \right)^a \quad (0 \leq a \leq j). \quad (164.3)$$

If  $d/(j + d)$  is bounded below by a positive constant, the maximum part is  $O_{\mathbb{P}}(\log(j + d + 1))$ .

*Proof.* Subtract  $a$  from the chosen part and count compositions. The ratio is  $(j)_a/(j + d - 1)_a$ , whose successive factors are bounded by the first one. A union bound over the  $d$  parts proves the second assertion. The case  $j = 0$  is immediate.  $\square$

**Theorem 164.3** (Diameter and joint fluctuations). *For fixed  $0 < p < 1$ , conditional on stability,*

$$\text{diam}(T_n) = H_n + O_{\mathbb{P}}(\log n), \quad \frac{\text{diam}(T_n)}{n} \xrightarrow{\mathbb{P}} p. \quad (164.4)$$

The joint central limit theorem (163.6) remains valid if  $H_n$  is replaced by  $\text{diam}(T_n)$ .

*Proof.* Theorem 163.2 gives  $K_n/n \rightarrow p^2$  and  $H_n/n \rightarrow p$  in conditional probability. Given  $n, k, \ell$ , the skeleton, and the subdivision data, the fringe forest is uniform on the prescribed  $\ell$  roots by the exact counting construction. Lemma 164.1 therefore makes its height  $O_{\mathbb{P}}(\log n)$ .

Given the skeleton and hull size, its  $j = \ell - k$  subdivision vertices are distributed uniformly among the  $d = k - 1$  edge lists. Since  $d/(j + d) \rightarrow p$ , Lemma 164.2 bounds every subdivided edge length by  $O_{\mathbb{P}}(\log n)$ .

By Corollary 163.3, with probability tending to one the skeleton is a path, a long  $D$  tree, or a double fork. In these cases it has a path omitting at most two pendant edges and their leaf endpoints. After subdivision, the hull has a path containing all but at most twice the maximum edge length of its vertices. Its diameter consequently differs from  $\ell$  by  $O_{\mathbb{P}}(\log n)$ . Attaching a forest can increase the diameter by at most twice its maximum root distance. This proves (164.4); since  $\log n = o(\sqrt{n})$ , Slutsky's theorem gives the final assertion.  $\square$

The error estimate in (164.4) is probabilistic. It does not claim a constant-order expansion for the mean diameter, nor replace the exact constant corrections for  $H_n$  in (163.7) by unproved corrections for diameter. It also does not extend automatically to the critical window, where  $H_n$  is only of order  $\sqrt{n}$  and the root fraction in Lemma 164.1 vanishes.

## 165 A deterministic diameter estimate and annealed–quenched separation

**Lemma 165.1** (Two marks away from a path). *For every stable marking  $S$  on any finite tree  $T$ , there is a host path  $P$  such that  $|S \setminus P| \leq 2$ . Conversely, every marking contained in a path is stable.*

*Proof.* The empty marking is immediate. Inspection of Table 20 shows that each allowed skeleton has a path omitting at most two vertices. For a one-branch skeleton, choose the two longest arms; the third arm has length at most two in every listed case. For the four-leaf star, choose two leaves; for a double fork, choose one leaf at each end. The lifted path in the hull omits precisely those marked skeleton vertices, regardless of how long the unmarked subdivisions are. This proves the first claim. A marking contained in a host path has a path hull and path skeleton, so Theorem 154.1 proves the converse.  $\square$

[Merge note: the converse is the lower-bound half of the proof of Part VIII's Theorem 123.3, and the first claim refines its upper-bound inspection  $|V(H)| \leq \text{diam}(H) + 3$ : Part VIII bounds the number of marks by the diameter, Lemma 165.1 places all but two of them on one host path. It is kept as a second, stronger route, because Theorem 165.2 needs the path.]

**Theorem 165.2** (Quenched probability within a polynomial factor). *Let  $T$  be any tree on  $n$  vertices, with diameter  $d$ , and let  $0 < p < 1$ . Then*

$$q^{n-d-1} \leq \pi_p(T) \leq \min \left\{ 1, \frac{n(n+1)}{2} \left( 1 + nr + \binom{n}{2} r^2 \right) q^{n-d-1} \right\}. \quad (165.1)$$

Consequently, for fixed  $p$ ,

$$\log \pi_p(T) = (n - d - 1) \log q + O_p(\log(n + 1)), \quad (165.2)$$

uniformly over all host trees. The subscript on  $O_p$  here denotes dependence on the fixed parameter  $p$ , not stochastic order.

*Proof.* Choose a longest path. The event that all marks lie on it has probability  $q^{n-d-1}$ , with no restriction on marks along the path, and is stable. This proves the lower bound.

There are exactly  $n(n+1)/2$  unoriented nonempty paths, allowing a single vertex. By Lemma 165.1, stability implies that at least one such path has at most two marks outside it. If that path has  $a$  vertices, this event has probability

$$q^{n-a} \left( 1 + (n-a)r + \binom{n-a}{2} r^2 \right).$$

Since  $a \leq d + 1$ , this is at most  $q^{n-d-1}(1 + nr + \binom{n}{2}r^2)$ . The union bound proves (165.1). Its multiplicative factor grows at most polynomially in  $n$  at fixed  $p$ , proving (165.2).  $\square$

This bound holds for paths, stars, balanced trees, and arbitrary labelled or unlabelled hosts. It supplies an exponential-scale description without running the source's exact endpoint algorithm. It is not a relative-error approximation scheme: the uncontrolled factor can still be polynomially large. [Merge note: "the source's exact endpoint algorithm" is Part VIII's formula (124.7) with Theorem 124.2, which evaluates  $\pi_p(T)$  exactly for one given host in  $O(n^5)$  arithmetic operations but gives no asymptotics; Theorem 165.2 gives its exponential order for every host without evaluating it.]

### 165.1 An elementary diameter large-deviation calculation

The following standard Cayley-forest calculation is included as an independent check of the exponential rate; first priority for this random-tree diameter rate is not claimed.

**Lemma 165.3** (Counting long paths). *Let  $N_d(T_n)$  be the number of unoriented paths of length  $d \geq 1$  in a uniform Cayley tree. Then*

$$\mathbb{E}N_d(T_n) = \frac{(n)_{d+1}(d+1)}{2n^d}. \quad (165.3)$$

For any fixed  $0 < \delta < 1$  and integers  $d_n/n \rightarrow \delta$ ,

$$\frac{1}{n} \log \mathbb{P}(\text{diam}(T_n) \geq d_n) \longrightarrow -J(\delta), \quad J(\delta) = \delta + (1 - \delta) \log(1 - \delta). \quad (165.4)$$

In particular  $\text{diam}(T_n)/n \rightarrow 0$  in probability.

*Proof.* A fixed length- $d$  labelled path is contained in  $(d+1)n^{n-d-2}$  Cayley trees: contract the path to a component of weight  $d+1$ , keeping all other vertices as weight-one components, and use the weighted tree identity. This identity follows directly from a Prüfer code: for component weights  $w_i$ , summing edge-target choices gives  $(\prod_i w_i)(\sum_i w_i)^{m-2}$  for  $m$  components. Thus the probability of the fixed path is  $(d+1)/n^d$ . There are  $(n)_{d+1}/2$  candidate unoriented paths, proving (165.3).

A tree has diameter at least  $d$  exactly when  $N_d \geq 1$ , and  $N_d \leq n(n-1)/2$ . Hence its probability is between  $\mathbb{E}N_d/n^2$  and  $\mathbb{E}N_d$ . The logarithm of the factorial ratio in (165.3), divided by  $n$ , tends to

$$\int_0^\delta \log(1-x) dx = -\delta - (1-\delta) \log(1-\delta).$$

All other factors have logarithm  $O(\log n)$ . Since  $J(\delta) > 0$  for  $\delta > 0$ , the last assertion follows.  $\square$

**Theorem 165.4** (Strict annealed–quenched gap). *For fixed  $0 < p < 1$  and a uniform Cayley host,*

$$\frac{1}{n} \log \pi_p(T_n) \xrightarrow{\mathbb{P}} \log(1-p), \quad \frac{1}{n} \log \mathbb{E}[\pi_p(T_n)] \longrightarrow -p. \quad (165.5)$$

*These rates differ strictly, since  $\log(1-p) < -p$ . More precisely, uniformly over hosts,*

$$\log \pi_p(T) - n \log q = -(\text{diam}(T) + 1) \log q + O_p(\log(n+1)). \quad (165.6)$$

*Proof.* Combine Theorem 165.2 with  $\text{diam}(T_n)/n \rightarrow 0$  for the first limit. The second follows from Theorem 160.1 and  $\mathbb{E}\pi_p(T_n) = \pi_n(p)$ . The inequality follows from the power series of  $\log(1-p)$  or its strict concavity. Equation (165.6) is a rearrangement of (165.2).  $\square$



There is also a variational explanation. The polynomial-factor sandwich reduces the annealed exponential rate to the exponential moment of the diameter. The tail estimate (165.4), upper bounds on a finite mesh, and the lower bound from any one tail event give

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1}{n} \log \pi_n(p) &= \log q + \sup_{0 \leq \delta \leq 1} \{-\delta \log q - J(\delta)\} \\ &= -p. \end{aligned} \quad (165.7)$$

Here  $J(0) = 0, J(1) = 1$  by continuity. To justify the upper bound, partition  $D/n$  into mesh intervals, bound the increasing weight  $q^{-D}$  by its right endpoint and the probability by the left-endpoint tail, then let the mesh tend to zero. For the lower bound, on  $D \geq \delta n$  the weight is at least  $q^{-\delta n}$ . Polynomial factors disappear after dividing logarithms by  $n$ .

Differentiation gives the unique maximizing diameter fraction  $\delta = p$ . Substitution gives  $\log q - p \log q - p - q \log q = -p$ . Thus the exponential gain in the average is supplied by atypically long trees, consistent with the stronger conditioned-diameter theorem. This argument recovers the exponential rate but neither the factor  $\frac{9}{8} \sqrt{2\pi} p^3 n^{5/2}$  nor the Gaussian covariance.

## 166 A general eventual-coefficient universality principle

The calculation is not tied to the names of the ADE diagrams. It uses an invariant class of terminal skeletons whose labelled coefficients are eventually constant.

**Theorem 166.1** (Constant-tail skeleton universality). *Let  $\mathcal{H}$  be an isomorphism-invariant class of finite trees, with  $c_k^{\mathcal{H}}$  equal to its number of labelled  $k$ -vertex members divided by  $k!$ . Suppose  $c_k^{\mathcal{H}} = c > 0$  for all sufficiently large  $k$ . Define admissible markings by requiring a branch-complete hull and a terminal skeleton in  $\mathcal{H}$ , allowing the empty marking. Then for  $p\sqrt{n} \rightarrow \infty$ ,*

$$\mathbb{P}(\mathcal{H}\text{-admissible}) \sim c \sqrt{2\pi} p^3 n^{5/2} e^{-pn}. \quad (166.1)$$

*At fixed  $0 < p < 1$ , the conditional joint Gaussian law for marks and hull size has exactly the covariance (163.5), and the constant-order moment corrections (163.7)–(163.9) are unchanged. In the critical window, replace  $c_k$  by  $c_k^{\mathcal{H}}$  in (162.1); the resulting function gives the same type of critical limit.*

*Proof.* The exact skeleton decomposition remains valid with  $A_{\mathcal{H}}(y) = \sum c_k^{\mathcal{H}} y^k$ . Write it as  $cy^2/(1-y)$  plus a polynomial (allowing the singleton coefficient to differ). The Poisson formula then has finitely many fixed- $k$  corrections; Theorem 158.1 bounds each one. The proof of Theorem 160.1 applies with a finite, possibly larger polynomial in  $p\sqrt{n}$  multiplying  $e^{-nI(p)}$ .

At fixed density the moving pole is unchanged, and the amplitude is (163.2) with  $c_*$  replaced by  $c$ . Its constant multiplier cancels in the conditional generating function. In the critical window, the same finite-correction argument or the nonnegative kernel gives the stated replacement of  $\Psi$ . Possible changes to the singleton term vanish after normalization by  $nq^n$ .  $\square$

The geometric results do *not* follow merely from eventual coefficient constancy. The diameter theorem used the concrete path-like shape of the three infinite ADE families; the deterministic sandwich used the stronger fact that every allowed skeleton is within two vertices of a path. These hypotheses must not be dropped when transferring the probabilistic algebra to another skeleton class.

## 167 Computational verification and reproducibility

### 167.1 Exact checks and their scope

[Merge note: source 12's programs and records are shipped with the prefix 12-random-trees-: code/verify.py and code/extra\_checks.py are code/12-random-trees-verify.py and code/12-random-trees-extra\_checks.py; every file of its data/ directory keeps its name behind the same prefix in data/; and its requirements.txt and build.sh are data/12-random-trees-requirements.txt and code/12-random-trees-build.sh. The text below keeps the delivered names.] The accompanying program code/verify.py independently generates labelled trees by Prüfer codes, enumerates all markings, prunes unmarked leaves to obtain their hulls, rejects unmarked hull branches, and tests the explicit ADE shape list. Its resulting histogram of  $(K, H)$  is compared to (156.2) using exact integers.

The completed exhaustive run covers all 18,249 labelled trees of orders one through seven and all 2,238,526 marked-tree pairs. Every one of the 70 nonzero joint bins agrees. It also checks 306 high-precision probability and conditional-moment identities against exact rational histograms at orders two through eighteen and three rational marking probabilities. These tests exercise enumeration, not a numerical stable-polynomial oracle.

The independent code/extra\_checks.py performs 170 exact rational checks of (159.2), at orders two through thirty-five and five marking probabilities. It enumerates all 435 nonisomorphic trees of orders two through eleven, compares the shape criterion with an exact positive-semidefiniteness test for  $2I - A$ , and sums inverse automorphism orders to recover Table 21 and the 9/8 tail at orders ten and eleven. The spectral test uses exact characteristic-polynomial coefficients, not floating-point eigenvalues. Since the matrix is real symmetric, all coefficients of  $\det(tI + 2I - A)$  are nonnegative exactly when it has no positive root, equivalently when  $2I - A \geq 0$ . Ten symbolic derivatives check the Gaussian covariance and the constant-order cumulant corrections.

| $n$ | Labelled trees | Marked-tree pairs | Stable pairs by $K = 0, 1, \dots, n$ |
|-----|----------------|-------------------|--------------------------------------|
| 1   | 1              | 2                 | 1, 1                                 |
| 2   | 1              | 4                 | 1, 2, 1                              |
| 3   | 3              | 24                | 3, 9, 9, 3                           |
| 4   | 16             | 256               | 16, 64, 96, 60, 16                   |
| 5   | 125            | 4000              | 125, 625, 1250, 1110, 560, 125       |

Table 22: Small exact checks. The machine-readable verification file also contains the larger orders six and seven and all joint-bin comparisons.

None of these finite tests proves the imported all-size equivalence between polynomial stability and the ADE criterion. They also do not replace the analytic arguments for limits. Their role is to detect normalization mistakes, double counting, wrong exceptional coefficients, and algebraic errors in identities derived in the manuscript.

### 167.2 Numerical asymptotics

Table 23 compares the exact finite kernel with the leading equivalent in (160.1). At large  $n$  the ratio is visibly governed by the first Stirling correction. The larger error for  $p = 0.2, n = 100$  is not a failure of the equivalent: that instance remains close enough to the crossover for finite corrections to matter.

| $n$  | $p$ | $\pi_n(p)$                | Ratio to (160.1) | $\mathbb{E}[K \mid E]$ | $\mathbb{E}[H \mid E]$ |
|------|-----|---------------------------|------------------|------------------------|------------------------|
| 100  | .2  | $6.14309 \cdot 10^{-6}$   | 1.32112609       | 7.024910               | 25.433451              |
| 500  | .2  | $4.71640 \cdot 10^{-39}$  | 1.00531355       | 22.327837              | 103.788321             |
| 2000 | .2  | $7.72917 \cdot 10^{-168}$ | 1.00004167       | 82.400000              | 404.000000             |
| 100  | .5  | $6.80496 \cdot 10^{-18}$  | 1.00091538       | 26.498531              | 50.998162              |
| 500  | .5  | $5.26053 \cdot 10^{-103}$ | 1.00016668       | 126.500000             | 251.000000             |
| 2000 | .8  | $3.47484 \cdot 10^{-687}$ | 1.00004167       | 1280.600000            | 1600.250000            |

Table 23: Finite-kernel evaluations, not Monte Carlo estimates. Expectations are conditional on stability. Full precision and additional rows are in the CSV files.

| $\lambda$ | $\Psi(\lambda)$ | $\pi_n(p)/(nq^n\Psi(\lambda))$ | $\mathbb{E}[K \mid E]$ |
|-----------|-----------------|--------------------------------|------------------------|
| 0.5       | 0.2774707592    | 1.01426855                     | 2.936290               |
| 1         | 3.5933595528    | 0.99547405                     | 4.580012               |
| 2         | 222.66178789    | 1.02695628                     | 7.407558               |
| 3         | 8469.3489971    | 1.10010587                     | 11.175896              |

Table 24: Critical-window check at  $n = 10,000$  and  $p = \lambda/\sqrt{n}$ . Convergence is visibly slower at  $\lambda = 3$ ; no uniform finite- $n$  error bound in  $\lambda$  is asserted by Theorem 162.2.

### 167.3 How to reproduce the package

The main verifier needs Python 3.10 or later and `mpmath` for high-precision evaluations. The extra verifier also uses `networkx` and `sympy`. From the package directory run

```
python code/verify.py --max-n 7 --tables
python code/extra_checks.py
```

The first script’s work grows rapidly with the exhaustive order; its finite scope should not be increased inadvertently. Both scripts write machine-readable records in `data/`. The shipped scripts and results were actually run for the bounds reported above. Rebuilding the PDF uses `build.sh`, which invokes `pdflatex` twice. No external figures or bibliography database are required. [Merge note: in this report the two programs cannot be run in place. `code/12-random-trees-extra_checks.py` imports the main verifier by its delivered module name `verify`, and both programs write unprefixed files (`verification.json`, `fixed_density.csv`, `critical_window.csv`, `extra_checks.json`) into the `data/` directory beside `code/`; no shipped file has those names, but they would be added there unprefixed. The main verifier’s default is `--max-n 6`; the recorded run used `--max-n 7`, which must be passed. The report’s README gives commands that restore the delivered layout in a scratch copy. `code/12-random-trees-build.sh` builds source 12’s own `article.tex`, which is not shipped; this report is built from its own `article.tex` instead.]

## 168 Further research questions and proposed approaches

The source question is now answered for the uniform labelled host and independent common-parameter marking law, with precise scaling limits. The following questions concern genuinely additional conclusions rather than restatements of the proved results. [Merge note: “the source question” is Part VIII’s Research question 63. Source 12’s ten questions continue the report’s numbering as Research questions 74–83; source 12 did not label them, and the labels are new. Research question 81 restates Part VIII’s Research question 59 (itself Part VI’s Research question 47) and adds a random-tree layer; Research question 83 overlaps Part VIII’s Research

question 65. Both are numbered, with pointers, because their formulations are broader. The other eight are new.]

**Research question 74** (The polynomial factor in the quenched estimate). Determine an asymptotic equivalent for  $\pi_p(T_n)$  conditional on a Cayley host, beyond the diameter-based estimate (165.2). Which combination of near-diameter path multiplicities and allowed one- or two-mark exceptions supplies the prefactor?

The exact endpoint formulas in the source (Section 124) and Lemma 165.1 suggest grouping stable markings by a canonical near-longest path. A naive union bound over all paths overcounts heavily; a canonical decomposition is the missing step.

**Research question 75** (Quenched fluctuations from continuum diameter). Use (165.6) with a precisely normalized continuum-random-tree diameter theorem to identify the limit law of  $(\log \pi_p(T_n) - n \log q)/\sqrt{n}$ .

The error in our deterministic estimate is negligible on this scale. The substantive task is an accurate comparison of normalization conventions in the continuum-tree literature [32], followed by any desired stronger coupling or tail refinement. No unexplained factor of two should be inserted into such a transfer.

**Research question 76** (Uniform conditional limits when  $p \downarrow 0$ ). Extend the fixed-density conditional Gaussian laws to sequences  $p \rightarrow 0$  with  $p\sqrt{n} \rightarrow \infty$ . Determine the correct centering, scaling, and uniform remainder across the overlap with the critical law.

The moving pole approaches the rooted-tree branch point, so the fixed complex neighbourhood in Lemma 163.1 collapses. The exact Poisson identity already controls the probability; derivatives of a suitable uniformly deformed identity are a concrete route to the conditional laws.

**Research question 77** (Critical corrections and local limits). Compute the first  $n^{-1/2}$  correction to (162.5) for a specified expansion of  $p_n$ , and prove a local limit theorem for the joint discrete pair  $(K_n, H_n)$  in the window.

Expand the falling factorial in (156.5) or use an Edgeworth expansion of the Poisson term together with the finite fixed- $k$  corrections. The dependence on the second term of  $p_n$  must be retained; Theorem 162.2 deliberately avoids hiding it in the normalization.

**Research question 78** (Fringe structure and extreme heights under stability). Identify the empirical distribution of fringe components in the fixed-density conditioned model, and sharpen the  $O_{\mathbb{P}}(\log n)$  bounds for the tallest fringe and longest subdivision chain.

The exact conditional forest law and uniform compositions reduce these to classical-looking occupancy and branching calculations, but their conditioning and dependence must be tracked. This would also refine the difference  $\text{diam}(T_n) - H_n$  beyond our logarithmic bound.

**Research question 79** (Other random-tree laws). Determine counterparts for simply generated trees, conditioned Galton–Watson trees, uniform unlabelled trees, and random trees with a prescribed degree sequence.

The structural classification still applies to each host. The attachment count and the relative weights of skeletons change, however; the factor  $9/8$  and the exponent  $-p$  should not be presumed universal across these models. A first tractable case is a degree-weighted labelled tree family with an analytic rooted-tree equation.

**Research question 80** (Inhomogeneous and dependent markings). Replace the common Bernoulli parameter by vertex-dependent activities or by marks correlated through a tree-indexed process. Determine which parts of the diameter sandwich and critical scaling survive.

The event that all marks lie on a path remains a useful lower bound, and the two-outside-marks necessity remains deterministic. Their probabilities no longer reduce to a function of path length alone; weighted path optimization becomes part of the problem. [Merge note: for independent vertex-dependent probabilities  $p_v$  on one given host, Part VIII's formula (124.7) already evaluates the stability probability exactly; the question asks for the asymptotic laws.]

**Research question 81** (Several apices). For two or more independently marked apex elements over the same tree incidence graph, obtain a structural stability criterion and then a random-tree probability theory.

This is not covered by the single-apex covariance or skeleton theorem. It requires a new stability analysis before enumerative substitutions are justified. Small exact Rayleigh certificates and covariance-rank constraints are plausible initial probes. [Merge note: the structural half is Part VIII's Research question 59, which is Part VI's Research question 47 with a request for a terminal-geometry criterion; source 12 adds the random-tree probability theory.]

**Research question 82** (More general skeleton singularities). Classify attainable labelled skeleton coefficient sequences and derive probability laws when their generating functions have a higher-order pole or a non-polar dominant singularity.

The substitution in Theorem 157.1 remains valid for a specified class. Its singularity controls both the prefactor and the conditional parameter laws. Theorem 166.1 settles only the eventual constant-coefficient case; geometric consequences need separate shape hypotheses.

**Research question 83** (Exact conditioned sampling and formal verification). Build an exact sampler for stable marked Cayley trees, and a proof-assistant verification of the decomposition and its connection to polynomial stability.

The joint counts choose  $(k, \ell)$ ; then choose an allowed labelled skeleton, a uniform weak composition, subdivision labels, and a rooted forest. Correctness follows from the bijection once the component samplers are verified. A formal development should separate the imported analytic stability theorem, the finite combinatorial bijection, and the later asymptotic analysis; accepting a tested shape classifier alone would not prove the polynomial statement. [Merge note: the formal half overlaps Part VIII's Research question 65 (a verified compiler from trees to stability certificates); the exact sampler is new.]

## 169 Conclusion

The marked-tree classification turns a multivariate zero-free condition into a finite list of terminal shapes. Labelled decomposition then turns that list into a rational skeleton series. Its eventual coefficient  $9/8$  produces both an exact Poisson-tail identity and a uniform rare-event equivalent. The critical window retains all finite skeleton sizes through one explicit entire function, while fixed-density conditioning selects three long, path-like families with Gaussian fluctuations.

The geometric conclusion is sharper than a probability estimate: stability at positive density is realized by host trees with diameter asymptotic to  $pn$ . A typical unconditioned host has sublinear diameter and a different exponential stability rate. The deterministic polynomial-factor sandwich identifies diameter as the quantity that separates the two perspectives. Together

these results answer the pinned random-tree question for a concrete model and expose several precise next problems, without confusing an ordinary mathematical proof with a formal proof certificate or a priority claim.

## 170 Proof dependencies and claim audit

(Source 12's Appendix A. "Pinned ProveIt Part VIII, theorem `mts:thm:main`" is Theorem 118.2, and the tree proof of the ADE list is Part VI's Theorem 88.1, which Part VIII cites as Proposition 122.4.)

| Claim or ingredient                                             | Dependency and status                                                                                                                   |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Stability iff branch-complete ADE skeleton                      | Imported from pinned ProveIt Part VIII, theorem <code>mts:thm:main</code> . No new formal verification is claimed.                      |
| ADE tree list                                                   | Imported spectral classification; the source supplies a tree proof. Classical spectral context is credited separately.                  |
| Automorphism factors and $A(y)$                                 | Derived here by counting the explicitly listed shapes. Independently checked through eleven vertices.                                   |
| Rooted-forest formula and long-path count                       | Classical matrix-tree/Prüfer tools, with derivations included. Not claimed as new random-tree enumeration theorems.                     |
| Exact $(K, H)$ count, generating function, Poisson identity     | Proved here from a reversible skeleton-subdivision-forest decomposition. Exact finite verification accompanies the proofs.              |
| Fixed- $k$ , sparse, critical, and uniform supercritical limits | Proved here from the exact identities, Gaussian Riemann sums, elementary Poisson bounds, and Stirling's formula.                        |
| Conditional cumulants and Gaussian law                          | Proved here by a uniformly isolated moving pole and explicit characteristic-function expansion. Fixed $p$ only.                         |
| Conditioned diameter and quenched sandwich                      | Proved here using the ADE path structure, a rooted-forest height count, and a uniform-composition bound.                                |
| Finite computations                                             | Executed Python checks. Neither exhaustive at all sizes nor proof-assistant certificates.                                               |
| Publication novelty or independent verification                 | Not established. The claim boundary is relative to the inspected repository question and the precise statements proved in this article. |

## 171 A compact map of parameter regimes

(Source 12's Appendix B.) For  $q_n = 1 - p_n$ , the following summary keeps the normalizations distinct.

| Regime                                               | Proved conclusion                                                                                             |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| $np_n \rightarrow \mu < \infty$                      | $\pi_n \rightarrow e^{-\mu}(1 + \mu + \mu^2/2)$ ; conditional marks are Poisson truncated to $0, 1, 2$ .      |
| $np_n \rightarrow \infty, p_n\sqrt{n} \rightarrow 0$ | $\pi_n \sim \binom{n}{2}p_n^2q_n^{n-2}$ ; conditional mark count tends to two.                                |
| $p_n\sqrt{n} \rightarrow \lambda \in (0, \infty)$    | $\pi_n \sim nq_n^n\Psi(\lambda)$ ; discrete mark-count law $w_k(\lambda)$ and a chi hull mixture.             |
| $p_n\sqrt{n} \rightarrow \infty$                     | $\pi_n \sim \frac{9}{8}\sqrt{2\pi}p_n^3n^{5/2}e^{-np_n}$ , uniformly along every such sequence.               |
| Fixed $p \in (0, 1)$ , conditioned on stability      | $K_n/n \rightarrow p^2, H_n/n \rightarrow p, \text{diam}(T_n)/n \rightarrow p$ ; explicit joint Gaussian law. |
| Fixed $p \in (0, 1)$ , typical unconditioned host    | $n^{-1} \log \pi_p(T_n) \rightarrow \log(1 - p)$ in probability.                                              |
| $p_n = 1, n \geq 10$                                 | Exact identity $\pi_n(1) = \frac{9}{8} n!/n^{n-2}$ .                                                          |

The critical and supercritical probability results do not by themselves imply uniform Gaussian or diameter limits when  $p_n$  varies. Such extensions are explicitly left among the further questions.



## A A hypertree proof of the known root identity

This appendix reconstructs Proposition 2.2 without using the support-enumerator theorem of [2]. It is included for conceptual clarity and as an independent route to the same normalization. Its one substantial imported result is the Kálmán–Postnikov theorem [3].

### A.1 Hypertrees and internal inactivity

For a connected bipartite graph with left shore  $L$ , a left hypertree is a vector  $a \in \mathbb{Z}_{\geq 0}^L$  realized by a spanning tree  $T$  through  $a_i = \deg_T(i_L) - 1$ . Distinct trees realizing the same vector contribute only one hypertree. Order the left shore. A coordinate  $i$  is internally inactive when there is a smaller coordinate  $j$  such that replacing  $a$  by  $a - \mathbf{e}_i + \mathbf{e}_j$  gives another hypertree. The interior polynomial is the sum of  $t^{\text{number of internally inactive coordinates}}$  over hypertrees.

The Kálmán–Postnikov theorem identifies this interior polynomial with the Ehrhart numerator of the bipartite root polytope. Their paper uses a reversed convention for the triangulation  $h$ -polynomial in its Theorem 1.1; the Ehrhart-basis formulation immediately following that theorem, equation (1.2), identifies the coefficients with the usual  $h^*$ -numerator used here. Thus there is no additional reversal of  $t$  in the statement we apply.

**Lemma A.1.** *The left hypertrees of  $\widehat{G}_\tau$  are exactly*

$$a = (n - x(E), x_1, \dots, x_n), \quad x \in P_\tau. \quad (\text{A.1})$$

*Proof.* A spanning tree of  $\widehat{G}_\tau$  has  $2n + 1$  edges, so the sum of its left degree-minus-one vector is  $(2n + 1) - (n + 1) = n$ . For a nonempty  $A \subseteq [n]$ , the tree edges incident with  $A_L$  form a forest on  $A_L \cup N(A)$ . Their number is  $a(A) + |A|$ , and a forest on this nonempty vertex set has at most  $|A| + |N(A)| - 1$  edges. Hence

$$a(A) \leq |N(A)| - 1 = |\downarrow A|.$$

Lemma 2.1 now shows that  $x = (a_1, \dots, a_n)$  belongs to  $P_\tau$ , and the sum condition determines  $a_0$ .

Conversely, let  $x \in P_\tau$ . Replace the left element  $i$  by  $x_i$  distinct clones, each adjacent to the right set  $\downarrow i \subseteq [n]$ . A set of clones whose labels lie in  $A$  has at most  $x(A) \leq |\downarrow A|$  members. Thus Hall's matching condition holds, and all clones can be matched to distinct right vertices. This is also a constructive statement: if an augmenting-path matching leaves a clone unmatched, the alternating paths reachable from it produce a set of clones with fewer neighbors, contradicting Hall's condition.

Build a tree by first taking all edges  $i_L 0_R$  for  $0 \leq i \leq n$ . Each right vertex  $j_R$ ,  $j > 0$ , is then attached to the left label of its matched clone, or to  $0_L$  if it is unmatched. The first edges form a star; every subsequent edge attaches a new leaf. Hence the result is a spanning tree. Its left degrees are  $x_i + 1$  for  $i > 0$  and  $n - x(E) + 1$  for  $i = 0$ , exactly as required.  $\square$

**Lemma A.2.** *Order the left vertices with 0 first. For the hypertree in (A.1), internal inactivity is  $|\text{supp}(x)|$ .*

*Proof.* The first coordinate is never internally inactive. A zero coordinate cannot transfer a unit, so it is also not internally inactive. If  $x_i > 0$ , then  $x - \mathbf{e}_i \in P_\tau$  because all the defining inequalities are upper bounds and the coordinates remain nonnegative. By Lemma A.1, the vector obtained by transferring one unit from coordinate  $i$  to coordinate 0 is another hypertree. Thus every positive nonzero-index coordinate is internally inactive, and there are no others.  $\square$

Summing over (A.1) gives an interior polynomial equal to  $h_\tau(t)$ . The Kálmán–Postnikov theorem identifies it with  $h^*(B_\tau, t)$ , proving Proposition 2.2 by this second route. Notice that the argument counts hypertree *vectors*, not spanning trees; counting all realizing trees would introduce incorrect multiplicities.

## B Proof dependencies and normalization audit

| Claim or input              | Where it comes from                                                                                                                                       |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| $h^*(B_\tau) = h_\tau$      | Known: [2, Theorem 1.2], specialized using Lemma 2.1.<br>Alternatively Appendix A plus [3].                                                               |
| Ideal Hall presentation     | Proposition 3.2; transitivity is used explicitly.                                                                                                         |
| Integer fiber factorization | Theorem 4.1; all coordinates and inverse parameters are displayed.                                                                                        |
| $h^*(C_\tau) = h_\tau$      | Corollary 4.2; a consequence of the fiber factorization and the known root identity.                                                                      |
| Unimodularity               | Leaf elimination in Lemma 5.1, followed by Proposition 5.3.                                                                                               |
| Polytopal pulling boundary  | The established pulling lemma [4, Lemma 2.1].                                                                                                             |
| Palindromicity, unimodality | Dehn–Sommerville and the $g$ -theorem applied to the constructed simplicial polytope; an independent reciprocity proof of palindromicity is in Section 6. |
| Alternative polytopal route | The explicit special simplex and [4, Theorem 3.5].                                                                                                        |

There are several easy places to make an otherwise plausible argument wrong. The support polynomial of  $Q_\tau$  is not asserted to be  $h^*(Q_\tau)$ . It is the numerator of the different polytopes  $B_\tau$  and  $C_\tau$ . The larger denominator has exponent  $2n + 1$  and the smaller one exponent  $n + 1$ . Both auxiliary vertices, and the edge between them, are required. The edge orientation convention is  $j \leq_\tau i \iff i_L j_R$  is present. The integer-fiber parameters consist of  $n + 1$  nonnegative entries; the exponent  $n$  in (4.9) results only after dividing by the additional slack factor in the Ehrhart series of  $C_\tau$ .

No argument substitutes a rational equivalence for an integral one. The fiber base is integral because  $a_\tau(y)$  is an integer for integral  $y$ . The determinant statements refer to the full relevant lattices. Finally, regularity or shellability alone is not used to claim that a sphere is polytopal: the specific pulling-boundary theorem supplies that step.

## C Reproducing the accompanying computations

The archive contains `article.tex`, `article.pdf`, a build script, `code/verify.py`, the full CSV enumeration, a JSON result report, and status notes. From the extracted directory, run

```
python3 code/verify.py --output data
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The verifier requires Python 3.10 or later and no third-party Python packages. Do not use Python’s `-O` option, since assertions are part of the checks. The LaTeX document uses standard TeX Live packages, including `newpxtext`, `newpxmath`, `tcolorbox`, and `hyperref`.

The CSV uses zero-based bit positions: the row value 5 represents the set  $\{0, 2\}$ . Row  $i$  records the principal ideal of element  $i$ , not its filter. A polynomial is stored as coefficients from

degree zero upward. The JSON report also records explicit edge sets for matching certificates and complete Ehrhart counts for the independently tested instances.

[Added 28 September 2026, batch 39: this appendix describes Part I’s package; Part II’s programs and how to rerun them are in Section 22.4.]

The calculations in the supplied report completed without failed assertions. Their runtime is recorded separately as an observation from this environment, not as a performance guarantee. No claim of formal verification is made.

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